

USAMO

§1 USAMO 1972

1. (*USAMO 1972*) The symbols $(a, b, \dots, g)$ and $[a, b, \dots, g]$ denote the greatest common divisor and least common multiple, respectively, of the positive integers $a, b, \dots, g$. For example, $(3, 6, 18) = 3$ and $[6, 15] = 30$. Prove that

$$\frac{[a, b, c]^2}{[a, b][b, c][c, a]} = \frac{(a, b, c)^2}{(a, b)(b, c)(c, a)}.$$

Solution 1

Consider an arbitrary prime p . Let p^α , p^β , and p^γ be the greatest powers of p that divide a , b , and c . WLOG let $\alpha \leq \beta \leq \gamma$.

Examining each factor in the equation, we see that the largest power of p that divides the left hand side is $2\gamma - (\beta + \gamma + \gamma) = -\beta$, and the largest power of p that divides the right hand side is $2\alpha - (\alpha + \beta + \alpha) = -\beta$. Since every prime has the same power in both expressions, the expressions are equal. ■

Solution 2

Let $p = (a, b, c)$, $pq = (a, b)$, $pr = (b, c)$, and $ps = (c, a)$. Then it follows that q, r, s are pairwise coprime and $a = pqsa'$, $b = pqr b'$, and $c = prs c'$, with a', b', c' pairwise coprime as well. Then, we wish to show

$$\frac{(pqrsa'b'c')^2}{(pqrsa'b')(pqr b'c')(prsc'a')} = \frac{p^2}{(pq)(pr)(ps)},$$

which can be checked fairly easily.

Solution 3 (very long, but detailed)

Note: This solution is more of a "bashy" solution, and is easier to think of than the first 2 solutions.

Dividing both sides of the equation by $(a, b, c)^2$, and then multiplying both sides by $[a, b][b, c][c, a]$ gives us

$$\left(\frac{[a, b, c]}{(a, b, c)} \right)^2 = \frac{[a, b][b, c][c, a]}{(a, b)(b, c)(c, a)}.$$

Now, we look at the prime factorisations of a, b, c . Let

$$a = 2^{x_1} \cdot 3^{x_2} \cdot 5^{x_3} \dots,$$

$$b = 2^{y_1} \cdot 3^{y_2} \cdot 5^{y_3} \dots,$$

$$c = 2^{z_1} \cdot 3^{z_2} \cdot 5^{z_3} \dots,$$

where all of the x_1, y_1, z_1 are nonnegative integers (they could be 0). Thus, we have

$$[a, b, c] = 2^{\max(x_1, y_1, z_1)} \cdot 3^{\max(x_2, y_2, z_2)} \cdot 5^{\max(x_3, y_3, z_3)} \dots$$

and

$$(a, b, c) = 2^{\min(x_1, y_1, z_1)} \cdot 3^{\min(x_2, y_2, z_2)} \cdot 5^{\min(x_3, y_3, z_3)} \dots$$

Now, we see that $\left(\frac{[a, b, c]}{(a, b, c)}\right) = 2^{\max(x_1, y_1, z_1) - \min(x_1, y_1, z_1)} \cdot 3^{\max(x_2, y_2, z_2) - \min(x_2, y_2, z_2)} \cdot 5^{\max(x_3, y_3, z_3) - \min(x_3, y_3, z_3)} \dots$. Squaring the LHS will just double all the exponents on the RHS.

Also, we have $[a, b] = 2^{\max(x_1, y_1)} \cdot 3^{\max(x_2, y_2)} \cdot 5^{\max(x_3, y_3)} \dots$. We can also build similar equations for $[b, c]$, $[c, a]$, (a, b) , (b, c) , (c, a) , using 2 of x, y, z and either the minimum of the exponents or the maximum. We see that when we multiply $[a, b]$, $[b, c]$, and $[c, a]$ together, the exponents of the prime factorization will be $\max(x_i, y_i) + \max(y_i, z_i) + \max(z_i, x_i)$, where i is chosen for the i 'th prime. When we multiply (a, b) , (b, c) , and (c, a) together, the exponents of the prime factorization will be $\min(x_i, y_i) + \min(y_i, z_i) + \min(z_i, x_i)$, where i is chosen for the i 'th prime.

Thus, when we divide the former by the latter, we have $\max(x_i, y_i) + \max(y_i, z_i) + \max(x_i, z_i) - (\min(x_i, y_i) + \min(y_i, z_i) + \min(x_i, z_i))$. We wish to prove that this is equal to the exponents we got from $([a, b, c]/(a, b, c))^2$.

In other words, this problem is now down to proving that

$$2(\max(x_i, y_i, z_i) - \min(x_i, y_i, z_i)) = \max(x_i, y_i) + \max(y_i, z_i) + \max(x_i, z_i) - (\min(x_i, y_i) + \min(y_i, z_i) + \min(x_i, z_i))$$

for any nonnegative integers x_i, y_i, z_i . WLOG, let $x_i \leq y_i \leq z_i$. Thus, computing maximums and minimums, this equation turns into

$$2(z_i - x_i) = y_i + z_i + z_i - x_i - y_i - x_i.$$

Finally, canceling out the y_i 's and collecting like terms, we see the RHS is $2z_i - 2x_i$, which is equal to the LHS. Thus, this equation is true, and we are done. $\square$

Solution 4

Let

$$a = dxzA$$

$$b = dxyB$$

$$c = dyzC$$

where $\gcd(A, B, C) = \gcd(x, y, z) = 1$. (Existence proof missing.) Then, $\text{lcm}(a, b, c)^2 = (dxyzABC)^2$ and $\text{lcm}(a, b) = dxyzAB$. Using this cyclically, we have

$$\text{LHS} = \frac{d^2 x^2 y^2 z^2 A^2 B^2 C^2}{(dxyzAB)(dxyzBC)(dxyzCA)} = \frac{1}{dxyz}.$$

Now, note that $\gcd(a, b, c)^2 = d^2$ and $\gcd(a, b) = dx$. Using this cyclically, we have

$$\text{RHS} = \frac{d^2}{(dx)(dy)(dz)} = \frac{1}{dxyz}.$$

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Note If one thinks of the problem from a set theory vantage point, it is quite intuitive to recognize from PIE:

$$\text{lcm}(a, b, c) = \frac{a \cdot b \cdot c \cdot \gcd(a, b, c)}{\gcd(a, b) \cdot \gcd(b, c) \cdot \gcd(c, a)}$$

where $\text{lcm}(a, b, c)$ can be thought of as the union, $a \cdot b \cdot c$ can be thought of as the raw addition of sets, and the $\gcd$ function can be thought of as intersections.

2. (USAMO 1972) A given tetrahedron $ABCD$ is isosceles, that is, $AB = CD$, $AC = BD$, $AD = BC$. Show that the faces of the tetrahedron are acute-angled triangles.

Solution 1

Suppose $\triangle ABD$ is fixed. By the equality conditions, it follows that the maximal possible value of BC occurs when the four vertices are coplanar, with C on the opposite side of $\overline{AD}$ as B . In this case, the tetrahedron is not actually a tetrahedron, so this maximum isn't actually attainable.

For the sake of contradiction, suppose $\angle ABD$ is non-acute. Then, $(AD)^2 \geq (AB)^2 + (BD)^2$. In our optimal case noted above, $ACDB$ is a parallelogram, so

$$\begin{aligned} 2(BD)^2 + 2(AB)^2 &= (AD)^2 + (CB)^2 \\ &= 2(AD)^2 \\ &\geq 2(BD)^2 + 2(AB)^2. \end{aligned}$$

However, as stated, equality cannot be attained, so we get our desired contradiction.

Solution 2

It's not hard to see that the four faces are congruent from SSS Congruence. Without loss of generality, assume that $AB \leq BC \leq CA$. Now assume, for the sake of contradiction, that each face is non-acute; that is, right or obtuse. Consider triangles $\triangle ABC$ and $\triangle ABD$. They share side AB . Let k and l be the planes passing through A and B , respectively, that are perpendicular to side AB . We have that triangles ABC and ABD are non-acute, so C and D are not strictly between planes k and l . Therefore the length of CD is at least the distance between the planes, which is AB . However, if $CD = AB$, then the four points A, B, C , and D are coplanar, and the volume of $ABCD$ would be zero. Therefore $CD > AB$. However, we were given that $CD = AB$ in the problem, which leads to a contradiction. Therefore the faces of the tetrahedron must all be acute.

Solution 3

Let $\vec{a} = \overrightarrow{DA}$, $\vec{b} = \overrightarrow{DB}$, and $\vec{c} = \overrightarrow{DC}$. The conditions given translate to

$$\begin{aligned} \vec{a} \cdot \vec{a} &= \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{c} - 2(\vec{b} \cdot \vec{c}) \\ \vec{b} \cdot \vec{b} &= \vec{c} \cdot \vec{c} + \vec{a} \cdot \vec{a} - 2(\vec{c} \cdot \vec{a}) \\ \vec{c} \cdot \vec{c} &= \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} - 2(\vec{a} \cdot \vec{b}) \end{aligned}$$

We wish to show that $\vec{a} \cdot \vec{b}$, $\vec{b} \cdot \vec{c}$, and $\vec{c} \cdot \vec{a}$ are all positive. WLOG, $\vec{a} \cdot \vec{a} \geq \vec{b} \cdot \vec{b}$, $\vec{c} \cdot \vec{c} > 0$, so it immediately follows that $\vec{a} \cdot \vec{b}$ and $\vec{a} \cdot \vec{c}$ are positive. Adding all three equations,

$$\vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{c} = 2(\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c})$$

In addition,

$$(\vec{a} - \vec{b} - \vec{c}) \cdot (\vec{a} - \vec{b} - \vec{c}) \geq 0$$

$$\begin{aligned}
\vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{c} &\geq 2(\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} - \vec{b} \cdot \vec{c}) \\
2(\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}) &\geq 2(\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} - \vec{b} \cdot \vec{c}) \\
\vec{b} \cdot \vec{c} &\geq 0
\end{aligned}$$

Equality could only occur if $\vec{a} = \vec{b} + \vec{c}$, which requires the vectors to be coplanar and the original tetrahedron to be degenerate.

Solution 4

Suppose for the sake of contradiction that $\angle BAC$ is not acute. Since all three sides of triangles BAC and CDB are congruent, those two triangles are congruent, meaning $\angle BDC = \angle BAC > 90^\circ$. Construct a sphere with diameter BC . Since angles BAC and BDC are both not acute, A and D both lie on or inside the sphere. We seek to make $AD = BC$ to satisfy the conditions of the problem. This can only occur when AD is a diameter of the sphere, since both points lie on or inside the sphere. However, for AD to be a diameter, all four points must be coplanar, as all diameters intersect at the center of the sphere. This would make tetrahedron $ABCD$ degenerate, creating a contradiction. Thus, all angles on a face of an isosceles tetrahedron are acute.

Solution 5

Proof by contradiction: Assume at least one of the tetrahedron's faces are obtuse. WLOG, assume $\angle BAC$ is an obtuse angle. Using SSS congruence to prove that all four faces of the tetrahedron are congruent also shows that the angles surrounding point A are congruent to angles in triangle ABC ; namely, $\angle BAD$ is congruent to $\angle ABC$, and $\angle CAD$ is congruent to $\angle ACB$. Since the internal angles of triangle ABC must add to 180 degrees, so do the angles surrounding point A . Now lay triangle ABC on a flat surface. A diagram would make it clear that from the perspective of an aerial view, the "apparent" measures of $\angle BAD$ and $\angle CAD$ (which are most likely distorted visually, assuming these angles stick up vertically from the flat surface on which triangle ABC lies) can never exceed the true measures of those angles (equality happens when these angles also lie flat on top of triangle ABC). This means that these two angles can never join to form side AD (because $\angle BAC$ is more than the sum of $\angle BAD$ and $\angle CAD$ - a direct consequence of the facts that $\angle BAC$ is obtuse and all three angles add up to 180 degrees), so the tetrahedron with obtuse triangle faces is impossible.

Solution 6

Lemma: given triangle ABC and the midpoint of BC , which we will call M , we can say that if $AM > \frac{BC}{2}$, then $\angle A < 90^\circ$. Proof: Since M is the midpoint of BC , $BM = MC = \frac{BC}{2}$. Since it is given that $AM > \frac{BC}{2}$, we can substitute $\frac{BC}{2}$ to get two inequalities:

$$AM > CM, \quad AM > BM.$$

The above inequalities imply that $\angle C > \angle CAM$ and $\angle B > \angle BAM$. Adding these inequalities and simplifying the RHS, we have that $\angle C + \angle B > \angle A$. Adding $\angle A$ to both sides, replacing the LHS with 180 and dividing by 2 gets us that $\angle A < 90^\circ$. This is our desired inequality, so we are done. Note that all faces of this tetrahedron are congruent, by SSS. In particular, we will use that $\triangle ABD \cong \triangle BAC$. WLOG, assume that $\angle ADB$ is the largest angle in triangle ABD . Because $\triangle ABD \cong \triangle BAC$, the median from D to AB is equal length to the median from C to AB . These points meet at E , the midpoint of AB . By the triangle inequality,

$$DE + CE > CD.$$

By substituting CD with AB (this is a given in the problem) and CE with DE , and then dividing by 2, we get that

$$DE > \frac{AB}{2}.$$

By the lemma we showed at the start, this implies that $\angle ADB < 90$, and since we said that $\angle ADB$ was the largest angle, triangle ADB must be acute. Since all of the faces of this tetrahedron are congruent, then, all of the faces must be acute.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

3. (*USAMO 1972*) A random number selector can only select one of the nine integers 1, 2, ..., 9, and it makes these selections with equal probability. Determine the probability that after n selections ($n > 1$), the product of the n numbers selected will be divisible by 10.

Solution 1

For the product to be divisible by 10, there must be a factor of 2 and a factor of 5 in there.

The probability that there is no 5 is $\left(\frac{8}{9}\right)^n$.

The probability that there is no 2 is $\left(\frac{5}{9}\right)^n$.

The probability that there is neither a 2 nor 5 is $\left(\frac{4}{9}\right)^n$, which is included in both previous cases.

The only possibility left is getting a 2 and a 5, making the product divisible by 10. By complementarity and principle of inclusion-exclusion, the probability of that is $1 - \left(\left(\frac{8}{9}\right)^n + \left(\frac{5}{9}\right)^n - \left(\frac{4}{9}\right)^n\right) = \boxed{1 - (8/9)^n - (5/9)^n + (4/9)^n}$.

Solution 2 (Recursion)

Define a_n as the probability that the product is divisible by 10 after selection n . Likewise, define b_n and c_n with divisibility by 2 and 5, respectively. Define d_n to be the chance of dividing neither 2 nor 5 (and thus not 10 either) similarly.

It is clear that $d_n = \left(\frac{4}{9}\right)^n$. Now we can define other recursive formulas:

We are able to reach a b state via selecting a non-5 from a b state and selecting an even number from a d state. Thus its formula is $b_n = \frac{8}{9}b_{n-1} + \frac{4}{9}d_{n-1}$.

We are able to reach a c state via selecting a non-even number from a c state and selecting a 5 from a d state. Thus its formula is $c_n = \frac{5}{9}c_{n-1} + \frac{1}{9}d_{n-1}$.

Finally, to reach an a state, we can select a 5 from a b state and select an even number from a c state. We can also reach a_n from a_{n-1} because of the fact that once the product is divisible by 10, it will always be divisible by 10 regardless of the following selections. Thus its formula is $a_n = a_{n-1} + \frac{1}{9}b_{n-1} + \frac{4}{9}c_{n-1}$.

For our formula for b_n , we can substitute to find that $b_n = \frac{8}{9}b_{n-1} + \left(\frac{4}{9}\right)^n$. Solving this recursion yields $b_n = \left(\frac{8}{9}\right)^n - \left(\frac{4}{9}\right)^n$.

For our formula for c_n , we can substitute to find that $c_n = \frac{5}{9}c_{n-1} + \frac{1}{9} \cdot \left(\frac{4}{9}\right)^{n-1}$. Solving this recursion yields $c_n = \left(\frac{5}{9}\right)^n - \left(\frac{4}{9}\right)^n$.

Finally, substituting the values into the a_n formula yields $a_n = a_{n-1} + \frac{1}{9} \left(\left(\frac{8}{9} \right)^{n-1} + 4 \cdot \left(\frac{5}{9} \right)^{n-1} - 5 \cdot \left(\frac{4}{9} \right)^{n-1} \right)$

Solving this recursion yields the final answer, $\boxed{1 - \left(\frac{8}{9} \right)^n - \left(\frac{5}{9} \right)^n + \left(\frac{4}{9} \right)^n}$.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

4. (*USAMO 1972*) Let R denote a non-negative rational number. Determine a fixed set of integers a, b, c, d, e, f , such that for *every* choice of R ,

$$\left| \frac{aR^2 + bR + c}{dR^2 + eR + f} - \sqrt[3]{2} \right| < |R - \sqrt[3]{2}|$$

Solution

Note that when R approaches $\sqrt[3]{2}$, $\frac{aR^2 + bR + c}{dR^2 + eR + f}$ must also approach $\sqrt[3]{2}$ for the given inequality to hold. Therefore

$$\lim_{R \rightarrow \sqrt[3]{2}} \frac{aR^2 + bR + c}{dR^2 + eR + f} = \sqrt[3]{2}$$

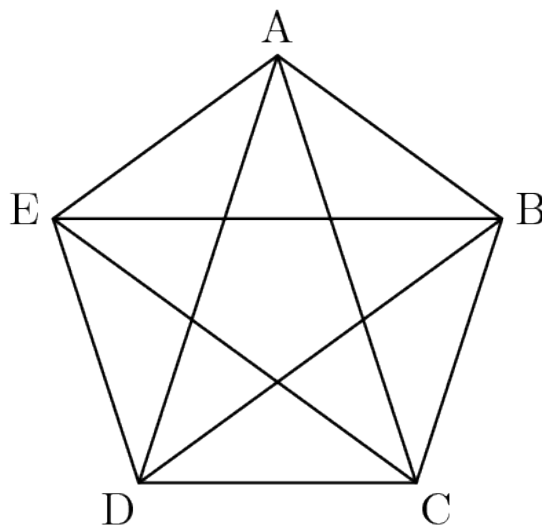
which happens if and only if

$$\frac{a\sqrt[3]{4} + b\sqrt[3]{2} + c}{d\sqrt[3]{4} + e\sqrt[3]{2} + f} = \sqrt[3]{2}$$

We cross multiply to get $a\sqrt[3]{4} + b\sqrt[3]{2} + c = 2d + e\sqrt[3]{4} + f\sqrt[3]{2}$. It's not hard to show that, since a, b, c, d, e , and f are integers, then $a = e$, $b = f$, and $c = 2d$.

Note, however, that this is a necessary but insufficient condition. For example, we must also have $a^2 < 2bc$ to ensure the function does not have any vertical asymptotes (which would violate the desired property). A simple search shows that $a = 0$, $b = 2$, and $c = 2$ works.

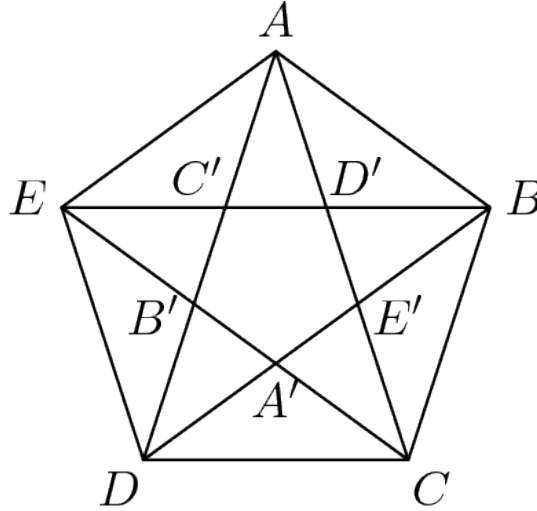
5. (*USAMO 1972*) A given convex pentagon $ABCDE$ has the property that the area of each of the five triangles ABC , BCD , CDE , DEA , and EAB is unity. Show that all pentagons with the above property have the same area, and calculate that area. Show, furthermore, that there are infinitely many non-congruent pentagons having the above area property.



Solution

Lemma: Convex pentagon $A_0A_1A_2A_3A_4$ has the property that $[A_0A_1A_2] = [A_1A_2A_3] = [A_2A_3A_4] = [A_3A_4A_0] = [A_4A_0A_1]$ if and only if $\overline{A_{n-1}A_{n+1}} \parallel \overline{A_{n-2}A_{n+2}}$ for $n = 0, 1, 2, 3, 4$ (indices taken mod 5).

Proof: For the "only if" direction, since $[A_0A_1A_2] = [A_1A_2A_3]$, A_0 and A_3 are equidistant from $\overline{A_1A_2}$, and since the pentagon is convex, $\overline{A_0A_3} \parallel \overline{A_1A_2}$. The other four pairs of parallel lines are established by a symmetrical argument, and the proof for the other direction is just this, but reversed.



Let $A'B'C'D'E'$ be the inner pentagon, labeled so that A and A' are opposite each other, and let a, b, c, d, e be the side lengths of the inner pentagon labeled opposite their corresponding vertices. From the lemma, pentagon $A'B'C'D'E'$ is similar to pentagon $ABCDE$ with $AB = m(A'B')$ and parallelogram $ABCB'$ (and cyclic) has area 2. Supposing

$$\begin{aligned} P &= [ABCDE] \\ Q &= [A'B'C'D'E'] \\ R &= \sum_{\text{cyc}} [AC'D'] \\ S &= \sum_{\text{cyc}} [ABD'], \end{aligned}$$

we have

$$\begin{aligned} \sum_{\text{cyc}} [ABCB'] &= 5Q + 3R + 2S \\ 10 &= 3Q + R + 2P. \end{aligned}$$

Since $BCDC'$ and $CDED'$ are parallelograms, $BD' = EC' = (m-1)a$. Triangle $EB'C'$ is similar to triangle ECB , so $me = BC = \left(\frac{2m-1}{m-1}\right)e$, and with the requirement that $m > 1$,

$$m = \frac{2m-1}{m-1} \Rightarrow m = \frac{3+\sqrt{5}}{2}.$$

Now, we compute that $[AC'D'] = \frac{1}{2m-1} = \sqrt{5} - 2$, and similar computation for the other four triangles gives $R = 5\sqrt{5} - 10$. From the aforementioned pentagon similarity, $Q = P/m^2 = \left(\frac{7-3\sqrt{5}}{2}\right)P$. Solving for P , we have

$$\begin{aligned} 10 &= 3 \left(\frac{7-3\sqrt{5}}{2} \right) P + (5\sqrt{5} - 10) + 2P \\ 20 - 5\sqrt{5} &= \left(\frac{25-9\sqrt{5}}{2} \right) P \\ P &= \frac{40-10\sqrt{5}}{25-9\sqrt{5}} \\ &= \frac{5+\sqrt{5}}{2}. \end{aligned}$$

To show that there are infinitely many pentagons with the given property, we start with triangle $A'B'C'$ and construct the original pentagon using the required length ratios. This triangle has three continuous degrees of freedom, and the condition that the equal areas be specifically unity means that there are two continuous degrees of freedom in which the configuration can vary, which implies the desired result.

Alternatively, one can shear a regular pentagon, a transformation which preserves areas and the property of a pair of lines being parallel.

§2 USAMO 1973

1. (USAMO 1973) Two points P and Q lie in the interior of a regular tetrahedron $ABCD$. Prove that angle $PAQ < 60^\circ$.

Solution 1

Let the side length of the regular tetrahedron be a . Link and extend AP to meet the plane containing triangle BCD at E ; link AQ and extend it to meet the same plane at F . We know that E and F are inside triangle BCD and that $\angle PAQ = \angle EAF$.

Now let's look at the plane containing triangle BCD with points E and F inside the triangle. Link and extend EF on both sides to meet the sides of the triangle BCD at I and J , I on BC and J on DC . We have $\angle EAF < \angle IAJ$.

But since E and F are interior of the tetrahedron, points I and J cannot be both at the vertices and $IJ < a$, $\angle IAJ < \angle BAD = 60$. Therefore, $\angle PAQ < 60$.

Solution with graphs posted at

<http://www.cut-the-knot.org/wiki-math/index.php?n=MathematicalOlympiads.USA1973Problem1>

Solution 2 (similar to solution 1, but full proof)

Let the side length of the regular tetrahedron $ABCD$ be s .

Extend AP and AQ to intersect the plane BCD at points P' and Q' inside $\triangle BCD$, respectively. We can observe that $\angle PAQ = \angle P'AQ'$.

Extend $P'Q'$ to intersect the sides of $\triangle BCD$ at P'' and Q'' . We can observe that $\angle P'AQ' < \angle P''AQ''$. We assume that, WLOG, P'' is on BD and Q'' is on BC .

Let the ratios $\frac{BP''}{BD} = \mu$ and $\frac{BQ''}{BC} = \lambda$. We can see that $\mu, \lambda \in (0, 1)$. It follows from Stewart's theorem that $AP''^2 = AD^2\mu + AB^2(1-\mu) - \mu(1-\mu)BD^2 = (\mu^2 - \mu + 1)s^2$
 $AQ''^2 = AC^2\lambda + AB^2(1-\lambda) - \lambda(1-\lambda)BC^2 = (\lambda^2 - \lambda + 1)s^2$

Furthermore, we have $P''C^2 = CD^2\mu + CB^2(1-\mu) - \mu(1-\mu)BD^2 = (\mu^2 - \mu + 1)s$
 Using this, we can conclude that $P''Q''^2 = P''C^2\lambda + (BD\mu)^2(1-\lambda) - \lambda(1-\lambda)BC^2$
 $= (\mu\lambda - \mu - \lambda + 2)s^2$

By cosine law, we have $\cos \angle P''AQ'' = \frac{P''A^2 + Q''A^2 - P''Q''^2}{2P''A \cdot Q''A}$
 $= \frac{\mu\lambda - \mu - \lambda + 2}{2\sqrt{(\mu^2 - \mu + 1)(\lambda^2 - \lambda + 1)}}$ for $\mu, \lambda \in (0, 1)$.

We first find a lower bound of the numerator. We write $\mu\lambda - \mu - \lambda + 2 = (\lambda - 1)\mu - \lambda + 1$ as a linear function of μ by letting λ be constant. Since $\lambda \in (0, 1)$, the coefficient $\lambda - 1$ of μ in this linear function is negative, meaning this expression is least when μ is least "at" 1. Hence, $\mu\lambda - \mu - \lambda + 2 < 1$.

Now we find an upper bound of the denominator. We complete the square in the expression $2\sqrt{(\mu^2 - \mu + 1)(\lambda^2 - \lambda + 1)} = 2\sqrt{((\mu - \frac{1}{2})^2 + \frac{3}{4})((\lambda - \frac{1}{2})^2 + \frac{3}{4})}$ which is least when λ and μ "are" both 1. Hence, $2\sqrt{(\mu^2 - \mu + 1)(\lambda^2 - \lambda + 1)} > 2$.

Hence, $\cos \angle P''AQ'' < \frac{1}{2}$, implying that $\angle P''AQ'' < 60^\circ$.

Therefore, $\angle PAQ = \angle P'AQ' < \angle P''AQ'' < 60^\circ$.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

hurdler: Remark on solution 1: This proof is not rigorous, in the very last step. The last step needs more justification.

2. (*USAMO 1973*) Let $\{X_n\}$ and $\{Y_n\}$ denote two sequences of integers defined as follows:

$$X_0 = 1, X_1 = 1, X_{n+1} = X_n + 2X_{n-1} \quad (n = 1, 2, 3, \dots),$$

$$Y_0 = 1, Y_1 = 7, Y_{n+1} = 2Y_n + 3Y_{n-1} \quad (n = 1, 2, 3, \dots).$$

Thus, the first few terms of the sequences are:

$$X : 1, 1, 3, 5, 11, 21, \dots,$$

$$Y : 1, 7, 17, 55, 161, 487, \dots$$

Prove that, except for the "1", there is no term which occurs in both sequences.

Solution

We can look at each sequence mod 8:

$$X: 1, 1, 3, 5, 3, 5, \dots,$$

$$Y: 1, 7, 1, 7, 1, 7, \dots$$

- Proof that X repeats mod 8:

The third and fourth terms are 3 and 5 mod 8. Plugging into the formula, we see that the next term is $11 \equiv 3 \pmod{8}$, and plugging 5 and 3, we get that the next term is $13 \equiv 5 \pmod{8}$. Thus the sequence X repeats, and the pattern is 3, 5, 3, 5, ...

- Proof that Y repeats mod 8:

The first and second terms are 1 and 7 mod 8. Plugging into the formula, we see that the next term is $17 \equiv 1 \pmod{8}$, and plugging 7 and 1, we get that the next term is $23 \equiv 7 \pmod{8}$. Thus the sequence Y repeats, and the pattern is 1, 7, 1, 7, ...

Combining both results, we see that X_i and Y_j are not congruent mod 8 when $i \geq 3$ and $j \geq 2$. Thus after the "1", the terms of each sequence are not equal.

Solution 2

We can solve this problem by finding a particular solution for each linear recurrence.

$X_{n+1} - X_n - 2X_{n-1} = 0$ with characteristic polynomial

$$X^2 - X - 2 = (X - 2)(X + 1), \text{ so } X = -1, 2$$

$$X_n = A(2)^n + B(-1)^n$$

After plugging in to find the particular solution:

$$X_0 = 1 = A + B$$

$$X_1 = 1 = 2A - B$$

$$A = \frac{1}{3} \text{ and } B = \frac{2}{3}, \text{ so}$$

$$X_n = \frac{1}{3}(2)^n + \frac{2}{3}(-1)^n$$

Doing the same for Y_n , we get

$$Y_n = 2(3)^n - (-1)^n$$

We know they're equal at $n = 0$, so let's set them equal and compare.

$$2(3)^n - (-1)^n = \frac{1}{3}((2)^n + 2(-1)^n)$$

$$6(3)^n - 3(-1)^n = (2)^n + 2(-1)^n$$

$$6(3)^n - 2^n = 5(-1)^n$$

By induction, we know in general that $(\alpha + 1)^n > \alpha^n$ for all $\alpha, n > 0$, so $6(3)^n > 2^n$ for all $n > 1$.

Therefore, the left hand side is always increasing and will never equal 5 again, so equality between X_n and Y_n only holds when $n = 0$, so they only share 1 term.

Solution 3 (simple)

First, X_n and Y_n are both positive for all nonnegative integers n . This is since X_1, X_2, Y_1 , and Y_2 are all positive, and if X_n and X_{n-1} is positive then X_{n+1} as a sum of two positive numbers is positive, and similarly for Y_n . Simple.

Next, consider $Z_n = Y_n - X_n$. We have $Z_{n+1} = Y_{n+1} - X_{n+1} = Y_n + Z_n + 3Z_{n-1} + X_{n-1}$. Another easy inductive argument like above shows that $Z_n > 0$ if $n > 0$. But if $Z_n > 0$ then $Z_n \neq 0$, so $X_n \neq Y_n$ as desired.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

3. (*USAMO 1973*) Three distinct vertices are chosen at random from the vertices of a given regular polygon of $(2n + 1)$ sides. If all such choices are equally likely, what is the probability that the center of the given polygon lies in the interior of the triangle determined by the three chosen random points?

Solution

There are $\binom{2n+1}{3}$ ways how to pick the three vertices. We will now count the ways where the interior does NOT contain the center. These are obviously exactly the ways where all three picked vertices lie among some $n + 1$ consecutive vertices of the polygon. We will count these as follows: We will go clockwise around the polygon. We can pick the first vertex arbitrarily ($2n + 1$ possibilities). Once we pick it, we have to pick 2 out of the next n vertices ($\binom{n}{2}$ possibilities).

Then the probability that our triangle does NOT contain the center is

$$p = \frac{(2n+1)\binom{n}{2}}{\binom{2n+1}{3}} = \frac{(1/2)(2n+1)(n)(n-1)}{(1/6)(2n+1)(2n)(2n-1)} = \frac{3n-3}{4n-2}$$

And then the probability we seek is

$$1 - p = \frac{(4n-2) - (3n-3)}{(4n-2)} = \boxed{\frac{n+1}{4n-2}}$$

Perplexing Edge Case However, for $n = 1$, (a rare edge case), the circumcenter of the triangle must lie in the center of the triangle, which is notably difficult to prove if not assumed, resulting in a probability of 1.

Observe that the probability tends to $\frac{1}{4}$ as n approaches infinity. Why does this occur?

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

4. (*USAMO 1973*) Determine all the **roots**, **real** or **complex**, of the system of simultaneous **equations**

$$\begin{aligned} x + y + z &= 3, \quad x^2 + y^2 + z^2 = 3, \\ x^3 + y^3 + z^3 &= 3. \end{aligned}$$

Solution 1

Let x , y , and z be the **roots** of the **cubic polynomial** $t^3 + at^2 + bt + c$. Let $S_1 = x + y + z = 3$, $S_2 = x^2 + y^2 + z^2 = 3$, and $S_3 = x^3 + y^3 + z^3 = 3$. From this, $S_1 + a = 0$, $S_2 + aS_1 + 2b = 0$, and $S_3 + aS_2 + bS_1 + 3c = 0$. Solving each of these, $a = -3$, $b = 3$, and $c = -1$. Thus x , y , and z are the roots of the polynomial $t^3 - 3t^2 + 3t - 1 = (t - 1)^3$. Thus $x = y = z = 1$, and there are no other solutions.

Solution 2

Let $P(t) = t^3 - at^2 + bt - c$ have roots x , y , and z . Then

$$0 = P(x) + P(y) + P(z) = 3 - 3a + 3b - 3c$$

using our system of equations, so $P(1) = 0$. Thus, at least one of x , y , and z is equal to 1; without loss of generality, let $x = 1$. Then we can use the system of equations to find that $y = z = 1$ as well, and so $\boxed{(1, 1, 1)}$ is the only solution to the system of equations.

Solution 3

Let $a = x - 1$, $b = y - 1$ and $c = z - 1$. Then

$$\begin{aligned} a + b + c &= 0, \\ a^2 + b^2 + c^2 &= 0, \\ a^3 + b^3 + c^3 &= 0. \end{aligned}$$

We have

$$\begin{aligned} 0 &= (a + b + c)^3 \\ &= (a^3 + b^3 + c^3) + 3a^2(b + c) + 3b^2(a + c) + 3c^2(a + b) + 6abc \end{aligned}$$

$$\begin{aligned}
&= 0 - 3a^3 - 3b^3 - 3c^3 + 6abc \\
&= 6abc.
\end{aligned}$$

Then one of a, b and c has to be 0, and easy to prove the other two are also 0. So $(1, 1, 1)$ is the only solution to the system of equations.

J.Z.

Solution 4

We are going to use Intermediate Algebra Techniques to solve this equation.

Let's start with the first one: $x + y + z = 3$. This will be referred as the FIRST equation.

We are going to use the first equation to relate to the SECOND one ($x^2 + y^2 + z^2 = 3$) and the THIRD one ($x^3 + y^3 + z^3 = 3$).

Squaring this equation: $x^2 + y^2 + z^2 + 2xy + 2yz + 2xz = 9$

Subtracting this equation from the 2nd equation in the problem, we have $2xy + 2yz + 2xz = 6$, so $xy + xz + yz = 3$.

Now we try the same idea with the cubed terms. Cube the first equation:

$x^3 + y^3 + z^3 + 3xy^2 + 3xz^2 + 3yx^2 + 3yz^2 + 3zx^2 + 3zy^2 = 27$. Plug in $x^3 + y^3 + z^3 = 3$ and factor partially:

$$3 + 3(x^2(y + z) + y^2(x + z) + z^2(x + y)) + 6xyz = 27.$$

Now here is the key step. Note that $z = 3 - x - y, y = 3 - x - z, x = 3 - y - z$. So we are going to substitute $y + z, x + z, x + y$ for each of the expressions and we get: $-x^3 + 3x^2 - y^3 + 3y^2 - z^3 + 3z^2 = 8 - 2xyz$ (I rearranged it a bit).

Resubstituting in the second and third equation: $-3 + 3(3) = 8 - 2xyz$. So $xyz = 1$.

So now we have three equations for the elementary symmetric sums of x, y, z :

Equation 4: $x + y + z = 3$ (this is also equation 1)

Equation 5: $xy + yz + xz = 3$

Equation 6: $xyz = 1$.

If we call the solutions of $t^3 - 3t^2 + 3t - 1 = 0$ (Equation 7) a, b, c , then x, y, z are the three roots a, b, c but in some order. Notice that Equation 7 can be factored as $(t - 1)^3 = 0$, which means that $t = 1$. Therefore (x, y, z) are permutations of $(1, 1, 1)$ in some order, which can be only $(1, 1, 1)$. (This step uses Vieta's formulas)

Therefore, the only solution is $(x, y, z) = (1, 1, 1)$.

Solution 5

From the question

Equation 1 : $x + y + z = 3$

Equation 2 : $x^2 + y^2 + z^2 = 3$

Equation 3 : $x^3 + y^3 + z^3 = 3$

From Eq(1) and Eq(2),

$$\text{Equation 4 : } (x + y + z)^2 = 9 \implies x^2 + y^2 + z^2 + 2\sum xy = 9 \implies \sum xy = 3$$

From Eq(1), Eq(3), and the fact that $(x + y + z)^3 = x^3 + y^3 + z^3 + 3\prod(x + y)$,

$$\text{Equation 5 : } (x + y + z)^3 = x^3 + y^3 + z^3 + 3\prod(x + y) = 27 \implies \prod(x + y) = 8$$

As from Eq(1), $x + y = 3 - z$, $y + z = 3 - x$, $z + x = 3 - y$

Equation 6 : $\prod(3 - x) = 8$

Now let there be a cubic function $f(t)$ with the roots x, y, z so from the relation between roots and coefficients,

Equation 7 : $f(t) = t^3 - 3t^2 + 3t - d$ (As we don't know the product let the constant be d)

As $f(t) = (t - x)(t - y)(t - z)$ then at $t = 3$,

Equation 8 : $f(3) = 9 - d = (3 - x)(3 - y)(3 - z)$

And from Eq(6),

Equation 9 : $d = -1$

So finally,

Equation 10 : $f(t) = t^3 - 3t^2 + 3t - 1 = (t - 1)^3$

So, $(x, y, z) = (1, 1, 1)$

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

5. (*USAMO 1973*) Show that the cube roots of three distinct prime numbers cannot be three terms (not necessarily consecutive) of an arithmetic progression.

Solution

Assume that the cube roots of three distinct prime numbers can be three terms of an arithmetic progression. Let the three distinct prime numbers be p , q , and r . WLOG, let $p < q < r$.

Then,

$$\frac{1}{q^{\frac{1}{3}}} = \frac{1}{p^{\frac{1}{3}}} + md$$

$$\frac{1}{r^{\frac{1}{3}}} = \frac{1}{p^{\frac{1}{3}}} + nd$$

where m, n are distinct integers, and d is the common difference in the progression. Then we have

$$\begin{aligned}
 \frac{1}{nq\sqrt[3]{3}} - \frac{1}{mr\sqrt[3]{3}} &= \frac{1}{(n-m)p\sqrt[3]{3}} \\
 n^3q - 3n^2mq\sqrt[3]{3}r\sqrt[3]{3} + 3nm^2q\sqrt[3]{3}r\sqrt[3]{3} - m^3r &= (n-m)^3p \\
 3nm^2q\sqrt[3]{3}r\sqrt[3]{3} - 3n^2mq\sqrt[3]{3}r\sqrt[3]{3} &= (n-m)^3p + m^3r - n^3q \\
 (3nmq\sqrt[3]{3}r\sqrt[3]{3})(mr\sqrt[3]{3} - nq\sqrt[3]{3}) &= (n-m)^3p + m^3r - n^3q \\
 (3nmq\sqrt[3]{3}r\sqrt[3]{3})((m-n)p\sqrt[3]{3}) &= (n-m)^3p + m^3r - n^3q \\
 \frac{1}{q\sqrt[3]{3}r\sqrt[3]{3}p\sqrt[3]{3}} &= \frac{(n-m)^3p + m^3r - n^3q}{(3mn)(m-n)} \\
 \frac{1}{(pqr)\sqrt[3]{3}} &= \frac{(n-m)^3p + m^3r - n^3q}{(3mn)(m-n)}
 \end{aligned}$$

Because p, q, r are distinct primes, pqr is not a perfect cube. Thus, the LHS is irrational but the RHS is rational, which is a contradiction. So, the cube roots of three distinct prime numbers cannot be three terms of an arithmetic progression.

Solution 2

Assume the contrary and suppose that the cube roots of three distinct prime numbers, $p_1^{\frac{1}{3}}, p_2^{\frac{1}{3}},$ and $p_3^{\frac{1}{3}},$ form an increasing arithmetic progression. Then we can write $p_2^{\frac{1}{3}} = p_1^{\frac{1}{3}} + md$ and $p_3^{\frac{1}{3}} = p_1^{\frac{1}{3}} + nd$ for some positive integral d, m, n with $m < n$. Thus,

$$\frac{p_3^{\frac{1}{3}} - p_1^{\frac{1}{3}}}{p_2^{\frac{1}{3}} - p_1^{\frac{1}{3}}} = \frac{nd}{md} = \frac{n}{m} = r,$$

for some real r . Because m and n are integral, r is rational.

This equation rearranges to

$$p_3^{\frac{1}{3}} = rp_2^{\frac{1}{3}} - (r-1)p_1^{\frac{1}{3}}.$$

Note that if $a - b = x$, then

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b) = a^3 - b^3 - 3abx$$

from the binomial theorem. As a result, cubing both sides of the equation gives

$$p_3 = r^3p_2 - (r-1)^3p_1 - 3r(r-1)(p_1p_2p_3)^{\frac{1}{3}}. (*)$$

Lemma: The sum, product, and difference of two rational numbers are all rational, and that of a rational and irrational number are all irrational.

Proof: The first part is trivial if we write the fractions as $\frac{m}{n}$ and $\frac{p}{q}$, and find the common denominators or simplify fractions as needed. The second part follows by contradiction: if the sum of a rational number and irrational number is rational, then the irrational number must be the difference of two rational numbers, which is rational. This is a contradiction; likewise for difference and product.

The LHS (*left-hand side*) of (*) is clearly rational (in fact, integral), and so the RHS must also be rational. This is the case for the first two terms $r^3 p_2$ and $-(r-1)^3 p_1$. But it is not the case for the third term. The number $(p_1 p_2 p_3)^{\frac{1}{3}}$ is irrational because $p_1 p_2 p_3$, the product of three distinct primes, cannot be a perfect cube. Hence, $-3r(r-1)(p_1 p_2 p_3)^{\frac{1}{3}}$, the product of a rational and an irrational number, is irrational, and the RHS is therefore the sum of two rational terms and one irrational term. This is irrational by the lemma, and hence we have a rational equalling an irrational, contradiction. It follows that the cube roots of three distinct prime numbers cannot form an arithmetic sequence.

§3 USAMO 1974

1. (USAMO 1974) Let a , b , and c denote three distinct integers, and let P denote a polynomial having all integral coefficients. Show that it is impossible that $P(a) = b$, $P(b) = c$, and $P(c) = a$.

Solution

It suffices to show that if a, b, c are integers such that $P(a) = b$, $P(b) = c$, and $P(c) = a$, then $a = b = c$.

We note that

$$a - b \mid P(a) - P(b) = b - c \mid P(b) - P(c) = c - a \mid P(c) - P(a) = a - b,$$

so the quantities $(a - b)$, $(b - c)$, $(c - a)$ must be equal in absolute value. In fact, two of them, say $(a - b)$ and $(b - c)$, must be equal. Then

$$0 = |(a - b) + (b - c) + (c - a)| = |2(a - b) + (c - a)| \geq 2|a - b| - |c - a| = |a - b|,$$

so $a = b = P(a)$, and $c = P(b) = P(a) = b$, so a , b , and c are equal, as desired. ■

Solution 1b

Let k be the value $(a - b)$, $(b - c)$, $(c - a)$ are equal to in absolute value. Assume k is nonzero. Then each of $(a - b)$, $(b - c)$, $(c - a)$ is equal to k or $-k$, so

$$0 = (a - b) + (b - c) + (c - a) = mk,$$

where m is one of -3 , -1 , 1 , or 3 . In particular, neither m nor k is zero, contradiction. Hence, $k = 0$, and a, b, c are equal, yielding a final contradiction of the existence of these variables. ■

In fact, this approach generalizes readily. Suppose that n is an odd integer, and that there exist n distinct integers $a_1, a_2, \dots, a_n$ such that $P(a_i) = a_{i+1}$ and $P(a_n) = 1$, for $1 \leq i \leq n - 1$. Then we have

$$a_1 - a_2 \mid a_2 - a_3 \mid \dots \mid a_n - a_1 \mid a_1 - a_2,$$

so the differences are all equal in absolute value and thus equal to k or $-k$ for some integer k . Adding the differences (in the order they are written) gives $0 = mk$ for

some odd integer m , so m is nonzero and hence $k = 0$. Thus, all the a_i are equal, a contradiction of their distinctness, so the sequence a_i cannot exist.

Solution 2

Consider the polynomial $Q(x) = (b-a)P(x) - (c-b)x - b^2 + ac$. By using the facts that $P(a) = b$ and $P(b) = c$, we find that $Q(a) = Q(b) = 0$. Thus, the polynomial $Q(x)$ has a and b as roots, and we can write $Q(x) = (x-a)(x-b)R(x)$ for some polynomial $R(x)$. Because $Q(x)$ and $(x-a)(x-b)$ are monic polynomials with integral coefficients, their quotient, $R(x)$, must also have integral coefficients, as can be demonstrated by simulating the long division. Thus, $Q(c)$, and hence $Q(c) - (x-a)(x-b)$, must be divisible by $(c-a)(c-b)$. But if $x = c-a$ and $y = c-b$, then we must have, after rearranging terms and substitution, that $(x-y)^2$ is divisible by xy . Equivalently, $S = x^2 + y^2$ is divisible by xy (after canceling the $2xy$ which is clearly divisible by xy). Because S must be divisible by both x and y , we quickly deduce that x is divisible by y and y is divisible by x . Thus, x and y are equal in absolute value. A similar statement holds for a cyclic permutation of the variables, and so x , y , and z are all equal in absolute value. The conclusion follows as in the first solution.

Solution 3

Assume for the sake of contradiction that is possible for unique integers a, b, c . Let $P(x) = d_1x^n + d_2x^{n-1} + \dots + d_n$. Note that

$$b = P(a) = d_1a^n + d_2a^{n-1} + \dots + d_n$$

$$c = P(b) = d_1b^n + d_2b^{n-1} + \dots + d_n$$

$$a = P(c) = d_1c^n + d_2c^{n-1} + \dots + d_n$$

Subtracting the second from the first, third from second, and first from third gives

$$b - c = P(a) - P(b) = d_1(a^n - b^n) + d_2(a^{n-1} - b^{n-1}) + \dots + d_{n-1}(a - b)$$

$$c - a = P(b) - P(c) = d_1(b^n - c^n) + d_2(b^{n-1} - c^{n-1}) + \dots + d_{n-1}(b - c)$$

$$a - b = P(c) - P(a) = d_1(c^n - a^n) + d_2(c^{n-1} - a^{n-1}) + \dots + d_{n-1}(c - a)$$

By the RHS, note that $a - b \mid P(a) - P(b)$ so $|a - b| \leq |b - c|$. Similarly, $|b - c| \leq |c - a|$ and $|c - a| \leq |a - b|$. Hence, $|a - b| \leq |b - c| \leq |c - a| \leq |a - b|$ so $|a - b| = |b - c| = |c - a|$. Assume WLOG that $a > b > c$ so $a - b = a - c$ and $b - c = c - a$. From the first equation, we get $b = c$ and substituting this in the second gives $c = a$. Hence, $a = b = c$, contradicting the uniqueness of a, b, c . ■

Solution 4

Assume that a, b and c are distinct integers which satisfy the given conditions. Then we have $P(c) = P(P(b)) = P(P(P(a))) = a$. Similarly $P(P(P(b))) = b$ and $P(P(P(c))) = c$. Since the polynomial has integer coefficients, this implies that the polynomial P is $P(x) = x$. But this is a contradiction since then $P(a) = b$ can only be true if $a = b$ which contradicts the fact that a, b and c are distinct integers.

Solution 5

We get that $a - b \mid b - c, b - c \mid c - a, c - a \mid a - b$. Thus, we can say that $(b - c) = (a - b)k, (c - a) = (b - c)j, (a - b) = (c - a)m$, for non zero integers, not necessarily positive, k, j , and m . Multiplying these equations and canceling, we get that $kjm=1$. This means that either $k=1, m=1, j=1$, or two of the variables are equal to 1 and

the other is equal to -1. If all of the variables are equal to zero, plugging in gives us that $a = b = c$, contradictory to the conditions given in the problem statement. If two of the variables are equal to -1, and the other is equal to 1, then WLOG let $k = m = -1, j = 1$. But this implies that the distinct integer condition is not satisfied, and hence, this case is invalid. Thus, it is impossible. ■
 Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

2. (USAMO 1974) Prove that if a, b , and c are positive real numbers, then

$$a^a b^b c^c \geq (abc)^{(a+b+c)/3}$$

Solution 1

Consider the function $f(x) = x \ln x$. $f''(x) = \frac{1}{x} > 0$ for $x > 0$; therefore, it is a convex function and we can apply [Jensen's Inequality](#):

$$\frac{a \ln a + b \ln b + c \ln c}{3} \geq \left(\frac{a+b+c}{3} \right) \ln \left(\frac{a+b+c}{3} \right)$$

Apply [AM-GM](#) to get

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

which implies

$$\frac{a \ln a + b \ln b + c \ln c}{3} \geq \left(\frac{a+b+c}{3} \right) \ln \left(\sqrt[3]{abc} \right)$$

Rearranging,

$$a \ln a + b \ln b + c \ln c \geq \left(\frac{a+b+c}{3} \right) \ln (abc)$$

Because $f(x) = e^x$ is an increasing function, we can conclude that:

$$e^{a \ln a + b \ln b + c \ln c} \geq e^{\ln(abc)(a+b+c)/3}$$

which simplifies to the desired inequality.

Solution 2

Note that $(a^a b^b c^c)(a^b b^c c^a)(a^c b^a c^b) = \left((abc)^{(a+b+c)/3} \right)^3$.

So if we can prove that $a^a b^b c^c \geq a^b b^c c^a$ and $a^a b^b c^c \geq a^c b^a c^b$, then we are done.

WLOG let $a \geq b \geq c$.

Note that $(a^a b^b c^c) \cdot \left(\frac{c}{a} \right)^{a-b} \cdot \left(\frac{c}{b} \right)^{b-c} = a^b b^c c^a$. Since $\frac{c}{a} \leq 1$, $\frac{c}{b} \leq 1$, $a - b \geq 0$, and $b - c \geq 0$, it follows that $a^a b^b c^c \geq a^b b^c c^a$.

Note that $(a^a b^b c^c) \cdot \left(\frac{b}{a} \right)^{a-b} \cdot \left(\frac{c}{a} \right)^{b-c} = a^c b^a c^b$. Since $\frac{b}{a} \leq 1$, $\frac{c}{a} \leq 1$, $a - b \geq 0$, and $b - c \geq 0$, it follows that $a^a b^b c^c \geq a^c b^a c^b$.

Thus, $(a^a b^b c^c)^3 \geq (a^a b^b c^c)(a^b b^c c^a)(a^c b^a c^b) = \left((abc)^{(a+b+c)/3} \right)^3$, and cube-rooting both sides gives $a^a b^b c^c \geq (abc)^{(a+b+c)/3}$ as desired.

Solution 3

WLOG let $a \geq b \geq c$. Let $b = ax$ and $c = ay$, where $x \geq 1$ and $y \geq 1$.

We want to prove that $(a)^a (ax)^{ax} (ay)^{ay} \geq (a \cdot ax \cdot ay)^{\frac{a+ax+ay}{3}}$.

Simplifying and combining terms on each side, we get $a^{a+ax+ay} x^{ax} y^{ay} \geq a^{a+ax+ay} (xy)^{\frac{a+ax+ay}{3}}$.

Since $a > 0$, we can divide out $a^{a+ax+ay}$ to get $x^{ax} y^{ay} \geq (xy)^{\frac{a+ax+ay}{3}}$.

Take the a th root of each side and then cube both sides to get $x^{3x} y^{3y} \geq (xy)^{1+x+y}$.

This simplifies to $x^{2x-1}y^{2y-1} \geq x^y y^x$.

Since $2x - 1 \geq x$ and $2y - 1 \geq y$, we only need to prove $x^x y^y \geq x^y y^x$ for our given x, y .

WLOG, let $y \geq x$ and $y = kx$ for $k \geq 1$. Then our expression becomes

$$x^x (xk)^{xk} \geq x^{xk} (xk)^x$$

$$x^{x+xk} k^{xk} \geq x^{x+xk} k^x$$

$$k^{xk} \geq k^x$$

$$k^k \geq k$$

This is clearly true for $k \geq 1$.

Solution 4

WLOG let $a \geq b \geq c$. Then sequence (a, b, c) majorizes $(\frac{a+b+c}{3}, \frac{a+b+c}{3}, \frac{a+b+c}{3})$. Thus by Muirhead's Inequality, we have $\sum_{sym} a^a b^b c^c \geq \sum_{sym} a^{\frac{a+b+c}{3}} b^{\frac{a+b+c}{3}} c^{\frac{a+b+c}{3}}$, so $a^a b^b c^c \geq (abc)^{\frac{a+b+c}{3}}$.

Solution 5

Let $x = \frac{a}{\sqrt[3]{abc}}$, $y = \frac{b}{\sqrt[3]{abc}}$ and $z = \frac{c}{\sqrt[3]{abc}}$. Then $xyz = 1$ and a straightforward calculation reduces the problem to

$$x^x y^y z^z \geq 1.$$

WLOG, assume $x \geq y \geq z$. Then $x \geq 1$, $z \leq 1$ and $xy = \frac{1}{z} \geq 1$. Therefore,

$$x^x y^y z^z = x^{x-y} (xy)^{y-z} (xyz)^z \geq 1.$$

J.Z.

Solution 6

Cubing both sides of the given inequality gives

$$a^{3a} b^{3b} c^{3c} \geq (abc)^{a+b+c}$$

If we take $a^{3a} b^{3b} c^{3c}$ as the product of $3a$ a 's, $3b$ b 's, and $3c$ c 's, we get that

$$\sqrt[3(a+b+c)]{a^{3a} b^{3b} c^{3c}} \geq \frac{3(a+b+c)}{\frac{3a}{a} + \frac{3b}{b} + \frac{3c}{c}} = \frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

by GM-HM, as desired.

Solution 7

Replacing a with ak , b with bk , and c with ck , for some positive real k we get: $a^{ak} k^{ak} b^{bk} k^{bk} c^{ck} k^{ck} = a^{ak} b^{bk} c^{ck} k^{ak+bk+ck} \geq k^3 abc^{ak+bk+ck/3} = k^{ak+bk+ck} abc^{ak+bk+ck/3}$. This means that this inequality is homogeneous since both sides have the same power of k as a factor. Since the inequality is homogeneous, we can scale abc so that their product is 1, i.e. $abc = 1$. This makes the inequality turn into something much more nicer to deal with. Now we have to prove: $a^a b^b c^c \geq 1$ given that $abc = 1$. Note that $a^a \geq a$ even if $a \leq 1$. Therefore $a^a \geq a$, $b^b \geq b$, and $c^c \geq c$. Multiplying these together we get: $a^a b^b c^c \geq abc = 1$. This proves the desired result. Equality holds when $a = b = c = 1$.

Solution 8 (Rearrangement)

Let $\log_2 a = x$, $\log_2 b = y$ and $\log_2 c = z$, and WLOG $a \geq b \geq c > 0$. Then we have both $a \geq b \geq c$ and $x \geq y \geq z$. By the rearrangement inequality, $ax + by + cz \geq$

$ay + bz + cx$ and $ax + by + cz \geq az + bx + cy$. Summing, $2(ax + by + cz) \geq a(y+z) + b(x+z) + c(x+y)$. Adding $ax + by + cz$, we get $ax + by + cz \geq \frac{(a+b+c)(x+y+z)}{3}$. Now we substitute back for x, y, z to get: $\log_2 a^a b^b c^c \geq \log_2 (abc)^{\frac{a+b+c}{3}}$. Raising 2 to the power of each side, we get $a^a b^b c^c \geq (abc)^{\frac{a+b+c}{3}}$.

Solution 9

Assume without loss of generality that $a \leq b \leq c$. Then, we have

$$b/a \geq 1, c/a \geq 1, c/b \geq 1, b-a \geq 0, c-a \geq 0, c-b \geq 0.$$

It follows that

$$(b/a)^{(b-a)/3} \cdot (c/a)^{(c-a)/3} \cdot (c/b)^{(c-b)/3} \geq 1.$$

We can "distribute" the exponents, which gives

$$\frac{b^{(b-a)/3}}{a^{(b-a)/3}} \cdot \frac{c^{(c-a)/3}}{a^{(c-a)/3}} \cdot \frac{c^{(c-b)/3}}{b^{(c-b)/3}} \geq 1.$$

Notice that in the $b^{(c-b)/3}$ in the denominator of the third fraction can be brought out of the denominator by negating its exponent. This gives

$$\frac{b^{(b-a)/3}}{a^{(b-a)/3}} \cdot \frac{c^{(c-a)/3}}{a^{(c-a)/3}} \cdot c^{(c-b)/3} \cdot b^{(b-c)/3} \geq 1.$$

We can move around terms on the left to get

$$\left(\frac{b^{(b-a)/3} \cdot b^{(b-c)/3}}{a^{(b-a)/3}} \right) \left(\frac{c^{(c-a)/3} \cdot c^{(c-b)/3}}{a^{(c-a)/3}} \right) \geq 1.$$

Now we combine the exponents in the numerators. This gives

$$\left(\frac{b^{(2b-c-a)/3}}{a^{(b-a)/3}} \right) \left(\frac{c^{(2c-b-a)/3}}{a^{(c-a)/3}} \right) \geq 1.$$

We multiply both sides by $\left(a^{(b-a)/3} \right) \left(a^{(c-a)/3} \right)$, which gives

$$\left(b^{(2b-c-a)/3} \right) \left(c^{(2c-b-a)/3} \right) \geq \left(a^{(b-a)/3} \right) \left(a^{(c-a)/3} \right).$$

We then multiply both sides by $\left(b^{(c-a)/3} \right) \left(c^{(b-a)/3} \right)$:

$$\left(b^{(2b-2a)/3} \right) \left(c^{(2c-2a)/3} \right) \geq \left(a^{(b-a)/3} \right) \left(c^{(b-a)/3} \right) \left(a^{(c-a)/3} \right) \left(b^{(c-a)/3} \right).$$

Multiplying both sides by $\left(b^{(b-a)/3} \right) \left(c^{(c-a)/3} \right)$, then simplifying both sides with exponent laws, gives

$$b^{b-a} \cdot c^{c-a} \geq (abc)^{(b+c-2a)/3}.$$

Finally, we multiply both sides by $(abc)^a$. Combining exponents one last time gives the desired

$$a^a b^b c^c \geq (abc)^{(a+b+c)/3}.$$

–MenuThreeOne

Solution 10

As in solution 2:

$$a^a b^b c^c \cdot a^b b^c c^a \cdot a^c b^a c^b = (abc)^{\left(\frac{a+b+c}{3}\right)^3}$$

We show

$$a^a b^b c^c \geq a^b b^c c^a$$

and

$$a^a b^b c^c \geq a^c b^a c^b$$

.

By definition, the first inequality is trivial via the Rearrangement Inequality:

$$a \ln(a) + b \ln(b) + c \ln(c) \geq b \ln(a) + c \ln(b) + a \ln(c)$$

.

Similarly, the second:

$$a \ln(a) + b \ln(b) + c \ln(c) \geq c \ln(a) + a \ln(b) + b \ln(c)$$

Hence the two results are proved. As in solution 2:

$$(a^a b^b c^c)^3 \geq (a^a b^b c^c)(a^b b^c c^a)(a^c b^a c^b) = \left((abc)^{(a+b+c)/3}\right)^3$$

and through cube roots the result $a^a b^b c^c \geq (abc)^{(a+b+c)/3}$ is obtained.

3. (*USAMO 1974*) Two boundary points of a ball of radius 1 are joined by a curve contained in the ball and having length less than 2. Prove that the curve is contained entirely within some hemisphere of the given ball.

Solution

Question: Why is the curve necessarily on a great circle? The curve is arbitrary—in space. Also, there is only one great circle through the two points, as three points determine a plane—a counterexample to this claim would then be readily found. Draw a **Great Circle** containing the two points and the curve. Then connect the two points with a chord in the circle. The circle has radius 1, so the circle has diameter 2, so the two points are all but diametrically opposite. Therefore we can draw a diameter EF parallel to the chord but not on it.

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Now take the plane through EF perpendicular to AG . This creates two hemispheres, and the curve is in one and only one of them.

Solution 2

Let the curve intersect the sphere at points A and B . Let B' be the point diametrically opposite A , and let Q be the plane through the center O of the sphere perpendicular to BB' and passing through the midpoint of BB' . We claim that the curve must be in the hemisphere divided by Q containing points A and B .

Assume the contrary and suppose that the curve intersects Q at point C . Reflect the portion of the curve from C to B to obtain a same-length curve from C to B' . Then the length of the curve equals the length of the new curve from A to B' , which is at least the distance from A to B' , or 2. This is a contradiction to the

claim that the length of the curve is less than 2, so the curve must be contained in the hemisphere divided by Q containing points A and B.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

4. (*USAMO 1974*) A father, mother and son hold a family tournament, playing a two person board game with no ties. The tournament rules are:
- (i) The weakest player chooses the first two contestants.
 - (ii) The winner of any game plays the next game against the person left out.
 - (iii) The first person to win two games wins the tournament.

The father is the weakest player, the son the strongest, and it is assumed that any player's probability of winning an individual game from another player does not change during the tournament. Prove that the father's optimal strategy for winning the tournament is to play the first game with his wife.

Solution

There are three possible strategies for the father:

- 1) Sit out the first game
- 2) Play the first game against the weaker of the other two players (the mother). This person is referred to throughout the solution as "the weaker player".
- 3) Play the first game against the stronger of the other two players (the son). This person is referred to throughout the solution as "the stronger player".

Strategy 1 is dominated by both strategy 2 and strategy 3. If strategy 1 is used and the weaker player wins the first game, it becomes strategy 2, but with the father starting behind 0-1. Likewise, if strategy 1 is used and the stronger player wins the first game, it becomes strategy 3, but with the father again starting behind 0-1.

Thus, the father must play the first game.

By the pigeonhole principle, there may be no more than 4 games; once the 4th game is finished, someone will have won two games. Let W denote a game the father wins, let L denote a game the father loses, and let x denote a game the father does not participate in. Then, with the knowledge that the father plays the first game, the following ways for the father to win two games are possible:

- (A) WW
- (B) $WLxW$
- (C) $LxWW$

If the father wins the first game, then he must either win the second game (A), or sit out the third game to win the fourth game (B). If the father loses the first game, then he must sit out the second game and win the third and fourth games (C).

Now we must determine which gives the largest probability of the father winning: playing the first game against the weaker player or the stronger player. We define the following probabilities:

x is the chance of the father defeating the weaker player. $(1 - x)$ is the chance of the weaker player defeating the father.

y is the chance of the father defeating the stronger player. $(1 - y)$ is the chance of the stronger player defeating the father.

z is the chance of the weaker player defeating the stronger player. $(1 - z)$ is the chance of the stronger player defeating the weaker player.

Note that $x > y$, and all of x, y and z are less than 0.5.

If the father plays the weaker player first, we split the probability of winning the tournament in the three mutually exclusive cases above:

$P(A) = xy$ because the father must defeat the weaker player, then the father must defeat the stronger player.

$P(B) = x(1 - y)zx$ because the father must defeat the weaker player, then stronger player must defeat the father. Third, the weaker player must defeat the stronger player (because if the stronger player wins a second game, the tournament is over). Fourth, the father plays the weaker player a second time and must win.

$P(C) = (1 - x)(1 - z)yx$ because the weaker player must defeat the father, followed by the stronger player defeating the weaker player (because if the weaker player wins two in a row, he wins the tournament). The father must then defeat the stronger player and the weaker player in the third and fourth game, respectively.

From this, the total probability of the father winning by playing the weaker player first is:

$$P(TOTAL) = P(A) + P(B) + P(C) = xy + x(1 - y)zx + (1 - x)(1 - z)yx$$

Likewise, we compute the probability of the father winning if he plays the stronger player first. We do this by symmetry:

All x will turn into y , because whenever the father defeated the weaker player before, he must defeat the stronger player now.

All y will turn into x , because whenever the father defeated the stronger player before, he must defeat the weaker player now.

All z will turn into $(1 - z)$, because whenever the weaker player defeated the stronger player, now the stronger player must defeat the weaker player.

All $(1 - z)$ will turn into z , because whenever the stronger player defeated the weaker player, now the weaker player must defeat the stronger player.

Applying these transformations and calling the new probabilities P' , we find that:

$$P'(TOTAL) = P'(A) + P'(B) + P'(C) = yx + y(1 - x)(1 - z)y + (1 - y)zxy$$

We compare these probabilities.

$P(A) = P'(A)$. This makes sense, as the probability of the father winning the first two games is the same, regardless of the order in which they are played.

$P(B) > P'(C)$: Calculating $P(B)/P'(C)$, we get $x^2(1 - y)z/xy(1 - y)z$, which simplifies to x/y .

Since $x > y$

$$x/y > 1$$

$$P(B)/P'(C) > 1$$

$$\text{so } P(B) > P'(C).$$

$P(C) > P'(B)$: Calculating $P(C)/P'(B)$, we get $(1 - x)(1 - z)yx/(1 - x)(1 - z)y^2$, which simplifies again to x/y . Thus, $P(C) > P'(B)$

Adding the above equations and inequalities, we see that $P(TOTAL) > P'(TOTAL)$, and thus the father is better off playing the weaker of the two players (ie the mother) first.

TL;DR Let the probability of the father winning against the son be a , the father winning against the mother be b , and the son winning against the mother be c . Here $a < b < \frac{1}{2}$ and $c > \frac{1}{2}$.

We define three first game cases as above:

1. Father vs Mother
2. Father vs Son
3. Mother vs Son

The probability that the father wins (1) (by casework or tree) is $P(1) = ab + b^2(1 - a)(1 - c) + (1 - b)abc$.

The probability that the father wins (2) is $P(2) = ab + a^2c(1 - b) + ab(1 - a)(1 - c)$.

The probability that the father wins (3) is $P(3) = ab$.

Evidently $P(1) > P(3)$, so all that is left is to prove that $P(1) > P(2)$.

$P(1) > P(2)$

$$ab + b^2(1 - a)(1 - c) + (1 - b)abc > ab + a^2c(1 - b) + ab(1 - a)(1 - c)$$

$$b^2(1 - a)(1 - c) + (1 - b)abc > a^2c(1 - b) + ab(1 - a)(1 - c)$$

$$b(b - a)(1 - a)(1 - c) > ac(a - b)(1 - b)$$

Using the fact that $a < b$, we easily find that the LHS is positive while the RHS is negative, proving the inequality.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

5. (*USAMO 1974*) Consider the two triangles $\triangle ABC$ and $\triangle PQR$ shown in Figure 1. In $\triangle ABC$, $\angle ADB = \angle BDC = \angle CDA = 120^\circ$. Prove that $x = u + v + w$.

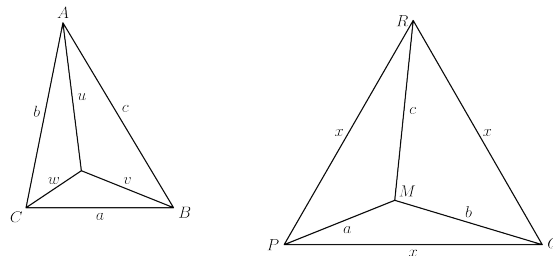
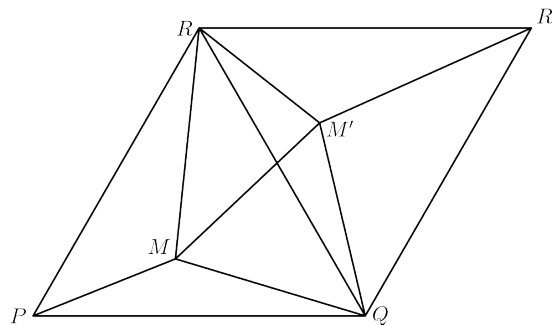


Figure 1

Solution 1

We rotate figure $PRQM$ by a clockwise angle of $\pi/3$ about Q to obtain figure $RR'QM'$:



Evidently, $MM'Q$ is an equilateral triangle, so triangles MRM' and ABC are congruent. Also, triangles PMQ and $RM'Q$ are congruent, since they are images of each other under rotations. Then

$$[ABC] + \frac{b^2\sqrt{3}}{4} = [MRM'] + [MM'Q] = [QMR] + [RM'Q] = [QMR] + [PMQ].$$

Then by symmetry,

$$3[ABC] + \frac{(a^2 + b^2 + c^2)\sqrt{3}}{4} = 2([PMQ] + [QMR] + [RMP]) = 2[PRQ].$$

But ABC is composed of three smaller triangles. The one with sides w, v, a has area $\frac{1}{2}wv \sin 120^\circ = \frac{wv\sqrt{3}}{4}$. Therefore, the area of ABC is

$$\frac{(wv + vu + uw)\sqrt{3}}{4}.$$

Also, by the **Law of Cosines** on that small triangle of ABC , $a^2 = w^2 + wv + v^2$, so by symmetry,

$$\frac{(a^2 + b^2 + c^2)\sqrt{3}}{4} = \frac{[2(u^2 + w^2 + v^2) + wv + vu + uw]\sqrt{3}}{4}.$$

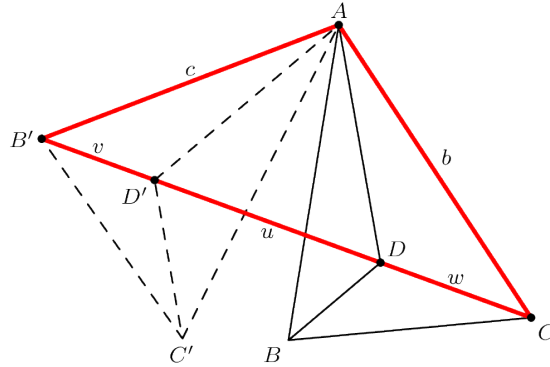
Therefore

$$\begin{aligned} \frac{(u + v + w)^2\sqrt{3}}{2} &= 3\frac{(wv + vu + uw)\sqrt{3}}{4} + \frac{[2(u^2 + w^2 + v^2) + wv + vu + uw]\sqrt{3}}{4} \\ &= 3[ABC] + \frac{(a^2 + b^2 + c^2)\sqrt{3}}{4} = 2[PRQ]. \end{aligned}$$

But the area of triangle PQR is $x^2\sqrt{3}/4$. It follows that $u + v + w = x$, as desired. ■

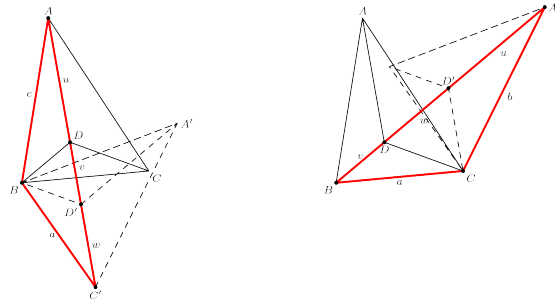
Solution 2

Rotate $\triangle ABC$ 60 degrees clockwise about A to get $\triangle AB'C'$. Observe that $\triangle ADD'$ is equilateral, which means $D'D = AD = u$. Also, B', D', D, C are collinear because $\angle B'D'A + \angle AD'D = 120 + 60 = 180$ and $\angle CDA + \angle DD'A = 120 + 60 = 180$. The resulting $\triangle B'AC$ has side lengths $b, c, u + v + w$ and the angle opposite side $u + v + w$ has magnitude $A + 60$.



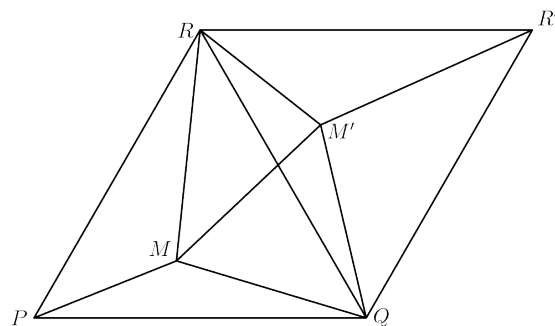
If we perform the rotation about points B and C , we get two triangles. One has side lengths $a, c, u + v + w$ and the angle opposite side $u + v + w$ has magnitude

$B + 60$, and the other has side lengths $b, c, u + v + w$ and the angle opposite side $u + v + w$ has magnitude $C + 60$.



These three triangles fit together because $(A + 60) + (B + 60) + (C + 60) = 360$. The result is an equilateral triangle of side length $u + v + w$.

Solution 3



As in the first solution, we rotate and establish that $\triangle MRM' \cong \triangle ABC$.

Let X and Y be points on $\overline{RQ}$ and $\overline{PQ}$, respectively, such that M lies on $\overline{XY}$ and $\overline{XY} \parallel \overline{PR}$. We note that $m\angle RXM = 120^\circ$. The rotation then takes Y to X , so $m\angle RXM' = m\angle PYM = 120^\circ$. It follows that $RX = BD = v$, $MX = AD = u$, $M'X = CD = w$.

Since $m\angle MXM' = 120^\circ$ and $m\angle MQM' = 60^\circ$, $MXM'Q$ is cyclic. By Ptolemy's theorem,

$$\begin{aligned} (MX)(M'Q) + (XM')(QM) &= (MM')(XQ) \\ XQ &= MX + M'X \\ &= u + w. \end{aligned}$$

Finally, $RQ = RX + XQ = u + v + w$, as desired.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

§4 USAMO 1975

- (USAMO 1975) (a) Prove that

$$[5x] + [5y] \geq [3x + y] + [3y + x],$$

where $x, y \geq 0$ and $[u]$ denotes the greatest integer $\leq u$ (e.g., $[\sqrt{2}] = 1$).

(b) Using (a) or otherwise, prove that

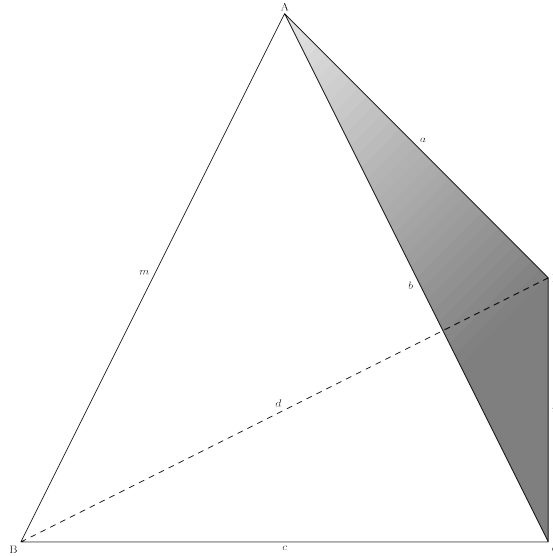
$$\frac{(5m)!(5n)!}{m!n!(3m+n)!(3n+m)!}$$

is integral for all positive integral m and n .

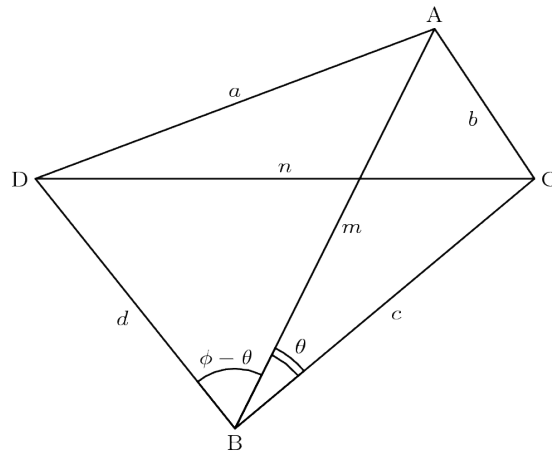
2. (USAMO 1975) Let A, B, C, D denote four points in space and AB the distance between A and B , and so on. Show that

$$AC^2 + BD^2 + AD^2 + BC^2 \geq AB^2 + CD^2.$$

Solution 1



If we project points A, B, C, D onto the plane parallel to $\overline{AB}$ and $\overline{CD}$, AB and CD stay the same but BC, AC, AD, BD all decrease, making the inequality sharper. Thus, it suffices to prove the inequality when A, B, C, D are coplanar:



Let $AD = a, AC = b, BC = c, BD = d, AB = m, CD = n$. We wish to prove that $a^2 + b^2 + c^2 + d^2 \geq m^2 + n^2$. Let us fix $\triangle BCD$ and the length AB and let A vary on the circle centered at B with radius m . If we find the minimum value of $a^2 + b^2$, which is the only variable quantity, and prove that it is larger than $m^2 + n^2 - c^2 - d^2$, we will be done.

First, we express $a^2 + b^2$ in terms of c, d, m, θ, ϕ , using the **Law of Cosines**:

$$\begin{aligned} a^2 + b^2 &= c^2 + d^2 + 2m^2 - 2cm \cos(\theta) - 2dm \cos(\phi - \theta) \\ (a^2 + b^2 - c^2 - d^2 - 2m^2)^2 &= 4m^2(c^2 \cos^2(\theta) + d^2 \cos^2(\phi - \theta) + 2cd \cos(\theta) \cos(\phi - \theta)) \end{aligned}$$

$a^2 + b^2$ is a function of θ , so we take the derivative with respect to θ and obtain that $a^2 + b^2$ takes a minimum when

$$\begin{aligned} c \sin(\theta) - d \sin(\phi - \theta) &= 0 \\ c^2 \sin^2(\theta) + d^2 \sin^2(\phi - \theta) - 2cd \sin(\theta) \sin(\phi - \theta) &= 0 \\ (a^2 + b^2 - c^2 - d^2 - 2m^2)^2 &= 4m^2(c^2 + d^2 + 2cd(\cos(\theta) \cos(\phi - \theta) - \sin(\theta) \sin(\phi - \theta))) \\ &= 4m^2(c^2 + d^2 + 2cd \cos \phi) \\ &= 4m^2(2c^2 + 2d^2 - n^2) \end{aligned}$$

Define $p = a^2 + b^2$ and $q = c^2 + d^2$:

$$\begin{aligned} (p - q - 2m^2)^2 &= 4m^2(2q - n^2) \\ p^2 + q^2 + 4m^4 - 4m^2p + 4m^2q - 2pq &= 8m^2q - 4m^2n^2 \\ p^2 + q^2 + 4m^4 - 4m^2p - 4m^2q - 2pq &= -4m^2n^2 \\ p^2 - 2pq + q^2 - 4m^2(p + q) &= -4m^2(m^2 + n^2) \\ \frac{(p - q)^2}{m^2} &= p + q - m^2 - n^2 \geq 0 \\ a^2 + b^2 + c^2 + d^2 &\geq m^2 + n^2 \end{aligned}$$

Solution 2

Let

$$\begin{aligned} A &= (0, 0, 0) \\ B &= (1, 0, 0) \\ C &= (a, b, c) \\ D &= (x, y, z). \end{aligned}$$

It is clear that every other case can be reduced to this. Then, with the distance formula and expanding,

$$\begin{aligned} AC^2 + BD^2 + AD^2 + BC^2 - AB^2 - CD^2 &= x^2 - 2x + 1 + y^2 + z^2 + a^2 - 2a + b^2 + c^2 + 2ax + 2by + 2cz \\ &= (x + a - 1)^2 + (y + b)^2 + (z + c)^2 \\ &\geq 0, \end{aligned}$$

which rearranges to the desired inequality.

Solution 3

Because the distances are all squared, we must only prove the result in one dimension, and then we can just add up the three individual inequalities for the x , y , and z dimension to get the desired result. Let x_a , x_b , x_c , and x_d be the positions of A , B , C , and D respectively. Then we must show that,

$$\begin{aligned}(x_a - x_c)^2 + (x_b - x_d)^2 + (x_a - x_d)^2 + (x_b - x_c)^2 &\geq (x_a - x_b)^2 + (x_c - x_d)^2 \\ x_a^2 + x_b^2 + x_c^2 + x_d^2 &\geq 2x_ax_c + 2x_bx_d + 2x_ax_d + 2x_bx_c - 2x_ax_b - 2x_cx_d \\ (x_a + x_b)^2 + (x_c + x_d)^2 &\geq 2(x_a + x_b)(x_c + x_d) \\ (x_a + x_b - x_c - x_d)^2 &\geq 0.\end{aligned}$$

So we are done.

Solution 4 (Vector bash)

Let a , b , c , d correspond to the position vectors of points A , B , C , and D , respectively, with respect to an arbitrary origin O . Let us also for simplicity define $a^2 = a \cdot a = ||a||^2$, where $||a||$ is the magnitude of vector a . Because squares are non-negative, a^2 is non-negative for all vectors a . Thus,

$$(a + b - c - d)^2 \geq 0$$

Because dot product is linear, we expand to obtain

$$a^2 + b^2 + c^2 + d^2 + 2a \cdot b + 2c \cdot d - 2a \cdot c - 2a \cdot d - 2b \cdot c - 2c \cdot d \geq 0,$$

from which we add $a^2 + b^2 + c^2 + d^2$ to both sides, rearrange, and complete the square to get

$$(a - c)^2 + (a - d)^2 + (b - c)^2 + (b - d)^2 \geq (a - b)^2 + (c - d)^2.$$

As $(a - b)^2 = ||a - b||^2 = ||AB||^2 = AB^2$ and likewise for the others,

$$AC^2 + AD^2 + BC^2 + BD^2 \geq AB^2 + CD^2,$$

which is what we wanted to prove.

NOTES:

- Equality holds when the vector equality $a + b = c + d$ holds, which occurs when A , B , C , and D are the vertices of a (planar) parallelogram, in that order.
- The algebra employed here is almost identical to the algebra used in Solution 3, which means that Solution 3 is just a simplification of this solution.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

- (USAMO 1975) If $P(x)$ denotes a polynomial of degree n such that $P(k) = k/(k+1)$ for $k = 0, 1, 2, \dots, n$, determine $P(n+1)$.

Solution 1

Let $Q(x) = (x+1)P(x) - x$, and clearly, $Q(x)$ has a degree of $n+1$.

Then, for $k = 0, 1, 2, \dots, n$, $Q(k) = (k+1)P(k) - k = (k+1) \cdot \frac{k}{k+1} - k = 0$.

Thus, $k = 0, 1, 2, \dots, n$ are the roots of $Q(x)$.

Since these are all $n + 1$ of the roots of the $n + 1^{\text{th}}$ degree polynomial, by the **Factor Theorem**, we can write $Q(x)$ as

$$Q(x) = c(x)(x - 1)(x - 2) \cdots (x - n)$$

where c is a constant.

Thus,

$$(x + 1)P(x) - x = c(x)(x - 1)(x - 2) \cdots (x - n).$$

We plug in $x = -1$ to cancel the $(x + 1)P(x)$ and find c :

$$\begin{aligned} -(-1) &= c(-1)(-1 - 1)(-1 - 2) \cdots (-1 - n) \\ 1 &= c(-1)^{n+1}(1)(2) \cdots (n + 1) \\ c &= (-1)^{n+1} \frac{1}{(n + 1)!} \end{aligned}$$

Finally, plugging in $x = n + 1$ to find $P(n + 1)$ gives:

$$\begin{aligned} Q(n + 1) &= (n + 2)P(n + 1) - (n + 1) \\ (-1)^{n+1} \frac{1}{(n + 1)!} \cdot (n + 1)! &= (n + 2)P(n + 1) - (n + 1) \\ (-1)^{n+1} &= (n + 2)P(n + 1) - (n + 1) \\ (-1)^{n+1} + (n + 1) &= (n + 2)P(n + 1) \\ P(n + 1) &= \frac{(-1)^{n+1} + (n + 1)}{n + 2} \end{aligned}$$

If n is even, this simplifies to $P(n + 1) = \frac{n}{n + 2}$. If n is odd, this simplifies to $P(n + 1) = 1$. $\square$

Solution 2

It is fairly natural to use Lagrange's Interpolation Formula on this problem:

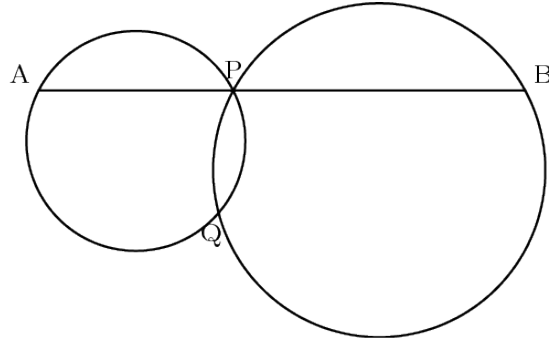
$$\begin{aligned} P(n + 1) &= \sum_{k=0}^n \frac{k}{k + 1} \prod_{j \neq k} \frac{n + 1 - j}{k - j} \\ &= \sum_{k=0}^n \frac{k}{k + 1} \cdot \frac{\frac{(n+1)!}{n+1-k}}{k(k-1)(k-2) \cdots 1 \cdot (-1)(-2) \cdots (k-n)} \\ &= \sum_{k=0}^n \frac{k}{k + 1} (-1)^{n-k} \cdot \frac{(n + 1)!}{k!(n + 1 - k)!} \\ &= \sum_{k=0}^n (-1)^{n-k} \binom{n + 1}{k} - \sum_{k=0}^n \frac{(n + 1)!(-1)^{n-k}}{(k + 1)!(n + 1 - k)!} \end{aligned}$$

$$\begin{aligned}
&= - \left(\sum_{k=0}^{n+1} (-1)^{n+1-k} \binom{n+1}{k} - 1 \right) + \frac{1}{n+2} \cdot \sum_{k=0}^n (-1)^{n+1-k} \binom{n+2}{k+1} \\
&= 1 + \frac{1}{n+2} \left(\sum_{k=-1}^{n+1} (-1)^{n+2-(k+1)} \binom{n+2}{k+1} - (-1)^{n+2} - 1 \right) \\
&= \boxed{1 - \frac{(-1)^n + 1}{n+2}}
\end{aligned}$$

through usage of the Binomial Theorem. $\square$

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

4. (*USAMO 1975*) Two given circles intersect in two points P and Q . Show how to construct a segment AB passing through P and terminating on the two circles such that $AP \cdot PB$ is a maximum.



Solution

A maximum $AP \cdot PB$ cannot be attained if AB intersects segment O_1O_2 because a larger value can be attained by making one of A or B diametrically opposite P , which (as is easily checked) increases the value of both AP and PB . Thus, assume AB does not intersect O_1O_2 .

Let E and F be the centers of the small and big circles, respectively, and r and R be their respective radii.

Let M and N be the feet of E and F to AB , and $\alpha = \angle APE$ and $\epsilon = \angle BPF$

We have:

$$AP \times PB = 2r \cos \alpha \times 2R \cos \epsilon = 4rR \cos \alpha \cos \epsilon$$

$AP \times PB$ is maximum when the product $\cos \alpha \cos \epsilon$ is a maximum.

We have $\cos \alpha \cos \epsilon = \frac{1}{2}[\cos(\alpha + \epsilon) + \cos(\alpha - \epsilon)]$

But $\alpha + \epsilon = 180^\circ - \angle EPF$ and is fixed, so is $\cos(\alpha + \epsilon)$.

So its maximum depends on $\cos(\alpha - \epsilon)$ which occurs when $\alpha = \epsilon$. To draw the line AB :

Draw a circle with center P and radius PE to cut the radius PF at H . Draw the line parallel to EH passing through P . This line meets the small and big circles at A and B , respectively.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

5. (*USAMO 1975*) A deck of n playing cards, which contains three aces, is shuffled at random (it is assumed that all possible card distributions are equally likely). The cards are then turned up one by one from the top until the second ace appears. Prove that the expected (average) number of cards to be turned up is $(n + 1)/2$.

Solution 1

We begin by induction. Our base case is naturally when $n = 3$, as there can be no less than 3 cards in the deck. The only way to turn up the second ace is to turn up the first, and then turn up the second, which requires 2 moves. This indeed is equal to $(3 + 1)/2$. Next, we assume that the expected number of cards required to pull the second ace from a deck of n cards is in fact $(n + 1)/2$. Now, we see that to simulate the case with $n + 1$ cards, we simply must add a card to our deck. By our assumption, we can expect to turn up $(n + 1)/2$ cards to be turned up, including our second ace. Thus ahead of our second ace, we expect $(n - 1)/2$ cards, and after it $(n - 1)/2$ as well. When we add our card, it is ahead of the second ace half the time, and behind it half the time. Therefore we expect to add one card to the number of cards drawn half the time, and add nothing for half the time. Hence we expect to add half a card, and the expected value is now $((n + 1)/2 + 1/2)$, or $(n + 2)/2$, the desired outcome for the $n + 1$ case.

Solution 2

Although we can induct as in the situation above, we can use a more direct approach. Consider a deck. The second ace is in the k th place. If we draw the deck backwards (which has equal probability as drawing it forwards), then we will draw it in the $n + 1 - k$ th place, since the n th place will become the first, the $n - 1$ th the second, etc.

Also, note that when second ace is still the second ace when we draw the deck backwards. (we need to note this, else this argument holds for the first ace and third ace as well)

Now, this implies that the expected location of the second ace is in the $(k + n + 1 - k)/2 = (n + 1)/2$ place.

Solution 3

Like Solution 2, this solution is also a direct solution.

The three aces divide the deck into four piles with a non-negative number of cards. Let a, b, c , and d be the number of cards in the first, second, third, and fourth piles, respectively, given a random deck D . Also, let $E(a), E(b), E(c), E(d)$ denote the expected value of a, b, c , and d over all decks D . Then $a + b + c + d = n - 3$ (the deck minus the three aces).

For every deck D , there must exist other decks that permute the values of a, b, c, d . (i.e. if there is a deck with $a = 2, b = 3, c = 1, d = 4$, there exists a deck with $a = 1, b = 2, c = 3, d = 4$.) Therefore, the value of a over all decks D is a permutation of the value of b over all such decks, and as a result $E(a) = E(b)$. A similar result holds for b, c , and d , and so $E(a) = E(b) = E(c) = E(d)$. Because

$$E(a) + E(b) + E(c) + E(d) = E(a + b + c + d) = E(n - 3) = n - 3,$$

$E(a) = E(b) = \frac{n-3}{4}$. We want to compute the number of cards up to and including the second ace, which is just

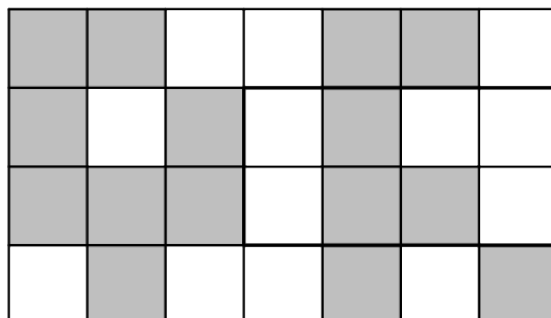
$$E(a) + 1 + E(b) + 1 = \frac{n-3}{2} + 2 = \frac{n+1}{2},$$

as desired.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

§5 USAMO 1976

1. (*USAMO 1976*)



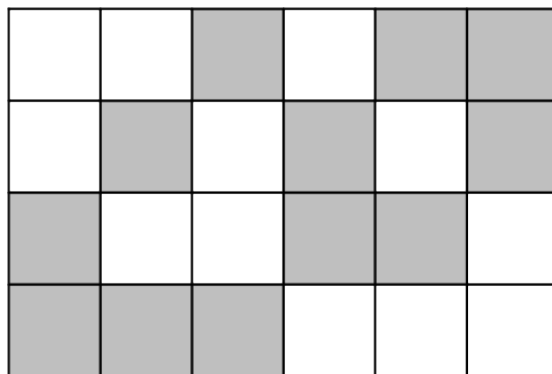
- (a) Suppose that each square of a 4×7 chessboard, as shown above, is colored either black or white. Prove that with *any* such coloring, the board must contain a rectangle (formed by the horizontal and vertical lines of the board such as the one outlined in the figure) whose four distinct unit corner squares are all of the same color.
- (b) Exhibit a black-white coloring of a 4×6 board in which the four corner squares of every rectangle, as described above, are not all of the same color.

Solution 1

There are many ways to prove the first part, we will show one.

Consider the first row. It contains at least four cells of the same color, say white. Pick any four such cells. Let's now consider these four columns only. If any of the remaining three rows contains two white cells in some of these columns, we are done. Otherwise, we have the following situation: three rows times four columns, and in each row there is at most one white cell. We can now pick any two rows and then find two columns such that all four cells where they intersect are black, and we are done. (Note that this proof in fact shows that already in a 3×7 board there must be such a rectangle.)

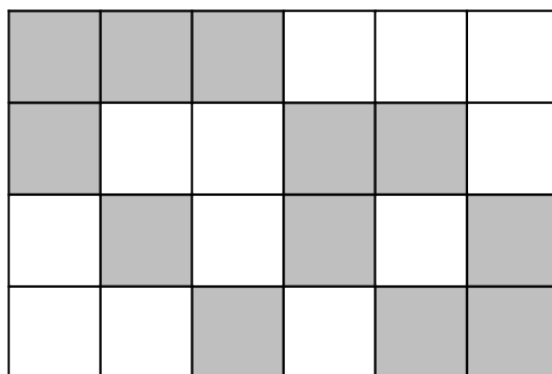
For the second part, consider the board shown below. Its six columns contain the six different permutations of two white and two black squares. From this observation it is immediately obvious that no two columns agree on two cells of the same color.



Solution 2

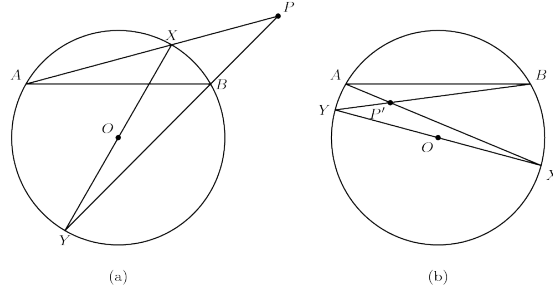
We will prove the first part by the pigeonhole principle. To form a rectangle, two squares have in one column have to be colored the same color as the corresponding squares in another column. In the example given, the two second and third squares of the fourth and seventh columns are all white. When coloring this rectangle, every column can be entirely one color, have 3 squares be of one color, or have 2 squares be of one color. Coloring each column with two white squares and two black squares will minimize the number of squares, since a smaller fraction of the column is one color (if more than 2 squares in the same column are the same color, then there is more than one coloring in another column that can create a rectangle). There are $\binom{4}{2}$, or 6 different ways a column can be colored with two white and two black squares. We need the same coloring in two different columns to create a rectangle, and this is guaranteed with 7 columns by the pigeonhole principle.

For the second part, we can draw a rectangle with each column having a different coloring with 2 black and 2 white squares. One such rectangle is shown below:



<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

2. (*USAMO 1976*) If A and B are fixed points on a given circle and XY is a variable diameter of the same circle, determine the locus of the point of intersection of lines AX and BY . You may assume that AB is not a diameter.



Solution

WLOG, assume that the circle is the unit circle centered at the origin. Then the points A and B have coordinates $(-a, b)$ and (a, b) respectively and X and Y have coordinates (r, s) and $(-r, -s)$. Note that these coordinates satisfy $a^2 + b^2 = 1$ and $r^2 + s^2 = 1$ since these points are on a unit circle. Now we can find equations for the lines:

$$\begin{aligned} AX &\longrightarrow y = \frac{(s - b)x + rb + sa}{r + a} \\ BY &\longrightarrow y = \frac{(s + b)x + rb - sa}{r + a}. \end{aligned}$$

Solving these simultaneous equations gives coordinates for P in terms of a, b, r , and s : $P = \left(\frac{as}{b}, \frac{1-ar}{b}\right)$. These coordinates can be parametrized in Cartesian variables as follows:

$$\begin{aligned} x &= \frac{as}{b} \\ y &= \frac{1 - ar}{b}. \end{aligned}$$

Now solve for r and s to get $r = \frac{1-by}{a}$ and $s = \frac{bx}{a}$. Then since $r^2 + s^2 = 1$, $\left(\frac{bx}{a}\right)^2 + \left(\frac{1-by}{a}\right)^2 = 1$ which reduces to $x^2 + (y - 1/b)^2 = \frac{a^2}{b^2}$. This equation defines a circle and is the locus of all intersection points P . In order to define this locus more generally, find the slope of this circle function using implicit differentiation:

$$\begin{aligned} 2x + 2(y - 1/b)y' &= 0 \\ (y - 1/b)y' &= -x \\ y' &= \frac{-x}{y - 1/b}. \end{aligned}$$

Now note that at points A and B , this slope expression reduces to $y' = \frac{-b}{a}$ and $y' = \frac{b}{a}$ respectively, values which are identical to the slopes of lines AO and BO . Thus we conclude that the complete locus of intersection points is the circle tangent to lines AO and BO at points A and B respectively.

Solution 2

Notice that $m\angle AYP = \frac{1}{2}m\widehat{AB}$ (Inscribed angle theorem) and that $m\angle XAY = 90$ since XY is a diameter, and thus subtends an arc of 180 . This will hold for all X and all Y , and so by AA similarity, the angle APY will be constant for all P , thus implying that the points A, B , and all P will be concyclic. If we assume that the center of this circle is H , we know that $m\angle AHB = 2m\angle APB = 2(90 - \frac{1}{2}m\widehat{AB}) = 180 - m\widehat{AB}$. We can assume that $\overline{AH}$ and $\overline{BH}$ intersect the original circle at points M ,

and N respectively. This will give us that $m\angle AHB = \frac{1}{2}|\widehat{m6.0ptMN} - \widehat{m6.0ptAB}|$ (Since H lies on the perpendicular bisector of AB , we know that M, N will be on the same side of AB .) Now we also know that $\widehat{m6.0ptMN} \leq \widehat{m6.0ptAB}$ or $\widehat{m6.0ptMN} \leq 360 - \widehat{m6.0ptAB}$. The only case where $\widehat{m6.0ptMN}$ satisfies the measure of $\angle AHB$, is when $M = A$ and $N = B$, implying that AH and BH are tangents, and so H is the intersection of the tangents from A and B to the original circle.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

3. (*USAMO 1976*) Determine all integral solutions of $a^2 + b^2 + c^2 = a^2b^2$.

Solution

Either $a^2 = 0$ or $a^2 > 0$. If $a^2 = 0$, then $b^2 = c^2 = 0$. Symmetry applies for b as well. If $a^2, b^2 \neq 0$, then $c^2 \neq 0$. Now we look at $a^2 \pmod 4$:

$a^2 \equiv 0 \pmod 4$: Since a square is either 1 or 0 mod 4, then all the other squares are 0 mod 4. Let $a = 2a_1$, $b = 2b_1$, and $c = 2c_1$. Thus $a_1^2 + b_1^2 + c_1^2 = 4a_1^2b_1^2$. Since the LHS is divisible by four, all the variables are divisible by 4, and we must do this over and over again, and from infinite descent, there are no non-zero solutions when $a^2 \equiv 0 \pmod 4$.

$a^2 \equiv 1 \pmod 4$: Since $b^2 \not\equiv 0 \pmod 4$, $b^2 \equiv 1 \pmod 4$, and $2 + c^2 \equiv 1 \pmod 4$. But for this to be true, $c^2 \equiv 3 \pmod 4$, which is an impossibility. Thus there are no non-zero solutions when $a^2 \equiv 1 \pmod 4$.

Thus the only solution is the solution above: $(a, b, c) = 0$.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

4. (*USAMO 1976*) If the sum of the lengths of the six edges of a trirectangular tetrahedron $PABC$ (i.e., $\angle APB = \angle BPC = \angle CPA = 90^\circ$) is S , determine its maximum volume.

Solution

Let the side lengths of AP , BP , and CP be a , b , and c , respectively. Therefore $S = a + b + c + \sqrt{a^2 + b^2} + \sqrt{b^2 + c^2} + \sqrt{c^2 + a^2}$. Let the volume of the tetrahedron be V . Therefore $V = \frac{abc}{6}$.

Note that $(a-b)^2 \geq 0$ implies $\frac{a^2-2ab+b^2}{2} \geq 0$, which means $\frac{a^2+b^2}{2} \geq ab$, which implies $a^2 + b^2 \geq ab + \frac{a^2+b^2}{2}$, which means $a^2 + b^2 \geq \frac{(a+b)^2}{2}$, which implies $\sqrt{a^2 + b^2} \geq \frac{1}{\sqrt{2}} \cdot (a + b)$. Equality holds only when $a = b$. Therefore

$$S \geq a + b + c + \frac{1}{\sqrt{2}} \cdot (a + b) + \frac{1}{\sqrt{2}} \cdot (c + b) + \frac{1}{\sqrt{2}} \cdot (a + c) \\ = (a + b + c)(1 + \sqrt{2}).$$

$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$ is true from AM-GM, with equality only when $a = b = c$. So $S \geq (a + b + c)(1 + \sqrt{2}) \geq 3(1 + \sqrt{2})\sqrt[3]{abc} = 3(1 + \sqrt{2})\sqrt[3]{6V}$. This means that $\frac{S}{3(1+\sqrt{2})} = \frac{S(\sqrt{2}-1)}{3} \geq \sqrt[3]{6V}$, or $6V \leq \frac{S^3(\sqrt{2}-1)^3}{27}$, or $V \leq \frac{S^3(\sqrt{2}-1)^3}{162}$, with equality only when $a = b = c$. Therefore the maximum volume is $\frac{S^3(\sqrt{2}-1)^3}{162}$.

Solution 2

Note that by AM-GM

$$S = a+b+c+\sqrt{a^2+b^2}+\sqrt{b^2+c^2}+\sqrt{c^2+a^2} \geq 3\sqrt[3]{abc}+\sqrt{2ab}+\sqrt{2bc}+\sqrt{2ac} \geq 3\sqrt[3]{abc}+3\sqrt{2}\sqrt[3]{abc},$$

so

$$\sqrt[3]{abc} \leq \frac{S}{3+3\sqrt{2}} = \frac{S}{3} \cdot (\sqrt{2}-1).$$

Proceed as before.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

5. (*USAMO 1976*) If $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ are all **polynomials** such that

$$P(x^5) + xQ(x^5) + x^2R(x^5) = (x^4 + x^3 + x^2 + x + 1)S(x),$$

prove that $x-1$ is a factor of $P(x)$.

Solution 1

In general we will show that if m is an integer less than n and $P_0, \dots, P_{m-1}$ and S are polynomials satisfying

$$P_0(x^n) + xP_1(x^n) + \dots + x^{m-1}P_{m-1}(x^n) = \sum_{k=0}^{n-1} x^k S(x),$$

then $P_k(1) = 0$, for all integers $0 \leq k \leq m-1$. For the problem, we may set $n = 5$, $m = 3$, and then note that since $P(1) = 0$, $(x-1)$ is a factor of $P(x)$.

Indeed, let $\omega_1, \dots, \omega_{n-1}$ be the n th **roots of unity** other than 1. Then for all integers $1 \leq i \leq n-1$,

$$\sum_{k=0}^{m-1} \omega_i^k P_k(\omega_i^n) = \sum_{k=0}^{m-1} \omega_i^k P_k(1) = \sum_{k=0}^{n-1} \omega_i^k S(\omega_i) = \frac{\omega_i^{n-1} - 1}{\omega_i - 1} S(\omega_i) = 0,$$

for all integers $1 \leq i \leq n$. This means that the $(m-1)$ th degree polynomial

$$\sum_{k=0}^{m-1} x^k P_k(1)$$

has $n-1 > m-1$ distinct roots. Therefore all its coefficients must be zero, so $P_k = 0$ for all integers $0 \leq k \leq m-1$, as desired. ■

Solution 2

Let ζ, ξ, ρ be three distinct primitive fifth **roots of unity**. Setting $x = \zeta, \xi$, we have

$$\begin{aligned} P(1) + \zeta Q(1) + \zeta^2 R(1) &= \frac{\zeta^5 - 1}{\zeta - 1} S(\zeta) = 0, \\ P(1) + \xi Q(1) + \xi^2 R(1) &= \frac{\xi^5 - 1}{\xi - 1} S(\xi) = 0. \end{aligned}$$

These equations imply that

$$\zeta Q(1) + \zeta^2 R(1) = \xi Q(1) + \xi^2 R(1),$$

or

$$Q(1) = -(\zeta + \xi)R(1).$$

But by symmetry,

$$Q(1) = -(\zeta + \rho)R(1).$$

Since $\xi \neq \rho$, it follows that $Q(1) = R(1) = 0$. Then, as noted above,

$$P(1) = P(1) + \zeta Q(1) + \zeta^2 R(1) = 0,$$

so $(x - 1)$ is a factor of $P(x)$, as desired. ■

Solution 3

Let z, z^2, z^3 be three of the 5th roots of unity not equal to one that satisfy $1 + z + z^2 + z^3 + z^4 = 0$ as a result. Plugging them into the equation gives the linear system of equations in $(A(1), B(1), C(1))$:

$$A(1) + zB(1) + z^2C(1) = 0$$

$$A(1) + z^2B(1) + z^4C(1) = 0$$

$$A(1) + z^3B(1) + z^6C(1) = 0$$

Direct observation gives $(A(1), B(1), C(1)) = (0, 0, 0)$ as a solution, and there are no others because the system of equations is linear. Hence, $A(1) = 0$, and so $(x - 1)$ is a factor of $A(x)$, as desired. ■

Note: We can generalize this approach to prove the claim in Solution 1 in an equally fast, concise, and readily understandable fashion.

§6 USAMO 1977

1. (USAMO 1977) Determine all pairs of positive integers (m, n) such that $(1 + x^n + x^{2n} + \dots + x^{mn})$ is divisible by $(1 + x + x^2 + \dots + x^m)$.

Solution 1

Denote the first and larger polynomial to be $f(x)$ and the second one to be $g(x)$. In order for $f(x)$ to be divisible by $g(x)$ they must have the same roots. The roots of $g(x)$ are the $(m+1)$ th roots of unity, except for 1. When plugging into $f(x)$, the root of unity is a root of $f(x)$ if and only if the terms $x^n, x^{2n}, x^{3n}, \dots, x^{mn}$ all represent a different $(m+1)$ th root of unity not equal to 1.

Note that if

$\gcd(m+1, n) = 1$, the numbers $n, 2n, 3n, \dots, mn$ represent a complete set of residues minus 0 modulo $m+1$. However, if $\gcd(m+1, n) = a$ not equal to 1, then $\frac{(m+1)(n)}{a}$ is congruent to 0 (mod $m+1$) and thus a complete set is not formed.

Therefore, $f(x)$ divides $g(x)$ if and only if $\boxed{\gcd(m+1, n) = 1}$. ■

Solution 2

We could instead consider $f(x)$ modulo $g(x)$. Notice that $x^{m+1} = 1 \pmod{g(x)}$, and thus we can reduce the exponents of $f(x)$ to their equivalent modulo $m+1$. We want the resulting $h(x)$ with degree less than $m+1$ to be equal to $g(x)$ (of degree m), which implies that the exponents of $f(x)$ must be all different modulo $m+1$. This can only occur if and only if $\gcd(m+1, n) = 1$, and this is our answer, as shown in Solution 1.

Solution 3

Notice that $1 + x^n + x^{2n} + \dots + x^{mn} = \frac{x^{mn+n}-1}{x^n-1}$ and $1 + x + x^2 + \dots + x^m = \frac{x^{m+1}-1}{x-1}$, so it remains to prove that $(x^n - 1)(x^{m+1} - 1) \mid (x^{mn+n} - 1)(x - 1)$. It is clear that $(x^n - 1) \mid (x^{mn+n} - 1)$ and $(x^{m+1} - 1) \mid (x^{mn+n} - 1)$. For any $k \in \mathbb{N}$, we

can use the fact that $x^k - 1 = \prod_{d|k} \Phi_d(x)$, where $\Phi_d(x)$ is the d th cyclotomic polynomial. If $\gcd(m+1, n) = d > 1$, then $x^{m+1} - 1$ and $x^n - 1$ share a common cyclotomic polynomial; namely, $\Phi_d(x)$. But since all the factors of $x^{mn+n} - 1$ are distinct, $(x^{mn+n} - 1)(x - 1)$ cannot be divisible by $(x^n - 1)(x^{m+1} - 1)$. We find that $\gcd(m+1, n) = 1$ must be the solution, since the only shared polynomial is $\Phi_1(x) = x - 1$, and we are done.

2. (USAMO 1977) ABC and $A'B'C'$ are two triangles in the same plane such that the lines AA', BB', CC' are mutually parallel. Let $[ABC]$ denote the area of triangle ABC with an appropriate $\pm$ sign, etc.; prove that

$$3([ABC] + [A'B'C']) = [AB'C'] + [BC'A'] + [CA'B'] + [A'BC] + [B'CA] + [C'AB].$$

3. (USAMO 1977) If a and b are two of the roots of $x^4 + x^3 - 1 = 0$, prove that ab is a root of $x^6 + x^4 + x^3 - x^2 - 1 = 0$.

Solution

Given the roots a, b, c, d of the equation $x^4 + x^3 - 1 = 0$.

First, Vieta's relations give $a + b + c + d = -1, ab + ac + ad + bc + bd + cd = 0, abc + abd + acd + bcd = 0, abcd = -1$.

Then $cd = -\frac{1}{ab}$ and $c + d = -1 - (a + b)$.

The other coefficients give $ab + (a + b)(c + d) + cd = 0$ or $ab + (a + b)[-1 - (a + b)] - \frac{1}{ab} = 0$.

Let $a + b = s$ and $ab = p$.

Thus, $0 = ab + ac + ad + bc + bd + cd = p + s(-1 - s) - \frac{1}{p}$. (1)

Also, $0 = abc + abd + acd + bcd = p(-1 - s) - s/p$.

Solving this equation for s , $s = \frac{-p^2}{p^2 + 1}$.

Substituting into (1): $\frac{p^6 + p^4 + p^3 - p^2 - 1}{p(p^2 + 1)^2} = 0$.

Conclusion: $p = ab$ is a root of $x^6 + x^4 + x^3 - x^2 - 1 = 0$.

4. (USAMO 1977) Prove that if the opposite sides of a skew (non-planar) quadrilateral are congruent, then the line joining the midpoints of the two diagonals is perpendicular to these diagonals, and conversely, if the line joining the midpoints of the two diagonals of a skew quadrilateral is perpendicular to these diagonals, then the opposite sides of the quadrilateral are congruent.

Solution

We first prove that if the opposite sides are congruent, then the line joining the midpoints of the two diagonals is perpendicular to these diagonals.

Let the vertices of the quadrilateral be A, B, C, D , in that order. Thus, the opposite sides congruent condition translates to (after squaring):

$$AB^2 = CD^2, BC^2 = AD^2$$

Let the position vectors of A, B, C, D be a, b, c, d with respect to an arbitrary origin O . Define the square of a vector $a^2 = a \cdot a$. Thus, $(a - b)^2 = (c - d)^2$ and so

$$a^2 - 2ab + b^2 = c^2 - 2cd + d^2.$$

Similarly,

$$b^2 - 2bc + c^2 = a^2 - 2ad + d^2.$$

Subtracting the second equation from the first eventually results in

$$a^2 - ab + cd + bc - ad - c^2 = 0,$$

which factors as

$$(a - b + c - d)(a - c) = 0.$$

Let M and N be the midpoints of AC and BD , with position vectors $m = \frac{a+c}{2}$ and $n = \frac{b+d}{2}$, respectively. Then notice that the perpendicularity condition for vectors gives MN perpendicular to AC . Similarly (by adding the equations), we find that MN is perpendicular to BD , completing the first part of the proof.

Proving the converse is straightforward also: indeed, MN perpendicular to both diagonals gives

$$\begin{aligned} a^2 - ab + cd + bc - ad - c^2 &= 0 \\ b^2 - ab + cd - bc + ad - d^2 &= 0 \end{aligned}$$

Adding and subtracting the equations eventually simplifies to

$$\begin{aligned} (a - b)^2 &= (c - d)^2 \\ (a - d)^2 &= (b - c)^2, \end{aligned}$$

or $AB^2 = CD^2$ and $AD^2 = BC^2$. This reduces to $AB = CD$ and $AD = BC$, as desired. ■

Solution 2

Let A, B, C, D be position vectors rather than the specified points. We're given that $|A - B| = |C - D| \rightarrow |A - B|^2 = |C - D|^2 \rightarrow (A - B) \cdot (A - B) = (C - D) \cdot (C - D) \rightarrow A \cdot A - 2A \cdot B + B \cdot B - C \cdot C + 2C \cdot D - D \cdot D = 0$. We're also given that $|A - D| = |C - B| \rightarrow |A - D|^2 = |C - B|^2 \rightarrow (A - D) \cdot (A - D) = (C - B) \cdot (C - B) \rightarrow A \cdot A - 2A \cdot D + D \cdot D - C \cdot C + 2C \cdot B - B \cdot B = 0$. Adding the two equations gives $2A \cdot A - 2A \cdot B - 2A \cdot D - 2C \cdot C + 2C \cdot D + 2C \cdot B = 0$. Dividing by 4 and rearranging gives $\frac{A+C-B-D}{2} \cdot (A - C) = 0$. On the left hand side of this equation, one of the two factors is the vector between the midpoints of the two diagonals, and the other factor is one of the two diagonals. Their dot product is 0, so they are perpendicular. The same methods can be used to show that the vector between the two midpoints is perpendicular to the other diagonal. We can employ this process in reverse to get the converse (this isn't completely simple, but I'm lazy so it's left as an exercise i guess).

5. (USAMO 1977) If a, b, c, d, e are positive numbers bounded by p and q , i.e, if they lie in $[p, q], 0 < p$, prove that

$$(a + b + c + d + e) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{1}{e} \right) \leq 25 + 6 \left(\sqrt{\frac{p}{q}} - \sqrt{\frac{q}{p}} \right)^2$$

and determine when there is equality.

Solution

Fix four of the variables and allow the other to vary. Suppose, for example, we fix all but x . Then the expression on the LHS has the form $(r+x)(s+\frac{1}{x}) = (rs+1) + sx + \frac{r}{x}$, where r and s are fixed. But this is convex. That is to say, as x increases it first decreases, then increases. So its maximum must occur at $x = p$ or $x = q$. This is true for each variable.

Suppose all five are p or all five are q , then the LHS is 25, so the inequality is true and strict unless $p = q$. If four are p and one is q , then the LHS is $17 + 4(\frac{p}{q} + \frac{q}{p})$. Similarly if four are q and one is p . If three are p and two are q , then the LHS is $13 + 6(\frac{p}{q} + \frac{q}{p})$. Similarly if three are q and two are p .

$\frac{p}{q} + \frac{q}{p} \geq 2$ with equality iff $p = q$, so if $p < q$, then three of one and two of the other gives a larger LHS than four of one and one of the other. Finally, we note that the RHS is in fact $13 + 6(\frac{p}{q} + \frac{q}{p})$, so the inequality is true with equality iff either (1) $p = q$ or (2) three of v, w, x, y, z are p and two are q or vice versa.

§7 USAMO 1978

1. (USAMO 1978) Given that a, b, c, d, e are real numbers such that

$$a + b + c + d + e = 8,$$

$$a^2 + b^2 + c^2 + d^2 + e^2 = 16.$$

Determine the maximum value of e .

Solution 1

By Cauchy Schwarz, we can see that $(1+1+1+1)(a^2+b^2+c^2+d^2) \geq (a+b+c+d)^2$ thus $4(16-e^2) \geq (8-e)^2$ Finally, $e(5e-16) \geq 0$ which means $\frac{16}{5} \geq e \geq 0$ so the maximum value of e is $\frac{16}{5}$.

from: [Image from Gon Mathcenter.net](#)

Solution 2

Seeing as we have an inequality with constraints, we can use Lagrange multipliers to solve this problem. We get the following equations:

- (1) $a + b + c + d + e = 8$
- (2) $a^2 + b^2 + c^2 + d^2 + e^2 = 16$
- (3) $0 = \lambda + 2a\mu$
- (4) $0 = \lambda + 2b\mu$
- (5) $0 = \lambda + 2c\mu$
- (6) $0 = \lambda + 2d\mu$
- (7) $1 = \lambda + 2e\mu$

If $\mu = 0$, then $\lambda = 0$ according to (6) and $\lambda = 1$ according to (7), so $\mu \neq 0$. Setting the right sides of (3) and (4) equal yields $\lambda + 2a\mu = \lambda + 2b\mu \implies 2a\mu = 2b\mu \implies a = b$. Similar steps yield that $a = b = c = d$. Thus, (1) becomes $4d + e = 8$ and (2) becomes $4d^2 + e^2 = 16$. Solving the system yields $e = 0, \frac{16}{5}$, so the maximum possible value of e is $\frac{16}{5}$.

Solution 3

A re-writing of Solution 1 to avoid the use of Cauchy Schwarz. We have

$$(a + b + c + d)^2 = (8 - e)^2,$$

and

$$a^2 + b^2 + c^2 + d^2 = 16 - e^2.$$

The second equation times 4, then minus the first equation,

$$(a-b)^2 + (a-c)^2 + (a-d)^2 + (b-c)^2 + (b-d)^2 + (c-d)^2 = 4(16 - e^2) - (8 - e)^2.$$

The rest follows.

J.Z.

Solution 4

By the **Principle of Insufficient Reasons**, since a, b, c, d are indistinguishable variables, the maximum of e is achieved when $a = b = c = d$, so we have

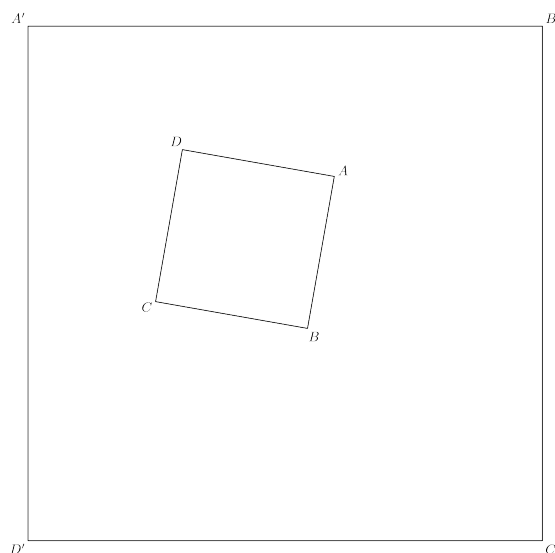
$$4a + \max e = 8$$

$$4a^2 + (\max e)^2 = 16$$

$$\implies \max e = \boxed{\frac{16}{5}}$$

. □ **Ddk001**

- Note: For some reason I think this solution is missing something.
2. (USAMO 1978) $ABCD$ and $A'B'C'D'$ are square maps of the same region, drawn to different scales and superimposed as shown in the figure. Prove that there is only one point O on the small map that lies directly over point O' of the large map such that O and O' each represent the same place of the country. Also, give a Euclidean construction (straight edge and compass) for O .



Solution 1

The point is obviously unique, because the two maps have different scales (but if P and Q where two fixed points the distance between them would be the same on both maps).

Let the small map square be $A'B'C'D'$ and the large be $ABCD$, where X and X' are corresponding points. We deal first with the special case where $A'B'$ is parallel

to AB . In this case let AA' and BB' meet at O . Then triangles OAB and $OA'B'$ are similar, so O must represent the same point. So assume $A'B'$ is not parallel to AB .

Let the lines $A'B'$ and AB meet at W , the lines $B'C'$ and BC meet at X , the lines $C'D'$ and CD meet at Y , and the lines $D'A'$ and DA meet at Z . We claim that the segments WY and XZ meet at a point O inside the smaller square. W cannot lie between A' and B' (or one of the vertices A' , B' of the smaller square would lie outside the larger square). If it lies on the opposite side of A' to B' , then Y must lie on the opposite side of C' to D' . Thus the segment WY must cut the side $A'D'$ at some point Z' and the side $B'C'$ at some point X' . The same conclusion holds if W lies on the opposite side of B' to A' , because then Y must lie on the opposite side of D' to C' . Similarly, the segment XZ must cut the side $A'B'$ at some point W' and the side $C'D'$ at some point Y' . But now the segments $X'Z'$ and $W'Y'$ join pairs of points on opposite sides of the small square and so they must meet at some point O inside the small square.

Now the triangles WOW' and YOY' are similar (WW' and YY' are parallel). Hence $OW/OY = OW'/OY'$. So if we set up coordinate systems with AB as the x -axis and AD as the y -axis (for the large square) and $A'B'$ as the x' -axis and $A'D'$ as the y' -axis (for the small square) so that corresponding points have the same coordinates, then the y coordinate of O equals the y' coordinate of O . Similarly, XOX' and ZOZ' are similar, so $OX/OZ = OX'/OZ'$, so the x -coordinate of O equals its x' -coordinate. In other words, O represents the same point on both maps.

Solution 2

The point is obviously unique because of the different scale and angle, which can also be seen as the point can and must be the unique center of spiral similarity taking $ABCD$ to $A'B'C'D'$, as under the spiral similarity the center does not move. (Recall that a spiral similarity is a dilation combined with a rotation, and that there is a unique spiral similarity taking any segment to another segment, provided the four points do not make a parallelogram.) To construct this center of spiral similarity, reference EGMO Lemma 10.1. Draw line DD' and AA' , and let them intersect at point X . Draw circles through points X, A, D and X, A', D' (using the perpendicular bisector constructions to find the center of the circle using two chords). The intersection point of the two circles other than X is the center of spiral similarity, and hence is O , which represents the same point on both maps.

3. (USAMO 1978) An integer n will be called *good* if we can write

$$n = a_1 + a_2 + \cdots + a_k,$$

where $a_1, a_2, \dots, a_k$ are positive integers (not necessarily distinct) satisfying

$$\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_k} = 1.$$

Given the information that the integers 33 through 73 are good, prove that every integer ≥ 33 is good.

Solution

Lemma: If g is a good integer, then so is $2g + 2$ and $2g + 9$.

Proof: Let $g = a_1 + a_2 + \cdots + a_k$ such that

$$\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_k} = 1,$$

where the a_i are positive integers. Then, notice that

$$1 = \frac{1}{2} + \frac{1}{2} = \frac{1}{2} + \frac{1}{2a_1} + \frac{1}{2a_2} + \cdots + \frac{1}{2a_k},$$

so $g_1 = 2 + 2a_1 + 2a_2 + \dots + 2a_k = 2 + 2g$ is also a good integer. Furthermore,

$$1 = \frac{1}{3} + \frac{1}{6} + \frac{1}{2} = \frac{1}{3} + \frac{1}{6} + \frac{1}{2a_1} + \frac{1}{2a_2} + \cdots + \frac{1}{2a_k},$$

so $g_2 = 3 + 6 + 2a_1 + 2a_2 + \dots + a_k = 9 + 2g$ is also good.

This completes the lemma. Now, we use induction to show that the integers from 33 to n are all good, where $n \geq 33$.

Base case: Our given takes care of $33 \leq n \leq 73$. For the inductive step, we assume that the integers k between 33 and 73, inclusive, are all good. Now, we do some casework on the parity of $k + 1$.

If $k + 1$ is odd, then write $k + 1 = 2g + 9$ for some integer g .

It follows that because $k \geq 74$, we must have

$$g = \frac{k}{2} - 4 \geq 37 - 4 = 33.$$

Thus, $33 \leq g < \frac{k}{2}$, and so g is good by the inductive hypothesis. Applying the lemma gives $2g + 9 = k + 1$ as a good integer.

If $k + 1$ is even, write $k + 1 = 2g + 2$ for some integer g . Therefore,

$$g = \frac{k - 1}{2} > \frac{73 - 1}{2} = 36 > 33,$$

so that $33 < g < \frac{k}{2}$, meaning that g is a good integer by the inductive hypothesis. From the lemma, $k + 1$ must be good as well.

This completes the inductive step and thus the induction. Therefore, all integers $n \geq 33$ are good integers.

4. (USAMO 1978) (a) Prove that if the six dihedral (i.e. angles between pairs of faces) of a given tetrahedron are congruent, then the tetrahedron is regular.
 (b) Is a tetrahedron necessarily regular if five dihedral angles are congruent?

Solution

(a) Let $ABCD$ be the said tetrahedron, and let the inscribed sphere of $ABCD$ touch the faces at W, X, Y, Z . Then, OW, OX, OY, OZ are normals to the respective faces. We know that the angle between any two normals is equal, so we have $|OW| = |OX| = |OY| = |OZ|$ at equal angles. Now, since

$$WX = 2 \cdot OW \sin \frac{\angle WOX}{2},$$

and similar for the other sides, we have that $WXYZ$ is a regular tetrahedron. Now, the faces of $ABCD$ are the tangent planes at W, X, Y , and Z . Then, consider a 120° rotation about OW . The rotation sends $X \mapsto Y, Y \mapsto Z$, and $Z \mapsto X$. Thus we have $AB = AC, AC = AD, AD = AB$, and $BC = CD = DB$. Performing rotations about the other axes yields that $ABCD$ has equal edges, so it is regular.

- (b) Consider the normals OW, OX, OY, OZ . We can perform a transformation in which we slightly shift X, Y , and Z closer such that $\triangle XYZ$ is still equilateral, and such that all angles between every pair of normals is less than 180° . Then, shift W such that $WX = WY = XY$. Then five of the distances are equal but the sixth is not.
5. (*USAMO 1978*) Nine mathematicians meet at an international conference and discover that among any three of them, at least two speak a common language. If each of the mathematicians speak at most three languages, prove that there are at least three of the mathematicians who can speak the same language.

Solution

Suppose no three mathematicians speak the same language. Then person A can share some language with at most 3 other delegates, because if he shared some language with 4 delegates there would be 3 with the same language. So there are 5 delegates who do not share a language with A. Let one of them be B. Using the same logic, one the remaining 4, let it be C, does not share a language with B. Now A, B, C do not have any common languages. Contradiction.

§8 USAMO 1979

1. (*USAMO 1979*) Determine all non-negative integral solutions $(n_1, n_2, \dots, n_{14})$ if any, apart from permutations, of the Diophantine Equation $n_1^4 + n_2^4 + \dots + n_{14}^4 = 1599$.

Solution 1

Recall that $n_i^4 \equiv 0, 1 \pmod{16}$ for all integers n_i . Thus the sum we have is anything from 0 to 14 modulo 16. But $1599 \equiv 15 \pmod{16}$, and thus there are no integral solutions to the given Diophantine equation.

Solution 2

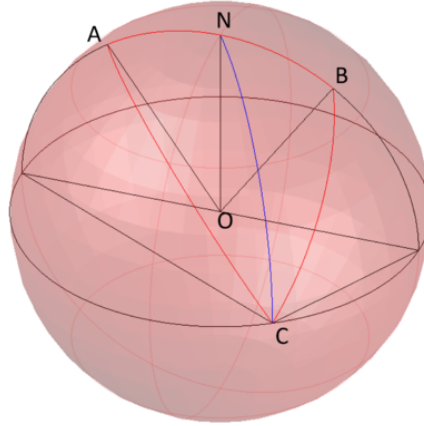
In base 16, this equation would look like:

$$n_1^4 + n_2^4 + \dots + n_{14}^4 = 63F_{16}$$

We notice that the unit digits of the LHS of this equation should equal to F_{16} . In base 16, the only unit digits of fourth powers are 0 and 1. Thus, the maximum of these 14 terms is 14 1's or E_{16} . Since E_{16} is less than F_{16} , there are no integral solutions for this equation.

2. (*USAMO 1979*) N is the north pole. A and B are points on a great circle through N equidistant from N . C is a point on the equator. Show that the great circle through C and N bisects the angle ACB in the spherical triangle ABC (a spherical triangle has great circle arcs as sides).

Solution



Since N is the north pole, we define the Earth with a sphere of radius one in space with $N = (0, 0, 1)$ and sphere center $O = (0, 0, 0)$. We then pick point N on the sphere and define the xz -plane as the plane that contains great circle points A , B , and N with the x -axis perpendicular to the z -axis and in the direction of A .

Using this coordinate system and x , y , and z axes $A = (\cos(\phi), 0, \sin(\phi))$ where ϕ is the angle from the xy -plane to A or latitude on this sphere with $-\frac{\pi}{2} < \phi < \frac{\pi}{2}$.

Since A and B are points on a great circle through N equidistant from N , then $B = (-\cos(\phi), 0, \sin(\phi))$.

Since C is a point on the equator, then $C = (\cos(\theta), \sin(\theta), 0)$ where θ is the angle on the xy -plane from the origin to C or longitude on this sphere with $-\pi < \theta \leq \pi$.

We note that vectors from the origin to points N , A , B , and C are all unit vectors because all those points are on the unit sphere.

So, we're going to define points N , A , B , and C as unit vectors with their coordinates.

We also define the following vectors as follows:

Vector $\overrightarrow{V_{CN}}$ is the unit vector in the direction of arc CN and tangent to the great circle of CN at C .

Vector $\overrightarrow{V_{CA}}$ is the unit vector in the direction of arc CA and tangent to the great circle of CA at C .

Vector $\overrightarrow{V_{CB}}$ is the unit vector in the direction of arc CB and tangent to the great circle of CB at C .

To calculate each of these vectors we shall use the cross product as follows:

$$\overrightarrow{V_{CN}} = (\vec{C} \times \vec{N}) \times \vec{C}$$

$$\overrightarrow{V_{CN}} = (\langle \cos(\theta), \sin(\theta), 0 \rangle \times \langle 0, 0, 1 \rangle) \times \langle \cos(\theta), \sin(\theta), 0 \rangle$$

$$\overrightarrow{V_{CN}} = \langle \sin(\theta), -\cos(\theta), 0 \rangle \times \langle \cos(\theta), \sin(\theta), 0 \rangle$$

$$\overrightarrow{V_{CN}} = \langle 0, 0, \sin^2(\theta) + \cos^2(\theta) \rangle$$

$$\overrightarrow{V_{CN}} = \langle 0, 0, 1 \rangle$$

Vector $\overrightarrow{V_{CA}}$:

$$\overrightarrow{V_{CA}} = (\vec{C} \times \vec{A}) \times \vec{C}$$

$$\overrightarrow{V_{CA}} = (\langle \cos(\theta), \sin(\theta), 0 \rangle \times \langle \cos(\phi), 0, \sin(\phi) \rangle) \times \langle \cos(\theta), \sin(\theta), 0 \rangle$$

$$\overrightarrow{V_{CA}} = \langle \sin(\theta)\sin(\phi), -\cos(\theta)\sin(\phi), -\sin(\theta)\cos(\phi) \rangle \times \langle \cos(\theta), \sin(\theta), 0 \rangle$$

Since we're only interested in the z component of the vector

$$\overrightarrow{V_{CA}} = \langle V_{CA_x}, V_{CA_y}, \sin^2(\theta)\sin(\phi) + \cos^2(\theta)\sin(\phi) \rangle$$

$$\overrightarrow{V_{CA}} = \langle V_{CA_x}, V_{CA_y}, \sin(\phi) \rangle$$

Vector $\overrightarrow{V_{CB}}$:

$$\overrightarrow{V_{CB}} = (\vec{C} \times \vec{b}) \times \vec{C}$$

$$\overrightarrow{V_{CB}} = (\langle \cos(\theta), \sin(\theta), 0 \rangle \times \langle -\cos(\phi), 0, \sin(\phi) \rangle) \times \langle \cos(\theta), \sin(\theta), 0 \rangle$$

$$\overrightarrow{V_{CB}} = \langle \sin(\theta)\sin(\phi), -\cos(\theta)\sin(\phi), \sin(\theta)\cos(\phi) \rangle \times \langle \cos(\theta), \sin(\theta), 0 \rangle$$

Since we're only interested in the z component of the vector

$$\overrightarrow{V_{CB}} = \langle V_{CB_x}, V_{CB_y}, \sin^2(\theta)\sin(\phi) + \cos^2(\theta)\sin(\phi) \rangle$$

$$\overrightarrow{V_{CB}} = \langle V_{CB_x}, V_{CB_y}, \sin(\phi) \rangle$$

Since we're working with unit vectors, then we can use dot products on the vectors with their angles as follows:

$$\cos(\angle ACN) = \overrightarrow{V_{CA}} \cdot \overrightarrow{V_{CN}}$$

$$\cos(\angle ACN) = \langle V_{CA_x}, V_{CA_y}, \sin(\phi) \rangle \cdot \langle 0, 0, 1 \rangle = 0 * V_{CA_x} + 0 * V_{CA_y} + 1 * \sin(\phi) = \sin(\phi)$$

Likewise,

$$\cos(\angle BCN) = \overrightarrow{V_{CB}} \cdot \overrightarrow{V_{CN}}$$

$$\cos(\angle BCN) = \langle V_{CB_x}, V_{CB_y}, \sin(\phi) \rangle \cdot \langle 0, 0, 1 \rangle = 0 * V_{CB_x} + 0 * V_{CB_y} + 1 * \sin(\phi) = \sin(\phi)$$

Therefore,

$$\cos(\angle ACN) = \cos(\angle BCN) \text{ and thus } \angle ACN = \angle BCN$$

Since those angles are equal, it proves that the great circle through C and N bisects the $\angle ACB$ in the spherical triangle ABC

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

3. (*USAMO 1979*) $a_1, a_2, \dots, a_n$ is an arbitrary sequence of positive integers. A member of the sequence is picked at random. Its value is a . Another member is picked at random, independently of the first. Its value is b . Then a third value, c . Show that the probability that $a + b + c$ is divisible by 3 is at least $\frac{1}{4}$.

Solution

Let x equal the probability of picking an element of the sequence equivalent to 0 mod 3, y equal those 1 mod 3, and z equal those 2 mod 3. Then, considering that $0+0+0$, $1+1+1$, $2+2+2$, and $0+1+2$ are divisible by 3, we obtain the equivalent inequality in the First Hint. This simplifies to $x^3 + y^3 + z^3 + 6xyz \geq x^2(y+z) + y^2(x+z) + z^2(x+y)$. Now, we attempt to prove that $x^3 + y^3 + z^3 + 3xyz \geq x^2(y+z) + y^2(x+z) + z^2(x+y)$. for all nonnegative real numbers x, y, z ; our inequality will follow from this. But this is just Schur's Inequality with $r = 1$! This completes the proof.

Note: Schur's Inequality states that $x^r(x-y)(x-z) + y^r(y-z)(y-x) + z^r(z-x)(z-y) \geq 0$. This can be proven by setting WLOG $x \geq y \geq z$ and bunching

together terms that are nonnegative. Challenge: when does equality hold? When does equality hold in the problem?

4. (USAMO 1979) P lies between the rays OA and OB . Find Q on OA and R on OB collinear with P so that $\frac{1}{PQ} + \frac{1}{PR}$ is as large as possible.

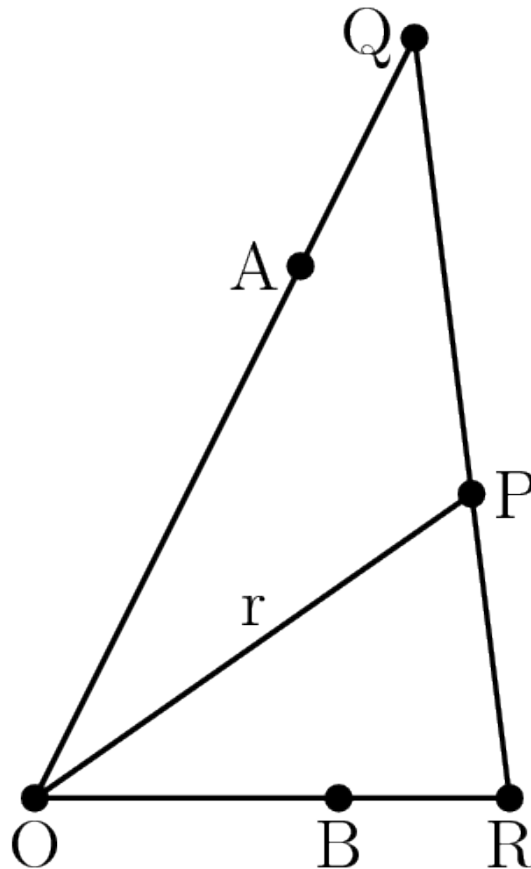
Solution (inversions)

Perform the inversion with center P and radius $\overline{PO}$. Lines OA, OB go to the circles $(O_1), (O_2)$ passing through P, O and the line QR cuts $(O_1), (O_2)$ again at the inverses Q', R' of Q, R . Hence

$$\frac{1}{PQ} + \frac{1}{PR} = \frac{PQ' + PR'}{PO^2} = \frac{Q'R'}{PO^2}$$

Thus, it suffices to find the line through P that maximizes the length of the segment $\overline{Q'R'}$. If M, N are the midpoints of PQ', PR' , i.e. the projections of O_1, O_2 onto QR , then from the right trapezoid O_1O_2NM , we deduce that $O_1O_2 \geq MN = \frac{1}{2}Q'R'$. Consequently, $2 \cdot O_1O_2$ is the greatest possible length of $Q'R'$, which obviously occurs when O_1O_2NM is a rectangle. Hence, Q, R are the intersections of OA, OB with the perpendicular to PO at P .

Solution (trig bash)



Let $r = OP, x = \angle OPR, a = \angle POR$, and $b = \angle POQ$. Then $\angle ORP = \pi - x - a$ and $\angle OQP = x - b$. Using the Law of Sines on $\triangle OPR$ gives

$$PR = \sin a * \frac{r}{\sin(\pi - x - a)} = \sin a * \frac{r}{\sin(x + a)},$$

and using the Law of Sines on $\triangle OPQ$ gives

$$PQ = \sin b * \frac{r}{\sin(x-b)}.$$

Note that r, a , and b are given constants. Hence,

$$\frac{1}{PR} + \frac{1}{PQ} = \frac{1}{r} \left(\frac{\sin(x+a)}{\sin a} + \frac{\sin(x-b)}{\sin b} \right) = \frac{\sin(x+a) \sin b + \sin(x-b) \sin a}{r \sin a \sin b} = \frac{\sin x \cos a \sin b + \sin a \cos x \sin b}{r \sin a \sin b}$$

Clearly, this quantity is maximized when $\sin x = 1$. Because x must be less than π , $\frac{1}{PQ} + \frac{1}{PR}$ is as large as possible when $x = \frac{\pi}{2}$, or when line QR is perpendicular to line PO .

5. (USAMO 1979) Let $A_1, A_2, \dots, A_{n+1}$ be distinct subsets of $[n]$ with $|A_1| = |A_2| = \dots = |A_{n+1}| = 3$. Prove that $|A_i \cap A_j| = 1$ for some pair $\{i, j\}$. Note that $[n] = \{1, 2, 3, \dots, n\}$, or, alternatively, $\{x : 1 \leq x \leq n\}$.

Solution

Suppose the problem statement does not hold. It is clear that $n \geq 4$. Choose the smallest n such that $|A_i \cap A_j| \neq 1$ for each $\{i, j\}$.

First, the $(n+1)$ subsets have $3(n+1)$ elements (some repeated) in conjunction. Because there are n elements of $[n]$ total, by the Pigeonhole Principle one element of $[n]$, say k , is in at least four subsets. Let these subsets be A_1, A_2, A_3, A_4 , without loss of generality, and let A_1 have elements k, m, n . Then without loss of generality let A_2 have elements k, m, p . If set A_3 has elements k, n, p , then simple casework shows that it is impossible to create A_4 without having A_4 intersect one of A_1, A_2, A_3 at exactly one element. Thus assume A_3 has elements k, m, q . Then A_4 has elements k, m, r . Consider each remaining set A_i . Then A_i either contains both k, m or none of them. Because there are $(n-2)$ distinct elements of $[n]$ apart from k, m , at most $(n-2)$ subsets can contain k, m . Then at least 3 subsets do not contain k, m , and it is easy to see that they are disjoint from those subsets that do contain k, m . Thus, we can partition $[n]$ into two subsets, one of which is the union of the t subsets that do contain k, m , and the other is the union of the $(n+1) - t$ subsets that do not contain k, m . Because the latter subset has $(n-t-2) < (n+1) - t - 1$ elements, we may use infinite descent to contradict the minimality of n . The proof is complete.

§9 USAMO 1980

1. (USAMO 1980) A balance has unequal arms and pans of unequal weight. It is used to weigh three objects. The first object balances against a weight A , when placed in the left pan and against a weight a , when placed in the right pan. The corresponding weights for the second object are B and b . The third object balances against a weight C , when placed in the left pan. What is its true weight?

Solution

The effect of the unequal arms and pans is that if an object of weight x in the left pan balances an object of weight y in the right pan, then $x = hy + k$ for some constants h and k . Thus if the first object has true weight x , then $x = hA + k, a = hx + k$.

So $a = h^2A + (h+1)k$.

Similarly, $b = h^2B + (h+1)k$. Subtracting gives $h^2 = \frac{a-b}{A-B}$ and so

$$(h+1)k = a - h^2A = \frac{bA - aB}{A - B}$$

The true weight of the third object is thus:

$$hC + k = \sqrt{\frac{a-b}{A-B}}C + \frac{bA - aB}{(A-B)(\sqrt{\frac{a-b}{A-B}} + 1)}$$

More readably:

$$h = \sqrt{\frac{a-b}{A-B}}; \text{weight} = hC + \frac{bA - aB}{(A-B)(h+1)}$$

Credit: John Scholes <https://prase.cz/kalva/usa/usoln/usol801.html>

2. (USAMO 1980) Find the maximum possible number of three term arithmetic progressions in a monotone sequence of n distinct reals.

Solution

Consider the first few cases for n with the entire n numbers forming an arithmetic sequence

$$(1, 2, 3, \dots, n)$$

If $n = 3$, there will be one ascending triplet (123). Let's only consider the ascending order for now. If $n = 4$, the first 3 numbers give 1 triplet, the addition of the 4 gives one more, for 2 in total. If $n = 5$, the first 4 numbers give 2 triplets, and the 5th number gives 2 more triplets (135 and 345). Repeating a few more times, we can quickly see that if n is even, the n th number will give

$$\frac{n}{2} - 1$$

more triplets in addition to all the prior triplets from the first $n - 1$ numbers. If n is odd, the n th number will give

$$\frac{n-1}{2}$$

more triplets. Let $f(n)$ denote the total number of triplets for n numbers. The above two statements are summarized as follows: If n is even,

$$f(n) = f(n-1) + \frac{n}{2} - 1$$

If n is odd,

$$f(n) = f(n-1) + \frac{n-1}{2}$$

Let's obtain the closed form for when n is even:

$$\begin{aligned} f(n) &= f(n-2) + n - 2 \\ f(n) &= f(n-4) + (n-2) + (n-4) \\ f(n) &= \sum_{i=1}^{n/2} n - 2i \end{aligned}$$

$$f(n \text{ even}) = \frac{n^2 - 2n}{4}$$

Now obtain the closed form when n is odd by using the previous result for when n is even:

$$\begin{aligned} f(n) &= f(n-1) + \frac{n-1}{2} \\ f(n) &= \frac{(n-1)^2 - 2(n-1)}{4} + \frac{n-1}{2} \end{aligned}$$

$$f(n \text{ odd}) = \frac{(n-1)^2}{4}$$

Note the ambiguous wording in the question! If the "arithmetic progression" is allowed to be a disordered subsequence, then every progression counts twice, both as an ascending progression and as a descending progression. Double the expression to account for the descending versions of each triple, to obtain:

$$\begin{aligned} f(n \text{ even}) &= \frac{n^2 - 2n}{2} \\ f(n \text{ odd}) &= \frac{(n-1)^2}{2} \\ f(n) &= \left\lfloor \frac{(n-1)^2}{2} \right\rfloor \end{aligned}$$

Solution 2

Unlike in the previous solution, I am interpreting "arithmetic progressions" to be in the same direction as the monotone sequence of real numbers. So, for example, if the sequence of reals is $1 < 4 < 5 < 6$, then $(4, 5, 6)$ counts but $(6, 5, 4)$ doesn't.

The first solution proves that there is a sequence with $\lfloor (n-1)^2/4 \rfloor$ such arithmetic progressions. In this solution, I prove that this is in fact the upper bound.

Count the arithmetic progressions by their center values. Let x_k be the k th greatest number. The number x_k is the center of at most $\min(k-1, n-k)$ arithmetic progressions, because any such progression includes a number less than x_k and a number greater than x_k . There are $k-1$ numbers less than x_k and $n-k$ numbers greater than x_k , and none of these can be in two separate arithmetic progressions with the same middle term x_k .

Thus, the total number of arithmetic progressions is at most

$$\sum_{k=1}^n \min(k-1, n-k).$$

This equals

$$1 + 2 + \cdots + \frac{n-3}{2} + \frac{n-1}{2} + \frac{n-3}{2} + \cdots + 2 + 1 = \frac{(n-1)^2}{4}$$

if n is odd, and

$$1 + 2 + \cdots + \frac{n-2}{2} + \frac{n}{2} + \frac{n}{2} + \frac{n-2}{2} + \cdots + 2 + 1 = \frac{(n-1)^2 - 1}{4}$$

if n is even. This can be summarized as

$$\left\lfloor \frac{(n-1)^2}{4} \right\rfloor.$$

3. (USAMO 1980) $A + B + C$ is an integral multiple of π . x, y , and z are real numbers. If $x \sin(A) + y \sin(B) + z \sin(C) = x^2 \sin(2A) + y^2 \sin(2B) + z^2 \sin(2C) = 0$, show that $x^n \sin(nA) + y^n \sin(nB) + z^n \sin(nC) = 0$ for any positive integer n .

Solution

Let $a = xe^{iA}$, $b = ye^{iB}$, $c = ze^{iC}$ be numbers in the complex plane.

Note that $A + B + C = k\pi$ implies $abc = xyz(e^{ik\pi}) = \pm xyz$ which is real. Also note that $x \sin(A), y \sin(B), z \sin(C)$ are the imaginary parts of a, b, c and that $x^2 \sin(2A), y^2 \sin(2B), z^2 \sin(2C)$ are the imaginary parts of a^2, b^2, c^2 by de Moivre's Theorem. Therefore, $a + b + c$ and $a^2 + b^2 + c^2$ are real because their imaginary parts sum to zero.

Finally, note that $\frac{1}{2}((a + b + c)^2 - (a^2 + b^2 + c^2)) = ab + bc + ac$ is real as well.

It suffices to show that $P_n = a^n + b^n + c^n$ is real for all positive integer n , which can be shown by induction.

Newton Sums gives the following relationship between sums of the form $P_k = a^k + b^k + c^k$

$$P_k - S_1 P_{k-1} + S_2 P_{k-2} - S_3 P_{k-3} = 0$$

Where $S_1 = a + b + c$, $S_2 = ab + bc + ac$, and $S_3 = abc$. It is given that P_0, P_1, P_2 are real. Note that if $P_{k-1}, P_{k-2}, P_{k-3}$ are real, then clearly P_k is real because all other parts of the above equation are real, completing the induction.

4. (USAMO 1980) The insphere of a tetrahedron touches each face at its centroid. Show that the tetrahedron is regular.

Solution

Let $ABCD$ be the tetrahedron, and let P and Q be the points at which the insphere touches faces ABC and ABD respectively (and therefore the centroids of those faces). Looking at the plane containing P , A , and Q , we see that the intersection of the sphere and the plane is a circle, and that AP and AQ are both tangent to said circle. $[AP]$ and $[AQ]$ are tangents to the circle from the same point and thus have the same length. The same goes for $[BP]$ and $[BQ]$. Thus, triangles ABP and ABQ are congruent, and $\angle PAB = \angle QAB$.

Let the intersection of AP and BC be M , and let the intersection of AQ and BD be N . Then AM and AN are medians of ABC and ABD , and thus $|AM| = \frac{3}{2}|AP| = \frac{3}{2}|AQ| = |AN|$. We already know from the previous congruence that $\angle MAB = \angle PAB = \angle QAB = \angle NAB$, and $|AB|$ is equal to itself. Thus, MAB and NAB are also congruent to each other. Finally, $|BC| = 2|BM| = 2|BN| = |BD|$ (Because M and N are midpoints of BC and BD respectively), and from the congruence of MAB and NAB we have $\angle CBA = \angle MBA = \angle NBA = \angle DBA$, and again $|AB|$ is equal to itself. Thus ABC and ABD are congruent, thus $|AC| = |AD|$ and $|BC| = |BD|$.

By applying the same logic to faces BCD and ACD we get $|AC| = |BC|$ and $|AD| = |BD|$. Finally, applying the same logic to faces ACB and ACD we get $|AB| = |AD|$ and $|CB| = |CD|$. Putting all these equalities together, we get that all edges of the tetrahedron are equal, and thus the tetrahedron is regular. ■

5. (USAMO 1980) If x, y, z are reals such that $0 \leq x, y, z \leq 1$, show that $\frac{x}{y+z+1} + \frac{y}{z+x+1} + \frac{z}{x+y+1} \leq 1 - (1-x)(1-y)(1-z)$

Solution

Rewrite the given inequality so that 1 is isolated on the right side. Set the left side to be $f(x, y, z)$. Now a routine computation shows

$$\frac{\partial^2 f}{\partial x^2} = \frac{2y}{(x+z+1)^3} + \frac{2z}{(x+y+1)^3} \geq 0$$

which shows that f is convex (concave up) in all three variables. Thus the maxima can only occur at the endpoints, i.e. if and only if $x, y, z \in \{0, 1\}$. Checking all eight cases shows that the value of the expression cannot exceed 1.

§10 USAMO 1981

1. (USAMO 1981) Prove that if n is not a multiple of 3, then the angle $\frac{\pi}{n}$ can be trisected with ruler and compasses.

Solution

Let $n = 3k + 1$. Multiply throughout by $\pi/3n$. We get

$$\frac{\pi}{3} = \frac{\pi \times k}{n} + \frac{\pi}{3n}$$

Re-arranging, we get

$$\frac{\pi}{3} - \frac{\pi \times k}{n} = \frac{\pi}{3n}$$

A way to interpret it is that if we know the value k , then the remainder angle of subtracting k times the given angle from $\frac{\pi}{3}$ gives us $\frac{\pi}{3n}$, the desired trisected angle.

This can be extended to the case when $n = 3k + 2$ where now, the equation becomes $\frac{\pi}{3} - \frac{\pi \times k}{n} = \frac{2\pi}{3n}$

Hence in this case, we will have to subtract k times the original angle from $\frac{\pi}{3}$ to get twice the the trisected angle. We can bisect it after that to get the trisected angle.

Generalization If regular polygons of m sides and n sides can be constructed, where m and n are relatively prime integers greater than or equal to three, then regular polygons of mn sides can be constructed. Indeed, such a polygon can be constructed by first constructing an m -gon, and then creating m distinct n -gons with at least one vertex being a vertex of the m -gon.

2. (USAMO 1981) What is the largest number of towns that can meet the following criteria. Each pair is directly linked by just one of air, bus or train. At least one pair is linked by air, at least one pair by bus and at least one pair by train. No town has an air link, a bus link and a train link. No three towns, A, B, C are such that the links between AB, AC and BC are all air, all bus or all train.

Solution

Assume AB, AC , and AD are all rail.

None of BC, CD , or BD can be rail, as those cities would form a rail triangle with A .

If BC is bus, then BD is bus as well, as otherwise B has all three types.

However, CD cannot be rail (as $\triangle ACD$ would be a rail triangle), bus (as BCD would be a bus triangle), or ferry (as C and D would have all three types).

Therefore, no city can have three connections of the same type.

Assume there are 5 towns - A , B , C , D , and E .

Two connections from A must be of one type, and two of another; otherwise there would be at least three connections of the same type from A , which has been shown to be impossible.

Let AB and AC be rail connections, and AD and AE be bus.

Assume CD is air.

BC cannot be rail ($\triangle ABC$ would be a rail triangle) or bus (C would have all three types), so BC must be air.

DE cannot be bus ($\triangle ADE$ would be a bus triangle) or rail (D would have all three types), so DE must be air.

BE cannot be rail (E would have all three types) or bus (B would have all three types), so BE must be air.

However, BD cannot be rail (D would have all three types), bus (B would have all three types), or air (D would have three air connections).

Therefore, the assumption that CD is air is false.

CD can equally be rail or bus; assume it is bus.

BC cannot be rail ($\triangle ABC$ would be a rail triangle) or air (C would have all three types), so BC must be bus.

BD cannot be air (B would have all three types) or bus (D would have three bus connections), so BD must be rail.

DE cannot be air (D would have all three types) or bus (D would have three bus connections), so DE must be rail.

CE cannot be air (C would have all three types) or bus (E would have three bus connections), so CE must be rail.

The only connection remaining is BE , which cannot be orange as both B and D would have all three types, but this means there are no air connections.

Therefore, it is impossible with five (or more) towns.

A four-town mapping is possible:

AB , BC , CD , and DA are connected by bus.

AC is connected by rail.

BD is connected by air.

Therefore, the maximum number of towns is 4.

3. (USAMO 1981) Show that for any triangle, $\frac{3\sqrt{3}}{2} \geq \sin(3A) + \sin(3B) + \sin(3C) \geq -2$.

When does the equality hold?

Solution

Given three angles that add to 180° , one can construct a triangle from them. However, its angles must all be nonnegative; thus the constraints on the angles of a triangle are $0^\circ \leq A, B, C \leq 180^\circ$ and $A + B + C = 180^\circ$.

In fact, at this point, we only care about $3A$, $3B$, and $3C$. Let us call them x , y , and z . We have:

$$0 \leq x, y, z \leq 540^\circ, x + y + z = 540^\circ$$

and we must prove that

$$\frac{3\sqrt{3}}{2} \geq \sin(x) + \sin(y) + \sin(z) \geq -2$$

Without loss of generality, assume $x \leq y \leq z$. It follows that $x \leq 180^\circ$ and $z \geq 180^\circ$ (otherwise, $x + y + z$ would be strictly greater than 180° or strictly less than 180° , respectively).

Since $\sin(u)$ is nonnegative over the interval $[0^\circ, 180^\circ]$ and $-1 \leq \sin(u) \leq 1$ for all real u , we can immediately use the $x \leq 180^\circ$ result to show that

$$\sin(x) + \sin(y) + \sin(z) \leq 0 + -1 + -1 = -2$$

This proves the lower bound; equality occurs when $\sin(x) = 0$ and $\sin(y) = \sin(z) = -1$, and this is reachable only when $y = z = 270^\circ$ and $x = 0^\circ$, which translates into $A = 0^\circ$ and $B = C = 90^\circ$.

Now for the upper bound. It is true that $\sin(u)$ is non-positive over the interval $[180^\circ, 360^\circ]$. Also, $\frac{3\sqrt{3}}{2} > 2$ (by a simple squaring argument). Then $z \geq 180^\circ$. If $z \leq 360^\circ$, then $\sin(z) \leq 0$, and then

$$\sin(x) + \sin(y) + \sin(z) \leq 1 + 1 + 0 = 2 < \frac{3\sqrt{3}}{2}$$

Therefore, we need to handle the case where $z \geq 360^\circ$. $\sin(u + 360^\circ) = \sin(u)$, so we can subtract 360° from z due to periodicity. Then the constraints become $0^\circ \leq x, y, z \leq 180^\circ$ and $x + y + z = 180^\circ$.

Let us suppose that (x, y, z) is the selection of x, y, z given the above constraints with the largest value of $\sin(x) + \sin(y) + \sin(z)$. We shall see that, given these constraints, this value increases when the distance between two of these variables decreases, and it is maximized when $x = y = z$.

Lemma: If $f(x)$ is well-behaved and $f'(x)$ is strictly decreasing over an open interval (a, b) , then, for any x, y selected from that interval with $x \neq y$, then $2f\left(\frac{x+y}{2}\right) > f(x) + f(y)$.

Proof of lemma: It is true that, for any well-behaved function $f(x)$, $f(x + d) = f(x) + \int_x^{x+d} f'(u) du$, and likewise, $f(x - d) = f(x) - \int_{x-d}^x f'(u) du$. Without loss of generality, suppose $x < y$; let $d = \frac{1}{2}(y - x)$. Since $f'(u)$ is given to be strictly decreasing over the relevant interval, we have $f'(u) > f'(x + d)$ for all $u \in (x, x + d)$; therefore

$$f(x + d) = f(x) + \int_x^{x+d} f'(u) du > f(x) + \int_x^{x+d} f'(x + d) du = f(x) + d \cdot f'(x + d)$$

Likewise, $f'(u) < f'(y - d)$ for all $u \in (y - d, y)$; therefore

$$f(y - d) = f(y) - \int_{y-d}^y f'(u) du > f(y) - \int_{y-d}^y f'(y - d) du = f(y) - d \cdot f'(y - d)$$

Therefore, adding our inequalities together,

$$f(x + d) + f(y - d) > f(x) + d \cdot f'(x + d) + f(y) - d \cdot f'(y - d)$$

But $x + d = y - d = \frac{x+y}{2}$. In that case, the $d \cdot f'(x + d)$ and $-d \cdot f'(y - d)$ terms cancel, and we get our desired result: $2f\left(\frac{x+y}{2}\right) > f(x) + f(y)$.

Since $\sin(u)$ is well-behaved, and $\sin'(u) = \cos(u)$, which is strictly decreasing over the open interval $(0^\circ, 180^\circ)$, then we can apply our lemma. Suppose for the sake of contradiction that x, y, z are not all the same value, and this produces a maximum. Then, without loss of generality, suppose $x \neq y$. Let $a = \frac{x+y}{2}$. Then the selection (a, a, z) also satisfies the constraints $0^\circ \leq a, a, z \leq 180^\circ$ and $a + a + z \leq 180^\circ$. Furthermore, by the above lemma, $2 \cdot \sin(a) > \sin(x) + \sin(y)$, so this new selection (a, a, z) has a larger corresponding value $\sin(a) + \sin(a) + \sin(z)$ than the selection (x, y, z) . This contradicts our original assumption that (x, y, z) was chosen to have the maximum. Therefore, $x = y = z$.

Then, since $x + y + z = 180^\circ$, we conclude $x = y = z = 60^\circ$. Then

$$\sin(x) + \sin(y) + \sin(z) = 3 \cdot \sin(60^\circ) = 3 \cdot \frac{\sqrt{3}}{2}$$

So $\frac{\sqrt{3}}{2}$ is indeed the maximum value of $\sin(x) + \sin(y) + \sin(z)$, proving the bound. Translating back into the previous problem, the maximum occurs when $x = y = 60^\circ$ and $z = 420^\circ$; translating into the original problem, we have proven the upper bound, and equality with the upper bound occurs when $A = B = 20^\circ$ and $C = 140^\circ$.

4. (*USAMO 1981*) The sum of the measures of all the face angles of a given complex polyhedral angle is equal to the sum of all its dihedral angles. Prove that the polyhedral angle is a trihedral angle.

Solution

$n = 3$ is certainly possible. For example, take $\angle APB = \angle APC = \angle BPC = 90^\circ$ (so that PA, PB, PC are mutually perpendicular). Then the three planes through P are also mutually perpendicular, so the two sums are both 270° .

We show that $n > 3$ is not possible.

It is almost obvious that the sum of the n dihedral angles at P is less than 360° . Take another plane which meets the lines PA, PB, PC etc at $A', B', C', \dots$ and so that the foot of the perpendicular from P to the plane lies inside the n -gon $A'B'C' \dots$ then as we move P down the perpendicular the angles $A'PB'$ etc all increase. But when it reaches the plane their sum is 360° . Take any point O inside the n -gon $ABC \dots$. We have $\angle PBA + \angle PBC > \angle ABC$. Adding the n such equations we get $\sum (180^\circ - \angle APB) > \sum \angle ABC = (n - 2)180^\circ$. So, $\sum \angle APB \dots < 360^\circ$. The sum of the n angles between the planes is at least $180(n - 2)^\circ$. If we take a sphere center P , then the lines PA, PB intersect it at $A'', B'', \dots$ which form a spherical polygon. The angles of this polygon are the angles between the planes. We can divide the polygon into $n - 2$ triangles. The angles in a spherical triangle sum to at least 180° . So the angles in the spherical polygon are at least $180(n - 2)^\circ$. So we have $(n - 2)180^\circ < 360^\circ$ and hence $n < 4$.

5. (*USAMO 1981*) Show that for any positive real x ,

$$[nx] \geq \sum_{k=1}^n \left(\frac{[kx]}{k} \right),$$

where $[x]$ denotes the greatest integer not exceeding x .

Solution(due to Pavel Zatitskiy)

First of all we write $[kx] = kx - \{kx\}$. So, we need to prove that $\sum_1^n \left(\frac{\{kx\}}{k}\right) \geq \{nx\}$. Let's denote $a_k = \{kx\}$. It is easy to see that $a_k + a_m \geq a_{k+m}$. We need to prove $\sum_1^n \left(\frac{a_k}{k}\right) \geq a_n$.

We will prove it by induction by n . The base is obvious, so we need to make a step.

Let's take m such that $\frac{a_m}{m}$ is minimal. If $m = n$ then our inequality is obvious. So, $m < n$. Then, by induction, $\sum_1^{n-m} \left(\frac{a_k}{k}\right) \geq a_{n-m}$ and $\sum_{n-m+1}^n \left(\frac{a_k}{k}\right) \geq m \cdot \frac{a_m}{m} = a_m$. Now we can add these two inequalities and get

$$\sum_1^n \left(\frac{a_k}{k}\right) \geq a_{n-m} + a_m \geq a_n$$

§11 USAMO 1982

1. (*USAMO 1982*) In a party with 1982 persons, among any group of four there is at least one person who knows each of the other three. What is the minimum number of people in the party who know everyone else?

Solution

We induct on n to prove that in a party with n people, there must be at least $(n-3)$ people who know everyone else. (Clearly this is achievable by having everyone know everyone else except three people A, B, C , who do not know each other.)

Base case: $n = 4$ is obvious.

Inductive step: Suppose in a party with k people (with $k \geq 4$), at least $(k-3)$ people know everyone else. Consider a party with $(k+1)$ people. Take k of the people (leaving another person, A , out) and apply the inductive step to conclude that at least $(k-3)$ people know everyone else in the k -person group, G .

Now suppose that everyone in the group G knows each other. Then take 3 of these people and A to deduce that A knows a person $B \in G$, which means B knows everyone else. Then apply the inductive step on the remaining k people (excluding B) to find $(k-3)$ people out of them that know everyone else (including B , of course). Then these $(k-3)$ people and B , which enumerate $(k-2)$ people, know everyone else.

Suppose that there exist two people $B, C \in G$ who do not know each other. Because $k-3 \geq 1$, there exist at least one person in G , person D , who knows everyone else in G . Now, take A, B, C, D and observe that because B, C do not know each other, either A or D knows everyone else of A, B, C, D (by the problem condition), so in particular A and D know each other. Then apply the inductive step on the remaining k people (excluding D) to find $(k-3)$ people out of them that know everyone else (including D , of course). Then these $(k-3)$ people and D , which enumerate $(k-2)$ people, know everyone else.

This completes the inductive step and thus the proof of this stronger result, which easily implies that at least $1982 - 3 = \boxed{1979}$ people know everyone else.

2. (*USAMO 1982*) Let $S_r = x^r + y^r + z^r$ with x, y, z real. It is known that if $S_1 = 0$,

$$(*) \quad \frac{S_{m+n}}{m+n} = \frac{S_m}{m} \frac{S_n}{n}$$

for $(m, n) = (2, 3), (3, 2), (2, 5)$, or $(5, 2)$. Determine *all* other pairs of integers (m, n) if any, so that $(*)$ holds for all real numbers x, y, z such that $S_1 = 0$.

Solution 1

Claim Both m, n can not be even.

Proof $x + y + z = 0 \implies x = -(y + z)$.

Since $\frac{S_{m+n}}{m+n} = \frac{S_m S_n}{mn}$,

by equating coefficient of y^{m+n} on LHS and RHS ,get

$$\begin{aligned}\frac{2}{m+n} &= \frac{4}{mn} \\ \implies \frac{m}{2} + \frac{n}{2} &= \frac{m \cdot n}{2 \cdot 2}.\end{aligned}$$

So we have, $\frac{m}{2} \mid \frac{n}{2}$ and $\frac{n}{2} \mid \frac{m}{2}$.

$$\implies m = n = 4.$$

So we have $S_8 = 2(S_4)^2$.

Now since it will true for all real $x, y, z, x + y + z = 0$. So choose $x = 1, y = -1, z = 0$.

$S_8 = 2$ and $S_4 = 2$ so $S_8 \neq 2S_4^2$.

This is contradiction. So, at least one of m, n must be odd. WLOG assume n is odd and m is even. The coefficient of y^{m+n-1} in $\frac{S_{m+n}}{m+n}$ is $\frac{\binom{m+n}{1}}{m+n} = 1$

The coefficient of y^{m+n-1} in $\frac{S_m \cdot S_n}{m \cdot n}$ is $\frac{2}{m}$.

Therefore, $\boxed{m = 2}$.

Now choose $x = y = \frac{1}{2}z = (-2)$. (sic)

Since $\frac{S_{n+2}}{2+n} = \frac{S_2 S_n}{2 \cdot n}$ holds for all real x, y, z such that $x + y + z = 0$.

We have $\frac{2^{n+2}-2}{n+2} = 3 \cdot \frac{2^n-2}{n}$. Therefore,

$$\frac{2^{n+1}-1}{n+2} = 3 \cdot \frac{2^{n-1}-1}{n} \dots \quad (**)$$

Clearly $(**)$ holds for $n \in \{5, 3\}$. And one can say that for $n \geq 6$, RHS of $(**)$ < LHS of $(**)$.

So our answer is $(m, n) = (5, 2), (2, 5), (3, 2), (2, 3)$.

-ftheftics (edited by integralarefun)

3. (USAMO 1982) If a point A_1 is in the interior of an equilateral triangle ABC and point A_2 is in the interior of $\triangle A_1BC$, prove that

$$I.Q.(A_1BC) > I.Q.(A_2BC),$$

where the *isoperimetric quotient* of a figure F is defined by

$$I.Q.(F) = \frac{\text{Area}(F)}{[\text{Perimeter}(F)]^2}$$

Solution

First, an arbitrary triangle ABC has isoperimetric quotient (using the notation $[ABC]$ for area and $s = \frac{a+b+c}{2}$):

$$\begin{aligned}\frac{[ABC]}{4s^2} &= \frac{[ABC]^3}{4s^2[ABC]^2} = \frac{r^3 s^3}{4s^2 \cdot s(s-a)(s-b)(s-c)} = \frac{r^3}{4(s-a)(s-b)(s-c)} \\ &= \frac{1}{4} \cdot \frac{r}{s-a} \cdot \frac{r}{s-b} \cdot \frac{r}{s-c} = \frac{1}{4} \tan A/2 \tan B/2 \tan C/2.\end{aligned}$$

Lemma. $\tan x \tan(A - x)$ is increasing on $0 < x < \frac{A}{2}$, where $0 < A < 90^\circ$.

Proof.

$$\begin{aligned}\tan x \tan(A - x) &= \tan x \cdot \frac{\tan A - \tan x}{1 + \tan A \tan x} = 1 - \frac{1}{\cos^2 x (1 + \tan A \tan x)} \\ &= 1 - \frac{2}{1 + \cos 2x + \tan A \sin 2x} = 1 - \frac{2}{1 + \sec A \cos(A - 2x)}\end{aligned}$$

is increasing on the desired interval, because $\cos(A - 2x)$ is increasing on $0 < x < \frac{A}{2}$.

Let x_1, y_1, z_1 and x_2, y_2, z_2 be half of the angles of triangles A_1BC and A_2BC in that order, respectively. Then it is immediate that $30^\circ > y_1 > y_2$, $30^\circ > z_1 > z_2$, and $x_1 + y_1 + z_1 = x_2 + y_2 + z_2 = 90^\circ$. Hence, by Lemma it follows that

$$\begin{aligned}\tan x_1 \tan y_1 \tan z_1 &= \tan(90^\circ - y_1 - z_1) \tan y_1 \tan z_1 > \tan(90^\circ - y_1 - z_2) \tan y_1 \tan z_2 \\ &> \tan(90^\circ - y_2 - z_2) \tan y_2 \tan z_2 = \tan x_2 \tan y_2 \tan z_2.\end{aligned}$$

Multiplying this inequality by $\frac{1}{4}$ gives that $I.Q[A_1BC] > I.Q[A_2BC]$, as desired.

4. (*USAMO 1982*) Prove that there exists a positive integer k such that $k \cdot 2^n + 1$ is composite for every positive integer n .

Solution

Indeed, $\boxed{k = 2935363331541925531}$ has the requisite property.

To see why, consider the primes 3, 5, 17, 257, 65537, 6700417, 641, and observe that

$$\begin{aligned}k &\equiv 1 \pmod{3} \\ k &\equiv 1 \pmod{5} \\ k &\equiv 1 \pmod{17} \\ k &\equiv 1 \pmod{257} \\ k &\equiv 1 \pmod{65537} \\ k &\equiv 1 \pmod{6700417} \\ k &\equiv -1 \pmod{641}\end{aligned}$$

Moreover,

$$\begin{aligned}\text{ord}_3(2) &= 2 \\ \text{ord}_5(2) &= 4 \\ \text{ord}_{17}(2) &= 8 \\ \text{ord}_{257}(2) &= 16 \\ \text{ord}_{65537}(2) &= 32 \\ \text{ord}_{6700417}(2) &= 64 \\ \text{ord}_{641}(2) &= 64\end{aligned}$$

We conclude that

$$\begin{aligned}n \equiv 1 \pmod{2} &\implies k \cdot 2^n - 1 \equiv k \cdot (-1) + 1 \equiv 0 \pmod{3} \\ n \equiv 2 \pmod{4} &\implies k \cdot 2^n - 1 \equiv k \cdot (-1) + 1 \equiv 0 \pmod{5}\end{aligned}$$

$$\begin{aligned}
 n \equiv 4 \pmod{8} &\implies k \cdot 2^n - 1 \equiv k \cdot (-1) + 1 \equiv 0 \pmod{17} \\
 n \equiv 8 \pmod{16} &\implies k \cdot 2^n - 1 \equiv k \cdot (-1) + 1 \equiv 0 \pmod{257} \\
 n \equiv 16 \pmod{32} &\implies k \cdot 2^n - 1 \equiv k \cdot (-1) + 1 \equiv 0 \pmod{65537} \\
 n \equiv 32 \pmod{64} &\implies k \cdot 2^n - 1 \equiv k \cdot (-1) + 1 \equiv 0 \pmod{6700417} \\
 n \equiv 0 \pmod{64} &\implies k \cdot 2^n - 1 \equiv k \cdot 1 + 1 \equiv 0 \pmod{641}
 \end{aligned}$$

And $k \gg 3, 5, 17, 257, 65537, 6700417, 641$ so the relevant values will, in fact, always be composite. ■

5. (USAMO 1982) A, B , and C are three interior points of a sphere S such that AB and AC are perpendicular to the diameter of S through A , and so that two spheres can be constructed through A, B , and C which are both tangent to S . Prove that the sum of their radii is equal to the radius of S .

Solution

Let the two tangent spheres be S_1 and S_2 , and let O, O_1, O_2 and R, R_1, R_2 be the origins and radii of S, S_1, S_2 respectively. Then AO stands normal to the plane P through $\triangle ABC$. Because both spheres go through A, B , and C , the line O_1O_2 also stands normal to P , meaning AO and O_1O_2 are both coplanar and parallel. Therefore the problem can be flattened to the plane P' through A, O, O_1 and O_2 .

Let X, Y, Z, M, N be points on P' such that $X = S \cap OO_1$, $Y = S_1 \cap S_2 \neq A$, $Z = S \cap OO_2$, $M = O_1Y \cap AO$, $N = O_2Y \cap AO$

Let J be the three circles radical center, meaning JX and JZ are tangent segments to S and $J \in AY$.

Because $Y \in P \iff \angle NAY = \angle YAM = 90^\circ$, we have that $\overline{YN}$ and $\overline{YM}$ are diameters.

This means that $\angle OZN = \angle ZNY = \angle JZY$ and $\angle MXO = \angle YMX = \angle YXJ$.

And because $\angle ZNO = 180^\circ - \angle AYZ = \angle ZYJ$ and $\angle OMX = 180^\circ - \angle XYA = \angle JYX$, we have that $\triangle JZY \sim \triangle OZN$ and $\triangle JYX \sim \triangle OMX$.

We then conclude that $\overline{OM} = \overline{OX} \cdot \frac{\overline{JY}}{\overline{JX}} = \overline{OZ} \cdot \frac{\overline{JY}}{\overline{JZ}} = \overline{ON}$

Let $a \top b$ denote line a bisecting line segment b .

Since $O_1O_2 \parallel MN$ it follows that $OY \top \overline{MN} \Rightarrow OY \top \overline{O_1O_2}$. Similarly we have that $O_1O_2 \top \overline{YM} \Rightarrow O_1O_2 \top \overline{OY}$.

And so O_1OO_2Y is a parallelogram because $\overline{OY}$ and $\overline{O_1O_2}$ bisect each other, meaning $R = \overline{OZ} = \overline{OO_2} + \overline{O_2Z} = \overline{O_1Y} + R_2 = R_1 + R_2$

Quod Erat Demonstrandum

§12 USAMO 1983

1. (USAMO 1983) On a given circle, six points A, B, C, D, E , and F are chosen at random, independently and uniformly with respect to arc length. Determine the probability that the two triangles ABC and DEF are disjoint, i.e., have no common points.

Solution

First we give the circle an orientation (e.g., letting the circle be the unit circle in polar coordinates). Then, for any set of six points chosen on the circle, there

are exactly $6!$ ways to label them one through six. Also, this does not affect the probability we wish to calculate. This will, however, make calculations easier.

Note that, for any unordered set of six points chosen from the circle boundary, the number of ways to number them such that they satisfy this disjoint-triangle property is constant: there are six ways to choose which triangle will be numbered with the numbers one through three, and there are $(3!)^2$ ways to arrange the numbers one through three and four through six on these two triangles. Therefore, for any given configuration of points, there are $6^3 = 216$ ways to label them to have this disjoint-triangle property. There are, however, $6! = 720$ ways to label the points in all, so given any six unordered points, the probability that when we inflict an ordering we produce the disjoint-triangle property is $216/720 = 3/10$.

Since this probability is constant for any configuration of six unordered points we choose, we must have that $3/10$ is the probability that we produce the disjoint-triangle property if we choose the points as detailed in the problem statement.

2. (USAMO 1983) Prove that the roots of

$$x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0$$

cannot all be real if $2a^2 < 5b$.

Solution

We prove the contrapositive: if the polynomial in question has the five real roots x_1, x_2, x_3, x_4, x_5 , then $5b \leq 2a^2$.

Because $a = -(x_1 + x_2 + x_3 + x_4 + x_5)$ and $b = x_1x_2 + x_1x_3 + \dots + x_4x_5$ by Vieta's Formulae, we have

$$\begin{aligned} 2b &= 2x_1x_2 + 2x_1x_3 + \dots + 2x_4x_5 = (x_1 + x_2 + x_3 + x_4 + x_5)^2 - (x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2) \\ &= a^2 - \frac{(1 + 1 + 1 + 1 + 1)(x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2)}{5} \\ &\leq a^2 - \frac{(x_1 + x_2 + x_3 + x_4 + x_5)^2}{5} \end{aligned}$$

(by Cauchy-Schwarz)

$$= \frac{4a^2}{5},$$

so $5b \leq 2a^2$, as desired.

Solution 2

Lemma:

For all real numbers $x_1, x_2, \dots, x_5$,

$$2(x_1^2 + x_2^2 + \dots + x_5^2) \geq$$

$$x_1x_2 + x_1x_3 + \dots + x_4x_5$$

By the trivial inequality,

$$x^2 + y^2 \geq 2xy \Rightarrow \frac{x^2}{2} + \frac{y^2}{2} \geq xy$$

Making such an inequality for all the variable pairs and summing them, we find the lemma is true.

Now, let our roots be $x_1, x_2, \dots, x_5$. By Vieta's, $a = x_1 + x_2 + \dots + x_5$ and $b = x_1x_2 + x_1x_3 + \dots + x_4x_5$

If we show that for all real $x_1, x_2, \dots, x_5$ that $2a^2 \geq 5b$, then we have a contradiction and all of $x_1, x_2, \dots, x_5$ cannot be real. We start by rewriting $2a^2 \geq 5b$ as

$$2(x_1 + x_2 + \dots + x_5)^2 \geq 5(x_1x_2 + x_1x_3 + \dots + x_4x_5)$$

We divide by 2 and find

$$(x_1 + x_2 + \dots + x_5)^2 \geq \frac{5}{2}(x_1x_2 + x_1x_3 + \dots + x_4x_5)$$

Expanding the LHS, we have

$$x_1^2 + x_2^2 + \dots + x_5^2 + 2(x_1x_2 + x_1x_3 + \dots + x_4x_5) \geq \frac{5}{2}(x_1x_2 + x_1x_3 + \dots + x_4x_5)$$

We subtract the sum in brackets, and then multiply by 2 to find

$$2x_1^2 + 2x_2^2 + \dots + 2x_5^2 \geq x_1x_2 + x_1x_3 + \dots + x_4x_5$$

which is true by our lemma.

3. (*USAMO 1983*) Each set of a finite family of subsets of a line is a union of two closed intervals. Moreover, any three of the sets of the family have a point in common. Prove that there is a point which is common to at least half the sets of the family.

Solution

Let us first see that this works for anything less than three sets.

Obviously, due to the second condition given, they all share at least one point in common.

Now, let us see that it works for three or more sets.

First, take any three of the subsets in the family of sets.

Let them be A, B, and C. We know that they share at least one point in common. Let us take the pair of segments that have the shortest length when you take the union of the pair. Let us denote this segment D. Evidently, if you take any other set of points, it must intersect D at least at one point. In order to minimize the number of intersections between the other varying segments, we must have at close to 1/2 of the remaining segments as possible intersecting each other. However, if we note, putting them that way forces at least 1/2 of the segments in each section.

4. (*USAMO 1983*) Six segments S_1, S_2, S_3, S_4, S_5 , and S_6 are given in a plane. These are congruent to the edges AB, AC, AD, BC, BD , and CD , respectively, of a tetrahedron $ABCD$. Show how to construct a segment congruent to the altitude of the tetrahedron from vertex A with straight-edge and compasses.

Solution

Throughout this solution, we denote the length of a segment S by $|S|$.

In this solution, we employ several lemmas. Two we shall take for granted: given any point A and a line ℓ not passing through A , we can construct a line ℓ' through A parallel to ℓ ; and given any point A on a line ℓ , we can construct a line ℓ' through A perpendicular to ℓ .

Lemma 1: If we have two segments S and T on the plane with non-zero length, we may construct a circle at either endpoint of S whose radius is $|T|$.

Proof: We can construct arbitrarily many copies of T by drawing a circle about one of its endpoints through its other endpoint, and then connecting the center of this circle with any other point on the circle. We can construct copies of T like this until we create a circle of radius $|T|$ and center P_1 that intersects segment S . We can then take this intersection point P_2 and draw a line ℓ through it perpendicular to S , and draw a circle with center P_2 passing through P_1 , and consider its intersection P_3 with ℓ . Note that $P_2P_3 \perp S$ and $|P_2P_3| = |T|$. Take an endpoint P_4 of S : then draw a line through P_3 parallel to S , and a line through P_4 parallel to P_2P_3 . Let these two lines intersect at P_5 . Then $P_2P_3P_5P_4$ is a rectangle, so $|P_4P_5| = |T|$. Our desired circle is then a circle centered at P_4 through P_5 .

Lemma 2: Given three collinear points A, B, C in this order, if $|AB| = a$ and $|BC| = b$ with $a > b$, then we can construct a segment of length $\sqrt{a^2 - b^2}$.

Proof: From Lemma 1, we can construct a circle through C with radius a , and then construct a perpendicular through B to AC : these two objects intersect at D and E . Both BD and DE have length $\sqrt{a^2 - b^2}$, from the Pythagorean Theorem.

Proof of the original statement: Note that we can construct a triangle $B'C'D'$ congruent to triangle BCD by applying Lemma 1 to segments S_4, S_5 , and S_6 . Similarly, we can construct A_B and A_C outside triangle $B'C'D'$ such that $A_B C' D' \cong ACD$ and $A_C B' D' \cong ABD$.

Let A' be a point outside of the plane containing S_1 through S_6 such that $A'B'C'D' \cong ABCD$. Then the altitudes of triangles $A_B C' D'$ and $A' C' D'$ to segment $C' D'$ are congruent, as are the altitudes of triangles $A_C B' D'$ and $A' B' D'$ to segment $B' D'$. However, if we project the altitudes of $A' C' D'$ and $A' B' D'$ from A' onto the plane, their intersection is the base of the altitude of tetrahedron $A'B'C'D'$ from A' . In addition, these altitude projections are collinear with the altitudes of triangles $A_B C' D'$ and $A_C B' D'$. Therefore, the altitudes of $A_B C' D'$ and $A_C B' D'$ from A_B and A_C intersect at the base X' of the altitude of $A'B'C'D'$ from A' . In summary, we can construct X' by constructing the perpendiculars from A_B and A_C to $C' D'$ and $B' D'$ respectively, and taking their intersection.

Let Y' be the intersection of $A_B X'$ with $C' D'$. Then the altitude length we seek to construct is, from the Pythagorean Theorem, $\sqrt{|A_B Y'|^2 - |X' Y'|^2}$. We can directly apply Lemma 2 to segment $A_B Y' X'$ to obtain this segment. This shows how to construct a segment of length $A' X'$.

5. (*USAMO 1983*) Consider an open interval of length $1/n$ on the real number line,

where n is a positive integer. Prove that the number of irreducible fractions p/q , with $1 \leq q \leq n$, contained in the given interval is at most $(n+1)/2$.

Solution

Let I be an open interval of length $1/n$ and F_n the set of fractions $p/q \in I$ with $p, q \in \mathbb{Z}$, $\gcd(p, q) = 1$ and $1 \leq q \leq n$.

Assume that $\frac{p}{q} \in F_n$. If $k \in \mathbb{Z}$ is such that $1 \leq kq \leq n$, and $p' \in \mathbb{Z}$ is such that $\gcd(p', kq) = 1$, then

$$\left| \frac{p}{q} - \frac{p'}{kq} \right| \geq \frac{1}{kq} \geq \frac{1}{n}$$

Therefore $\frac{p'}{kq} \notin I \supset F_n$. This means that $\frac{p}{q}$ is the only fraction in F_n with denominator q or multiple of q .

Therefore, from each of the pairs in $P = \left\{ (k, 2k) : 1 \leq k \leq \left\lfloor \frac{n+1}{2} \right\rfloor \right\}$ at most one element from each can be a denominator of a fraction in F_n .

Hence $|F_n| \leq |P| \leq \frac{n+1}{2}$

§13 USAMO 1984

1. (USAMO 1984) The product of two of the four roots of the quartic equation $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ is -32 . Determine the value of k .

Solution 1 (ingenious)

Using Vieta's formulas, we have:

$$\begin{aligned} a + b + c + d &= 18, \\ ab + ac + ad + bc + bd + cd &= k, \\ abc + abd + acd + bcd &= -200, \\ abcd &= -1984. \end{aligned}$$

From the last of these equations, we see that $cd = \frac{abcd}{ab} = \frac{-1984}{-32} = 62$. Thus, the second equation becomes $-32 + ac + ad + bc + bd + 62 = k$, and so $ac + ad + bc + bd = k - 30$. The key insight is now to factor the left-hand side as a product of two binomials: $(a+b)(c+d) = k - 30$, so that we now only need to determine $a+b$ and $c+d$ rather than all four of a, b, c, d .

Let $p = a + b$ and $q = c + d$. Plugging our known values for ab and cd into the third Vieta equation, $-200 = abc + abd + acd + bcd = ab(c+d) + cd(a+b)$, we have $-200 = -32(c+d) + 62(a+b) = 62p - 32q$. Moreover, the first Vieta equation, $a + b + c + d = 18$, gives $p + q = 18$. Thus we have two linear equations in p and q , which we solve to obtain $p = 4$ and $q = 14$.

Therefore, we have $\underbrace{(a+b)}_4 \underbrace{(c+d)}_{14} = k - 30$, yielding $k = 4 \cdot 14 + 30 = \boxed{86}$.

Solution 2 (cool)

We start as before: $ab = -32$ and $cd = 62$. We now observe that a and b must be the roots of a quadratic, $x^2 + rx - 32$, where r is a constant (secretly, r is just

-(a+b)=-p from Solution #1). Similarly, c and d must be the roots of a quadratic $x^2 + sx + 62$.

Now

$$\begin{aligned} x^4 - 18x^3 + kx^2 + 200x - 1984 &= (x^2 + rx - 32)(x^2 + sx + 62) \\ &= x^4 + (r + s)x^3 + (62 - 32 + rs)x^2 \\ &\quad + (62s - 32r)x - 1984. \end{aligned}$$

Equating the coefficients of x^3 and x with their known values, we are left with essentially the same linear equations as in Solution #1, which we solve in the same way. Then we compute the coefficient of x^2 and get $k = \boxed{86}$.

Solution 3 AM-GM

Let r_1, r_2, r_3, r_4 be the roots of the polynomial $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$. We are given that $r_1r_2 = -32$.

By Vieta's formulas, we have:

$$r_1 + r_2 + r_3 + r_4 = 18$$

$$r_1r_2 + r_1r_3 + r_1r_4 + r_2r_3 + r_2r_4 + r_3r_4 = k$$

$$r_1r_2r_3 + r_1r_2r_4 + r_1r_3r_4 + r_2r_3r_4 = -200$$

$$r_1r_2r_3r_4 = -1984$$

Since $r_1r_2 = -32$, we have $(-32)r_3r_4 = -1984$, so $r_3r_4 = \frac{-1984}{-32} = 62$.

Also, $r_1r_2r_3 + r_1r_2r_4 + r_1r_3r_4 + r_2r_3r_4 = -200$, so $r_1r_2(r_3 + r_4) + r_3r_4(r_1 + r_2) = -200$. Substituting $r_1r_2 = -32$ and $r_3r_4 = 62$, we have $-32(r_3 + r_4) + 62(r_1 + r_2) = -200$.

Let $A = r_1 + r_2$ and $B = r_3 + r_4$. Then $A + B = 18$, so $B = 18 - A$. Substituting this into the equation, we have $-32B + 62A = -200$, so $-32(18 - A) + 62A = -200$. $-576 + 32A + 62A = -200$, so $94A = 376$, which means $A = \frac{376}{94} = 4$. Then $B = 18 - A = 18 - 4 = 14$.

So we have $r_1 + r_2 = 4$ and $r_1r_2 = -32$. Then r_1 and r_2 are roots of $x^2 - 4x - 32 = 0$, so $(x - 8)(x + 4) = 0$, which means $r_1 = 8$ and $r_2 = -4$ (or vice versa).

Also we have $r_3 + r_4 = 14$ and $r_3r_4 = 62$. Then r_3 and r_4 are roots of $x^2 - 14x + 62 = 0$.

Using the quadratic formula, $x = \frac{14 \pm \sqrt{14^2 - 4(62)}}{2} = \frac{14 \pm \sqrt{196 - 248}}{2} = \frac{14 \pm \sqrt{-52}}{2} = 7 \pm i\sqrt{13}$. Then $r_3 = 7 + i\sqrt{13}$ and $r_4 = 7 - i\sqrt{13}$.

We want to find $k = r_1r_2 + r_1r_3 + r_1r_4 + r_2r_3 + r_2r_4 + r_3r_4$. $k = r_1r_2 + r_3r_4 + (r_1 + r_2)(r_3 + r_4) = -32 + 62 + (4)(14) = 30 + 56 = 86$.

Final Answer: The final answer is $\boxed{86}$

Solution 4 (Alcumus)

Since two of the roots have product -32 , the equation can be factored in the form

$$x^4 - 18x^3 + kx^2 + 200x - 1984 = (x^2 + ax - 32)(x^2 + bx + c).$$

Expanding, we get

$$x^4 - 18x^3 + kx^2 + 200x - 1984 = x^4 + (a+b)x^3 + (ab+c-32)x^2 + (ac-32b)x - 32c = 0.$$

Matching coefficients, we get

$$\begin{aligned}a + b &= -18, \\ab + c - 32 &= k, \\ac - 32b &= 200, \\-32c &= -1984.\end{aligned}$$

Then $c = \frac{-1984}{-32} = 62$, so $62a - 32b = 200$. With $a + b = -18$, we can solve to find $a = -4$ and $b = -14$. Then

$$k = ab + c - 32 = \boxed{86}.$$

Solution 5 (If You Don't Know Vieta)

We know that any given quartic polynomial can be factored into two quadratic polynomials. We see that for any given quadratic, the product of its roots is:

$$\frac{-b - \sqrt{b^2 - 4ac}}{2a} \cdot \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \frac{b^2 - b^2 + 4ac}{4a^2} = \frac{c}{a}$$

We see that both quadratics must have leading coefficient of 1, so $a = 1$

We first try a quadratic where the product of its roots is -32

$$x^2 + b_1x - 32$$

We then see that the other quadratic has the form

$$x^2 + b_2x + 62$$

since the constant term in both quadratics must multiply to -1984

To get our general form quartic from these, we multiply

$$(x^2 + b_1x - 32)(x^2 + b_2x + 62)$$

Expanding, then combining coefficients, we get

$$\begin{aligned}x^4 + b_2x^3 + 62x^2 + b_1x^3 + b_1b_2x^2 + 62b_1x - 32x^2 - 32b_2x - 1984 \\x^4 + (b_1 + b_2)x^3 + (30 + b_1b_2)x^2 + (62b_1 - 32b_2)x - 1984\end{aligned}$$

Looking back at our original polynomial, we extract the system

$$\begin{cases} b_1 + b_2 = -18 \\ 62b_1 - 32b_2 = 200 \\ 30 + b_1b_2 = k \end{cases}$$

To solve it, we see

$$\begin{aligned}b_1 &= -b_2 - 18 \\ 62(-b_2 - 18) - 32b_2 &= 200 \\ 31(-b_2 - 18) - 16b_2 &= 100 \\ -47b_2 &= 658 \\ b_2 &= -14 \\ b_1 &= -4 \\ k &= 30 + 56 \\ k &= \boxed{86}\end{aligned}$$

shockfront99

2. (USAMO 1984) The geometric mean of any set of m non-negative numbers is the m -th root of their product.

(i) For which positive integers n is there a finite set S_n of n distinct positive integers such that the geometric mean of any subset of S_n is an integer?

(ii) Is there an infinite set S of distinct positive integers such that the geometric mean of any finite subset of S is an integer?

Solution

a) We claim that for any numbers $p_1, p_2, \dots, p_n, p_1^{n!}, p_2^{n!}, \dots, p_n^{n!}$ will satisfy the condition, which holds for any number n .

Since $\sqrt[k]{ab} = \sqrt[k]{a} * \sqrt[k]{b}$, we can separate each geometric mean into the product of parts, where each part is the k th root of each member of the subset and the subset has k members.

Assume our subset has k members. Then, we know that the k th root of each of these members is an integer (namely $p^{n!/k}$), because $k \leq n$ and thus $k|n!$. Since each root is an integer, the geometric mean will also be an integer.

b) If we define q as an arbitrarily large number, and x and y as numbers in set S , we know that $\sqrt[q]{\frac{x}{y}}$ is irrational for large enough q , meaning that it cannot be expressed as the fraction of two integers. However, both the geometric mean of the set of x and $q-1$ other arbitrary numbers in S and the set of y and the same other $q-1$ numbers are integers, so since the other numbers cancel out, the geometric means divided, or $\sqrt[q]{\frac{x}{y}}$, must be rational. This is a contradiction, so no such infinite S is possible.

-aops111 (first solution dont bully me)

Solution 2

(i) The solution is the same as in the first solution

(ii) Let p and q be some prime numbers, and let $v_p(n)$ be the largest number such that $p^{v_p(n)}$ divides n . $v_p(n)$ is also the exponent of p in the prime factorization of n (or 0, if p isn't a divisor of n). Then, because S is infinite, we can choose $q-1$ numbers $a_1, a_2, \dots, a_{q-1}$ in S such that $v_p(a_i)$ all give the same remainder when divided by q . (Otherwise, because $v_p(n) \bmod q$ has q distinct values, S could contain at most $q(p-2)$ numbers, which is finite.) Call that remainder r .

Now take some integer b in S , not part of the other $q-1$ values. Together, their product must be a q th power, meaning $q|v_p(a_1 a_2 \dots a_{q-1} b)$. Thus, $v_p(a_1 a_2 \dots a_{q-1} b) \equiv 0 \pmod{q}$. It can be seen easily that for two positive integers x and y we have $v_p(xy) = v_p(x) + v_p(y)$, and by definition $v_p(a_i) \equiv r \pmod{q}$, so we can write $v_p(b) + (q-1)r \equiv 0 \pmod{q}$. Adding r to both sides we have $v_p(b) \equiv v_p(b) + qr \equiv r \pmod{q}$.

The choice of b was arbitrary, meaning for all numbers n in S , $v_p(n)$ modulo q is the constant r , whether part of $a_1, a_2, \dots, a_{q-1}$ or not. The choice of q was also arbitrary, thus for two integers m and n in S , $v_p(m) = v_p(n)$. Otherwise, we could pick q larger than both values, and get a contradiction. Thus for all numbers n in S , $v_p(n)$ is constant. The choice of p was also arbitrary, meaning any two distinct numbers in S have the same prime factorization. By the fundamental theorem of arithmetic, this means the two numbers are the same, contradicting with them being distinct. Thus, no such S exists.

-Circling

3. (USAMO 1984) P, A, B, C , and D are five distinct points in space such that $\angle APB = \angle BPC = \angle CPD = \angle DPA = \theta$, where θ is a given acute angle. Determine the greatest and least values of $\angle APC + \angle BPD$.

Solution

Greatest value is achieved when all the points are as close as possible to all being on a plane.

Since $\theta < \frac{\pi}{2}$, then $\angle APC + \angle BPD < \pi$

Smallest value is achieved when point P is above and the remaining points are as close as possible to colinear when $\theta > 0$, then $\angle APC + \angle BPD > 0$

and the inequality for this problem is:

$$0 < \angle APC + \angle BPD < \pi$$

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

4. (USAMO 1984) A difficult mathematical competition consisted of a Part I and a Part II with a combined total of 28 problems. Each contestant solved 7 problems altogether. For each pair of problems, there were exactly two contestants who solved both of them. Prove that there was a contestant who, in Part I, solved either no problems or at least four problems.

Solution

First, notice that for any given problem, among the participants who solved that problem, exactly two of them must have also solved each of the other 27 problems. Since each person solved exactly 6 of the other problems, the number of participants who solved any given problem must be $\frac{2 \cdot 27}{6} = 9$. This also allows us to easily calculate the number of participants as $\frac{9 \cdot 28}{7} = 36$.

Now, let n be the number of problems in part I, and assume that no participant solved 0 or more than 4 problems in part I. In other words, assume that all participants solved either 1, 2 or 3 problems in part I. Let m_1, m_2, m_3 be the number of participants who solved exactly 1, 2 and 3 problems in part I, respectively.

Clearly, this implies $m_1 + m_2 + m_3 = 36$.

Also, since we know 9 participants solved each problem, $m_1 + 2m_2 + 3m_3 = 9n$, as both sides must count the total number of solves in part I.

Clearly n is at least 4, as 36 participants all solved at least 1 of the problems in part I, but only 9 could have solved each of them. Now consider that exactly two participants must have solved each of the $\binom{n}{2}$ pairs of problems within part I. Since each participant who solved 2 problems in part I solved 1 pair of part I problems, and each participant who solved 3 problems in part I solved 3 pairs of part I problems, we can write $m_2 + 3m_3 = 2\binom{n}{2}$.

We can now simplify this system of three equations to solve for m_3, m_2, m_1 in terms of n . If we first solve for m_3 we get $m_3 = 2\binom{n}{2} + 36 - 9n = n^2 - 10n + 36$. If we then solve for m_2 we get $9n - 36 = m_2 + 2m_3 = m_2 + 2(n^2 - 10n + 36)$ and so $m_2 = 29n - 2n^2 - 108$, but the right hand side is negative for all values of n , and clearly m_2 cannot be negative. Thus we have reached a contradiction.

5. (USAMO 1984) $P(x)$ is a polynomial of degree $3n$ such that

$$\begin{aligned}
P(0) = P(3) = \cdots &= P(3n) = 2, \\
P(1) = P(4) = \cdots &= P(3n-2) = 1, \\
P(2) = P(5) = \cdots &= P(3n-1) = 0, \quad \text{and} \\
&P(3n+1) = 730.
\end{aligned}$$

Determine n .

Solution

By Lagrange Interpolation Formula $f(x) = 2 \sum_{p=0}^n \left(\prod_{0 \leq r \neq 3p \leq 3n} \frac{x-r}{3p-r} \right) + \sum_{p=1}^n \left(\prod_{0 \leq r \neq 3p-2 \leq 3n} \frac{x-r}{3p-2-r} \right)$

and hence $f(3n+1) = 2 \sum_{p=0}^n \left(\prod_{0 \leq r \neq 3p \leq 3n} \frac{3n+1-r}{3p-r} \right) + \sum_{p=1}^n \left(\prod_{0 \leq r \neq 3p-2 \leq 3n} \frac{3n+1-r}{3p-2-r} \right)$

after some calculations we get $f(3n+1) = \left(\binom{3n+1}{0} - \binom{3n+1}{3} + \binom{3n+1}{6} - \cdots \right) (2 \cdot (-1)^{3n} - 1) + 1$

Given $f(3n+1) = 730$ so we have to find n such that $\left(\binom{3n+1}{0} - \binom{3n+1}{3} + \binom{3n+1}{6} - \cdots \right) (2 \cdot (-1)^{3n} - 1) = 729$

Lemma: If p is even $\binom{p}{0} - \binom{p}{3} + \binom{p}{6} - \cdots = \frac{2^{p+1} \sin^p\left(\frac{\pi}{3}\right) (i)^p \left(\cos\left(\frac{p\pi}{3}\right)\right)}{3}$

and if p is odd $\binom{p}{0} - \binom{p}{3} + \binom{p}{6} - \cdots = \frac{-2^{p+1} \sin^p\left(\frac{\pi}{3}\right) (i)^{p+1} \left(\sin\left(\frac{p\pi}{3}\right)\right)}{3}$

i is $\sqrt{-1}$ Using above lemmas we do not get any solution when n is odd, but when n is even $3n+1 = 13$ satisfies the required condition, hence $n = 4$

§14 USAMO 1985

- (USAMO 1985) Determine whether or not there are any positive integral solutions of the simultaneous equations

$$x_1^2 + x_2^2 + \cdots + x_{1985}^2 = y^3, \quad x_1^3 + x_2^3 + \cdots + x_{1985}^3 = z^2$$

with distinct integers $x_1, x_2, \dots, x_{1985}$.

Solution

Lemma: For a positive integer n , $1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$ (Also known as Nicomachus's theorem)

Proof by induction: The identity holds for 1. Suppose the identity holds for a number n . It is well known that the sum of first n positive integers is $\frac{n(n+1)}{2} = \frac{n^2+n}{2}$.

Thus its square is $\frac{n^4+2n^3+n^2}{4}$. Adding $(n+1)^3 = n^3 + 3n^2 + 3n + 1$ to this we get $\frac{n^4+6n^3+13n^2+12n+4}{4}$, which can be rewritten as $\frac{(n^4+4n^3+6n^2+4n+1)+2(n^3+3n^2+3n+1)+(n^2+2n+1)}{4}$

This simplifies to $\frac{(n+1)^4+2(n+1)^3+(n+1)^2}{4} = \left(\frac{(n+1)^2+(n+1)}{2}\right)^2 = (1+2+\cdots+n+(n+1))^2$.

The induction is complete.

Let j be the sum $1 + 2 + \cdots + 1985$, and let k be the sum $1^2 + 2^2 + \cdots + 1985^2$. Then assign x_i the value ik^4 for each $i = 1, 2, \dots, 1985$. Then:

$$\begin{aligned}
x_1^2 + x_2^2 + \cdots + x_{1985}^2 &= 1^2 k^8 + 2^2 k^8 + \cdots + 1985^2 k^8 = k^8 (1^2 + 2^2 + \cdots + 1985^2) = k^9 = (k^3)^3 \\
x_1^3 + x_2^3 + \cdots + x_{1985}^3 &= 1^3 k^{12} + 2^3 k^{12} + \cdots + 1985^3 k^{12} = k^{12} (1^3 + 2^3 + \cdots + 1985^3) = k^{12} j^2 = (k^6 j)^2
\end{aligned}$$

Thus, a positive integral solution exists.

-Circling

2. (*USAMO 1985*) Determine each real root of
 $x^4 - (2 \cdot 10^{10} + 1)x^2 - x + 10^{20} + 10^{10} - 1 = 0$
 correct to four decimal places.

Solution

The equation can be re-written as

$$(x + 10^5)^2(x - 10^5)^2 - (x + 10^5)(x - 10^5) - x - 1 = 0. \quad (1)$$

We first prove that the equation has no negative roots. Let $x \leq 0$. The equation above can be further re-arranged as

$$[(x + 10^5)(x - 10^5) + 1][(x + 10^5)(x - 10^5) - 2] = x - 1.$$

The right hand side of the equation is negative. Therefore

$$[(x + 10^5)(x - 10^5) + 1][(x + 10^5)(x - 10^5) - 2] < 0,$$

and we have $-1 < (x + 10^5)(x - 10^5) < 2$. Then the left hand side of the equation is bounded by

$$|[(x + 10^5)(x - 10^5) + 1][(x + 10^5)(x - 10^5) - 2]| \leq 3 \times 3.$$

However, since $|(x + 10^5)(x - 10^5)| \leq 2$ and $x < 0$, it follows that $|x + 10^5| < \frac{2}{|x - 10^5|} < 2 \times 10^{-5}$ for negative x . Then $x < 2 \times 10^{-5} - 10^5$. The right hand side of the equation is then a large negative number. It cannot be equal to the left hand side which is bounded by 9.

Now let $x > 0$. When $x = 10^5$, the left hand side of equation (1) is negative. Therefore the equation has real roots on both side of 10^5 , as its leading coefficient is positive. We will prove that $x = 10^5$ is a good approximation of the roots (within 10^{-2}). In fact, we can solve the "quadratic" equation (1) for $(x + 10^5)(x - 10^5)$:

$$(x + 10^5)(x - 10^5) = \frac{1 \pm \sqrt{1 + 4(x + 1)}}{2}.$$

Then

$$x - 10^5 = \frac{1 \pm \sqrt{1 + 4(x + 1)}}{2(x + 10^5)}.$$

Easy to see that $|x - 10^5| < 1$ for positive x . Therefore, $10^5 - 1 < x < 10^5 + 1$. Then

$$\begin{aligned} |x - 10^5| &= \left| \frac{1 \pm \sqrt{1 + 4(x + 1)}}{2(x + 10^5)} \right| \\ &\leq \left| \frac{1}{2(x + 10^5)} \right| + \left| \frac{\sqrt{1 + 4(x + 1)}}{2(x + 10^5)} \right| \\ &\leq \frac{1}{2(10^5 - 1 + 10^5)} + \frac{\sqrt{1 + 4(10^5 + 1 + 1)}}{2(10^5 - 1 + 10^5)} \\ &< 10^{-2}. \end{aligned}$$

Let x_1 be a root of the equation with $x_1 < 10^5$. Then $0 < 10^5 - x_1 < 10^{-2}$ and

$$x_1 - 10^5 = \frac{1 - \sqrt{1 + 4(x_1 + 1)}}{2(x_1 + 10^5)}.$$

An approximation of x_1 is defined as follows:

$$\tilde{x}_1 = 10^5 + \frac{1 - \sqrt{1 + 4(10^5 + 1)}}{2(10^5 + 10^5)}.$$

We check the error of the estimate:

$$\begin{aligned} |\tilde{x}_1 - x_1| &= \left| \frac{1 - \sqrt{1 + 4(10^5 + 1)}}{2(10^5 + 10^5)} - \frac{1 - \sqrt{1 + 4(x_1 + 1)}}{2(x_1 + 10^5)} \right| \\ &\leq \left| \frac{1}{2(10^5 + 10^5)} - \frac{1}{2(x_1 + 10^5)} \right| + \left| \frac{\sqrt{1 + 4(10^5 + 1)}}{2(10^5 + 10^5)} - \frac{\sqrt{1 + 4(x_1 + 1)}}{2(x_1 + 10^5)} \right|. \end{aligned}$$

The first absolute value

$$\left| \frac{1}{2(10^5 + 10^5)} - \frac{1}{2(x_1 + 10^5)} \right| = \frac{|x_1 - 10^5|}{2(10^5 + 10^5)(x_1 + 10^5)} < 10^{-12}.$$

The second absolute value

$$\begin{aligned} &\left| \frac{\sqrt{1 + 4(10^5 + 1)}}{2(10^5 + 10^5)} - \frac{\sqrt{1 + 4(x_1 + 1)}}{2(x_1 + 10^5)} \right| \\ &\leq \left| \frac{\sqrt{1 + 4(10^5 + 1)}}{2(10^5 + 10^5)} - \frac{\sqrt{1 + 4(10^5 + 1)}}{2(10^5 + 10^5)} \right| + \left| \frac{\sqrt{1 + 4(x_1 + 1)}}{2(10^5 + 10^5)} - \frac{\sqrt{1 + 4(x_1 + 1)}}{2(x_1 + 10^5)} \right| \\ &\leq 10^{-7} + 10^{-9}, \end{aligned}$$

through a rationalized numerator. Therefore $|\tilde{x}_1 - x_1| \leq 10^{-6}$.

For a real root x_2 with $x_2 > 10^5$, we choose

$$\tilde{x}_2 = 10^5 + \frac{1 + \sqrt{1 + 4(10^5 + 1)}}{2(10^5 + 10^5)}.$$

We can similarly prove it has the desired approximation.

Notes Another round of iteration can increase the accuracy to more than 10 decimal places:

$$\begin{aligned} \tilde{x}'_1 &= 10^5 + \frac{1 - \sqrt{1 + 4(\tilde{x}_1 + 1)}}{2(\tilde{x}_1 + 10^5)}, \\ \tilde{x}'_2 &= 10^5 + \frac{1 + \sqrt{1 + 4(\tilde{x}_2 + 1)}}{2(\tilde{x}_2 + 10^5)}. \end{aligned}$$

J.Z.

3. (*USAMO 1985*) Let A, B, C, D denote four points in space such that at most one of the distances AB, AC, AD, BC, BD, CD is greater than 1. Determine the maximum value of the sum of the six distances.

Solution

Suppose that AB is the length that is more than 1. Let spheres with radius 1 around A and B be S_A and S_B . C and D must be in the intersection of these spheres, and they must be on the circle created by the intersection to maximize the distance. We have $AC + BC + AD + BD = 4$.

In fact, CD must be a diameter of the circle. This maximizes the five lengths AC , BC , AD , BD , and CD . Thus, quadrilateral $ACBD$ is a rhombus.

Suppose that $\angle CAD = 2\theta$. Then, $AB + CD = 2\sin\theta + 2\cos\theta$. To maximize this, we must maximize $\sin\theta + \cos\theta$ on the range 0° to 90° . However, note that we really only have to solve this problem on the range 0° to 45° , since $\theta > 45$ is just a symmetrical function.

For $\theta < 45$, $\sin\theta \leq \cos\theta$. We know that the derivative of $\sin\theta$ is $\cos\theta$, and the derivative of $\cos\theta$ is $-\sin\theta$. Thus, the derivative of $\sin\theta + \cos\theta$ is $\cos\theta - \sin\theta$, which is nonnegative between 0° and 45° . Thus, we can conclude that this is an increasing function on this range.

It must be true that $2\sin\theta \leq 1$, so $\theta \leq 30^\circ$. But, because $\sin\theta + \cos\theta$ is increasing, it is maximized at $\theta = 30^\circ$. Thus, $AB = \sqrt{3}$, $CD = 1$, and our sum is $5 + \sqrt{3}$.

4. (*USAMO 1985*) There are n people at a party. Prove that there are two people such that, of the remaining $n - 2$ people, there are at least $\lfloor n/2 \rfloor - 1$ of them, each of whom knows both or else knows neither of the two. Assume that "know" is a symmetrical relation; $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x .

Solution

Consider the number of pairs (X, Y, Z) , where X, Y, Z are distinct points such that X is joined to just one of Y, Z . If X is joined to just k points, then there are just $k(n - 1 - k) \leq (n - 1)2/4$ such pairs (X, Y, Z) . Hence in total there are at most $\frac{n(n-1)^2}{4}$ such pairs. But there are $\frac{n(n-1)}{2}$ possible pairs Y, Z . So we must be able to find one of them A, B which belongs to at most $\lfloor \frac{n-1}{2} \rfloor$ such pairs.

Hence there are at least $n - 2 - \lfloor \frac{n-1}{2} \rfloor = \lfloor \frac{n}{2} \rfloor - 1$ points X which are joined to both of A and B or to neither of A and B . (If confused by the floor, consider $n = 2m$ and $n = 2m+1$ separately!)

5. (*USAMO 1985*) Let $a_1, a_2, a_3, \dots$ be a non-decreasing sequence of positive integers. For $m \geq 1$, define $b_m = \min\{n : a_n \geq m\}$, that is, b_m is the minimum value of n such that $a_n \geq m$. If $a_{19} = 85$, determine the maximum value of $a_1 + a_2 + \dots + a_{19} + b_1 + b_2 + \dots + b_{85}$.

Solution

We create an array of dots like so: the array shall go out infinitely to the right and downwards, and at the top of the i th column we fill the first a_i cells with one dot each. Then the 19th row shall have 85 dots. Now consider the first 19 columns of this array, and consider the first 85 rows. In row j , we see that the number of blank cells is equal to $b_j - 1$. Therefore the number of filled cells in the first 19 columns of row j is equal to $20 - b_j$.

We now count the number of cells in the first 19 columns of our array, but we do it in two different ways. First, we can sum the number of dots in each column: this is simply $a_1 + \dots + a_{19}$. Alternatively, we can sum the number of dots in each row: this is $(20 - b_1) + \dots + (20 - b_{85})$. Since we have counted the same number in two different ways, these two sums must be equal. Therefore

$$a_1 + \dots + a_{19} + b_1 + \dots + b_{85} = 20 \cdot 85 = \boxed{1700}.$$

Note that this shows that the value of the desired sum is constant.

§15 USAMO 1986

1. (*USAMO 1986*) (a) Does there exist 14 consecutive positive integers each of which is divisible by one or more primes p from the interval $2 \leq p \leq 11$?
 (b) Does there exist 21 consecutive positive integers each of which is divisible by one or more primes p from the interval $2 \leq p \leq 13$?

Solution

(a) To solve part (a), we first note that for any 14 consecutive positive integers, exactly 7 are even (divisible by 2) and therefore satisfy the criteria. We can remove these from the problem, and simplify it to the following question, which is equivalent to part (a): "Do there exist 7 consecutive positive odd integers each of which is divisible by one or more primes p from the interval $3 \leq p \leq 11$?" Among any 7 consecutive positive odd integers, the following holds:

Either 2 or 3 are divisible by 3

Either 1 or 2 are divisible by 5

Exactly 1 is divisible by 7

Either 0 or 1 are divisible by 11

For every one of these seven integers to be divisible by one of 3, 5, 7, or 11, there must be 3 multiples of 3, 2 multiples of 5, 1 multiple of 7, and 1 multiple of 11. Additionally, none of these integers may be a multiple of any two of the four aforementioned primes. Otherwise, the Pigeonhole Principle dictates that at least one integer is not divisible by any of the four, thus failing to meet the criteria. But this cannot be. Calling the consecutive odd integers $a_1, a_2, \dots, a_7$, we note that a_1, a_4 , and a_7 must be multiples of 3, and therefore cannot be multiples of 5, 7, or 11. But for there to be two multiples of 5, they must be one of the two pairs (a_1, a_6) and (a_2, a_7) . But each of these pairs contains a multiple of 3, and so at least one of the 7 odd integers is divisible by none of the primes 3, 5, 7, or 11. Therefore the answer to part (a) is no.

(b) To solve part (b), we use a strategy similar to the one used in part (a), reducing part (b) to this equivalent question: "Do there exist 10 consecutive positive odd integers each of which is divisible by one or more primes p from the interval $3 \leq p \leq 13$?" (We will ignore the case where the first of the 21 integers is odd, resulting in 11 odd integers instead of 10, as it is only true if the weaker, 10-integer argument is also true.) Among any 10 consecutive positive odd integers, the following holds:

Either 3 or 4 are divisible by 3

Exactly 2 are divisible by 5

Either 1 or 2 are divisible by 7

Either 0 or 1 are divisible by 11

Either 0 or 1 are divisible by 13

For every one of these ten integers to be divisible by one of 3, 5, 7, 11, or 13, there must be 4 multiples of 3, 2 multiples of 5, 2 multiples of 7, 1 multiple of 11, and 1 multiple of 13. As before, none of these integers may be a multiple of any two of the five primes. Calling the consecutive odd integers $a_1, a_2, \dots, a_{10}$, we note that

a_1, a_4, a_7 , and a_{10} must be multiples of 3, and therefore cannot be multiples of 5, 7, 11, or 13. Unlike part (a), however, this stipulation is not a dead end. Let the multiples of 5 be a_3 and a_8 and let the multiples of 7 be a_2 and a_9 . The multiples of 11 and 13 are a_5 and a_6 , in some order (it doesn't really matter which). An example of a sequence of 21 consecutive positive integers satisfying part (b) is the integers from 9440 to 9460 (inclusive), which can be obtained by solving modular equations that result from these statements. So the answer to part (b) is yes.

2. (*USAMO 1986*) During a certain lecture, each of five mathematicians fell asleep exactly twice. For each pair of mathematicians, there was some moment when both were asleep simultaneously. Prove that, at some moment, three of them were sleeping simultaneously.

Solution

We shall assume to the contrary that there was never a time when three mathematicians were sleeping simultaneously, and derive a contradiction.

As a subtle but logically necessary note, we will assume without loss of generality that no two events (an event is one mathematician either falling asleep or waking up) happen at the same time.

- (i) Rationale: Consider a group of events happening simultaneously—i.e., there is a moment before any of the events have happened, and there is a moment after all the events have happened, but there are no moments in which at least one but not all of the events have happened. If there is a valid timeline—i.e. one in which every pair of mathematicians was simultaneously asleep at some moment, but in which no three mathematicians are simultaneously asleep at any moment—containing a group of simultaneous events, then there is also a valid timeline in which that group of events is broken up into a sequence; you can construct the second timeline by putting the "mathematician wakes up" events first and the "mathematician falls asleep" events last. (In the first timeline, there were n mathematicians asleep in the moments immediately before the group of events, and m mathematicians asleep immediately afterwards; in the second timeline, there will be additional moments when $n-1, n-2, \dots, k, k+1, \dots, m-1$ mathematicians were asleep. If the first timeline is valid, then n and m are both less than 3, so the additional intermediate moments will not violate the "no 3 asleep at once" rule; and since we did not remove any moments, if the "all sleeping-pairs" rule was satisfied in the first timeline, then it is satisfied in the second timeline.) So the existence of any valid timeline implies the existence of a sequential valid timeline.

We have $(5 \text{ choose } 2) = 10$ sleeping-pairs of mathematicians to account for. Also, since 5 mathematicians each fell asleep 2 times, we have a total of 10 occasions on which a mathematician fell asleep. Now, if two mathematicians are ever asleep simultaneously, then one of them fell asleep while the other was already asleep. However, because of the "no three asleep at once" rule, if a mathematician falls asleep, then at most one other mathematician could have been asleep already. Therefore, each occasion when a mathematician falls asleep can account for at *most* one sleeping-pair. It would appear that we have just enough to make it. However, when the first mathematician falls asleep, no other mathematicians are asleep...

Hey, wait a minute. Couldn't one mathematician, or even two, have been already asleep when the lecture started? In fact, the wording of the problem does not forbid this, and this permits us to construct an easy counterexample:

Call the mathematicians A,B,C,D,E. A starts out asleep. B sleeps; A wakes; C sleeps; B wakes; D sleeps; C wakes; E sleeps; D wakes; A sleeps; E wakes; then, C sleeps; A wakes; E sleeps; C wakes; B sleeps; E wakes; D sleeps; B wakes; A sleeps; D wakes. This creates, in order, the sleeping-pairs AB, BC, CD, DE, EA, then AC, CE, EB, BD, DA; each mathematician falls asleep and wakes up exactly twice, and at no time are three mathematicians asleep.

Well, this must be considered a hole in the problem as written. However, if we add the (nontrivial) assumption that the mathematicians were all awake when the lecture began, then it follows that the first occasion when a mathematician falls asleep can only account for zero sleeping-pairs, and the remaining 9 occasions can only account for 9 of 10 sleeping-pairs, and so there must be some pair of mathematicians that are not simultaneously asleep during the lecture. So our assumption to the contrary implies an impossible sequence of events. Therefore, that assumption must be wrong, and there must be a moment when three mathematicians are asleep.

3. (*USAMO 1986*) What is the smallest integer n , greater than one, for which the root-mean-square of the first n positive integers is an integer?

Note. The root-mean-square of n numbers $a_1, a_2, \dots, a_n$ is defined to be

$$\left[\frac{a_1^2 + a_2^2 + \dots + a_n^2}{n} \right]^{1/2}$$

Solution 1

We use the fact that $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$. Thus the root mean square is an integer if and only if $\frac{n(n+1)(2n+1)}{6n} = \frac{(n+1)(2n+1)}{6}$ is an integer. Since $\gcd(n+1, 2n+1) = 1$ and $2n+1$ is not even, $n+1$ is even, so n is odd. Thus, $2n+1 \equiv 3 \pmod{4}$. Since perfect squares are only 0 or 1 modulo 4, $2n+1$ cannot be a perfect square. Therefore, there are nonnegative integers a and b with $n+1 = 2a^2$ and $2n+1 = 3b^2$, which means that $n \equiv 1 \pmod{3}$. Since b is odd, $b^2 \equiv 1 \pmod{8} \Rightarrow n \equiv 1 \pmod{4}$. Thus a must be an odd integer not divisible by 3. We test $a = 5, 7, 11$, all of which do not work. Testing $a = 13$ gives $(n, b) = (337, 15)$, therefore our answer is 337.

Solution 2

Let's first obtain an algebraic expression for the root mean square of the first n integers, which we denote I_n . By repeatedly using the identity $(x+1)^3 = x^3 + 3x^2 + 3x + 1$, we can write

$$1^3 + 3 \cdot 1^2 + 3 \cdot 1 + 1 = 2^3,$$

$$1^3 + 3 \cdot (1^2 + 2^2) + 3 \cdot (1 + 2) + 1 + 1 = 3^3,$$

and

$$1^3 + 3 \cdot (1^2 + 2^2 + 3^2) + 3 \cdot (1 + 2 + 3) + 1 + 1 + 1 = 4^3.$$

We can continue this pattern indefinitely, and thus for any positive integer n ,

$$1 + 3 \sum_{j=1}^n j^2 + 3 \sum_{j=1}^n j^1 + \sum_{j=1}^n j^0 = (n+1)^3.$$

Since $\sum_{j=1}^n j = n(n+1)/2$, we obtain

$$\sum_{j=1}^n j^2 = \frac{2n^3 + 3n^2 + n}{6}.$$

Therefore,

$$I_n = \left(\frac{1}{n} \sum_{j=1}^n j^2 \right)^{1/2} = \left(\frac{2n^2 + 3n + 1}{6} \right)^{1/2}.$$

Requiring that I_n be an integer, we find that

$$(2n + 1)(n + 1) = 6k^2,$$

where k is an integer. Using the Euclidean algorithm, we see that $\gcd(2n + 1, n + 1) = \gcd(n + 1, n) = 1$, and so $2n + 1$ and $n + 1$ share no factors greater than 1. The equation above thus implies that $2n + 1$ and $n + 1$ is each proportional to a perfect square. Since $2n + 1$ is odd, there are only two possible cases:

Case 1: $2n + 1 = 3a^2$ and $n + 1 = 2b^2$, where a and b are integers.

Case 2: $2n + 1 = a^2$ and $n + 1 = 6b^2$.

In Case 1, $2n + 1 = 4b^2 - 1 = 3a^2$. This means that $(4b^2 - 1)/3 = a^2$ for some integers a and b . We proceed by checking whether $(4b^2 - 1)/3$ is a perfect square for $b = 2, 3, 4, \dots$ (The solution $b = 1$ leads to $n = 1$, and we are asked to find a value of n greater than 1.) The smallest positive integer b greater than 1 for which $(4b^2 - 1)/3$ is a perfect square is $b = 13$, which results in $n = 337$.

In Case 2, $2n + 1 = 12b^2 - 1 = a^2$. Note that $a^2 = 2n + 1$ is an odd square, and hence is congruent to 1 (mod 4). But $12b^2 - 1 \equiv 3 \pmod{4}$ for any b , so Case 2 has no solutions.

Alternatively, one can proceed by checking whether $12b^2 - 1$ is a perfect square for $b = 1, 2, 3, \dots$. We find that $12b^2 - 1$ is not a perfect square for $b = 1, 2, 3, \dots, 7, 8$, and $n = 383$ when $b = 8$. Thus the smallest positive integers a and b for which $12b^2 - 1 = a^2$ result in a value of n exceeding the value found in Case 1, which was 337.

In summary, the smallest value of n greater than 1 for which I_n is an integer is 337.

4. (USAMO 1986) Two distinct circles K_1 and K_2 are drawn in the plane. They intersect at points A and B , where AB is the diameter of K_1 . A point P on K_2 and inside K_1 is also given.

Using only a "T-square" (i.e. an instrument which can produce a straight line joining two points and the perpendicular to a line through a point on or off the line), find a construction for two points C and D on K_1 such that CD is perpendicular to AB and $\angle CPD$ is a right angle.

Solution

<i>This problem needs a solution. If you have a solution for it, please help us out by [adding it](#).</i>

5. (USAMO 1986) By a partition π of an integer $n \geq 1$, we mean here a representation of n as a sum of one or more positive integers where the summands must be put in nondecreasing order. (E.g., if $n = 4$, then the partitions π are $1 + 1 + 1 + 1$, $1 + 1 + 2$, $1 + 3$, $2 + 2$, and 4).

For any partition π , define $A(\pi)$ to be the number of 1's which appear in π , and define $B(\pi)$ to be the number of distinct integers which appear in π . (E.g., if $n = 13$ and π is the partition $1 + 1 + 2 + 2 + 2 + 5$, then $A(\pi) = 2$ and $B(\pi) = 3$).

Prove that, for any fixed n , the sum of $A(\pi)$ over all partitions of π of n is equal to the sum of $B(\pi)$ over all partitions of π of n .

Solution

Let $S(n) = \sum_{\pi} A(\pi)$ and let $T(n) = \sum_{\pi} B(\pi)$. We will use generating functions to approach this problem – specifically, we will show that the generating functions of $S(n)$ and $T(n)$ are equal.

Let us start by finding the generating function of $S(n)$. This function counts the total number of 1's in all the partitions of n . Another way to count this is by counting the number of partitions of n that contain x 1's and multiplying this by x , then summing for $1 \leq x \leq n$. However, the number of partitions of n that contain x 1's is the same as the number of partitions of $n - x$ that contain no 1's, so

$$S(n) = \sum_{k=1}^n k \cdot (\# \text{ of partitions of } n - k \text{ with no 1's}).$$

The number of partitions of m with no 1's is the coefficient of x^m in

$$\begin{aligned} F(x) &= (1 + x^2 + x^4 + \dots)(1 + x^3 + x^6 + \dots)(1 + x^4 + x^8 + \dots) \dots \\ &= \prod_{i=2}^{\infty} \frac{1}{1 - x^i} \end{aligned}$$

Note that there is no $(1 + x + x^2 + x^3 + \dots)$ term in $F(x)$ because we cannot have any 1's in the partition.

Let c_m be the coefficient of x^m in the expansion of $F(x)$, so we can rewrite it as $F(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + \dots$. We wish to compute $S(n) = 1 \cdot c_{n-1} + 2 \cdot c_{n-2} + \dots + n \cdot c_0$.

Consider the power series $G(x) = (x + 2x^2 + 3x^3 + 4x^4 + \dots)F(x)$.

$$\begin{aligned} G(x) &= (x + 2x^2 + 3x^3 + 4x^4 + \dots)(c_0 + c_1x + c_2x^2 + c_3x^3 + \dots) \\ &= x(1 + 2x + 3x^2 + 4x^3 + \dots) \prod_{i=2}^{\infty} \frac{1}{1 - x^i} \\ &= x \cdot \frac{1}{(1 - x)^2} \prod_{i=2}^{\infty} \frac{1}{1 - x^i} \\ &= \frac{x}{1 - x} \prod_{i=1}^{\infty} \frac{1}{1 - x^i} \end{aligned}$$

If we expand the first line, we see that the coefficient of x^n in $G(x)$ for any n is $1 \cdot c_{n-1} + 2 \cdot c_{n-2} + \dots + n \cdot c_0$, which is exactly $S(n)$! So by definition, $G(x) = \frac{x}{1-x} \prod_{i=1}^{\infty} \frac{1}{1-x^i}$ is the generating function of $S(n)$.

Now let's find the generating function of $T(n)$. Notice that counting the number of distinct elements in each partition and summing over all partitions is equivalent to counting how many partitions of n contain i and then summing for all $1 \leq i \leq n$.

So,

$$T(n) = \sum_{k=1}^n k \cdot (\# \text{ of partitions of } n \text{ that contain a } k).$$

However, the number of partitions of n that contain a i is the same as the total number of partitions of $n - i$, so

$$T(n) = \sum_{k=1}^n k \cdot (\# \text{ of partitions of } n - k).$$

The generating function for the number of partitions is

$$\begin{aligned} P(x) &= (1 + x + x^2 + \dots)(1 + x^2 + x^4 + \dots)(1 + x^3 + x^6 + \dots) \dots \\ &= \prod_{i=1}^{\infty} \frac{1}{1 - x^i}. \end{aligned}$$

Let's write the expansion of $P(x)$ as $P(x) = d_0 + d_1x + d_2x^2 + d_3x^3 + \dots$, so we wish to find $T(n) = d_0 + d_1 + d_2 + \dots + d_{n-1}$.

Consider the power series $H(x) = (x + x^2 + x^3 + x^4 + \dots)P(x)$.

$$\begin{aligned} H(x) &= (x + x^2 + x^3 + x^4 + \dots)(d_0 + d_1x + d_2x^2 + d_3x^3 + \dots) \\ &= x(1 + x + x^2 + x^3 + \dots) \prod_{i=1}^{\infty} \frac{1}{1 - x^i} \\ &= x \cdot \frac{1}{1 - x} \prod_{i=1}^{\infty} \frac{1}{1 - x^i} \\ &= \frac{x}{1 - x} \prod_{i=1}^{\infty} \frac{1}{1 - x^i} \end{aligned}$$

If we expand the first line, we see that the coefficient of x^n in $H(x)$ for any n is $d_{n-1} + d_{n-2} + \dots + d_0$, which is precisely $T(n)$. This means $H(x)$ is the generating function of $T(n)$.

Thus, the generating functions of $S(n)$ and $T(n)$ are the same, so $S(n) = T(n)$ for all n and we are done.

Solution 2

First of all, the generating function for all partitions of n is

$$\frac{1}{(1 - x)(1 - x^2)(1 - x^3) \dots}$$

Let us define

$$s(n) = \{\# \text{ of } \Pi \text{ for } n \text{ with one } 1 \text{ in the partition}\} + 2 \cdot \{\# \text{ of } \Pi \text{ for } n \text{ with two } 1\text{'s}\} + \dots$$

Now, we make a great discovery.

$$\{\# \text{ of } \Pi \text{ with for } n \text{ with one } 1\} = \{\# \text{ of } \Pi \text{ for } n - 1 \text{ with no } 1\text{'s}\}$$

This is because when n has one 1 in a partition, then if we remove that 1, then we get the number $n - 1$, and our partition no longer has any 1s.

So, we can rewrite $s(n)$ as

$$s(n) = \{\# \text{ of } \Pi \text{ for } n-1 \text{ with no } 1 \text{ in the partition}\} + 2 \cdot \{\# \text{ of } \Pi \text{ for } n-2 \text{ with no } 1\text{s}\} + \dots$$

Let us define

$$a_n = \{\# \text{ of } \Pi \text{ for } n \text{ with no } 1\text{s}\}$$

Then,

$$s(n) = a_{n-1} + 2a_{n-2} + \dots$$

The generating function for a_n (call it $\alpha(n)$) is just

$$\frac{1}{(1-x^2)(1-x^3)\dots}$$

We removed the $(1-x)$ because $\Pi \in a_n$ cannot have any 1s. Now, let us try to find the generating function for $s(n)$.

We also know that

$$\alpha(n) = a_1x + a_2x^2 + \dots$$

By the definition of $s(n)$, we know that its generating function (call it $\sigma(n)$) looks something like this:

$$\sigma(n) = a_1x^2 + (2a_1 + a_2)x^3 + (3a_1 + 2a_2 + a_3)x^4 + \dots$$

Let's try to write this in terms of $\alpha(n)$. We know that $x\alpha(n)$ is

$$x\alpha(n) = a_1x^2 + a_2x^3 + \dots$$

Now, when we do $\sigma(n) - x\alpha(n)$, we will get that $a_1x^2, a_2x^3, a_3x^4, \dots$ cancel out.

However, we still need more to cancel out.

Let's focus on that $2a_1x^3$ that didn't cancel out and is in $\sigma(n)$. To make that cancel out, we'd have to do $\sigma(n) - x\alpha(n) - 2x^2\alpha(n)$, where

$$2x^2\alpha(n) = 2a_1x^3 + 2a_2x^4 + \dots$$

See the $2a_1x^3$ in both $\sigma(n)$ and $2x^2\alpha(n)$?

Great, now we focus on $3a_1x^4$. Just like last time, we'd have to do $\sigma(n) - x\alpha(n) - 2x^2\alpha(n) - 3x^3\alpha(n)$, where

$$3x^3\alpha(n) = 3a_1x^4 + 3a_2x^5 + \dots$$

In order to make all of that stuff in $\sigma(n)$ cancel out, following that pattern, we would do

$$\sigma(n) - \alpha(n) \cdot (x + 2x^2 + 3x^3 + 4x^4 + \dots) = 0$$

or

$$\sigma(n) = \alpha(n) \cdot (x + 2x^2 + 3x^3 + 4x^4 + \dots)$$

So what's a better way to write $x + 2x^2 + 3x^3 + 4x^4 + \dots$?

Well, we already know that

$$\frac{x}{1-x} = x + x^2 + x^3 + \dots$$

And also

$$\frac{x^2}{1-x} = x^2 + x^3 + x^4 + \dots$$

So adding these up:

$$\begin{array}{cccccc} & x & +x^2 & +x^3 & +x^4 & +x^5 & +\dots \\ + & & x^2 & +x^3 & +x^4 & +x^5 & +\dots \\ \hline & x & +2x^2 & +2x^3 & +2x^4 & +2x^5 & +\dots \end{array}$$

We already have the first few terms matching! Now, if we add on $\frac{x^3}{1-x} = x^3 + x^4 + x^5 + \dots$, we get

$$\begin{array}{cccccc} & x & +x^2 & +x^3 & +x^4 & +x^5 & +x^6 & +\dots \\ + & & x^2 & +x^3 & +x^4 & +x^5 & +x^6 & +\dots \\ + & & & x^3 & +x^4 & +x^5 & +x^6 & +\dots \\ \hline & x & +2x^2 & +3x^3 & +3x^4 & +3x^5 & +3x^6 & +\dots \end{array}$$

We see that in order to get our wanted polynomial, following the pattern, we'll have to do

$$x + 2x^2 + 3x^3 + 4x^4 + \dots = \frac{x}{1-x} + \frac{x^2}{1-x} + \frac{x^3}{1-x} + \dots$$

However, this can be written as the infinite geometric series with $a = \frac{x}{1-x}$ and $r = x$. So,

$$x + 2x^2 + 3x^3 + \dots = \frac{\frac{x}{1-x}}{1-x}$$

Thus,

$$\sigma(n) = \frac{1}{(1-x^2)(1-x^3)\dots} [\text{because of } \alpha(n)] \cdot \frac{x}{(1-x)^2} [\text{because of the thing we found above}]$$

Now, we go to $t(n)$. $t(n)$ is $\sum_{\Pi \text{ is a partition}} g(\Pi)$, where $g(\Pi)$ is the number of distinct numbers in a partition.

Let $r(1, n)$ be the number of Π that have a 1 in them $r(2, n)$ be the number of Π that have a 2 in them...

Then,

$$t(n) = r(1, n) + r(2, n) + \dots + r(n, n)$$

by the definition of $t(n)$.

Now, we notice that the generating function for $r(1, n)$ is x -normal partition gen function. because $r(1, n)$ is the number of partitions of $n-1$ (p_{n-1}) (it is a normal partion of $n-1$ with a 1 added on). $r(2, n)$ is the number of partitions of $n-2$ (p_{n-2})

So now, we see that

$$t(n) = p_{n-1} + p_{n-2} + \dots$$

We want to write down the generating function for $t(n)$, call it $\tau(x)$. When $n = 2$: $t(2) = p_1$ $t(3) = p_1 + p_2$

Similarly, by choosing some values of n , we get

$$\tau(x) = p_1 x^2 + (p_1 + p_2) x^3 + \dots$$

If $\rho(x)$ is the partition generating function, then

$$\rho(x) = p_1x + p_2x^2 + \cdots$$

And, in that case, we can substitute in $\rho(x)$, into $\tau(x)$, to get

$$\tau(x) = \rho(x)(x + x^2 + \cdots)$$

The second term in this expression $(x + x^2 + \cdots)$ can be written as the infinite sum $\frac{x}{1-x}$, so we can simplify:

$$\tau(x) = \frac{1}{(1-x)(1-x^2)\cdots} \cdot \frac{x}{1-x}$$

From above, we found that

$$\sigma(x) = \frac{1}{(1-x^2)(1-x^3)\cdots} \cdot \frac{x}{(1-x)^2}$$

If we can prove that $\tau(x) = \sigma(x)$, then we are done! They are equal, so we have proved the given statement.

Q.E.D

§16 USAMO 1987

1. (USAMO 1987) Find all solutions to $(m^2 + n)(m + n^2) = (m - n)^3$, where m and n are non-zero integers.

Solution

Expanding both sides,

$$m^3 + mn + m^2n^2 + n^3 = m^3 - 3m^2n + 3mn^2 - n^3$$

Note that m^3 can be canceled and as $n \neq 0$, n can be factored out. Writing this as a quadratic equation in n :

$$2n^2 + (m^2 - 3m)n + (3m^2 + m) = 0$$

.The discriminant $b^2 - 4ac$ equals

$$\begin{aligned} & (m^2 - 3m)^2 - 8(3m^2 + m) \\ &= m^4 - 6m^3 - 15m^2 - 8m \end{aligned}$$

, which we want to be a perfect square. Miraculously, this factors as $m(m-8)(m+1)^2$. This is square iff $m^2 - 8m$ is square or $m + 1 = 0$. It can be checked that the only nonzero m that work are $-1, 8, 9$. Finally, plugging this in and discarding extraneous roots gives all possible ordered pairs (m, n) as

$$\{(-1, -1), (8, -10), (9, -6), (9, -21)\}$$

.

2. (*USAMO 1987*) The feet of the angle bisectors of $\triangle ABC$ form a right-angled triangle. If the right-angle is at X , where AX is the bisector of $\angle A$, find all possible values for $\angle A$.

Solution

Let two new segments FD and BE intersect at point X and let AD and CF intersect at Y . Consider quadrilateral $AFDC$. Then we get that the cross ratio $(B, Y; X, E) = -1$. Thus the condition $\angle FDE = 90^\circ$ implies $\angle FDA = \angle FDB$. Hence F is the A -excenter of the triangle ADC . Thus we get $\angle TAF = \angle FAD = \angle DAC$, where T is intersection of FD and AC . Then we get $\angle BAC = 120^\circ$.

3. (*USAMO 1987*) X is the smallest set of polynomials $p(x)$ such that:

- (i) 1. $p = x$ belongs to X .
- (ii) 2. If r belongs to X , then $x \cdot r$ and $(x + (1 - x) \cdot r)$ both belong to X .

Show that if $r(x)$ and $s(x)$ are distinct elements of X , then $r(x) \neq s(x)$ for any $0 < x < 1$.

Solution

Let $s(x)$ be an arbitrary polynomial in X . Then $0 < s(x) < 1$ when $0 < x < 1$. Define $X_1 = \{s(x) \in X : s(x) = x \cdot s_1(x) \text{ for some } s_1(x) \in X\}$, and $X_2 = \{t(x) \in X : t(x) = x + (1 - x)t_1(x) \text{ for some } t_1(x) \in X\}$.

If $s(x) \in X_1$ and $t(x) \in X_2$, we have $s(x) < x < t(x)$ for all x with $0 < x < 1$. Therefore $s(x) \neq t(x)$ for any $0 < x < 1$.

For any $s(x), t(x) \in X_1$, Let $s(x) = x \cdot s_1(x)$ and $t(x) = x \cdot t_1(x)$ for $s_1(x), t_1(x) \in X$. If $s_1(x) \neq t_1(x)$ for $0 < x < 1$, then $s(x) - t(x) = x(s_1(x) - t_1(x)) \neq 0$ for $0 < x < 1$.

Similarly, for any $s(x), t(x) \in X_2$, Let $s(x) = x + (1 - x)s_1(x)$ and $t(x) = x + (1 - x)t_1(x)$ for $s_1(x), t_1(x) \in X$. If $s_1(x) \neq t_1(x)$ for $0 < x < 1$, then $s(x) - t(x) = (1 - x)(s_1(x) - t_1(x)) \neq 0$ for $0 < x < 1$.

The proof is done by an induction.

J.Z.

4. (*USAMO 1987*) Three circles C_i are given in the plane: C_1 has diameter AB of length 1; C_2 is concentric and has diameter k ($1 < k < 3$); C_3 has center A and diameter $2k$. We regard k as fixed. Now consider all straight line segments XY which have one endpoint X on C_2 , one endpoint Y on C_3 , and contain the point B .

For what ratio $\frac{XB}{BY}$ will the segment XY have minimal length?

Solution

<i>This problem needs a solution. If you have a solution for it, please help us out by [adding it](#).</i>

5. (*USAMO 1987*) $a_1, a_2, \dots, a_n$ is a sequence of 0's and 1's. T is the number of triples (a_i, a_j, a_k) with $i < j < k$ which are not equal to $(0, 1, 0)$ or $(1, 0, 1)$. For $1 \leq i \leq n$, $f(i)$ is the number of $j < i$ with $a_j = a_i$ plus the number of $j > i$ with $a_j \neq a_i$. Show that $T = \sum_{i=1}^n f(i) \cdot \left(\frac{f(i)-1}{2}\right)$. If n is odd, what is the smallest value of T ?

Solution

Let $g(i)$ be the number of $j < i$ with $a_j = a_i$ and $h(i)$ be the number of $j > i$ with $a_j \neq a_i$, so that $f(i) = g(i) + h(i)$. Then

$$\begin{aligned} f(i) \left(\frac{f(i) - 1}{2} \right) &= (g(i) + h(i)) \left(\frac{g(i) + h(i) - 1}{2} \right) \\ &= \frac{g(i)^2 + h(i)^2 + 2g(i)h(i) - g(i) - h(i)}{2} \\ &= \frac{g(i)(g(i) - 1)}{2} + \frac{h(i)(h(i) - 1)}{2} + g(i)h(i) \\ &= \binom{g(i)}{2} + \binom{h(i)}{2} + \binom{g(i)}{1} \binom{h(i)}{1} \end{aligned}$$

The first term counts the number of triples of the form $(0, 0, 0)$ or $(1, 1, 1)$ with a_i as the last member. The second term counts the number of triples of the form $(0, 1, 1)$ or $(1, 0, 0)$ with a_i as the first member. The third term counts the number of triples of the form $(0, 0, 1)$ or $(1, 1, 0)$ with a_i as the middle member. Thus $\sum_{i=1}^n f(i) \cdot \left(\frac{f(i)-1}{2} \right)$ exhaustively and uniquely counts all triples except for those of the forms $(1, 0, 1)$ and $(0, 1, 0)$.

Finding the smallest value of T is trickier. First, consider what happens when we reverse the order of the sequence. For a given a_i , the less than/greater than relation between a_i and a_j , which determines whether f counts $a_i = a_j$ or $a_i \neq a_j$, is reversed for all $i \neq j$, so in the new sequence a_i will correspond to a function value of $n - 1 - f(i)$. However, notice that T does not change when we reverse the order of the sequence, as any triple $(1, 0, 1)$ or $(0, 1, 0)$ will also be present in the reversed sequence. We'll set T from the original sequence equal to T from the reversed sequence

$$\begin{aligned} \frac{1}{2} \sum_{i=1}^n f(i)(f(i) - 1) &= \frac{1}{2} \sum_{i=1}^n (n - 1 - f(i)) \cdot ((n - 2 - f(i))) \\ \sum_{i=1}^n (f(i)^2 - f(i)) &= \sum_{i=1}^n (n^2 - 2n - nf(i) - n + 2 + f(i) - nf(i) + 2f(i) + f(i)^2) \\ &= - \sum_{i=1}^n f(i) = \sum_{i=1}^n ((3 - 2n)f(i) + n^2 - 3n + 2) \\ &= - \sum_{i=1}^n f(i) = (3 - 2n) \sum_{i=1}^n (f(i)) + n^3 - 3n^2 + 2n \\ &= - \sum_{i=1}^n f(i) = (3 - 2n) \sum_{i=1}^n (f(i)) + n(n - 1)(n - 2) \\ &= (2n - 4) \sum_{i=1}^n f(i) = n(n - 1)(n - 2) \\ &= \sum_{i=1}^n f(i) = \frac{n(n - 1)}{2} \end{aligned}$$

We find that for all sequences, $\sum_{i=1}^n f(i) = \frac{n(n-1)}{2}$. We'll use the method of Lagrange multipliers to minimize $\frac{1}{2} \sum_{i=1}^n f(i)(f(i) - 1)$ subject to $\sum_{i=1}^n f(i) = \frac{n(n-1)}{2}$. Note

that this only establishes a sufficient condition for a minimum, we will show later that it is possible to define a sequence which satisfies that condition.

$$\mathcal{L}(f(1), \dots, f(n), \lambda) = \frac{1}{2} \sum_{i=1}^n f(i)(f(i) - 1) + \lambda \left(\sum_{i=1}^n (f(i)) - \frac{n(n-1)}{2} \right)$$

For all i , we get $\frac{d\mathcal{L}}{df(i)} = \frac{1}{2}(2f(i) - 1) + \lambda$. Setting the gradient to 0 and solving quickly yields $f(i) = \frac{1}{2} - \lambda$. Then we take the derivative with respect to λ : $\frac{d\mathcal{L}}{d\lambda} = \sum_{i=1}^n (f(i)) - \frac{n(n-1)}{2}$. Setting the gradient to 0 and substituting our expression for $f(i)$ gives $0 = n(\frac{1}{2} - \lambda) - \frac{n(n-1)}{2}$, which simplifies to $\lambda = \frac{2-n}{2}$. Finally, substituting this back into the expression for $f(i)$ gives $f(i) = \frac{n-1}{2}$.

This tells us that if we can define a sequence for any odd n such that $f(i) = \frac{n-1}{2}$ for all i , this would certainly be a sequence which minimizes T . In fact, an alternating sequence $0, 1, 0, 1, \dots, 0$ satisfies this, because for any i , the number of $j < i$ such that $a_j = a_i$ is $\lfloor \frac{i-1}{2} \rfloor$ and the number of $j > i$ such that $a_j \neq a_i$ is $\lceil \frac{n-i}{2} \rceil$. For odd i , $\lfloor \frac{i-1}{2} \rfloor + \lceil \frac{n-i}{2} \rceil = \frac{i-1}{2} + \frac{n-i}{2} = \frac{n-1}{2}$ and for even i , $\lfloor \frac{i-1}{2} \rfloor + \lceil \frac{n-i}{2} \rceil = \frac{i-2}{2} + \frac{n-i+1}{2} = \frac{n-1}{2}$. Thus T is minimized by this alternating sequence, where it is equal to $\sum_{i=1}^n \frac{n-1}{2} \frac{n-3}{4} = \frac{n(n-1)(n-3)}{8}$.

§17 USAMO 1988

1. (USAMO 1988) The repeating decimal $0.ab \cdots k \overline{pq \cdots u} = \frac{m}{n}$, where m and n are relatively prime integers, and there is at least one decimal before the repeating part. Show that n is divisible by 2 or 5 (or both). (For example, $0.0113\overline{6} = 0.01136363636 \cdots = \frac{1}{88}$, and 88 is divisible by 2.)

Solution 1

First, split up the nonrepeating parts and the repeating parts of the decimal, so that the nonrepeating parts equal to $\frac{a}{b}$ and the repeating parts of the decimal is equal to $\frac{c}{d}$.

Suppose that the length of $0.ab \cdots k$ is p digits. Then $\frac{a}{b} = \frac{0.ab \cdots k}{10^{p+1}}$. Since $0.ab \cdots k < 10^{p+1}$, after reducing the fraction, there MUST be either a factor of 2 or 5 remaining in the denominator. After adding the fractions $\frac{a}{b} + \frac{c}{d}$, the simplified denominator n will be

$\text{lcm}(b, d)$ and since b has a factor of 2 or 5, n must also have a factor of 2 or 5.

■

Solution 2

It is well-known that $0.ab \cdots k \overline{pq \cdots u} = \frac{ab \cdots u - ab \cdots k}{99 \cdots 900 \cdots 0}$, where there are a number of 9s equal to the count of digits in $pq \cdots u$, and there are a number of 0s equal to the count of digits c in $ab \cdots k$. Obviously $ab \cdots k$ is different from $pq \cdots u$ (which is itself the repeating part), so the numerator cannot have c consecutive terminating zeros, and hence the denominator still possesses a factor of 2 or 5 not canceled out. This completes the proof.

2. (USAMO 1988) The cubic polynomial $x^3 + ax^2 + bx + c$ has real coefficients and three real roots $r \geq s \geq t$. Show that $k = a^2 - 3b \geq 0$ and that $\sqrt{k} \leq r - t$.

Solution 1

By Vieta's Formulas, $a = -r - s - t$, $b = rs + st + rt$, and $c = -rst$. Now we know $k = a^2 - 3b$; in terms of r , s , and t , then,

$$k = (-r - s - t)^2 - 3(rs + st + rt)$$

$$k = r^2 + s^2 + t^2 - rs - st - rt$$

Now notice that we can multiply both sides by 2, and rearrange terms to get $2k = (r - s)^2 + (s - t)^2 + (r - t)^2$. But since $r, s, t \in \mathbb{R}$, the three terms of the RHS are all non-negative (as the square of a real number is always non-negative), and therefore their sum is also non-negative – that is, $2k \geq 0 \Rightarrow k \geq 0$.

Now, we will show that $\sqrt{k} \leq r - t$. We can square both sides, and the inequality will hold since they are both non-negative (it is given that $r \geq t$, therefore $r - t \geq 0$). This gives $k \leq r^2 - 2rt + t^2$. Now we already have $k = r^2 + s^2 + t^2 - rs - st - rt$, so substituting this for k gives

$$r^2 + s^2 + t^2 - rs - st - rt \leq r^2 - 2rt + t^2$$

$$s^2 - rs - st + rt \leq 0$$

$$s^2 - (r + t)s + rt \leq 0$$

Note that this is a quadratic. Since its leading coefficient is positive, its value is less than 0 when s is between the two roots. Using the quadratic formula:

$$s = \frac{r + t \pm \sqrt{(-r - t)^2 - 4rt}}{2}$$

$$s = \frac{r + t \pm \sqrt{r^2 - 2rt + t^2}}{2}$$

$$s = \frac{r + t \pm (r - t)}{2}$$

$$s \in \{r, t\}$$

The quadratic is 0 when s is equal to r or t , and the inequality holds when its value is less than or equal to 0 – that is, $r \geq s \geq t$. (Its value is less than or equal to 0 when s is between the roots, since the graph of the quadratic opens upward.) In fact, the problem tells us this is true. Q.E.D.

Solution 2

From Vieta's Formula (which tells us that $a = -(r + s + t)$ and $b = rs + st + rt$), we have that

$$k = a^2 - 3b = r^2 + s^2 + t^2 - rs - st - rt = \frac{1}{2}((r - s)^2 + (s - t)^2 + (r - t)^2),$$

clearly non-negative. To prove $\sqrt{k} \leq r - t$, it suffices to prove the square of this relation, or

$$r^2 + s^2 + t^2 - rs - st - rt \leq r^2 - 2rt + t^2.$$

This in turn simplifies to

$$rs + st - rt - s^2 \geq 0,$$

or

$$(r - s)(s - t) \geq 0,$$

which is clearly true as $r \geq s \geq t$. This completes the proof.

Solution 3

By Vieta's Formulas, $a = -(r + s + t)$ and $b = rs + st + rt$. $k = (r + s + t)^2 - 3(rs + st + rt) = r^2 + s^2 + t^2 - rs - st - rt = \frac{(r-s)^2 + (s-t)^2 + (r-t)^2}{2}$.

To show that $k \geq 0$, simply note that by the trivial inequality, all three squares are greater than 0 as they are the squares of real numbers.

To show that $\sqrt{k} \leq r - t$, since both are positive, it is sufficient to show that $k \leq (r - t)^2$. $\frac{(r-s)^2 + (s-t)^2 + (r-t)^2}{2} \leq (r-t)^2$ implies that $k \leq (r-t)^2$. $\frac{(r-s)^2 + (s-t)^2 + (r-t)^2}{2} \leq 0$.

Let $y = r - s$ and $z = s - t$. We then have $\frac{y^2 + z^2 - (y+z)^2}{2} \leq 0 \implies -2yz \leq 0$, which is clearly true as both y and z are positive.

3. (USAMO 1988) Let X be the set $\{1, 2, \dots, 20\}$ and let P be the set of all 9-element subsets of X . Show that for any map $f : P \mapsto X$ we can find a 10-element subset Y of X , such that $f(Y - \{k\}) \neq k$ for any k in Y .

Solution

For a given function f , consider the set of elements $Q \in P$ paired with elements $t \in S$ such that $f(Q) \neq t$. Call the set of pairs (Q, t) set R .

To employ the Pigeonhole Principle, we need a lower bound on $|R|$. There are $\binom{20}{9}$ elements Q in P . If $f(Q) \in Q$, then there are 11 choices for t so that $f(Q) \neq t$. If $f(Q) \notin Q$, then there are 10 choices for t so that $f(Q) \neq t$. Therefore $|R| \geq 10 \cdot \binom{20}{9}$. These pairs (Q, t) must be dispersed over all the 10-element subsets of S . By the Pigeonhole Principle, some 10-element subset of S must contain at least

$$\lceil \frac{|R|}{\binom{20}{10}} \rceil = \lceil \frac{10 \cdot \binom{20}{9}}{\binom{20}{10}} \rceil = \lceil \frac{100}{11} \rceil = 10$$

pairs of a 9-element subset Q with the tenth element t of that subset such that $f(Q) \neq t$, which satisfies the conditions of the problem.

4. (USAMO 1988) $\triangle ABC$ is a triangle with incenter I . Show that the circumcenters of $\triangle IAB$, $\triangle IBC$, and $\triangle ICA$ lie on a circle whose center is the circumcenter of $\triangle ABC$.

Solution 1

Let the circumcenters of $\triangle IAB$, $\triangle IBC$, and $\triangle ICA$ be O_c , O_a , and O_b , respectively. It then suffices to show that A , B , C , O_a , O_b , and O_c are concyclic.

We shall prove that quadrilateral ABO_aC is cyclic first. Let $\angle BAC = \alpha$, $\angle CBA = \beta$, and $\angle ACB = \gamma$. Then $\angle ICB = \gamma/2$ and $\angle IBC = \beta/2$. Therefore minor arc BIC in the circumcircle of IBC has a degree measure of $\beta + \gamma$. This shows that $\angle CO_aB = \beta + \gamma$, implying that $\angle BAC + \angle BO_aC = \alpha + \beta + \gamma = 180^\circ$. Therefore quadrilateral ABO_aC is cyclic.

This shows that point O_a is on the circumcircle of $\triangle ABC$. Analogous proofs show that O_b and O_c are also on the circumcircle of ABC , which completes the proof. ■

Solution 2

Let M denote the midpoint of arc AC . It is well known that M is equidistant from A , C , and I (to check, prove $\angle IAM = \angle AIM = \frac{\angle BAC + \angle ABC}{2}$), so that M is the circumcenter of AIC . Similar results hold for BIC and CIA , and hence O_c , O_a , and O_b all lie on the circumcircle of ABC .

Solution 3

Extend CI to point L on (ABC) . By The Incenter-Excenter Lemma, B, I, A are all concyclic. Thus, L is the circumcenter of triangle IAB . In other words, $L = O_c$, so O_c is on (ABC) . Similarly, we can show that O_a and O_b are on (ABC) , and thus, A, B, C, O_a, O_b, O_c are all concyclic. It follows that the circumcenters are equal.

Solution 4

Let the centers be T, R, S . We want to show that ABC and TRS have the same circumcircle. By Fact 5 we know that $CATB$ lie on a circle and similarly with the others. Thus the two triangles have the same circumcircle. coolmath_2018

5. (USAMO 1988) Let $p(x)$ be the polynomial $(1-x)^a(1-x^2)^b(1-x^3)^c \cdots (1-x^{32})^k$, where $a, b, \dots, k$ are integers. When expanded in powers of x , the coefficient of x^1 is -2 and the coefficients of $x^2, x^3, \dots, x^{32}$ are all zero. Find k .

Solution 1

First, note that if we reverse the order of the coefficients of each factor, then we will obtain a polynomial whose coefficients are exactly the coefficients of $p(x)$ in reverse order. Therefore, if

$$p(x) = (1-x)^{a_1}(1-x^2)^{a_2}(1-x^3)^{a_3} \cdots (1-x^{32})^{a_{32}},$$

we define the polynomial $q(x)$ to be

$$q(x) = (x-1)^{a_1}(x^2-1)^{a_2}(x^3-1)^{a_3} \cdots (x^{32}-1)^{a_{32}},$$

noting that if the polynomial has degree n , then the coefficient of x^{n-1} is -2 , while the coefficients of x^{n-k} for $k = 2, 3, \dots, 32$ are all 0.

Let P_n be the sum of the n th powers of the roots of $q(x)$. In particular, by Vieta's formulas, we know that $P_1 = 2$. Also, by Newton's Sums, as the coefficients of x^{n-k} for $k = 2, 3, \dots, 32$ are all 0, we find that

$$\begin{aligned} P_2 - 2P_1 &= 0 \\ P_3 - 2P_2 &= 0 \\ P_4 - 2P_3 &= 0 \\ &\vdots \\ P_{32} - 2P_{31} &= 0. \end{aligned}$$

Thus $P_n = 2^n$ for $n = 1, 2, \dots, 32$. Now we compute P_{32} . Note that the roots of $(x^n - 1)^{a_n}$ are all n th roots of unity. If $\omega = e^{2\pi i/n}$, then the sum of 32nd powers of these roots will be

$$a_n(1 + \omega^{32} + \omega^{32 \cdot 2} + \cdots + \omega^{32 \cdot (n-1)}).$$

If $\omega^{32} \neq 1$, then we can multiply by $(\omega^{32} - 1)/(\omega^{32} - 1)$ to obtain

$$\frac{a_n(1 - \omega^{32n})}{1 - \omega^{32}}.$$

But as $\omega^n = 1$, this is just 0. Therefore the sum of the 32nd powers of the roots of $q(x)$ is the same as the sum of the 32nd powers of the roots of

$$(x-1)^{a_1}(x^2-1)^{a_2}(x^4-1)^{a_4}(x^8-1)^{a_4}(x^{16}-1)^{a_{16}}(x^{32}-1)^{a_{32}}.$$

The 32nd power of each of these roots is just 1, hence the sum of the 32nd powers of the roots is

$$P_{32} = 2^{32} = a_1 + 2a_2 + 4a_4 + 8a_8 + 16a_{16} + 32a_{32}. \quad (1)$$

On the other hand, we can use the same logic to show that

$$P_{16} = 2^{16} = a_1 + 2a_2 + 4a_4 + 8a_8 + 16a_{16}. \quad (2)$$

Subtracting (2) from (1) and dividing by 32, we find

$$a_{32} = \frac{2^{32} - 2^{16}}{2^5}.$$

Therefore, $a_{32} = 2^{27} - 2^{11}$.

Solution 2

By a limiting process, we can extend the problem to that of finding a sequence $b_1, b_2, \dots$ of integers such that

$$(1 - z)^{b_1}(1 - z^2)^{b_2}(1 - z^3)^{b_3} \dots = 1 - 2z.$$

(The notation comes from the Alcumus version of this problem.)

If we take logarithmic derivatives on both sides, we get

$$\sum_{n=1}^{\infty} \frac{b_n \cdot (-nz^{n-1})}{1 - z^n} = \frac{-2}{1 - 2z},$$

and upon multiplying both sides by $-z$, this gives us the somewhat simple form

$$\sum_{n=1}^{\infty} nb_n \cdot \frac{z^n}{1 - z^n} = \frac{2z}{1 - 2z}.$$

Expanding all the fractions as geometric series, we get

$$\sum_{n=1}^{\infty} nb_n \sum_{k=1}^{\infty} z^{nk} = \sum_{n=1}^{\infty} 2^n z^n.$$

Comparing coefficients, we get

$$\sum_{d|n} db_d = 2^n$$

for all positive integers n . In particular, as in Solution 1, we get

$$\begin{aligned} b_1 + 2b_2 + 4b_4 + 8b_8 + 16b_{16} + 32b_{32} &= 2^{32}, \\ b_1 + 2b_2 + 4b_4 + 8b_8 + 16b_{16} &= 2^{16}, \end{aligned}$$

from which the answer $b_{32} = 2^{27} - 2^{11}$ follows.

Remark: To avoid the question of what an infinite product means in the context of formal power series, we could instead view the problem statement as saying that

$$(1 - z)^{b_1}(1 - z^2)^{b_2} \dots (1 - z^{32})^{b_{32}} \equiv 1 - 2z \pmod{z^{33}};$$

modular arithmetic for polynomials can be defined in exactly the same way as modular arithmetic for integers. Uniqueness of the b_n 's comes from the fact that we have

$$(1 - z)^{b_1} \cdots (1 - z^{n-1})^{b_{n-1}} \equiv 1 - 2z \pmod{z^n}$$

for all $n \leq 33$ by further reduction modulo z^n (as $z^n \mid z^{33}$ for $n \leq 33$), so we could uniquely solve for the b_n 's one at a time. (This idea can be pushed further to explain why it's fine to pass to the infinite product version of the problem.)

To convert the above solution to one that works with polynomials modulo z^{33} , note that the derivative is not well-defined, as for instance, 1 and $1 + z^{33}$ are equivalent modulo z^{33} , but their derivatives, 0 and $33z^{32}$, are not. However, the operator $f(z) \mapsto zf'(z)$ is well-defined. The other key idea is that for any n , modulo z^n , polynomials of the form $1 - zf(z)$ are invertible, with inverse

$$\frac{1}{1 - zf(z)} \equiv \frac{1 - (zf(z))^n}{1 - zf(z)} = 1 + zf(z) + \cdots + (zf(z))^{n-1}.$$

Therefore, for the polynomial in the problem, call it $g(z)$, we can still form the expression $zg'(z)/g(z)$, which is what we originally got by taking the logarithmic derivative and multiplying by z , and expand it to eventually get

$$\sum_{n=1}^{32} nb_n \sum_{k=1}^{32} z^{nk} \equiv \sum_{n=1}^{32} 2^n z^n \pmod{z^{33}},$$

which gets us the same relations (for $n \leq 32$).

Solution 3

From the starting point of Solution 2,

$$(1 - z)^{b_1} (1 - z^2)^{b_2} (1 - z^3)^{b_3} \cdots = 1 - 2z,$$

taking reciprocals and expanding with geometric series gives us

$$\prod_{n=1}^{\infty} \left(\sum_{k=0}^{\infty} z^{kn} \right)^{b_n} = \sum_{n=0}^{\infty} 2^n z^n.$$

On the right, we have the generating function for the number of monic polynomials of degree n over the field $\mathbb{F}_2$ of two elements, and on the left, we have the factorisation of this generating function that considers the breakdown of any given monic polynomial into monic irreducible factors. As such, we have the interpretation

$$b_n = \text{number of monic irreducible polynomials of degree } n \text{ over } \mathbb{F}_2.$$

From here, to determine b_n , we analyse the elements of $\mathbb{F}_{2^n}$, of which there are 2^n in total. Given $\alpha \in \mathbb{F}_{2^n}$, if the minimal polynomial f_α of α has degree d , then $d \mid n$ and all other roots of f_α appear in $\mathbb{F}_{2^n}$. Moreover, if $d \mid n$ and f is an irreducible polynomial of degree d , then all roots of f appear in $\mathbb{F}_{2^n}$. (These statements are all well-known in the theory of finite fields.) As such, for each $d \mid n$, there are precisely db_d elements of $\mathbb{F}_{2^n}$ of degree d , and we obtain the same equation as in Solution 2,

$$\sum_{d \mid n} db_d = 2^n.$$

The rest is as before.

§18 USAMO 1989

1. (*USAMO 1989*) For each positive integer n , let

$$S_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

$$T_n = S_1 + S_2 + S_3 + \cdots + S_n$$

$$U_n = \frac{T_1}{2} + \frac{T_2}{3} + \frac{T_3}{4} + \cdots + \frac{T_n}{n+1}.$$

Find, with proof, integers $0 < a, b, c, d < 1000000$ such that $T_{1988} = aS_{1989} - b$ and $U_{1988} = cS_{1989} - d$.

Solution

We note that for all integers $n \geq 2$,

$$\begin{aligned} T_{n-1} &= 1 + \left(1 + \frac{1}{2}\right) + \left(1 + \frac{1}{2} + \frac{1}{3}\right) + \cdots + \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1}\right) \\ &= \sum_{i=1}^{n-1} \left(\frac{n-i}{i}\right) = n \left(\sum_{i=1}^{n-1} \frac{1}{i}\right) - (n-1) = n \left(\sum_{i=1}^n \frac{1}{i}\right) - n \\ &= n \cdot S_n - n. \end{aligned}$$

It then follows that

$$\begin{aligned} U_{n-1} &= \sum_{i=2}^n \frac{T_{i-1}}{i} = \sum_{i=2}^n (S_i - 1) = T_{n-1} + S_n - (n-1) - S_1 \\ &= (nS_n - n) + S_n - n = (n+1)S_n - 2n. \end{aligned}$$

If we let $n = 1989$, we see that $(a, b, c, d) = (1989, 1989, 1990, 2 \cdot 1989)$ is a suitable solution. ■

Notice that it is also possible to use induction to prove the equations relating T_n and U_n with S_n .

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

2. (*USAMO 1989*) The 20 members of a local tennis club have scheduled exactly 14 two-person games among themselves, with each member playing in at least one game. Prove that within this schedule there must be a set of 6 games with 12 distinct players.

Solution 1

Consider a graph with 20 vertices and 14 edges. The sum of the degrees of the vertices is 28; by the **Pigeonhole Principle** at least 12 vertices have degrees of 1 and at most 8 vertices have degrees greater than 1. If we keep deleting edges of vertices with degree greater than 1 (a maximum of 8 such edges), then we are left with at least 6 edges, and all of the vertices have degree either 0 or 1. These 6 edges represent the 6 games with 12 distinct players.

Solution 2

Let a slot be a place we can put a member in a game, so there are two slots per game, and 28 slots total. We begin by filling exactly 20 slots each with a distinct member since each member must play at least one game. Let there be m games with both slots filled and n games with only one slot filled, so $2m + n = 20$. Since there

are only 14 games, $m + n \leq 14 \implies 2m + n \leq 14 + m \iff 20 \leq 14 + m \implies m \geq 6$, so there must be at least 6 games with two distinct members each, and we must have our desired set of 6 games.

Solution 3

Assume the contrary.

Consider the largest set of disjoint edges E . By assumption it has less than 6 edges, i.e. maximum 10 vertices. Call it a vertex set V .

10 vertices remain outside V and each has to be attached to at least one edge. Now, if any two vertices outside V are connected by, say, edge e , we could have included e in E and gotten a larger disjoint set, so - a contradiction. Therefore the only option would be that all vertices outside V are connected each by one edge to some vertices inside V . That would take 10 edges, but E already includes 5 - again a contradiction.

All possibilities yield a contradiction, so our assumption can not be correct.

(Cases when largest set E is smaller than 6 are equivalent and weaker)

Solution 4

Assume, for the sake of contradiction, that there is no set of 6 games with 12 distinct players. Then the largest possible collection of games with all distinct players must consist of at most 5 games, involving at most 10 players. So suppose we have 5 such games with 10 distinct players. This leaves 10 other players who must still appear in the schedule. Since there are a total of 14 games, there are $14 - 5 = 9$ games remaining to cover these 10 players.

Each game has 2 players, so the 9 games account for 18 "player slots. *If every one of the 10 unused players is to appear at least once, then by the pigeonhole principle, some game among these 9 must contain two of the 10 unused players.* But then that game, together with the original 5 disjoint games, would give 6 games with 12 distinct players. This contradicts our assumption that no such collection exists. Therefore, there must in fact exist a set of 6 games with 12 distinct players.

3. (USAMO 1989) Let $P(z) = z^n + c_1 z^{n-1} + c_2 z^{n-2} + \cdots + c_n$ be a polynomial in the complex variable z , with real coefficients c_k . Suppose that $|P(i)| < 1$. Prove that there exist real numbers a and b such that $P(a + bi) = 0$ and $(a^2 + b^2 + 1)^2 < 4b^2 + 1$.

Solution

Let $z_1, \dots, z_n$ be the (not necessarily distinct) roots of P , so that

$$P(z) = \prod_{j=1}^n (z - z_j).$$

Since all the coefficients of P are real, it follows that if w is a root of P , then $P(\bar{w}) = \overline{P(w)} = 0$, so $\bar{w}$, the **complex conjugate** of w , is also a root of P .

Since

$$|i - z_1| \cdot |i - z_2| \cdots |i - z_n| = |P(i)| < 1,$$

it follows that for some (not necessarily distinct) conjugates z_i and z_j ,

$$|z_i - i| \cdot |z_j - i| < 1.$$

Let $z_i = a + bi$ and $z_j = a - bi$, for real a, b . We note that

$$(a + b + 1)^2 - (a + b - 1)^2 = 4a + 4b.$$

Thus

$$\begin{aligned}
 (a^2 + b^2 + 1)^2 &= (a^2 + b^2 - 1)^2 + 4a^2 + 4b^2 = |a^2 + b^2 - 1 - 2ai|^2 + 4b^2 \\
 &= |(a - i)^2 - (bi)^2|^2 + 4b^2 \\
 &= (|a + bi - i| \cdot |a - bi - i|)^2 + 4b^2 \\
 &= (|z_i - i| \cdot |z_j - i|)^2 + 4b^2 < 1 + 4b^2.
 \end{aligned}$$

Since $P(a + bi) = P(z_i) = 0$, these real numbers a, b satisfy the problem's conditions. ■

4. (USAMO 1989) Let ABC be an acute-angled triangle whose side lengths satisfy the inequalities $AB < AC < BC$. If point I is the center of the inscribed circle of triangle ABC and point O is the center of the circumscribed circle, prove that line IO intersects segments AB and BC .

Solution 1

Proceed with barycentrics. First, note that the desired conclusion is equivalent to showing that $X = \overleftrightarrow{IO} \cap \overleftrightarrow{AC}$ is not on segment AC .

Let $X = (a : 0 : t)$, so

$$0 = \begin{vmatrix} a & 0 & t \\ a & b & c \\ a^2 S_A & b^2 S_B & c^2 S_C \end{vmatrix} = abc \begin{vmatrix} 1 & 0 & \frac{t}{c} \\ 1 & 1 & 1 \\ a S_A & b S_B & c S_C \end{vmatrix} \implies t = \frac{c(c S_C - b S_B)}{a S_A - b S_B}$$

But,

$$t = \frac{c(c S_C - b S_B)}{a S_A - b S_B} = \frac{c(c(a^2 + b^2 - c^2) - b(a^2 + c^2 - b^2))}{a(b^2 + c^2 - a^2) - b(a^2 + c^2 - b^2)} = \frac{c(c - b)(a^2 + b^2 + c^2)}{(a - b)(a^2 + b^2 + c^2)} = \frac{c(c - b)}{a - b} < 0$$

as needed.

-sillybone

Solution 2

Consider the lines that pass through the circumcenter O . Extend AO, BO, CO to D, E, F on a, b, c , respectively.

We notice that IO passes through sides a and c if and only if I belongs to either regions AOF or COD .

Since $AO = BO = CO = R$, we let $\alpha = \angle OAC = \angle OCA$, $\beta = \angle BAO = \angle ABO$, $\gamma = \angle BCO = \angle CBO$.

We have $c < b < a \implies C < B < A \implies \gamma + \alpha < \beta + \gamma < \alpha + \beta \implies \gamma < \alpha < \beta$

Since IA divides angle A into two equal parts, it must be in the region marked by the β of angle A , so I is in ABD .

Similarly, I is in ACF and ABE . Thus, I is in their intersection, AOF . From above, we have IO passes through a and c . ■

5. (USAMO 1989) Let u and v be real numbers such that

$$(u + u^2 + u^3 + \cdots + u^8) + 10u^9 = (v + v^2 + v^3 + \cdots + v^{10}) + 10v^{11} = 8.$$

Determine, with proof, which of the two numbers, u or v , is larger.

Solution

The answer is v .

We define real functions U and V as follows:

$$U(x) = (x + x^2 + \cdots + x^8) + 10x^9 = \frac{x^{10} - x}{x - 1} + 9x^9$$

$$V(x) = (x + x^2 + \cdots + x^{10}) + 10x^{11} = \frac{x^{12} - x}{x - 1} + 9x^{11}.$$

We wish to show that if $U(u) = V(v) = 8$, then $u < v$.

We first note that when $x \leq 0$, $x^{12} - x \geq 0$, $x - 1 < 0$, and $9x^9 \leq 0$, so

$$U(x) = \frac{x^{10} - x}{x - 1} + 9x^9 \leq 0 < 8.$$

Similarly, $V(x) \leq 0 < 8$.

We also note that if $x \geq 9/10$, then

$$U(x) = \frac{x - x^{10}}{1 - x} + 9x^9 \geq \frac{9/10 - 9^9/10^9}{1/10} + 9 \cdot \frac{9^9}{10^9}$$

$$= 9 - 10 \cdot \frac{9^9}{10^9} + 9 \cdot \frac{9^9}{10^9} = 9 - \frac{9^9}{10^9} > 8.$$

Similarly $V(x) > 8$. It then follows that $u, v \in (0, 9/10)$.

Now, for all $x \in (0, 9/10)$,

$$V(x) = U(x) + V(x) - U(x) = U(x) + 10x^{11} + x^{10} - 9x^9$$

$$= U(x) + x^9(10x - 9)(x + 1) < U(x).$$

Since V and U are both strictly increasing functions over the nonnegative reals, it then follows that

$$V(u) < U(u) = 8 = V(v),$$

so $u < v$, as desired. ■

§19 USAMO 1990

1. (*USAMO 1990*) A certain state issues license plates consisting of six digits (from 0 through 9). The state requires that any two plates differ in at least two places. (Thus the plates 027592 and 020592 cannot both be used.) Determine, with proof, the maximum number of distinct license plates that the state can use.

Solution 1

Consider license plates of n digits, for some fixed n , issued with the same criteria.

We first note that by the pigeonhole principle, we may have at most 10^{n-1} distinct plates. Indeed, if we have more, then there must be two plates which agree on the first $n - 1$ digits; these plates thus differ only on one digit, the last one.

We now show that it is possible to issue 10^{n-1} distinct license plates which satisfy the problem's criteria. Indeed, we issue plates with all 10^{n-1} possible combinations for the first $n - 1$ digit, and for each plate, we let the last digit be the sum of the preceding digits taken mod 10. This way, if two plates agree on the first $n - 1$

digits, they agree on the last digit and are thus the same plate, and if two plates differ in only one of the first $n - 1$ digits, they must differ as well in the last digit.

It then follows that 10^{n-1} is the greatest number of license plates the state can issue. For $n = 6$, as in the problem, this number is 10^5 . ■

Solution 2

Constructive Induction for $n = 6$

$[n=1 \quad \text{1 element}] \quad n=2 \quad \text{10 elements}]$ Suppose for n ,
 $[S_{n+1} = \left\{ \left(a_1, a_2, \dots, a_{n-k} \right) \mid k = 1, 2, \dots, n \right\}]$
 $[a_1 = b_1, \dots, a_{n-1} = b_{n-1}, a_n = b_n \Rightarrow a_{n+k} \neq b_{n+k}]$
 $[a_1 = b_1, \dots, a_n = b_n \quad (\text{removing } i)]$ Contradiction! If $n \geq 1$,
 $[\text{1 dimension: 1 element}]$
 $[\text{2 dimensions: 10 elements}]$
 $[\text{3 dimensions: 100 elements}]$
 $[10^6 \text{ points can be covered by } 10^5 \text{ lines}] \Rightarrow \text{At most } 10^5 \text{ lines}$

2. (USAMO 1990) A sequence of functions $\{f_n(x)\}$ is defined recursively as follows:

$$f_1(x) = \sqrt{x^2 + 48}, \quad \text{and} \\ f_{n+1}(x) = \sqrt{x^2 + 6f_n(x)} \quad \text{for } n \geq 1.$$

(Recall that $\sqrt{}$ is understood to represent the positive square root.) For each positive integer n , find all real solutions of the equation $f_n(x) = 2x$.

Solution 1

We define $f_0(x) = 8$. Then the recursive relation holds for $n = 0$, as well.

Since $f_n(x) \geq 0$ for all nonnegative integers n , it suffices to consider nonnegative values of x .

We claim that the following set of relations hold true for all natural numbers n and nonnegative reals x :

$$\begin{aligned} f_n(x) &< 2x \text{ if } x > 4; \\ f_n(x) &= 2x \text{ if } x = 4; \\ f_n(x) &> 2x \text{ if } x < 4. \end{aligned}$$

To prove this claim, we induct on n . The statement evidently holds for our base case, $n = 0$.

Now, suppose the claim holds for n . Then

$$\begin{aligned} f_{n+1}(x) &= \sqrt{x^2 + 6f_n(x)} < \sqrt{x^2 + 12x} < \sqrt{4x^2} = 2x, \text{ if } x > 4; \\ f_{n+1}(x) &= \sqrt{x^2 + 6f_n(x)} = \sqrt{x^2 + 12x} = \sqrt{4x^2} = 2x, \text{ if } x = 4; \\ f_{n+1}(x) &= \sqrt{x^2 + 6f_n(x)} > \sqrt{x^2 + 12x} > \sqrt{4x^2} = 2x, \text{ if } x < 4. \end{aligned}$$

The claim therefore holds by induction. It then follows that for all nonnegative integers n , $x = 4$ is the unique solution to the equation $f_n(x) = 2x$. ■

Solution 2

We claim that the only solution is $\boxed{x = 4}$ for all such n . To show this, we consider all x and find solutions n .

Before we consider solutions, we show that for all x , $f_n(x)$ is positive for all positive integers n by induction. For our base case:

$$f_1(x) = \sqrt{x^2 + 48} \geq \sqrt{48} > 0$$

so it is positive. Next, for our inductive step, assume for some $n = k - 1$ that $f_{k-1}(x)$ is positive; thus $f_{k-1}(x) > 0$, and we show that $f_k(x)$ is positive:

$$f_k(x) = \sqrt{x^2 + 6f_{k-1}(x)} \geq \sqrt{0 + 6f_{k-1}(x)} > \sqrt{0 + 0} = 0$$

thus $f_k(x) > 0$, so we have proven the claim by induction.

First, we consider negative x . We know that for negative x , if the equation were to have a solution, we would have $f_n(x) = 2x < 0$ for some n . However, $f_n(x)$ is always nonnegative since $f_n(x)$ is always the square root of some number, which can never be negative; thus there are no solutions in this case.

Next, we consider $x = 0$. Solutions would have to satisfy $f_n(0) = 0$; we have previously shown that all $f_n(x)$ are positive, so there are no such n .

Finally, we consider positive x . We divide this set into three groups: $x < 4$, $x = 4$, and $x > 4$. We claim that for increasing n , $f_n(x)$ is decreasing, constant, and increasing, respectively, and we prove this using induction. First, we analyze $x < 4$. Then

$$f_1(x) < \sqrt{4^2 + 48} = \sqrt{64} = 8$$

For our base case:

$$f_2(x) = \sqrt{x^2 + 6f_1(x)} < \sqrt{x^2 + 6 \cdot 8} = f_1(x)$$

Thus $f_2(x) < f_1(x)$. For our inductive step, if for some $n = k - 1$ we have $f_k(x) < f_{k-1}(x)$, we show that $f_{k+1}(x) < f_k(x)$:

$$f_{k+1}(x) = \sqrt{x^2 + 6f_k(x)} < \sqrt{x^2 + 6f_{k-1}(x)} = f_k(x)$$

Thus $f_1(x) > f_2(x) > f_3(x) > \dots$ and the conclusion follows.

Next, for $x = 4$, $f_1(4) = 8$ by substituting. Then, if $f_{n-1}(4) = 8$, we show that $f_n(4) = 8$ as well: $f_n(4) = \sqrt{4^2 + 6f_{n-1}(4)} = \sqrt{16 + 6 \cdot 8} = \sqrt{64} = 8$ so the sequence is constant. Notice that in this case, the equation we must solve becomes $f_n(4) = 8$, which is true for all positive integers n .

Finally, we analyze $x > 4$. Then

$$f_1(x) > \sqrt{4^2 + 48} = \sqrt{64} = 8$$

For our base case:

$$f_2(x) = \sqrt{x^2 + 6f_1(x)} > \sqrt{x^2 + 6 \cdot 8} = f_1(x)$$

Thus $f_2(x) > f_1(x)$. For our inductive step, if for some $n = k - 1$ we have $f_k(x) > f_{k-1}(x)$, we show that $f_{k+1}(x) > f_k(x)$:

$$f_{k+1}(x) = \sqrt{x^2 + 6f_k(x)} > \sqrt{x^2 + 6f_{k-1}(x)} = f_k(x)$$

Thus $f_1(x) < f_2(x) < f_3(x) < \dots$ and the conclusion follows.

If for some positive integer n we have $f_n(x) = 2x$, then:

$$\begin{aligned} f_n(x) &= \sqrt{x^2 + 6f_{n-1}(x)} \\ 2x &= \sqrt{x^2 + 6f_{n-1}(x)} \\ 4x^2 &= x^2 + 6f_{n-1}(x) \\ 3x^2 &= 6f_{n-1}(x) \\ \frac{1}{2}x^2 &= f_{n-1}(x) \end{aligned}$$

However, if $0 < x < 4$ is a solution, then we must have $f_{n-1}(x) > f_n(x)$; substituting values yields $\frac{1}{2}x^2 > 2x$, which implies $x > 4$; this is a contradiction. Similarly, if $x > 4$ is a solution, then we must have $f_{n-1}(x) < f_n(x)$; substituting values yields $\frac{1}{2}x^2 < 2x$, which implies $x < 4$; this is also a contradiction.

As a result, since we have already established that $x = 4$ is a solution for all n and no other x can be solutions, we conclude that for each positive integer n , the only real solution of the equation is $x = 4$.

3. (USAMO 1990) Suppose that necklace A has 14 beads and necklace B has 19. Prove that for any odd integer $n \geq 1$, there is a way to number each of the 33 beads with an integer from the sequence

$$\{n, n+1, n+2, \dots, n+32\}$$

so that each integer is used once, and adjacent beads correspond to relatively prime integers. (Here a "necklace" is viewed as a circle in which each bead is adjacent to two other beads.)

Solution

Lemma. For every positive odd integer n , there exists a nonnegative integer $k \leq 17$ such that $n+k$ is relatively prime to $n+k+15$, $n+k+1$ is relatively prime to $n+k+14$.

Proof. Consider the positive integers $n, n+1, n+2$. Note that at most one of these is divisible by 3, and at most one is divisible by 5. Therefore one of these, say $n+a$, is divisible by neither, and is therefore relatively prime to 15. Furthermore, $n+a+1$ and $n+(a+15)+1$ have different residues mod 13, so one of them, say $n+b+1$, is relatively prime to 13. Since $n+b \equiv n+a \pmod{15}$, $n+b$ is relatively prime to 15. But

$$\gcd(n+b, n+b+15) = \gcd[n+b, (n+b+15) - (n+b)] = \gcd(n+b, 15) = 1,$$

so $n+b$ is relatively prime to $n+b+15$. Also,

$$\gcd(n+b+1, n+b+14) = \gcd[n+b+1, (n+b+14) - (n+b+1)] = \gcd(n+b+1, 13),$$

so $n+b+1$ is relatively prime to $n+b+14$. Finally, $b \leq a+15 \leq 17$. It follows that setting $k = b$ satisfies the lemma. ■

Let k be an integer as described in the lemma. We place the integers $n, \dots, n+k, n+k+15, \dots, n+32$ on the necklace with 19 beads, and the integers $n+k+$

$1, \dots, n + k + 14$ on the necklace with 14 beads, in those orders. Since n is odd, n is relatively prime to $n + 32 = n + 2^5$. By definition, $n + k$ and $n + k + 15$ are relatively prime, as are $n + k + 1$ and $n + k + 14$; finally, a and $a + 1$ are relatively prime for all integers a . It follows that each bead in this arrangement is relatively prime to its neighbors. ■

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

4. (*USAMO 1990*) Find, with proof, the number of positive integers whose base- n representation consists of distinct digits with the property that, except for the leftmost digit, every digit differs by ± 1 from some digit further to the left. (Your answer should be an explicit function of n in simplest form.)

Solution

Let a k -good sequence be a sequence of distinct integers $\{a_i\}_{i=1}^k$ such that for all integers $2 \leq i \leq k$, a_i differs from some preceding term by ± 1 .

Lemma. Let a be an integer. Then there are 2^{k-1} k -good sequences starting on a , and furthermore, the terms of each of these sequences constitute a permutation of k consecutive integers.

Proof. We induct on k . For $k = 1$, the lemma is trivially true. Now, suppose the lemma holds for k . If $\{a_i\}_{i=1}^{k+1}$ is a $(k+1)$ -good sequence, then $\{a_i\}_{i=1}^k$ is a k -good sequence which starts on a , so it is a permutation of k consecutive integers, say $m, \dots, M$. Then the only possibilities for a_{k+1} are $m - 1$ and $M + 1$; either way, $\{a_i\}_{i=1}^{k+1}$ constitutes a permutation of $k + 1$ consecutive integers. Since there are 2^k possible sequences $\{a_i\}_{i=1}^k$, and 2 choices of a_{k+1} for each of these sequences, it also follows that there are $2^k \cdot 2 = 2^{k+1}$ $(k+1)$ -good sequences which start on a . Thus the lemma holds by induction. ■

We now consider the number of desired positive integers with k digits. Evidently, k must be less than or equal to n . We also note that the digits of such an integer must constitute a k -good sequence. Since the minimum of this sequence can be any of the digits $0, \dots, n - k$, unless the minimum is 0 and is the first digit (in which case the only possible sequence is an increasing arithmetic sequence), and there are 2^{k-1} k -good sequences up to translation, it follows that there are $(n - k + 1)2^{k-1} - 1$ desired positive integers with k digits. Thus the total number of desired positive integers is

$$\begin{aligned} \sum_{k=1}^n [(n - k + 1)2^{k-1} - 1] &= -n + \sum_{k=1}^n \sum_{j=k}^n 2^{k-1} = -n + \sum_{j=1}^n \sum_{k=1}^j 2^{k-1} \\ &= -n + \sum_{j=1}^n (2^j - 1) = -2n - 1 + \sum_{j=0}^n 2^j, \end{aligned}$$

which is equal to $2^{n+1} - 2(n + 1)$, our answer. ■

The lemma in the above solution is wrong. A 3-term sequence starting on 1 will not comply, for instance. My solution is shown as follows. We call all qualified numbers 'good numbers', with k digits, $1 \leq k \leq n$. See any good number as a k -term sequence, m being the minimal term, a being the first term. Note that $0 \leq m \leq n - k$. Lemma. It is trivially true that any good number consists of consecutive digits. It can be proven with induction easily. Now pick any m , let $f(i)$ be the number of sequences when $a = i$, we prove that $f(i) = C(k-1, i-m)$, $m \leq i \leq m+k-1$. Per Lemma, a

sequence consists of consecutive digits: $m, m+1, \dots, i, \dots, m+k-1$. The first term $a=i$ determined, there are $k-1$ remaining spaces to be filled. The digits smaller than i are $m, m+1, \dots, i-1$, their only link to i is $i-1$, then $i-2$, so on so forth, eventually we can know they can only enter the sequence in one order which is descending: $i-1, i-2, \dots, m$. Likewise, the digits bigger than i can only enter the sequence in one order which is ascending: $i+1, i+2, \dots, m+k-1$. Therefore, the number of sequences is equivalent to the number of ways in which the $i-m$ digits smaller than i occupy the $k-1$ remaining spaces, i.e., $C(k-1, i-m)$. So, for any given m , the total number of sequences is $\sum_{i=m}^{m+k-1} C(k-1, i-m) = \sum_{i=0}^{k-1} C(k-1, i) = 2^{k-1}$. Considering that there are $n-k+1$ possibilities for m , and deducting the one special case with $m=0$ and $a=0$ (because the first digit cannot be 0), we now can have the total number of sequences for any given k : $(n-k+1)2^{k-1} - 1$. Now we can arrive at the grand total: $\sum_{k=1}^n ((n-k+1)2^{k-1} - 1)$. It is trivial that the final sum is $2^{n+1} - 2(n+1)$. Done. Tmath2024 2026.5.18

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

5. (*USAMO 1990*) An acute-angled triangle ABC is given in the plane. The circle with diameter AB intersects altitude CC' and its extension at points M and N , and the circle with diameter AC intersects altitude BB' and its extensions at P and Q . Prove that the points M, N, P, Q lie on a common circle.

Solution 1

Let A' be the intersection of the two circles (other than A). AA' is perpendicular to both BA' , CA' implying B, C, A' are collinear. Since A' is the foot of the altitude from A : A, H, A' are concurrent, where H is the orthocentre.

Now, H is also the intersection of BB' , CC' which means that AA', MN, PQ are concurrent. Since A, M, N, A' and A, P, Q, A' are cyclic, M, N, P, Q are cyclic by the radical axis theorem.

Solution 2

Define A' as the foot of the altitude from A to BC . Then, $AA' \cap BB' \cap CC'$ is the orthocenter. We will denote this point as H . Since $\angle AA'C$ and $\angle AA'B$ are both 90° , A' lies on the circles with diameters AC and AB .

Now we use the Power of a Point theorem with respect to point H . From the circle with diameter AB we get $AH \cdot A'H = MH \cdot NH$. From the circle with diameter AC we get $AH \cdot A'H = PH \cdot QH$. Thus, we conclude that $PH \cdot QH = MH \cdot NH$, which implies that P, Q, M , and N all lie on a circle.

Solution 3 (Radical Lemma)

Let ω_1 be the circumcircle with diameter AB and ω_2 be the circumcircle with diameter AC . We claim that the second intersection of ω_1 and ω_2 other than A is A' , where A' is the feet of the perpendicular from A to segment BC . Note that

$$\angle AA'B = 90^\circ = \angle AB'B$$

so A' lies on ω_1 . Similarly, A' lies on ω_2 . Hence, AA' is the radical axis of ω_1 and ω_2 . By the Radical Lemma, it suffices to prove that the intersection of lines MN and PQ lie on AA' . But, MN is the same line as CC' and PQ is the same line as BB' . Since AA', BB' , and CC' intersect at the orthocenter H , H lies on the radical axis AA' and we are done. ■

Solution 4

We know that $BCB'C'$ is a cyclic quadrilateral. Hence,

$$HB \cdot HB' = HC \cdot HC'$$

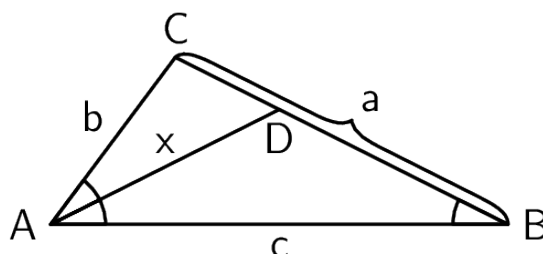
$$\implies Pow_{\omega_1} = Pow_{\omega_2}$$

$$\implies HM \cdot HN = HP \cdot HQ$$

$$\implies MPNQ \text{ is cyclic } \blacksquare.$$

§20 USAMO 1991

- (USAMO 1991) In triangle ABC , angle A is twice angle B , angle C is obtuse, and the three side lengths a, b, c are integers. Determine, with proof, the minimum possible perimeter.

Solution 1


(diagram by integralarefun)

After drawing the triangle, also draw the **angle bisector** of $\angle A$, and let it intersect $\overline{BC}$ at D . Notice that $\triangle ADC \sim \triangle BAC$, and let $AD = x$. Now from similarity,

$$x = \frac{bc}{a}$$

However, from the angle bisector theorem, we have

$$BD = \frac{ac}{b+c}$$

but $\triangle ABD$ is isosceles, so

- (USAMO 1991) For any nonempty set S of numbers, let $\sigma(S)$ and $\pi(S)$ denote the sum and product, respectively, of the elements of S . Prove that

$$\sum \frac{\sigma(S)}{\pi(S)} = (n^2 + 2n) - \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}\right)(n+1),$$

where " Σ " denotes a sum involving all nonempty subsets S of $\{1, 2, 3, \dots, n\}$.

Solution

Let $N(m)$ denote the set $\{1, \dots, m\}$. Since $\sigma(\emptyset)$, the empty sum, is equal to zero, and $\pi(\emptyset)$, the empty product, is equal to 1, the equation

$$\sum_{S \subseteq N(n)} \frac{\sigma(S)}{\pi(S)} = (n+1)^2 - 1 - (n+1) \sum_{k=1}^n \frac{1}{k}$$

is equivalent to the desired equation when $n > 0$. We will prove this equation, but first, we prove a lemma.

Lemma. For all nonnegative integers n , $\sum_{S \subseteq N(n)} \frac{1}{\pi(S)} = n + 1$.

Proof. Evidently,

$$\sum_{S \subseteq N(n)} \frac{1}{\pi(S)} = \sum_{k=0}^n \sum_{S \subseteq N(n), |S|=k} \frac{1}{\pi(S)}.$$

But the terms of this sum are the coefficients of the polynomial $P(x) = \prod_{k=1}^n (x + 1/k)$, and the sum of the coefficients of this polynomial is

$$P(1) = \prod_{k=1}^n (1 + 1/k) = \prod_{k=1}^n \frac{k+1}{k} = \frac{n+1}{1} = n+1,$$

as desired. ■

We now prove our equation by induction on n . For $n = 0$, we have the simple equation $0=0$.

Suppose the equation holds when $n = a$. Then

$$\begin{aligned} \sum_{S \subseteq N(a+1)} \frac{\sigma(S)}{\pi(S)} &= \sum_{S \subseteq N(a)} \frac{\sigma(S)}{\pi(S)} + \sum_{S \subseteq N(a)} \frac{\sigma(S \cup \{a+1\})}{\pi(S \cup \{a+1\})} \\ &= \sum_{S \subseteq N(a)} \frac{\sigma(S)}{\pi(S)} + \sum_{S \subseteq N(a)} \frac{\sigma(S) + (a+1)}{(a+1) \cdot \pi(S)} \\ &= \frac{a+2}{a+1} \sum_{S \subseteq N(a)} \frac{\sigma(S)}{\pi(S)} + \sum_{S \subseteq N(a)} \frac{1}{\pi(S)}. \end{aligned}$$

By inductive hypothesis,

$$\begin{aligned} \frac{a+2}{a+1} \sum_{S \subseteq N(a)} \frac{\sigma(S)}{\pi(S)} &= \frac{a+2}{a+1} \left[(a+1)^2 - 1 - (a+1) \sum_{k=1}^a \frac{1}{k} \right] \\ &= (a+2)(a+1) - \frac{a+2}{a+1} - (a+2) \sum_{k=1}^a \frac{1}{k} \\ &= (a+2)(a+1) - (a+2) \sum_{k=1}^{a+1} \frac{1}{k}, \end{aligned}$$

and by the lemma,

$$\sum_{S \subseteq N(a)} \frac{1}{\pi(S)} = a+1.$$

Since $(a+2)(s+1) = (a+3)(a+1) = (a+2)^2 - 1$, our equation thus holds by induction. Thus the problem statement is proven. ■

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

3. (*USAMO 1991*) Show that, for any fixed integer $n \geq 1$, the sequence

$$2, 2^2, 2^{2^2}, 2^{2^{2^2}}, \dots \pmod{n}$$

is eventually constant.

[The tower of exponents is defined by $a_1 = 2$, $a_{i+1} = 2^{a_i}$. Also $a_i \pmod n$ means the remainder which results from dividing a_i by n .]

Solution 1

Suppose that the problem statement is false for some integer $n \geq 1$. Then there is a least n , which we call b , for which the statement is false.

Since all integers are equivalent mod 1, $b \neq 1$.

Note that for all integers b , the sequence $2^0, 2^1, 2^2, \dots$ eventually becomes cyclic mod b . Let k be the period of this cycle. Since there are $k - 1$ nonzero residues mod b , $1 \leq k \leq b - 1 < b$. Since

$$2, 2^2, 2^{2^2}, 2^{2^{2^2}}, \dots$$

does not become constant mod b , it follows the sequence of exponents of these terms, i.e., the sequence

$$1, 2, 2^2, 2^{2^2}, \dots$$

does not become constant mod k . Then the problem statement is false for $n = k$. Since $k < b$, this is a contradiction. Therefore the problem statement is true. ■

Note that we may replace 2 with any other positive integer, and both the problem and this solution remain valid.

Solution 2

We'll prove by strong induction that for every natural number n , the sequence $a_1, a_2, \dots$ is eventually constant. Since every term of the sequence is 0 mod 1, the claim is true when $n = 1$. Assuming that it's true for $1, \dots, n$, we'll now show that it's true for $n + 1$ as well.

Suppose first that $n + 1$ is odd. Since $\varphi(n + 1) < n + 1$, by our inductive hypothesis there exists an m such that

$$a_m = a_{m+1} = a_{m+2} = \dots \pmod{\varphi(n + 1)}.$$

Since $n + 1$ is coprime to powers of 2, it follows by Euler's Totient theorem that

$$2^{a_m} = 2^{a_{m+1}} = 2^{a_{m+2}} = \dots \pmod{n + 1},$$

or equivalently

$$a_{m+1} = a_{m+2} = a_{m+3} = \dots \pmod{n + 1},$$

which is what we wanted to show.

Now suppose that $n + 1$ is even. Write $n + 1 = 2^k \cdot s$, where $1 \leq s < n + 1$ is odd. The series must eventually be constant mod 2^k , since $a_m = 0 \pmod{2^k}$ for large enough m . And by our inductive hypothesis, the series must also eventually be constant mod s . So for large enough m ,

$$a_m = a_{m+1} = a_{m+2} = \dots \pmod{2^k},$$

$$a_m = a_{m+1} = a_{m+2} = \dots \pmod{s}.$$

Since 2^k and s are coprime, these equations are also true modulo $2^k \cdot s = n + 1$. So

$$a_m = a_{m+1} = a_{m+2} = \cdots \pmod{n+1},$$

which completes the proof.

4. (USAMO 1991) Let $a = (m^{m+1} + n^{n+1})/(m^m + n^n)$, where m and n are positive integers. Prove that $a^m + a^n \geq m^m + n^n$.

[You may wish to analyze the ratio $(a^N - N^N)/(a - N)$, for real $a \geq 0$ and integer $N \geq 1$.]

Solution

Let us assume without loss of generality that $m \geq n$. We then note that

$$m - a = \frac{m^{m+1} + m \cdot n^n}{m^m + n^n} - \frac{m^{m+1} - n^{n+1}}{m^m + n^n} = n^n \frac{m - n}{m^m + n^n} \quad (*).$$

Similarly,

$$a - n = m^m \frac{m - n}{m^m + n^n} \quad (**).$$

We note that equations (*) and (**) imply that $n \leq a \leq m$. Then $a/m \leq 1 \leq a/n$, so

$$\frac{1}{m} \sum_{i=0}^{m-1} (a/m)^i \leq 1 \leq \frac{1}{n} \sum_{i=0}^{n-1} (a/n)^i.$$

Multiplying this inequality by $m^m n^n (m - n)/(m^m + n^n)$, we have

$$n^n \frac{(m - n)}{m^m + n^n} \sum_{i=0}^{m-1} a^i m^{m-1-i} \leq m^m \frac{(m - n)}{m^m + n^n} \sum_{i=0}^{n-1} a^i n^{n-1-i}.$$

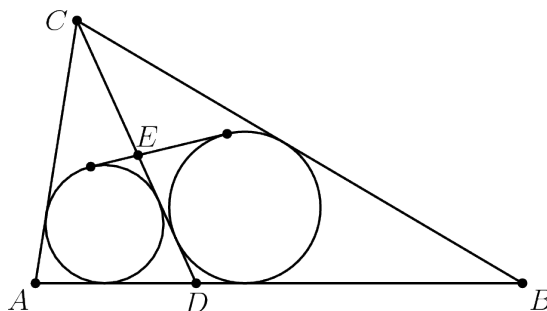
It then follows that

$$\begin{aligned} m^m - a^m &= (m - a) \sum_{i=0}^{m-1} a^i m^{m-1-i} = n^n \frac{m - n}{m^m + n^n} \sum_{i=0}^{m-1} a^i m^{m-1-i} \\ &\leq m^m \frac{m - n}{m^m + n^n} \sum_{i=0}^{n-1} a^i n^{n-1-i} = (a - n) \sum_{i=0}^{n-1} a^i n^{n-1-i} \\ &= a^n - n^n. \end{aligned}$$

Rearranging this inequality, we find that $a^m + a^n \geq m^m + n^n$, as desired. ■

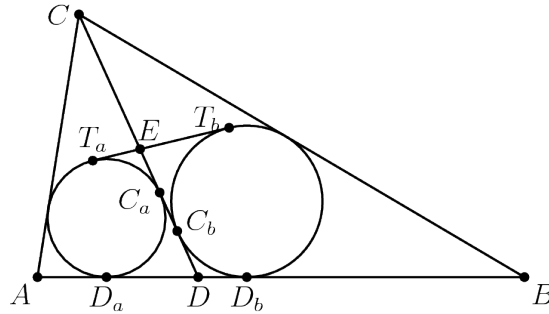
<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

5. (USAMO 1991) Let D be an arbitrary point on side AB of a given triangle ABC , and let E be the interior point where CD intersects the external common tangent to the incircles of triangles ACD and BCD . As D assumes all positions between A and B , prove that the point E traces the arc of a circle.



Solution 1

Let the incircle of ACD and the incircle of BCD touch line AB at points D_a, D_b , respectively; let these circles touch CD at C_a, C_b , respectively; and let them touch their common external tangent containing E at T_a, T_b , respectively, as shown in the diagram below.



We note that

$$CE = CC_a - EC_a = CC_b - EC_b = \frac{CC_a + CC_b - (EC_a + EC_b)}{2}.$$

On the other hand, since EC_a and ET_a are tangents from the same point to a common circle, $EC_a = T_aE$, and similarly $EC_b = ET_b$, so

$$EC_a + EC_b = T_aE + ET_b = T_aT_b.$$

On the other hand, the segments T_aT_b and D_aD_b evidently have the same length, and $D_aD_b = D_aD + DD_b$, so $EC_a + EC_b = D_aD + DD_b$. Thus

$$CE = \frac{CC_a + CC_b - (EC_a + EC_b)}{2} = \frac{CC_a + CC_b - D_aD - DD_b}{2}.$$

If we let s_a be the semiperimeter of triangle ACD , then $CC_a = s_a - AD$, and $D_aD = s_a - AC$, so

$$CC_a - D_aD = (s_a - AD) - (s_a - AC) = AC - AD.$$

Similarly,

$$CC_b - DD_b = BC - DB,$$

so that

$$\begin{aligned} CE &= \frac{CC_a + CC_b - D_aD - DD_b}{2} = \frac{AC + BC - (AD + DB)}{2} \\ &= \frac{AC + BC - AB}{2}. \end{aligned}$$

Thus E lies on the arc of the circle with center C and radius $(AB + BC - AB)/2$ intercepted by segments CA and CB . If we choose an arbitrary point X on this arc and let D be the intersection of lines CX and AB , then X becomes point E in the diagram, so every point on this arc is in the locus of E . ■

Solution 2

Define the same points as in the first solution. First extend T_aT_b to intersect AB at a point P ; without loss of generality let A lie in between B and P . Then the

incircle of $\triangle ACD$ is also the incircle of $\triangle PED$, while the incircle of $\triangle BCD$ is the P -excircle of $\triangle PED$. It follows that $EC_a = C_bD$; denote this equality by $(*)$.

Now remark that

$$CC_a + CC_b = \frac{AC + AD - CD}{2} + \frac{BC + DC - BD}{2} = \frac{AC + BC - AB}{2} + CD.$$

Hence

$$CC_a + CC_b - CD = CC_a - C_bD \stackrel{(*)}{=} CC_a - C_aE = CE$$

is a constant equal to $r := \frac{AC+BC-AB}{2}$, and so C lies on the circle with center C and radius r .

§21 USAMO 1992

1. (*USAMO 1992*) Find, as a function of n , the sum of the digits of

$$9 \times 99 \times 9999 \times \cdots \times (10^{2^n} - 1),$$

where each factor has twice as many digits as the previous one.

Solution

The answer is $9 \cdot 2^n$.

Let us denote the quantity $\prod_{k=0}^n (10^{2^k} - 1)$ as P_n . We wish to find the sum of the digits of P_n .

We first note that

$$P_{n-1} < \prod_{k=0}^{n-1} 10^{2^k} = 10^{2^n-1},$$

so P_{n-1} is a number of at most 2^n digits. We also note that the units digit is not equal to zero. We may thus represent P_{n-1} as

$$\sum_{k=0}^{2^n-1} 10^k d_k,$$

where the d_k are digits and $d_0 \neq 0$. Then

$$\begin{aligned} P_n &= (10^{2^n} - 1)P_{n-1} = \sum_{k=0}^{2^n-1} 10^k d_k + \sum_{k=0}^{2^n-1} 10^{2^n+k} d_k \\ &= (10 - d_0) + \sum_{k=1}^{2^n-1} 10^k (9 - d_k) + 10^{2^n} (d_0 - 1) + \sum_{k=1}^{2^n-1} 10^{2^n+k} d_k. \end{aligned}$$

Thus the digits of P_n are

$$10 - d_0, 9 - d_1, 9 - d_2, \dots, 9 - d_{2^n-1}, d_0 - 1, d_1, d_2, \dots, d_{2^n-1},$$

and the sum of these is evidently $9 \cdot 2^n$, as desired. ■

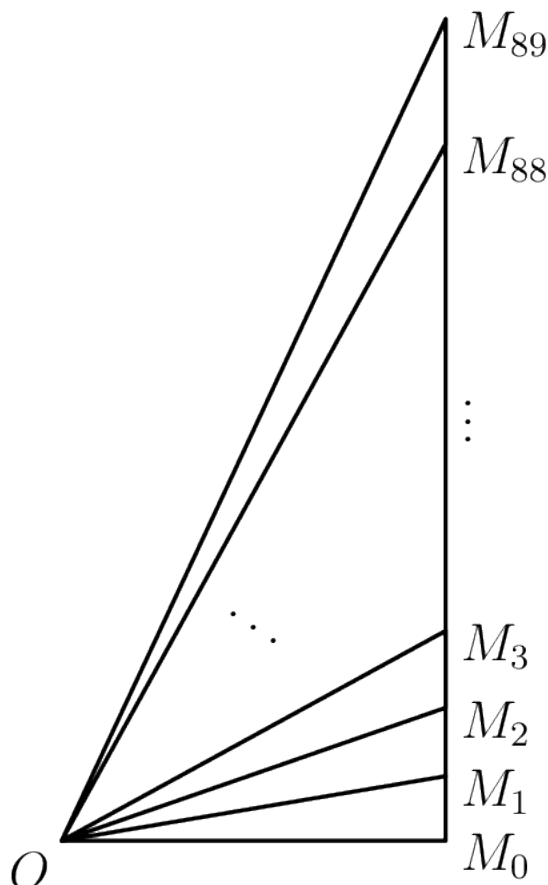
<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

2. (USAMO 1992) Prove

$$\frac{1}{\cos 0^\circ \cos 1^\circ} + \frac{1}{\cos 1^\circ \cos 2^\circ} + \cdots + \frac{1}{\cos 88^\circ \cos 89^\circ} = \frac{\cos 1^\circ}{\sin^2 1^\circ}.$$

Solution 1

Consider the points $M_k = (1, \tan k^\circ)$ in the coordinate plane with origin $O = (0, 0)$, for integers $0 \leq k \leq 89$.



Evidently, the angle between segments OM_a and OM_b is $(b - a)^\circ$, and the length of segment OM_a is $1/\cos a^\circ$. It then follows that the area of triangle M_aOM_b is $\frac{1}{2} \sin(b - a)^\circ \cdot OM_a \cdot OM_b = \frac{1}{2} \sin(b - a)^\circ / (\cos a^\circ \cdot \cos b^\circ)$. Therefore

$$\begin{aligned} \sum_{k=0}^{88} \frac{\frac{1}{2} \sin 1^\circ}{\cos k^\circ \cos (k+1)^\circ} &= \sum_{k=0}^{88} [M_k OM_{k+1}] \\ &= [M_0 OM_{89}] \\ &= \frac{\frac{1}{2} \sin 89^\circ}{\cos 89^\circ} = \frac{\frac{1}{2} \cos 1^\circ}{\sin 1^\circ}, \end{aligned}$$

so

$$\sum_{k=0}^{88} \frac{1}{\cos k^\circ \cos (k+1)^\circ} = \frac{\cos 1^\circ}{\sin^2 1^\circ},$$

as desired. ■

Solution 2

First multiply both sides of the equation by $\sin 1$, so the right hand side is $\frac{\cos 1}{\sin 1}$. Now by rewriting $\sin 1 = \sin((k+1) - k) = \sin(k+1)\cos(k) + \sin(k)\cos(k+1)$, we can derive the identity $\tan(n+1) - \tan(n) = \frac{\sin 1}{\cos(n)\cos(n+1)}$. Then the left hand side of the equation simplifies to $\tan 89 - \tan 0 = \tan 89 = \frac{\sin 89}{\cos 89} = \frac{\cos 1}{\sin 1}$ as desired.

Solution 3

Multiply by $\sin 1$. We get:

$$\frac{\sin 1}{\cos 0 \cos 1} + \frac{\sin 1}{\cos 1 \cos 2} + \dots + \frac{\sin 1}{\cos 88 \cos 89} = \frac{\cos 1}{\sin 1}$$

we can write this as:

$$\frac{\sin 1 - 0}{\cos 0 \cos 1} + \frac{\sin 2 - 1}{\cos 1 \cos 2} + \dots + \frac{\sin 89 - 88}{\cos 88 \cos 89} = \frac{\cos 1}{\sin 1}$$

$$\text{This is an identity } \tan a - \tan b = \frac{\sin(a-b)}{\cos a \cos b}$$

Therefore;

$$\sum_{i=1}^{89} [\tan k - \tan(k-1)] = \tan 89 - \tan 0 = \cot 1, \text{ because of telescoping.}$$

but since we multiplied $\sin 1$ in the beginning, we need to divide by $\sin 1$. So we get that:

$$\frac{1}{\cos 0 \cos 1} + \frac{1}{\cos 1 \cos 2} + \frac{1}{\cos 2 \cos 3} + \dots + \frac{1}{\cos 88 \cos 89} = \frac{\cos 1}{\sin^2 1} \text{ as desired. QED}$$

Solution 4

$$\text{Let } S = \frac{1}{\cos 0^\circ \cos 1^\circ} + \frac{1}{\cos 1^\circ \cos 2^\circ} + \dots + \frac{1}{\cos 88^\circ \cos 89^\circ}.$$

Multiplying by $\sin 1^\circ$ gives

$$S \sin 1^\circ = \frac{\sin(1^\circ - 0^\circ)}{\cos 0^\circ \cos 1^\circ} + \dots + \frac{\sin(89^\circ - 88^\circ)}{\cos 88^\circ \cos 89^\circ}$$

Notice that $\frac{\sin((x+1^\circ)-x)}{\cos 0^\circ \cos 1^\circ} = \tan(x+1^\circ) - \tan x$ after expanding the sine, and so

$$S \sin 1^\circ = (\tan 1^\circ - \tan 0^\circ) + \dots + (\tan 89^\circ - \tan 88^\circ) = \tan 89^\circ - \tan 0^\circ = \cot 1^\circ = \frac{\cos 1^\circ}{\sin 1^\circ},$$

so

$$S = \frac{\cos 1^\circ}{\sin^2 1^\circ}.$$

3. (*USAMO 1992*) For a nonempty set S of integers, let $\sigma(S)$ be the sum of the elements of S . Suppose that $A = \{a_1, a_2, \dots, a_{11}\}$ is a set of positive integers with $a_1 < a_2 < \dots < a_{11}$ and that, for each positive integer $n \leq 1500$, there is a subset S of A for which $\sigma(S) = n$. What is the smallest possible value of a_{10} ?

Solution

Let's a n - p set be a set Z such that $Z = \{a_1, a_2, \dots, a_n\}$, where $\forall i < n, i \in \mathbb{Z}^+, a_i < a_{i+1}$, and for each $x \leq p, x \in \mathbb{Z}^+, \exists Y \subseteq Z, \sigma(Y) = x, \nexists \sigma(Y) = p+1$.

(For Example $\{1, 2\}$ is a 2-3 set and $\{1, 2, 4, 10\}$ is a 4-8 set)

Furthermore, let call a n - p set a n - p good set if $a_n \leq p$, and a n - p bad set if $a_n \geq p+1$ (note that $\nexists \sigma(Y) = p+1$ for any n - p set. Thus, we can ignore the case where $a_n = p+1$).

Furthermore, if you add any amount of elements to the end of a n - p bad set to form another n - p set (with a different n), it will stay as a n - p bad set because $a_{n+x} > a_n > p+1$ for any positive integer x and $\nexists \sigma(Y) = p+1$.

Lemma) If Z is a n - p set, $p \leq 2^n - 1$.

For $n = 1$, $p = 0$ or 1 because $a_1 = 1 \rightarrow p = 1$ and $a_1 \neq 1 \rightarrow p = 0$.

Assume that the lemma is true for some n , then 2^n is not expressible with the n - p set. Thus, when we add an element to the end to form a $n + 1$ - r set, a_{n+1} must be $\leq p + 1$ if we want $r > p$ because we need a way to express $p + 1$. Since $p + 1$ is not expressible by the first n elements, $p + 1 + a_{n+1}$ is not expressible by these $n + 1$ elements. Thus, the new set is a $n + 1$ - r set, where $r \leq p + 1 + a_{n+1} \leq 2^{n+1} - 1$.

Lemma Proven

The answer to this question is $\max(a_{10}) = 248$.

The following set is a 11-1500 set:

$$\{1, 2, 4, 8, 16, 32, 64, 128, 247, 248, 750\}$$

Note that the first 8 numbers are power of 2 from 0 to 7, and realize that any 8 or less digit binary number is basically sum of a combination of the first 8 elements in the set. Thus, $\exists Y \subseteq \{1, 2, 4, 8, 16, 32, 64, 128\}$, $\sigma(Y) = x \forall 1 \leq x \leq 255$.

$248 \leq \sigma(y) + a_9 \leq 502$ which implies that $\exists A \subseteq \{1, 2, 4, 8, 16, 32, 64, 128, 247\}$, $\sigma(A) = x \forall 1 \leq x \leq 502$.

Similarly $\exists B \subseteq \{1, 2, 4, 8, 16, 32, 64, 128, 247, 248\}$, $\sigma(A) = x \forall 1 \leq x \leq 750$ and $\exists C \subseteq \{1, 2, 4, 8, 16, 32, 64, 128, 247, 248, 750\}$, $\sigma(A) = x \forall 1 \leq x \leq 1500$.

Thus, $\{1, 2, 4, 8, 16, 32, 64, 128, 247, 248, 750\}$ is a 11-1500 set.

Now, let's assume for contradiction that $\exists a_{10} \leq 247$ such that $a_1, a_2, \dots, a_{11}$ is a 11- q set where $q \geq 1500$

$a_1, a_2, \dots, a_8$ is a 8- a set where $a \leq 255$ (lemma).

$$\max(a_9) = a_{10} - 1 \leq 246$$

Let $a_1, a_2, \dots, a_{10}$ be a 10- b set where the first 8 elements are the same as the previous set. Then, $256 + a_9 + a_{10}$ is not expressible as $\sigma(Y)$. Thus, $b \leq 255 + a_9 + a_{10} \leq 748$.

In order to create a 11- d set with $d > 748$ and the first 10 elements being the ones on the previous set, $a_{11} \leq 749$ because we need to make 749 expressible as $\sigma(Y)$. Note that $b + 1 + a_{11}$ is not expressible, thus $d < b + 1 + a_{11} \leq 1498$.

Done but not elegant...

4. (USAMO 1992) Chords AA' , BB' , and CC' of a sphere meet at an interior point P but are not contained in the same plane. The sphere through A , B , C , and P is tangent to the sphere through A' , B' , C' , and P . Prove that $AA' = BB' = CC'$.

Solution 1

Consider the plane through A, A', B, B' . This plane, of course, also contains P . We can easily find the $\triangle APB$ is isosceles because the base angles are equal. Thus, $AP = BP$. Similarly, $A'P = B'P$. Thus, $AA' = BB'$. By symmetry, $BB' = CC'$ and $CC' = AA'$, and hence $AA' = BB' = CC'$ as desired.

QED.

Add-on By another person $\hat{v}$

The person that came up with the solution did not prove that $\triangle APB$ is isosceles nor the base angles are congruent. I will add on to the solution.

There is a common tangent plane that pass through P for the 2 spheres that are tangent to each other.

Since any cross section of sphere is a circle. It implies that A, A', B, B' be on the same circle (ω_1), A, B, P be on the same circle (ω_2), and A', B', P be on the same circle (ω_3).

$m\angle APB = m\angle A'PB'$ because they are vertical angles. By power of point, $(AP)(A'P) = (BP)(B'P) \rightarrow \frac{AP}{B'P} = \frac{BP}{A'P}$

By the SAS triangle similarity theory, $\triangle APB \sim \triangle B'PA'$. That implies that $\angle ABP \cong \angle B'PA'$.

Let's call the interception of the common tangent plane and the plane containing A, A', B, B', P , line l .

l must be the common tangent of ω_2 and ω_3 .

The acute angles form by l and $\overline{AA'}$ are congruent to each other (vertical angles) and by the tangent-chord theorem, the central angle of chord $\overline{AP}$ and $\overline{A'P}$ are equal.

Similarly the central angle of chord $\overline{BP}$ and $\overline{B'P}$ are equal.

The length of any chord with central angle 2θ and radius r is $2r \sin(\theta)$, which can easily be seen if we drop the perpendicular from the center to the chord.

Thus, $\frac{AP}{A'P} = \frac{BP}{B'P}$.

By the SAS triangle similarity theory, $\triangle APB \sim \triangle A'PB'$. That implies that $\angle ABP \cong \angle B'PA'$.

That implies that $\angle ABP \cong \angle A'PB' \cong \angle B'PA'$. Thus, $\triangle A'PB'$ is an isosceles triangle and since $\triangle APB \sim \triangle A'PB'$, then $\triangle APB$ must be an isosceles triangle too.

Solution 2

Call the large sphere O_1 , the one containing A O_2 , and the one containing A' O_3 . The centers are c_1, c_2 , and c_3 .

Since two spheres always intersect in a circle, we know that A, B , and C must lie on a circle (w_1) completely contained in O_1 and O_2 .

Similarly, A', B' , and C' must lie on a circle (w_2) completely contained in O_1 and O_3 .

So, we know that 3 lines connecting a point on w_1 and P hit a point on w_2 . This implies that w_1 projects through P to w_2 , which in turn means that w_1 is in a plane parallel to that of w_2 . Then, since w_1 and w_2 lie on the same sphere, we know that they must have the same central axis, which also must contain P (since the center projects through P to the other center).

So, all line from a point on w_1 to P are of the same length, as are all lines from a point on w_2 to P . Since AA', BB' , and CC' are all composed of one of each type of line, they must all be equal.

5. (USAMO 1992) Let $P(z)$ be a polynomial with complex coefficients which is of degree 1992 and has distinct zeros. Prove that there exists complex numbers $a_1, a_2, \dots, a_{1991}$ such that $P(z)$ divides the polynomial

$$\left(\dots \left((z - a_1)^2 - a_2 \right)^2 \dots - a_{1991} \right)^2 - a_{1992}$$

Solution

Since the zeros $z_1, z_2, \dots, z_{1992}$ of the polynomial $P(z)$ are distinct, the polynomial $P(z)$ divides the polynomial

$$Q(z) = \left(\dots \left((z - a_1)^2 - a_2 \right)^2 \dots - a_{1991} \right)^2 - a_{1992}$$

if and only if $Q(z_i) = 0$ for every $i \in \{1; 2; \dots; 1992\}$. So it is enough to show that for any 1992 complex numbers $z_1, z_2, \dots, z_{1992}$, there exist 1992 complex numbers $a_1, a_2, \dots, a_{1992}$, such that the polynomial

$$Q(z) = \left(\dots \left((z - a_1)^2 - a_2 \right)^2 \dots - a_{1991} \right)^2 - a_{1992}$$

satisfies $Q(z_i) = 0$ for every $i \in \{1; 2; \dots; 1992\}$. This is a special case of the following lemma.

Lemma: Let n be a positive integer, and $z_1, z_2, \dots, z_n$ be n complex numbers. Then, there exist n complex numbers $a_1, a_2, \dots, a_n$ such that the polynomial $Q(z) = \left(\dots \left((z - a_1)^2 - a_2 \right)^2 \dots - a_{n-1} \right)^2 - a_n$ satisfies $Q(z_i) = 0$ for every $i \in \{1; 2; \dots; n\}$.

Proof: We use induction over n .

For $n = 1$, the lemma is trivial, since $Q(z) = z - a_1$, so we can take $a_1 = z_1$, and then the polynomial $Q(z) = z - z_1$ clearly satisfies $Q(z_1) = 0$.

Let k be a positive integer. Assume that for $n = k$, the lemma is true. Therefore, there exist k complex numbers $a_1, a_2, \dots, a_k$ such that the polynomial $Q(z) = \left(\dots \left((z - a_1)^2 - a_2 \right)^2 \dots - a_{k-1} \right)^2 - a_k$ satisfies $Q(z_i) = 0$ for every $i \in \{1; 2; \dots; k\}$. Then we wish to prove that for any $k + 1$ complex numbers $z_1, z_2, \dots, z_{k+1}$, there exist $k + 1$ complex numbers $b_1, b_2, \dots, b_{k+1}$ such that the polynomial $R(z) = \left(\dots \left((z - b_1)^2 - b_2 \right)^2 \dots - b_k \right)^2 - b_{k+1}$ satisfies $R(z_i) = 0$ for every $i \in \{1; 2; \dots; k + 1\}$. We will construct this polynomial R from the polynomial Q .

Let $b_i = a_i$ for every $i \leq k - 1$, and let $b_k = a_k + \frac{1}{2}Q(z_{k+1})$ and $b_{k+1} = \frac{1}{4}(Q(z_{k+1}))^2$. Then,

$$\begin{aligned} R(z) &= \left(\dots \left((z - b_1)^2 - b_2 \right)^2 \dots - b_k \right)^2 - b_{k+1} \\ &= \left(\dots \left((z - a_1)^2 - a_2 \right)^2 \dots - a_{k-1} \right)^2 - \left(a_k + \frac{1}{2}Q(z_{k+1}) \right)^2 - \frac{1}{4}(Q(z_{k+1}))^2 \\ &= \left(\left(\dots \left((z - a_1)^2 - a_2 \right)^2 \dots - a_{k-1} \right)^2 - a_k \right) - \frac{1}{2}Q(z_{k+1}) - \frac{1}{4}(Q(z_{k+1}))^2 \\ &= \left(Q(z) - \frac{1}{2}Q(z_{k+1}) \right)^2 - \frac{1}{4}(Q(z_{k+1}))^2 = Q(z)(Q(z) - Q(z_{k+1})). \end{aligned}$$

Then as $Q(z_i) = 0$ for $i \in \{1; 2; \dots; k\}$, we have

$$R(z_i) = Q(z_i)(Q(z_i) - Q(z_{k+1})) = 0$$

for all $i \in \{1; 2; \dots; k\}$. Also,

$$R(z_{k+1}) = Q(z_{k+1})(Q(z_{k+1}) - Q(z_{k+1})) = 0.$$

Thus, $R(z_i) = 0$ holds for every $i \in \{1; 2; \dots; k+1\}$, and this proves the lemma for $n = k+1$. Hence, the induction step is complete, and the lemma is proven.

As the problem is a special case of the lemma ($n = 1992$), we are done.

§22 USAMO 1993

1. (USAMO 1993) For each integer $n \geq 2$, determine, with proof, which of the two positive real numbers a and b satisfying

$$a^n = a + 1, \quad b^{2n} = b + 3a$$

is larger.

Solution 1

First, note that $a, b > 1$, since otherwise $a^n - a$ or $b^{2n} - b$ would be nonpositive. Then, let $f(x) = x^{2n} - x$. Since f is strictly increasing on the interval $(1, \infty)$, we have

$$\begin{aligned} (a-1)^2 > 0 &\implies a^2 + 2a + 1 > 4a \implies a^{2n} > 4a \\ \implies a^{2n} - a > 3a &\implies f(a) > 3a = f(b) \implies a > b \quad \square \end{aligned}$$

Solution 2

Square and rearrange the first equation and also rearrange the second.

$$a^{2n} - a = a^2 + a + 1 \tag{2}$$

$$b^{2n} - b = 3a \tag{3}$$

It is **trivial** that

$$(a-1)^2 > 0 \tag{3}$$

since $a-1$ clearly cannot equal 0 (Otherwise $a^n = 1 \neq 1+1$). Thus

$$a^2 + a + 1 > 3a \tag{4}$$

$$a^{2n} - a > b^{2n} - b \tag{5}$$

where we substituted in equations (1) and (2) to achieve (5). Notice that from $a^n = a + 1$ we have $a > 1$. Thus, if $b > a$, then $b^{2n-1} - 1 > a^{2n-1} - 1$. Since $a > 1 \implies a^{2n-1} - 1 > 0$, multiplying the two inequalities yields $b^{2n} - b > a^{2n} - a$, a contradiction, so $a > b$. However, when n equals 0 or 1, the first equation becomes meaningless, so we conclude that for each integer $n \geq 2$, we always have $a > b$.

Solution 3

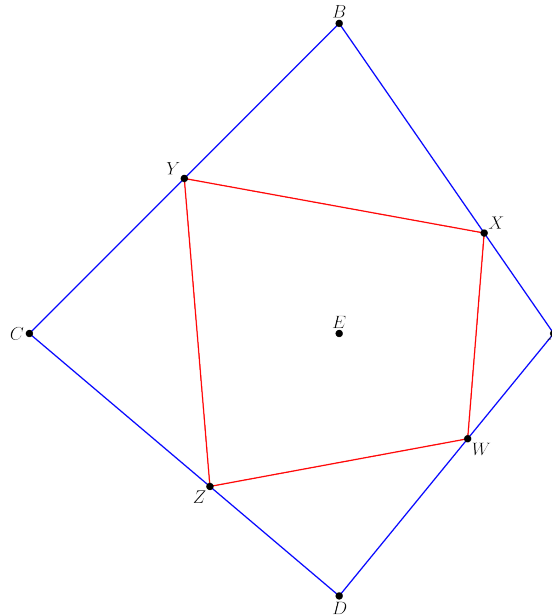
Define $f(x) = x^n - x - 1$ and $g(x) = x^{2n} - x - 3a$. By Descartes's Rule of Signs, both polynomials' only positive roots are a and b , respectively. With the Intermediate Value Theorem and the fact that $f(1) = -1$ and $f(2) = 2^n - 3 > 0$, we have $a \in (1, 2)$. Thus, $-3a \in (-6, -3)$, which means that $g(1) = -3a < 0$. Also, we find that $g(a) = a^{2n} - 4a$. All that remains to prove is that $g(a) > 0$, or $a^{2n} - 4a > 0$. We can then conclude that b is between 1 and a from the Intermediate Value Theorem. From the first equation given, $a^{2n} = (a+1)^2 = a^2 + 2a + 1$. Subtracting $4a$ gives us $a^2 - 2a + 1 > 0$, which is clearly true, as $a \neq 1$. Therefore, we conclude that $1 < b < a < 2$.

2. (USAMO 1993) Let $ABCD$ be a convex quadrilateral such that diagonals AC and BD intersect at right angles, and let E be their intersection. Prove that the reflections of E across AB , BC , CD , DA are concyclic.

Solution 1

Diagram

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Work Let X, Y, Z, W be the foot of the altitude from point E of $\triangle AEB$, $\triangle BEC$, $\triangle CED$, $\triangle DEA$.

Note that reflection of E over all the points of $XYZW$ is similar to $XYZW$ with a scale of 2 with center E . Thus, if $XYZW$ is cyclic, then the reflections are cyclic.

$\angle EWA$ is right angle and so is $\angle EXA$. Thus, $EXAW$ is cyclic with EA being the diameter of the circumcircle.

Follow that, $\angle EWX \cong \angle EAX \cong \angle EAB$ because they inscribe the same angle.

Similarly $\angle EWZ \cong \angle EDC$, $\angle EYX \cong \angle EBA$, $\angle EYZ \cong \angle ECD$.

Futhermore, $m\angle XYZ + m\angle XWZ = m\angle EWX + m\angle EYX + m\angle EYZ + m\angle EWZ = 360^\circ - m\angle CED - m\angle AEB = 180^\circ$.

Thus, $\angle XYZ$ and $\angle XWZ$ are supplementary and follows that, $XYZW$ is cyclic.

Q.E.D

Solution 2

Suppose the reflection of E over AB is W , and similarly define X, Y , and Z . $\triangle BEA \cong \triangle BWA$ by reflection gives $BE = BW$ $\triangle BEC \cong \triangle BXC$ by reflection gives $BE = BX$ These two tell us that E, W , and X belong to a circle with center B . Similarly, we can get that: E, Z , and W belong to a circle with center A , E, X , and Y belong to a circle with center C , E, Y , and Z belong to a circle with center D .

To prove that W, X, Y, Z are concyclic, we want to prove $\angle XWZ + \angle XYZ = 180^\circ$
 $\angle XWZ + \angle XYZ = \angle XWE + \angle EWZ + \angle XYE + \angle EYZ = \frac{1}{2}\angle XBE + \frac{1}{2}\angle EAZ + \frac{1}{2}\angle XCE + \frac{1}{2}\angle EDZ = \frac{1}{2}(\angle XBE + \angle XCE) + \frac{1}{2}(\angle EAZ + \angle EDZ)$

$\angle AED = 90^\circ$ and $\angle AED = \angle AZD$ tells us that $\angle EAZ + \angle EDZ = 180^\circ$ Similarly, $\angle XBE + \angle XCE = 180^\circ$ Thus, $\angle XWZ + \angle XYZ = \frac{1}{2} \cdot 180^\circ + \frac{1}{2} \cdot 180^\circ = 180^\circ$, and we are done. – Lucas.xue (someone pls help with a diagram)

Solution 3

E lies on the isoptic cubic of ABCD, so it has an isogonal conjugate in ABCD.

3. (USAMO 1993) Consider functions $f : [0, 1] \rightarrow \mathbb{R}$ which satisfy

(i)	$f(x) \geq 0$ for all x in $[0, 1]$,
(ii)	$f(1) = 1$,
(iii)	$f(x) + f(y) \leq f(x + y)$ whenever x, y , and $x + y$ are all in $[0, 1]$.

Find, with proof, the smallest constant c such that

$$f(x) \leq cx$$

for every function f satisfying (i)-(iii) and every x in $[0, 1]$.

Solution

My claim: $c \geq 2$

Lemma 1) $f\left(\left(\frac{1}{2}\right)^n\right) \leq \left(\frac{1}{2}\right)^n$ for $n \in \mathbb{Z}, n \geq 0$

For $n = 0$, $f(1) = 1$ (ii)

Assume that it is true for $n - 1$, then $f\left(\left(\frac{1}{2}\right)^n\right) + f\left(\left(\frac{1}{2}\right)^n\right) \leq f\left(\left(\frac{1}{2}\right)^{n-1}\right) \leq \left(\frac{1}{2}\right)^{n-1}$

$$f\left(\left(\frac{1}{2}\right)^n\right) \leq \left(\frac{1}{2}\right)^n$$

By principle of induction, **lemma 1 is proven.**

Lemma 2) For any x , $\left(\frac{1}{2}\right)^{n+1} < x \leq \left(\frac{1}{2}\right)^n \leq 1$ and $n \in \mathbb{Z}$, $f(x) \leq \left(\frac{1}{2}\right)^n$.

$$f(x) + f\left(\left(\frac{1}{2}\right)^n - x\right) \leq f\left(\left(\frac{1}{2}\right)^n\right) \leq \left(\frac{1}{2}\right)^n \text{ (lemma 1 and (iii))}$$

$$f(x) \leq \left(\frac{1}{2}\right)^n \text{ (because } f\left(\left(\frac{1}{2}\right)^n - x\right) \geq 0 \text{ (i))}$$

$$\forall 0 \leq x \leq 1, \left(\frac{1}{2}\right)^{n-1} \geq 2x \geq \left(\frac{1}{2}\right)^n \geq f(x). \text{ Thus, } c = 2 \text{ works.}$$

$$\text{Let's look at a function } g(x) = \begin{cases} 0 & 0 \leq x \leq \frac{1}{2}; \\ 1 & \frac{1}{2} < x \leq 1; \end{cases}$$

It clearly have property (i) and (ii). For $0 \leq x \leq \frac{1}{2}$ and WLOG let $x \leq y$, $f(x) + f(y) = 0 + f(y) \leq f(y)$

For $\frac{1}{2} < x \leq 1$, $x + y > 1$. Thus, property (iii) holds too. Thus $g(x)$ is one of the legit function.

$$\lim_{x \rightarrow \frac{1}{2}^+} cx \geq \lim_{x \rightarrow \frac{1}{2}^+} g(x) = 1$$

$$\frac{1}{2}c > 1$$

$c > 2$ but approach to 2 when x is extremely close to $\frac{1}{2}$ from the right side.

Q.E.D

4. (USAMO 1993) Let a, b be odd positive integers. Define the sequence (f_n) by putting $f_1 = a$, $f_2 = b$, and by letting f_n for $n \geq 3$ be the greatest odd divisor of $f_{n-1} + f_{n-2}$. Show that f_n is constant for n sufficiently large and determine the eventual value as a function of a and b .

Solution

Part 1) Prove that f_n is constant for sufficiently large n .

Note that if there is some $f_n = f_{n-1}$ for any n , then $\frac{f_n + f_{n-1}}{2} = f_n$, which is odd. Thus, $f_{n+1} = f_n = f_{n-1}$ and by induction, all f_p is constant for $p \geq n$.

Also note that $f_n > 0$ since average of 2 positive number is always positive.

Thus, assume for contradiction, $\nexists n, f_n = f_{n-1}$.

Then, $f_{n+1} \leq \frac{f_n + f_{n-1}}{2} < \max(f_n, f_{n-1})$, $f_{n+2} \leq \frac{f_{n+1} + f_n}{2} < \max(f_n, f_{n+1}) \leq \max(f_n, f_{n-1})$

Thus, $\max(f_n, f_{n-1}) > \max(f_{n+1}, f_{n+2})$ and that means that $\max(f_{2n}, f_{2n+1})$ is a strictly decreasing function and it must reach 0 as $n \rightarrow \infty$, which contradict with the fact that $f_n > 0$.

Part 1 proven.

Part 2) Show that the constant is $\gcd(a, b)$.

For any f_n where $\gcd(a, b) = d \neq 1$. $f_n = dg_n$ for g_n with the same property except with $g_1 = \frac{a}{d}$ and $g_2 = \frac{b}{d}$.

Therefore, if I prove that the constant for any f_n with relatively prime a, b is 1, then I have shown that **part 2** is true.

Lemma) If $\gcd(f_n, f_{n-1}) = 1$, then $\gcd(f_n, f_{n+1}) = 1$.

Assume for contradiction that $\gcd(f_n, f_{n+1}) = d \neq 1$, since both f_n and f_{n+1} are odd, d is not divisible by 2.

$f_{n+1} = \frac{f_n + f_{n-1}}{2^n}$ for some $n \in \mathbb{Z}^+$ such that f_{n+1} is odd.

$$(2^n)f_{n+1} - f_n = f_{n-1}$$

$d(p+q) = f_{n-1}$, where p and q is another integer.

Thus, f_{n-1} is divisible by d which contradicts with the assumption that $\gcd(f_n, f_{n-1}) = 1$.

Lemma proven

By induction, $\gcd(f_n, f_{n-1}) = 1$ since $\gcd(a, b) = \gcd(f_1, f_2) = 1$.

Since there must exist some n where $f_n = f_{n+1}$ (part 1), $\gcd(f_n, f_{n+1}) = f_n = 1$.

Q.E.D

5. (USAMO 1993) Let $a_0, a_1, a_2, \dots$ be a sequence of positive real numbers satisfying $a_{i-1}a_{i+1} \leq a_i^2$ for $i = 1, 2, 3, \dots$. (Such a sequence is said to be *log concave*.) Show that for each $n > 1$,

$$\frac{a_0 + \dots + a_n}{n+1} \cdot \frac{a_1 + \dots + a_{n-1}}{n-1} \geq \frac{a_0 + \dots + a_{n-1}}{n} \cdot \frac{a_1 + \dots + a_n}{n}.$$

Solution

Notice that because

$$\begin{aligned} (a_0 + a_1 + \dots + a_{n-1})(a_1 + a_2 + \dots + a_n) &= (a_0 + (a_1 + \dots + a_{n-1}))(-a_0 + (a_0 + a_1 + \dots + a_n)) \\ &= -a_0^2 + a_0((a_0 + a_1 + \dots + a_n) - (a_1 + a_2 + \dots + a_{n-1})) + (a_1 + a_2 + \dots + a_{n-1})(a_0 + a_1 + \dots + a_n) \\ &= a_0a_n + (a_1 + a_2 + \dots + a_{n-1})(a_0 + a_1 + \dots + a_n), \end{aligned}$$

we may subtract $\frac{(a_1 + a_2 + \cdots + a_{n-1})(a_0 + a_1 + \cdots + a_n)}{n^2}$ from both sides of the inequality and observe that it is sufficient to prove that

$$\frac{(a_1 + a_2 + \cdots + a_{n-1})(a_0 + a_1 + \cdots + a_n)}{(n^2 - 1)n^2} \geq \frac{a_0 a_n}{n^2},$$

or

$$(a_1 + a_2 + \cdots + a_{n-1})(a_0 + a_1 + \cdots + a_n) \geq (n^2 - 1)a_0 a_n.$$

Fortunately, this is an easy inequality. Indeed, from AM-GM applied on each group of terms we have

$$(a_1 + a_2 + \cdots + a_{n-1})(a_0 + a_1 + \cdots + a_n) \geq (n^2 - 1) {}^{n-1}\sqrt{a_1 a_2 \cdots a_{n-1}} {}^{n+1}\sqrt{a_0 a_1 \cdots a_n},$$

and so it suffices to prove

$${}^{n-1}\sqrt{a_1 a_2 \cdots a_{n-1}} {}^{n+1}\sqrt{a_0 a_1 \cdots a_n} \geq a_0 a_n,$$

or, after taking both sides to the $(n^2 - 1)$ power, simplifying, and taking the n -th root of both sides, to prove

$$a_1^2 a_2^2 a_3^2 \cdots a_{n-1}^2 \geq a_0^{n-1} a_n^{n-1}.$$

This easily follows from the Fact that $a_0 a_n \leq a_i a_{n-i}$ for $1 \leq i \leq n-1$. Indeed, we are given that

$$a_0 a_2 \leq a_1^2$$

$$a_1 a_3 \leq a_2^2$$

$$a_2 a_4 \leq a_3^2$$

...

$$a_{n-3} a_{n-1} \leq a_{n-2}^2$$

$$a_{n-2} a_n \leq a_{n-1}^2.$$

Multiply all inequalities together and cancel $a_1, a_2^2, a_3^2, \dots, a_{n-2}^2, a_{n-1}$ to give $a_0 a_n \leq a_1 a_{n-1}$. Similarly, by multiplying all inequalities except the first and the last, we deduce that $a_2 a_{n-2} \geq a_1 a_{n-1} \geq a_0 a_n$, and a simple induction argument proves the verity of the Fact for $1 \leq i \leq \frac{n}{2}$, and so by the Commutative Property the Fact is true for all $1 \leq i \leq n-1$, as desired. Now multiply each inequality of the Fact for $i = 1, 2, 3, \dots, n-1$ to give the desired result.

Note: The Fact can be generalized into a Lemma: $a_x a_w \leq a_y a_z$ whenever $x \leq y \leq z \leq w$ and $x + w = y + z$. The proof is similar to that of the Fact and is left as an exercise to the reader.

–User sul1, April 15, 2015.

§23 USAMO 1994

1. (*USAMO 1994*) Let $k_1 < k_2 < k_3 < \dots$ be positive integers, no two consecutive, and let $s_m = k_1 + k_2 + \dots + k_m$ for $m = 1, 2, 3, \dots$. Prove that, for each positive integer n , the interval $[s_n, s_{n+1})$ contains at least one perfect square.

Solution

We want to show that the distance between s_n and s_{n+1} is greater than the distance between s_n and the next perfect square following s_n .

Given $s_n = \sum_{i=1}^n k_i$, where no k_i are consecutive, we can put a lower bound on k_n . This occurs when all $k_{i+1} = k_i + 2$:

$$\begin{aligned} s_n &= (k_{n,\min}) + (k_{n,\min} - 2) + (k_{n,\min} - 4) + \dots + (k_{n,\min} - 2n + 2) \\ &= nk_{n,\min} - \sum_{i=1}^{n-1} 2i \\ &= nk_{n,\min} - 2 \sum_{i=1}^n i + 2n \\ &= nk_{n,\min} - n(n+1) + 2n \\ &= nk_{n,\min} - n^2 + n \end{aligned}$$

Rearranging, $k_{n,\min} = \frac{s_n}{n} + n - 1$. So, $k_n \geq \frac{s_n}{n} + n - 1$, and the distance between s_n and s_{n+1} is $k_{n+1} \geq k_n + 2 \geq \frac{s_n}{n} + n + 1$.

Also, let $d(s_n)$ be the distance between s_n and the next perfect square following s_n . Let's look at the function $d(x)$ for all positive integers x .

When x is a perfect square, it is easy to see that $d(x) = 2\sqrt{x} + 1$. Proof: Choose $x = m^2$. $d(m^2) = (m+1)^2 - m^2 = 2m + 1 = 2\sqrt{m^2} + 1$.

When x is not a perfect square, $d(x) < 2\sqrt{x} + 1$. Proof: Choose $x = m^2 + p$ with $0 < p < 2m + 1$. $d(m^2 + p) = (m+1)^2 - m^2 - p = 2m + 1 - p < 2m + 1 = 2\sqrt{m^2} + 1 < 2\sqrt{m^2 + p} + 1$.

So, $d(x) \leq 2\sqrt{x} + 1$ for all x and $d(s_n) \leq 2\sqrt{s_n} + 1$ for all s_n .

Now, it suffices to show that $k_{n+1} \geq d(s_n)$ for all n .

$$\begin{aligned} k_{n+1} - d(s_n) &\geq \frac{s_n}{n} + n + 1 - 2\sqrt{s_n} - 1 \\ &= \frac{1}{n}(s_n + n^2 - 2n\sqrt{s_n}) \\ &= \frac{s_n^2 + n^4 + 2n^2s_n - 4n^2s_n}{n(s_n + n^2 + 2n\sqrt{s_n})} \\ &= \frac{(s_n - n^2)^2}{n(s_n + n^2 + 2n\sqrt{s_n})} \\ &\geq 0 \end{aligned}$$

So, $k_{n+1} \geq d(s_n)$ and all intervals between s_n and s_{n+1} will contain at least one perfect square.

Solution 2

We see that by increasing n by some amount, we simply shift our interval by a finite amount. It suffices to consider the case $n = 1$ (since this can be inducted across all positive integers). Let $k_1 = x$. We want the smallest interval, so we have $[x, 2x + 2]$. Simple induction reveals that the ration of consecutive squares grows slower than our linear bound. We now consider sufficiently small x (where $\frac{(n+1)^2}{n^2} < 2$). This first happens at $n = 3$. By simple casework, our answer is as desired ■.

Solution 3

We will first prove by Induction on $n \in \mathbb{N}$ that $(k_n + 1)^2 \geq 4(k_1 + k_2 + \cdots + k_n)$. Denote this statement by $P(n)$.

For the Base Case let $n = 1$, we know that;

$$\begin{aligned} (k_1 - 1)^2 &\geq 0 \\ \Rightarrow k_1^2 - 2k_1 + 1 &\geq 0 \\ \Rightarrow k_1^2 + 2k_1 + 1 &\geq 4k_1 \\ \Rightarrow (k_1 + 1)^2 &\geq 4k_1. \end{aligned}$$

Hence, the Base Case holds.

For the Inductive Step, suppose that $P(m)$ holds for some $k \in \mathbb{N}$, we will prove that $P(m + 1)$ holds as well.

Assume for contradiction that $P(m + 1)$ doesn't hold, then we know that;

$$\begin{aligned} (k_m + 1)^2 &< 4(k_1 + k_2 + \cdots + k_m) \\ \Rightarrow k_m^2 + 2k_m + 1 &< 4(k_1 + k_2 + \cdots + k_{m-1}) + 4k_m \\ \Rightarrow k_m^2 - 2k_m + 1 &< 4(k_1 + k_2 + \cdots + k_{m-1}) \\ \Rightarrow (k_m - 1)^2 &< 4(k_1 + k_2 + \cdots + k_{m-1}). \end{aligned}$$

We know that since k_m and k_{m-1} are not consecutive, we have;

$$\begin{aligned} k_{m-1} &\geq k_m - 2 \\ \Rightarrow k_{m-1} + 1 &\leq k_m - 1 \\ \Rightarrow (k_{m-1} + 1)^2 &\leq (k_m - 1)^2 \\ \Rightarrow (k_{m-1} + 1)^2 &\leq (k_m - 1)^2 < 4(k_1 + k_2 + \cdots + k_{m-1}). \end{aligned}$$

But this contradicts the Inductive Hypothesis that

$$\Rightarrow (k_{m-1} + 1)^2 \geq 4(k_1 + k_2 + \cdots + k_{m-1}),$$

thus our assumption was false and the Inductive Step is complete.

Hence, we have proved that $P(n)$ holds, and we will use $P(n)$ to solve the original problem.

Now suppose that for some positive integer n , the interval $[s_n, s_{n+1})$ does not contain any perfect square, then we know that there must exist two perfect squares of consecutive integers, such that the smaller one is lesser than s_n and the larger one greater than or equal to s_{n+1} .

We thus know that there exists some $x \in \mathbb{N}$ such that;

$$x^2 < s_n \quad (4)$$

$$\text{and } (x+1)^2 \geq s_{n+1} \quad (5)$$

By Inequality 1;

$$\begin{aligned} x &< \sqrt{s_n} \\ &= \sqrt{k_1 + k_2 + \cdots + k_n}. \end{aligned}$$

Hence, we know that;

$$\begin{aligned} (x+1)^2 &< (\sqrt{k_1 + k_2 + \cdots + k_n} + 1)^2 \\ &= k_1 + k_2 + \cdots + k_n + 2\sqrt{k_1 + k_2 + \cdots + k_n} + 1. \end{aligned}$$

Combining this with Inequality 2 gives;

$$\begin{aligned} s_{n+1} &\leq (x+1)^2 < k_1 + k_2 + \cdots + k_n + 2\sqrt{k_1 + k_2 + \cdots + k_n} + 1 \\ \Rightarrow k_1 + k_2 + \cdots + k_n + k_{n+1} &< k_1 + k_2 + \cdots + k_n + 2\sqrt{k_1 + k_2 + \cdots + k_n} + 1 \\ \Rightarrow k_{n+1} &< 2\sqrt{k_1 + k_2 + \cdots + k_n} + 1 \\ \Rightarrow k_{n+1} - 1 &< 2\sqrt{k_1 + k_2 + \cdots + k_n}. \end{aligned}$$

We know that since k_{n+1} are not consecutive, $k_n + 1 \leq k_{n+1} - 1$, hence, we have;

$$\begin{aligned} k_{n+1} &\leq k_{n+1} - 1 < 2\sqrt{k_1 + k_2 + \cdots + k_n} \\ k_{n+1} &< 2\sqrt{k_1 + k_2 + \cdots + k_n} \\ (k_{n+1}^2) &< 4(k_1 + k_2 + \cdots + k_n). \end{aligned}$$

But this contradicts the statement $P(n)$ which was proved earlier. $\square$

2. (*USAMO 1994*) The sides of a 99-gon are initially colored so that consecutive sides are red, blue, red, blue, ... red, blue, yellow. We make a sequence of modifications in the coloring, changing the color of one side at a time to one of the three given colors (red, blue, yellow), under the constraint that no two adjacent sides may be the same color. By making a sequence of such modifications, is it possible to arrive at the coloring in which consecutive sides are red, blue, red, blue, red, blue, ... red, yellow, blue?

Solution

We proceed by representing the colors as numbers, i.e. Red = 0, Blue = 1, Yellow = 2. Thus, we start with some sequence 0101...012 and are trying to end up with the sequence 1010...0102. Generate a second sequence of terms by subtracting each term by the following term and taking it modulus 3, i.e. (1-0, 0-1, 1-0, 0-1, ... 0-1, 1-0, 0-2, 2-1) = 1212...2111 is the sequence generated using the start sequence. The sequence generated using the end sequence is (0-1, 1-0, 0-1, 1-0, ... 1-0, 0-1, 1-2, 2-0) = 2121...2122. Observe that the sum of terms generated by the start sequence is different from the sum of terms generated by the end sequence.

We must now prove that changing the color of a side does not change the sum of the generated sequence. We do this by noting that we can only change the color of a side if its adjacent sides have the same color as each other. Thus, changing a side does not change the sum of terms in the generated sequence, as the term generated by the changing side and the side to its left is simply the "negative" of the term generated by the changing side and the side to its right (i.e. 212 to 202 changes 12 to 21).

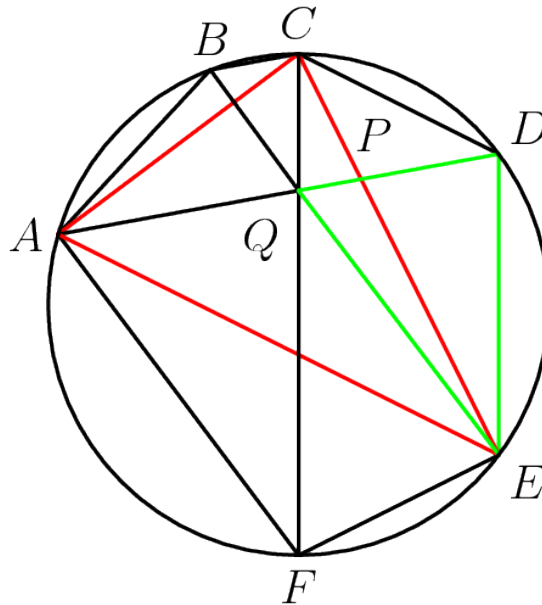
Since it is impossible to change the sum of terms in the generated sequence, and the sequences generated by the start and end sequences have two different sums, it is impossible to perform a series of such modifications that change the start sequence to the end sequence.

3. (USAMO 1994) A convex hexagon $ABCDEF$ is inscribed in a circle such that $AB = CD = EF$ and diagonals AD , BE , and CF are concurrent. Let P be the intersection of AD and CE . Prove that $CP/PE = (AC/CE)^2$.

Solution

Let the diagonals AD , BE , CF meet at Q .

First, let's show that the triangles $\triangle AEC$ and $\triangle QED$ are similar.

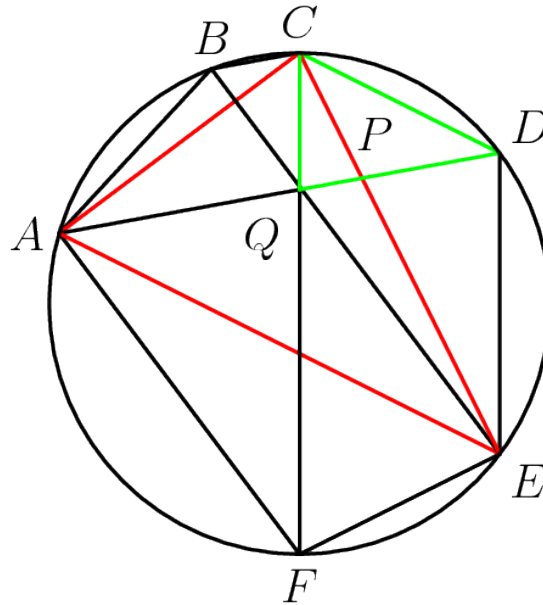


$\angle ACE = \angle ADE$ because A, C, D and E all lie on the circle, and $\angle ADE = \angle QDE$.
 $\angle AEB = \angle CED$ because $AB = CD$, and A, B, C, D and E all lie on the circle.
 Then,

$$\angle AEB = \angle CED \rightarrow \angle AEB + \angle BEC = \angle CED + \angle BEC \rightarrow \angle AEC = \angle QED$$

Therefore, $\triangle AEC$ and $\triangle QED$ are similar, so $AC/CE = QD/DE$.

Next, let's show that $\triangle AEC$ and $\triangle CDQ$ are similar.

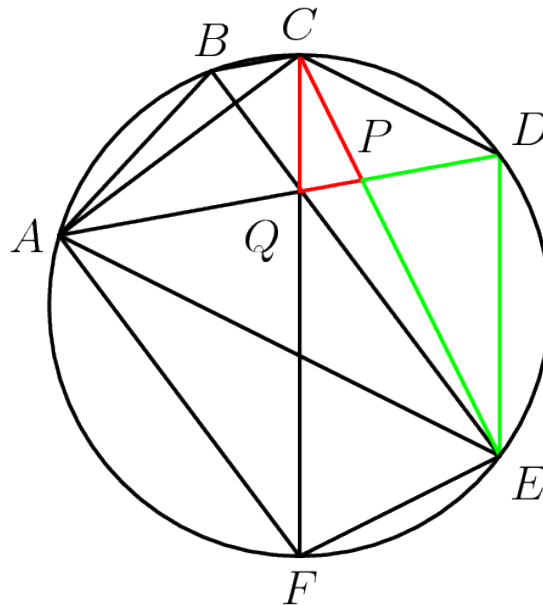


$\angle AEC = \angle ADC$ because A, C, D and E all lie on the circle, and $\angle ADC = \angle CDQ$.
 $\angle EAD = \angle ECD$ because A, C, D and E all lie on the circle. $\angle DAC = \angle ECF$
 because $CD = EF$, and A, C, D, E and F all lie on the circle. Then,

$$\angle EAC = \angle EAD + \angle DAC = \angle ECD + \angle ECF = \angle DCQ$$

Therefore, $\triangle AEC$ and $\triangle CDQ$ are similar, so $AC/CE = CQ/QD$.

Lastly, let's show that $\triangle CPQ$ and $\triangle EPD$ are similar.



Because $CD = EF$ and C, D, E and F all lie on the circle, CF is parallel to DE .
 So, $\triangle CPQ$ and $\triangle EPD$ are similar, and $CQ/DE = CP/PE$.

Putting it all together, $\frac{CP}{PE} = \frac{CQ}{DE} = \frac{AC}{CE} \cdot \frac{QD}{DE} = \left(\frac{AC}{CE}\right)^2$.

Borrowed from <https://mks.mff.cuni.cz/kalva/usa/usoln/usol943.html>

4. (USAMO 1994) Let $a_1, a_2, a_3, \dots$ be a sequence of positive real numbers satisfying $\sum_{j=1}^n a_j \geq \sqrt{n}$ for all $n \geq 1$. Prove that, for all $n \geq 1$,

$$\sum_{j=1}^n a_j^2 > \frac{1}{4} \left(1 + \frac{1}{2} + \dots + \frac{1}{n} \right).$$

Solution 1

Note that if we try to minimize $(a_j)^2$, we would try to make the a_j as equal as possible. However, by the condition given in the problem, this isn't possible, the a_j 's have to be an increasing sequence. Thinking of minimizing sequences, we realize that the optimal equation is $a_n = \sqrt{n} - \sqrt{n-1} = 1/(\sqrt{n} + \sqrt{n-1})$. Note that this is strictly greater than $1/(2\sqrt{n})$. So it is greater than the sum of $(1/\sqrt{4n})^2$ over all n from 1 to ∞ . So the sum we are looking to minimize is strictly greater than the sum of $1/4n$ over all n from 1 to ∞ , which is what we wanted to do.

—Dear the critical link here is missing: to prove that the optimal equation is $a(n) = \sqrt{n} - \sqrt{n-1}$. It is therefore incomplete. T

Solution 2

Consider a sequence $b_n = \sqrt{n} - \sqrt{n-1}$.

Lemma: b_n minimizes $\sum_{k=1}^n a_k^2$ over the term $S_k = a_1 + \dots + a_k \geq \sqrt{k}$.

Proof of the Lemma:

Note $b_k = \sqrt{k} - \sqrt{k-1}$. We show that if $S_k \geq \sqrt{k}$, then $\sum a_k^2 \geq \sum b_k^2$. Define $c_k = S_k - \sqrt{k} \geq 0$ the extra amount over equality. Then $a_k - b_k = (S_k - S_{k-1}) - (\sqrt{k} - \sqrt{k-1}) = c_k - c_{k-1}$. Hence in other words, a_k is just the equality sequence b_k plus c_k . Now, $\sum a_k^2 = \sum (b_k + c_k - c_{k-1})^2 = \sum b_k^2 + \sum (c_k - c_{k-1})^2 + 2 \sum b_k (c_k - c_{k-1})$. Now we must show that $\sum (c_k - c_{k-1})^2 + 2 \sum b_k (c_k - c_{k-1}) \geq 0$. The first term is trivial. The second one, rewrite as $b_n c_n + \sum_{k=1}^{n-1} (b_k - b_{k+1}) c_k$ in which $c_k \geq 0, b_n \geq 0$, and $b_k \geq b_{k+1}$, as it is concave. Hence, each term, when summed, yields ≥ 0 . Hence we are done. ■

USAMO 1994/4:

Consider a_j . Notice that $\sqrt{j} - \sqrt{j-1} = \frac{j-(j-1)}{\sqrt{j}+\sqrt{j-1}} = \frac{1}{\sqrt{j}+\sqrt{j-1}}$. Squaring, $(\sqrt{j} - \sqrt{j-1})^2 = \frac{1}{(\sqrt{j}+\sqrt{j-1})^2}$. Also, $\sqrt{j} + \sqrt{j-1} < 2\sqrt{j}$ is trivial by $\sqrt{j-1} < \sqrt{j}$ hence $(\sqrt{j} + \sqrt{j-1})^2 < 4j$. Also, $(\sqrt{j} - \sqrt{j-1})^2 = \frac{1}{(\sqrt{j}+\sqrt{j-1})^2}$ and thus $\frac{1}{(\sqrt{j}+\sqrt{j-1})^2} > \frac{1}{4j}$. Therefore, $(\sqrt{j} - \sqrt{j-1})^2 > \frac{1}{4j}$. Taking the sum 1 through n gives the desired result. □

5. (USAMO 1994) Let $|U|$, $\sigma(U)$ and $\pi(U)$ denote the number of elements, the sum, and the product, respectively, of a finite set U of positive integers. (If U is the empty set, $|U| = 0$, $\sigma(U) = 0$, $\pi(U) = 1$.) Let S be a finite set of positive integers. As usual, let $\binom{n}{k}$ denote $\frac{n!}{k!(n-k)!}$. Prove that

$$\sum_{U \subseteq S} (-1)^{|U|} \binom{m - \sigma(U)}{|S|} = \pi(S)$$

for all integers $m \geq \sigma(S)$.

Solution

Let, for $|S| \geq k \geq 1$, A_k be a collection of sets such that $|\bigcap_{k \in U} A_k| = \binom{m-\sigma(U)}{|S|}$. We'll try to find such a collection to prove $|\bigcup_{k \in U} A_k| = \binom{m}{|S|} - \pi(S)$, since we can use PIE to show that $|\bigcup_{k \in U} A_k| = \sum_{\emptyset \neq U \subseteq S} (-1)^{|U|+1} |\bigcap_{k \in U} A_k|$.

Let Z be the set of binary sequences with size m , which we'll divide into blocks, in particular, with $S = \{x_1, x_2, \dots, x_n\}$, the first block has size x_1 , the next has size x_2 and so on, until we reach $\sigma(S)$, the last few we shall consider as an extra block with size $m - \sigma(S) = R$.

Now we perceive that the $\sigma(U)$ function has the property, that, for $U \subseteq S$, $\sigma(S) - \sigma(U) = \sigma(S \setminus U)$, so, $m - \sigma(U) = \sigma(S \setminus U) + R$.

Let A_k be the number of binary sequences with exactly $|S|$ times 1, but no appearances of 1 in the k -th block. It is easy to see that $\bigcap_{k \in U} A_k$ is all of the binary sequences with exactly $|S|$ times 1, but no appearances of 1 in any of the k -th blocks, for $k \in U$. So, we can simply eliminate those blocks, with is equivalent to eliminating U from S , so we are left with $\sigma(S \setminus U) + R$ places to put the 1's $|S|$ times. This is $\binom{\sigma(S \setminus U) + R}{|S|}$, and we saw that is equal to $\binom{m-\sigma(U)}{|S|}$. So, we've found a collection that fits the description.

The number of ways to pick 1's $|S|$ times from m places is $\binom{m}{|S|}$. But we don't want for the 1's to fill every block, since this is not in the union. We have $\pi(S)$ ways to put the 1's $|S|$ times, where each 1 is in exacty one block (x_1 places $\times$ x_2 places and so on). So, we find that $|\bigcup_{k \in U} A_k| = \binom{m}{|S|} - \pi(S)$. Which implies, by the Principle of Inclusion Exclusion, $\binom{m}{|S|} - \pi(S) = \sum_{\emptyset \neq U \subseteq S} (-1)^{|U|+1} \binom{m-\sigma(U)}{|S|} \iff \pi(S) = \sum_{U \subseteq S} (-1)^{|U|} \binom{m-\sigma(U)}{|S|}$

§24 USAMO 1995

1. (USAMO 1995) Let p be an odd prime. The sequence $(a_n)_{n \geq 0}$ is defined as follows: $a_0 = 0$, $a_1 = 1$, $\dots$, $a_{p-2} = p-2$ and, for all $n \geq p-1$, a_n is the least positive integer that does not form an arithmetic sequence of length p with any of the preceding terms. Prove that, for all n , a_n is the number obtained by writing n in base $p-1$ and reading the result in base p .

Solution

I have to make the assumption that $a_{n+1} > a_n$ (without this assumption, I can have the sequence

$$1, \dots, p-2, 1, 1, \dots, 1, 2, 1, \dots)$$

All of the following work are in base p otherwise stated

Lemma 1: for an arithmetic sequence of length p to exist, there must be a number in the sequence with $(p-1)$ as a digit.

A arithmetic sequence of length p can be represented as

$$a, a+d, a+2d, \dots, a+(p-1)d$$

Since no number repeats, $d \neq 0$. Thus, d must have a rightmost non-zero digits, and every term in the sequence $0, d, 2d, \dots, (p-1)d$ have the same number of trailing zeros, let's say there are x trailing zeros.

$$\text{and let } \frac{0, d, 2d, \dots, (p-1)d}{p^x} = S$$

$S \equiv 1, 2, \dots, (p-1) \pmod{p}$ since p is an odd prime and the operation divide by p^x has removed all factors of p in S

Thus, there must be a number with $(p-1)$ as a digit if a length- p sequence exists.

Now, I'm going to prove the statement by strong induction.

I'm going to assume that for all k less than n

a_k is the number obtained by writing k in base $p-1$ and reading the result in base p .

(which is true for $n = p-2$ already)

Or in another word, the terms precede a_n contains all number less than n written in base $p-1$ and reading the result in base p .

Lemma 2: Any term containing the digit $p-1$ will form an arithmetic sequence of length- p with preceding terms

let $p-1$ be the x^{th} digit from the right (There may be more than 1 $(p-1)$)

a = the number with all $(p-1)$ replaced by 0 (e.g: $p = 7$, the number = 1361264, then $a = 1301204$)

$d = \sum p^{x-1}$ (for all x 's)

$a, a+d, a+2d, \dots, a+(p-2)d$ all precede a_n , thus, $a+(p-1)d$ can't be in the sequence $(a_n)_{n \geq 0}$

Therefore, any term containing the digit $p-1$ can't be a term in the sequence $(a_n)_{n \geq 0}$

Since the digit $p-1$ won't appear (lemma 2) and as long as it doesn't appear, the arithmetic sequence of length p won't be formed (lemma 1), and a_n must be as small as possible,

Therefore, for all n , a_n is the number obtained by writing n in base $p-1$ and reading the result in base p .

2. (USAMO 1995) A calculator is broken so that the only keys that still work are the $\sin$, $\cos$, $\tan$, $\sin^{-1}$, $\cos^{-1}$, and $\tan^{-1}$ buttons. The display initially shows 0. Given any positive rational number q , show that pressing some finite sequence of buttons will yield q . Assume that the calculator does real number calculations with infinite precision. All functions are in terms of radians.

Solution

We will prove the following, stronger statement : If m and n are relatively prime nonnegative integers such that $n > 0$, then the some finite sequence of buttons will yield $\sqrt{m/n}$.

To prove this statement, we induct strongly on $m+n$. For our base case, $m+n = 1$, we have $n = 1$ and $m = 0$, and $\sqrt{m/n} = 0$, which is initially shown on the screen. For the inductive step, we consider separately the cases $m = 0$, $0 < m \leq n$, and $n < m$.

If $m = 0$, then $n = 1$, and we have the base case.

If $0 < m \leq n$, then by inductive hypothesis, $\sqrt{(n-m)/m}$ can be obtained in finitely many steps; then so can

$$\cos \tan^{-1} \sqrt{(n-m)/m} = \sqrt{m/n}.$$

If $n < m$, then by the previous case, $\sqrt{n/m}$ can be obtained in finitely many steps. Since $\cos \tan^{-1} \sqrt{n/m} = \sin \tan^{-1} \sqrt{m/n}$, it follows that

$$\tan \sin^{-1} \cos \tan^{-1} \sqrt{n/m} = \sqrt{m/n}$$

can be obtained in finitely many steps. Thus the induction is complete. ■

Alternate Solution (using infinite descent): This solution is similar to the above solution. We can invert a number with

$$\tan \sin^{-1} \cos \tan^{-1} \sqrt{n/m} = \sqrt{m/n}$$

and we can get

$$\tan \sin^{-1} \cos \tan^{-1} \cos \tan^{-1} \sqrt{x} = \sqrt{x+1}.$$

Then we do $\sqrt{x}$ to $\sqrt{x+1}$ in reverse, trying to get 1. Subtracting one inside the radical for a fraction $\sqrt{m/n}$ will eventually result in a fraction less than or equal to 1. Taking the reciprocal of this will result in a fraction greater than or equal to 1, and we can keep subtracting one inside the radical.

We do this repeatedly, and this should eventually result in 1 in a finite number of steps. This is because each time we do this, $m+n$ is smaller. This is true even if $\sqrt{m/n}$ can be simplified, as simplifying will only result in a smaller value. The smallest possible value of the fraction is $1/1$, so this will be achieved after finite steps.

Finally, we know that 1 is achievable as $\cos(0) = 1$.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

3. (*USAMO 1995*) Given a nonisosceles, nonright triangle ABC , let O denote the center of its circumscribed circle, and let A_1 , B_1 , and C_1 be the midpoints of sides BC , CA , and AB , respectively. Point A_2 is located on the ray OA_1 so that $\triangle OAA_1$ is similar to $\triangle OA_2A$. Points B_2 and C_2 on rays OB_1 and OC_1 , respectively, are defined similarly. Prove that lines AA_2 , BB_2 , and CC_2 are concurrent, i.e. these three lines intersect at a point.

Solution 1

LEMMA 1: In $\triangle ABC$ with circumcenter O , $\angle OAC = 90 - \angle B$.

PROOF of Lemma 1: The arc AC equals $2\angle B$ which equals $\angle AOC$. Since $\triangle AOC$ is isosceles we have that $\angle OAC = \angle OCA = 90 - \angle B$. QED

Define $H \in BC$ s.t. $AH \perp BC$. Since $OA_1 \perp BC$, $AH \parallel OA_1$. Let $\angle AA_2O = \angle A_1AO = x$ and $\angle AA_1O = \angle A_2AO = y$. Since we have $AH \parallel OA_1$, we have that $\angle HAA_2 = x$. Also, we have that $\angle A_2AA_1 = y - x$. Furthermore, $\angle BAH = 90 - \angle B = \angle OAC$, by lemma 1. Therefore, $\angle A_1AC = 90 - \angle B + x = \angle BAA_2$. Since A_1 is the midpoint of BC , AA_1 is the median. However $\angle A_1AC = \angle BAA_2$ tells us that AA_2 is just AA_1 reflected across the internal angle bisector of A . By definition, AA_2 is the A -symmedian. Likewise, BB_2 is the B -symmedian and CC_2 is the C -symmedian. Since the symmedians concur at the symmedian point, we are done.

QED

Solution 2

Let AH be the altitude of $\triangle ABC \implies$

$$\begin{aligned}\angle BAH &= 90^\circ - \angle ABC, \angle OAC = \angle OCA = \\ &= \frac{180^\circ - \angle AOC}{2} = \frac{180^\circ - 2\angle ABC}{2} = \angle BAH.\end{aligned}$$

Hence AH and AO are isogonals with respect to the angle $\angle BAC$.

$$\triangle OAA_1 \sim \triangle OA_2A, AH \parallel A_1O \implies \angle AA_1O = \angle A_2AO = \angle A_1AH.$$

AA_2 and AA_1 are isogonals with respect to the angle $\angle BAC$.

Similarly BB_2 and BB_1 are isogonals with respect to $\angle ABC$.

Similarly CC_2 and CC_1 are isogonals with respect to $\angle ACB$.

Let $G = AA_1 \cap BB_1$ be the centroid of $\triangle ABC$, $L = AA_2 \cap BB_2$.

L is the isogonal conjugate of a point G with respect to a triangle $\triangle ABC$.

$$G \in CC_1 \implies L \in CC_2.$$

<i>Corollary</i>

If median and symmedian start from any vertex of the triangle, then the angle formed by the symmedian and the angle side has the same measure as the angle between the median and the other side of the angle.

AA_1, BB_1, CC_1 are medians, therefore AA_2, BB_2, CC_2 are symmedians, so the three symmedians meet at a point which is triangle center called the Lemoine point.

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4. (USAMO 1995) Suppose $q_0, q_1, q_2, \dots$ is an infinite sequence of integers satisfying the following two conditions:

- (i) $m - n$ divides $q_m - q_n$ for $m > n \geq 0$,
- (ii) there is a polynomial P such that $|q_n| < P(n)$ for all n .

Prove that there is a polynomial Q such that $q_n = Q(n)$ for all n .

Solution

Step 1: Suppose P has degree d . Let Q be the polynomial of degree at most d with $Q(x) = q_x$ for $0 \leq x \leq d$. Since the q_x are all integers, Q has rational coefficients, and there exists k so that kQ has integer coefficients. Then $m - n \mid kQ(m) - kQ(n)$ for all $m, n \in \mathbb{N}_0$.

Step 2: We show that Q is the desired polynomial.

Let $x > n$ be given. Now

$$kq_x \equiv kq_m \pmod{x - m} \text{ for all integers } m \in [0, d]$$

Since $kQ(x)$ satisfies these relations as well, and $kq_m = kQ(m)$,

$$kq_x \equiv kQ(x) \pmod{x - m} \text{ for all integers } m \in [0, d]$$

and hence

$$kq_x \equiv kQ(x) \pmod{\text{lcm}(x, x - 1, \dots, x - d)}. \quad (1)$$

Now

$$\begin{aligned}
 \text{lcm}(x, x-1, \dots, x-i-1) &= \text{lcm}[\text{lcm}(x, x-1, \dots, x-i), x-i-1] \\
 &= \frac{\text{lcm}(x, x-1, \dots, x-i)(x-i-1)}{\gcd[\text{lcm}(x, x-1, \dots, x-i), (x-i-1)]} \\
 &\geq \frac{\text{lcm}(x, x-1, \dots, x-i)(x-i-1)}{\gcd[x(x-1)\cdots(x-i), (x-i-1)]} \\
 &\geq \frac{\text{lcm}(x, x-1, \dots, x-i)(x-i-1)}{(i+1)!}
 \end{aligned}$$

so by induction $\text{lcm}(x, x-1, \dots, x-d) \geq \frac{x(x-1)\cdots(x-d)}{d!(d-1)!\cdots 1!}$. Since $P(x), Q(x)$ have degree d , for large enough x (say $x > L$) we have $\left|Q(x) \pm \frac{x(x-1)\cdots(x-d)}{kd!(d-1)!\cdots 1!}\right| > P(x)$. By (1) kq_x must differ by a multiple of $\text{lcm}(x, x-1, \dots, x-d)$ from $kQ(x)$; hence q_x must differ by a multiple of $\frac{x(x-1)\cdots(x-d)}{kd!(d-1)!\cdots 1!}$ from $Q(x)$, and for $x > L$ we must have $q_x = Q(x)$.

Now for any y we have $kQ(y) \equiv kQ(x) \equiv kq_x \equiv kq_y \pmod{x-y}$ for any $x > L$. Since $x-y$ can be arbitrarily large, we must have $Q(y) = q_y$, as needed.

5. (USAMO 1995) Suppose that in a certain society, each pair of persons can be classified as either *amicable* or *hostile*. We shall say that each member of an amicable pair is a *friend* of the other, and each member of a hostile pair is a *foe* of the other. Suppose that the society has n persons and q amicable pairs, and that for every set of three persons, at least one pair is hostile. Prove that there is at least one member of the society whose foes include $q(1 - 4q/n^2)$ or fewer amicable pairs.

Solution

Consider the graph with two people joined if they are friends.

For each person X , let $A(X)$ be the set of its friends and $B(X)$ the set of its foes. Note that any edge goes either: from X to $A(X)$ (type 1), from $A(X)$ to $B(X)$ (type 2) or from a point of $B(X)$ to another (type 3), but there's no edge joining two points of $A(X)$ (since they would form a triangle with X). Let the number of type 1, type 2, type 3 edges of X be x_1, x_2, x_3 respectively, so that x_1 is the degree of X and we want to show that for some X , we have $x_3 \leq q(1 - 4q/n^2)$.

Since each edge is of one of those types, we have $x_1 + x_2 + x_3 = q$. Thus

$$x_3 \leq q(1 - 4q/n^2) \iff x_3 \leq q - 4q^2/n^2 \iff q - x_1 - x_2 \leq q - 4q^2/n^2 \iff x_1 + x_2 \geq 4q^2/n^2.$$

That is, what we want is equivalent to proving that for some vertex X , the set of edges touching either X or a vertex joined to X is at least $4q^2/n^2$. Obviously now we'll sum $x_1 + x_2$ over all vertices X . In the resulting sum, an edge joining X, Y is counted once for each edge that X, Y have, that is, it is counted $D(X) + D(Y)$ times, where $D(X) = x_1$ is the degree of X . Thus each vertex X contributes to the overall sum with $D(X)$ for each edge it has, and since it has $D(X)$ edges, it contributes with $D(X)^2$. Thus the considered sum is equal to

$$\sum_{X \in G} D(X)^2 \geq \frac{(\sum_{X \in G} D(X))^2}{n} = \frac{(2q)^2}{n} = 4q^2/n.$$

(That's Cauchy.) Since we are summing over n vertices, one of the summands is at least $4q^2/n^2$ by pigeonhole, which is what we wanted to prove.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

§25 USAMO 1996

1. (*USAMO 1996*) Prove that the average of the numbers $n \sin n^\circ$ ($n = 2, 4, 6, \dots, 180$) is $\cot 1^\circ$.

Solution 1

First, as $180 \sin 180^\circ = 0$, we omit that term. Now, we multiply by $\sin 1^\circ$ to get, after using product to sum, $(\cos 1^\circ - \cos 3^\circ) + 2(\cos 3^\circ - \cos 5^\circ) + \dots + 89(\cos 177^\circ - \cos 179^\circ)$. This simplifies to $\cos 1^\circ + \cos 3^\circ + \cos 5^\circ + \cos 7^\circ + \dots + \cos 177^\circ - 89 \cos 179^\circ$. Since $\cos x = -\cos(180 - x)$, this simplifies to $90 \cos 1^\circ$. We multiplied by $\sin 1^\circ$ in the beginning, so we must divide by it now, and thus the sum is just $90 \cot 1^\circ$, so the average is $\cot 1^\circ$, as desired.

□

Solution 2

Notice that for every $n \sin n^\circ$ there exists a corresponding pair term $(180^\circ - n) \sin 180^\circ - n = (180^\circ - n) \sin n^\circ$, for n not 90° . Pairing gives the sum of all $n \sin n^\circ$ terms to be $90(\sin 2^\circ + \sin 4^\circ + \dots + \sin 178^\circ)$, and thus the average is

$$S = (\sin 2^\circ + \sin 4^\circ + \dots + \sin 178^\circ).(*)$$

We need to show that $S = \cot 1^\circ$. Multiplying $(*)$ by $2 \sin 1^\circ$ and using sum-to-product and telescoping gives $2 \sin 1^\circ S = \cos 1^\circ - \cos 179^\circ = 2 \cos 1^\circ$. Thus, $S = \frac{\cos 1^\circ}{\sin 1^\circ} = \cot 1^\circ$, as desired.

□

Solution 3 -hashtagmath

We know that the average of a list of numbers is the sum of all the terms, divided by the number of terms. So we will set up an average. This average will average to $\cot 1^\circ$. So we can set it equal to $\frac{\cos 1^\circ}{\sin 1^\circ}$. Doing so, gives us

$$\frac{2 \sin 2^\circ + 4 \sin 4^\circ + 6 \sin 6^\circ + 8 \sin 8^\circ + 10 \sin 10^\circ + 12 \sin 12^\circ \dots + 180 \sin 180^\circ}{90} = \frac{\cos 1^\circ}{\sin 1^\circ}.(1)$$

We should try to simplify the numerator as it looks pretty messy.

Now we know that $\sin x = \sin(180 - x)$, so maybe we can use this to clear things up a bit. Applying this to (1) gives us

$$\frac{2 \sin 178^\circ + 4 \sin 176^\circ + 6 \sin 174^\circ + 8 \sin 172^\circ + 10 \sin 170^\circ + 12 \sin 168^\circ \dots + 180 \sin 0^\circ}{90} = \frac{\cos 1^\circ}{\sin 1^\circ}.(2)$$

Now, we know that $\sin 2^\circ = \sin 178^\circ$, and $\sin 4^\circ = \sin 176^\circ$ and so on. We also know that $\sin 0^\circ = 0$, so we can omit the last term. Also, we know that $\sin 90 = 1$. Thus $90 \sin 90 = 90$. So we will use all these when we further simplify. Another thing that we can simplify is to notice that when we reach the term $92 \sin 88^\circ$, we can rewrite it as $92 \sin 92^\circ$. Thus we can rewrite everything in the form $x \sin y^\circ$ as $x \sin x^\circ$, starting with the term $92 \sin 88^\circ$, then $94 \sin 86^\circ$, and so on.

Thus we can further simplify our equation. Doing so, gives us

$$\frac{2 \sin 178^\circ + 4 \sin 176^\circ + 6 \sin 174^\circ + 8 \sin 172^\circ + 10 \sin 170^\circ + 12 \sin 168^\circ \dots + 176 \sin 176^\circ + 178 \sin 178^\circ}{90}$$

Then, to make things easier, we can rewrite all the numbers with degrees less than 90° and combine like terms. Doing so gives us

$$\frac{180 \sin 2^\circ + 180 \sin 4^\circ + 180 \sin 6^\circ + 180 \sin 8^\circ + 180 \sin 10^\circ \dots + 180 \sin 88^\circ + (90)}{90} = \frac{\cos 1^\circ}{\sin 1^\circ}. (4)$$

Now we can factor out 180 to get

$$\frac{180(\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \sin 8^\circ + \sin 10^\circ \dots + \sin 88^\circ) + 90}{90} = \frac{\cos 1^\circ}{\sin 1^\circ}. (5)$$

Then we can simplify it by getting rid of the denominator to get to

$$2(\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \sin 8^\circ + \sin 10^\circ \dots + \sin 88^\circ) + 1 = \frac{\cos 1^\circ}{\sin 1^\circ}. (6)$$

Now we notice that we can multiply both sides by $\sin 1^\circ$. Doing so gives us

$$\sin 1^\circ(2(\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \sin 8^\circ + \sin 10^\circ \dots + \sin 88^\circ) + 1) = \cos 1^\circ. (7)$$

Now, simplifying $\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \sin 8^\circ + \sin 10^\circ \dots + \sin 88^\circ$ doesn't look too promising. So maybe if we expand again, we can maybe somehow use our product to sum formulas.

Doing so, gives us

$$(\sin 1^\circ)(2 \sin 2^\circ) + (\sin 1^\circ)(2 \sin 4^\circ) + (\sin 1^\circ)(2 \sin 6^\circ) + (\sin 1^\circ)(2 \sin 8^\circ) + (\sin 1^\circ)(2 \sin 10^\circ) \dots + (\sin 1^\circ)(2 \sin 88^\circ)$$

Now we can use our product of sines to sum formula and see if we can find a pattern.

We will start with expanding $(\sin 1^\circ)(2 \sin 2^\circ)$. Because the format of our formula for the product of sines is $\sin \alpha + \sin \beta$, we can factor out the 2 and find the product, then multiply by 2. So, we are at $2(\sin 1^\circ)(\sin 2^\circ)$. Our formula for the product of sines is $\sin \alpha + \sin \beta = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$. Plugging in our values into the formula and simplifying gives us $2\left(\frac{1}{2}(\cos 1^\circ - \cos 3^\circ)\right)$. We know that $2\left(\frac{1}{2}\right)$ will just cancel out to 1. So we are left with $\cos 1^\circ - \cos 3^\circ$. (9)

Next we expand $(\sin 1^\circ)(2 \sin 4^\circ)$. We use the same steps or strategy as we did above and get $\cos 3^\circ - \cos 5^\circ$. (10)

We may be noticing a pattern. Just to make sure it is true, we will expand $(\sin 1^\circ)(2 \sin 6^\circ)$. When we expand this using the same strategy above, we get $\cos 5^\circ - \cos 7^\circ$. (11) So our pattern is that $(\sin 1^\circ)(2 \sin \mu^\circ) = \cos(\mu - 1)^\circ - \cos(\mu + 1)^\circ$. (12)

Now we notice that when we add (9) and (10), $-\cos 3^\circ$ and $+\cos 3^\circ$ cancel themselves out to just $\cos 1^\circ - \cos 5^\circ$. Then, when we add (11) to it, $+\cos 5^\circ$ and $-\cos 5^\circ$ also cancel themselves out.

Now, we need to figure out when we should stop canceling out. We can use formula (12) and replace 88 with μ since 88 is the last one in (8). Applying this formula and simplifying gives us $\cos 87^\circ - \cos 89^\circ$, so the last term would be $-\cos 89^\circ$.

Thus we know that everything we cancel themselves out except the first and the last term. So, we are left with

$$(\cos 1^\circ - \cos 89^\circ) + \sin 1^\circ = \cos 1^\circ. (13)$$

Now we notice that we have $\cos 1^\circ$ on both sides. Thus we can subtract $\cos 1^\circ$ from both sides to get $-\cos 89^\circ + \sin 1^\circ = 0$. Now we can add $\cos 89^\circ$ to both sides and get $\sin 1^\circ = \cos 89^\circ$. Now we also know that $\sin(\omega)^\circ = \cos(90 - \omega)^\circ$. Thus we know that $\sin 1^\circ = \cos 89^\circ$. Hence, our proof is complete. $\square$

2. (USAMO 1996) For any nonempty set S of real numbers, let $\sigma(S)$ denote the sum of the elements of S . Given a set A of n positive integers, consider the collection of all distinct sums $\sigma(S)$ as S ranges over the nonempty subsets of A . Prove that this collection of sums can be partitioned into n classes so that in each class, the ratio of the largest sum to the smallest sum does not exceed 2.

Solution 1

Let set A consist of the integers $a_1 \leq a_2 \leq a_3 \leq \dots \leq a_n$. For $k \geq 2$, call a_k greedy if $\sum_{i=1}^{k-1} a_i < a_k$. Also call a_1 greedy. Now put all elements of A into groups of consecutive terms in such a way that each group G begins with a greedy term, call it a_p , and ends on the term a_{q-1} just before the next greedy term after a_p . (If a_p is the last greedy term, let $q-1 = n$.) We introduce some more terminology. A sum $\sigma(S)$ is said to "belong to" a group G if $\max(S) \in G$. Denote by $\mathcal{S}(G)$ the set of all sums belonging to G .

We now show that we can divide $\mathcal{S}(G)$ into $|G|$ (the cardinality of G) classes in such a way that the maximum and minimum sum belonging to each class differ by no more than a factor of 2. Using the previous notation, we first prove that $\frac{\max \mathcal{S}(G)}{\min \mathcal{S}(G)} < 2^{|G|} = 2^{q-p}$. Note that

$$\max \mathcal{S}(G) = \sum_{i=1}^{q-1} a_i$$

and

$$\min \mathcal{S}(G) = a_p.$$

Taking note of that fact that $a_{p+1}, a_{p+2}, \dots, a_{q-1}$ are not greedy numbers, we write:

$$\begin{aligned} \max \mathcal{S} &= \sum_{i=1}^{q-1} a_i \\ &= a_{q-1} + \sum_{i=1}^{q-2} a_i \\ &\leq 2 \sum_{i=1}^{q-2} a_i \\ &\leq 2^2 \sum_{i=1}^{q-3} a_i \\ &\vdots \\ &\leq 2^{q-p-1} \sum_{i=1}^p a_i \end{aligned}$$

$$\begin{aligned}
&= 2^{q-p-1} \left(\sum_{i=1}^{p-1} a_i + a_p \right) \\
&< 2^{q-p-1} (2a_p) \\
&= 2^{q-p} \min \mathcal{S}
\end{aligned}$$

where the inequalities after the ellipses result from the fact that a_p is a greedy number (which implies by definition that $\sum_{i=1}^{p-1} a_i < a_p$). This proves the desired inequality. Now we can prove the result in bold above. Divide $\mathcal{S}(G)$ into $|G| = q - p$ classes by taking those terms in $[a_p, 2a_p)$ and placing them in the first class, taking those terms in $[2a_p, 2^2a_p)$ and placing them in another, and so on, until we reach $[2^{q-p-1}a_p, 2^{q-p}a_p)$. The inequality we proved above shows that all of the sums in $\mathcal{S}(G)$ will fall in one of these classes, as the intervals into which the classes fall form a continuous range bounded by $\min \mathcal{S}(G) = a_p$ on the bottom and $2^{q-p}a_p > \max \mathcal{S}(G)$ on the top. This proves the result in bold.

However, that clearly implies the desired conclusion, as every sum belongs to a group, and every sum belonging to a group is a member of a class. Moreover, there will be precisely n classes, because every term a_i belongs to a group and for each group G , there are as many classes for $\mathcal{S}(G)$ as there are terms in G .

Solution 2

We proceed to solve this problem by induction. Our base case is when A has 1 integer - there is 1 class: itself.

Let A now have n elements, and assume it works for this case. We add k , another element, to the set. WLOG, let k be the maximum element. Note that $\sigma(A)$ has to be the maximum element of a class.

We always get a new class that works: $\sigma(A + \{k\})$ being the maximum element. This way all elements including k will be in that class if $k > \sigma(A)$. When $k < \sigma(A)$ it also works since k must be the maximum element from our WLOG condition above, so our induction step is done.

3. (USAMO 1996) Let ABC be a triangle. Prove that there is a line ℓ (in the plane of triangle ABC) such that the intersection of the interior of triangle ABC and the interior of its reflection $A'B'C'$ in ℓ has area more than $\frac{2}{3}$ the area of triangle ABC .

Solution

Let the triangle be ABC . Assume A is the largest angle. Let AD be the altitude. Assume $AB \leq AC$, so that $BD \leq BC/2$. If $BD > \frac{BC}{3}$, then reflect in AD . If B' is the reflection of B , then $B'D = BD$ and the intersection of the two triangles is just ABB' . But $BB' = 2BD > \frac{2}{3}BC$, so ABB' has more than $\frac{2}{3}$ the area of ABC .

If $BD < BC/3$, then reflect in the angle bisector of C . The reflection of A' is a point on the segment BD and not D . (It lies on the line BC because we are reflecting in the angle bisector. $A'C > DC$ because $\angle CAD < \angle CDA = 90^\circ$. Finally, $A'C \leq BC$ because we assumed $\angle B$ does not exceed $\angle A$). The intersection is at least $AA'C$. But $\frac{[AA'C]}{[ABC]} = \frac{CA'}{CB} > CD/CB \geq 2/3$.

4. (USAMO 1996) An n -term sequence $(x_1, x_2, \dots, x_n)$ in which each term is either 0 or 1 is called a *binary sequence of length n* . Let a_n be the number of binary sequences of length n containing no three consecutive terms equal to 0, 1, 0 in

that order. Let b_n be the number of binary sequences of length n that contain no four consecutive terms equal to 0, 0, 1, 1 or 1, 1, 0, 0 in that order. Prove that $b_{n+1} = 2a_n$ for all positive integers n .

Solution

Given any binary sequence $B = (b_1, b_2, b_3, \dots, b_k)$, define $f(B) = (|b_2 - b_1|, |b_3 - b_2|, \dots, |b_k - b_{k-1}|)$. The operator f basically takes pairs of consecutive terms and returns 0 if the terms are the same and 1 otherwise. Note that for every sequence S of length n there exist exactly two binary sequences B of length $n + 1$ such that $f(B) = S$.

If $f(B)$ does not contain the string 0, 1, 0, B cannot contain either of the strings 0, 0, 1, 1 or 1, 1, 0, 0. Conversely, if B does not contain the sequences 0, 0, 1, 1 or 1, 1, 0, 0, $f(B)$ cannot contain 0, 1, 0. There are a_n such $f(B)$ and b_{n+1} such B . Since each S corresponds with two B , there are twice as many such B as such S ; thus, $b_{n+1} = 2a_n$.

5. (USAMO 1996) Let ABC be a triangle, and M an interior point such that $\angle MAB = 10^\circ$, $\angle MBA = 20^\circ$, $\angle MAC = 40^\circ$ and $\angle MCA = 30^\circ$. Prove that the triangle is isosceles.

Solution 1

Clearly, $\angle AMB = 150^\circ$ and $\angle AMC = 110^\circ$. Now by the Law of Sines on triangles ABM and ACM , we have

$$\frac{AB}{\sin 150^\circ} = \frac{AM}{\sin 20^\circ}$$

and

$$\frac{AC}{\sin 110^\circ} = \frac{AM}{\sin 30^\circ}.$$

Combining these equations gives us

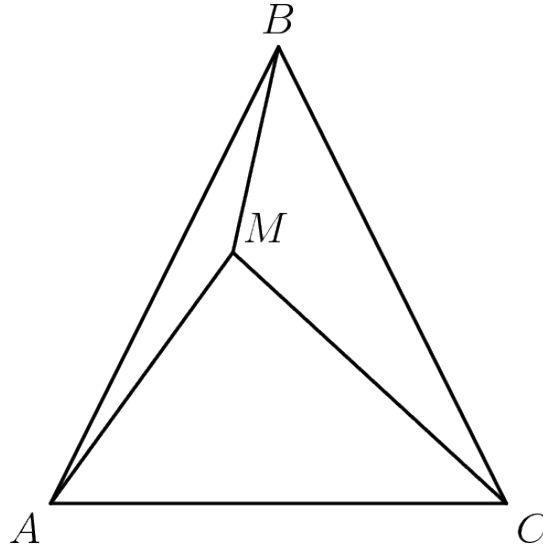
$$\frac{AB}{AC} = \frac{\sin 150^\circ \sin 30^\circ}{\sin 20^\circ \sin 110^\circ}.$$

Without loss of generality, let $AB = \sin 150^\circ \sin 30^\circ = \frac{1}{4}$ and $AC = \sin 20^\circ \sin 110^\circ$. Then by the Law of Cosines, we have

$$\begin{aligned} BC^2 &= AB^2 + AC^2 - 2(AB)(AC) \cos \angle BAC \\ &= \frac{1}{16} + \sin^2 20^\circ \sin^2 110^\circ - 2 \left(\frac{1}{4} \right) \sin 20^\circ \sin 110^\circ \cos 50^\circ \\ &= \frac{1}{16} + \sin^2 20^\circ \sin^2 110^\circ - \frac{1}{2} \sin 20^\circ \sin 110^\circ \sin 40^\circ \\ &= \frac{1}{16} + \sin^2 20^\circ \sin^2 110^\circ - \sin 20^\circ \sin 110^\circ \sin 20^\circ \cos 20^\circ \\ &= \frac{1}{16} \end{aligned}$$

Thus, $AB = BC$, our desired conclusion.

Solution 2



By the law of sines, $\frac{BM}{\sin(10^\circ)} = \frac{AM}{\sin(20^\circ)}$ and $\frac{CM}{\sin(40^\circ)} = \frac{AM}{\sin(30^\circ)}$, so $\frac{BM}{CM} = \frac{\sin(10^\circ)\sin(30^\circ)}{\sin(20^\circ)\sin(40^\circ)}$.

Let $\angle MBC = x$. Then, $\angle MCB = 80^\circ - x$. By the law of sines, $\frac{BM}{CM} = \frac{\sin(80^\circ - x)}{\sin(x)}$.

Combining, we have $\frac{\sin(80^\circ - x)}{\sin(x)} = \frac{\sin(10^\circ)\sin(30^\circ)}{\sin(20^\circ)\sin(40^\circ)}$. From here, we can use the given trigonometric identities at each step:

$$\begin{aligned}
 \frac{\sin(80^\circ - x)}{\sin(x)} &= \frac{\sin(10^\circ)\sin(30^\circ)}{\sin(20^\circ)\sin(40^\circ)} \\
 \sin(80^\circ - x)\sin(20^\circ)\sin(40^\circ) &= \sin(10^\circ)\sin(30^\circ)\sin(x) \\
 \sin(80^\circ - x)\sin(20^\circ)\sin(40^\circ) &= \frac{1}{2}\sin(10^\circ)\sin(x) \\
 \sin(80^\circ - x)\sin(30^\circ - 10^\circ)\sin(30^\circ + 10^\circ) &= \frac{1}{2}\sin(10^\circ)\sin(x) \\
 \sin(80^\circ - x)(\cos^2(10^\circ) - \cos^2(30^\circ)) &= \frac{1}{2}\sin(10^\circ)\sin(x) & [\sin(A - B)\sin(A + B) = \sin^2 A - \sin^2 B] \\
 \sin(80^\circ - x)(\cos^2(10^\circ) - \frac{3}{4}) &= \frac{1}{2}\sin(10^\circ)\sin(x) \\
 \sin(80^\circ - x)\frac{4\cos^3(10^\circ) - 3\cos(10^\circ)}{4\cos(10^\circ)} &= \frac{1}{2}\sin(10^\circ)\sin(x) \\
 \sin(80^\circ - x)\frac{\cos(30^\circ)}{4\cos(10^\circ)} &= \frac{1}{2}\sin(10^\circ)\sin(x) & [\cos(3A) = 4\cos^3 A - 3\cos A] \\
 \sin(80^\circ - x)\cos(30^\circ) &= 2\sin(10^\circ)\cos(10^\circ)\sin(x) \\
 \sin(80^\circ - x)\cos(30^\circ) &= \sin(20^\circ)\sin(x) & [2\sin A \cos A = \sin(2A)] \\
 \sin(80^\circ - x)\sin(60^\circ) &= \sin(20^\circ)\sin(x) & [\sin A \cos B = \frac{1}{2}(\sin(A+B) - \sin(A-B))] \\
 \frac{1}{2}(\cos(20^\circ - x) - \cos(140^\circ - x)) &= \frac{1}{2}(\cos(20^\circ - x) - \cos(20^\circ + x)) & [\sin A \sin B = \frac{1}{2}(\cos(A-B) - \cos(A+B))] \\
 \cos(140^\circ - x) &= \cos(20^\circ + x)
 \end{aligned}$$

The only acute angle satisfying this equality is $x = 60^\circ$. Therefore, $\angle ACB = 80^\circ - x + 30^\circ = 50^\circ$ and $\angle BAC = 10^\circ + 40^\circ = 50^\circ$. Thus, $\triangle ABC$ is isosceles.

Solution 3

If $\angle MBC = x$ then by Angle Sum in a Triangle we have $\angle MCB = 80^\circ - x$. By Trig Ceva we have

$$\sin 10^\circ \sin x \sin 30^\circ = \sin(80^\circ - x) \sin 40^\circ \sin 20^\circ.$$

Because $\frac{\sin x}{\sin(80^\circ - x)}$ is monotonic increasing over $(0, \frac{\pi}{2})$, there is only one solution $0 \leq x \leq \frac{\pi}{2}$ to the equation. We claim it is $x = 60^\circ$, which will make ABC isosceles with $\angle A = \angle C$.

Notice that

$$\begin{aligned} \sin 20^\circ \sin 20^\circ \sin 40^\circ &= 2 \sin 10^\circ \cos 10^\circ \sin 20^\circ \sin 40^\circ \\ &= \sin 10^\circ \left(\sin 10^\circ + \frac{1}{2} \right) \sin 40^\circ \\ &= \sin 10^\circ \left(\frac{1}{2} \sin 40^\circ + \frac{1}{2} (\cos 30^\circ - \cos 50^\circ) \right) \\ &= \sin 10^\circ \frac{1}{2} \cos 30^\circ \\ &= \sin 10^\circ \sin 30^\circ \sin 60^\circ, \end{aligned}$$

as desired.

6. (*USAMO 1996*) Determine (with proof) whether there is a subset X of the integers with the following property: for any integer n there is exactly one solution of $a + 2b = n$ with $a, b \in X$.

Solution

Start with an incomplete subset $S = (S_1, S_2, S_3, \dots, S_m)$, such that for any integer n , there is exactly zero or one solutions to $a + 2b = n$ with $a, b \in S$. Let N be the smallest integer such that for any S_i , $|S_i| < N$. Note that $|S_i + 2S_j| < 3N$ for any S_i and S_j

Suppose M is the smallest non-negative integer without a solution in S yet. Clearly, $0 \leq M \leq 3N$. Generate S_{m+1} and S_{m+2} such that $S_{m+1} = -10N - M$, and $S_{m+2} = 5N + M$. Thus, we now have the solution $S_{m+1} + 2S_{m+2} = M$.

Note: The values 10 and 5 can be replaced by any sufficiently large values such that the first is twice the second.

Now, we must prove that the addition of these two terms to S does not result in an integer n that has two solutions. Of course, $S_{m+1} + 2S_{m+2} = M$ which previously had no solutions. Furthermore, $S_{m+1} + 2S_{m+2} = -15N - M$.

- (i) For any S_i , $-12N - M < S_{m+1} + 2S_i < -8N - M$, and $-21N - 2M < S_i + 2S_{m+1} < -19N - 2M$.

<dd>

- (i) Since $0 \leq M \leq 3N$, we get that $-18N < S_{m+1} + 2S_i < -8N$ and $-27N < S_i + 2S_{m+1} < -19N$

</dd>

- (i) Similarly, $3N + M < S_{m+2} + 2S_i < 5N + M$, and $9N + 2M < S_i + 2S_{m+2} < 11N + 2M$.

<dd>

- (i) Since $0 \leq M \leq 3N$, we get that $3N < S_{m+2} + 2S_i < 8N$ and $9N < S_i + 2S_{m+2} < 17N$.

</dd>

Since all of these sums (other than M) are either greater than $3N$ or less than $-3N$, they are all sums that previously had no solutions. Furthermore, none of these sums are duplicated, as sums of different forms are contained in disjoint ranges of integers.

Thus, we have proved that we can generate a subset S such that all non-negative integers n have a unique solution $a + 2b = n$.

For negative integers M that have no solutions in S a similar proof holds, but instead generating the terms $S_{m+1} = 10N - M$ and $S_{m+2} = -5N + M$.

For any integer M that currently has no solution in S , we can always add two terms S_{m+1} and S_{m+2} such that $S_{m+1} + 2S_{m+2} = M$ that do not result in duplicated sums.

Thus, there does exist a subset X of the integers such that for any integer n there is exactly one solution to $a + 2b = n$ with $a, b \in X$.

§26 USAMO 1997

1. (*USAMO 1997*) Let $p_1, p_2, p_3, \dots$ be the prime numbers listed in increasing order, and let x_0 be a real number between 0 and 1. For positive integer k , define

$$x_k = \begin{cases} 0 & \text{if } x_{k-1} = 0 \\ \left\{ \frac{p_k}{x_{k-1}} \right\} & \text{if } x_{k-1} \neq 0 \end{cases}$$

where $\{x\}$ denotes the fractional part of x . (The fractional part of x is given by $x - [x]$ where $[x]$ is the greatest integer less than or equal to x .) Find, with proof, all x_0 satisfying $0 < x_0 < 1$ for which the sequence $x_0, x_1, x_2, \dots$ eventually becomes 0.

Solution

All rational numbers between 0 and 1 inclusive will eventually yield some $x_k = 0$. To begin, note that by definition, all rational numbers can be written as a quotient of coprime integers. Let $x_0 = \frac{m}{n}$, where m, n are coprime positive integers. Since $0 < x_0 < 1$, $0 < m < n$. Now

$$x_1 = \left\{ \frac{p_1}{\frac{m}{n}} \right\} = \left\{ \frac{np_1}{m} \right\}.$$

From this, we can see that applying the iterative process will decrease the value of the denominator, since $m < n$. Moreover, the numerator is always smaller than the denominator, thanks to the fractional part operator. So we have a strictly decreasing denominator that bounds the numerator. Thus, the numerator will eventually become 0.

On the other hand, if x_0 is irrational, then a simple induction will show that x_k will always be irrational. Indeed, the base case has been established, and, if x_k is irrational, then $\frac{p_{k+1}}{x_k}$ must be too, and likewise for its fractional portion, which differs from it by an integer. Hence x_{k+1} is irrational, completing the proof.

2. (USAMO 1997) Let ABC be a triangle, and draw isosceles triangles BCD, CAE, ABF externally to ABC , with BC, CA, AB as their respective bases. Prove that the lines through A, B, C perpendicular to the lines $\overleftrightarrow{EF}, \overleftrightarrow{FD}, \overleftrightarrow{DE}$, respectively, are concurrent.

Solution 1

Let the perpendicular from A meet FE at A' . Define B' and C' similarly. By **Carnot's Theorem**, The three lines are **concurrent** if

$$FA'^2 - EA'^2 + EC'^2 - DC'^2 + DB'^2 - FB'^2 = AF^2 - AE^2 + CE^2 - CD^2 + BD^2 - BF^2 = 0$$

But this is clearly true, since D lies on the perpendicular bisector of BC , $BD = DC$.

QED

Solution 2

We split this into two cases:

Case 1: D, E, F are non-collinear

Observe that since D, E, F lie on perpendicular bisectors, then we get that $DC = DB$, $EC = EA$, and $FA = FB$. This motivates us to construct a circle ω_1 centered at D with radius DC , and similarly construct ω_2 and ω_3 respectively for E and F .

Now, clearly ω_1 and ω_2 intersect at C and some other point. Now, we know that DE is the line containing the two centers. So, the line perpendicular to DE and passing through C must be the radical axis of ω_1 and ω_2 , which is exactly the line that the problem describes! We do this similarly for the others pairs of circles.

Now by the radical lemma, the pairwise radical axes of $\omega_1, \omega_2, \omega_3$ are concurrent, as desired, and they intersect at the radical center.

Case 2: D, E, F are collinear

Now, we are drawing perpendicular lines from A, B , and C onto the single line DEF . Clearly, these lines are parallel and concur at the point of infinity.

Solution 3

Notice that DEF and ABC are orthologic, so the perpendiculars concur at the opposite center of orthology.

3. (USAMO 1997) Prove that for any integer n , there exists a unique polynomial Q with coefficients in $\{0, 1, \dots, 9\}$ such that $Q(-2) = Q(-5) = n$.

Solution

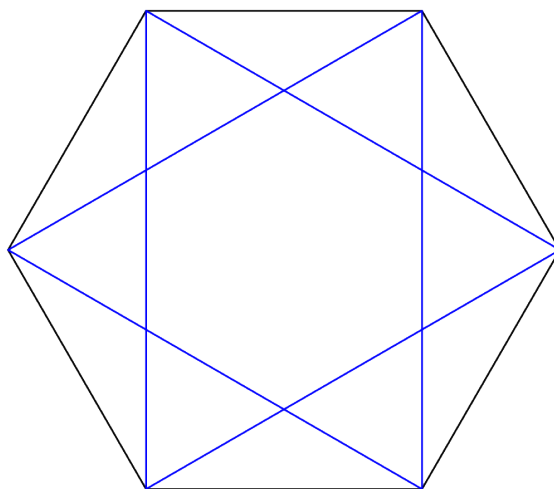
The problem implies that $Q(x) = (x+2)(x+5) \cdot H_y(x) + n$. We define $H_y(x) = a_0 + a_1x + \dots + a_nx^n$. Then, after expanding, the coefficient of x^k is $10a_k + 7a_{k-1} + a_{k-2}$ where $2 \leq k \leq n$. The constant term is $10a_0 + n$. Since this has to be within the set $0, 1, \dots, 9$, a_0 must be unique according to n . The linear term is determined by $10a_1 + 7a_0$, which also must be within the set $0, 1, \dots, 9$. Since a_0 is determined uniquely by n , in order to keep the linear coefficient in the set $0, 1, \dots, 9$, a_1 is also determined uniquely. With a_0 and a_1 determined uniquely, we move on to the general coefficient.

Note that $10a_k + 7a_{k-1} + a_{k-2}$ is in the set $0, 1, \dots, 9$. We define $J_h(x)$ to be a function determining the remainder when x is divided by 10. Thus, for all x , $0 \leq J_h(x) \leq 9$. Note also that $J_h(10a_k + 7a_{k-1} + a_{k-2}) = 10a_k + 7a_{k-1} + a_{k-2}$, since $10a_k + 7a_{k-1} + a_{k-2}$ is in the range of $J_h(x)$. Therefore, $J_h(7a_{k-1} + a_{k-2}) =$

$10a_k + 7a_{k-1} + a_{k-2}$, which results in $a_k = \lfloor \frac{7a_{k-1} + a_{k-2}}{10} \rfloor$ for all $2 \leq k \leq n$. Thus, a_k is determined uniquely, since all coefficients a_i where $0 \leq i < k$ are uniquely determined, which in turn determines a unique polynomial $Q(x)$.

4. (USAMO 1997) To *clip* a convex n -gon means to choose a pair of consecutive sides AB, BC and to replace them by three segments AM, MN , and NC , where M is the midpoint of AB and N is the midpoint of BC . In other words, one cuts off the triangle MBN to obtain a convex $(n+1)$ -gon. A regular hexagon P_6 of area 1 is clipped to obtain a heptagon P_7 . Then P_7 is clipped (in one of the seven possible ways) to obtain an octagon P_8 , and so on. Prove that no matter how the clippings are done, the area of P_n is greater than $\frac{1}{3}$, for all $n \geq 6$.

Solution



Claim: It is impossible to choose two non-adjacent sides and clip a whole part of it off.

Proof: If you clip adjacent sides, you can cut off at most up to the blue lines; Clipping more is impossible due to the degrees getting larger and larger and more and more circular.

Thus, after infinitely many clips, the hexagon bounded by the blue lines is left, so after finitely many clips, the area left is more than that hexagon.

The side length of that hexagon is $\frac{\sqrt{3}}{3}$ of the large one, because of 30-30-120 triangles. Thus, the area is $\frac{1}{3}$ of the larger one, so we are done.

5. (USAMO 1997) Prove that, for all positive real numbers a, b, c ,

$$\frac{1}{a^3+b^3+abc} + \frac{1}{b^3+c^3+abc} + \frac{1}{a^3+c^3+abc} \leq \frac{1}{abc}.$$

Solution 1

Because the inequality is homogenous (i.e. (a, b, c) can be replaced with (ka, kb, kc) without changing the inequality other than by a factor of k^n for some n), without loss of generality, let $abc = 1$.

Lemma:

$$\frac{1}{a^3 + b^3 + 1} \leq \frac{c}{a + b + c}.$$

Proof: Rearranging gives $(a^3 + b^3)c + c \geq a + b + c$, which is a simple consequence of $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ and

$$(a^2 - ab + b^2)c \geq (2ab - ab)c = abc = 1.$$

Thus, by $abc = 1$:

$$\begin{aligned} & \frac{1}{a^3 + b^3 + abc} + \frac{1}{b^3 + c^3 + abc} + \frac{1}{c^3 + a^3 + abc} \\ & \leq \frac{c}{a + b + c} + \frac{a}{a + b + c} + \frac{b}{a + b + c} = 1 = \frac{1}{abc}. \end{aligned}$$

Solution 2

Rearranging the AM-HM inequality, we get $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \leq \frac{9}{x+y+z}$. Letting $x = a^3 + b^3 + abc$, $y = b^3 + c^3 + abc$, and $z = c^3 + a^3 + abc$, we get

$$\frac{1}{a^3 + b^3 + abc} + \frac{1}{b^3 + c^3 + abc} + \frac{1}{c^3 + a^3 + abc} \leq \frac{9}{2a^3 + 2b^3 + 2c^3 + 3abc}.$$

By AM-GM on a^3 , b^3 , and c^3 , we have

$$a^3 + b^3 + c^3 \geq 3abc \Rightarrow 2a^3 + 2b^3 + 2c^3 + 3abc \geq 9abc \Rightarrow \frac{9}{2a^3 + 2b^3 + 2c^3 + 3abc} \leq \frac{1}{abc}.$$

So, $\frac{1}{a^3 + b^3 + abc} + \frac{1}{b^3 + c^3 + abc} + \frac{1}{c^3 + a^3 + abc} \leq \frac{1}{abc}$. -Tigerzhang

WARNING:

This solution doesn't work because $(x+y+z) \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right) \geq 9$, so $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq \frac{9}{x+y+z}$

Solution 3

If we multiply each side by abc , we get that we must just prove that

$$\frac{abc}{a^3 + b^3 + abc} + \frac{abc}{b^3 + c^3 + abc} + \frac{abc}{c^3 + a^3 + abc} \leq 1$$

If we divide our LHS equation, we get that

$$\begin{aligned} & \frac{1}{\frac{a^3 + b^3 + abc}{abc}} + \frac{1}{\frac{b^3 + c^3 + abc}{abc}} + \frac{1}{\frac{c^3 + a^3 + abc}{abc}} \\ & = \frac{1}{\frac{a^2}{bc} + \frac{b^2}{ac} + 1} + \frac{1}{\frac{b^2}{ac} + \frac{c^2}{ab} + 1} + \frac{1}{\frac{c^2}{ab} + \frac{a^2}{bc} + 1} \end{aligned}$$

Make the astute observation that by Titu's Lemma,

$$\begin{aligned} & \frac{a^2}{bc} + \frac{b^2}{ac} \geq \frac{(a+b)^2}{bc+ac} \\ & \sum_{cyc} \frac{(a+b)^2}{bc+ac} \leq \sum_{cyc} \frac{a^2}{bc} + \frac{b^2}{ac} \end{aligned}$$

Therefore:

$$\sum_{cyc} \frac{1}{\frac{(a+b)^2}{c(a+b)} + 1} \geq \sum_{cyc} \frac{1}{\frac{a^2}{bc} + \frac{b^2}{ac} + 1}$$

If we expand it out, we get that

$$\sum_{cyc} \frac{1}{\frac{(a+b)^2}{c(a+b)} + 1} = \frac{a+b+c}{a+b+c} = 1$$

Since our original equation is less than this, we get that

$$\sum_{cyc} \frac{1}{\frac{a^2}{bc} + \frac{b^2}{ac} + 1} \leq 1$$

$$\sum_{cyc} \frac{abc}{a^3 + b^3 + abc} \leq 1$$

$$\sum_{cyc} \frac{1}{a^3 + b^3 + abc} \leq \frac{1}{abc}$$

-KEVIN_LIU

6. (USAMO 1997) Suppose the sequence of nonnegative integers $a_1, a_2, \dots, a_{1997}$ satisfies

$$a_i + a_j \leq a_{i+j} \leq a_i + a_j + 1$$

for all $i, j \geq 1$ with $i + j \leq 1997$. Show that there exists a real number x such that $a_n = \lfloor nx \rfloor$ (the greatest integer $\leq x$) for all $1 \leq n \leq 1997$.

Solution

Given nonnegative integers $a_1, a_2, \dots, a_{1997}$ satisfying the given inequalities, let I_n be the set of all x such that $a_n = \lfloor nx \rfloor$. Therefore,

$$I_n = \{x : a_n \leq nx < a_n + 1\}.$$

We can rewrite this as

$$I_n = \left[\frac{a_n}{n}, \frac{a_n + 1}{n} \right). \quad (1)$$

So we know that each I_n is an interval. We start by proving that if n and k are positive integers, then $I_n \cap I_k \neq \emptyset$. To prove this, we need the following lemma.

Lemma 1: If n and k are positive integers, then

$$na_k < k(a_n + 1) \quad (2)$$

Proof: We prove this by induction. If $k = 1$, then we wish to show that $na_1 < a_n + 1$. By the given lower bound, we know that

$$a_n \geq a_{n-1} + a_1 \geq (a_{n-2} + a_1) + a_1 \geq \dots \geq na_1.$$

Therefore, $na_1 \leq a_n < a_n + 1$, so the hypothesis is true for $k = 1$. If $n = 1$, then we wish to prove that $a_k < k(a_1 + 1)$. By the given upper bound, we know that

$$a_k \leq a_{k-1} + (a_1 + 1) \leq (a_{k-2} + (a_1 + 1)) + (a_1 + 1) \leq \dots \leq ka_1 + (k - 1).$$

Therefore, $a_k < ka_1 + k = k(a_1 + 1)$, so the hypothesis is true for $n = 1$.

Now suppose that (2) holds for all $k, n < j$. If $k = n = j$, then (2) is equivalent to $0 < j$, which is obviously true. If $n = j > k$, then by the division algorithm, we can write $n = qk + r$ for nonnegative integers q and r with $0 \leq r < k$. Thus by applying the lower bound repeatedly (with one of the indices equal to 1), we find

$$k(a_n + 1) \geq k(qa_k + a_r + 1) = kqa_k + k(a_r + 1). \quad (3)$$

But then as $k, r < j$, we can apply the inductive hypothesis with $n = r$ and $k = k$ to find $k(a_r + 1) > ra_k$. Substituting this into (3), we find

$$k(a_n + 1) > kqa_k + ra_k = (kq + r)a_k = na_k.$$

This proves the hypothesis for $n = j > k$.

Suppose that $k = j > n$. Then by the division algorithm, we can write $k = qn + r$ for nonnegative integers q and r with $0 \leq r < n$. Then by applying the upper bound repeatedly (with one of the indices equal to 1), we find

$$na_k \leq n(qa_n + a_r + q) = nqa_n + na_r + nq. \quad (4)$$

But as $n, r < j$, we can apply the inductive hypothesis with $n = n$ and $k = r$, finding that $na_r < r(a_n + 1)$. Substituting this into (4), we find

$$na_k < nqa_n + r(a_n + 1) + nq = (nq + r)(a_n + 1) = k(a_n + 1).$$

This proves the hypothesis for $k = j > n$.

Therefore, if the hypothesis is true for all $k, n < j$, then it must be true for all $k, n \leq j$. Hence by induction, it must be true for all positive integers k and n . $\square$

Now suppose for the sake of contradiction that I_n and I_k are disjoint intervals. Without loss of generality, we may assume that I_n precedes I_k on the number line. Hence the upper bound of I_n is less than or equal to the minimum value of I_k , or rather

$$\frac{a_n + 1}{n} \leq \frac{a_k}{k}.$$

This simplifies to $na_k \geq k(a_n + 1)$. But this contradicts the statement of Lemma 1. Therefore, $I_k \cap I_n \neq \emptyset$.

Now we claim that $I_1 \cap I_2 \cap \cdots \cap I_{1997} \neq \emptyset$. We prove this by induction. By the above, we know that $I_1 \cap I_2 \neq \emptyset$. Now suppose that $I_1 \cap I_2 \cap \cdots \cap I_k \neq \emptyset$. The intersection of two overlapping intervals of the form $[a, b)$ and $[c, d)$ is an interval of the form $[e, f)$, where $e = \max\{a, c\}$ and $f = \min\{b, d\}$. Therefore, by induction, we know that if the intersection of k overlapping intervals is nonempty, then it must also be an interval, say

$$I_1 \cap I_2 \cap \cdots \cap I_k = [x_k, y_k).$$

If I_{k+1} does not intersect $[x_k, y_k)$, then as an interval, it must appear either completely before x_k or completely after y_k . If I_{k+1} appears completely before x_k , then it has a nonempty intersection with each of $I_1, I_2, \dots, I_k$. But we also know that x_k is a lower bound of one of the intervals, hence I_{k+1} cannot intersect that interval, a contradiction. A similar contradiction arises if I_{k+1} appears completely after y_k . Therefore,

$$I_1 \cap I_2 \cap \cdots \cap I_{k+1} \neq \emptyset.$$

Thus by induction, $I_1 \cap I_2 \cap \cdots \cap I_{1997} \neq \emptyset$. So let $x \in I_1 \cap I_2 \cap \cdots \cap I_{1997}$. Then by the definition of I_n , we know that $a_n = \lfloor nx \rfloor$ for all $1 \leq n \leq 1997$, and we are done.

§27 USAMO 1998

1. (USAMO 1998) Suppose that the set $\{1, 2, \dots, 1998\}$ has been partitioned into disjoint pairs $\{a_i, b_i\}$ ($1 \leq i \leq 999$) so that for all i , $|a_i - b_i|$ equals 1 or 6. Prove that the sum

$$|a_1 - b_1| + |a_2 - b_2| + \dots + |a_{999} - b_{999}|$$

ends in the digit 9.

Solution

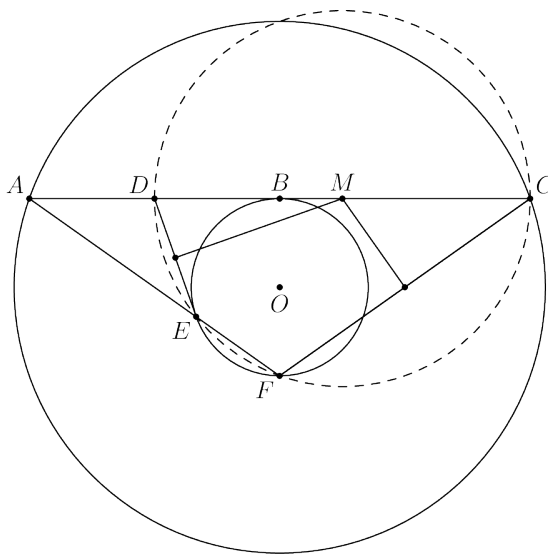
Notice that $|a_i - b_i| \equiv 1 \pmod{5}$, so $S = |a_1 - b_1| + |a_2 - b_2| + \dots + |a_{999} - b_{999}| \equiv 1 + 1 + \dots + 1 \equiv 999 \equiv 4 \pmod{5}$.

Also, for integers M, N we have $|M - N| \equiv M - N \equiv M + N \pmod{2}$.

Thus, we also have $S \equiv a_1 + b_1 + a_2 + b_2 + \dots + a_{999} + b_{999} \equiv 1 + 2 + \dots + 1998 \equiv 999 * 1999 \equiv 1 \pmod{2}$ also, so by the Chinese Remainder Theorem $S \equiv 9 \pmod{10}$. Thus, S ends in the digit 9, as desired.

2. (USAMO 1998) Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be concentric circles, with $\mathcal{C}_2$ in the interior of $\mathcal{C}_1$. From a point A on $\mathcal{C}_1$ one draws the tangent AB to $\mathcal{C}_2$ ($B \in \mathcal{C}_2$). Let C be the second point of intersection of AB and $\mathcal{C}_1$, and let D be the midpoint of AB . A line passing through A intersects $\mathcal{C}_2$ at E and F in such a way that the perpendicular bisectors of DE and CF intersect at a point M on AB . Find, with proof, the ratio AM/MC .

Solution 1



First, $AD = \frac{AB}{2} = \frac{AC}{4}$. Because E, F and B all lie on a circle, $AE \cdot AF = AB \cdot AB = \frac{AB}{2} \cdot 2AB = AD \cdot AC$. Therefore, $\triangle ACF \sim \triangle AED$, so $\angle ACF = \angle AED$. Thus, quadrilateral $CFED$ is cyclic, and M must be the center of the circumcircle of $CFED$, which implies that $MC = \frac{CD}{2}$. Putting it all together,

$$\frac{AM}{MC} = \frac{AC - MC}{MC} = \frac{AC - \frac{CD}{2}}{\frac{CD}{2}} = \frac{AC - \frac{AC - AD}{2}}{\frac{AC - AD}{2}} = \frac{AC - \frac{3AC}{8}}{\frac{3AC}{8}} = \frac{\frac{5AC}{8}}{\frac{3AC}{8}} = \boxed{\frac{5}{3}}.$$

Borrowed from <https://mks.mff.cuni.cz/kalva/usa/usoln/usol982.html>

Solution 2

We will use signed lengths. WLOG let $AC = 4$, $AM = x$, and $AE = y > 0$. Then $AB = 2$ and $AD = 1$. Therefore, $DM = x - 1$ and $MC = 4 - x$. By Power of a Point, $AE \cdot AF = AB^2 = 4$, so $AF = \frac{4}{AE} = \frac{4}{y}$, and $EF = \frac{4}{y} - y = \frac{4-y^2}{y}$. Since M is on the perpendicular bisectors of DE and CF , we have $|DM| = |EM| = |x - 1|$ and $|CM| = |FM| = |4 - x|$.

By Stewart's Theorem on $\triangle MFA$ and cevian DM ,

$$\begin{aligned} \frac{4}{y}(y(\frac{4-y^2}{y}) + (x-1)^2) &= y(4-x)^2 + \frac{4-y^2}{y} \cdot x^2; \\ 4(4-y^2 + (x-1)^2) &= y^2(4-x)^2 + (4-y^2)x^2; \\ 16 - 4y^2 + 4x^2 - 8x + 4 &= x^2y^2 - 8xy^2 + 16y^2 + 4x^2 - x^2y^2; \\ 8xy^2 - 8x - 20y^2 + 20 &= (8x-20)(y^2-1) = 0. \end{aligned}$$

So either $8x - 20 = 0$ or $y^2 - 1 = 0$. If $y^2 - 1 = 0$, then $AE = y = 1 = AD$ and $AF = AC = 4$, so the perpendicular bisectors of CF and DE are the same line, and they do not intersect at a point. Therefore, $AM = x = \frac{5}{2}$ and $MC = \frac{3}{2}$, so

$$\frac{AM}{MC} = \boxed{\frac{5}{3}}.$$

3. (USAMO 1998) Let $a_0, a_1, \dots, a_n$ be numbers from the interval $(0, \pi/2)$ such that

$$\tan\left(a_0 - \frac{\pi}{4}\right) + \tan\left(a_1 - \frac{\pi}{4}\right) + \dots + \tan\left(a_n - \frac{\pi}{4}\right) \geq n - 1.$$

Prove that

$$\tan a_0 \tan a_1 \dots \tan a_n \geq n^{n+1}.$$

Solution

Let $y_i = \tan\left(a_i - \frac{\pi}{4}\right)$, where $0 \leq i \leq n$. Then we have:

- $y_0 + y_1 + \dots + y_n \geq n - 1$,
- $1 + y_i \geq \sum_{j \neq i} (1 - y_j)$,
- $\frac{1+y_i}{n} \geq \frac{1}{n} \sum_{j \neq i} (1 - y_j)$.

By AM-GM:

$$\begin{aligned} \frac{1}{n} \sum_{j \neq i} (1 - y_j) &\geq \prod_{j \neq i} (1 - y_j)^{\frac{1}{n}} \\ \frac{1+y_i}{n} &\geq \prod_{j \neq i} (1 - y_j)^{\frac{1}{n}} \\ \prod_{i=0}^n \frac{1+y_i}{n} &\geq \prod_{i=0}^n \prod_{j \neq i} (1 - y_j)^{\frac{1}{n}} \\ \prod_{i=0}^n \frac{1+y_i}{n} &\geq \prod_{i=0}^n (1 - y_i) \end{aligned}$$

$$\prod_{i=0}^n \frac{1+y_i}{1-y_i} \geq \prod_{i=0}^n n$$

$$\prod_{i=0}^n \frac{1+y_i}{1-y_i} \geq n^{n+1}.$$

Note that by the addition formula for tangents:

$$\tan(a_i) = \tan\left[\left(a_i - \frac{\pi}{4}\right) + \frac{\pi}{4}\right] = \frac{1 + \tan\left(a_i - \frac{\pi}{4}\right)}{1 - \tan\left(a_i - \frac{\pi}{4}\right)} = \frac{1 + y_i}{1 - y_i}.$$

So $\prod_{i=0}^n \frac{1+y_i}{1-y_i} = \tan(a_0) \tan(a_1) \cdots \tan(a_n) \geq n^{n+1}$, as desired. ■

4. (*USAMO 1998*) A computer screen shows a 98×98 chessboard, colored in the usual way. One can select with a mouse any rectangle with sides on the lines of the chessboard and click the mouse button: as a result, the colors in the selected rectangle switch (black becomes white, white becomes black). Find, with proof, the minimum number of mouse clicks needed to make the chessboard all one color.

Solution

Answer: 98.

There are $4 \cdot 97$ adjacent pairs of squares in the border and each pair has one black and one white square. Each move can fix at most 4 pairs, so we need at least 97 moves. However, we start with two corners one color and two another, so at least one rectangle must include a corner square. But such a rectangle can only fix two pairs, so at least 98 moves are needed.

It is easy to see that 98 moves suffice: take $49 \cdot 1 \times 98$ rectangles (alternate rows), and $49 \cdot 98 \times 1$ rectangles (alternate columns).

credit: <https://mks.mff.cuni.cz/kalva/usa/usoln/usol984.html>

editor: Brian Joseph

second editor: integralarefun

5. (*USAMO 1998*) Prove that for each $n \geq 2$, there is a set S of n integers such that $(a-b)^2$ divides ab for every distinct $a, b \in S$.

Solution

Proof by induction. For $n=2$, the proof is trivial, since $S = (1, 2)$ satisfies the condition. Assume now that there is such a set S of n elements, $a_1, a_2, \dots, a_n$ which satisfy the condition. The key is to note that if $m = a_1 a_2 \dots a_n$, then if we define $b_i = a_i + km$ for all $i \leq n$, where k is a positive integer, then $a_i \mid b_i$ and $b_i - b_j = a_i - a_j$, and so $(b_i - b_j)^2 = (a_i - a_j)^2 \mid a_i a_j \mid b_i b_j$.

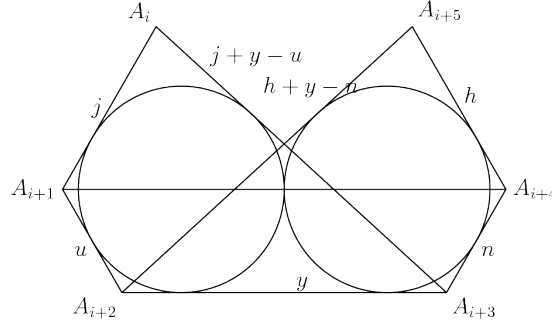
Let $b_{n+1} = m + km$. Consider the set $T = (b_1, b_2, \dots, b_n, b_{n+1})$. To finish the proof, we simply need to choose a k such that $(b_{n+1} - b_i)^2 \mid b_{n+1} b_i$ for all $i \leq n$. Since $(b_{n+1} - b_i)^2 = (m - a_i)^2$, simply choose k so that $k+1 = (m - a_1)^2 (m - a_2)^2 \dots (m - a_n)^2$.

6. (*USAMO 1998*) Let $n \geq 5$ be an integer. Find the largest integer k (as a function of n) such that there exists a convex n -gon $A_1 A_2 \dots A_n$ for which exactly k of the quadrilaterals $A_i A_{i+1} A_{i+2} A_{i+3}$ have an inscribed circle. (Here $A_{n+j} = A_j$.)

Solution

Lemma: If quadrilaterals $A_i A_{i+1} A_{i+2} A_{i+3}$ and $A_{i+2} A_{i+3} A_{i+4} A_{i+5}$ in an equiangular n -gon are tangential, and $A_i A_{i+3}$ is the longest side quadrilateral $A_i A_{i+1} A_{i+2} A_{i+3}$ for all i , then quadrilateral $A_{i+1} A_{i+2} A_{i+3} A_{i+4}$ is not tangential.

Proof:



If quadrilaterals $A_i A_{i+1} A_{i+2} A_{i+3}$ and $A_{i+2} A_{i+3} A_{i+4} A_{i+5}$ are tangential, then $A_i A_{i+3}$ must have side length of $j + y - u$, and $A_{i+2} A_{i+5}$ must have side length of $h + y - n$ (One can see this from what is known as walk-around). Suppose quadrilateral $A_{i+1} A_{i+2} A_{i+3} A_{i+4}$ is tangential. Then, again, we see that $A_{i+1} A_{i+4}$ must have side length $u + n - y$. We assumed by lemma that $A_i A_{i+3} > A_i A_{i+1}$ for all i , so we have $A_i A_{i+3} > j$, $A_{i+1} A_{i+4} > y$, and $A_{i+2} A_{i+5} > h$. If we add up the side lengths $A_i A_{i+3} + A_{i+1} A_{i+4} + A_{i+2} A_{i+5}$, we get:

$$A_i A_{i+3} + A_{i+1} A_{i+4} + A_{i+2} A_{i+5} = j + y - u + h + y - n + u + n - y$$

$$A_i A_{i+3} + A_{i+1} A_{i+4} + A_{i+2} A_{i+5} = j + h + y$$

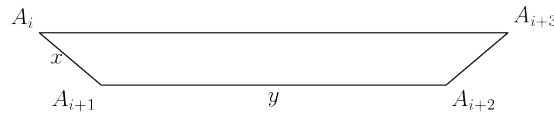
However, by the lemma, we assumed that $A_i A_{i+3} > j$, $A_{i+1} A_{i+4} > y$, and $A_{i+2} A_{i+5} > h$. Adding these up, we get:

$$A_i A_{i+3} + A_{i+1} A_{i+4} + A_{i+2} A_{i+5} > j + h + y,$$

which is a contradiction. Thus, quadrilateral $A_{i+1} A_{i+2} A_{i+3} A_{i+4}$ is not tangential, proving the lemma.

By lemma, the maximum number of quadrilaterals in a n -gon occurs when the tangential quadrilaterals alternate, giving us $k = \lfloor \frac{n}{2} \rfloor$.

Note that one can find the ratio of side of an equiangular n -gon in order for alternating quadrilaterals to be tangential.



Since exterior angles of a equiangular n -gon have degree measure $\frac{2\pi}{n}$, one can write the equation:

$$\begin{aligned} 2x &= y + y + 2x \cos \frac{2\pi}{n} \\ y &= x \left(1 - \cos \frac{2\pi}{n} \right) \end{aligned}$$

$$\frac{y}{x} = \frac{1}{1 - \cos \frac{2\pi i}{n}}$$

Thus, we can find the ratio of sides of an equiangular n -gon which fits the maximum to be $1 : 1 - \cos \frac{2\pi}{n}$. Note that if n is even, we can easily alternate them, but if n is odd, we must have two adjacent sides be the same length, and that length must be the larger side in the ratio of adjacent sides. The proof is left as an exercise for the reader.

§28 USAMO 1999

1. (USAMO 1999) Some checkers placed on an $n \times n$ checkerboard satisfy the following conditions:

- (a) every square that does not contain a checker shares a side with one that does;
- (b) given any pair of squares that contain checkers, there is a sequence of squares containing checkers, starting and ending with the given squares, such that every two consecutive squares of the sequence share a side.

Prove that at least $(n^2 - 2)/3$ checkers have been placed on the board.

Solution 1

For the proof let's look at the checkers board as a graph G , where the checkers are the vertices and the edges are every pair of checkers sharing a side.

Define R as the circuit rank of G (the minimum number of edges to remove from a graph to remove all its cycles).

Define x as the number of checkers placed on the board.

1. From (a) for every square containing a checker there can be up to 4 distinct squares without checkers; thus, $4 \times x + x = 5 \times x \geq n^2$ (this is the most naive upper bound).
2. From (b) this graph has to be connected. which means that no square which contains a checker can share 4 sides with squares that doesn't contain a checker (because it has to share at least one side with a square that contains a checker).
3. Now it is possible to improve the upper bound. Since every edge in G represents a square with a checker that shares a side with a square with another checker, the new upper bound is $5 \times x - [\text{sum of degrees in } G] \geq n^2$.
4. Thus, $5x - (2x - 2 + 2R) \geq n^2$ (see lemma below).
5. Thus, $3x + 2 - 2R \geq n^2$.
6. Thus, $x \geq \frac{n^2 - 2 + 2R}{3}$, where $R \geq 0$ and is equal to 0 if G is a tree.

Clarification/Lemma: The sum of degrees of a connected graph $G = (V, E)$ is $2V - 2 + 2R = 2E$, where R is the circuit rank of G .

Proof. $2 \times V - 2$ is the sum of ranks of the spanning tree created by decircuiting the graph G . Since the circuit rank of G is R , $2R$ is the sum of ranks removed from G . Thus, $2V - 2 + 2R = 2E$. ■

Solution 2

Call a square of the checkerboard “good” if it either has a checker on it or shares a side with a square with a checker on it.

Say there are k checkers on the board, sitting on squares $s_1, \dots, s_k$. We can assume WLOG that $s_1, \dots, s_k$ is ordered in such a way that if $1 \leq i \leq k$, then s_i shares a side with at least one of $s_1, \dots, s_{i-1}$. (If such an ordering were impossible, then there would be an s_i which isn't connected to s_1 by a sequence of squares with checkers on them, violating condition (b).)

Now imagine removing all the checkers and then putting them back onto the squares $s_1, \dots, s_k$ in that order, counting the number of good squares after each step. When we put a checker on s_1 , we increase the number of good squares by at most 5: the square s_1 becomes good, and each square it shares a side with (at most 4) becomes good. When we put a checker onto s_i , where $i > 1$, we increase the number of good squares by at most 3, since s_i is already good, and at least 1 of its at most 4 neighbors already has a checker on it.

So the total number of good squares when we've put back every checker is at most $5 + 3(k-1) = 3k + 2$. In order to satisfy condition (a), we have to make each of the n^2 squares good. In other words, we need $3k + 2 \geq n^2$, or

$$k \geq \frac{n^2 - 2}{3}.$$

2. (USAMO 1999) Let $ABCD$ be a cyclic quadrilateral. Prove that

$$|AB - CD| + |AD - BC| \geq 2|AC - BD|.$$

Solution

Let arc AB of the circumscribed circle (which we assume WLOG has radius 0.5) have value $2x$, BC have $2y$, CD have $2z$, and DA have $2w$. Then our inequality reduces to, for $x + y + z + w = 180^\circ$:

$$|\sin x - \sin z| + |\sin y - \sin w| \geq 2|\sin(x + y) - \sin(y + z)|.$$

This is equivalent to by sum-to-product and use of $\cos x = \sin(90^\circ - x)$:

$$\left| \sin \frac{x-z}{2} \sin \frac{y+w}{2} \right| + \left| \sin \frac{y-w}{2} \sin \frac{x+z}{2} \right| \geq 2 \left| \sin \frac{x-z}{2} \sin \frac{y-w}{2} \right|.$$

Clearly $90^\circ \geq \frac{x+z}{2} > \frac{x-z}{2} \geq 0^\circ$. As sine is increasing over $[0, \pi/2]$, $|\sin \frac{x+z}{2}| > |\sin \frac{x-z}{2}|$.

Similarly, $|\sin \frac{y+w}{2}| > |\sin \frac{y-w}{2}|$. The result now follows after multiplying the first inequality by $|\sin \frac{x-z}{2}|$, the second by $|\sin \frac{y-w}{2}|$, and adding. (Equality holds if and only if $x = z$ and $y = w$, ie. $ABCD$ is a parallelogram.)

–Suli 11:23, 5 October 2014 (EDT)

3. (USAMO 1999) Let $p > 2$ be a prime and let a, b, c, d be integers not divisible by p , such that

$$\left\{ \frac{ra}{p} \right\} + \left\{ \frac{rb}{p} \right\} + \left\{ \frac{rc}{p} \right\} + \left\{ \frac{rd}{p} \right\} = 2$$

for any integer r not divisible by p . Prove that at least two of the numbers $a + b$, $a + c$, $a + d$, $b + c$, $b + d$, $c + d$ are divisible by p . (Note: $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of x .)

Solution

We see that $\left\{ \frac{ra+rb+rc+rd}{p} \right\} = 0$ means that $p|r(a+b+c+d)$. Now, since p does not divide r and p is prime, their GCD is 1 so $p \mid a+b+c+d$.

Since $\left\{ \frac{ra}{p} \right\} + \left\{ \frac{rb}{p} \right\} + \left\{ \frac{rc}{p} \right\} + \left\{ \frac{rd}{p} \right\} = 2$, then we see that they have to represent mods mod p , and thus, our possible values of p are all such that $k^4 \equiv 1 \pmod{p}$ for all k that are relatively prime to p . This happens when $p = 3$ or 5 .

When $p = 3$ then r is not divisible by 3, thus two are 1, and the other two are 2. Thus, four pairwise sums sum to 3.

When $p = 5$ then r is not divisible by 5 so a, b, c, d are 1, 2, 3, and 4, so two pairwise sums sum to 5.

All three possible cases work so we are done.

(This solution makes absolutely no sense. Why is $k^4 \equiv 1$? And how do we know that only 3 and 5 work!?) why can't you have $a, b, c, d = 2, 2, 2, 4$ $p = 5$ $r=1$? it doesn't say distinct integers

4. (USAMO 1999) Let $a_1, a_2, \dots, a_n$ ($n > 3$) be real numbers such that

$$a_1 + a_2 + \dots + a_n \geq n \quad \text{and} \quad a_1^2 + a_2^2 + \dots + a_n^2 \geq n^2.$$

Prove that $\max(a_1, a_2, \dots, a_n) \geq 2$.

Solution 1

First, suppose all the a_i are positive. Then

$$\max(a_1, \dots, a_n) \geq \sqrt{\frac{a_1^2 + \dots + a_n^2}{n}} \geq \sqrt{n} \geq 2.$$

Suppose, on the other hand, that without loss of generality,

$$a_1 \geq a_2 \geq \dots \geq a_k \geq 0 > a_{k+1} \geq \dots \geq a_n,$$

with $1 \leq k < n$. If $a_1 > 2$ we are done, so suppose that $a_1 \leq 2$. Then $\sum_{i=1}^k a_i \leq 2k$, so

$$\sum_{i=k+1}^n -a_i \leq 2k - n.$$

Since $-a_i$ is a positive real for all $k+1 \leq i \leq n$, it follows that

$$\sum_{i=k+1}^n a_i^2 \leq \left(\sum_{i=k+1}^n -a_i \right)^2 \leq (2k - n)^2.$$

Then

$$\begin{aligned} \max(a_1, \dots, a_n)^2 &\geq \sum_{i=1}^k a_i^2 / k \\ &\geq \left(n^2 - \sum_{i=k+1}^n a_i^2 \right) / k \\ &\geq \frac{n^2 - (2k - n)^2}{k} = 4(n - k). \end{aligned}$$

Since $k < n$,

5. (*USAMO 1999*) The Y2K Game is played on a 1×2000 grid as follows. Two players in turn write either an S or an O in an empty square. The first player who produces three consecutive boxes that spell SOS wins. If all boxes are filled without producing SOS then the game is a draw. Prove that the second player has a winning strategy.

Solution

Label the squares 1 through 2000. The first player writes a letter in any square k . The second player now selects any empty square l such that $\min(|l-1|, |l-2000|, |l-k|) > 3$, and fills in a S. The first player may again play anywhere, say m . If the second player can now complete SOS, he does so; otherwise, if $l > m$ he fills an S in $l+3$ and if $l < m$ he fills an S in $l-3$.

We thus have two Ss separated by two empty squares. Note that if any player plays any letter between the two Ss, the other player can always complete a SOS.

Now the second player repeats the following algorithm:

- If he can complete SOS on his turn, he does so.
- Otherwise, since there are an odd number of empty squares at the beginning of his turn, there must be either one empty square surrounded by two filled squares or one empty square surrounded by two empty squares. In both cases, the player places an O in that square. In both situations, it is easy to verify that the first player cannot win on the next turn.

Eventually, assuming the second player has not won yet (and since the first player does not have a winning opportunity yet), this algorithm must stop. It follows that all of the empty squares are now in pairs. Since there are an even number of empty squares, it is the first player's turn. If the first player plays on one of any pair of empty squares that does not have Ss on both its sides, then the second player plays any letter in the other square of the pair. Then, we will only have pairs of empty squares with Ss on both sides remaining (since from the first moves we have constructed at least one, and no one has played between them otherwise the game will have ended), and the first player will have to play between them, so that the second player will win.

6. (*USAMO 1999*) Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. The inscribed circle ω of triangle BCD meets CD at E . Let F be a point on the (internal) angle bisector of $\angle DAC$ such that $EF \perp CD$. Let the circumscribed circle of triangle ACF meet line CD at C and G . Prove that the triangle AFG is isosceles.

Solution

Quadrilateral $ABCD$ is cyclic since it is an isosceles trapezoid. $AD = BC$. Triangle ADC and triangle BCD are reflections of each other with respect to diameter which is perpendicular to AB . Let the incircle of triangle ADC touch DC at K . The reflection implies that $DK = CE$, which then implies that the excircle of triangle ADC is tangent to DC at E . Since EF is perpendicular to DC which is tangent to the excircle, this implies that EF passes through center of excircle of triangle ADC .

We know that the center of the excircle lies on the angular bisector of DAC and the perpendicular line from DC to E . This implies that F is the center of the excircle.

Now $\angle GFA = \angle GCA = \angle DCA$. $\angle ACF = 90^\circ + \frac{\angle DCA}{2}$. This means that $\angle AGF = 90^\circ - \frac{\angle ACD}{2}$. (due to cyclic quadrilateral $ACFG$ as given). Now $\angle FAG = 180^\circ - (\angle AFG + \angle FGA) = 90^\circ - \frac{\angle ACD}{2} = \angle AGF$.

Therefore $\angle FAG = \angle AGF$. QED.

§29 USAMO 2000

1. (USAMO 2000) Call a real-valued function f very convex if

$$\frac{f(x) + f(y)}{2} \geq f\left(\frac{x+y}{2}\right) + |x-y|$$

holds for all real numbers x and y . Prove that no very convex function exists.

Solution 1

Let $y \geq x$, and substitute $a = x, 2b = y - x$. Then a function is very convex if $\frac{f(a) + f(a+2b)}{2} \geq f(a+b) + 2b$, or rearranging,

$$\left[\frac{f(a+2b) - f(a+b)}{b} \right] - \left[\frac{f(a+b) - f(a)}{b} \right] \geq 4$$

Let $g(a) = \frac{f(a+b) - f(a)}{b}$, which is the slope of the secant between $(a, f(a))$ and $(a+b, f(a+b))$. Let b be arbitrarily small; then it follows that $g(a+b) - g(a) > 4$, $g(a+2b) - g(a+b) > 4$, $\dots$, $g(a+kb) - g(a+[k-1]b) > 4$. Summing these inequalities yields $g(a+kb) - g(a) > 4k$. As $k \rightarrow \infty$ (but $b < \frac{1}{k}$, so $bk < \epsilon$ is still arbitrarily small), we have $\lim_{k \rightarrow \infty} g(a+kb) - g(a) = g(a+\epsilon) - g(a) > \lim_{k \rightarrow \infty} 4k = \infty$. This implies that in the vicinity of any a , the function becomes vertical, which contradicts the definition of a function. Hence no very convex function exists.

Solution 2

Suppose, for the sake of contradiction, that there exists a very convex function f . Notice that $f(x)$ is convex if and only if $f(x) + C$ is convex, where C is a constant. Thus, we may set $f(0) = 0$ for convenience.

Suppose that $f(1) = A$ and $f(-1) = B$. By the very convex condition,

$$\frac{f(0) + f(2^{-n})}{2} \geq f\left(2^{-(n+1)}\right) + \frac{1}{2^n}.$$

A straightforward induction shows that:

$$f(2^{-n}) \leq \frac{A - 2n}{2^n}$$

for all nonnegative integers n . Using a similar line of reasoning as above,

$$f(-2^{-n}) \leq \frac{B - 2n}{2^n}.$$

Therefore, for every nonnegative integer n ,

$$f(2^{-n}) + f(-2^{-n}) \leq \frac{A + B - 4n}{2^n}.$$

Now, we choose n large enough such that $n > \frac{A+B}{4} - 1$. This is always possible because A and B are fixed for any particular function f . It follows that:

$$f(2^{-n}) + f(-2^{-n}) < \frac{1}{2^{n-2}}.$$

However, by the very convex condition,

$$f(2^{-n}) + f(-2^{-n}) \geq \frac{1}{2^{n-2}}.$$

This is a contradiction. It follows that no very convex function exists.

Solution 3

Define the symmetric average

$$F(x, y) = \frac{f(x) + f(y)}{2}.$$

Then the inequality becomes

$$F(x, y) \geq f\left(\frac{x+y}{2}\right) + |x-y|.$$

Because $|x-y|$ is not differentiable along the line $x=y$, the surface defined by $F(x, y)$ must have a permanent kink along the diagonal. A kink is a sharp crease where the surface fails to have a well-defined tangent plane.

If f were a true function on $\mathbb{R}$ satisfying the inequality at all scales, then this kink would have to persist no matter how closely x and y are chosen. That means the graph of F never becomes smooth near the diagonal.

But every differentiable real-valued function f produces a smooth symmetric average F . Therefore, the presence of a permanent kink contradicts the assumption that f is differentiable at any scale. The kink cannot be removed by local smoothing.

Hence, the geometric shape of the inequality forces a kink that no function f can produce, so no function can be very convex.

Solution 4

Assume, for contradiction, that $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$\frac{f(x) + f(y)}{2} \geq f\left(\frac{x+y}{2}\right) + |x-y| \quad \text{for all } x, y \in \mathbb{R}.$$

Set $x = a - h$ and $y = a + h$. Then

$$f(a+h) + f(a-h) - 2f(a) \geq 4|h| \quad \text{for all } a, h \in \mathbb{R}.$$

Thus

$$\Delta_h^2 f(a) := f(a+h) + f(a-h) - 2f(a) \geq 0 \quad \text{for all } a, h \in \mathbb{R},$$

so f is convex. Let $\mu = f''$ in the distributional sense. Then μ is a nonnegative locally finite Borel measure.

For $h > 0$, the second difference can be written as

$$\Delta_h^2 f(a) = \int_{\mathbb{R}} 2(h - |x-a|)_+ d\mu(x),$$

where

$$(t)_+ = \max\{t, 0\}.$$

Since

$$(h - |x - a|)_+ \leq h \mathbf{1}_{[a-h, a+h]}(x),$$

we get

$$\Delta_h^2 f(a) \leq 2h \mu([a - h, a + h]).$$

On the other hand, the hypothesis gives

$$\Delta_h^2 f(a) \geq 4h.$$

Hence

$$\mu([a - h, a + h]) \geq 2 \quad \text{for every } h > 0.$$

Letting $h \downarrow 0$, continuity from above yields

$$\mu(\{a\}) \geq 2.$$

Since a was arbitrary, every point of $\mathbb{R}$ is an atom of μ of mass at least 2, contradicting local finiteness of μ .

Therefore no function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$\frac{f(x) + f(y)}{2} \geq f\left(\frac{x + y}{2}\right) + |x - y| \quad \text{for all } x, y \in \mathbb{R}.$$

2. (USAMO 2000) Let S be the set of all triangles ABC for which

$$5 \left(\frac{1}{AP} + \frac{1}{BQ} + \frac{1}{CR} \right) - \frac{3}{\min\{AP, BQ, CR\}} = \frac{6}{r},$$

where r is the inradius and P, Q, R are the points of tangency of the incircle with sides AB, BC, CA , respectively. Prove that all triangles in S are isosceles and similar to one another.

Solution

We let $x = AP = s - a, y = BQ = s - b, z = CR = s - c$, and **without loss of generality** let $x \leq y \leq z$. Then $x + y + z = 3s - (a + b + c) = s$, so $r = \frac{A}{s} = \frac{\sqrt{(x+y+z)xyz}}{x+y+z} = \sqrt{\frac{xyz}{x+y+z}}$. Thus,

$$6\sqrt{\frac{x+y+z}{xyz}} = \frac{2yz + 5xy + 5xz}{xyz}$$

Squaring and simplifying yields (after much grueling work)

$$0 = 4y^2z^2 + 25x^2y^2 + 25x^2z^2 - 16xy^2z - 16xyz^2 + 14x^2yz$$

We claim that the inequality

$$0 \leq 4y^2z^2 + 25x^2y^2 + 25x^2z^2 - 16xy^2z - 16xyz^2 + 14x^2yz \quad (*)$$

holds true, with equality iff $4x = y = z$.

Note that $(*)$ is homogeneous in x, y, z , so without loss of generality, scale so that $x = 1$. Then

$$0 \leq y^2(4z^2 - 16z + 25) + y(14z - 16z^2) + 25z^2$$

which is a quadratic in y . As $4z^2 - 16z + 25 = 4(z - 4)^2 + 9 \geq 0$, it suffices to show that the quadratic cannot have more than one root, or the discriminant $\Delta \leq 0$. Then,

$$\Delta = z^2(14 - 16z)^2 - 4(4z^2 - 16z + 25)(25z^2) = -144(z - 4)^2 \leq 0$$

as desired. Equality comes when $z = 4$; since $(*)$ is symmetric in y and z , it follows that $y = 4$ is also necessary for equality. Reversing our scaling, it follows that $x : y : z = 1 : 4 : 4$.

Then $s = x + y + z = 9x$, and $s - a = x$, $s - b = 4x$, $s - c = 4x$ yields $a : b : c = 8 : 5 : 5$. Thus, we have proved that all possible ABC are isosceles (as $b = c$), and that they are similar to a 5-5-8 triangle. ■

Solution 2

Let A, B, C be the three angles of the triangle, then $r/AP = \tan(A/2)$, $r/BQ = \tan(B/2)$, and $r/CR = \tan(C/2)$. Without loss of generality, assume $A \leq B \leq C$, then $CR = \min(AP, BQ, CR)$. Denote $x = \tan(A/2)$, $y = \tan(B/2)$, $z = \tan(C/2)$, then we have,

$$5(x + y + z) - 3z = 6 \quad (1)$$

Meanwhile,

$$z = \tan(C/2) = \tan(90 - (A + B)/2) = \frac{1 - xy}{x + y}$$

Substitute z in (1), we get

$$5x^2 + 8xy + 5y^2 - 6x - 6y + 2 = 0 \quad (2)$$

Treating (2) as a quadratic equation for x , the discriminant is:

$$\Delta = (8y - 6)^2 - 4 * 5(5y^2 - 6y + 2) = -36y^2 + 24y - 4 = -(6y - 2)^2 \leq 0$$

For x, y to be both real numbers, Δ must not be negative, so $\Delta = 0$, which yields $y = 1/3$. Solving for x , we get $x = \frac{-(8y-6)}{2*5} = 1/3$. Using the formula $\tan(A) = \frac{2 \tan(A/2)}{1 - \tan^2(A/2)}$, we have $\tan(A) = \tan(B) = 3/4$. Therefore $A = B = \arctan(3/4)$. This proves that all possible triangle ABC are isosceles with the same set of angles, i.e., they are similar to each other.

It's inconsequential to the proof, but it can be easily found that, since $\tan(A) = 3/4$, the altitude to the base of the isosceles triangle divides the triangle into two 3-4-5 right triangles with 4 on the base; therefore, the entire triangle is 5-5-8. □.

Solution by *Mathdummy*.

3. (*USAMO 2000*) A game of solitaire is played with R red cards, W white cards, and B blue cards. A player plays all the cards one at a time. With each play he accumulates a penalty. If he plays a blue card, then he is charged a penalty which is the number of white cards still in his hand. If he plays a white card, then he is charged a penalty which is twice the number of red cards still in his hand. If he plays a red card, then he is charged a penalty which is three times the number of blue cards still in his hand. Find, as a function of R, W , and B , the minimal total penalty a player can amass and all the ways in which this minimum can be achieved.

Solution

We claim (inductively) that the minimum is just going to be $\min(BW, 2WR, 3RB)$. We'll start our induction with the case where one of the three quantities is zero, in which case we verify that we can indeed get away without any penalty by, for example, discarding blue if we are out of white.

Now, for the inductive step, let $f(B, W, R)$ be the minimum we seek. Note that

$$f(B, W, R) = \min(W + f(B - 1, W, R), 2R + f(B, W - 1, R), 3B + f(B, W, R - 1))$$

By our inductive hypothesis, $f(B - 1, W, R) = \min((B - 1)W, 2WR, 3R(B - 1))$. In order for this to cause our inductive step not to hold, we would require that $W + \min((B - 1)W, 2WR, 3R(B - 1)) < \min(BW, 2WR, 3RB)$. It is evident that the first two entries in the $\min$ expression cannot cause this to happen, so that we need only consider $W + 3R(B - 1) < \min(BW, 2WR, 3RB)$. So $W + 3R(B - 1) < BW$, whence $3R < W$. But $W + 3R(B - 1) < 3RB$, so that $W < 3R$, a contradiction.

For the other two cases, we can get similar contradictions, so that our inductive step must hold, and so $f(B, W, R)$ is indeed $\min(BW, 2WR, 3RB)$.

We now need only establish how many ways to do this. If one of these quantities is smaller, our induction and the fact that it is eventually zero guarantees that it will continue to be the smallest quantity as cards are discarded. (For example, if it is currently optimal to discard a blue card, it will continue to be so until we run out of blue cards.) Therefore, assuming that there is currently only one best choice of card to discard, this will continue to be so in the future, whence if $BW \neq 2WR \neq 3RB$, there is only 1 optimal strategy.

Suppose, now, that $BW = 2WR$. It is thus optimal to discard either a B or W card. If we ever discard a blue card, then we will cause $BW < 2WR$, whence there is only one possible strategy from then on. However, if we discard a white card, then we will still have $BW = 2WR$, meaning that we continue to have the choice of discarding a white or blue card. Since we can discard a white card at most W times, there are $W + 1$ choices for how many W cards to discard (0 to W), meaning that there are $W + 1$ optimal strategies.

By similar logic, we get $R + 1$ optimal strategies if $2WR = 3RB$, and $B + 1$ optimal strategies if $3RB = BW$.

The final case, then, is if $BW = 2WR = 3RB$. In this case, if we first discard a white card, we are left with the $BW = 2WR$ case, and similarly for a blue and red card. The total number of optimal strategies in this case is then the sum of the optimal strategies in those cases, or, in other words, $B + W + R$.

To summarize: The minimum penalty is $\min(BW, 2WR, 3RB)$. If $BW \neq 2WR \neq 3RB$, there is 1 optimal strategy. If $BW = 2WR < 3RB$, there are $W + 1$ strategies. If

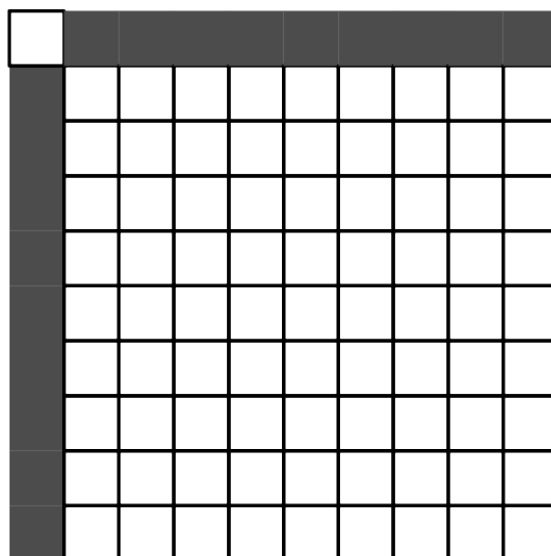
$2WR = 3RB < BW$, there are $R + 1$ strategies. If $3RB = BW < 2WR$, there are $B + 1$ strategies. If $BW = 2WR = 3RB$, there are $R + B + W$ strategies.

By J Steinhardt, from AoPS Community

4. (*USAMO 2000*) Find the smallest positive integer n such that if n squares of a 1000×1000 chessboard are colored, then there will exist three colored squares whose centers form a right triangle with legs parallel to the edges of the board.

Solution 1

We claim that $n = 1999$ is the smallest such number. For $n \leq 1998$, we can simply color any of the 1998 squares forming the top row and the left column, but excluding the top left corner square.



We now show that no configuration with no colored right triangles exists for $n = 1999$. We call a row or column *filled* if all 1000 of its squares are colored. Then any of the remaining 999 colored squares must share a column or row, respectively, with one of the colored squares in a filled row or column. These two squares, and any other square in the filled row or column, form a colored right triangle, giving us a contradiction. Hence, no filled row or column may exist.

Let m be the number of columns with 1 colored square. Then there are $1999 - m$ colored squares in the remaining columns, and in each of these $< 1999 - m$ columns that have a colored square must have at least two colored squares in them. These two colored squares will form a triangle with any other colored square in either of the rows containing the colored squares. Hence, each of the $1999 - m$ colored squares must be placed in different rows, but as there are only 1000 rows, the inequality $1999 - m \leq 1000 \implies m \geq 999$ holds. If $m = 1000$, then each column only has 1 colored square, leaving no place for the remaining 999, contradiction. If $m = 999$, then each of the 1000 rows has 1 black square, leaving no place for the other 999, contradiction. Hence $n = \boxed{1999}$ is the minimal value.

Solution 2

I claim that we can color at minimality 1999 squares and end up with one triangle whose legs are parallel to the sides of the chessboard.

Lemma 1: $n \leq 1998$ is impossible for all configurations of the n pieces.

Suppose one colors at maximum 1998 squares. From any configuration, there must never exist a configuration for which no such right triangle exists. However, one can color the $999 + 999 = 1998$ edge grids, exclusion of the corner, and now we show no right triangle is possible. Because the edges of the chessboard intersect at 90 degrees, the conjoined centres of the squares must also intersect at 90 degrees. Therefore, we can only form such a right triangle by using the corner piece. However, this was excluded from our original coloring. Hence, $n = 1998$ does not work. Then, one may arrange all the other $n \leq 1997$ colored squares in the same configuration such that the corner piece is not colored whereas the edges are, and no such triangle can be formed by the centres.

Lemma 2: $n = 1999$ is possible for all configurations of the 1999 pieces.

We show that three pieces must form a right triangle whose sides are parallel to the grid. Suppose each distinct row obtains a colored piece. If we want to refute the claim, we put each of the 1000 squares on the diagonal, so now no two are shared. But now, no matter where we place the last colored piece, by the Pigeonhole Principle, this piece lies in the same row as another, and the same column as another. If we don't place the original 1000 squares on the diagonal, then at minimum two squares will be collinear. From there, we can place the 999 remaining squares anywhere, and by the Pigeonhole Principle, at least two pairs of squares share a common square that form a 90 degree triangle.

2000 USAMO/Q4:

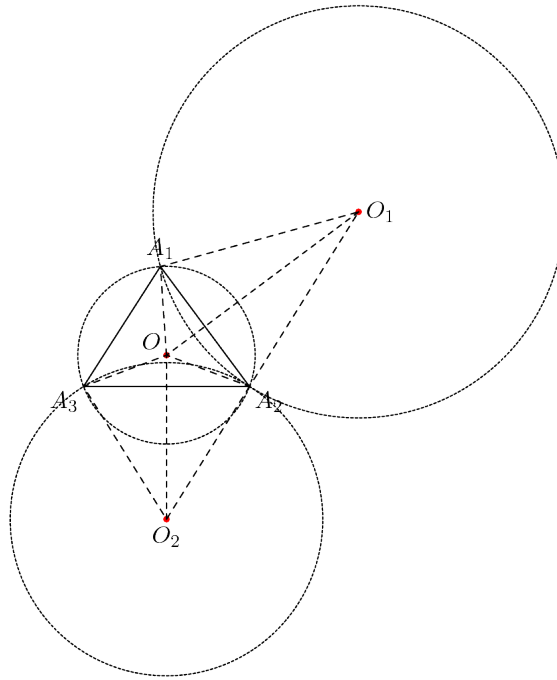
We must color a minimum of $n = \boxed{1999}$ squares.

5. (USAMO 2000) Let $A_1A_2A_3$ be a triangle and let ω_1 be a circle in its plane passing through A_1 and A_2 . Suppose there exist circles $\omega_2, \omega_3, \dots, \omega_7$ such that for $k = 2, 3, \dots, 7$, ω_k is externally tangent to ω_{k-1} and passes through A_k and A_{k+1} , where $A_{n+3} = A_n$ for all $n \geq 1$. Prove that $\omega_7 = \omega_1$.

Solution 1

Let the circumcenter of $\triangle ABC$ be O , and let the center of ω_k be O_k . ω_k and ω_{k-1} are externally tangent at the point A_k , so O_k, A_k, O_{k-1} are collinear.

O is the intersection of the perpendicular bisectors of $\overline{A_1A_2}$, $\overline{A_2A_3}$, $\overline{A_3A_1}$, and each of the centers O_k lie on the perpendicular bisector of the side of the triangle that determines ω_k . It follows from $OA_k = OA_{k+1}$, $O_kA_k = O_kA_{k+1}$, $OO_k = OO_k \implies \triangle OA_kO_k \cong \triangle OA_{k+1}O_k$ that $\angle OA_kO_k = \angle OA_{k+1}O_k$.



Since O , A_k , and the perpendicular bisector of $\overline{A_k A_{k+1}}$ are fixed, the angle $OA_k O_k$ determines the position of O_k (since O_k lies on the perpendicular bisector). Let $\theta_k = m\angle OA_k O_k$; then, $\theta_i = \theta_j$ and $i \equiv j \pmod{3}$ together imply that $O_i \equiv O_j$.

Now $\theta_1 = \angle OA_1 O_1 = \angle OA_2 O_1 = 180 - \angle OA_2 O_2 = 180 - \theta_2$ (due to collinearity). Hence, we have the recursion $\theta_k = 180 - \theta_{k-1}$, and so $\theta_k = \theta_{k-2}$. Thus, $\theta_1 = \theta_7$.

$\theta_1 = \theta_7$ implies that $O_1 \equiv O_7$, and circles ω_1 and ω_7 are the same circle since they have the same center and go through the same two points.

Solution 2

Using the collinearity of certain points and the fact that $A_k A_{k+1} O_k$ is isosceles, we quickly deduce that

$$\angle A_1 A_2 O_1 = 180^\circ - \angle A_2 - (180^\circ - \angle A_3) + (180^\circ - \angle A_1) - (180^\circ - \angle A_2) + (180^\circ - \angle A_3) - (180^\circ - \angle A_1) + \angle A_7$$

From ASA Congruence we deduce that $A_1 A_2 O_1$ and $A_7 A_8 O_7$ are congruent triangles, and so $O_1 A_1 = O_7 A_7$, that is $\omega_1 = \omega_7$.

6. (USAMO 2000) Let $a_1, b_1, a_2, b_2, \dots, a_n, b_n$ be nonnegative real numbers. Prove that

$$\sum_{i,j=1}^n \min\{a_i a_j, b_i b_j\} \leq \sum_{i,j=1}^n \min\{a_i b_j, a_j b_i\}.$$

Solution 1

Credit for this solution goes to Ravi Boppana.

Lemma 1: If $r_1, r_2, \dots, r_n$ are non-negative reals and $x_1, x_2, \dots, x_n$ are reals, then

$$\sum_{i,j} \min(r_i, r_j) x_i x_j \geq 0.$$

Proof: Without loss of generality assume that the sequence $\{r_i\}$ is increasing. For convenience, define $r_0 = 0$. The LHS of our inequality becomes

$$\sum_i r_i x_i^2 + 2 \sum_{i < j} r_i x_i x_j.$$

This expression is equivalent to the sum

$$\sum_i (r_i - r_{i-1}) \left(\sum_{j=i}^n x_j \right)^2.$$

Each term in the summation is non-negative, so the sum itself is non-negative. ■

We now define $r_i = \frac{\max(a_i, b_i)}{\min(a_i, b_i)} - 1$. If $\min(a_i, b_i) = 0$, then let r_i be any non-negative number. Define $x_i = \operatorname{sgn}(a_i - b_i) \min(a_i, b_i)$.

Lemma 2: $\min(a_i b_j, a_j b_i) - \min(a_i a_j, b_i b_j) = \min(r_i, r_j) x_i x_j$

Proof: Switching the signs of a_i and b_i preserves inequality, so we may assume that $a_i > b_i$. Similarly, we can assume that $a_j > b_j$. If $b_i b_j = 0$, then both sides are zero, so we may assume that b_i and b_j are positive. We then have from the definitions of r_i and x_i that

$$\begin{aligned} r_i &= \frac{a_i}{b_i} - 1 \\ r_j &= \frac{a_j}{b_j} - 1 \\ x_i &= b_i \\ x_j &= b_j. \end{aligned}$$

This means that

$$\begin{aligned} \min(r_i, r_j) x_i x_j &= \min\left(\frac{a_i}{b_i} - 1, \frac{a_j}{b_j} - 1\right) b_i b_j \\ &= \min(a_i b_j, a_j b_i) - b_i b_j \\ &= \min(a_i b_j, a_j b_i) - \min(a_i a_j, b_i b_j). \end{aligned}$$

This concludes the proof of Lemma 2. ■

We can then apply Lemma 2 and Lemma 1 in order to get that

$$\begin{aligned} \sum_{i,j} \min(a_i b_j, a_j b_i) - \sum_{i,j} \min(a_i a_j, b_i b_j) &= \sum_{i,j} [\min(a_i b_j, a_j b_i) - \min(a_i a_j, b_i b_j)] \\ &= \sum_{i,j} \min(r_i, r_j) x_i x_j \\ &\geq 0. \end{aligned}$$

This implies the desired inequality.

Solution 2

WLOG let both sequences be increasing. Furthermore allow the sequence $a_i \leq b_i$. Hence, it is sufficient to show that

$$\sum_{i,j=1}^n a_i a_j \leq \sum_{i,j=1}^n \min(a_i b_j, a_j b_i)$$

Lemma 1: $\min(a_i b_j, a_j b_i) = a_i b_j$ for $i \leq j$.

Proof of Lemma 1:

Through the Rearrangement Inequality,

$$\begin{aligned} a_i b_i + a_j b_j &\geq a_i b_j + a_j b_i \\ a_i b_i + a_j b_j - a_i b_j - a_j b_i &\geq 0 \\ (a_j - a_i)(b_j - b_i) &\geq 0 \end{aligned}$$

As both sequences are increasing, the property holds.

(Note: A proof by contradiction of assuming that the other inequality is true derives the fact that a_i, b_i is decreasing which goes against our WLOG).

Lemma 2: $\min(a_i b_j, a_j b_i) = a_j b_i$ for $i > j$

For space reasons, the proof is similar to that of Lemma 1.

2000 USAMO Q6:

Rewrite the sum as

$$\sum_{i,j=1}^n \min(a_i b_j, a_j b_i) = \sum_{i=j} \min(a_i b_i, a_i b_i) + \sum_{i < j} \min(a_i b_j, a_j b_i) + \sum_{i > j} \min(a_i b_j, a_j b_i)$$

Obviously, $2n$ is even hence form the $n \times n$ matrix. The RHS sum $\sum_{i \leq j} a_i b_j + \sum_{i > j} a_j b_i$ then satisfies $\sum_{i > j} a_j b_i = \sum_{i < j} a_i b_j$. This is trivial by considering all values $a_j b_i$ below the diagonal of the matrix, $a_i b_j$ above the diagonal, and all $a_i b_i$ in the diagonal itself. It is then clear that the RHS simplifies to

$$\sum_i a_i b_i + 2 \sum_{i < j} a_i b_j$$

Note that

$$\sum_{i,j=1}^n a_i a_j = \sum_i a_i^2 + 2 \sum_{i < j} a_i a_j$$

Because $a_i \leq b_i$, we have $a_i^2 \leq a_i b_i$ via Rearrangement Inequality. For $i < j$, $a_i a_j \leq a_i b_j$ by the Lemma's. Hence, through inequality equivalencies,

$$\begin{aligned} \sum_i a_i^2 + 2 \sum_{i < j} a_i a_j &\leq \sum_i a_i b_i + 2 \sum_{i < j} a_i b_j \\ \iff \sum_{i,j=1}^n a_i a_j &\leq \sum_{i,j=1}^n \min(a_i b_j, a_j b_i) \end{aligned}$$

as desired. $\square$

§30 USAMO 2001

1. (*USAMO 2001*) Each of eight boxes contains six balls. Each ball has been colored with one of n colors, such that no two balls in the same box are the same color, and no two colors occur together in more than one box. Determine, with justification, the smallest integer n for which this is possible.

Solution 1

We claim that $n = 23$ is the minimum. Consider the following construction (replacing colors with numbers) which fulfills this:

$$\begin{bmatrix} 1 & 1 & 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 7 & 12 & 7 & 8 & 9 & 10 & 11 \\ 3 & 8 & 13 & 12 & 13 & 14 & 15 & 16 \\ 4 & 9 & 14 & 17 & 17 & 17 & 18 & 19 \\ 5 & 10 & 15 & 18 & 20 & 22 & 20 & 21 \\ 6 & 11 & 16 & 19 & 21 & 23 & 22 & 23 \end{bmatrix}$$

Suppose a configuration exists with $n \leq 22$.

Suppose a ball appears 5 or more times. Then the remaining balls of the 5 boxes must be distinct, so that there are at least $n \geq 5 \cdot 5 + 1 = 26$ balls, contradiction. If a ball appears 4 or more times, the remaining balls of the 4 boxes must be distinct, leading to $5 \cdot 4 + 1 = 21$ balls. The fifth box can contain at most four balls from the previous boxes, and then the remaining two balls must be distinct, leading to $n \geq 2 + 21 = 23$, contradiction.

However, by the **Pigeonhole Principle**, at least one ball must appear 3 times. **Without loss of generality** suppose that 1 appears three times, and let the boxes that contain these have balls with colors $\{1, 2, 3, 4, 5, 6\}$, $\{1, 7, 8, 9, 10, 11\}$, $\{1, 12, 13, 14, 15, 16\}$. Each of the remaining five boxes can have at most 3 balls from each of these boxes. Thus, each of the remaining five boxes must have 3 additional balls > 16 . Thus, it is necessary that we use $\leq 22 - 16 = 6$ balls to fill a 3×5 grid by the same rules.

Again, no balls may appear ≥ 4 times, but by Pigeonhole, one ball must appear 3 times. **Without loss of generality**, let this ball have color 17; then the three boxes containing 17 must have at least $2 \cdot 3 + 1 = 7$ balls, contradiction.

Therefore, $n = 23$ is the minimum.

Solution 2

Similar to the above solution, no color can appear 4 times or more.

We can use PIE. Let $S_1, S_2, \dots, S_8$ be the colors in each of the 8 boxes. Then $n = |\cup_{i=1}^8 S_i|$. By PIE we know that

$$|\cup_{i=1}^8 S_i| = \sum_{i=1}^8 |S_i| - \sum_{1 \leq i < j \leq 8} |S_i \cap S_j| + \sum_{1 \leq i < j < k \leq 8} |S_i \cap S_j \cap S_k| - \sum_{1 \leq i < j < k < l \leq 8} |S_i \cap S_j \cap S_k \cap S_l| \cdots$$

Note however that no color can appear 4 times or more, so all the items after $\sum_{1 \leq i < j < k \leq 8} |S_i \cap S_j \cap S_k|$ must be equal to 0, so

$$|\cup_{i=1}^8 S_i| = \sum_{i=1}^8 |S_i| - \sum_{1 \leq i < j \leq 8} |S_i \cap S_j| + \sum_{1 \leq i < j < k \leq 8} |S_i \cap S_j \cap S_k|.$$

Now note that for each i , $|S_i| = 6$, so $\sum_{i=1}^8 |S_i| = 48$. Also no pair of colors appears twice, so for any i, j , $|S_i \cap S_j| \leq 1$, and so $\sum_{1 \leq i < j \leq 8} |S_i \cap S_j| \leq \binom{8}{2} = 28$. Thus, $n \geq 48 - 28 + \sum_{1 \leq i < j < k \leq 8} |S_i \cap S_j \cap S_k|$. Given that there are n colors, to minimize the number of colors in 3 boxes means each color is in either 2 or 3 boxes, and since there are 48 total spots, this means there are $48 - 2n$ colors in 3 boxes and $3n - 48$ colors in 2. Thus $n \geq 20 + 48 - 2n$, so $3n \geq 68$, and $n \geq 23$ and we are done.

Solution 3

Let $m_{i,j}$ be the number of balls which are the same color as the j^{th} ball in box i (including that ball). For a fixed box i , $1 \leq i \leq 8$, consider the sums

$$S_i = \sum_{j=1}^6 m_{i,j} \quad \text{and} \quad s_i = \sum_{j=1}^6 \frac{1}{m_{i,j}}.$$

For each fixed i , since no pair of colors is repeated, each of the remaining seven boxes can contribute at most one ball to S_i . Thus, $S_i \leq 13$. It follows by the convexity of $f(x) = 1/x$ that s_i is minimized when one of the $m_{i,j}$ is equal to 3 and the other five equal to 2. Hence $s_i \geq 17/6$. Note that

$$n = \sum_{i=1}^8 \sum_{j=1}^6 \frac{1}{m_{i,j}} \geq 8 \cdot \frac{17}{6} = \frac{68}{3}.$$

Hence there must be 23 colors. The construction for $n = 23$ is above.

2. (USAMO 2001) Let ABC be a triangle and let ω be its incircle. Denote by D_1 and E_1 the points where ω is tangent to sides BC and AC , respectively. Denote by D_2 and E_2 the points on sides BC and AC , respectively, such that $CD_2 = BD_1$ and $CE_2 = AE_1$, and denote by P the point of intersection of segments AD_2 and BE_2 . Circle ω intersects segment AD_2 at two points, the closer of which to the vertex A is denoted by Q . Prove that $AQ = D_2P$.

Solution 1

It is well known that the excircle opposite A is tangent to $\overline{BC}$ at the point D_2 . (Proof: let the points of tangency of the excircle with the lines BC, AB, AC be D_3, F, G respectively. Then $AB + BD_3 = AB + BF = AF = AG = AC + CG = AC + CD_3$. It follows that $2CD_3 = AB + BC - AC$, and $CD_3 = s - b = BD_1 = CD_2$, so $D_3 \equiv D_2$.)

Now consider the homothety that carries the incircle of $\triangle ABC$ to its excircle. The homothety also carries Q to D_2 (since A, Q, D_2 are collinear), and carries the tangency points E_1 to G . It follows that $\frac{AQ}{QD_2} = \frac{AE_1}{E_1G} = \frac{s-a}{E_1C+CD_2} = \frac{s-a}{CD_1+BD_1} = \frac{s-a}{a}$.

Solution 2

Lemma: Segment D_1Q is a diameter of circle ω .

Let l be the line tangent to circle ω at Q' , and let l intersect the segments AB and AC at B_1 and C_1 , respectively. Then ω is an excircle of triangle AB_1C_1 . Let $\mathbf{H}_1$ denote the dilation with its center at A and ratio AD'/AQ' . Since $l \perp D_1Q'$ and $BC \perp D_1Q'$, $l \parallel BC$. Hence $AB/AB_1 = AC/AC_1 = AD'/AQ'$. Thus $\mathbf{H}_1(Q') = D'$, $\mathbf{H}_1(B_1) = B$, and $\mathbf{H}_1(C_1) = C$. It also follows that an excircle Ω of triangle ABC is tangent to the side BC at D' .

$$CD_1 = \frac{1}{2}(BC + CA - AB).$$
$$AX = AY = \frac{1}{2}(AX + AY) = \frac{1}{2}(AB + BX + YC + CA) = \frac{1}{2}(AB + BC + CA).$$
$$BD' = BX = AX - AB = \frac{1}{2}(BC + CA - AB).$$
$$BD_2 = BD_1 - D_2D_1 = D_2C - D_2D_1 = CD_1 = BD',$$
$$QD_2 = 2IM_1$$

and $AD_2 \parallel M_1I$. Similarly, we can prove that $BE_2 \parallel M_2I$.

Let G be the centroid of triangle ABC . Thus segments AM_1 and BM_2 intersect at G . Define transformation $\mathbf{H}_2$ as the dilation with its center at G and ratio $-1/2$. Then $\mathbf{H}_2(A) = M_1$ and $\mathbf{H}_2(B) = M_2$. Under the dilation, parallel lines go to parallel lines and the intersection of two lines goes to the intersection of their images. Since $AD_2 \parallel M_1I$ and $BE_2 \parallel M_2I$, $\mathbf{H}_2$ maps lines AD_2 and BE_2 to lines M_1I and M_2I , respectively. It also follows that $\mathbf{H}_2(I) = P$ and

$$\frac{IM_1}{AP} = \frac{GM_1}{AG} = \frac{1}{2}$$

or

$$AP = 2IM_1.$$

This yields

$$AQ = AP - QP = 2IM_1 - QP = QD_2 - QP = PD_2,$$

as desired.

Note: We used directed lengths in our calculations to avoid possible complications caused by the different shapes of triangle ABC .

Solution 3

Here is a rather nice solution using barycentric coordinates:

Let A be $(1, 0, 0)$, B be $(0, 1, 0)$, and C be $(0, 0, 1)$. Let the side lengths of the triangle be a, b, c and the semi-perimeter s .

Now,

$$CD_1 = s - c, BD_1 = s - b, AE_1 = s - a, CE_1 = s - c.$$

Thus,

$$CD_2 = s - b, BD_2 = s - c, AE_2 = s - c, CE_2 = s - a.$$

Therefore, $D_2 = (0 : s - b : s - c)$ and $E_2 = (s - a : 0 : s - c)$. Clearly then, the non-normalized coordinates of $P = (s - a : s - b : s - c)$.

Normalizing, we have that

$$D_2 = \left(0, \frac{s-b}{s}, \frac{s-c}{s}\right), E_2 = \left(\frac{s-a}{s}, 0, \frac{s-c}{s}\right), P = \left(\frac{s-a}{s}, \frac{s-b}{s}, \frac{s-c}{s}\right).$$

Now, we find the point Q' inside the triangle on the line AD_2 such that $AQ' = D_2P$. It is then sufficient to show that this point lies on the incircle.

P is the fraction $\frac{s-a}{s}$ of the way "up" the line segment from D_2 to A . Thus, we are looking for the point that is $\frac{s-a}{s}$ of the way "down" the line segment from A to D_2 , or, the fraction $1 - \frac{s-a}{s}$ of the way "up".

Thus, Q' has normalized x -coordinate $1 - \frac{s-a}{s} = \frac{a}{s}$.

As the line AD_2 has equation $(s-c)y = (s-b)z$, it can easily be found that

$$Q' = \left(\frac{a^2}{as}, \frac{(s-a)(s-b)}{as}, \frac{(s-a)(s-c)}{as}\right) = (a^2 : (s-a)(s-b) : (s-a)(s-c)).$$

Recalling that the equation of the incircle is

$$a^2yz + b^2xz + c^2xy + (x+y+z)[(s-a)^2x + (s-b)^2y + (s-c)^2z] = 0.$$

We must show that this equation is true for Q' 's values of x, y, z .

Plugging in our values, this means showing that

$$a^2(s-a)^2(s-b)(s-c) + a^2(s-a)[b^2(s-c) + c^2(s-b)] + as[a^2(s-a)^2 + (s-a)(s-b)^3 + (s-a)(s-c)^3] = 0.$$

Dividing by $a(s-a)$, this is just

$$a(s-a)(s-b)(s-c) + a[b^2(s-c) + c^2(s-b)] + a^2s(s-a) + s(s-b)^3 + s(s-c)^3 = 0.$$

Plugging in the value of s :

$$\frac{a(-a+b+c)(a-b+c)(a+b-c)}{8} + \frac{ab^2(a+b-c)}{2} + \frac{ac^2(a-b+c)}{2} + \frac{a^2(a+b+c)(-a+b+c)}{4} + \frac{(a+b+c)(-a+b+c)^3}{4} + \frac{(a+b+c)(a-b+c)^3}{4}$$

$$2a(-a+b+c)(a-b+c)(a+b-c) + 8ab^2(a+b-c) + 8ac^2(a-b+c) + 4a^2(a+b+c)(-a+b+c) + (a+b+c)(a-b+c)^3 + (a+b+c)(-a+b+c)^3$$

$$2a[(-a+b+c)(a-b+c)(a+b-c) + 4b^2(a+b-c) + 4c^2(a-b+c)] + (a+b+c)[4a^2(-a+b+c) + (a-b+c)^3 + (-a+b+c)^3]$$

The first bracket is just

$$\begin{aligned} & -a^3 - b^3 - c^3 + a^2b + a^2c + b^2c + b^2a + c^2a + c^2b - 2abc + 4ab^2 + 4b^3 - 4b^2c + 4ac^2 - 4bc^2 + 4c^3 \\ & = -a^3 + 3b^3 + 3c^3 + a^2b + a^2c - 3b^2c + 5b^2a + 5c^2a - 3c^2b - 2abc \end{aligned}$$

and the second bracket is

$$-2a^3 + 4a^2b + 4a^2c + 6ab^2 + 6ac^2 - 12abc.$$

Dividing everything by $2a$ gives

$$-a^3 + 3b^3 + 3c^3 + a^2b + a^2c - 3b^2c + 5b^2a + 5c^2a - 3c^2b - 2abc + (a+b+c)(-a^2 + 2ab + 2ac + 3b^2 + 3c^2 - 6bc),$$

which is 0, as desired.

As Q' lies on the incircle and AD_2 , $Q' = Q$, and our proof is complete.

Solution 4

We again use Barycentric coordinates. As before, let A be $(1, 0, 0)$, B be $(0, 1, 0)$, and C be $(0, 0, 1)$. Also

$$D_1 = \left(0, \frac{s-c}{a}, \frac{s-b}{a}\right), D_2 = \left(0, \frac{s-b}{a}, \frac{s-c}{a}\right), E_2 = \left(\frac{s-a}{b}, 0, \frac{s-c}{b}\right), P = \left(\frac{s-a}{s}, \frac{s-b}{s}, \frac{s-c}{s}\right).$$

Now, consider a point Q' for which $AQ' = PD_2$. Then working component-wise, we get $Q' + P = A + D_2$ from which we can easily get the coordinates of Q' as;

$$Q' = \left(\frac{a}{s}, \frac{(s-a)(s-b)}{sa}, \frac{(s-a)(s-c)}{sa}\right)$$

It suffices to show that $Q' = Q$.

Let $I = \left(\frac{a}{2s}, \frac{b}{2s}, \frac{c}{2s}\right)$ be the incenter of triangle ABC . We claim that I is the midpoint of $Q'D_1$. Indeed,

$$\frac{1}{2} \left(0 + \frac{a}{s}\right) = \frac{a}{2s}$$

$$\frac{1}{2} \left(\frac{(s-a)(s-b)}{sa} + \frac{s-c}{a} \right) = \frac{(s-a)(s-b) + s(s-c)}{2sa} = \frac{ab}{2sa} = \frac{b}{2s}$$

$$\frac{1}{2} \left(\frac{(s-a)(s-c)}{sa} + \frac{s-b}{a} \right) = \frac{c}{2s}$$

Hence the claim has been proved.

Since I is the center of ω and the midpoint of $Q'D_1$, thus Q' is the point diametrically opposite to D_1 , and hence it lies on ω and closer to A off the 2 points. Thus $Q = Q'$ as desired.

3. (USAMO 2001) Let $a, b, c \geq 0$ and satisfy

$$a^2 + b^2 + c^2 + abc = 4.$$

Show that

$$0 \leq ab + bc + ca - abc \leq 2.$$

Solution 1

First we prove the lower bound.

Note that we cannot have a, b, c all greater than 1. Therefore, suppose $a \leq 1$. Then

$$ab + bc + ca - abc = a(b + c) + bc(1 - a) \geq 0.$$

Note that, by the **Pigeonhole Principle**, at least two of a, b, c are either both greater than or less than 1. **Without loss of generality**, let them be b and c . Therefore, $(b-1)(c-1) \geq 0$. From the given equation, we can express a in terms of b and c as

$$a = \frac{\sqrt{(4-b^2)(4-c^2)} - bc}{2}$$

Thus,

$$ab + bc + ca - abc = -a(b-1)(c-1) + a + bc \leq a + bc = \frac{\sqrt{(4-b^2)(4-c^2)} + bc}{2}$$

From the **Cauchy-Schwarz Inequality**,

$$\frac{\sqrt{(4-b^2)(4-c^2)} + bc}{2} \leq \frac{\sqrt{(4-b^2+b^2)(4-c^2+c^2)}}{2} = 2.$$

This completes the proof.

Solution 2

The proof for the lower bound is the same as in the first solution.

Now we prove the upper bound. Let us note that at least two of the three numbers a, b , and c are both greater than or equal to 1 or less than or equal to 1. Without loss of generality, we assume that the numbers with this property are b and c . Then we have

$$(1-b)(1-c) \geq 0.$$

The given equality $a^2 + b^2 + c^2 + abc = 4$ and the inequality $b^2 + c^2 \geq 2bc$ imply that

$$a^2 + 2bc + abc \leq 4,$$

or

$$bc(2+a) \leq 4-a^2.$$

Dividing both sides of the last inequality by $2+a$ yields

$$bc \leq 2-a.$$

Thus,

$$ab+bc+ca-abc \leq ab+2-a+ac(1-b) = 2-a(1+bc-b-c) = 2-a(1-b)(1-c) \leq 2,$$

as desired.

The last equality holds if and only if $b=c$ and $a(1-b)(1-c)=0$. Hence equality for the upper bound holds if and only if (a,b,c) is one of the triples $(1,1,1)$, $(0,\sqrt{2},\sqrt{2})$, $(\sqrt{2},0,\sqrt{2})$, and $(\sqrt{2},\sqrt{2},0)$. Equality for the lower bound holds if and only if (a,b,c) is one of the triples $(2,0,0)$, $(0,2,0)$ and $(0,0,2)$.

Solution 3

The proof for the lower bound is the same as in the first solution.

Now we prove the upper bound. It is clear that $a,b,c \leq 2$. If $abc=0$, then the result is trivial. Suppose that $a,b,c > 0$. Solving for a yields

$$a = \frac{-bc + \sqrt{b^2c^2 - 4(b^2 + c^2 - 4)}}{2} = \frac{-bc + \sqrt{(4-b^2)(4-c^2)}}{2}.$$

This asks for the trigonometric substitution $b = 2\sin u$ and $c = 2\sin v$, where $0^\circ < u, v < 90^\circ$. Then

$$a = 2(-\sin u \sin v + \cos u \cos v) = 2\cos(u+v),$$

and if we set $u = B/2$ and $v = C/2$, then $a = 2\sin(A/2)$, $b = 2\sin(B/2)$, and $c = 2\sin(C/2)$, where A, B , and C are the angles of a triangle. We have

$$\begin{aligned} ab &= 4 \sin \frac{A}{2} \sin \frac{B}{2} \\ &= 2\sqrt{\sin A \tan \frac{A}{2} \sin B \tan \frac{B}{2}} = 2\sqrt{\sin A \tan \frac{B}{2} \sin B \tan \frac{A}{2}} \\ &\leq \sin A \tan \frac{B}{2} + \sin B \tan \frac{A}{2} \\ &= \sin A \cot \frac{A+C}{2} + \sin B \cot \frac{B+C}{2}, \end{aligned}$$

where the inequality step follows from AM-GM. Likewise,

$$\begin{aligned} bc &\leq \sin B \cot \frac{B+A}{2} + \sin C \cot \frac{C+A}{2}, \\ ca &\leq \sin A \cot \frac{A+B}{2} + \sin C \cot \frac{C+B}{2}. \end{aligned}$$

Therefore

$$\begin{aligned} ab+bc+ca &\leq (\sin A + \sin B) \cot \frac{A+B}{2} + (\sin B + \sin C) \cot \frac{B+C}{2} + (\sin C + \sin A) \cot \frac{C+A}{2} \\ &= 2 \left(\cos \frac{A-B}{2} \cos \frac{A+B}{2} + \cos \frac{B-C}{2} \cos \frac{B+C}{2} + \cos \frac{C-A}{2} \cos \frac{C+A}{2} \right) \end{aligned}$$

$$\begin{aligned}
 &= 2(\cos A + \cos B + \cos C) \\
 &= 6 - 4 \left(\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} \right) \\
 &= 6 - (a^2 + b^2 + c^2) = 2 + abc,
 \end{aligned}$$

as desired.

4. (*USAMO 2001*) Let P be a point in the plane of triangle ABC such that the segments PA , PB , and PC are the sides of an obtuse triangle. Assume that in this triangle the obtuse angle opposes the side congruent to PA . Prove that $\angle BAC$ is acute.

Solution 1

We know that $PB^2 + PC^2 < PA^2$ and we wish to prove that $AB^2 + AC^2 > BC^2$. It would be sufficient to prove that

$$PB^2 + PC^2 + AB^2 + AC^2 \geq PA^2 + BC^2.$$

Set $A(0, 0)$, $B(1, 0)$, $C(x, y)$, $P(p, q)$. Then, we wish to show

$$\begin{aligned}
 (p-1)^2 + q^2 + (p-x)^2 + (q-y)^2 + 1 + x^2 + y^2 &\geq p^2 + q^2 + (x-1)^2 + y^2 \\
 2p^2 + 2q^2 + 2x^2 + 2y^2 - 2p - 2px - 2qy + 2 &\geq p^2 + q^2 + x^2 + y^2 - 2x + 1 \\
 p^2 + q^2 + x^2 + y^2 + 2x - 2p - 2px - 2qy + 1 &\geq 0 \\
 (x-p)^2 + (q-y)^2 + 2(x-p) + 1 &\geq 0 \\
 (x-p+1)^2 + (q-y)^2 &\geq 0,
 \end{aligned}$$

which is true by the trivial inequality.

Solution 2

Let A be the origin. For a point Q , denote by q the vector $\overrightarrow{AQ}$, and denote by $|q|$ the length of q . The given conditions may be written as

$$|p-b|^2 + |p-c|^2 < |p|^2,$$

or

$$p \cdot p + b \cdot b + c \cdot c - 2p \cdot b - 2p \cdot c < 0.$$

Adding $2b \cdot c$ on both sides of the last inequality gives

$$|p-b-c|^2 < 2b \cdot c.$$

Since the left-hand side of the last inequality is nonnegative, the right-hand side is positive. Hence

$$\cos \angle BAC = \frac{b \cdot c}{|b||c|} > 0,$$

that is, $\angle BAC$ is acute.

Solution 3

For the sake of contradiction, let's assume to the contrary that $\angle BAC$. Let $AB = c$, $BC = a$, and $CA = b$. Then $a^2 \geq b^2 + c^2$. We claim that the quadrilateral

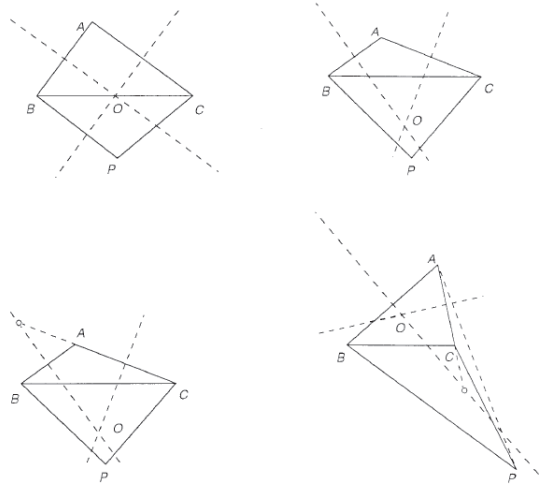
$ABPC$ is convex. Now applying the generalized Ptolemy's Theorem to the convex quadrilateral $ABPC$ yields

$$a \cdot PA \leq b \cdot PB + c \cdot PC \leq \sqrt{b^2 + c^2} \sqrt{PB^2 + PC^2} \leq a \sqrt{PB^2 + PC^2},$$

where the second inequality is by Cauchy-Schwarz. This implies $PA^2 \leq PB^2 + PC^2$, in contradiction with the facts that PA , PB , and PC are the sides of an obtuse triangle and $PA > \max\{PB, PC\}$.

We present two arguments to prove our claim.

First argument: Without loss of generality, we may assume that A , B , and C are in counterclockwise order. Let lines l_1 and l_2 be the perpendicular bisectors of segments AB and AC , respectively. Then l_1 and l_2 meet at O , the circumcenter of triangle ABC . Lines l_1 and l_2 cut the plane into four regions and A is in the interior of one of these regions. Since $PA > PB$ and $PA > PC$, P must be in the interior of the region that opposes A . Since $\angle BAC$ is not acute, ray AC does not meet l_1 and ray AB does not meet l_2 . Hence B and C must lie in the interiors of the regions adjacent to A . Let $\mathcal{R}_X$ denote the region containing X . Then $\mathcal{R}_A$, $\mathcal{R}_B$, $\mathcal{R}_P$, and $\mathcal{R}_C$ are the four regions in counterclockwise order. Since $\angle BAC \geq 90^\circ$, either O is on side BC or O and A are on opposite sides of line BC . In either case P and A are on opposite sides of line BC . Also, since ray AB does not meet l_2 and ray AC does not meet l_1 , it follows that $\mathcal{R}_P$ is entirely in the interior of $\angle BAC$. Hence B and C are on opposite sides of AP . Therefore $ABPC$ is convex.



Second argument: Since $PA > PB$ and $PA > PC$, A cannot be inside or on the sides of triangle PBC . Since $PA > PB$, we have $\angle ABP > \angle BAP$ and hence $\angle BAC \geq 90^\circ > \angle BAP$. Hence C cannot be inside or on the sides of triangle BAP . Symmetrically, B cannot be inside or on the sides of triangle CAP . Finally, since $\angle ABP > \angle BAP$ and $\angle ACP > \angle CAP$, we have

$$\angle ABP + \angle ACP > \angle BAC \geq 90^\circ \geq \angle ABC + \angle ACB.$$

Therefore P cannot be inside or on the sides of triangle ABC . Since this covers all four cases, $ABPC$ is convex.

Solution 4

Let P be the origin in vector space, and let a, b, c denote the position vectors of A, B, C respectively. Then the obtuse triangle condition, $PA^2 > PB^2 + PC^2$,

becomes $a^2 > b^2 + c^2$ using the fact that the square of a vector (the dot product of itself and itself) is the square of its magnitude. Now, notice that to prove $\angle BAC$ is acute, it suffices to show that $(a - b)(a - c) > 0$, or $a^2 - ab - ac + bc > 0$. But this follows from the observation that

$$(-a + b + c)^2 \geq 0,$$

which leads to

$$2a^2 - 2ab - 2ac + 2bc > a^2 + b^2 + c^2 - 2ab - 2ac + 2bc \geq 0$$

and therefore our desired conclusion.

Solution 5

Let M, N be midpoints of AP and BC , respectively. For the points A, B, P, C ; let's apply Euler's quadrilateral formula,

$$AB^2 + BP^2 + PC^2 + CA^2 = AP^2 + BC^2 + 4MN^2 \geq AP^2 + BC^2.$$

Given that $AP^2 > BP^2 + PC^2$. Thus,

$$AB^2 + AC^2 > BC^2.$$

and we get $\angle BAC$ is acute.

(Lokman GÖKÇE)

Solution 6

Without loss of generality, assume that in a Cartesian coordinate system, A is at the point $(0, 0)$ and C is at the point $(1, 0)$. Let B be at the point (b_x, b_y) and P be at the point (p_x, p_y) . Without loss of generality, also assume that $b_y > 0$.

Now, assume for contradiction that $\angle BAC$ is not acute. Since PA , PB , and PC are the sides of an obtuse triangle, with PA the longest side, it follows that $PA^2 > PB^2 + PC^2$, implying that $p_x^2 + p_y^2 > (p_x - b_x)^2 + (p_y - b_y)^2 + (p_x - 1)^2 + p_y^2$. This inequality simplifies to $b_x^2 - 2p_x b_x + b_y^2 - 2p_y b_y + p_x^2 - 2p_x + 1 + p_y^2 < 0$. Note that since $p_x^2 - 2p_x + 1$ and $b_y^2 - 2p_y b_y + p_y^2$ are both perfect squares, all terms of this inequality except for $-2p_x b_x$ are already guaranteed to be nonnegative.

If $p_x < 0$, then P would be closer to A than to C , but since $PA^2 = PB^2 + PC^2$, this is not possible. Therefore, $p_x \geq 0$. Since $\angle BAC$ not being acute implies that $b_x \leq 0$, it follows that $-2p_x b_x \geq 0$. But now since all terms of $b_x^2 - 2p_x b_x + b_y^2 - 2p_y b_y + p_x^2 - 2p_x + 1 + p_y^2 < 0$ are guaranteed to be nonnegative, this entire expression cannot be negative, leading to a contradiction. Therefore, $\angle BAC$ is acute.

Solution 7

To maximize $\angle BAC$, we try to maximize $\angle PAB$ and $\angle PAC$, which occurs when $PB = PA \sin \angle PAB$ and $PC = PA \sin \angle PAC$. Let $\angle PAB = \theta_1$ and let $\angle PAC = \theta_2$.

Since PC, PB , and PA form an obtuse triangle, $PC^2 + PB^2 < PA^2$.

We now have that $\sin^2(\theta_1) + \sin^2(\theta_2) = \frac{PB^2 + PC^2}{PA^2} < 1$. The premise is now obvious; $\sin^2(\theta_2) < \cos^2(\theta_1)$, which means that $\theta_2 < 90 - \theta_1$, so we're done.

5. (USAMO 2001) Let S be a set of integers (not necessarily positive) such that
- (a) there exist $a, b \in S$ with $\gcd(a, b) = \gcd(a - 2, b - 2) = 1$;
 - (b) if x and y are elements of S (possibly equal), then $x^2 - y$ also belongs to S .

Prove that S is the set of all integers.

Solution

In the solution below we use the expression S is stable under $x \mapsto f(x)$ to mean that if x belongs to S then $f(x)$ also belongs to S . If $c, d \in S$, then by (B), S is stable under $x \mapsto c^2 - x$ and $x \mapsto d^2 - x$, hence stable under $x \mapsto c^2 - (d^2 - x) = x + (c^2 - d^2)$. Similarly S is stable under $x \mapsto x + (d^2 - c^2)$. Hence S is stable under $x \mapsto x + n$ and $x \mapsto x - n$ whenever n is an integer linear combination of numbers of the form $c^2 - d^2$ with $c, d \in S$. In particular, this holds for $n = m$, where $m = \gcd\{c^2 - d^2 : c, d \in S\}$.

Since $S \neq \emptyset$ by (A), it suffices to prove that $m = 1$. For the sake of contradiction, assume that $m \neq 1$. Let p be a prime dividing m . Then $c^2 - d^2 \equiv 0 \pmod{p}$ for all $c, d \in S$. In other words, for each $c, d \in S$, either $d \equiv c \pmod{p}$ or $d \equiv -c \pmod{p}$. Given $c \in S$, $c^2 - c \in S$ by (B), so $c^2 - c \equiv c \pmod{p}$ or $c^2 - c \equiv -c \pmod{p}$. Hence

$$\text{For each } c \in S, \text{ either } c \equiv 0 \pmod{p} \text{ or } c \equiv 2 \pmod{p}. \quad (*)$$

By (A), there exist some a and b in S such that $\gcd(a, b) = 1$, that is, at least one of a or b cannot be divisible by p . Denote such an element of S by α : thus, $\alpha \not\equiv 0 \pmod{p}$. Similarly, by (A), $\gcd(a - 2, b - 2) = 1$, so p cannot divide both $a - 2$ and $b - 2$. Thus, there is an element of S , call it β , such that $\beta \not\equiv 2 \pmod{p}$. By (*), $\alpha \equiv 2 \pmod{p}$ and $\beta \equiv 0 \pmod{p}$. By (B), $\beta^2 - \alpha \in S$. Taking $c = \beta^2 - \alpha$ in (*) yields either $-2 \equiv 0 \pmod{p}$ or $-2 \equiv 2 \pmod{p}$, so $p = 2$. Now (*) says that all elements of S are even, contradicting (A). Hence our assumption is false and S is the set of all integers.

6. (USAMO 2001) Each point in the plane is assigned a real number such that, for any triangle, the number at the center of its inscribed circle is equal to the arithmetic mean of the three numbers at its vertices. Prove that all points in the plane are assigned the same number.

Solution

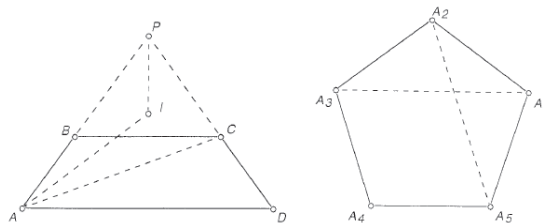
We label each upper case point with the corresponding lower case letter as its assigned number. The key step is the following lemma.

Lemma: If $ABCD$ is an isosceles trapezoid, then $a + c = b + d$.

Proof: Assume without loss of generality that $BC \parallel AD$, and that rays AB and DC meet at P . Let I be the incenter of triangle PAC , and let line l bisect $\angle APD$. Then I is on l , so reflecting everything across line l shows that I is also the incenter of triangle PDB . Therefore,

$$\frac{p + a + c}{3} = i = \frac{p + b + d}{3}.$$

Hence $a + c = b + d$, as desired. ■



For any two distinct points A_1 and A_2 in the plane, we construct a regular pentagon $A_1A_2A_3A_4A_5$. Applying the lemma to isosceles trapezoids $A_1A_3A_4A_5$ and $A_2A_3A_4A_5$ yields

$$a_1 + a_4 = a_3 + a_5 \quad \text{and} \quad a_2 + a_4 = a_3 + a_5.$$

Hence $a_1 = a_2$. Since A_1 and A_2 were arbitrary, all points in the plane are assigned the same number.

§31 USAMO 2002

1. (USAMO 2002) Let S be a set with 2002 elements, and let N be an integer with $0 \leq N \leq 2^{2002}$. Prove that it is possible to color every subset of S either blue or red so that the following conditions hold:
 - (a) the union of any two red subsets is red;
 - (b) the union of any two blue subsets is blue;
 - (c) there are exactly N red subsets.

Solution 1

Let a set colored in such a manner be *properly colored*. We prove that any set with n elements can be properly colored for any $0 \leq N \leq 2^n$. We proceed by induction. The base case, $n = 0$, is trivial.

Suppose that our claim holds for $n = k$. Let $s \in S$, $|S| = k + 1$, and let S' denote the set of all elements of S other than s .

If $N \leq 2^k$, then we may color all subsets of S which contain s blue, and we may properly color S' . This is a proper coloring because the union of any two red sets must be a subset of S' , which is properly colored, and any the union of any two blue sets either must be in S' , which is properly colored, or must contain s and therefore be blue.

If $N > 2^k$, then we color all subsets containing s red, and we color $N - 2^k$ elements of S' red in such a way that S' is colored properly. Then S is properly colored, using similar reasoning as before. Thus the induction is complete.

Solution 2

If $N = 0$, color every subset blue. If $N = 2^{2002}$, color every subset red. Otherwise, let S be $\{0, 1, \dots, 2001\}$. Write N in binary, i.e., let

$$N = \sum_{i=1}^k 2^{a_i},$$

where each of the a_i is an element of S . We color each of the a_i red and all the other elements of S blue. We color the empty set blue, and we color any other set the color of its largest element. This satisfies the problem's first two conditions, as the largest element of the union of two red (or blue) sets will have a red (or

blue) number as its largest element. In addition, for each integer $n \in S$, there are 2^n subsets of S with n as a maximal element, so $\sum_{i=1}^k 2^{a_i} = N$ subsets of S are colored red, as desired.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

2. (*USAMO 2002*) Let ABC be a triangle such that

$$\left(\cot \frac{A}{2}\right)^2 + \left(2 \cot \frac{B}{2}\right)^2 + \left(3 \cot \frac{C}{2}\right)^2 = \left(\frac{6s}{7r}\right)^2,$$

where s and r denote its [semiperimeter](#) and [inradius](#), respectively. Prove that triangle ABC is similar to a triangle T whose side lengths are all positive integers with no common divisor and determine those integers.

Solution

Let a, b, c denote BC, CA, AB , respectively. Since the line from a triangle's incenter to one of its vertices bisects the angle at the triangle's vertex, the condition of the problem is equivalent to the

$$\left(\frac{s-a}{r}\right)^2 + 4\left(\frac{s-b}{r}\right)^2 + 9\left(\frac{s-c}{r}\right)^2 = \left(\frac{6s}{7r}\right)^2,$$

or

$$\frac{(s-a)^2}{36} + \frac{(s-b)^2}{9} + \frac{(s-c)^2}{4} = \frac{s^2}{36+9+4}.$$

But by the [Cauchy-Schwarz Inequality](#), we know

$$\begin{aligned} (36+9+4) \left[\frac{(s-a)^2}{36} + \frac{(s-b)^2}{9} + \frac{(s-c)^2}{4} \right] &\geq [(s-a) + (s-b) + (s-c)]^2 \\ &= s^2 \\ \frac{(s-a)^2}{36} + \frac{(s-b)^2}{9} + \frac{(s-c)^2}{4} &\geq \frac{s^2}{36+9+4}, \end{aligned}$$

with equality only when $\frac{(s-a)^2}{36}, \frac{(s-b)^2}{9}, \frac{(s-c)^2}{4}$ are directly proportional to 36, 9, 4, respectively. Therefore (clearing denominators and taking square roots) our problem requires that $(s-a), (s-b), (s-c)$ be directly proportional to 36, 9, 4, and since $a = (s-b) + (s-c)$ etc., this is equivalent to the condition that a, b, c be in proportion with 13, 40, 45, Q.E.D.

Sidenote: A 13, 40, 45 obtuse triangle has an integer area of 252 and an inradius of $\frac{36}{7}$.

Solution 2 (Lagrange Multipliers)

Let D, E, F be the projections of the incenter onto the sides BC, AC , and AB , respectively, and let $x = AF, y = BD, z = CE$. We thus have

$$\cot \frac{A}{2} = \frac{x}{r}, \cot \frac{B}{2} = \frac{y}{r}, \cot \frac{C}{2} = \frac{z}{r}$$

and additionally,

$$\begin{aligned} x + y + z &= s \\ x^2 + 4y^2 + 9z^2 &= \left(\frac{6s}{7}\right)^2 \end{aligned}$$

We first determine the side lengths of the triangle similar to this triangle with semiperimeter 1. It is asked of us to show that such a triangle is unique, but we note that this is true if and only if the plane $x + y + z = 1$ and the ellipsoid

$x^2 + 4y^2 + 9z^2 = (\frac{6}{7})^2$ is tangent. This can be shown if we use the method of Lagrange Multipliers on the constraint $x + y + z = 1$ and function $x^2 + 4y^2 + 9z^2$ to prove that $(\frac{6}{7})^2$ is a extrema.

The Method of Lagrange Multipliers gives the system of

$$2x = \lambda$$

$$8y = \lambda$$

$$18z = \lambda$$

$$x + y + z = 1$$

from which we easily obtain $(x, y, z) = (\frac{36}{49}, \frac{9}{49}, \frac{4}{49})$. We calculate that this indeed satisfy $x^2 + 4y^2 + 9z^2 = (\frac{6}{7})^2$, so $(\frac{6}{7})^2$ is indeed an relative extrema. To check that it is a global extrema, one notes that as x , y , or z goes towards plus or minus infinity, the function $x^2 + 4y^2 + 9z^2$ goes to positive infinity, so that $(\frac{6}{7})^2$ is the global minima. Now, $AB = x + y = \frac{45}{49}$, $AC = x + z = \frac{40}{49}$, and $BC = y + z = \frac{13}{49}$, so we see that the desired side lengths are $AB = 45$, $AC = 40$, $BC = 13$. $\square$.

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

3. (*USAMO 2002*) Prove that any monic polynomial (a polynomial with leading coefficient 1) of degree n with real coefficients is the average of two monic polynomials of degree n with n real roots.

Solution 1

Lemma. For any set of ordered pairs of reals $\{(x_i, y_i)\}_{i=1}^n$ with $x_i \neq x_j$ for all $i \neq j$, there exists a unique monic real polynomial $P(x)$ of degree n such that $P(x_i) = y_i$ for all integers $i \in [1, n]$.

Proof 1. By the [Lagrange Interpolation Formula](#), there exists a unique real polynomial $Q(x)$ of degree less than n such that $Q(x_i) = y_i - x_i^n$. Then it is necessary and sufficient that $P(x) = x^n + Q(x)$.

Proof 2. Again by the Lagrange Interpolation Formula, there exists a unique real polynomial $Q(x)$ of degree less than n such that $Q(x_i) = y_i$. Then it is necessary and sufficient that $P(x) = Q(x) + \prod_{i=1}^n (x - x_i)$.

Let $F(x)$ be the monic polynomial of degree n given in the problem. Let $x_1, \dots, x_n$ be a sequence of strictly increasing reals, and choose $y_1, \dots, y_n$ such that for odd i , $y_i > \max(0, 2F(x_i))$, but for even i , $y_i < \min(0, 2F(x_i))$. Let $P(x)$ be the unique monic polynomial such that $P(x_i) = y_i$. We shall prove that $P(x)$ and $Q(x) = 2F(x) - P(x)$ satisfy the problem's conditions.

Note that for $i < n$, $P(x_i)$ and $P(x_{i+1})$ have different signs, so $P(x)$ has a real root between x_i and x_{i+1} . It follows that $P(x)$ has at least $n - 1$ real roots. But a polynomial with real coefficients must have an even number of non-real roots, so $P(x)$ must have n real roots. Similarly, $Q(x)$ must have n real roots. Q.E.D.

Solution 2

Let $F(x)$ be our polynomial. If $n = 1$, then we may let $F = x + a$, which is the average of the polynomials x and $x + 2a$, each of which has a real root. Otherwise, let

$$G(x) = \prod_{i=1}^{n-1} (x - 2i),$$

and let

$$P(x) = x^n - kG(x)$$

and

$$Q(x) = 2F(x) - x^n + kG(x).$$

We will prove that for sufficiently large k , $P(x)$ and $Q(x)$ satisfy the problem's conditions.

We note that for the values $1, 3, \dots, 2n-1$ of x , $G(x)$ alternates in sign, and always has magnitude at least 1 (since it is the product of several factors of magnitude at least one). On the other hand, there exists some c such that for all $x \in [1, 2n-1]$, $|x^n| < c$ and $|F(n) - x^n| < c$. Let k be greater than c . Then both $P(x)$ and $Q(x)$ alternate in sign for $x = 1, 3, \dots, 2n-1$. It follows that each has at least $n-1$ real roots. But since a polynomial with real coefficients can only have an even number of complex roots, this means that both P and Q must have n roots. Q.E.D.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

4. (USAMO 2002) Let $\mathbb{R}$ be the set of **real numbers**. Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 - y^2) = xf(x) - yf(y)$$

for all pairs of real numbers x and y .

Solution 1

We first prove that f is **odd**.

Note that $f(0) = f(x^2 - x^2) = xf(x) - xf(x) = 0$, and for nonzero y , $xf(x) + yf(-y) = f(x^2 - y^2) = xf(x) - yf(y)$, or $yf(-y) = -yf(y)$, which implies $f(-y) = -f(y)$. Therefore f is odd. Henceforth, we shall assume that all variables are non-negative.

If we let $y = 0$, then we obtain $f(x^2) = xf(x)$. Therefore the problem's condition becomes

$$f(x^2 - y^2) + f(y^2) = f(x^2).$$

But for any a, b , we may set $x = \sqrt{a}$, $y = \sqrt{b}$ to obtain

$$f(a - b) + f(b) = f(a).$$

(It is well known that the only **continuous** solutions to this functional equation are of the form $f(x) = kx$, but there do exist other solutions to this which are not solutions to the equation of this problem.)

We may let $a = 2t$, $b = t$ to obtain $2f(t) = f(2t)$.

Letting $x = t + 1$ and $y = t$ in the original condition yields

$$\begin{aligned} f(2t + 1) &= (t + 1)f(t + 1) - tf(t) \\ &= (t + 1)[f(t) + f(1)] - tf(t) \\ &= f(t) + (t + 1)f(1) \end{aligned}$$

But we know $f(2t + 1) = f(2t) + f(1) = 2f(t) + f(1)$, so we have $2f(t) + f(1) = f(t) + tf(1) + f(1)$, or

$$f(t) = tf(1).$$

Hence all solutions to our equation are of the form $f(x) = kx$. It is easy to see that real value of k will suffice.

Solution 2

As in the first solution, we obtain the result that f satisfies the condition

$$f(a) + f(b) = f(a + b).$$

We note that

$$f(x) = f\left[\left(\frac{x+1}{2}\right)^2 - \left(\frac{x-1}{2}\right)^2\right] = \frac{x+1}{2}f\left(\frac{x+1}{2}\right) - \frac{x-1}{2}f\left(\frac{x-1}{2}\right).$$

Since $f(2t) = 2f(t)$, this is equal to

$$\frac{(x+1)[f(x)+f(1)]}{4} - \frac{(x-1)[f(x)-f(1)]}{4} = \frac{xf(1)+f(x)}{2}$$

It follows that f must be of the form $f(x) = kx$.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

5. (*USAMO 2002*) Let a, b be integers greater than 2. Prove that there exists a positive integer k and a finite sequence $n_1, n_2, \dots, n_k$ of positive integers such that $n_1 = a$, $n_k = b$, and $n_i n_{i+1}$ is divisible by $n_i + n_{i+1}$ for each i ($1 \leq i < k$).

Solutions

We may say that two integers a and b are *connected* (and write $a \leftrightarrow b$) if there exists such a sequence of integers as described in the problem. For reference, we note that $\leftrightarrow$ is an **equivalence relation**: it is **reflexive** ($a \leftrightarrow a$), **symmetric** ($a \leftrightarrow b$ implies $b \leftrightarrow a$), and **transitive** ($a \leftrightarrow b \leftrightarrow c$ implies $a \leftrightarrow c$).

Solution 1

Note that for any divisor d of some n , $n(d-1) + n = nd \mid n^2(d-1)$, so $n \leftrightarrow n(d-1)$. It follows that in fact $n \leftrightarrow n(d-1)^k$, for any nonnegative integer k , and

$$n \leftrightarrow n(d-1) \leftrightarrow n(d-1)(d-2) \leftrightarrow \dots \leftrightarrow n \prod_{i=c}^{d-1} i$$

for any positive integer $c < d$.

For all integers $a > b > 2$, there exists some integer λ such that $(b-1)^\lambda > a$. Let

$$X = \prod_{i=b}^{(b-1)^\lambda} i.$$

We have

$$a \leftrightarrow \prod_{i=b}^a i \leftrightarrow (b-1)^\lambda \prod_{i=b}^a i \leftrightarrow (b-1)^\lambda \prod_{i=b}^a i \cdot \prod_{i=a+1}^{(b-1)^\lambda-1} i = X.$$

But we also have

$$b \leftrightarrow b(b-1)^\lambda \leftrightarrow b(b-1)^\lambda \prod_{i=b+1}^{(b-1)^\lambda-1} i = X.$$

Hence $a \leftrightarrow X \leftrightarrow b$, so $a \leftrightarrow b$, as desired.

Solution 2

We note that for any integer $n \geq 3$, $n \leftrightarrow 2n$, for

$$n \leftrightarrow n(n-1) \leftrightarrow n(n-1)(n-2) \leftrightarrow n(n-2) \leftrightarrow 2n$$

It follows that for $n \geq 4$,

$$n \leftrightarrow n(n-1) \leftrightarrow n(n-1)(n-2) \leftrightarrow n(n-1)(n-2)(n-3) \leftrightarrow 2(n-1)(n-2) \leftrightarrow (n-1)(n-2) \leftrightarrow (n-1).$$

Thus all integers greater than 2 are connected.

Solution 3

We note that $n_i + n_{i+1} \mid n_i n_{i+1}$ if and only if

$$(n_i + n_{i+1}) \mid [(n_i + n_{i+1})n_i - n_i n_{i+1}] = n_i^2.$$

Therefore for any divisor $d > n_i$ of n_i^2 , $d - n_i \leftrightarrow n_i$.

Now, for $a \geq 2$,

$$4a \leftrightarrow 4a^2 - 4a = 4a(a-1) \leftrightarrow 8a^2 - 4a(a-1) = 4a(a+1) \leftrightarrow 4(a+1)^2 - 4a(a+1) = 4(a+1).$$

Also,

$$4 \leftrightarrow 4^2 - 4 = 12.$$

This means that all positive multiples of 4 are connected.

Furthermore, for $a \geq 2$,

$$2a \leftrightarrow 2a(a-1),$$

which implies that every even number greater than 2 is connected to a multiple of 4.

Finally, for $a \geq 3$,

$$a \leftrightarrow a(a-1).$$

Since $a(a-1)$ is even, this means that all integers greater than or equal to 2 are connected to some even number.

Together, these imply that all integers greater than 2 are connected.

Solution 4

We note that if $d > 2$ is a divisor of n , then $n \leftrightarrow n(d-1)$.

We say a positive integer k is *safe* if for all integers $n \geq 3$, $n \leftrightarrow kn$. Note that the product of two safe numbers is also a safe number. Define $f(n)$ ($n > 2$) to be the smallest divisor of n that is greater than 2. We will prove that 2 is safe, by strong induction on $f(n)$. For the case $f(n) = 3$, we have $n \leftrightarrow (3-1)n = 2n$. If $f(n) > 3$, we note that $f[n(f(n)-1)] < f(n)$, since $f(n)-1$ must be less than all the divisors of n . Thus the inductive hypothesis gives us $[f(n)-1]n \leftrightarrow 2[f(n)-1]n$. Furthermore, we have $n \leftrightarrow [f(n)-1]n$ and $2n \leftrightarrow 2[f(n)-1]n$, both from our initial note. Thus $n \leftrightarrow 2n$.

We will now prove that every prime p is safe, by strong induction. We have already proven the base case $p = 2$. Now, for odd p , $p+1$ is the product of odd primes less than p , so $p+1$ is safe. Then we have

$$n \leftrightarrow (p+1)n \leftrightarrow p(p+1)n \leftrightarrow pn.$$

Thus the induction is complete, and all primes are safe.

We have now shown that all integers greater than 1 are safe. Specifically, a and b are safe, and $a \leftrightarrow ab \leftrightarrow b$.

Solution 5

Similarly to the first solution, we observe that for any integer $k < a$ and any positive integers $e_1, \dots, e_k$,

$$a \leftrightarrow a \prod_{i=1}^k (a-i)^{e_i}.$$

We also note that if $a \leftrightarrow b$, then for any positive integer k , $ka \leftrightarrow kb$, for $(a+b) \mid (ab)$ implies $k(a+b) \mid k^2(ab)$.

It is sufficient to prove that for any $a > 3$, $a \leftrightarrow (a-1)$. If a is composite, then there exist $m \geq n > 0$ such that $(a-1-m)(a-1-n) = a$, and

$$(a-1) \leftrightarrow (a-1-m)(a-1-n) \prod_{i=0}^{m-1} (a-1-i) = a \prod_{i=0}^{m-1} (a-1-i) \leftrightarrow a.$$

If, on the other hand, a is prime, then we use strong induction. For the base case, $a = 5$, we note $4 \leftrightarrow 3 \leftrightarrow 3(3-1) = 6 \leftrightarrow 5$. Now, assuming that $a > 5$ and that this holds for all primes less than a (we know it holds for all composites), it is sufficient to show that $(a+1) \leftrightarrow (a-1)$, since $(a+1)$ is composite and therefore $(a+1) \leftrightarrow a$. But since $(a+1)$ and $(a-1)$ are both even, it suffices to show that $\frac{a+1}{2} \leftrightarrow \frac{a-1}{2} = \frac{a+1}{2} - 1$. But this is true, since $\frac{a+1}{2}$ is either composite, or a prime less than a , and it is greater than 3, since $a > 5$. Thus the induction is complete.

6. (USAMO 2002) I have an $n \times n$ sheet of stamps, from which I've been asked to tear out blocks of three adjacent stamps in a single row or column. (I can only tear along the perforations separating adjacent stamps, and each block must come out of the sheet in one piece.) Let $b(n)$ be the smallest number of blocks I can tear out and make it impossible to tear out any more blocks. Prove that there are real constants c and d such that

$$\frac{1}{7}n^2 - cn \leq b(n) \leq \frac{1}{5}n^2 + dn$$

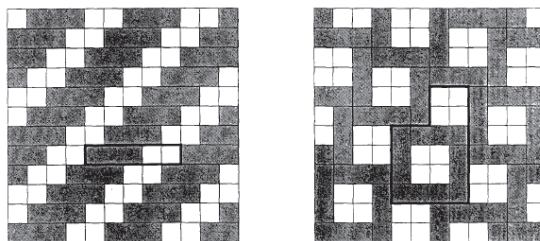
for all $n > 0$.

Solution

The upper bound requires an example of a set of $\frac{1}{5}n^2 + dn$ blocks whose removal makes it impossible to remove any further blocks. It suffices to show that we can tile the plane by tiles containing one block for every five stamps such that no more blocks can be chosen. Two such tilings are shown below with one tile outlined in heavy lines. Given an $n \times n$ section of the tiling take all blocks lying entirely within that section and add as many additional blocks as possible. If the basic tile is contained in an $(m+1) \times (m+1)$ square, then the $n \times n$ section is covered by tiles contained in a concentric $(n+2m) \times (n+2m)$ square. Hence there are at most $\frac{1}{5}(n+2m)^2$ blocks entirely within the section. For an $n \times n$ section of the tiling, there are at most $4n$ blocks which lie partially in and partially out of that section (hence these blocks contain at most $8n$ stamps in the $n \times n$ square) and each of the additional blocks must contain one of these stamps. Thus there are at most $8n$ additional blocks. Thus there are at most

$$\frac{1}{5}(n+2m)^2 + 8n \leq \frac{1}{5}n^2 + \frac{4m^2 + 4m + 40}{5}n$$

blocks total.



The lower bound requires an argument. Suppose that we have a set of $b(n)$ blocks whose removal makes removing any further blocks impossible.

Solution 1

There are $2n(n-2)$ potential blocks of three consecutive stamps in a row or column. Each of these must meet at least one of the $b(n)$ blocks removed. Conversely, each

of the $b(n)$ blocks removed meets at most 14 of these potential blocks (5 oriented the same way, including itself, and 9 oriented the orthogonal way). Therefore $14b(n) \geq 2n(n-2)$ or

$$b(n) \geq \frac{1}{7}n^2 - \frac{2}{7}n.$$

Solution 2

Call a stamp used if it belongs to one of the $b(n)$ removed blocks. Consider the $(n-2)^2$ five-stamp crosses centered at each stamp not on an edge of the sheet. Each cross must contain two used stamps. (One stamp not in the center is not enough to prevent another block from being torn out, and it is impossible to use one stamp in the center and use no other stamps in the cross.) In addition, each block not lying along an edge of the sheet lies entirely inside one cross, which thus contains three used stamps. There are at most $4n/3$ of the $b(n)$ blocks lying along the edges, hence there are at least $b(n) - 4n/3$ crosses containing three used stamps.

Now count the number of pairs of a used stamp and a cross containing that stamp, in two ways. First counting block by block, we get $3b(n)$ used stamps, and each used stamp is contained in at most five crosses (exactly five if it is not on an edge), for a total of at most $15b(n)$ pairs. Next, counting cross by cross, each of the $(n-2)^2$ crosses contains at least two used stamps and we have at least $b(n) - 4n/3$ crosses containing three used stamps, for a total of at least $2(n-2)^2 + b(n) - 4n/3$ pairs. Therefore

$$15b(n) \geq 2(n-2)^2 + b(n) - \frac{4n}{3},$$

or

$$b(n) \geq \frac{1}{7}n^2 - \frac{16}{21}n.$$

Solution 3

Call a stamp used if it belongs to one of the $b(n)$ removed blocks. Count the number of pairs consisting of a used stamp and an adjacent unused stamp, in two ways.

There are at least $(n-2)^2 - 3b(n)$ unused stamps which are not on an edge. Since no more blocks can be torn out, either the stamp to the left or right and either the stamp or below such an unused stamp must be used. Thus we have at least $2n^2 - 8n - 6b(n)$ such pairs.

Each block removed is adjacent to at most eight other stamps. However these eight stamps contain two blocks of three consecutive stamps. Hence at most six of these eight stamps can be unused. Thus each of the $b(n)$ blocks removed is involved in at most six pairs. Thus there are at most $6b(n)$ pairs.

Combining these we have

$$6b(n) \geq 2n^2 - 8n - 6b(n),$$

or

$$b(n) \geq \frac{1}{6}n^2 - \frac{2}{3}n.$$

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

§32 USAMO 2003

1. (*USAMO 2003*) Prove that for every positive integer n there exists an n -digit number divisible by 5^n all of whose digits are odd.

Solution 1

We proceed by induction. For our base case, $n = 1$, we have the number 5. Now, suppose that there exists some number $a \cdot 5^{n-1}$ with $n - 1$ digits, all of which are odd. It is then sufficient to prove that there exists an odd digit k such that $k \cdot 10^{n-1} + a \cdot 5^{n-1} = 5^{n-1}(k \cdot 2^{n-1} + a)$ is divisible by 5^n . This is equivalent to proving that there exists an odd digit k such that $k \cdot 2^{n-1} + a$ is divisible by 5, which is true when $k \equiv -3^{n-1}a \pmod{5}$. Since there is an odd digit in each of the residue classes mod 5, k exists and the induction is complete.

Solution 2

First, we note that there are 5^n n digit numbers with only odd digits. Now, we will prove that none of these numbers have the same residue mod 5^n , and therefore one of them must be $0 \pmod{5^n}$.

Proof by contradiction: Assume we have two distinct numbers $A_1A_2A_3\dots A_n$ and $B_1B_2B_3\dots B_n$ with only odd digits that leave the same residue mod 5^n . Then, subtracting the larger from the smaller would yield a new number that is a multiple of 5^n and has only even digits. We could then halve all of the digits in that number to get a second multiple of 5^n with at most n digits that only uses the digits 0 through 4.

Lemma: Every multiple of 5^n with n digits or less has a 5 as one of its digits.

All numbers of this type can be written as $k5^n$. Then, let k have x factors of 2 in it. ($x < n$, or else our number would have more than n digits). So, we have $k5^n = a2^x5^n = a \cdot 10^x \cdot 5^{n-x}$ for some odd a . Now $a \cdot 10^x \cdot 5^{n-x}$ is an odd multiple of 5 ($a5^{n-x}$) with x zeroes after it, and all multiples of 5 end in 5. Therefore, $a \cdot 10^x \cdot 5^{n-x}$ always contains a 5 as its $(x+1)^{st}$ digit, and we have proven our lemma.

By our lemma, our number with only the digits 0 through 4 cannot be a multiple of 5^n , and so we have reached a contradiction. QED

Note: Not only does this prove the desired claim that there exists such a number, but it also proves that there is exactly one such number.

Solution 3 (Construct digits of exemplar.)

$$5^n \pmod{10} \equiv 5.$$

$$2^k 5^n \pmod{10^{k+1}} \equiv 10^k 5^{n-k} \equiv 5 \cdot 10^k.$$

For any n , we can construct a $a = 5^n \cdot \sum_{0 \leq i < n} 2^{x_i}$:

$$x_0 = 1$$

$$a_i = 5^n \sum_{0 \leq j \leq i} 2^{x_j}$$

$x_i = \frac{a_{i-1}}{10^i} + 1 \pmod{2}$. (This makes the 10^i place of a_i odd, without changing the the smaller place digits).

$$a = a_{n-1} = 5^n \cdot \sum_{0 \leq j < n} 2^{x_j}, \text{ with digits in place } 10^i \text{ odd, for } 0 \leq i < n.$$

Since $\sum_{0 \leq j < n} 2^{x_j} < 2^n$, $a < 5^n 2^n = 10^n$.

By construction, the digits in all the places up through 10^{n-1} are odd, and since $a < 10^n$, there are no other digits.

In fact, if $A_m = a$ that solves the case $n = m$, then $A_{m+1} \bmod 10^m \equiv A_m$.

Note that in some cases (like $n = 5$), both of $x_{n-1} \in \{0, 1\}$ yield distinct numbers $a - 5 \cdot 10^{n-1}$ and a with all odd digits. $a - 5 \cdot 10^{n-1}$ has $n - 1 < n$ digits, and so $x_{n-1} = 1$ is needed for padding.

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

2. (*USAMO 2003*) A convex polygon $\mathcal{P}$ in the plane is dissected into smaller convex polygons by drawing all of its diagonals. The lengths of all sides and all diagonals of the polygon $\mathcal{P}$ are rational numbers. Prove that the lengths of all sides of all polygons in the dissection are also rational numbers.

Solution

When $\mathcal{P}$ is a triangle, the problem is trivial. Otherwise, it is sufficient to prove that any two diagonals of the polygon cut each other into rational lengths. Let two diagonals which intersect at a point P within the polygon be AC and BD . Since $ABCD$ is a convex quadrilateral with sides and diagonals of rational length, we consider it in isolation.

By the [Law of Cosines](#), $\cos BAC = \frac{AB^2 + CA^2 - BC^2}{2AB \cdot CA}$, which is rational. Similarly, $\cos CAD$ is rational, as well as $\cos BAD = \cos(BAC + CAD) = \cos BAC \cos CAD - \sin BAC \sin CAD$. It follows that $\sin BAC \sin CAD$ is rational. Since $\sin^2 CAD = 1 - \cos^2 CAD$ is rational, this means that $\frac{\sin BAC \sin CAD}{\sin^2 CAD} = \frac{\sin BAC}{\sin CAD} = \frac{\sin BAP}{\sin PAD}$ is rational. This implies that $\frac{AB \sin BAP}{AD \sin PAD} = \frac{\frac{1}{2} AB \cdot AP \sin BAP}{\frac{1}{2} AP \cdot AD \sin PAD} = \frac{[BAP]}{[PAD]} = \frac{BP}{PD}$ is rational.

Define r to be equal to $\frac{BP}{PD}$. We know that $\frac{BP}{PD}$ is rational; hence r is rational. We also have $(1 + r)PD = (1 + \frac{BP}{PD})PD = PD + BP = BD$, which is, of course, rational. It follows that BP and PD both have rational length, as desired.

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

3. (*USAMO 2003*) Let $n \neq 0$. For every sequence of integers

$$A = a_0, a_1, a_2, \dots, a_n$$

satisfying $0 \leq a_i \leq i$, for $i = 0, \dots, n$, define another sequence

$$t(A) = t(a_0), t(a_1), t(a_2), \dots, t(a_n)$$

by setting $t(a_i)$ to be the number of terms in the sequence A that precede the term a_i and are different from a_i . Show that, starting from any sequence A as above, fewer than n applications of the transformation t lead to a sequence B such that $t(B) = B$.

Solution

Consider some sequence $C = c_0, \dots, c_n$ as the image of A after t has been applied some finite number of times.

Lemma 1. If $t(c_k) = c_k = j$, then $c_j = \dots = c_{j+i} = \dots = c_k = j$ ($0 \leq i \leq k - j$).

Proof. Since the j terms $c_0, \dots, c_{j-1}$ are all less than j , no other terms that precede c_k can be unequal to c_k . The lemma follows.

Lemma 2. If $t(c_k) = c_k = j$, then $t^2(c_k) = t(c_k)$ (where $f^2(x) = f(f(x))$).

Proof. Since only i terms precede the term c_i , we will have $t^m(c_i) \leq i$, for any integer m . This means that we will always have $t^m(c_0), \dots, t^m(c_{j-1}) < j$. This means that $c_j = t(c_j) = \dots = c_k = t(c_k)$, and the lemma follows by iteration.

Thus we may regard a term c_k as *stable* if $t(c_k) = c_k$. We will call a sequence *stable* if all of its terms are stable.

Lemma 3. If $c_k = j$ is not stable, then $t(c_k) > c_k$.

Proof. We have $c_0, \dots, c_{j-1} < j$, so we always have $t(c_k) \geq c_k$. Equality implies that c_k is stable.

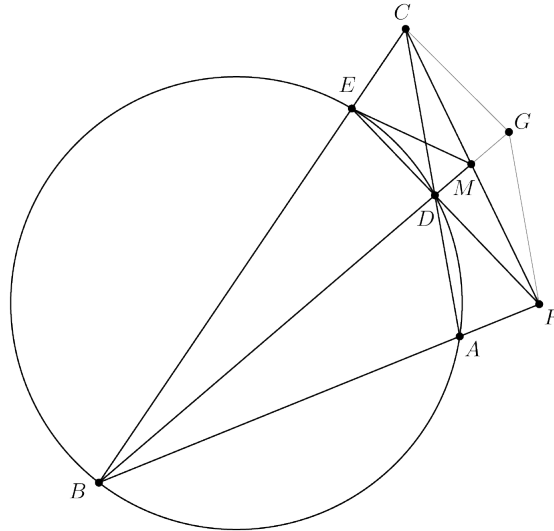
We will now prove the problem by induction. For the base case, $n = 1$, we have either 0,0 or 0,1, both of which are stable. Now, suppose that $t^{n-2}(a_0), \dots, t^{n-2}(a_{n-1})$ are stable. We must then have $t^{n-2}(a_n) \in \{n-2, n-1, n\}$. If $t^{n-2}(a_n) = n$, then it is already stable. If $t^{n-2}(a_n) = n-1$, then either it is already stable or $t^{n-1}(a_n) = n$, which is stable. If $t^{n-2}(a_n) = t^{n-2}(a_{n-1})$, then $t^{n-2}(a_n)$ must already be stable. The only possibilities remaining are $t^{n-2}(a_n) = n-2, t^{n-2}(a_{n-1}) = n-1$ and $t^{n-2}(a_n) = n-2, t^{n-2}(a_{n-1}) < n-2$. In the first case, we must have $t^{n-1}(a_n)$ equal to $n-1$ or n , both of which will make it stable. In the second case, we must have $t^{n-2}(a_{n-2}) = t^{n-2}(a_{n-1}) < n-2$, giving us $t^{n-1}(a_n) = n$, which will make it stable. Thus the induction is complete.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

4. (USAMO 2003) Let ABC be a triangle. A circle passing through A and B intersects segments AC and BC at D and E , respectively. Lines AB and DE intersect at F , while lines BD and CF intersect at M . Prove that $MF = MC$ if and only if $MB \cdot MD = MC^2$.

Solution 1

Extend segment DM through M to G such that $FG \parallel CD$.



Then $MF = MC$ if and only if quadrilateral $CDFG$ is a parallelogram, or, $FD \parallel CG$. Hence $MC = MF$ if and only if $\angle GCD = \angle FDA$, that is, $\angle FDA + \angle CGF = 180^\circ$.

Because quadrilateral $ABED$ is cyclic, $\angle FDA = \angle ABE$. It follows that $MC = MF$ if and only if

$$180^\circ = \angle FDA + \angle CGF = \angle ABE + \angle CGF,$$

that is, quadrilateral $CBFG$ is cyclic, which is equivalent to

$$\angle CBM = \angle CBG = \angle CFG = \angle DCF = \angle DCM.$$

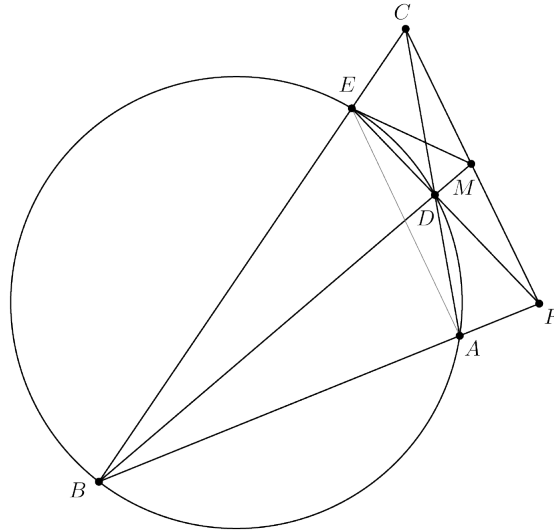
Because $\angle DMC = \angle CMB$, $\angle CBM = \angle DCM$ if and only if triangles BCM and CDM are similar, that is

$$\frac{CM}{BM} = \frac{DM}{CM},$$

or $MB \cdot MD = MC^2$.

Solution 2

We first assume that $MB \cdot MD = MC^2$. Because $\frac{MC}{MD} = \frac{MB}{MC}$ and $\angle CMD = \angle BMC$, triangles CMD and BMC are similar. Consequently, $\angle MCD = \angle MBC$. This exact condition can also be visualized through power of M with respect to the circumcircle of triangle BDC and getting the angle condition from the alternate segment theorem.



Because quadrilateral $ABED$ is cyclic, $\angle DAE = \angle DBE$. Hence

$$\angle FCA = \angle MCD = \angle MBC = \angle DBE = \angle DAE = \angle CAE,$$

implying that $AE \parallel CF$, so $\angle AEF = \angle CFE$. Because quadrilateral $ABED$ is cyclic, $\angle ABD = \angle AED$. Hence

$$\angle FBM = \angle ABD = \angle AED = \angle AEF = \angle CFE = \angle MFD.$$

Because $\angle FBM = \angle DFM$ and $\angle FMB = \angle DMF$, triangles BFM and FDM are similar. Consequently, $\frac{FM}{DM} = \frac{BM}{FM}$, or $FM^2 = BM \cdot DM = MC^2$. Therefore $MC^2 = MB \cdot MD$ implies $MC = MF$.

Now we assume that $MC = MF$. Applying Ceva's Theorem to triangle BCF and cevians BM, CA, FE gives

$$\frac{BA}{AF} \cdot \frac{FM}{MC} \cdot \frac{CE}{EB} = 1,$$

implying that $\frac{BA}{AF} = \frac{BE}{EC}$, so $AE \parallel CF$.

Consequently, $\angle DCM = \angle DAE$. Because quadrilateral $ABED$ is cyclic, $\angle DAE = \angle DBE$. Hence

$$\angle DCM = \angle DAE = \angle DBE = \angle CBM.$$

Because $\angle CBM = \angle DCM$ and $\angle CMB = \angle DMC$, triangles BCM and CDM are similar. Consequently, $\frac{CM}{DM} = \frac{BM}{CM}$, or $CM^2 = BM \cdot DM$.

Combining the above, we conclude that $MF = MC$ if and only if $MB \cdot MD = MC^2$.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

5. (*USAMO 2003*) Let a, b, c be positive real numbers. Prove that

$$\frac{(2a+b+c)^2}{2a^2+(b+c)^2} + \frac{(2b+c+a)^2}{2b^2+(c+a)^2} + \frac{(2c+a+b)^2}{2c^2+(a+b)^2} \leq 8.$$

Solution 1

Since all terms are homogeneous, we may assume WLOG that $a+b+c=3$.

Then the LHS becomes $\sum \frac{(a+3)^2}{2a^2+(3-a)^2} = \sum \frac{a^2+6a+9}{3a^2-6a+9} = \sum \left(\frac{1}{3} + \frac{8a+6}{3a^2-6a+9} \right)$.

Notice $3a^2 - 6a + 9 = 3(a-1)^2 + 6 \geq 6$, so $\frac{8a+6}{3a^2-6a+9} \leq \frac{8a+6}{6}$.

So $\sum \frac{(a+3)^2}{2a^2+(3-a)^2} \leq \sum \left(\frac{1}{3} + \frac{8a+6}{6} \right) = 1 + \frac{8(a+b+c)+18}{6} = 8$, as desired.

Solution 2

Note that

$$\begin{aligned} (2x+y)^2 + 2(x-y)^2 &= 4x^2 + 4xy + y^2 + 2x^2 - 4xy + 2y^2 \\ &= 3(2x^2 + y^2). \end{aligned}$$

Setting $x=a$ and $y=b+c$ yields

$$(2a+b+c)^2 + 2(a-b-c)^2 = 3(2a^2 + (b+c)^2).$$

Thus, we have

$$\frac{(2a+b+c)^2}{2a^2+(b+c)^2} = \frac{3(2a^2+(b+c)^2) - 2(a-b-c)^2}{2a^2+(b+c)^2} = 3 - \frac{2(a-b-c)^2}{2a^2+(b+c)^2},$$

and its analogous forms. Thus, the desired inequality is equivalent to

$$\frac{(a-b-c)^2}{2a^2+(b+c)^2} + \frac{(b-c-a)^2}{2b^2+(c+a)^2} + \frac{(c-a-b)^2}{2c^2+(a+b)^2} \geq \frac{1}{2}.$$

Because

6. (*USAMO 2003*) At the vertices of a regular hexagon are written six nonnegative integers whose sum is 2003. Bert is allowed to make moves of the following form: he may pick a vertex and replace the number written there by the absolute value of the difference between the numbers written at the two neighboring vertices. Prove that Bert can make a sequence of moves, after which the number 0 appears at all six vertices.

Solution

Assume the original numbers are a, b, c, d, e, f . Since $a+b+c+d+e+f$ is odd, either $a+c+e$ or $b+d+f$ must be odd. WLOG let $a+c+e$ be odd and $a \geq c \geq e \geq 0$.

Case 1 $a, c, e > 0$. Define *Operation A* as the sequence of moves from Step 1 to Step 3, shown below:

b	c	$a - c$	c	$a - c$	c
a	d	a	$c - e$	$c - e$	$c - e$
f	e	$a - e$	e	$a - e$	$a - c$
Step 1		Step 2		Step 3	

Notice that Operation A changes the numbers a, c, e to $c - e, c, a - c$ and they are all nonnegative, since $a \geq c \geq e$. Their sum changes from $a + c + e$ to $a + c - e$; it decreases as long as $e \neq 0$. If we repeat Operation A enough times, its sum will decrease and eventually we will arrive at a point where at least one of the numbers in the positions originally occupied by a, c, e has become a 0.

Case 2 $a, c > 0$ and $e = 0$. Define *Operation B* as the sequence of moves from Step 1 to Step 3, shown below:

b	c	$a - c$	c	$a - c$	$2c - a$
a	d	a	c	c	$a - 2c$
f	0	a	0	a	0
Step 1		Step 2		Step 3	

where in Step 3, we take the nonnegative choice of $2c - a$ or $a - 2c$. $a, c, 0$ is changed to either $c, 2c - a, 0$ or $c, a - 2c, 0$. If we have $c, 2c - a, 0$, their sum is $3c - a$ and this is less than $a + c + 0$ (the original sum) unless $a = c$, but $a \neq c$ since the original sum $a + c + 0$ is odd by assumption. If we have $c, a - 2c, 0$, their sum is $a - c$, which is less than $a + c$. Operation B applied repeatedly will cause either a or c to become 0.

Case 3 $a > 0$ and $c = e = 0$. Define *Operation C* as the sequence of moves from Step 1 to Step 4, shown below:

b	0	a	0	a	0	0	0
a	d	a	0	0	0	0	0
f	0	a	0	a	0	0	0
Step 1		Step 2		Step 3		Step 4	

§33 USAMO 2004

1. (USAMO 2004) Let $ABCD$ be a quadrilateral circumscribed about a circle, whose interior and exterior angles are at least 60° . Prove that

$$\frac{1}{3}|AB^3 - AD^3| \leq |BC^3 - CD^3| \leq 3|AB^3 - AD^3|$$

When does equality hold?

Solution 1

It suffices to show that

$$|BC^3 - CD^3| \leq 3|AB^3 - AD^3|,$$

because the other side of the inequality follows by symmetry. By Pitot's Theorem, we have $AB - AD = BC - CD$. WLOG, let $AB \geq AD$. Then we have that

$BC \geq CD$, so we must show $BC^3 - CD^3 \leq 3(AB^3 - AD^3)$. If $AB = AD$, we are done. Therefore, we will assume for the rest of the proof that $AB > AD$, $BC > CD$.

Then we would like to show that

$$BC^2 + BC \cdot CD + CD^2 \leq 3(AB^2 + AB \cdot AD + AD^2).$$

By the Law of Cosines, we have that

$$BD^2 = BC^2 + k_1(BC \cdot CD) + CD^2 = AB^2 + k_2(AB \cdot AD) + AD^2,$$

where $-1 \leq k_1, k_2 \leq 1$ (this is because $60^\circ \leq \angle BAD, \angle BCD \leq 120^\circ$). Therefore,

$$3(BC^2 - BC \cdot CD + CD^2) \leq 3(BC^2 + k_1(BC \cdot CD) + CD^2) = 3(AB^2 + k_2(AB \cdot AD) + AD^2) \leq 3(AB^2 + AB \cdot AD + AD^2)$$

Then we wish to show that

$$3(BC^2 - BC \cdot CD + CD^2) - (BC^2 + BC \cdot CD + CD^2) \geq 0 \rightarrow 2(BC - CD)^2 \geq 0,$$

and we are done. ■

Solution 2

By a well-known property of tangential quadrilaterals, the sum of the two pairs of opposite sides are equal; hence $a + c = b + d \Rightarrow a - b = d - c \Rightarrow |a - b| = |d - c|$. Now we factor the desired expression into $\frac{|d-c|(c^2+d^2+cd)}{3} \leq |a-b|(a^2+b^2+ab) \leq 3|d-c|(c^2+d^2+cd)$. Temporarily discarding the case where $a = b$ and $c = d$, we can divide through by the $|a - b| = |d - c|$ to get the simplified expression $(c^2 + d^2 + cd)/3 \leq a^2 + b^2 + ab \leq 3(c^2 + d^2 + cd)$.

Now, draw diagonal BD . By the law of cosines, $c^2 + d^2 + 2cd \cos A = BD^2$. Since each of the interior and exterior angles of the quadrilateral is at least 60 degrees, we have that $A \in [60^\circ, 120^\circ]$. Cosine is monotonically decreasing on this interval, so by setting A at the extreme values, we see that $c^2 + d^2 - cd \leq BD^2 \leq c^2 + d^2 + cd$. Applying the law of cosines analogously to a and b , we see that $a^2 + b^2 - ab \leq BD^2 \leq a^2 + b^2 + ab$; we hence have $c^2 + d^2 - cd \leq BD^2 \leq a^2 + b^2 + ab$ and $a^2 + b^2 - ab \leq BD^2 \leq c^2 + d^2 + cd$.

We wrap up first by considering the second inequality. Because $c^2 + d^2 - cd \leq BD^2 \leq a^2 + b^2 + ab$, $\text{RHS} \geq 3(a^2 + b^2 - ab)$. This latter expression is of course greater than or equal to $a^2 + b^2 + ab$ because the inequality can be rearranged to $2(a - b)^2 \geq 0$, which is always true. Multiply the first inequality by 3 and we see that it is simply the second inequality with the variables swapped; hence by symmetry it is true as well.

Equality occurs when $a = b$ and $c = d$, or when $ABCD$ is a kite.

Resources

2. (USAMO 2004) Suppose $a_1, \dots, a_n$ are integers whose greatest common divisor is 1. Let S be a set of integers with the following properties:

- For $i = 1, \dots, n$, $a_i \in S$.
- For $i, j = 1, \dots, n$ (not necessarily distinct), $a_i - a_j \in S$.
- For any integers $x, y \in S$, if $x + y \in S$, then $x - y \in S$.

Prove that S must be the set of all integers.

Solution

Suppose a_i has only one element; then for the greatest common divisor to be 1, 1 has to be the sole element. Then 1 is in S by (a), 0 is in S by (b), $0 + 1 = 1 \in S \Rightarrow 0 - 1 = -1 \in S$ by (c), and we can apply (c) analogously to get that $n \cdot 1 \in S$ for integers n and hence S is the set of all integers, as desired.

Lemma: If $x, y \in a_i$, then $ax + by \in S$ for integers a, b .

Proof: Assume a_i has at least two elements; x and y . By (b), $x - y$ is in S , and by the application of (c) above, we get that $n(x - y)$ for integers n is in S . Then apply (c) to $n(x - y)$ and ny or nx to get that $ax + by \in S$ for all $a, b \in \mathbb{Z}$.

Now let the terms be $a_1, a_2, \dots, a_n$. By applying our lemma many times, all numbers in the form $\sum c_i a_i$ for a sequence of integers c_i are attainable if the sequence is of a length which is a power of 2. If not, we "pad" the sequence with many copies of an existing element of the sequence until it does have a length which is a power of 2 - it is apparent that this will not change S .

By Schur's theorem (a generalization of the more well-known Chicken McNugget theorem), every integer greater than some integer n is attainable, and hence there are two members of S in the form $\sum c_i a_i$ which are consecutive integers. Furthermore, because such numbers are closed under addition, their sum is in S , and hence so is their difference; 1. Thus, by the argument at the beginning of this proof, S is the set of all integers, as desired.

Resources

3. (*USAMO 2004*) For what values of $k > 0$ is it possible to dissect a $1 \times k$ rectangle into two similar, but incongruent, polygons?

Solution

We will show that a dissection satisfying the requirements of the problem is possible if and only if $k \neq 1$.

We first show by contradiction that such a dissection is not possible when $k = 1$. Assume that we have such a dissection. The common boundary of the two dissecting polygons must be a single broken line connecting two points on the boundary of the square (otherwise either the square is subdivided in more than two pieces or one of the polygons is inside the other). The two dissecting polygons must have the same number of vertices. They share all the vertices on the common boundary, so they have to use the same number of corners of the square as their own vertices. Therefore, the common boundary must connect two opposite sides of the square (otherwise one of the polygons will contain at least three corners of the square, while the other at most two). However, this means that each of the dissecting polygons must use an entire side of the square as one of its sides, and thus each polygon has a side of length 1. A side of longest length in one of the polygons is either a side on the common boundary or, if all those sides have length less than 1, it is a side of the square. But this is also true of the polygon, which means that the longest side length in the two polygons is the same. This is impossible since they are similar but not congruent, so we have a contradiction.

We now construct a dissection satisfying the requirements of the problem when $k \neq 1$. Notice that we may assume that $k > 1$, because a $1 \times k$ rectangle is similar to a $1 \times \frac{1}{k}$ rectangle.

We first construct a dissection of an appropriately chosen rectangle (denoted by $ABCD$ below) into two similar noncongruent polygons. The construction depends on

two parameters (n and r below). By appropriate choice of these parameters we show that the constructed rectangle can be made similar to a $1 \times k$ rectangle, for any $k > 1$. The construction follows.

Let $r > 1$ be a real number. For any positive integer n , consider the following sequence of $2n + 2$ points:

$$A_0 = (0, 0), A_1 = (1, 0), A_2 = (1, r), A_3 = (1 + r^2, r), A_4 = (1 + r^2, r + r^3), A_5 = (1 + r^2 + r^4, r + r^3),$$

and so on, until

$$A_{2n+1} = (1 + r^2 + r^4 + \cdots + r^{2n}, r + r^3 + r^5 + \cdots + r^{2n-1}).$$

Define a rectangle $ABCD$ by

$$A = A_0, B = (1 + r^2 + \cdots + r^{2n}, 0), C = A_{2n+1}, \text{ and } D = (0, r + r^3 + \cdots + r^{2n-1}).$$

The sides of the $(2n + 2)$ -gon $A_1 A_2 \dots A_{2n+1} B$ have lengths

$$r, r^2, r^3, \dots, r^{2n}, r + r^3 + r^5 + \cdots + r^{2n-1}, r^2 + r^4 + r^6 + \cdots + r^{2n},$$

and the sides of the $(2n + 2)$ -gon $A_0 A_1 A_2 \dots A_{2n} D$ have lengths

$$1, r, r^2, \dots, r^{2n-1}, 1 + r^2 + r^4 + \cdots + r^{2n-2}, r + r^3 + r^5 + \cdots + r^{2n-1},$$

respectively. These two polygons dissect the rectangle $ABCD$ and, apart from orientation, it is clear that they are similar but noncongruent, with coefficient of similarity $r > 1$. The rectangle $ABCD$ and its dissection are thus constructed.

The rectangle $ABCD$ is similar to a rectangle of size $1 \times f_n(r)$, where

$$f_n(r) = \frac{1 + r^2 + \cdots + r^{2n}}{r + r^3 + \cdots + r^{2n-1}}.$$

It remains to show that $f_n(r)$ can have any value $k > 1$ for appropriate choices of n and r . Choose n sufficiently large so that $1 + \frac{1}{n} < k$. Since

$$f_n(1) = 1 + \frac{1}{n} < k < k \frac{1 + k^2 + \cdots + k^{2n}}{k^2 + k^4 + \cdots + k^{2n}} = f_n(k)$$

and $f_n(r)$ is a continuous function for positive r , there exists an r such that $1 < r < k$ and $f_n(r) = k$, so we are done.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

Resources

4. (*USAMO 2004*) Alice and Bob play a game on a 6 by 6 grid. On his or her turn, a player chooses a rational number not yet appearing on the grid and writes it in an empty square of the grid. Alice goes first and then the players alternate. When all squares have numbers written in them, in each row, the square with the greatest number is colored black. Alice wins if she can then draw a line from the top of the grid to the bottom of the grid that stays in black squares, and Bob wins if she can't. (If two squares share a vertex, Alice can draw a line from one to the other that stays in those two squares. Find, with proof, a winning strategy for one of the players.

Solution 1

Before the game, Bob may select three *useless* squares per row. He may then move according to the following rules:

- If Alice writes a number in a useless square, then Bob writes a higher number in a non-useless square in the same row on his next turn.
- If Alice writes a number in a non-useless square, then Bob writes a lower number in a useless square in the same row.

Bob can always do this because there are an even number of squares in each row. In this way, Bob can prevent any useless square from being colored black at the end. Then it will be sufficient for Bob to make the squares $(1,1)$, $(2,1)$, $(2,2)$, $(3,2)$, $(3,3)$, $(4,3)$, $(4,4)$, $(5,4)$, $(5,5)$, $(6,5)$, $(6,6)$ useless, where (m,n) is the square in the m th column and n th row. Thus Bob can always win.

Note that it is not necessary for Bob to move first; even if he moves second, he will always be able to ensure that at least two squares per row of his choosing are not colored black.

Solution 2

Bob may move according to the following rules:

- If Alice writes a number in rows 3, 4, 5, or 6, then Bob also writes a number in row 3, 4, 5, or 6.
- If Alice writes a number in row $a \in \{1, 2\}$, column c , then Bob writes a number in row $3 - a$, column $c + 3 \pmod{6}$. If Alice's number is the highest in its row, then so is Bob's; if it is not, then neither is Bob's.

In this way, there will be 2 columns in between the black squares in rows 1 and 2, so they cannot possibly touch, which means that Bob will win.

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

Resources

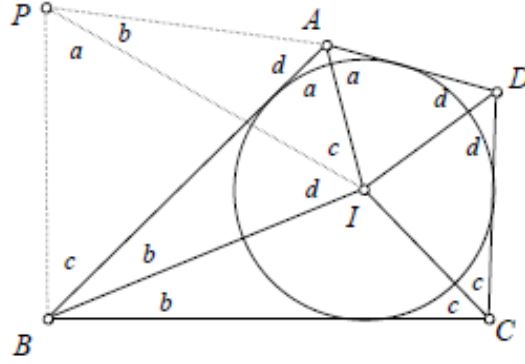
5. (*USAMO 2004*) Let a , b , and c be positive real numbers. Prove that $(a^5 - a^2 + 3)(b^5 - b^2 + 3)(c^5 - c^2 + 3) \geq (a + b + c)^3$.
6. (*USAMO 2004*) A circle ω is inscribed in a quadrilateral $ABCD$. Let I be the center of ω . Suppose that $(AI + DI)^2 + (BI + CI)^2 = (AB + CD)^2$. Prove that $ABCD$ is an [isosceles trapezoid](#).

Solution

Our proof is based on the following key Lemma.

Lemma: If a circle ω , centered at I , is inscribed in a quadrilateral $ABCD$, then

$$BI^2 + \frac{AI}{DI} \cdot BI \cdot CI = AB \cdot BC. \quad (1)$$



Proof: Since circle ω is inscribed in $ABCD$, we get $m\angle DAI = m\angle IAB = a$, $m\angle ABI = m\angle IBC = b$, $m\angle BCI = m\angle ICD = c$, $m\angle CDI = m\angle IDA = d$, and $a + b + c + d = 180^\circ$. Construct a point P outside of the quadrilateral such that $\triangle ABP$ is similar to $\triangle DCI$. We obtain

$$\begin{aligned} m\angle PAI + m\angle PBI &= m\angle PAB + m\angle BAI + m\angle PBA + m\angle ABI \\ &= m\angle IDC + a + m\angle ICD + b \\ &= a + b + c + d = 180^\circ, \end{aligned}$$

implying that the quadrilateral $PAIB$ is cyclic. By Ptolemy's Theorem, we have $AI \cdot BP + BI \cdot AP = AB \cdot IP$, or

$$BP \cdot \frac{AI}{IP} + BI \cdot \frac{AP}{IP} = AB. \quad (2)$$

Because $PAIB$ is cyclic, it is not difficult to see that, as indicated in the figure, $m\angle IPB = m\angle IAB = a$, $m\angle API = m\angle ABI = b$, $m\angle AIP = m\angle ABP = c$, and $m\angle PIB = m\angle PAB = d$. Note that $\triangle AIP$ and $\triangle ICB$ are similar, implying that

$$\frac{AI}{IP} = \frac{IC}{CB} \text{ and } \frac{AP}{IP} = \frac{IB}{CB}.$$

Substituting the above equalities into the identity (2), we arrive at

$$BP \cdot \frac{CI}{BC} + \frac{BI^2}{BC} = AB,$$

or

$$BP \cdot CI + BI^2 = AB \cdot BC. \quad (3)$$

Note also that $\triangle BIP$ and $\triangle IDA$ are similar, implying that $\frac{BP}{BI} = \frac{IA}{ID}$, or

$$BP = \frac{AI}{ID} \cdot IB.$$

Substituting the above identity back into (3) gives the desired relation (1), establishing the Lemma. ■

Now we prove our main result. By the Lemma and symmetry, we have

$$CI^2 + \frac{DI}{AI} \cdot BI \cdot CI = CD \cdot BC. \quad (4)$$

Adding the two identities (1) and (4) gives

$$BI^2 + CI^2 + \left(\frac{AI}{DI} + \frac{DI}{AI} \right) BI \cdot CI = BC(AB + CD).$$

By the AM-GM Inequality, we have $\frac{AI}{DI} + \frac{DI}{AI} \geq 2$. Thus,

$$BC(AB + CD) \geq IB^2 + IC^2 + 2IB \cdot IC = (BI + CI)^2,$$

where the equality holds if and only if $AI = DI$. Likewise, we have

$$AD(AB + CD) \geq (AI + DI)^2,$$

where the equality holds if and only if $BI = CI$. Adding the last two identities gives

$$(AI + DI)^2 + (BI + CI)^2 \leq (AD + BC)(AB + CD) = (AB + CD)^2,$$

because $AD + BC = AB + CD$. (The latter equality is true because the circle ω is inscribed in the quadrilateral $ABCD$.)

By the given condition in the problem, all the equalities in the above discussion must hold, that is, $AI = DI$ and $BI = CI$. Consequently, we have $a = d$, $b = c$, and so $\angle DAB + \angle ABC = 2a + 2b = 180^\circ$, implying that $AD \parallel BC$. It is not difficult to see that $\triangle AIB$ and $\triangle DIC$ are congruent, implying that $AB = CD$. Thus, $ABCD$ is an isosceles trapezoid.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

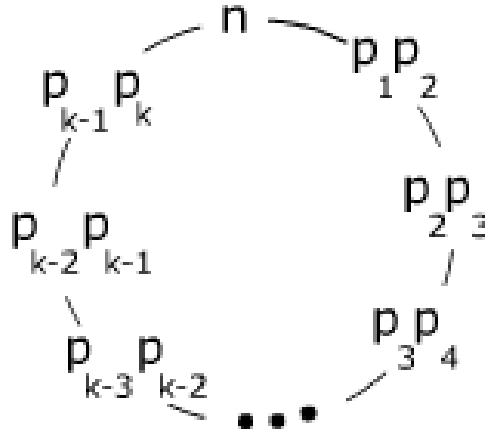
Resources

§34 USAMO 2005

1. (*USAMO 2005*) (*Zuming Feng*) Determine all composite positive integers n for which it is possible to arrange all divisors of n that are greater than 1 in a circle so that no two adjacent divisors are relatively prime.

Solution 1 (official solution)

No such circular arrangement exists for $n = pq$, where p and q are distinct primes. In that case, the numbers to be arranged are p ; q and pq , and in any circular arrangement, p and q will be adjacent. We claim that the desired circular arrangement exists in all other cases. If $n = p^e$ where $e \geq 2$, an arbitrary circular arrangement works. Henceforth we assume that n has prime factorization $p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, where $p_1 < p_2 < \cdots < p_k$ and either $k > 2$ or else $\max(e_1, e_2) > 1$. To construct the desired circular arrangement of $D_n := \{d : d|n \text{ and } d > 1\}$, start with the circular arrangement of $n, p_1 p_2, p_2 p_3 \dots, p_{k-1} p_k$ as shown.



Then between n and $p_1 p_2$, place (in arbitrary order) all other members of D_n that have p_1 as their smallest prime factor. Between $p_1 p_2$ and $p_2 p_3$, place all members of D_n other than $p_2 p_3$ that have p_2 as their smallest prime factor. Continue in this way, ending by placing $p_k, p_k^2, \dots, p_k^{e_k}$ between $p_{k-1} p_k$ and n . It is easy to see that each element of D_n is placed exactly one time, and any two adjacent elements have a common prime factor. Hence this arrangement has the desired property.

Note. In graph theory terms, this construction yields a Hamiltonian cycle in the graph with vertex set D_n in which two vertices form an edge if the two corresponding numbers have a common prime factor. The graphs below illustrate the construction for the special cases $n = p^2 q$ and $n = pqr$.

Solution 2

The proof that no arrangement exists for $n = pq$, where p, q are distinct primes follows from above. Apply induction to prove all other cases are possible

Base case:

- a) $n = p^r$, where p is a prime and r is a positive integer. Any arrangement suffices
- b) $n = pqr$, where p, q, r are distinct primes. The following configuration works

$$p, pq, pr, r, qr, q, pqr$$

Inductive step: Suppose the desired arrangement exists for a composite n , show the arrangement exists for np^r , where p is a prime relatively prime to n and r is a positive integer

Let $a_1, a_2, \dots, a_m$ be the arrangement of divisors of n , then $(a_i, a_{i+1}) > 1$ for $i = 1, 2, \dots, m$, where $a_{m+1} = a_1$. The divisors of np^r greater than 1 are of the form

$$a_i p^j, p^j \quad 1 \leq i \leq m, 1 \leq j \leq r$$

The following sequence works

$$a_1, \dots, a_{m-1}, a_m p, \text{other divisors in arbitrary order}, a_m p, a_m$$

since all other divisors are divisible by p .

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

2. (*USAMO 2005*) (*Răzvan Gelca*) Prove that the system

$$\begin{aligned}x^6 + x^3 + x^3y + y &= 147^{157} \\ x^3 + x^3y + y^2 + y + z^9 &= 157^{147}\end{aligned}$$

has no solutions in integers x , y , and z .

Solution 1

It suffices to show that there are no solutions to this system in the integers mod 19. We note that $152 = 8 \cdot 19$, so $157 \equiv -147 \equiv 5 \pmod{19}$. For reference, we construct a table of powers of five:

n	5^n	n	5^n
1	5	6	7
2	6	7	-3
3	-8	8	4
4	-2	9	1
5	9		

Evidently, the order of 5 is 9. Hence 5 is the square of a multiplicative generator of the nonzero integers mod 19, so this table shows all nonzero squares mod 19, as well.

It follows that $147^{157} \equiv (-5)^{13} \equiv -5^4 \equiv 2$, and $157^{147} \equiv 5^3 \equiv -8$. Thus we rewrite our system thus:

$$\begin{aligned}(x^3 + y)(x^3 + 1) &\equiv 2 \\ (x^3 + y)(y + 1) + z^9 &\equiv -8.\end{aligned}$$

Adding these, we have

$$(x^3 + y + 1)^2 - 1 + z^9 \equiv -6,$$

or

$$(x^3 + y + 1)^2 \equiv -z^9 - 5.$$

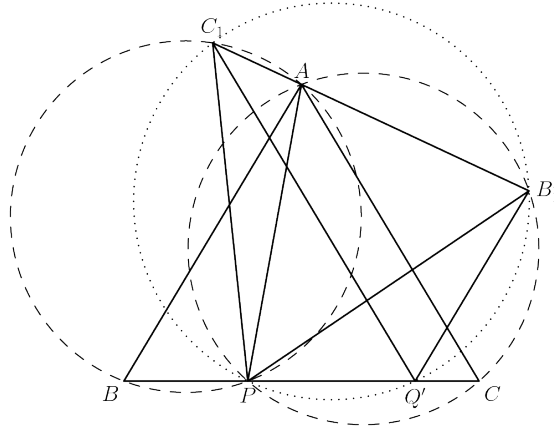
By **Fermat's Little Theorem**, the only possible values of z^9 are ± 1 and 0, so the only possible values of

3. (*USAMO 2005*) (*Zuming Feng*) Let ABC be an acute-angled triangle, and let P and Q be two points on side BC . Construct point C_1 in such a way that convex quadrilateral $APBC_1$ is cyclic, $QC_1 \parallel CA$, and C_1 and Q lie on opposite sides of line AB . Construct point B_1 in such a way that convex quadrilateral $APCB_1$ is cyclic, $QB_1 \parallel BA$, and B_1 and Q lie on opposite sides of line AC . Prove that points B_1, C_1, P , and Q lie on a circle.

Solution 1

Let B'_1 be the second intersection of the line C_1A with the circumcircle of APC , and let Q' be the second intersection of the circumcircle of B'_1C_1P and line BC . It

is enough to show that $B'_1 = B_1$ and $Q' = Q$. All our angles will be directed, and measured mod π .



Since points C_1, P, Q', B'_1 are concyclic and points C_1, A, B'_1 are collinear, it follows that

$$\angle C_1 Q' P \equiv \angle C_1 B'_1 P \equiv \angle A B'_1 P.$$

But since points A, B'_1, P, C are concyclic,

$$\angle A B'_1 P \equiv \angle A C P.$$

It follows that lines AC and $C_1 Q'$ are parallel. If we exchange C with B and C_1 with B'_1 in this argument, we see that lines AB and $B'_1 Q'$ are likewise parallel.

It follows that Q' is the intersection of BC and the line parallel to AC and passing through C_1 . Hence $Q' = Q$. Then B'_1 is the second intersection of the circumcircle of APC and the line parallel to AB passing through Q . Hence $B'_1 = B_1$, as desired. ■

Motivation: One can notice that if you take such a B'_1 then PQB'_1C_1 is cyclic, and that similarly $PQB_1C'_1$ is also cyclic. One gets the intuition that only one such circle should exist where the other chord passes through A , and so sets up a ghost point for Q , which works. cocohearts

Solution 2

Lemma. B_1, A, C_1 are collinear.

Suppose they are not collinear. Let line $B_1 A$ intersect circle ABP (i.e. the circumcircle of ABP) again at C_2 distinct from C_1 . Because $\angle C_2 B_1 Q = \angle C_2 A B = \angle C_2 P B = 180^\circ - \angle C_2 P Q$, we have that $B_1 C_2 P Q$ is cyclic. Hence $\angle C_2 Q P = \angle C_2 B_1 P = \angle P B_1 A = \angle C$, so $C_2 Q \parallel AC$. Then C_2 must be the other intersection of the parallel to AC through Q with circle ABP . Then C_2 is on segment $C_1 Q$, so C_2 is contained in triangle ABQ . However, line AB_1 intersects this triangle only at point A because B_1 lies on arc AC not containing P of circle APC , a contradiction. Hence, B_1, A, C_1 are collinear, as desired.

As a result, we have $\angle C_1 B_1 Q = \angle C_1 A B = \angle C_1 P B = 180^\circ - \angle C_1 P Q$, so $B_1 C_1 P Q$ is cyclic, as desired.

Solution 3

Due to the parallel lines, $m\angle C_1 Q B_1 = m\angle A$. Therefore, it suffices to prove that

$$m\angle C_1 P B_1 = m\angle C_1 B A + m\angle A C B_1 = m\angle A.$$

Note that $m\angle BC_1A + m\angle AB_1C = 180^\circ$ by the cyclic quadrilaterals. Now, the condition simplifies to proving C_1, A, B_1 collinear.

Use barycentric coordinates. Let

$$A = (1, 0, 0), B = (0, 1, 0), C = (0, 0, 1), P = (0, p, 1 - p), Q = (0, q, 1 - q).$$

By the parallel lines,

$$C_1 = (1 - q - z', q, z'), B_1 = (q - y', y', 1 - q)$$

for some y', z' . As A is a vertex of the reference triangle, we must prove that $q(1 - q) = y'z'$.

Now we find the circumcircle of $\triangle APC$. Let its equation be

$$-a^2yz - b^2xz - c^2xy + (x + y + z)(ux + vy + wz) = 0.$$

Substituting in A, C gives $u = w = 0$. Substituting in P yields

$$-a^2p(1 - p) + vp = 0 \implies v = a^2(1 - p).$$

Substituting in B_1 yields

$$0 = -a^2y'(1 - q) - b^2(q - y')(1 - q) - c^2y'(q - y') + a^2(1 - p)y' = 0.$$

$$c^2y'^2 + (a^2q - a^2p + b^2(1 - q) - c^2q)y' - b^2q(1 - q) = 0$$

Similarly, the circumcircle of $\triangle APB$ is

$$-a^2yz - b^2xz - c^2xy + (x + y + z)a^2pz.$$

Now, we substitute C_1 into the equation

$$0 = -a^2qz' - b^2z'(1 - q - z') - c^2q(1 - q - z') + a^2pz' = 0$$

$$b^2z'^2 + (a^2p - a^2q - b^2(1 - q) + c^2q)z' - c^2q(1 - q) = 0$$

Let $k = a^2p - a^2q - b^2(1 - q) + c^2q$, and note that both y', z' are negative. By the quadratic formula, we have

$$y' = \frac{k - \sqrt{k^2 + 4b^2c^2q(1 - q)}}{2c^2}, z' = \frac{-k - \sqrt{k^2 + 4b^2c^2q(1 - q)}}{2b^2}$$

Multiplying these two, we have

$$y'z' = \frac{4b^2c^2q(1 - q)}{4b^2c^2} = q(1 - q),$$

as desired. ■

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

4. (USAMO 2005) Legs L_1, L_2, L_3, L_4 of a square table each have length n , where n is a positive integer. For how many ordered 4-tuples (k_1, k_2, k_3, k_4) of nonnegative integers can we cut a piece of length k_i from the end of leg L_i ($i = 1, 2, 3, 4$) and still have a stable table?

(The table is stable if it can be placed so that all four of the leg ends touch the floor. Note that a cut leg of length 0 is permitted.)

Solution 1

First flip the table upside down and observe the plane that the ends of the legs form. Let this plane be $ax + by + c = z$. Assume WLOG that the table is the square with endpoints $(0, 0), (0, 1), (1, 0), (1, 1)$ on the xy plane. Thus the lengths of the legs will be $c, c + b, c + a, c + a + b$ respectively. In other words $k_1 + k_3 = k_2 + k_4 = x$ for some $0 \leq x \leq 2n$.

For $0 \leq x \leq n$, it is easy to see that the number of stable tables is $(x + 1)^2$.

For $n + 1 \leq x \leq 2n$, we can biject the stable tables of x to the stable of $2n - x$ by replacing all of the legs with the cut off portions.

So the total number of stable tables is $1^2 + 2^2 + \dots + n^2 + (n+1)^2 + n^2 + \dots + 2^2 + 1^2 = (1^2 + 2^2 + \dots + (n+1)^2) + (1^2 + 2^2 + \dots + n^2) = \frac{(n+1)(n+2)(2n+3)}{6} + \frac{n(n+1)(2n+1)}{6} =$

$$\boxed{\frac{2n^3 + 6n^2 + 7n + 3}{3}}.$$

Solution 2

Let C_i be the number of sets of cuts $\{k_1, k_2, k_3, k_4\}$ that result in the longest leg being of length i . Then $C_i = C_{i-1} + x$, and we will evaluate x to develop a recursion. Note that if a set of cuts has non-zero cuts, then if that set is counted in C_i , subtract 1 from each of the cuts to obtain a set of cuts that is counted in C_{i-1} . For example, if $\{2, 3, 4, 5\}$ was counted in C_5 , then $\{1, 2, 3, 4\}$ would have been counted in C_4 because subtracting the same length from each leg preserves the quality of the legs being coplanar, and there is a bijection.

Now we must evaluate those sets of cuts that don't fall in this bijection, which are clearly those where one leg is completely cut off. After some simple geometric reasoning that I won't include here, we see that one corner has a leg of length of zero, and the opposite leg, called the initial leg, will have length i . If the remaining legs have length y and z , then $(y, z) = (0, i), (1, i-1), \dots, (i-1, 1), (i, 0)$, so $i + 1$ options. The initial corner can be any of the four legs, but each of the four permutations of the set of cuts $0, 0, i, i$ is repeated twice, so we have $4(i+1) - 4 = 4i$ total additional sets of cuts. We conclude that $C_i = C_{i-1} + 4i$, and note that $C_0 = 1$. Now we can create generating functions. $F(x) = \sum_{i=0}^{\infty} C_i x^i$. Also, $G(x) = \sum_{i=1}^{\infty} 4i x^i$. From the recursion, we have

$$F(x) = C_0 + xF(x) + G(x) \implies F(x) = \frac{1 + G(x)}{1 - x}$$

This final equation is easy to analyze. Simply use $G(x)$ as the first term of a geometric series with constant multiplier of x . This gives $C_i = 2i^2 + 2i + 1$. The total number of sets of cuts for a table of legs of length n is the sum of all the C_i , $0 \leq i \leq n$, from which we deduce the answer $\frac{2n^3 + 6n^2 + 7n + 3}{3}$, or $\frac{(n+1)(2n^2 + 4n + 3)}{3}$.

Solution 3

The problem's condition is that the ends of the legs must be coplanar. So the plane formed by legs L_1 , L_2 , and L_3 is the same as the plane formed by the legs L_2 , L_3 , and L_4 iff the problem's condition is met.

We now assign coordinates to the ends of the legs. Let the table be on the xy -plane such that L_1 is at $(0, 0, k_1)$, L_2 is at $(s, 0, k_2)$, L_3 is at $(0, s, k_3)$, and L_4 is at (s, s, k_4) , where s is the side length of the table. (Note that we have turned the table over so that the top lies on the xy plane while the legs point in the $+z$ direction - this does not affect the result.) A plane in 3-space has the equation form $Ax + By + Cz = 1$. Plugging the coordinates of L_1 , L_2 , and L_3 into $Ax + By + Cz = 1$, we eventually get $A_1 = \frac{k_1 - k_2}{sk_1}$, $B_1 = \frac{k_1 - k_3}{sk_1}$, $C_1 = \frac{1}{k_1}$. Plugging the coordinates of L_2 , L_3 , and L_4 into $Ax + By + Cz = 1$, we get $A_2 = \frac{k_3 - k_4}{s(k_2 + k_3 - k_4)}$, $B_2 = \frac{k_2 - k_4}{s(k_2 + k_3 - k_4)}$, $C_2 = \frac{1}{k_2 + k_3 - k_4}$.

Since we need these to be coplanar, we set $A_1 = A_2$, $B_1 = B_2$, and $C_1 = C_2$. (Notice that the s 's cancel.) From $A_1 = A_2$, we get $(k_1 - k_2)(k_2 + k_3 - k_4) = k_1(k_3 - k_4)$ [1]. From $B_1 = B_2$, we get $(k_1 - k_3)(k_2 + k_3 - k_4) = k_1(k_2 - k_4)$ [2]. From $C_1 = C_2$, we get $k_1 = k_2 + k_3 - k_4$ [3]. Notice that [1] and [2] can both be obtained directly from [3] and are therefore redundant. The only condition we have left to ensure that the ends of the legs are coplanar is $k_1 + k_4 = k_2 + k_3$. Let $k_1 + k_4 = k_2 + k_3 = m$.

We proceed using casework by m . m is an integer between 0 and $2n$, inclusive. Furthermore, we note that k_i range from 0 to n , inclusive. For $m = 0$, each of (k_1, k_4) and (k_2, k_3) have one choice, $(0, 0)$. For $m = 1$, they each have two choices, $(0, 1)$ and $(1, 0)$. And so on, until $m = n$, where they each have $n + 1$ choices: $(0, n), (1, n - 1), \dots, (n - 1, 1), (n, 0)$. When $m = n + 1$, the restriction that $0 \leq k_i \leq n$ takes effect: there are only n choices for each ordered pair, which are $(1, n), (2, n - 1), \dots, (n - 1, 2), (n, 1)$. And so on, until $m = 2n$, where there is one choice for each ordered pair, (n, n) .

The number of possible 4-tuples (k_1, k_2, k_3, k_4) is thus $1^2 + 2^2 + \dots + n^2 + (n + 1)^2 + n^2 + \dots + 2^2 + 1^2$, which is equivalent to $(1^2 + 2^2 + \dots + (n + 1)^2) + (1^2 + 2^2 + \dots + n^2)$. Our final answer is $\frac{(n + 1)(n + 2)(2n + 3)}{6} + \frac{n(n + 1)(2n + 1)}{6} = \boxed{\frac{2n^3 + 6n^2 + 7n + 3}{3}}$.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

5. (*USAMO 2005*) Let n be an integer greater than 1. Suppose $2n$ points are given in the plane, no three of which are collinear. Suppose n of the given $2n$ points are colored blue and the other n colored red. A line in the plane is called a *balancing line* if it passes through one blue and one red point and, for each side of the line, the number of blue points on that side is equal to the number of red points on the same side.

Prove that there exist at least two balancing lines.

Solution 1

Consider the convex hull of the $2n$ points, or the points that would form the largest convex polygon. If the points in the convex hull contain both red and blue points, then there must be at least 2 edges of the graph of the convex hull such that the edge connects a blue and a red point. Drawing a line through those points

would give a balancing line, as we have $n-1$ blue points and $n-1$ red points on one side, and 0 points on the other.

Therefore it suffices to show that there exist at least 2 balancing lines when the convex hull is colored all the same color.

Pick a random point on the convex hull, and without loss of generality we can say it is blue (if there are no red we can change all the colors, and end up with an equivalent setup). Consider a line going through it and not any other points. As we rotate the line clockwise, we encounter the red points in some order. Let the i th point encountered be R_i . Let b_i and r_i be the number of points encountered before R_i . Then $r_i = i - 1$.

Define a sequence $f(i) = b_i - r_i$. Then $f(1) = b_1 - (1 - 1) = b_1 \geq 0$, $f(n) = b_n - (n - 1) = b_n - n + 1 \leq 0$, because we can only encounter up to $n-1$ blue points. Thus, $f(i)$ goes from negative to positive as i goes from 1 to n . We can also see that $f(i)$ can only increase by one for each change in i , so we know $f(i)$ must be 0 for some value of i . Take the first $f(i)$ where $f(i) = 0$. Since $f(i - 1)$ must have been positive, the i th point must have been red. Thus there is a balancing line for every point on the convex hull. Since a polygon must have at least 3 vertices, there must be at least $3 > 2$ balancing lines for the set of points when the convex hull is all the same color, and the statement is true as we desired.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

Solution 2

Case 1 One can notice That if any blue and red point have not points on one of it's sides this mean the line between them is a balancing line (as on the other side we will have $n - 1$ blue points and $n - 1$ red).

Case 2: Now we consider the case in which all the points are inside a polygon with all vertices being one color (assume WLOG it's blue

Now consider a line that is pivoting at any vertex and rotating till reaching another vertex of the polygon (convex hull) passing through every point in side the polygon. and let x be the number of red points.minus the number of blue points on one side of the line.In other words every time the line passes through a blue point x will increase by 1, and decrease by 1 if it passes through a red point.Firstly we have that $x = 1$ and when the line reaches the other vertex of the polygon it will be equal to -1 .So when x will be equal to zero, we have a balancing line.Now repeating this process on all the other vertices of the polygon (convex hull) then we will have balancing lines equal to the number of the vertices.

By Ywgh1

6. (USAMO 2005) For m a positive integer, let $s(m)$ be the sum of the digits of m . For $n \geq 2$, let $f(n)$ be the minimal k for which there exists a set S of n positive integers such that $s(\sum_{x \in X} x) = k$ for any nonempty subset $X \subset S$. Prove that there are constants $0 < C_1 < C_2$ with

$$C_1 \log_{10} n \leq f(n) \leq C_2 \log_{10} n.$$

Solution

For the upper bound, let p be the smallest integer such that $10^p \geq n(n+1)/2$ and let

$$S = \{10^p - 1, 2(10^p - 1), \dots, n(10^p - 1)\}.$$

The sum of any nonempty set of elements of S will have the form $k(10^p - 1)$ for some $1 \leq k \leq n(n+1)/2$. Write $k(10^p - 1) = [(k-1)10^p] + [(10^p - 1) - (k-1)]$. The second term gives the bottom p digits of the sum and the first term gives at most p top digits. Since the sum of a digit of the second term and the corresponding digit of $k-1$ is always 9, the sum of the digits will be $9p$. Since $10^{p-1} < n(n+1)/2$, this example shows that

$$f(n) \leq 9p \leq 9 \log_{10}(5n(n+1)).$$

Since $n \geq 2$, $5(n+1) < n^4$, and hence

$$f(n) < 9 \log_{10} n^5 = 45 \log_{10} n.$$

For the lower bound, let S be a set of $n \geq 2$ positive integers such that any nonempty $X \subset S$ has $s(\sum_{x \in X} x) = f(n)$. Since $s(m)$ is always congruent to m modulo 9, $\sum_{x \in X} x \equiv f(n) \pmod{9}$ for all nonempty $X \subset S$. Hence every element of S must be a multiple of 9 and $f(n) \geq 9$. Let q be the largest positive integer such that $10^q - 1 \leq n$. Lemma 1 below shows that there is a nonempty subset X of S with $\sum_{x \in X} x$ a multiple of $10^q - 1$, and hence Lemma 2 shows that $f(n) \geq 9q$.

Lemma 1. Any set of m positive integers contains a nonempty subset whose sum is a multiple of m .

Proof. Suppose a set T has no nonempty subset with sum divisible by m . Look at the possible sums mod m of nonempty subsets of T . Adding a new element a to T will give at least one new sum mod m , namely the least multiple of a which does not already occur. Therefore the set T has at least $|T|$ distinct sums mod m of nonempty subsets and $|T| < m$. ■

Lemma 2. Any positive multiple M of $10^q - 1$ has $s(M) \geq 9q$.

Proof. Suppose on the contrary that M is the smallest positive multiple of $10^q - 1$ with $s(M) < 9q$. Then $M \neq 10^q - 1$, hence $M > 10^q$. Suppose the most significant digit of M is the 10^m digit, $m \geq q$. Then $N = M - 10^{m-q}(10^q - 1)$ is a smaller positive multiple of $10^q - 1$ and has $s(N) \leq s(M) < 9q$, a contradiction. ■

Finally, since $10^{q+1} > n$, we have $q+1 > \log_{10} n$. Since $f(n) \geq 9q$ and $f(n) \geq 9$, we have

$$f(n) \geq \frac{9q+9}{2} > \frac{9}{2} \log_{10} n.$$

Weaker versions of Lemmas 1 and 2 are still sufficient to prove the desired type of lower bound.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>-mahthir marshad

§35 USAMO 2006

1. (*USAMO 2006*) Let p be a prime number and let s be an integer with $0 < s < p$. Prove that there exist integers m and n with $0 < m < n < p$ and

$$\left\{ \frac{sm}{p} \right\} < \left\{ \frac{sn}{p} \right\} < \frac{s}{p}$$

if and only if s is not a divisor of $p - 1$.

Note: For x a real number, let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x , and let $\{x\} = x - \lfloor x \rfloor$ denote the fractional part of x .

Solution 1

We proceed by contradiction. Assume that $s \mid (p - 1)$. Then for some positive integer k , $sk = p - 1$. The conditions given are equivalent to stating that $sm \bmod p < sn \bmod p < s \bmod p$. Now consider the following array modulo p :

Row 1	s	$2s$	$\dots$	ks
Row 2	$s - 1$	$2s - 1$	$\dots$	$ks - 1$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
Row s	1	$s + 1$	$\dots$	$(k - 1)s + 1$

Obviously, there are s rows and k columns. The first entry of each row is simply $((r - 1)k + 1)s \bmod p$. Since we wish for $sm \bmod p$ and $sn \bmod p$ to both be in the first column, while also satisfying the given conditions, we can easily see that sm must be in a row m_r with $m_r > n_r$, where n_r denotes the row $sn \bmod p$ is in. However, since the values of each entry decreases while $((r - 1)k + 1)s$ keeps increasing, we can see that the condition can never be satisfied and thus, $s \nmid (p - 1)$.

To prove the other direction, let $sj + r = p - 1$ for positive integers j and r with j being the largest integer such that $sj < p - 1$ and $r < s$. Since $s \nmid (p - 1)$, $1 < s < p - 1$.

Note that for $0 < l < j$, $ls \geq s \bmod p$. Thus, the first integer multiplied by s modulo p that will be less than s is $j + 1$. $(j + 1)s \equiv s - r - 1 \bmod p$. Since $\{s, 2s, \dots, js, (j + 1)s, \dots, xs, \dots, (p - 1)s\}$ is a complete residue system mod p with the exception of the 0 term, we can find an $x > j + 1$ such that $xs \equiv s - 1 \bmod p$.

Thus, choose $m = j + 1$ and $n = x$ to complete the proof.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

2. (USAMO 2006) For a given positive integer k find, in terms of k , the minimum value of N for which there is a set of $2k + 1$ distinct positive integers that has sum greater than N but every subset of size k has sum at most $N/2$.

Solution 1

Let one optimal set of integers be $\{a_1, \dots, a_{2k+1}\}$ with $a_1 > a_2 > \dots > a_{2k+1} > 0$.

The two conditions can now be rewritten as $a_1 + \dots + a_k \leq N/2$ and $a_1 + \dots + a_{2k+1} > N$. Subtracting, we get that $a_{k+1} + \dots + a_{2k+1} > N/2$, and hence $a_{k+1} + \dots + a_{2k+1} > a_1 + \dots + a_k$. In words, the sum of the $k + 1$ smallest numbers must exceed the sum of the k largest ones.

Let $a_{k+1} = C$. As all the numbers are distinct integers, we must have $\forall i \in \{1, \dots, k\} : a_{k+1-i} \geq C + i$, and also $\forall i \in \{1, \dots, k\} : a_{k+1+i} \leq C - i$.

Thus we get that $a_1 + \dots + a_k \geq kC + \frac{k(k+1)}{2}$, and $a_{k+1} + \dots + a_{2k+1} \leq (k+1)C - \frac{k(k+1)}{2}$.

As we want the second sum to be larger, clearly we must have $(k+1)C - \frac{k(k+1)}{2} > kC + \frac{k(k+1)}{2}$. This simplifies to $C > k(k+1)$.

Hence we get that:

$$\begin{aligned} N &\geq 2(a_1 + \cdots + a_k) \\ &\geq 2\left(kC + \frac{k(k+1)}{2}\right) \\ &= 2kC + k(k+1) \\ &\geq 2k(k^2 + k + 1) + k(k+1) \\ &= 2k^3 + 3k^2 + 3k \end{aligned}$$

On the other hand, for the set $\{k^2 + k + 1 + i \mid i \in \{-k, \dots, k\}\}$ the sum of the largest k elements is exactly $k^3 + k^2 + k + \frac{k(k+1)}{2}$, and the sum of the entire set is $(k^2 + k + 1)(2k + 1) = 2k^3 + 3k^2 + 3k + 1$, which is more than twice the sum of the largest set.

Hence the smallest possible N is $\boxed{N = 2k^3 + 3k^2 + 3k}$.

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

3. (*USAMO 2006*) For integral m , let $p(m)$ be the greatest prime divisor of m . By convention, we set $p(\pm 1) = 1$ and $p(0) = \infty$. Find all polynomials f with integer coefficients such that the sequence $\{p(f(n^2)) - 2n\}_{n \geq 0}$ is bounded above. (In particular, this requires $f(n^2) \neq 0$ for $n \geq 0$.)

Solution 1

Let $f(x)$ be a non-constant polynomial in x of degree d with integer coefficients, suppose further that no prime divides all the coefficients of f (otherwise consider the polynomial obtained by dividing f by the gcd of its coefficients). We further normalize f by multiplying by -1 , if necessary, to ensure that the leading coefficient (of x^d) is positive.

Let $g(n) = f(n^2)$, then $g(n)$ is a polynomial of degree $2d$ or more and $g(n) = g(-n)$. Let $g_1, \dots, g_k$ be the factorization of g into irreducible factors with positive leading coefficients. Such a factorization is unique. Let $d(g_i)$ denote the degree of g_i . Since $g(-n) = g(n)$ the factors g_i are either even functions of n or come in pairs (g_i, h_i) with $g_i(-n) = (-1)^{d(g_i)} h_i(n)$.

Let $P(0) = \infty$, $P(\pm 1) = 1$. For any other integer m let $P(m)$ be the largest prime factor of m .

Suppose that for some finite constant C and all $n \geq 0$ we have $P(g(n)) - 2n < C$. Since the polynomials g_i divide g , the same must be true for each of the irreducible polynomials g_i .

A theorem of T. Nagell implies that if $d(g_i) \geq 2$ the ratio $P(g_i(n))/n$ is unbounded for large values of n . Since in our case the $P(g_i(n))/n$ is asymptotically bounded above by 2 for large n , we conclude that all the irreducible factors g_i are linear. Since linear polynomials are not even functions of n , they must occur in pairs

$g_i(n) = a_i n + b_i, h_i(n) = a_i n - b_i$. Without loss of generality, $b_i \geq 0$. Since the coefficients of f are relatively prime, so are a_i and b_i , and since $P(0) = \infty$, neither polynomial can have any non-negative integer roots, so $a_i > 1$ and thus $b_i > 0$.

On the other hand, by Dirichlet's theorem, $a_i \leq 2$, since otherwise the sequence $a_i n + b_i$ would yield infinitely many prime values with $P(g_i(n)) = a_i n + b_i \geq 3n$. So $a_i = 2$ and therefore b_i is a positive odd integer. Setting $b_i = 2c_i + 1$, clearly $P(g_i(n)) - 2n < 2c_i + 2$. Since this holds for each factor g_i , it is true for the product g of all the factors with the bound determined by the factor with the largest value of c_i .

Therefore, for suitable non-negative integers c_i , $g(n)$ is a product of polynomials of the form $4n^2 - (2c_i + 1)^2$. Now, since $g(n) = f(n^2)$, we conclude that $f(n)$ is a product of linear factors of the form $4n - (2c_i + 1)^2$.

Since we restricted ourselves to non-constant polynomials with relatively prime coefficients, we can now relax this condition and admit a possibly empty list of linear factors as well as an arbitrary non-zero integer multiple M . Thus for a suitable non-zero integer M and $k \geq 0$ non-negative integers c_i , we have:

$$f(n) = M \cdot \prod_{i=1}^k (4n - (2c_i + 1)^2)$$

Solution 2

The polynomial f has the required properties if and only if

$$f(x) = c(4x - a_1^2)(4x - a_2^2) \cdots (4x - a_k^2), \quad (*)$$

where $a_1, a_2, \dots, a_k$ are odd positive integers and c is a nonzero integer. It is straightforward to verify that polynomials given by $(*)$ have the required property. If p is a prime divisor of $f(n^2)$ but not of c , then $p|(2n - a_j)$ or $p|(2n + a_j)$ for some $j \leq k$. Hence $p - 2n \leq \max\{a_1, a_2, \dots, a_k\}$. The prime divisors of c form a finite set and do not affect whether or not the given sequence is bounded above. The rest of the proof is devoted to showing that any f for which $\{p(f(n^2)) - 2n\}_{n \geq 0}$ is bounded above is given by $(*)$.

Let $\mathbb{Z}[x]$ denote the set of all polynomials with integer coefficients. Given $f \in \mathbb{Z}[x]$, let $\mathcal{P}(f)$ denote the set of those primes that divide at least one of the numbers in the sequence $\{f(n)\}_{n \geq 0}$. The solution is based on the following lemma.

Lemma. If $f \in \mathbb{Z}[x]$ is a nonconstant polynomial then $\mathcal{P}(f)$ is infinite.

Proof. Repeated use will be made of the following basic fact: if a and b are distinct integers and $f \in \mathbb{Z}[x]$, then $a - b$ divides $f(a) - f(b)$. If $f(0) = 0$, then p divides $f(p)$ for every prime p , so $\mathcal{P}(f)$ is infinite. If $f(0) = 1$, then every prime divisor p of $f(n!)$ satisfies $p > n$. Otherwise p divides $n!$, which in turn divides $f(n!) - f(0) = f(n!) - 1$. This yields $p|1$, which is false. Hence $f(0) = 1$ implies that $\mathcal{P}(f)$ is infinite. To complete the proof, set $g(x) = f(f(0)x)/f(0)$ and observe that $g \in \mathbb{Z}[x]$ and $g(0) = 1$. The preceding argument shows that $\mathcal{P}(g)$ is infinite, and it follows that $\mathcal{P}(f)$ is infinite. ■

Suppose $f \in \mathbb{Z}[x]$ is nonconstant and there exists a number M such that $p(f(n^2)) - 2n \leq M$ for all $n \geq 0$. Application of the lemma to $f(x^2)$ shows that there is an infinite sequence of distinct primes $\{p_j\}$ and a corresponding infinite sequence of nonnegative integers $\{k_j\}$ such that $p_j | f(k_j)^2$ for all $j \geq 1$. Consider the sequence

$\{r_j\}$ where $r_j = \min\{k_j \pmod{p_j}, p_j - k_j \pmod{p_j}\}$. Then $0 \leq r_j \leq (p_j - 1)/2$ and $p_j | f(r_j)^2$. Hence $2r_j + 1 \leq p_j \leq p(f(r_j^2)) \leq M + 2r_j$, so $1 \leq p_j - 2r_j \leq M$ for all $j \geq 1$. It follows that there is an integer a_1 such that $1 \leq a_1 \leq M$ and $a_1 = p_j - 2r_j$ for infinitely many j . Let $m = \deg f$. Then $p_j | 4^m f(((p_j - a_1)/2)^2)$ and $4^m f(((x - a_1)/2)^2) \in \mathbb{Z}[x]$. Consequently, $p_j | f((a_1/2)^2)$ for infinitely many j , which shows that $(a_1/2)^2$ is a zero of f . Since $f(n^2) \leq 0$ for $n \geq 0$, a_1 must be odd. Then $f(x) = (4x - a_1)^2 g(x)$, where $g \in \mathbb{Z}[x]$. (See the note below.) Observe that $\{p(g(n^2)) - 2n\}_{n \geq 0}$ must be bounded above. If g is constant, we are done. If g is nonconstant, the argument can be repeated to show that f is given by (*).

Note. The step that gives $f(x) = (4x - a_1^2)g(x)$ where $g \in \mathbb{Z}[x]$ follows immediately using a lemma of Gauss. The use of such an advanced result can be avoided by first writing $f(x) = r(4x - a_1^2)g(x)$ where r is rational and $g \in \mathbb{Z}[x]$. Then continuation gives $f(x) = c(4x - a_1^2) \cdots (4x - a_k^2)$ where c is rational and the a_i are odd. Consideration of the leading coefficient shows that the denominator of c is 2^s for some $s \geq 0$ and consideration of the constant term shows that the denominator is odd. Hence c is an integer.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

4. (*USAMO 2006*) Find all positive integers n for which there exists an integer $k \geq 2$ and positive rationals $a_1, a_2, \dots, a_k$ satisfying $a_1 + a_2 + \dots + a_k = a_1 \cdot a_2 \cdots a_k = n$.

Solution 1

First, consider composite numbers. We can then factor n into $p_1 p_2$. It is easy to see that $p_1 + p_2 \leq n$, and thus, we can add $(n - p_1 - p_2)$ 1s in order to achieve a sum and product of n . For $p_1 + p_2 = n$, which is only possible in one case, $n = 4$, we consider $p_1 = p_2 = 2$.

Secondly, let n be a prime. Then we can find the following procedure: Let $a_1 = \frac{n}{2}, a_2 = 4, a_3 = \frac{1}{2}$ and let the rest of the a_k be 1. The only numbers we now need to check are those such that $\frac{n}{2} + 4 + \frac{1}{2} > n \implies n < 9$. Thus, we need to check for $n = 1, 2, 3, 5, 7$. One is included because it is neither prime nor composite.

For $n = 1$, consider $a_1 a_2 \dots a_k = 1$. Then by AM-GM, $a_1 + a_2 + \dots + a_k \geq k \sqrt[k]{1} > 1$ for $k \geq 2$. Thus, $n = 1$ is impossible.

If $n = 2$, once again consider $a_1 a_2 \dots a_k = 2$. Similar to the above, $a_1 + a_2 + \dots \geq k \sqrt[k]{2} > 2$ for $k \geq 2$ since $\sqrt[k]{2} > 1$ and $k > 2$. Obviously, $n = 2$ is then impossible.

If $n = 3$, let $a_1 a_2 \dots a_k = 3$. Again, $a_1 + a_2 + \dots \geq k \sqrt[k]{3} > 3$. This is obvious for $k \geq 3$. Now consider $k = 2$. Then $2\sqrt{3} \approx 3.4$ is obviously greater than 3. Thus, $n = 3$ is impossible.

If $n = 5$, proceed as above and consider $k = 2$. Then $a_1 + a_2 = 5$ and $a_1 a_2 = 5$. However, we then come to the quadratic $a_1^2 - 5a_1 + 5 = 0 \implies a_1 = \frac{5 \pm \sqrt{5}}{2}$, which is not rational. For $k = 3$ and $k = 4$ we note that $\sqrt[3]{5} > \frac{5}{3}$ and $\sqrt[4]{5} > \frac{5}{4}$. This is trivial to prove. If $k \geq 5$, it is obviously impossible, and thus $n = 5$ does not work.

The last case, where $n = 7$, is possible using the following three numbers. $a_1 = \frac{9}{2}, a_2 = \frac{4}{3}, a_3 = \frac{7}{6}$ shows that $n = 7$ is possible.

Hence, n can be any positive integer greater than 3 with the exclusion of 5.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

5. (*USAMO 2006*) A mathematical frog jumps along the number line. The frog starts at 1, and jumps according to the following rule: if the frog is at integer n , then it can jump either to $n + 1$ or to $n + 2^{m_n+1}$ where 2^{m_n} is the largest power of 2 that is a factor of n . Show that if $k \geq 2$ is a positive integer and i is a nonnegative integer, then the minimum number of jumps needed to reach $2^i k$ is greater than the minimum number of jumps needed to reach 2^i .

Solution 1

For $i \geq 0$ and $k \geq 1$, let $x_{i,k}$ denote the minimum number of jumps needed to reach the integer $n_{i,k} = 2^i k$. We must prove that

$$x_{i,k} > x_{i,1} \quad (1)$$

for all $i \geq 0$ and $k \geq 2$. We prove this using the method of descent.

First note that (1) holds for $i = 0$ and all $k \geq 2$, because it takes 0 jumps to reach the starting value $n_{0,1} = 1$, and at least one jump to reach $n_{0,k} = k \geq 2$. Now assume that (1) is not true for all choices of i and k . Let i_0 be the minimal value of i for which (1) fails for some k , let k_0 be the minimal value of $k > 1$ for which $x_{i_0,k} \leq x_{i_0,1}$. Then it must be the case that $i_0 \geq 1$ and $k_0 \geq 2$.

Let J_{i_0,k_0} be a shortest sequence of $x_{i_0,k_0} + 1$ integers that the frog occupies in jumping from 1 to $2^{i_0} k_0$. The length of each jump, that is, the difference between consecutive integers in J_{i_0,k_0} , is either 1 or a positive integer power of 2. The sequence J_{i_0,k_0} cannot contain 2^{i_0} because it takes more jumps to reach $2^{i_0} k_0$ than it does to reach 2^{i_0} . Let 2^{M+1} , $M \geq 0$ be the length of the longest jump made in generating J_{i_0,k_0} . Such a jump can only be made from a number that is divisible by 2^M (and by no higher power of 2). Thus we must have $M < i_0$, since otherwise a number divisible by 2^{i_0} is visited before $2^{i_0} k_0$ is reached, contradicting the definition of k_0 .

Let 2^{m+1} be the length of the jump when the frog jumps over 2^{i_0} . If this jump starts at $2^m(2t - 1)$ for some positive integer t , then it will end at $2^m(2t - 1) + 2^{m+1} = 2^m(2t + 1)$. Since it goes over 2^{i_0} we see $2^m(2t - 1) < 2^{i_0} < 2^m(2t + 1)$ or $(2^{i_0-m} - 1)/2 < t < (2^{i_0-m} + 1)/2$. Thus $t = 2^{i_0-m-1}$ and the jump over 2^{i_0} is from $2^m(2^{i_0-m} - 1) = 2^{i_0} - 2^m$ to $2^m(2^{i_0-m} + 1) = 2^{i_0} + 2^m$.

Considering the jumps that generate J_{i_0,k_0} , let N_1 be the number of jumps from 1 to $2^{i_0} + 2^m$, and let N_2 be the number of jumps from $2^{i_0} + 2^m$ to $2^{i_0} k_0$. By definition of i_0 , it follows that 2^m can be reached from 1 in less than N_1 jumps. On the other hand, because $m < i_0$, the number $2^{i_0}(k_0 - 1)$ can be reached from 2^m in exactly N_2 jumps by using the same jump length sequence as in jumping from $2^m + 2^{i_0}$ to $2^{i_0} k_0 = 2^{i_0}(k_0 - 1) + 2^{i_0}$. The key point here is that the shift by 2^{i_0} does not affect any of divisibility conditions needed to make jumps of the same length. In particular, with the exception of the last entry, $2^{i_0} k_0$, all of the elements of J_{i_0,k_0} are of the form $2^p(2t + 1)$ with $p < i_0$, again because of the definition of k_0 . Because $2^p(2t + 1) - 2^{i_0} = 2^p(2t - 2^{i_0-p} + 1)$ and the number $2t - 2^{i_0-p} + 1$ is odd, a jump of size 2^{p+1} can be made from $2^p(2t + 1) - 2^{i_0}$ just as it can be made from $2^p(2t + 1)$.

Thus the frog can reach 2^m from 1 in less than N_1 jumps, and can then reach $2^{i_0}(k_0 - 1)$ from 2^m in N_2 jumps. Hence the frog can reach $2^{i_0}(k_0 - 1)$ from 1 in less than $N_1 + N_2$ jumps, that is, in fewer jumps than needed to get to $2^{i_0} k_0$ and hence in fewer jumps than required to get to 2^{i_0} . This contradicts the definition of k_0 .

Solution 2

Suppose $x_0 = 1, x_1, \dots, x_t = 2^i k$ are the integers visited by the frog on his trip from 1 to $2^i k$, $k \geq 2$. Let $s_j = x_j - x_{j-1}$ be the jump sizes. Define a reduced path y_j inductively by

$$y_j = \begin{cases} y_{j-1} + s_j & \text{if } y_{j-1} + s_j \leq 2^i, \\ y_{j-1} & \text{otherwise.} \end{cases}$$

Say a jump s_j is deleted in the second case. We will show that the distinct integers among the y_j give a shorter path from 1 to 2^i . Clearly $y_j \leq 2^i$ for all j . Suppose $2^i - 2^{r+1} < y_j \leq 2^i - 2^r$ for some $0 \leq r \leq i-1$. Then every deleted jump before y_j must have length greater than 2^r , hence must be a multiple of 2^{r+1} . Thus $y_j \equiv x_j \pmod{2^{r+1}}$. If $y_{j+1} > y_j$, then either $s_{j+1} = 1$ (in which case this is a valid jump) or $s_{j+1}/2 = 2^m$ is the exact power of 2 dividing x_j . In the second case, since $2^r \geq s_{j+1} > 2^m$, the congruence says 2^m is also the exact power of 2 dividing y_j , thus again this is a valid jump. Thus the distinct y_j form a valid path for the frog. If $j = t$ the congruence gives $y_t \equiv x_t \equiv 0 \pmod{2^{r+1}}$, but this is impossible for $2^i - 2^{r+1} < y_t \leq 2^i - 2^r$. Hence we see $y_t = 2^i$, that is, the reduced path ends at 2^i . Finally since the reduced path ends at $2^i < 2^i k$ at least one jump must have been deleted and it is strictly shorter than the original path.

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

6. (USAMO 2006) Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $AE/ED = BF/FC$. Ray FE meets rays BA and CD at S and T respectively. Prove that the circumcircles of triangles SAE , SBF , TCF , and TDE pass through a common point.

Solution 1

Let the intersection of the circumcircles of SAE and SBF be X , and let the intersection of the circumcircles of TCF and TDE be Y .

$BXF = BSF = AXE$ because BSF tends both arcs AE and BF . $BFX = XSB = XEA$ because XSB tends both arcs XA and XB . Thus, $XAE \sim XBF$ by AA similarity, and X is the center of spiral similarity for A, E, B , and F . $FCY = FTC = EYD$ because FTC tends both arcs ED and FC . $FCY = FTY = EDY$ because FTY tends both arcs YF and YE . Thus, $YED \sim YFC$ by AA similarity, and Y is the center of spiral similarity for E, D, F , and C .

From the similarity, we have that $XE/XF = AE/BF$. But we are given $ED/AE = CF/BF$, so multiplying the 2 equations together gets us $ED/FC = XE/XF$. DEX, CFX are the supplements of AEX, BFX , which are congruent, so $DEX = CFX$, and so $XED \sim XFC$ by SAS similarity, and so X is also the center of spiral similarity for E, D, F , and C . Thus, X and Y are the same point, which all the circumcircles pass through, and so the statement is true.

Solution 2

We will give a solution using complex coordinates. The first step is the following lemma.

Lemma. Suppose s and t are real numbers and x, y and z are complex. The circle in the complex plane passing through $x, x + ty$ and $x + (s+t)z$ also passes through the point $x + syz/(y - z)$, independent of t .

Proof. Four points z_1, z_2, z_3 and z_4 in the complex plane lie on a circle if and only

if the cross-ratio

$$cr(z_1, z_2, z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$$

is real. Since we compute

$$cr(x, x + ty, x + (s + t)z, x + syz/(y - z)) = \frac{s + t}{s}$$

the given points are on a circle. ■

Lay down complex coordinates with $S = 0$ and E and F on the positive real axis. Then there are real r_1, r_2 and R with $B = r_1A$, $F = r_2E$ and $D = E + R(A - E)$ and hence $AE/ED = BF/FC$ gives

$$C = F + R(B - F) = r_2(1 - R)E + r_1RA.$$

The line CD consists of all points of the form $sC + (1 - s)D$ for real s . Since T lies on this line and has zero imaginary part, we see from $\text{Im}(sC + (1 - s)D) = (sr_1R + (1 - s)R)\text{Im}(A)$ that it corresponds to $s = -1/(r_1 - 1)$. Thus

$$T = \frac{r_1D - C}{r_1 - 1} = \frac{(r_2 - r_1)(R - 1)E}{r_1 - 1}.$$

Apply the lemma with $x = E$, $y = A - E$, $z = (r_2 - r_1)E/(r_1 - 1)$, and $s = (r_2 - 1)(r_1 - r_2)$. Setting $t = 1$ gives

$$(x, x + y, x + (s + 1)z) = (E, A, S = 0)$$

and setting $t = R$ gives

$$(x, x + Ry, x + (s + R)z) = (E, D, T).$$

Therefore the circumcircles to SAE and TDE meet at

$$x + \frac{sy}{y - z} = \frac{AE(r_1 - r_2)}{(1 - r_1)E - (1 - r_2)A} = \frac{AF - BE}{A + F - B - E}.$$

This last expression is invariant under simultaneously interchanging A and B and interchanging E and F . Therefore it is also the intersection of the circumcircles of SBF and TCF .

Solution 3

Let M be the Miquel point of $ABCD$; then M is the center of the spiral similarity that takes AD to BC . Because $\frac{AE}{ED} = \frac{BF}{FC}$, the same spiral similarity also takes E to F , so M is the center of the spiral similarity that maps AE to BF and ED to FC . Then it is obvious that the circumcircles of SAE , SBF , TCF , and TDE pass through M .

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

§36 USAMO 2007

1. (*USAMO 2007*) Let n be a positive integer. Define a sequence by setting $a_1 = n$ and, for each $k > 1$, letting a_k be the unique integer in the range $0 \leq a_k \leq k - 1$ for which $a_1 + a_2 + \cdots + a_k$ is divisible by k . For instance, when $n = 9$ the obtained

sequence is $9, 1, 2, 0, 3, 3, 3, \dots$. Prove that for any n the sequence $a_1, a_2, a_3, \dots$ eventually becomes constant.

Solution 1

Let $S_k = a_1 + a_2 + \dots + a_k$ and $b_k = \frac{S_k}{k}$. Thus, because $S_{k+1} = S_k + a_{k+1}$,

$$b_{k+1} = \frac{b_k \cdot k + a_{k+1}}{k+1} = \left(\frac{k}{k+1} \right) \cdot b_k + \frac{a_{k+1}}{k+1}$$

$\frac{k}{k+1} < 1$, and by definition, $\frac{a_{k+1}}{k+1} < 1$. Thus, $b_{k+1} < b_k + 1$. Also, both b_k and b_{k+1} are integers, so $b_{k+1} \leq b_k$. As the b_k 's form a **non-increasing** sequence of positive integers, they must eventually become constant.

Therefore, $b_k = b_{k+1}$ for some sufficiently large value of k . Then $a_{k+1} = S_{k+1} - S_k = b_k(k+1) - b_k(k) = b_k$, so eventually the sequence a_k becomes constant.

Solution 2

Let $a_1 = n$. Since $a_k \leq k-1$, we have that

$$a_1 + a_2 + \dots + a_n \leq n + 1 + 2 + \dots + n - 1 = \frac{n(n+1)}{2}.$$

Since $a_1 + a_2 + \dots + a_n = nk$ for some integer k , we can keep adding k to satisfy the conditions, provided that $k \leq n$. This is true since $k \leq \frac{n+1}{2} \leq n$, so the sequence must eventually become constant.

Solution 3

Define $S_k = a_1 + a_2 + \dots + a_k$, and $b_k = \frac{S_k}{k}$. By the problem hypothesis, b_k is an integer valued sequence.

Lemma: There exists a k such that $b_k < k$.

Proof: Choose any k such that $k^2 + 3k - 2 > 2n$. Then

$$\begin{aligned} \frac{k^2 + 3k - 2}{2} &> n \\ k^2 &> \frac{k^2 - 3k + 2}{2} + n \\ k^2 &> (k-2) + (k-1) + \dots + 1 + n \\ k^2 &> a_{k-1} + a_{k-2} + \dots + a_2 + a_1 \\ k^2 &> S_k \\ k &> \frac{S_k}{k} \\ k &> b_k, \end{aligned}$$

as desired.

End Lemma

Let k be the smallest k such that $b_k < k$. Then $b_k = m < k$, and $S_k = km$. To make b_{k+1} an integer, $S_{k+1} = S_k + a_{k+1}$ must be divisible by $k+1$. Thus, because $km + m$ is divisible by $k+1$, $a_{k+1} \equiv m \pmod{k+1}$, and, because $0 \leq a_{k+1} < k$, $a_{k+1} = m$. Then $b_{k+1} = \frac{(k+1)m}{k+1} = m$ as well. Repeating the same process using $k+1$ instead of k gives $a_{k+2} = m$, and an easy induction can prove that for all $N > k+1$, $a_N = m$. Thus, a_k becomes a constant function for arbitrarily large values of k .

Solution 4

For $k \geq 1$, let

$$s_k = a_1 + a_2 + \cdots + a_k.$$

We claim that for some m we have $s_m = m(m-1)$. To this end, consider the sequence which computes the differences between s_k and $k(k-1)$, i.e., whose k -th term is $s_k - k(k-1)$. Note that the first term of this sequence is positive (it is equal to n) and that its terms are strictly decreasing since

$$(s_k - k(k-1)) - (s_{k+1} - (k+1)k) = 2k - a_{k+1} \geq 2k - k = k \geq 1.$$

Further, a negative term cannot immediately follow a positive term. Suppose otherwise, namely that $s_k > k(k-1)$ and $s_{k+1} < (k+1)k$. Since s_k and s_{k+1} are divisible by k and $k+1$, respectively, we can tighten the above inequalities to $s_k \geq k^2$ and $s_{k+1} \leq (k+1)(k-1) = k^2 - 1$. But this would imply that $s_k > s_{k+1}$, a contradiction. We conclude that the sequence of differences must eventually include a term equal to zero.

Let m be a positive integer such that $s_m = m(m-1)$. We claim that

$$m-1 = a_{m+1} = a_{m+2} = a_{m+3} = a_{m+4} = \cdots$$

This follows from the fact that the sequence $a_1, a_2, a_3, \dots$ is uniquely determined and choosing $a_{m+i} = m-1$, for $i \geq 1$, satisfies the range condition

$$0 \leq a_{m+i} = m-1 \leq m+i-1,$$

and yields

$$s_{m+i} = s_m + i(m-1) = m(m-1) + i(m-1) = (m+i)(m-1).$$

Solution 5

Let $S_k = \sum_{i=1}^k a_i$, where $S_k \equiv 0 \pmod{k}$ and $0 \leq a_k < k$. Define $m_k = S_k/k$, with $m_1 = n$.

First, for $k > 1$, since $S_{k-1} = (k-1) \cdot m_{k-1}$ and $S_k = S_{k-1} + a_k$, we need $S_k \equiv 0 \pmod{k}$. Set $a_k \equiv m_{k-1} \pmod{k}$. Then,

$S_k \pmod{k} \equiv ((k-1) \cdot m_{k-1} + (m_{k-1} \pmod{k})) \pmod{k}$. Since $(k-1) \cdot m_{k-1} \equiv -m_{k-1} \pmod{k}$, and $-m_{k-1} + (m_{k-1} \pmod{k}) \equiv 0 \pmod{k}$, this holds.

Next, compute $m_k = S_k/k = ((k-1) \cdot m_{k-1} + a_k)/k$. If $m_{k-1} \geq k$, let $m_{k-1} = qk + r$ with $0 \leq r < k$, so $a_k = r$, and $m_k = \frac{(k-1) \cdot m_{k-1} + r}{k} < m_{k-1}$, so $m_k < m_{k-1}$. Moreover, so long as $m_{k-1} \geq k$ then m_i is strictly decreasing while k is strictly increasing, until $m_i \leq k$, in which case the sequence terminates as $m_{k-1} = m_k$.

Solution 6 (like solution 2)

First, we may make an observation and say that for $n \equiv p \pmod{k}$, $\sum_{m=2}^k a_m \equiv k - p \pmod{k}$ must occur for the whole sum to be divisible by k . Thus, the following is apparent:

$$\sum_{m=2}^k a_m = (q+1)k - p, k < n$$

Then, we may make another observation that when $n = k$, the sum also has to be divisible by n . We may then explore when $n=k$:

$$\sum_{m=2}^n a_m \equiv 0 \pmod{n}, \sum_{m=2}^n a_m = rn$$

and

$$a_n = \sum_{m=2}^n a_m - \sum_{m=2}^{n-1} a_m$$

Then,

$$a_n = rn - \sum_{m=2}^{n-1} a_m \leq n - 1$$

Also,

$$a_{n+s} = r, r \leq n$$

for $s \geq 1$. This is because:

$$\sum_{m=2}^n a_m + r = rn + r = r(n+1)$$

This must be true since $r(n+1)$ will be divisible by $n+1$ and $k = n+1$, we may then generalize this to all $r(n+s), s \in \mathbb{Z}, s \geq 1$

$$rn + sr = r(n+s), \frac{r(n+s)}{r} = n+s = \frac{\sum_{m=2}^{n+s} a_m}{r}$$

Thus, we may say that the sequence $a_1, a_2, \dots, a_k$ must converge to some integer value $r \leq n$ when $k \geq n+1$.

Solution 7(A simpler version of Solution 1)

$$a_1 = 1 \cdot x_1, a_1 + a_2 = 2 \cdot x_2, a_1 + a_2 + a_3 = 3 \cdot x_3, \dots$$

Claim.

$$x_1 \geq x_2 \geq x_3 \geq \dots$$

Proof.

$$a_2 = 2 \cdot x_2 - 1 \cdot x_1, a_3 = 3 \cdot x_3 - 2 \cdot x_2, a_4 = 4 \cdot x_4 - 3 \cdot x_3, \dots$$

F.T.S.O.C suppose $x_{n-1} < x_n$ for some $n > 1$.

$$a_n = n \cdot x_n - (n-1) \cdot x_{n-1}.$$

Note that $n \cdot x_n > (n-1) \cdot (x_{n-1} + 1)$ as $x_n \geq x_{n-1} + 1$.

This directly implies $n \cdot x_n - (n-1) \cdot x_{n-1} = a_n > n-1$, a contradiction to the bound given on a_n by the problem.

Hence, it is impossible to have $x_{n-1} < x_n$ for any $n > 1$, implying $x_{n-1} \geq x_n$ for each $n > 1$, that is, $x_1 \geq x_2 \geq x_3 \geq \dots$, proving our claim.

As the sequence x_i only consists of positive integers, it can't be ever decreasing, so that it must become eventually constant,

that is, $x_k = x_{k+1} = \dots$ from some $k \geq 1$.

Then,

$$a_1 + \dots + a_k = k \cdot x_k, a_1 + \dots + a_k + a_{k+1} = (k+1) \cdot x_{k+1} = (k+1) \cdot x_k, \dots$$

giving us $x_k = a_{k+1} = a_{k+2} = \dots$ (by subtracting the 1st equation from the 2nd, the 2nd from the 3rd, and so on), completing the proof to the problem.

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

2. (*USAMO 2007*) A square grid on the Euclidean plane consists of all points (m, n) , where m and n are integers. Is it possible to cover all grid points by an infinite family of discs with non-overlapping interiors if each disc in the family has radius at least 5?

Solution 1

Lemma. Among 3 [tangent](#) circles with radius greater than or equal to 5, one can always fit a circle with radius greater than $\frac{1}{\sqrt{2}}$ between those 3 circles.

Proof. [Descartes' Circle Theorem](#) states that if a is the curvature of a circle ($a = \frac{1}{r}$, positive for [externally tangent](#), negative for [internally tangent](#)), then we have that

$$(a + b + c + d)^2 = 2(a^2 + b^2 + c^2 + d^2)$$

Solving for a , we get

$$a = b + c + d + 2\sqrt{bc + cd + db}$$

Take the positive root, as the negative root corresponds to internally tangent circle.

Now clearly, we have $b + c + d \leq \frac{3}{5}$, and $bc + cd + db \leq \frac{3}{25}$. Summing/[square root](#)/multiplying appropriately shows that $a \leq \frac{3+2\sqrt{3}}{5}$. Incidentally, $\frac{3+2\sqrt{3}}{5} < \sqrt{2}$, so $a < \sqrt{2}$, $r > \frac{1}{\sqrt{2}}$, as desired. ■

For sake of [contradiction](#), assume that we have a satisfactory placement of circles. Consider 3 circles, p , q , r where there are no circles in between. By [Appolonius' problem](#), there exists a circle t tangent to p , q , r externally that is between those 3 circles. Clearly, if we move p , q , r together, t must decrease in radius. Hence it is sufficient to consider 3 tangent circles. By lemma 1, there is always a circle of radius greater than $\frac{1}{\sqrt{2}}$ that lies between p , q , r . However, any circle with $r > \frac{1}{\sqrt{2}}$ must contain a [lattice point](#). (Consider placing an unit square parallel to the gridlines in the circle.) That is a contradiction. Hence no such tiling exists.

Solution 2

It is not possible. The proof is by contradiction. Suppose that such a covering family $\mathcal{F}$ exists. Let $D(P, \rho)$ denote the disc with center P and radius ρ . Start with an arbitrary disc $D(O, r)$ that does not overlap any member of $\mathcal{F}$. Then $D(O, r)$ covers no grid point. Take the disc $D(O, r)$ to be maximal in the sense that any further enlargement would cause it to violate the non-overlap condition. Then $D(O, r)$ is tangent to at least three discs in $\mathcal{F}$. Observe that there must be two of the three tangent discs, say $D(A, a)$ and $D(B, b)$ such that $\angle AOB \leq 120^\circ$. By the Law of Cosines applied to triangle ABO ,

$$(a + b)^2 \leq (a + r)^2 + (b + r)^2 + (a + r)(b + r),$$

which yields

$$ab \leq 3(a + b)r + 3r^2,$$

and thus

$$12r^2 \geq (a - 3r)(b - 3r).$$

Note that $r < 1/\sqrt{2}$ because $D(O, r)$ covers no grid point, and $(a - 3r)(b - 3r) \geq (5 - 3r)^2$ because each disc in $\mathcal{F}$ has radius at least 5. Hence $2\sqrt{3}r \geq 5 - 3r$, which gives $5 \leq (3 + 2\sqrt{3})r < (3 + 2\sqrt{3})/\sqrt{2}$ and thus $5\sqrt{2} < 3 + 2\sqrt{3}$. Squaring both sides of this inequality yields $50 < 21 + 12\sqrt{3} < 21 + 12 \cdot 2 = 45$. This contradiction completes the proof.

Remark: The above argument shows that no covering family exists where each disc has radius greater than $(3 + 2\sqrt{3})/\sqrt{2} \approx 4.571$. In the other direction, there exists a covering family in which each disc has radius $\sqrt{13}/2 \approx 1.802$. Take discs with this radius centered at points of the form $(2m + 4n + \frac{1}{2}, 3m + \frac{1}{2})$, where m and n are integers. Then any grid point is within $\sqrt{13}/2$ of one of the centers and the distance between any two centers is at least $\sqrt{13}$. The extremal radius of a covering family is unknown.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

3. (*USAMO 2007*) Let S be a set containing $n^2 + n - 1$ elements, for some positive integer n . Suppose that the n -element subsets of S are partitioned into two classes. Prove that there are at least n pairwise disjoint sets in the same class.

Solution 1

Claim: If we have instead $k(n + 1) - 1$ elements, then we must have k disjoint subsets in a class. We proceed by Induction

Base Case: $k = 1$ is trivial.

Case 1: If there exists an $n + 1$ element subset X s.t. X has 2 n -element subsets in different classes. Consider the $(k - 1)(n + 1) - 1$ elements that are not in X , from the Induction hypothesis, we have $k - 1$ disjoint subsets in one of the classes. Those $k - 1$ subsets paired with a subset of X in the same class forms k disjoint subsets of size n .

Case 2: All $n + 1$ element subsets have all of their n -element subsets in one class. Look at any 2 n -element subsets a_1 and a_p . Then we can create a path $a_1, a_2, \dots, a_p$ s.t. any 2 consecutive terms belong to the same $n + 1$ -element subset. (i.e. we change 1 element each time.) Thus, a_1 and a_p are in the same class and we see that all subsets are in one class. Obviously we can find k disjoint subsets of size n .

The question is asking for the case when we substitute $k = n$ which we have proved.

Solution 2

Call an $n + 1$ -element subset of S separable if it has a subset in each class of the partition. We [recursively](#) build a set Q of disjoint separable subsets of S : begin with Q empty and at each step if there is a separable subset which is disjoint from all sets in Q add that set to Q . The process terminates when every separable subset intersects a set in Q . Let T be the set of elements in S which are not in any set in Q . We claim that one class contains every n -element subset of T .

Suppose that $a_1, a_2, \dots, a_k$ are elements of T . Denote by A_i the set $\{a_i, a_{i+1}, \dots, a_{i+n-1}\}$. Note that for each i , $A_i \cup A_{i+1}$ is not separable, so that A_i and A_{i+1} are in the same class. But then A_i is in the same class for each $1 \leq i \leq k - n + 1$ — in particular, A_1 and A_{k-n+1} are in the same class. But for any two sets we may construct such a sequence with A_1 equal to one and A_{k-n+1} equal to the other.

We are now ready to construct our n disjoint sets. Suppose that $|Q| = q$. Then $|T| = (n + 1)(n - q) - 1 \geq n(n - q)$, so we may select $n - q$ disjoint n -element

subsets of T . Then for each of the q sets in Q , we may select a subset which is in the same class as all the subsets of T , for a total of n disjoint sets.

Solution 3

In order to apply induction, we generalize the result to be proved so that it reads as follows:

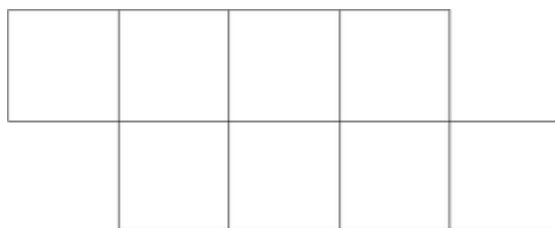
Proposition. If the n -element subsets of a set S with $(n+1)m - 1$ elements are partitioned into two classes, then there are at least m pairwise disjoint sets in the same class.

Proof. Fix n and proceed by induction on m . The case of $m = 1$ is trivial. Assume $m > 1$ and that the proposition is true for $m - 1$. Let $\mathcal{P}$ be the partition of the n -element subsets into two classes. If all the n -element subsets belong to the same class, the result is obvious. Otherwise select two n -element subsets A and B from different classes so that their intersection has maximal size. It is easy to see that $|A \cap B| = n - 1$. (If $|A \cap B| = k < n - 1$, then build C from B by replacing some element not in $A \cap B$ with an element of A not already in B . Then $|A \cap C| = k + 1$ and $|B \cap C| = n - 1$ and either A and C or B and C are in different classes.) Removing $A \cup B$ from S , there are $(n+1)(m-1) - 1$ elements left. On this set the partition induced by $\mathcal{P}$ has, by the inductive hypothesis, $m - 1$ pairwise disjoint sets in the same class. Adding either A or B as appropriate gives m pairwise disjoint sets in the same class. ■

Remark: The value $n^2 + n - 1$ is sharp. A set S with $n^2 + n - 2$ elements can be split into a set A with $n^2 - 1$ elements and a set B of $n - 1$ elements. Let one class consist of all n -element subsets of A and the other consist of all n -element subsets that intersect B . Then neither class contains n pairwise disjoint sets.

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

4. (USAMO 2007) An *animal* with n cells is a connected figure consisting of n equal-sized cells.¹ The figure below shows an 8-cell animal.



A *dinosaur* is an animal with at least 2007 cells. It is said to be *primitive* if its cells cannot be partitioned into two or more dinosaurs. Find with proof the maximum number of cells in a primitive dinosaur.

¹Animals are also called *polyominoes*. They can be defined inductively. Two cells are *adjacent* if they share a complete edge. A single cell is an animal, and given an animal with n cells, one with $n + 1$ cells is obtained by adjoining a new cell by making it adjacent to one or more existing cells.

Solution 1

Let a n -dino denote an animal with n or more cells.

We show by **induction** that an n -dino with $4n - 2$ or more animal cells is not primitive. (Note: if it had more, we could just take off enough until it had $4n - 2$, which would have a partition, and then add the cells back on.)

Base Case: If $n = 1$, we have two cells, which are clearly not primitive.

Inductive Step: Assume any $4n - 2$ cell animal can be partitioned into two or more n -dinos.

For a given $(4n + 2)$ -dino, take off any four cells (call them w, x, y, z) to get an animal with $4n - 2$ cells.

This can be partitioned into two or more n -dinos, let's call them A and B . This means that A and B are connected.

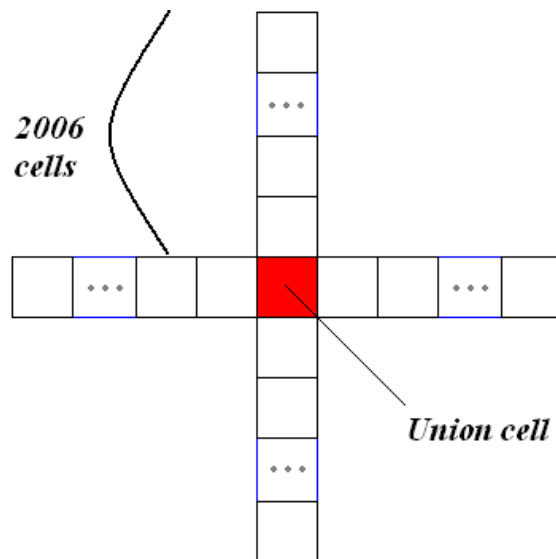
If both A and B are $(n + 1)$ -dinos or if w, x, y, z don't all attach to one of them, then we're done.

So assume A has n cells and thus B has at least $3n - 2$ cells, and that w, x, y, z are added to B . So B has $3n + 2$ cells total.

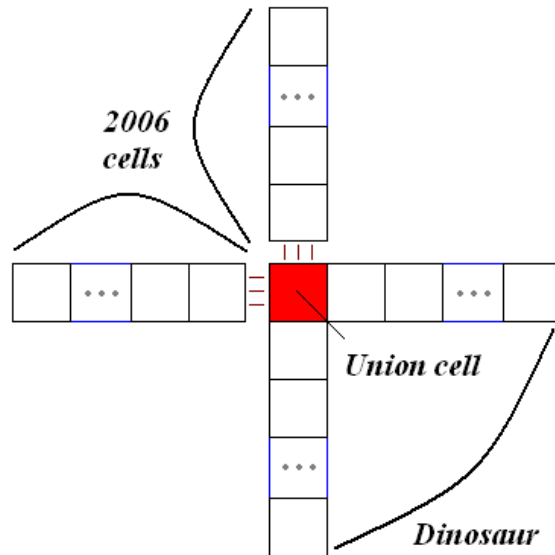
Let C denote the cell of B attached to A . There are $3n + 1$ cells on B besides C . Thus, of the three (or less) sides of C not attached to A , one of them must have $n + 1$ cells by the **pigeonhole principle**. It then follows that we can add A, C , and the other two sides together to get an $(n + 1)$ dino, and the side of C that has $n + 1$ cells is also an $n + 1$ -dino, so we can partition the animal with $4n + 2$ cells into two $(n + 1)$ -dinos and we're done.

Thus, our answer is $4(2007) - 3 = 8025$ cells.

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Example of a solution Attempting to partition solution into ≥ 2 dinosaurs

Solution 2

For simplicity, let $k = 2007$ and let n be the number of **squares**. Let the centers of the squares be **vertices**, and connect any centers of adjacent squares with edges. Suppose we have some **loops**. Just remove an edge in the loop. We are still connected since you can go around the other way in the loop. Now we have no loops. Each vertex can have at most 4 edges coming out of it. For each point, assign it the **quadruple**: (a, b, c, d) where a, b, c, d are the numbers of vertices on each branch, **WLOG** $a \geq b \geq c \geq d$. Note $a + b + c + d = n - 1$.

Claim: If $n = 4k - 2$, then we must be able to divide the animal into two dinosaurs. Chose a vertex, v , for which a is minimal (i.e. out of all maximal elements in a quadruple, choose the one with the least maximal element). We have that $4a \geq a + b + c + d = 4k - 3$, so $a \geq k$. Hence we can just cut off that branch, that forms a dinosaur.

But suppose the remaining vertices do not make a dinosaur. Then we have $b + c + d + 1 \leq k - 1 \iff n - a \leq k - 1 \iff a \geq 3k - 1$. Now move to the first point on the branch at a . We have a new quadruple $(p, q, r, b + c + d + 1)$ where $p + q + r = a - 1 \geq 3k - 2$.

Now consider the maximal element of that quadruple. We already have $b + c + d + 1 \leq k - 1$. **WLOG** $p \geq q \geq r \geq 0$, then $3p \geq p + q + r = a - 1 \geq 3k - 2 \implies p \geq k$ so $p > k - 1 = b + c + d + 1$, so p is the maximal element of that quadruple.

Also $a - 1 = p + q + r \geq p + 0 + 0$, so $p < a$. But that is a **contradiction** to the minimality of a . Therefore, we must have that $b + c + d + 1 \geq k$, so we have a partition of two dinosaurs.

Maximum: $n = 4k - 3$. Consider a cross with each branch having $k - 1$ vertices. Clearly if we take partition k vertices, we remove the center, and we are not connected.

So $k = 2007$: $4 \cdot 2007 - 3 = 8025$.

Solution 3 (Generalization)

Turn the dinosaur into a **graph** (cells are **vertices**, adjacent cells connected by an edge) and prove this result about graphs. A connected graph with V vertices, where each vertex has degree less than or equal to D , can be partitioned into connected components of sizes at least $\frac{V-1}{D}$. So then in this special case, we have $D = 4$, and so $V = 2006 \times 4 + 1$ (a possible configuration of this size that works consists of a center and 4 lines of cells each of size 2006 connected to the center). We next throw out all the **geometry** of this situation, so that we have a completely unconstrained graph. If we prove the above-mentioned result, we can put the geometry back in later by taking the connected components that our partition gives us, then filling back all edges that have to be there due to adjacent cells. This won't change any of the problem constraints, so we can legitimately do this.

Going, now, to the case of arbitrary graphs, we WOP on the number of edges. If we can remove any edge and still have a connected graph, then we have found a smaller graph that does not obey our theorem, a contradiction due to the minimality imposed by WOP. Therefore, the only case we have to worry about is when the graph is a tree. If it's a tree, we can root the tree and consider the size of subtrees. Pick the root such that the size of the largest subtree is minimized. This minimum must be at least $\frac{V-1}{D}$, otherwise the sum of the size of the subtrees is smaller than the size of the graph, which is a contradiction. Also, it must be at most $\frac{V}{2}$, or else pick the subtree of size greater than $\frac{V}{2}$ and you have decreased the size of the largest subtree if you root from that vertex instead, so you have some subtree with size between $\frac{V-1}{D}$ and $\frac{V}{2}$. Cut the edge connecting the root to that subtree, and use that as your partition.

It is easy to see that these partitions satisfy the contention of our theorem, so we are done.

Solution 4

Let s denote the minimum number of cells in a dinosaur; the number this year is $s = 2007$.

Claim: The maximum number of cells in a primitive dinosaur is $4(s - 1) + 1$.

First, a primitive dinosaur can contain up to $4(s - 1) + 1$ cells. To see this, consider a dinosaur in the form of a cross consisting of a central cell and four arms with $s - 1$ cells apiece. No connected figure with at least s cells can be removed without disconnecting the dinosaur.

The proof that no dinosaur with at least $4(s - 1) + 2$ cells is primitive relies on the following result.

Lemma. Let D be a dinosaur having at least $4(s - 1) + 2$ cells, and let R (red) and B (black) be two complementary animals in D , i.e. $R \cap B = \emptyset$ and $R \cup B = D$. Suppose $|R| \leq s - 1$. Then R can be augmented to produce animals $\tilde{R} \subset R$ and $\tilde{B} = D \setminus \tilde{R}$ such that at least one of the following holds:

- (i) $|\tilde{R}| \geq s$ and $|\tilde{B}| \geq s$,
- (ii) $|\tilde{R}| = |R| + 1$,
- (iii) $|R| < |\tilde{R}| \leq s - 1$.

Proof. If there is a black cell adjacent to R that can be made red without disconnecting B , then (ii) holds. Otherwise, there is a black cell c adjacent to R whose removal disconnects B . Of the squares adjacent to c , at least one is red, and at least one is black, otherwise B would be disconnected. Then there are at most

three resulting components $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ of B after the removal of c . Without loss of generality, $\mathcal{C}_3$ is the largest of the remaining components. (Note that $\mathcal{C}_1$ or $\mathcal{C}_2$ may be empty.) Now $\mathcal{C}_3$ has at least $\lceil (3s-2)/3 \rceil = s$ cells. Let $\tilde{B} = \mathcal{C}_3$. Then $|\tilde{R}| = |R| + |\mathcal{C}_1| + |\mathcal{C}_2| + 1$. If $|\tilde{B}| \leq 3s-2$, then $|\tilde{R}| \geq s$ and (i) holds. If $|\tilde{B}| \geq 3s-1$ then either (ii) or (iii) holds, depending on whether $|\tilde{R}| \geq s$ or not. ■

Starting with $|R| = 1$, repeatedly apply the Lemma. Because in alternatives (ii) and (iii) $|R|$ increases but remains less than s , alternative (i) eventually must occur. This shows that no dinosaur with at least $4(s-1) + 2$ cells is primitive.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

5. (*USAMO 2007*) Prove that for every nonnegative integer n , the number $7^{7^n} + 1$ is the product of at least $2n + 3$ (not necessarily distinct) primes.

Solution 1

The proof is by induction. The base is provided by the $n = 0$ case, where $7^{7^0} + 1 = 7^1 + 1 = 2^3$. To prove the inductive step, it suffices to show that if $x = 7^{2m-1}$ for some positive integer m then $(x^7 + 1)/(x + 1)$ is composite. As a consequence, $x^7 + 1$ has at least two more prime factors than does $x + 1$. To confirm that $(x^7 + 1)/(x + 1)$ is composite, observe that

$$\begin{aligned} \frac{x^7 + 1}{x + 1} &= \frac{(x + 1)^7 - ((x + 1)^7 - (x^7 + 1))}{x + 1} \\ &= (x + 1)^6 - \frac{7x(x^5 + 3x^4 + 5x^3 + 5x^2 + 3x + 1)}{x + 1} \\ &= (x + 1)^6 - 7x(x^4 + 2x^3 + 3x^2 + 2x + 1) \\ &= (x + 1)^6 - 7^{2m}(x^2 + x + 1)^2 \\ &= \{(x + 1)^3 - 7^m(x^2 + x + 1)\}\{(x + 1)^3 + 7^m(x^2 + x + 1)\}. \end{aligned}$$

Also each factor exceeds 1. It suffices to check the smaller one; $\sqrt{7x} \leq x$ gives

$$\begin{aligned} (x + 1)^3 - 7^m(x^2 + x + 1) &= (x + 1)^3 - \sqrt{7x}(x^2 + x + 1) \\ &\geq x^3 + 3x^2 + 3x + 1 - x(x^2 + x + 1) \\ &= 2x^2 + 2x + 1 \geq 113 > 1. \end{aligned}$$

Hence $(x^7 + 1)/(x + 1)$ is composite and the proof is complete.

Solution 2

We prove by induction on n that $7^{7^n} + 1$ is the product of at least $2n + 3$ (not necessarily distinct) primes.

For $n = 0$, we have $7^{7^0} + 1 = 7 + 1 = 8 = 2^3$, which has $3 = 2(0) + 3$ prime factors. Thus, the base case holds.

Assume that for some $n \geq 0$, the number $7^{7^n} + 1$ is the product of at least $2n + 3$ primes. We must show that $7^{7^{n+1}} + 1$ has at least $2n + 5$ prime factors.

Let $a = 7^n$. Since a is odd, we can apply the Aurifeuillian factorization: $7^{7^a} + 1 = (7^a + 1)(7^{3a} + 3 \cdot 7^{2a} + 3 \cdot 7^a + 1)(7^{\frac{3a}{2} + \frac{1}{2}} + 7^{\frac{3a}{2} + \frac{1}{2}} + 7^{\frac{a}{2} + \frac{1}{2}})$.

Each factor is an integer greater than 1, so the factorization is nontrivial.

The first factor is $7^a + 1 = 7^{7^n} + 1$, which by the inductive hypothesis has at least $2n + 3$ prime factors. It remains to show that the quotient $\frac{7^{7^a} + 1}{7^a + 1}$ is composite.

From the factorization above, we have $\frac{7^{7a}+1}{7^a+1} = (7^{3a} + 3 \cdot 7^{2a} + 3 \cdot 7^a + 1)(7^{\frac{5a}{2}+\frac{1}{2}} + 7^{\frac{3a}{2}+\frac{1}{2}} + 7^{\frac{a}{2}+\frac{1}{2}})$,

which is clearly the product of two integers greater than 1, and thus composite.

Therefore, $\frac{7^{7a}+1}{7^a+1}$ is always composite, and hence $7^{7a}+1$ has at least two more prime factors than 7^a+1 .

By the inductive hypothesis, 7^a+1 has at least $2n+3$ prime factors, so $7^{7a}+1$ has at least $(2n+3)+2=2n+5$ prime factors.

By induction, $7^{7^n}+1$ is the product of at least $2n+3$ (not necessarily distinct) primes for all nonnegative integers n .

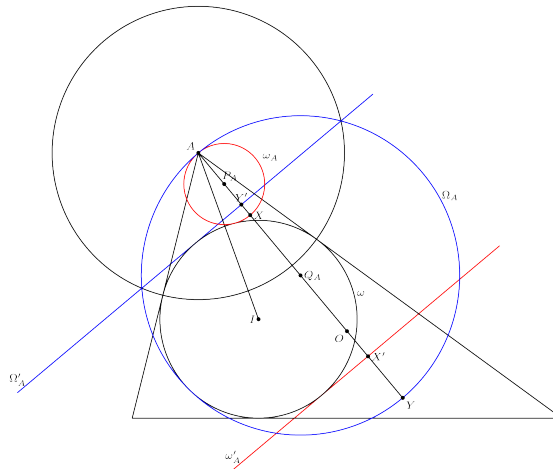
<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

6. (USAMO 2007) Let ABC be an acute triangle with ω , Ω , and R being its incircle, circumcircle, and circumradius, respectively. Circle ω_A is tangent internally to Ω at A and tangent externally to ω . Circle Ω_A is tangent internally to Ω at A and tangent internally to ω . Let P_A and Q_A denote the centers of ω_A and Ω_A , respectively. Define points P_B, Q_B, P_C, Q_C analogously. Prove that

$$8P_AQ_A \cdot P_BQ_B \cdot P_CQ_C \leq R^3,$$

with equality if and only if triangle ABC is equilateral.

Solution 1



Lemma.

$$P_AQ_A = \frac{4R^2(s-a)^2(s-b)(s-c)}{r^2b^2c^2}$$

Proof. Note P_A and Q_A lie on AO since for a pair of tangent circles, the point of tangency and the two centers are **collinear**.

Let ω touch BC , CA , and AB at D , E , and F , respectively. Note $AE = AF = s-a$. Consider an **inversion**, $\mathcal{I}$, centered at A , passing through E , F . Since $IE \perp AE$, ω is **orthogonal** to the inversion circle, so $\mathcal{I}(\omega) = \omega$. Consider $\mathcal{I}(\omega_A) = \omega'_A$. Note that ω_A passes through A and is tangent to ω_A , hence ω'_A is a line that is tangent to ω . Furthermore, $\omega'_A \perp AO$ because ω_A is symmetric about OA , so the inversion preserves that reflective symmetry. Since it is a line that is symmetric about AO ,

it must be perpendicular to AO . Likewise, $\mathcal{I}(\Omega_A) = \Omega'_A$ is the other line tangent to ω and **perpendicular** to AO .

Let $\omega_A \cap AO = X$ and $\omega'_A \cap AO = X'$ (second intersection).

Let $\Omega_A \cap AO = Y$ and $\Omega'_A \cap AO = Y'$ (second intersection).

Evidently, $AX = 2AP_A$ and $AY = 2AQ_A$. We want:

$$\star = AQ_A - AP_A = \frac{1}{2}(AY - AX) = \frac{(s-a)^2}{2} \left(\frac{1}{AY'} - \frac{1}{AX'} \right)$$

by inversion. Note that $\omega'_A \parallel \Omega'_A$, and they are tangent to ω , so the **distance** between those lines is $2r = AX' - AY'$. Drop a perpendicular from I to AO , touching at H . Then $AH = AI \cos \angle OAI = AI \cos \frac{1}{2}|\angle B - \angle C|$. Then $AX', AY' = AI \cos \frac{1}{2}|\angle B - \angle C| \pm r$. So $AX' \cdot AY' = AI^2 \cos^2 \frac{1}{2}|\angle B - \angle C| - r^2$.

$$\begin{aligned} \star &= \frac{(s-a)^2}{2} \cdot \frac{AX' - AY'}{AY' \cdot AX'} \\ &= \frac{r(s-a)^2}{AI^2 \cos^2 \frac{1}{2}|\angle B - \angle C| - r^2} \\ &= \frac{\frac{(s-a)^2}{r}}{\left(\frac{AI}{r}\right)^2 \cos^2 \frac{1}{2}|\angle B - \angle C| - 1}. \end{aligned}$$

Note that $\frac{AI}{r} = \frac{1}{\sin \frac{A}{2}}$. Applying the double angle formulas and $1 - \cos \angle A = \frac{2(s-b)(s-c)}{bc}$, we get

$$\begin{aligned} \star &= \frac{\frac{(s-a)^2}{r}}{\frac{1+\cos(\angle B - \angle C)}{1-\cos A} + 1} \\ &= \frac{\frac{(s-a)^2}{r} \cdot (1 - \cos \angle A)}{\cos(\angle B - \angle C) + \cos(\pi - \angle B - \angle C)} \\ &= \frac{(s-a)^2(1 - \cos \angle A)}{2r \sin \angle B \sin \angle C} \\ &= \frac{(s-a)^2(s-b)(s-c)}{rbc \sin \angle B \sin \angle C} \\ P_A Q_A &= \frac{4R^2(s-a)^2(s-b)(s-c)}{rb^2c^2} \end{aligned}$$

End Lemma

The problem becomes:

$$\begin{aligned} 8 \prod \frac{4R^2(s-a)^2(s-b)(s-c)}{rb^2c^2} &\leq R^3 \\ \frac{2^9 R^6 [\prod(s-a)]^4}{r^3 a^4 b^4 c^4} &\leq R^3 \\ \left(\frac{4AR}{abc}\right)^4 \cdot \left(\frac{A}{rs}\right)^4 2rR^2 &\leq R^3 \\ 2r &\leq R, \end{aligned}$$

which is true because $OI^2 = R(R - 2r)$, equality is when the circumcenter and incenter coincide. As before, $\angle OAI = \frac{1}{2}|\angle B - \angle C| = 0$, so, by symmetry, $\angle A = \angle B = \angle C$. Hence the inequality is true iff $\triangle ABC$ is equilateral.

Comment: It is much easier to determine AP_A by considering $\triangle IAP_A$. We have AI , $\angle IAO$, $IP_A = r + r(P_A)$, and $AP_A = r(P_A)$. However, the inversion is always nice to use. This also gives an easy construction for w_A because the tangency point is collinear with the intersection of w'_A and w .

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

§37 USAMO 2008

1. (*USAMO 2008*) (*Titu Andreescu*) Prove that for each positive integer n , there are pairwise relatively prime integers $k_0, k_1, \dots, k_n$, all strictly greater than 1, such that $k_0 k_1 \cdots k_n - 1$ is the product of two consecutive integers.

Solution 1

We will prove the problem for each nonnegative integer n . We wish to show that

$$k_0 k_1 \cdots k_n = a^2 + a + 1,$$

for some integer a . We induct on n . For our base case, $n = 0$, we may let a be positive integer.

For the inductive step, suppose that $k_0, \dots, k_{n-1}$ are pairwise relatively prime integers such that

$$k_0 \cdots k_{n-1} = a^2 + a + 1,$$

for some integer a . Let $k_n = a^2 + 3a + 3$. Evidently, $k_n > 1$. Also,

$$\gcd(a^2 + a + 1, k_n) = \gcd(a^2 + a + 1, k_n - (a^2 + a + 1)) = \gcd(a^2 + a + 1, 2(a + 1)).$$

Since $a^2 + a + 1 = a(a + 1) + 1$ is odd and relatively prime to $a + 1$, it follows that $a^2 + a + 1$ and k_n are relatively prime, so k_n is relatively prime to each of $k_0, \dots, k_{n-1}$. Finally,

$$\begin{aligned} k_0 k_1 \cdots k_n &= (a^2 + a + 1)(a^2 + 3a + 3) \\ &= [(a + 1)^2 - a] \cdot [(a + 1)^2 + a + 2] \\ &= (a + 1)^4 + 2(a + 1)^2 - a^2 - 2a \\ &= (a + 1)^4 + (a + 1)^2 + 1. \end{aligned}$$

This completes the induction.

Solution 2

Lemma. If p is prime such that $p \equiv 1 \pmod{3}$, there exists a residue r such that $p \mid r^2 + r + 1$.

Proof. Let g be a multiplicative generator of the nonzero integers mod p . Set $r \equiv g^{(p-1)/3}$. Then $r - 1 \not\equiv 0 \pmod{p}$, but $r^3 - 1 \equiv 0 \pmod{p}$, so $\frac{r^3 - 1}{r - 1} \equiv r^2 + r + 1 \equiv 0 \pmod{p}$.

End Lemma

By **Dirichlet's Theorem**, there are infinitely many primes congruent to 1 (mod 3). Let $p_0, \dots, p_n$ be $(n + 1)$ such primes, and let $r_0, \dots, r_n$ be respective residues as described in the lemma. By the **Chinese Remainder Theorem**, there is a positive integer a that satisfies the relation

$$a \equiv r_i \pmod{p_i}$$

for each integer $i \in [0, n]$. Then

$$p_0 \dots p_n \mid a^2 + a + 1.$$

Now, for $1 \leq i \leq n$, take k_i to be the greatest power of p_i that divides $a^2 + a + 1$, and let $k_0 = (a^2 + a + 1)/(k_1 \dots k_n)$. Since all the k_i are pairwise relatively prime and are greater than 1, we are done.

Solution 3

Firstly, we see that there are $n + 1$ numbers k_i . Since $n \geq 1$, there are at least 2 values of k_i . Define a relatively prime partition to be a set of relatively prime numbers such that their product is equal to some natural number.

Define P to be the greatest possible cardinality of a relatively prime partition for that number.

Lemma 1. All cardinalities of the relatively prime partitions of a number up to P can be attained.

Proof. $P = P - 0$ satisfies the properties. For any $P - n$ which satisfies the properties, we can take any of the 2^{P-n} numbers and multiply them together. Because they are both relatively prime to all the other numbers, their product is relatively prime to all the other numbers as well, and that results in $P - n - 1$ numbers which satisfy the conditions, unless $P - n = 1$, because there is only one number left. Therefore, all numbers of numbers of relatively prime factors from 1 to $P - 1$ are attainable, if P is attainable as well.

End Lemma

Lemma 2. $a^2 + a + 1$ can have arbitrarily many prime factors for some value of a .

Proof. Let $f(x) = x^2 + x + 1$. Let $a^2 + a + 1 = k$. Then $f(a + k) = a^2 + 2ak + k^2 + a + k + 1 = 2ak + k^2 + 2k$. After factoring, we get $k(k + 2 + 2a) = (a^2 + a + 1)(a^2 + 3a + 3)$. $a^2 + 3a + 3 \geq a^2 + a + 1$, because a is strictly positive. Let their GCD = g . $g \mid 2a + 2g \mid 2a^2 + 2ag \mid a^2 - 3a - 3 + a^2 + 3a + 3$, $g \mid 2a^2g \mid 2a$, and so $g \mid 2$, but $g \mid a^2 + a + 3$ which is strictly odd, so $g = 1$. $(a^2 + a + 1)(a^2 + 3a + 3)$ therefore must contain prime factors not in $a^2 + a + 1$. Upon repeating this an arbitrary number of times, we get a number of the form $a^2 + a + 1$ which has arbitrarily many distinct prime factors.

End Lemma

Then $a(a + 1) + 1 = a^2 + a + 1$ would not have $n + 1 = P$ elements in the relatively prime partition for some value of n , and any value of a . It is possible to choose a value of a such that $a^2 + a + 1$ has arbitrarily many unique prime factors, by our second Lemma, and so it is possible for P to be arbitrarily high. By our first Lemma, all numbers up to P are possible values for the cardinality of some relatively prime partition, and so there always exists some number that has an arbitrary number of elements in a relatively prime partition. Because $a^2 + a + 1 = a(a + 1) + 1$, $a(a + 1)$ is the product of 2 consecutive integers, we see that the given statement is true because if $k_0 k_1 \dots k_n = a(a + 1) + 1$, then $k_0 k_1 \dots k_n - 1 = a(a + 1)$.

Solution 4

Write the relation to be proved as

$$4k_0 k_1 \dots k_n = 4a(a + 1) + 4 = (2a + 1)^2 + 3.$$

There are infinitely many primes for which -3 is a quadratic residue. Let $2 < p_0 < p_1 < \dots < p_n$ be such primes. Using the Chinese Remainder Theorem to specify a modulo p_n , we can find an integer a such that $(2a+1)^2 + 3 = 4p_0p_1 \dots p_n m$ for some positive integer m . Grouping the factors of m appropriately with the p_i 's, we obtain $(2a+1)^2 + 3 = 4k_0k_1 \dots k_n$ with k_i pairwise relatively prime. We then have $k_0k_1 \dots k_n - 1 = a(a+1)$, as desired.

Solution 5

We are supposed to show that for every positive integer n , there is a positive integer x such that $x(x+1)+1 = x^2+x+1$ has at least n distinct prime divisors. We can actually prove a more general statement.

Claim. Let $P(x) = a_dx^d + \dots + a_1x + 1$ be a polynomial of degree $d \geq 1$ with integer coefficients. Then for every positive integer n , there is a positive integer x such that $P(x)$ has at least n distinct prime divisors.

The proof follows from the following two lemmas.

Lemma 1. The set

$$Q = \{p \mid p \text{ a prime for which there is an integer } x \text{ such that } p \text{ divides } P(x)\}$$

is infinite.

Proof. The proof is analogous to Euclid's proof that there are infinitely many primes. Namely, if we assume that there are only finitely many primes $p_1, p_2, \dots, p_k$ in Q , then for each integer m , $P(mp_1p_2 \dots p_k)$ is an integer with no prime factors, which must equal 1 or -1 . However, since P has degree $d \geq 1$, it takes each of the values 1 and -1 at most d times, a contradiction.

End Lemma

Lemma 2. Let $p_1, p_2, \dots, p_n$, $n \geq 1$ be primes in Q . Then there is a positive integer x such that $P(x)$ is divisible by $p_1p_2 \dots p_n$.

Proof. For $i = 1, 2, \dots, n$, since $p_i \in Q$ we can find an integer c_i such that $P(x)$ is divisible by p_i whenever $x \equiv c_i \pmod{p_i}$. By the Chinese Remainder Theorem, the system of n congruences $x \equiv c_i \pmod{p_i}$, $i = 1, 2, \dots, n$ has positive integer solutions. For every positive integer x that solves this system, $P(x)$ is divisible by $p_1p_2 \dots p_n$.

End Lemma

Solution 6

Very similar to some above solutions.

Let $k_0k_1 \dots k_n - 1 = a(a+1)$. We wish to show this equation has a solution $(a, k_0, \dots, k_n) \in \mathbb{Z}^{n+2}$ for any $n \in \mathbb{N}$. If $a^2 + a + 1$ has at least $n+1$ distinct prime factors, we can set $a^2 + a + 1 = p_1^{q_1} p_2^{q_2} \dots p_m^{q_m}$ for some $m \geq n$ where $p_1, \dots, p_m$ are distinct primes. Then, we let $k_i = p_{i+1}^{q_{i+1}}$ for all $0 \leq i \leq n-1$ and set $k_n = p_{n+1}^{q_{n+1}} \dots p_m^{q_m}$. Thus, it remains to show $a^2 + a + 1$ can have at least $n+1$ distinct prime factors. We will do this by showing it can have arbitrarily many.

Using an induction-type argument (it isn't exactly induction though), we first show that $a^2 + a + 1$ can have 1 prime factor by setting $a = 2$. Now, assume $a^2 + a + 1$ has n distinct prime factors. We wish to show there exists some b such that $b^2 + b + 1$ has more than n distinct prime factors (we need not show it has $n+1$, since we

only wish to demonstrate that the number of prime factors can be arbitrarily large rather than any integer). Set $b = a^2$. Then

$$\begin{aligned} b^2 + b + 1 &= (b + 1)^2 - b \\ &= (a^2 + 1)^2 - a^2 \\ &= (a^2 + a + 1)(a^2 - a + 1). \end{aligned}$$

Suppose p is a positive integer dividing both $a^2 + a + 1$ and $a^2 - a + 1$. Then $p \mid (a^2 + a + 1) - (a^2 - a + 1) = 2a$, and because $(a^2 + a + 1)$ is always odd, we have $2 \nmid p$ so $p \mid a$. But this yields $p \mid \gcd(a, a^2 + a + 1) = 1$, so $p = 1$ and thus $a^2 + a + 1$ and $a^2 - a + 1$ are coprime. Therefore, as one can make $a^2 - a + 1 > 1$, we see that $a^2 - a + 1$ has at least one prime factor which does not divide $a^2 + a + 1$ (which has n distinct prime factors). Therefore, $b^2 + b + 1$ has more than n distinct prime factors.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.</i>

2. (USAMO 2008) (Zuming Feng) Let ABC be an acute, **scalene** triangle, and let M , N , and P be the midpoints of $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$, respectively. Let the **perpendicular bisectors** of $\overline{AB}$ and $\overline{AC}$ intersect ray AM in points D and E respectively, and let lines BD and CE intersect in point F , inside of triangle ABC . Prove that points A , N , F , and P all lie on one circle.

Solution 1 (synthetic)

Without Loss of Generality, assume $AB > AC$. It is sufficient to prove that $\angle OFA = 90^\circ$, where O is the circumcenter of $\triangle ABC$, as this would immediately prove that A, P, O, F, N are concyclic. By applying the Menelaus' Theorem in the Triangle $\triangle BFC$ for the transversal E, M, D , we have (in magnitude)

$$\frac{FE}{EC} \cdot \frac{CM}{MB} \cdot \frac{BD}{DF} = 1 \iff \frac{FE}{EC} = \frac{DF}{BD}$$

Here, we used that $BM = MC$, as M is the midpoint of BC . Now, since $EC = EA$ and $BD = DA$, we have

$$\frac{FE}{EA} = \frac{DF}{DA} \iff \frac{DA}{AE} = \frac{DF}{FE} \iff AF \text{ bisects exterior } \angle EFD$$

Now, note that OE bisects the exterior $\angle FED$ and OD bisects exterior $\angle FDE$, making O the F -excentre of $\triangle FED$. This implies that OF bisects interior $\angle EFD$, making $OF \perp AF$, as was required.

Solution 2 (complex)

Let $A = 1, B = b, C = c$ where b, c all lie on the unit circle. Then O is 0. As noted earlier, $(FOBC)$ is cyclic. We will find the ghost point F' , the second intersection of OBC and ANP .

We know that these two circles already intersect at O so we can reflect over the line between their centers. The center of $ANPO$ is the midpoint of AO namely $\frac{1}{2}$. With the tangent formula and then taking the midpoint, we find that the center of OBC is $\frac{bc}{b+c}$. Then we want to find the reflection of 0 over the line through $\frac{1}{2}$ and $\frac{bc}{b+c}$. Then we get

$$\begin{aligned}
f' &= \frac{\left(\frac{bc}{b+c} - \frac{bc}{b+c}\right) \div 2}{(1/2) - \frac{bc}{b+c}} \\
&= \frac{\frac{bc-1}{2(b+c)}}{\frac{b+c-2}{2(b+c)}} \\
&= \frac{bc-1}{b+c-2}.
\end{aligned}$$

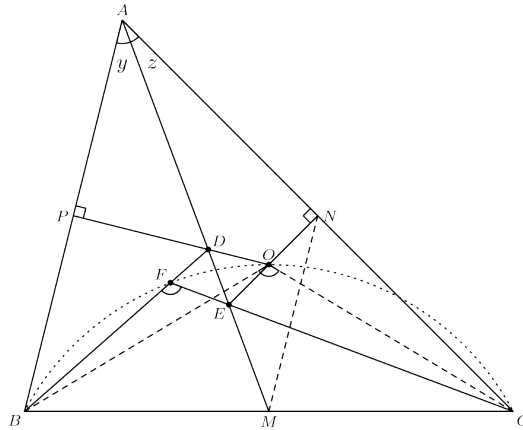
Now it remains to show $\angle F'BA = \angle ABM$; the other angle equality would follow by symmetry.

Then we get:

$$\begin{aligned}
\frac{f' - b}{b - a} \div \frac{b - a}{a - m} &= \frac{\frac{bc-1}{b+c-2} - b}{b - 1} \div \frac{b - 1}{1 - \frac{b+c}{2}} \\
&= \frac{\frac{bc-1-b(b+c-2)}{b+c-2} (2 - b + c)}{2(b - 1)^2} \\
&= \frac{b^2 - 2b + 1}{2(b - 1)^2} \\
&= \frac{1}{2}.
\end{aligned}$$

Thus $\angle F'BA = \angle BAM$, so $F' = F$ and we're done.

Solution 3 (synthetic)



Without loss of generality $AB < AC$. The intersection of NE and PD is O , the circumcenter of $\triangle ABC$.

Let $\angle BAM = y$ and $\angle CAM = z$. Note D lies on the perpendicular bisector of AB , so $AD = BD$. So $\angle FBC = \angle B - \angle ABD = B - y$. Similarly, $\angle FCB = C - z$, so $\angle BFC = 180 - (B + C) + (y + z) = 2A$. Notice that $\angle BOC$ intercepts the minor arc BC in the **circumcircle** of $\triangle ABC$, which is double $\angle A$. Hence $\angle BFC = \angle BOC$, so $BFOC$ is cyclic.

Lemma. $\triangle FEO$ is directly similar to $\triangle NEM$

Proof.

$$\angle OFE = \angle OFC = \angle OBC = \frac{1}{2} \cdot (180 - 2A) = 90 - A$$

since F, E, C are collinear, $BFOC$ is cyclic, and $OB = OC$. Also

$$\angle ENM = 90 - \angle MNC = 90 - A$$

because $NE \perp AC$, and MNP is the medial triangle of $\triangle ABC$ so $AB \parallel MN$. Hence $\angle OFE = \angle ENM$.

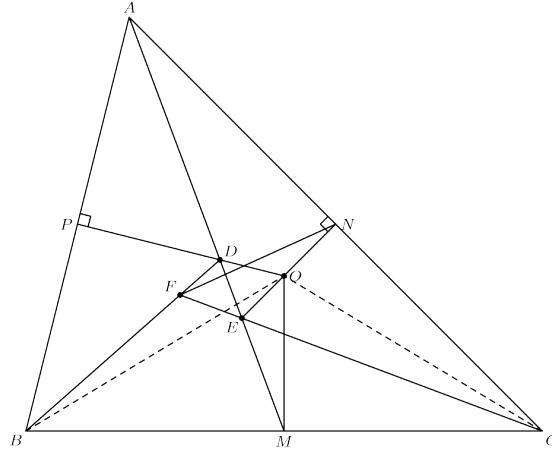
Notice that $\angle AEN = 90 - z = \angle CEN$ since $NE \perp AC$. $\angle FED = \angle MEC = 2z$. Then

$$\angle FEO = \angle FED + \angle AEN = \angle CEM + \angle CEN = \angle NEM$$

Hence $\angle FEO = \angle NEM$.

Hence $\triangle FEO$ is similar to $\triangle NEM$ by AA similarity. It is easy to see that they are oriented such that they are directly similar.

End Lemma



By the similarity in the Lemma, $FE : EO = NE : EM \implies FE : EN = OE : EM$. $\angle FEN = \angle OEM$ so $\triangle FEN \sim \triangle OEM$ by SAS similarity. Hence

$$\angle EMO = \angle ENF = \angle ONF$$

Using essentially the same angle chasing, we can show that $\triangle PDM$ is directly similar to $\triangle FDO$. It follows that $\triangle PDF$ is directly similar to $\triangle MDO$. So

$$\angle EMO = \angle DMO = \angle DPF = \angle OPF$$

Hence $\angle OPF = \angle ONF$, so $FONP$ is cyclic. In other words, F lies on the circumcircle of $\triangle PON$. Note that $\angle ONA = \angle OPA = 90$, so $APON$ is cyclic. In other words, A lies on the circumcircle of $\triangle PON$. A, P, N, O , and F all lie on the circumcircle of $\triangle PON$. Hence A, P, F , and N lie on a circle, as desired.

Solution 4 (synthetic)

This solution utilizes the *phantom point method*. Clearly, $APON$ are cyclic because $\angle OPA = \angle ONA = 90$. Let the circumcircles of triangles APN and BOC intersect at F' and O .

Lemma. If A, B, C are points on circle ω with center O , and the tangents to ω at B, C intersect at Q , then AP is the symmedian from A to BC .

End Lemma

Solution 5 (trigonometric)

Then $\angle FAB + \angle ABF = 180^\circ - \angle AFB = 180^\circ - \angle AFC = \angle FAC + \angle ACF$ and $\angle ABF + \angle ACF = \angle A = \angle FAB + \angle FAC$. Subtracting these two equations, $\angle FAB - \angle FCA = \angle FCA - \angle FAB$ so $\angle BAF = \angle ACF$. Therefore, $\triangle ABF \sim \triangle CAF$ (by AA similarity), so a spiral similarity centered at F takes B to A and A to C . Therefore, it takes the midpoint of $\overline{BA}$ to the midpoint of $\overline{AC}$, or P to N . So $\angle APF = \angle CNF = 180^\circ - \angle ANF$ and $APFN$ is cyclic.

Now by the **homothety** centered at A with ratio $\frac{1}{2}$, B is taken to P and C is taken to N . Thus O is taken to the circumcenter of $\triangle APN$ and is the midpoint of AO , which is also the circumcenter of $\triangle AFO$, so A, P, N, F, O all lie on a circle.

Solution 7 (symmedians)

Median AM of a triangle ABC implies $\frac{\sin BAM}{\sin CAM} = \frac{\sin B}{\sin C}$. Trig ceva for F shows that AF is a symmedian. Then FP is a median, use the lemma again to show that $AFP = C$, and similarly $AFN = B$, so you're done.

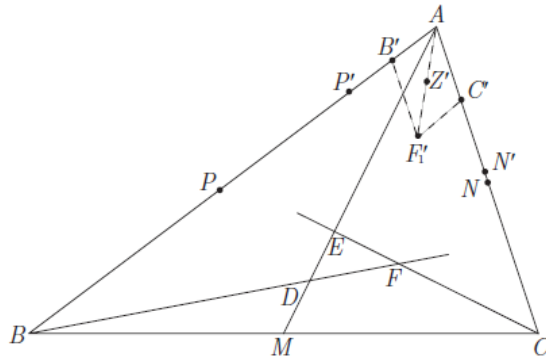
Solution 8 (inversion) (Official Solution 2)

Invert the figure about a circle centered at A , and let X' denote the image of the point X under this inversion. Find point F'_1 so that $AB'F'_1C'$ is a parallelogram and let Z' denote the center of this parallelogram. Note that $\triangle BAC \sim \triangle C'AB'$ and $\triangle BAD \sim \triangle D'AB'$. Because M is the midpoint of BC and Z' is the midpoint of $B'C'$, we also have $\triangle BAM \sim \triangle C'AZ'$. Thus

$$\angle AF'_1B' = \angle F'_1AC' = \angle Z'AC' = \angle MAB = \angle DAB = \angle DBA = \angle AD'B'.$$

Hence quadrilateral $AB'D'F'_1$ is cyclic and, by a similar argument, quadrilateral $AC'E'F'_1$ is also cyclic. Because the images under the inversion of lines BDF and CFE are circles that intersect in A and F' , it follows that $F'_1 = F'$.

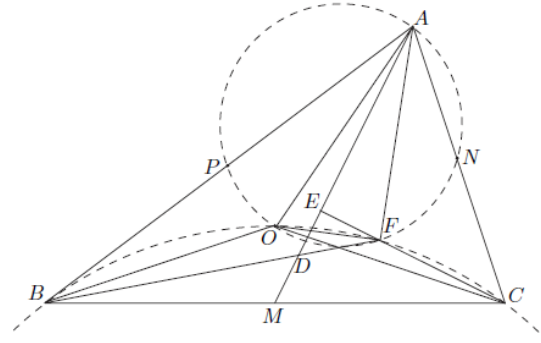
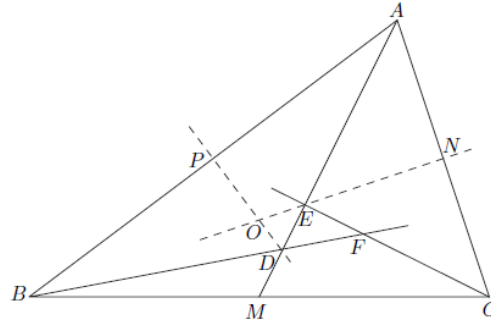
Next note that B' , Z' , and C' are collinear and are the images of P' , F' , and N' , respectively, under a homothety centered at A and with ratio $1/2$. It follows that P' , F' , and N' are collinear, and then that the points A , P , F , and N lie on a circle.

**Solution 9 (Official Solution 1)**

Let O be the circumcenter of triangle ABC . We prove that

$$\angle APO = \angle ANO = \angle AFO = 90^\circ. \quad (1)$$

It will then follow that A, P, O, F, N lie on the circle with diameter $\overline{AO}$. Indeed, the fact that the first two angles in (1) are right is immediate because $\overline{OP}$ and $\overline{ON}$ are the perpendicular bisectors of $\overline{AB}$ and $\overline{AC}$, respectively. Thus we need only prove that $\angle AFO = 90^\circ$.



We may assume, without loss of generality, that $AB > AC$. This leads to configurations similar to the ones shown above. The proof can be adapted to other configurations. Because $\overline{PO}$ is the perpendicular bisector of $\overline{AB}$, it follows that triangle ADB is an isosceles triangle with $AD = BD$. Likewise, triangle AEC is isosceles with $AE = CE$. Let $x = \angle ABD = \angle BAD$ and $y = \angle CAE = \angle ACE$, so $x + y = \angle BAC$.

Applying the Law of Sines to triangles ABM and ACM gives

$$\frac{BM}{\sin x} = \frac{AB}{\sin \angle BMA} \quad \text{and} \quad \frac{CM}{\sin y} = \frac{AC}{\sin \angle CMA}.$$

Taking the quotient of the two equations and noting that $\sin \angle BMA = \sin \angle CMA$, we find

$$\frac{BM \sin y}{CM \sin x} = \frac{AB \sin \angle CMA}{AC \sin \angle BMA} = \frac{AB}{AC}.$$

Because $BM = MC$, we have

$$\frac{\sin x}{\sin y} = \frac{AC}{AB}. \quad (2)$$

Applying the Law of Sines to triangles ABF and ACF , we find

$$\frac{AF}{\sin x} = \frac{AB}{\sin \angle AFB} \quad \text{and} \quad \frac{AF}{\sin y} = \frac{AC}{\sin \angle AFC}.$$

Taking the quotient of the two equations yields

$$\frac{\sin x}{\sin y} = \frac{AC \sin \angle AFB}{AB \sin \angle AFC},$$

so by (2),

$$\sin \angle AFB = \sin \angle AFC. \quad (3)$$

Because $\angle ADF$ is an exterior angle to triangle ADB , we have $\angle EDF = 2x$. Similarly, $\angle DEF = 2y$. Hence

$$\angle EFD = 180^\circ - 2x - 2y = 180^\circ - 2\angle BAC.$$

Thus $\angle BFC = 2\angle BAC = \angle BOC$, so $BOFC$ is cyclic. In addition,

$$\angle AFB + \angle AFC = 360^\circ - 2\angle BAC > 180^\circ,$$

and hence, from (3), $\angle AFB = \angle AFC = 180^\circ - \angle BAC$. Because $BOFC$ is cyclic and $\triangle BOC$ is isosceles with vertex angle $\angle BOC = 2\angle BAC$, we have $\angle OFB = \angle OCB = 90^\circ - \angle BAC$. Therefore,

$$\angle AFO = \angle AFB - \angle OFB = (180^\circ - \angle BAC) - (90^\circ - \angle BAC) = 90^\circ.$$

This completes the proof.

Solution 10 (Anti-Steiner point)

First, let O be the circumcenter of $\triangle ABC$. Note that since $AP \perp PO$ and $AN \perp NO$, then A, N, O and P , are concyclic.

Notice that $NM \parallel AB$ and $AB \perp PO$, which means $NM \perp PO$. It follows analogously that $PM \perp NO$. This means that M is the orthocenter of triangle NOP .

Now consider what happens when we reflect line AM over line PO . Since PO is just the perpendicular bisector of AB , point A will map to point B , and point D , which is both line AM and PO , will map to itself. Therefore, we get line BD from this reflection. It follows analogously that by reflecting line AM over line NO , we get CE .

Since line AM passes through M , the orthocenter of triangle NOP , its reflections over sides PO and NO , which correspond to BD and CE respectively, intersect on $\odot NOP$ (this is the Anti-Steiner point application). Therefore, F , the intersection of BD and CE , lies on $\odot NOP$.

Since A, N, O , and P are concyclic, then $\odot NOP$ is the same as $\odot ANP$, so F lies on $\odot ANP$, which is the same as saying A, N, F , and P are concyclic, as desired.

Solution 11 (Bary from EGMO)

First, we find the coordinates of D . As D lies on AM , we know $D = (t : 1 : 1)$ for some T . Now, we find

$$0 = b^2(t - 1) + (a^2 - c^2) \implies t = \frac{c^2 + b^2 - a^2}{b^2}.$$

Thus we obtain

$$D = (2S_A : c^2 : c^2).$$

(Note we are using Conway's notation here). Similarly, $E = (2S_A : b^2 : b^2)$, and it follows that

$$F = (2S_A : b^2 : c^2).$$

The sum of the coordinates of F is

$$(b^2 + c^2 - a^2) + b^2 + c^2 = 2b^2 + 2c^2 - a^2.$$

Hence, the reflection of A over F is simply

$$2F - A = (-a^2 : 2b^2 : 2c^2).$$

Then, it is obvious that F lies on $(ABC) : -a^2yz - b^2zx - c^2xy = 0$.

<i>Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please **add it** to this page.</i>

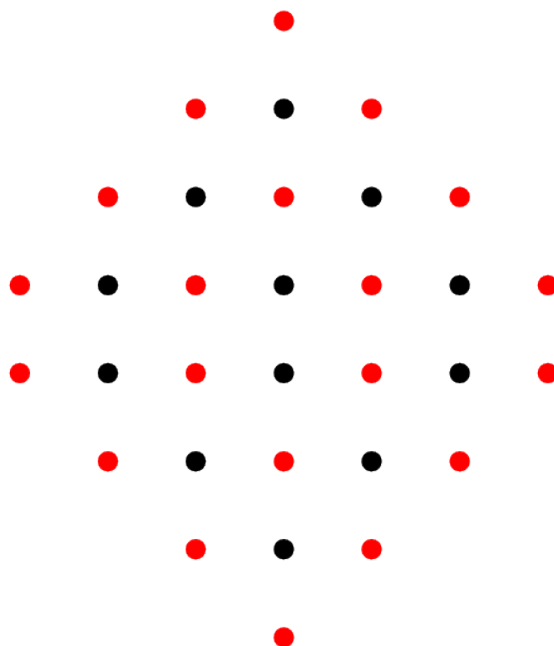
3. (*USAMO 2008*) (*Gabriel Carroll*) Let n be a positive integer. Denote by S_n the set of points (x, y) with integer coordinates such that

$$|x| + \left| y + \frac{1}{2} \right| < n$$

A path is a sequence of distinct points $(x_1, y_1), (x_2, y_2), \dots, (x_\ell, y_\ell)$ in S_n such that, for $i = 2, \dots, \ell$, the distance between (x_i, y_i) and (x_{i-1}, y_{i-1}) is 1 (in other words, the points (x_i, y_i) and (x_{i-1}, y_{i-1}) are neighbors in the lattice of points with integer coordinates). Prove that the points in S_n cannot be partitioned into fewer than n paths (a partition of S_n into m paths is a set $\mathcal{P}$ of m nonempty paths such that each point in S_n appears in exactly one of the m paths in $\mathcal{P}$).

Solution 1

Color all the points in S_n red or black such that each row of points starts and ends with a red point, and alternates between red and black for each point in the row. This creates a checkerboard pattern, except that the middle two rows are identical.



Suppose there is a partition of $m < n$ paths that works. For each of the m paths, we break a path into two separate paths wherever there are consecutive red points. This only happens only with the n red pairs in the middle two rows. We now have $m + l$, $l \leq n$ paths whose colors are strictly alternating. In a path with alternating colors, there can be at most one more red dot than the number of black dots (which occurs when both endpoints are red). Thus there can be no more than $m + l$ red dots than black dots in S_n .

$$2n < m + l < m + n$$

Solution 2

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If a path already has the point you're coming from and the point you're going to as adjacent in the path, do nothing.

If not the first case, and there is exactly one endpoint among the point you're coming from and the point you're going to, remove a segment from the one in the middle of a path (paths +1) and now you have two endpoints; join them (paths -1). If the point in the middle of the path is the one you're coming from, remove the segment that doesn't come from the point that came before it. The number of paths remains the same.

Thus, given any configuration we can create the path shown above without increasing the number of paths. The remaining points form S_{n-1} . Now suppose for some positive integer n , we have a partition into less than n paths. Apply the algorithm to make the path shown above. Then the remaining points are partitioned into less than $n - 1$ paths. Apply the algorithm again. Repeat until we show that S_1 is partitioned into less than 1 path, contradiction.

We induct on n to prove that an $n - 1$ -path partition is impossible. This can be restated as saying that we cannot fit $2n^2 - n + 1$ edges into the graph with each vertex having degree at most 2. The base case is trivial (2 edges when there only is 1). Suppose it was possible for $n + 1$, and we'll prove it's possible for n .

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edges in H_{n+1} out of $4n^2 + 4n + 1$ in G_{n+1} (not hard to prove), and each vertex has maximum order 2. Now consider the complement of H_{n+1} in G_{n+1} , that is, take all the edges in G_{n+1} that are not in H_{n+1} and of course all the vertices. Call this F_{n+1} . We will trim F_{n+1} down a bit to get H_n .

First, call the points in S_{n+1} but not S_n exterior points, and the points in S_n interior points. F_{n+1} has $2n^2 + n - 1$ edges. Since the interior points have degree 4 in G_{n+1} , they have degree at least 2 in F_{n+1} , for a total degree of $4n^2 + m$ where $0 \leq m \leq 2n - 2$ since the total overall degree is $2(2n^2 + n - 1) = 4n^2 + 2n - 2$. This makes the total degree of the exterior points $2n - 2 - m$. Since the exterior points are not connected in G_{n+1} , these $2n - 2 - m$ edges must connect on exterior point to an interior point.

I then take away $2n - 2$ edges and the exterior points, yielding $2n^2 - n + 1$ edges and the interior points, which is H_n , as follows: First take away those $2n - 2 - m$ interior-exterior edges and shave off all the exterior points. Then the total degree (now just of the interior points) must be $4n^2 - 2n + 2 + 2m$.

Let the number of 0-degree vertices be a , 1-degree vertices be b , 2-degree vertices be c , 3-degree vertices be d , and 4-degree vertices be e . Since all the vertices began as 2-degree or higher, $2a + b \leq 2n - 2 - m$. Moreover, the total degree is $b + 2c + 3d + 4e = 2(2n^2) + (d + 2e) - (b + 2a) = 4n^2 - 2n + 2 + 2m$. So $d + 2e = b + 2a - (2n - 2 + 2m) \leq (2n - 2 - m) - (2n - 2 - 2m) = m$. So we can use our last m edges to remove any excess degree on vertices with degree 3 or 4, yielding H_n .

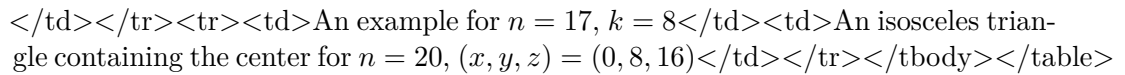
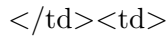
Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

4. (USAMO 2008) (Gregory Galparin) Let $\mathcal{P}$ be a **convex polygon** with n sides, $n \geq 3$. Any set of $n - 3$ diagonals of $\mathcal{P}$ that do not intersect in the interior of the polygon determine a *triangulation* of $\mathcal{P}$ into $n - 2$ triangles. If $\mathcal{P}$ is regular and there is a triangulation of $\mathcal{P}$ consisting of only isosceles triangles, find all the possible values of n .

Solution

We label the vertices of $\mathcal{P}$ as $P_0, P_1, P_2, \dots, P_n$. Consider a diagonal $d = \overline{P_a P_{a+k}}$, $k \leq n/2$ in the triangulation. We show that k must have the form 2^m for some nonnegative integer m .

This diagonal **partitions** $\mathcal{P}$ into two regions $Q, \mathcal{R}$, and is the side of an isosceles triangle in both regions. **Without loss of generality** suppose the area of Q is less than the area of $\mathcal{R}$ (so the **center** of $\mathcal{P}$ does not lie in the interior of Q); it follows that the lengths of the edges and diagonals in Q are all smaller than d . Thus d must be the base of the isosceles triangle in Q , from which it follows that the isosceles triangle is $\triangle P_a P_{a+k/2} P_{a+k}$, and so $2|k$. Repeating this process on the legs of isosceles triangle $(\overline{P_a P_{a+k/2}}, \overline{P_{a+k} P_{a+k/2}})$, it follows that $k = 2^m$ for some positive integer m (if we allow **degeneracy**, then we can also let $m = 0$).


$$n = 2 \cdot 2^a + 2^b = 2^{a+1} + 2^b,$$

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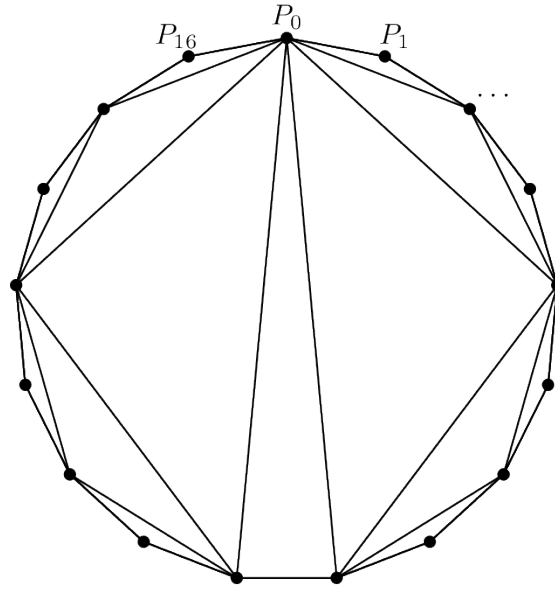
We now claim that this condition is sufficient. Suppose without loss of generality that $a + 1 \geq b$; then we rewrite this as

$$n = 2^b(2^{a-b+1} + 1).$$

- *Lemma 1:* All regular polygons with $n = 2^k + 1$ or $n = 4$ have triangulations that meet the conditions.

By **induction**, it follows that we can cover all the desired n .

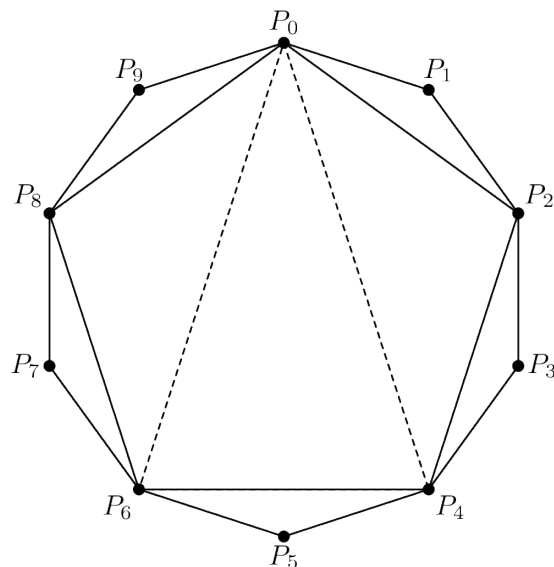
For $n = 3, 4$, this is trivial. For $k > 1$, we construct the diagonals of equal length $\overline{P_0P_{2^{k-1}}}$ and $\overline{P_{2^{k-1}+1}P_0}$. This partitions $\mathcal{P}$ into 3 regions: an isosceles $\triangle P_0P_{2^{k-1}}P_{2^{k-1}+1}$, and two other regions. For these two regions, we can recursively construct the isosceles triangles defined above in the second paragraph. It follows that we have constructed $2(2^{k-1} - 1) + (1) = 2^k - 1 = n - 2$ isosceles triangles with non-intersecting diagonals, as desired.



An example for $n = 17 = 2^4 + 1$

- *Lemma 2:* If a regular polygon with n sides has a working triangulation, then the regular polygon with $2n$ sides also has a triangulation that meets the conditions.

We construct the diagonals $\overline{P_0P_2}$, $\overline{P_2P_4}$, $\dots$, $\overline{P_{n-2}P_0}$. This partitions $\mathcal{P}$ into n isosceles triangles of the form $\triangle P_{2k}P_{2k+1}P_{2k+2}$, as well as a central regular polygon with n sides. However, we know that there exists a triangulation for the n -sided polygon that yields $n - 2$ isosceles triangles. Thus, we have created $(n) + (n - 2) = 2n - 2$ isosceles triangles with non-intersecting diagonals, as desired.



An example for $n = 10$, $n/2 = 5$

In summary, the answer is all n that can be written in the form $2^{a+1} + 2^b$, $a, b \geq 0$. Alternatively, this condition can be expressed as either $n = 2^k$, $k \geq 2$ (this is the case when $a + 1 = b$) or n is the sum of two distinct powers of 2, where $1 = 2^0$ is considered a power of 2.

5. (USAMO 2008) (Kiran Kedlaya) Three nonnegative real numbers r_1, r_2, r_3 are written on a blackboard. These numbers have the property that there exist integers a_1, a_2, a_3 , not all zero, satisfying $a_1 r_1 + a_2 r_2 + a_3 r_3 = 0$. We are permitted to perform the following operation: find two numbers x, y on the blackboard with $x \leq y$, then erase y and write $y - x$ in its place. Prove that after a finite number of such operations, we can end up with at least one 0 on the blackboard.

Solution

Every time we perform an operation on the numbers on the blackboard $R = \langle r_1, r_2, r_3 \rangle$, we perform the corresponding operation on the integers $A = \langle a_1, a_2, a_3 \rangle$ so that $R \cdot A = 0$ continues to hold. (For example, if we replace r_1 with $r_1 - r_2$ then we replace a_2 with $a_1 + a_2$.)

It's possible to show we can always pick an operation so that $|A|^2$ is strictly decreasing. **Without loss of generality**, let $r_3 > r_2 > r_1$ and a_3 be positive. Then it cannot be true that both a_1 and a_2 are at least $\frac{-a_3}{2}$, or else $a_1 r_1 + a_2 r_2 + a_3 r_3 > 0$. Without loss of generality, let $a_1 < \frac{-a_3}{2}$. Then we can replace a_1 with $a_1 + a_3$ and r_3 with $r_3 - r_1$ to make $|A|$ smaller. Since it is a strictly decreasing sequence of positive integers, after a finite number of operations we have $a_3 = 0$. We can now see that this result holds for $(r_1, r_2, 0)$ if and only if it holds for $(1, \frac{r_2}{r_1}, 0)$. We can see that $\frac{r_2}{r_1}$ is a rational number given that $a_3 = 0$. It is a well known result of the euclidean algorithm that if we continue to perform these operations, r_1 or r_2 will eventually be 0.

6. (USAMO 2008) (**Sam Vandervelde**) At a certain mathematical conference, every pair of mathematicians are either friends or strangers. At mealtime, every participant eats in one of two large dining rooms. Each mathematician insists upon eating in a room which contains an even number of his or her friends. Prove that

the number of ways that the mathematicians may be split between the two rooms is a power of two (i.e., is of the form 2^k for some positive integer k).

Solution 1 (linear algebra)

Make the obvious re-interpretation as a **graph**. Let $f : G \rightarrow \{0, 1\}$ be an **indicator function** with $f(v) = 0$ if a vertex is in the first **partition** and $f(v) = 1$ otherwise (this corresponds, in the actual problem, to putting a mathematician in the first or second room). Then look at f as a function into the **field** with two elements, F_2 . Let V be the **vector space** of all such functions. Define the linear operator $A : V \rightarrow V$ as

$$(Af)(v) = \sum_{v \sim w} f(v) - f(w)$$

where $\sim$ denotes adjacency. (Note that we can also think of A as a matrix, which is essentially the adjacency matrix where the diagonal is changed to be 1 whenever the degree is odd; in more technical terms, it is the Laplacian of G over F_2). Then f is a valid partition **iff** $(Af)(v) = d(v)$, where d is the degree of v , for all v (this is taken over F_2). So we want all solution to $Af = d$. Note that if $Af = d, Ag = d$, then $A(f - g) = 0$, so $f - g$ is in the **nullspace**. Thus in particular the number of solutions, if non-zero, is the size of the nullspace of A , which is $2^{\dim(\text{Null}(A))}$ by considering all linear combinations of any basis of $\text{Null}(A)$ over F_2 . Also let i be such that $i(v) = 1$ for all v . Then clearly $Ai = 0$, so $\dim(\text{Null}(A)) > 0$, establishing that the number of ways to do this is 2^k , $k > 0$. Thus we need only prove the existence of a solution.

Since we can add a new vertex connected to exactly one previously existing vertex without changing the problem, **without loss of generality** all vertices have odd degree. Then we want to show that $i \in \text{Col}(A)$. But it is a well-known fact in linear algebra that $\text{Col}(A) = \text{Null}(A^T)^\perp = \text{Null}(A)^\perp$ since A is symmetric. Thus we need to show that if $f \in \text{Null}(A)$, f is perpendicular to i and we will be done. So let $f \in \text{Null}(A)$. Take the submatrix of A consisting of the rows and columns v such that $f(v) = 1$. Then, since $f \in \text{Null}(A)$, the sum of each row in this submatrix must be 0 in F_2 . Thus the total number of 1s in this submatrix is even. But since it is symmetric, the total number of 1s off of the diagonal is even, so the total number of 1s on the diagonal is even. But since every vertex has odd degree, the entire diagonal of A consists of 1s, so this says that the size of the diagonal of this submatrix is even. But this is also the number of v such that $f(v) = 1$, so $f(v) = 1$ for an even number of v , thus is perpendicular to i , and we have our result.

Solution 2 (group theory)

Define an *order* to be a set of instructions, one instruction given to each mathematician. Each mathematician is told to either *move* or to *stay* (we can think of this as stay is 0, and move is 1). Now take some good configuration. Consider the set of all orders which, when performed on this configuration, give us another good configuration. Note the **identity** order, $(0, 0, \dots, 0)$, is in this set. We claim this set is an **abelian group** under **composition**.

Proof: Clearly each is its own **inverse**, there is the identity, and the operation is clearly associative and commutative (because it's equivalent to addition of n -dimensional vectors $\pmod{2}$). So it suffices to show this set of orders is closed under composition.

Consider any mathematician. If, in one of these orders, he is told to stay, then the number of his friends who are told to move must be even. Similarly, if he is told to move, then the number of his friends who are told to stay must be even. Now just consider two orders A and B and you can show that in $A \times B$ the same property will hold using **parity**.

Now that we've shown it is a group (which we will call G), we'll prove it has order two. Let I be the identity.

Let $\{I, T_1\} = H_1$, where T is some element of G . Now pick an element T_2 of G which is not in H_1 . Notice that because the elements of H_1 are distinct, the elements of $\{T_2X | X \in H_1\}$ are distinct (if two elements of that set were the same, multiply by each on the right by T_2^{-1} and you have a contradiction). Now notice that for any $A, B \in H_1$, if we were to have $T_2A = B$, then $T_2 = BA^{-1} \in H_1$. Therefore, H_1 and $\{T_2X | X \in H_1\}$ are disjoint and of the same size. Moreover, the product of any element in the first group and any element in the second group is a member of the second group. Therefore, these two groups together form a group of order 2^2 . Call this H_2 . You progressively build larger and larger subgroups of G until you get to G itself, whose order must then be a power of two. Therefore, the number of good configurations of the mathematicians was a power of two.

This, of course, was all assuming I existed and was in the group.

Solution 3

Let n be the number of participants at the conference. We proceed by induction on n .

If $n = 1$, then we have one participant who can eat in either room; that gives us a total of $2 = 2^1$ options.

Let $n \geq 2$. The case in which some participant, P , has no friends is trivial. In this case, P can eat in either of the two rooms, so the total number of ways to split n participants is twice as many as the number of ways to split $(n - 1)$ participants besides the participant P . By induction, the latter number is a power of two, 2^k , hence the number of ways to split n participants is $2 \times 2^k = 2^{k+1}$, also a power of two. So we assume from here on that every participant has at least one friend.

We consider two different cases separately: the case when some participant has an odd number of friends, and the case when each participant has an even number of friends.

Claim: Some participant, Z , has an odd number of friends.

Remove Z from consideration and for each pair (X, Y) of Z 's friends, reverse the relationship between X and Y (from friends to strangers or vice versa).

Claim. The number of possible seatings is unchanged after removing Z and reversing the relationship between X and Y in each pair (X, Y) of Z 's friends.

Proof of the claim. Suppose we have an arrangement prior to Z 's departure. By assumption, Z has an even number of friends in the room with him.

If this number is 0, the room composition is clearly still valid after Z leaves the room.

If this number is positive, let X be one of Z 's friends in the room with him. By assumption, person X also has an even number of friends in the same room. Remove Z from the room; then X will have an odd number of friends left in the room, and there will be an odd number of Z 's friends in this room besides X . Reversing the

relationship between X and each of Z 's friends in this room will therefore restore the parity to even.

The same reasoning applies to any of Z 's friends in the other dining room. Indeed, there will be an odd number of them in that room, hence each of them will reverse relationships with an even number of individuals in that room, preserving the parity of the number of friends present.

Moreover, a legitimate seating without Z arises from exactly one arrangement including Z , because in the case under consideration, only one room contains an even number of Z 's friends.

End Claim

Thus, we have to double the number of seatings for $(n - 1)$ participants which is, by the induction hypothesis, a power of 2. Consequently, for n participants we will get again a power of 2 for the number of different arrangements.

Case 2: Each participant has an even number of friends.

In this case, each valid split of participants in two rooms gives us an even number of friends in either room.

Let (A, B) be any pair of friends. Remove this pair from consideration and for each pair (C, D) , where C is a friend of A and D is a friend of B , change the relationship between C and D to the opposite; do the same if C is a friend of B and D is a friend of A . Note that if C and D are friends of both A and B , their relationship will be reversed twice, leaving it unchanged.

Consider now an arbitrary participant X different from A and B and choose one of the two dining rooms. [Note that in the case under consideration, the total number of participants is at least 3, so such a triplet (A, B, X) can be chosen.] Let A have m friends in this room and let B have n friends in this room; both m and n are even. When the pair (A, B) is removed, X 's relationship will be reversed with either n , or m , or $m + n - 2k$ (for k the number of mutual friends of A and B in the chosen room), or 0 people within the chosen room (depending on whether he/she is a friend of only A , only B , both, or neither). Since m and n are both even, the parity of the number of X 's friends in that room will be therefore unchanged in any case.

Again, a legitimate seating without A and B will arise from exactly one arrangement that includes the pair (A, B) : just add each of A and B to the room with an odd number of the other's friends, and then reverse all the relationships between a friend of A and a friend of B . In this way we create a one-to-one correspondence between all possible seatings before and after the (A, B) removal.

Since the number of arrangements for n participants is twice as many as that for $(n - 2)$ participants, and that number for $(n - 2)$ participants is, by the induction hypothesis, a power of 2, we get in turn a power of 2 for the number of arrangements for n participants. The problem is completely solved.

Solution 4 (proving only existence of solution)

By induction on the number of mathematicians, assuming solution exists for $n \leq k$ mathematicians. For $n = k + 1$ mathematicians, if each of them has even degree then we're done. Otherwise, let v be a node with odd degree, and denote its neighbors by $N(v)$. We form a new graph G' by deleting v and flipping all edges between members of $N(v)$. That is, for each pair of a and b in $N(v)$, they're

connected in G' if and only if they're not in G . By induction we know a solution S' exists for G' , and it is easily verifiable that the similar solution S exists in G by adding v to the side in S' with even number of neighbors in $N(v)$.

Alternate solutions are always welcome. If you have a different, elegant solution to this problem, please [add it](#) to this page.

§38 USAMO 2009

1. (USAMO 2009) Given circles ω_1 and ω_2 intersecting at points X and Y , let ℓ_1 be a line through the center of ω_1 intersecting ω_2 at points P and Q and let ℓ_2 be a line through the center of ω_2 intersecting ω_1 at points R and S . Prove that if P, Q, R and S lie on a circle then the center of this circle lies on line XY .

Solution 1

Let ω_3 be the **circumcircle** of $PQRS$, r_i to be the radius of ω_i , and O_i to be the center of the circle ω_i , where $i \in \{1, 2, 3\}$. Note that SR and PQ are the **radical axes** of O_1, O_3 and O_2, O_3 respectively. Hence, by **power of a point** (the power of O_1 can be expressed using circle ω_2 and ω_3 and the power of O_2 can be expressed using circle ω_1 and ω_3),

$$O_1O_2^2 - r_2^2 = O_1O_3^2 - r_3^2$$

$$O_2O_1^2 - r_1^2 = O_2O_3^2 - r_3^2$$

Subtracting these two equations yields that $O_1O_3^2 - r_1^2 = O_2O_3^2 - r_2^2$, so O_3 must lie on the **radical axis** of ω_1, ω_2 .

One needs to be careful of when O_1, O_2 , and O_3 lie on a line.

2. (USAMO 2009) Let n be a positive integer. Determine the size of the largest subset of $\{-n, -n+1, \dots, n-1, n\}$ which does not contain three elements a, b, c (not necessarily distinct) satisfying $a + b + c = 0$.

Solution 1

Let S be a subset of $\{-n, -n+1, \dots, n-1, n\}$ of largest size satisfying $a + b + c \neq 0$ for all $a, b, c \in S$. First, observe that $0 \notin S$. Next note that $|S| \geq \lceil n/2 \rceil$, by observing that the set of all the odd numbers in $\{-n, -n+1, \dots, n-1, n\}$ works. To prove that $|S| \leq \lceil n/2 \rceil$, it suffices to only consider even n , because the statement for $2k$ implies the statement for $2k-1$ as well. So from here forth, assume n is even.

For any two sets A and B , denote by $A + B$ the set $\{a + b \mid a \in A, b \in B\}$, and by $-A$ the set $\{-a \mid a \in A\}$. Also, let A_+ denote $A \cap \{1, 2, \dots\}$ and A_- denote $A \cap \{-1, -2, \dots\}$. First, we present a lemma:

Lemma 1: Let A and B be two sets of integers. Then $|A + B| \geq |A| + |B| - 1$.

Proof: Write $A = \{a_1, \dots, a_n\}$ and $B = \{b_1, \dots, b_m\}$ where $a_1 < \dots < a_n$ and $b_1 < \dots < b_m$. Then $a_1 + b_1, a_1 + b_2, \dots, a_1 + b_m, a_2 + b_m, \dots, a_n + b_m$ is a strictly increasing sequence of $n + m - 1$ integers in $A + B$.

Now, we consider two cases:

Case 1: One of $n, -n$ is not in S . Without loss of generality, suppose $-n \notin S$. Let $U = \{-n+1, -n+2, \dots, n-2, n-1, n\}$ (a set with $2n$ elements), so that $S \subseteq U$ by our assumption. Now, the condition that $a + b + c \neq 0$ for all $a, b, c \in S$ implies that $-S \cap (S_+ + S_-) = \emptyset$. Since any element of $S_+ + S_-$ has absolute value at most

$n - 1$, we have $S_+ + S_- \subseteq \{-n + 1, -n + 2, \dots, n - 2, n - 1\} \subseteq U$. It follows that $S_+ + S_- \subseteq U \setminus -S$, so $|S_+ + S_-| \leq 2n - |S|$. However, by Lemma 1, we also have $|S_+ + S_-| \geq |S_+| + |S_-| - 1 = |S| - 1$. Therefore, we must have $|S| - 1 \leq 2n - |S|$, or $2|S| \leq 2n + 1$, or $|S| \leq n$.

Case 2: Both n and $-n$ are in S . Then $n/2$ and $-n/2$ are not in S , and at most one of each of the pairs $\{1, n - 1\}, \{2, n - 2\}, \dots, \{\frac{n}{2} - 1, \frac{n}{2} + 1\}$ and their negatives are in S . This means S contains at most $2 + (n - 2) = n$ elements.

Thus we have proved that $|S| \leq n = \lceil n/2 \rceil$ for even n , and we are done.

Solution 2

Let Z_n be the set of subsets satisfying the $a + b + c \neq 0$ condition for n , and let s_n be the largest size of a set in Z_n . Let $n = 2k$ if n is even, and $n = 2k - 1$ if n is odd. We note that $s_n \geq 2k$ due to the following construction:

$$\{-2k + 1, -2k + 3, \dots, 2k - 3, 2k - 1\}$$

or all of the odd numbers in the set. Then the sum of any three will be odd and thus nonzero.

Lemma 1: $s_{n+1} \leq s_n + 2$. If $M_{n+1} \in Z_{n+1}$, $|M_{n+1}| = s_{n+1}$, then we note that $M_{n+1} \setminus \{-2n + 1, 2n - 1\} \in Z_n$, so $|M_{n+1}| - 2 \leq s_n$. $\square$

Lemma 2: $s_{2k} = s_{2k-1}$. Suppose, for sake of contradiction, that $M_{2k} \in Z_{n+1}$ and $M_{2k} > s_{2k-1}$. Remove $2k, -2k$ from M_{2k} , and partition the rest of the elements into two sets P_{2k-1}, Q_{2k-1} , where P and Q contain all of the positive and negative elements of M_{2k} , respectively. (obviously $0 \notin M_{2k}$, because $0 + 0 + 0 = 0$). WLOG, suppose $|P_{2k-1}| \geq |Q_{2k-1}|$. Then $|P_{2k-1}| + |Q_{2k-1}| \geq s_{2k-1} - 1 \geq 2k - 1$. We now show the following two sub-results:

- (i) *Sub-lemma (A):* if $|P_{2k-1}| \geq k$, $-2k \notin M_{2k}$ [and similar for Q]; and
- (i) *Sub-lemma (B):* we cannot have both $|P_{2k-1}| + |Q_{2k-1}| \geq 2k$ and $|Q_{2k-1}| < k$ simultaneously hold.

This is sufficient, because the only two elements that may be in M_{2k} that are not in P_{2k-1}, Q_{2k-1} are $2k$ and $-2k$; for $M_{2k} > s_{2k-1}$, we must either have $|P_{2k-1}| + |Q_{2k-1}| = s_{2k-1} - 1 \geq 2k - 1$ and both $2k, -2k \in M_{2k}$ [but by pigeonhole $|P_{2k-1}| \geq k$, see sub-lemma (A)], or $|P_{2k-1}| + |Q_{2k-1}| = s_{2k-1} \geq 2k$, and $2k \in M_{2k}$, in which case by (A) we must have $|Q_{2k-1}| < k$, violating (B).

(A): Partition $\{1, 2, \dots, 2k - 1\}$ into the k sets $\{k\}, \{1, 2k - 1\}, \{2, 2k - 2\}, \dots, \{k - 1, k + 1\}$. Because $(k + \delta) + (k - \delta) - 2k = 0$, then if any of those sets are within P_{2k-1} , $-2k \notin M_{2k}$. But by Pigeonhole at most $k - 1$ elements may be in P_{2k-1} , contradiction.

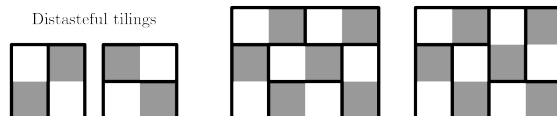
(B): We prove this statement with another induction. We see that the statement easily holds true for $k = 1$ or 2 , so suppose it is true for $k - 1$, but [for sake of contradiction] false for k . Let $P_{2k-3} = P_{2k-1} \setminus \{2k - 2, 2k - 1\}$, and similarly for Q_{2k-3} . Again WLOG $|P_{2k-3}| \geq |Q_{2k-3}|$. Then we have $|P_{2k-1}| + |Q_{2k-1}| \leq |P_{2k-3}| + |Q_{2k-3}| + 4$.

- If $|P_{2k-3}| + |Q_{2k-3}| \geq 2k - 2$, then by inductive hypothesis, we must have $|P_{2k-3}| = |Q_{2k-3}| = k - 1$. But (A) implies that we cannot add $2k - 2$ or $-2k + 2$. So to satisfy $|P_{2k-1}| = |Q_{2k-1}| \geq 2k$ we must have $2k - 1, -2k + 1$ added, but then $|P_{2k-1}| = |Q_{2k-1}| = k$ contradiction.

- If $|P_{2k-3}| + |Q_{2k-3}| = 2k - 3$, then at least three of $\{2k - 2, 2k - 1, -2k + 2, -2k + 1\}$ added. But $|P_{2k-3}| \geq k - 1$, and by (A) we have that $-2k + 2$ cannot be added. If $|P_{2k-3}| \geq k$, then another grouping similar to (A) shows that $-2k + 1$ cannot be added, contradiction. So $|P_{2k-3}| = k - 1$, $|Q_{2k-3}| = k - 2$, and adding the three remaining elements gives $|P_{2k-1}| = |Q_{2k-1}| = k$ contradiction.
- If $|P_{2k-3}| + |Q_{2k-3}| = 2k - 4$, then all four of $\{2k - 2, 2k - 1, -2k + 2, -2k + 1\}$ must be added, and furthermore $|P_{2k-3}| > |Q_{2k-3}|$. Then $|P_{2k-3}| \geq k - 1$, and by previous paragraph we cannot add $-2k + 2$. $\square$

So we have
$$\begin{cases} s_1 &= 2 \\ s_{2k} &= s_{2k-1} \\ s_{2k-1} &\leq s_{2k-2} + 2 \end{cases} \quad \text{and by induction, that } s_n \leq \boxed{2 \left\lceil \frac{n}{2} \right\rceil} = 2k, \text{ which}$$
 we showed is achievable above.

3. (USAMO 2009) We define a *chessboard polygon* to be a polygon whose sides are situated along lines of the form $x = a$ or $y = b$, where a and b are integers. These lines divide the interior into unit squares, which are shaded alternately grey and white so that adjacent squares have different colors. To tile a chessboard polygon by dominoes is to exactly cover the polygon by non-overlapping 1×2 rectangles. Finally, a *tasteful tiling* is one which avoids the two configurations of dominoes shown on the left below. Two tilings of a 3×4 rectangle are shown; the first one is tasteful, while the second is not, due to the vertical dominoes in the upper right corner.



a) Prove that if a chessboard polygon can be tiled by dominoes, then it can be done so tastefully.

b) Prove that such a tasteful tiling is unique.

Solution

<i>This problem needs a solution. If you have a solution for it, please help us out by [adding it](#).

4. (USAMO 2009) For $n \geq 2$ let $a_1, a_2, \dots, a_n$ be positive real numbers such that

$$(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \leq \left(n + \frac{1}{2} \right)^2$$

Prove that $\max(a_1, a_2, \dots, a_n) \leq 4 \min(a_1, a_2, \dots, a_n)$.

Solution 1

Assume without loss of generality that $a_1 \geq a_2 \geq \dots \geq a_n$. Now we seek to prove that $a_1 \leq 4a_n$.

By the [Cauchy-Schwarz Inequality](#),

$$\begin{aligned} (a_n + a_2 + a_3 + \dots + a_{n-1} + a_1) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) &\geq \left(\sqrt{\frac{a_n}{a_1}} + n - 2 + \sqrt{\frac{a_1}{a_n}} \right)^2 \\ (n + \frac{1}{2})^2 &\geq \left(\sqrt{\frac{a_n}{a_1}} + n - 2 + \sqrt{\frac{a_1}{a_n}} \right)^2 \end{aligned}$$

$$\begin{aligned}
 n + \frac{1}{2} &\geq n - 2 + \sqrt{\frac{a_n}{a_1}} + \sqrt{\frac{a_1}{a_n}} \\
 \frac{5}{2} &\geq \sqrt{\frac{a_n}{a_1}} + \sqrt{\frac{a_1}{a_n}} \\
 \frac{17}{4} &\geq \frac{a_n}{a_1} + \frac{a_1}{a_n} \\
 0 &\geq (a_1 - 4a_n) \left(a_1 - \frac{a_n}{4} \right)
 \end{aligned}$$

Since $a_1 \geq a_n$, clearly $(a_1 - \frac{a_n}{4}) > 0$, dividing yields:

$$0 \geq (a_1 - 4a_n) \implies 4a_n \geq a_1$$

as desired.

Solution 2

Assume without loss of generality that $a_1 \geq a_2 \geq \dots \geq a_n$. Using the Cauchy–Bunyakovsky–Schwarz inequality and the inequality given,

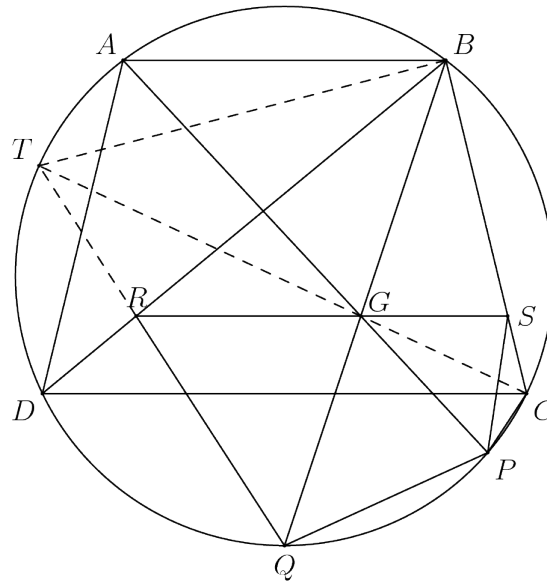
$$\begin{aligned}
 &\left(a_1 + a_2 + \dots + a_n + 3a_n - \frac{3a_1}{4} \right) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \\
 &= \left(\frac{a_1}{4} + a_2 + \dots + a_{n-1} + 4a_n \right) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \\
 &\geq \left(\frac{1}{2} + n - 2 + 2 \right)^2 \\
 &= \left(n + \frac{1}{2} \right)^2 \\
 &\geq (a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right).
 \end{aligned}$$

(Note that $n - 2 \geq 0$ since $n \geq 2$ as given!) This implies that $3a_n - \frac{3a_1}{4} \geq 0 \iff 4a_n \geq a_1$ as desired.

5. (*USAMO 2009*) Trapezoid $ABCD$, with $\overline{AB} \parallel \overline{CD}$, is inscribed in circle ω and point G lies inside triangle BCD . Rays AG and BG meet ω again at points P and Q , respectively. Let the line through G parallel to $\overline{AB}$ intersect $\overline{BD}$ and $\overline{BC}$ at points R and S , respectively. Prove that quadrilateral $PQRS$ is cyclic if and only if $\overline{BG}$ bisects $\angle CBD$.

Solution 1

We will use directed angles in this solution. Extend QR to T as follows:



If:

Note that

$$\begin{aligned}\angle GBT + \angle TRG &= \frac{m\widehat{TQ}}{2} + \angle TRB + \angle BRG \\ &= \frac{m\widehat{TQ} + m\widehat{DQ} + m\widehat{CB} + m\widehat{BT}}{2}.\end{aligned}$$

Thus, $BTRG$ is cyclic.

Also, note that $GSCP$ is cyclic because

$$\begin{aligned}\angle CSG + \angle GPC &= \angle CBA + \angle APC \\ &= 180^\circ \text{ or } 0^\circ,\end{aligned}$$

depending on the configuration.

Next, we have T, G, C are collinear since

$$\angle GTR = \angle GBR = \frac{m\widehat{DQ}}{2} = \frac{m\widehat{QC}}{2} = \angle CTQ.$$

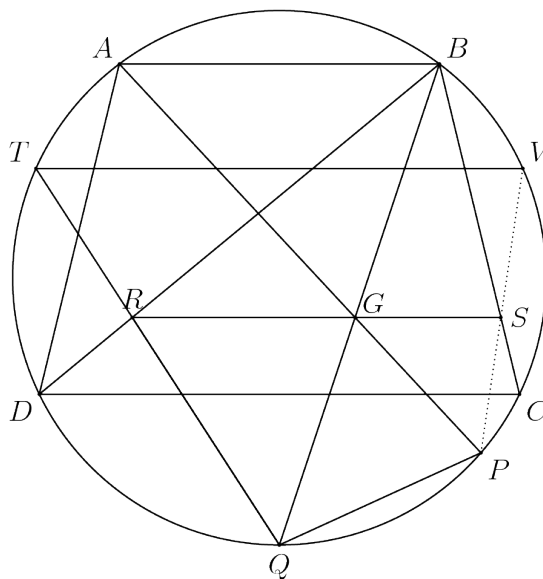
Therefore,

$$\begin{aligned}\angle RQP + \angle PSR &= \frac{m\widehat{PBT}}{2} + \angle PCG \\ &= \frac{m\widehat{PBT} + m\widehat{TDQ}}{2} \\ &= 180^\circ\end{aligned}$$

, so $PQRS$ is cyclic.

Only If: These steps can be reversed.

Solution 2 (Projective)



Suppose the s_i can be represented as $\frac{a_i}{b_i}$ for every i , and suppose t_i can be represented as $\frac{c_i}{d_i}$. Let's start with only the first two terms in the two sequences, s_1 and s_2 for sequence s and t_1 and t_2 for sequence t . Then by the conditions of the problem, we have $(s_2 - s_1)(t_2 - t_1)$ is an integer, or $(\frac{a_2}{b_2} - \frac{a_1}{b_1})(\frac{c_2}{d_2} - \frac{c_1}{d_1})$ is an integer. Now we can set $r = \frac{b_1 b_2}{d_1 d_2}$, because the least common denominator of $s_2 - s_1$ is $b_1 b_2$ and of $t_2 - t_1$ is $d_1 d_2$, and multiplying or dividing appropriately by $\frac{b_1 b_2}{d_1 d_2}$ will always give an integer.

Now suppose we kept adding s_i and t_i until we get to $s_m = \frac{a_m}{b_m}$ in sequence s and $t_m = \frac{c_m}{d_m}$ in sequence t so that $(t_m - t_i)(s_m - s_i)$ is an integer for all i with $1 \leq i < m$, where m is a positive integer. At this point, we will have $r = \frac{\prod_{n=1}^m b_n}{\prod_{n=1}^m d_n}$, because these are the least common denominators of the two sequences up to m . As we keep adding s_i and t_i , r will always have value $\frac{\prod_{n=1}^m b_n}{\prod_{n=1}^m d_n}$, and we are done.

§39 USAMO 2010

1. (USAMO 2010) Let $AXYZB$ be a convex pentagon inscribed in a semicircle of diameter AB . Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ , respectively. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle XOZ$, where O is the midpoint of segment AB .

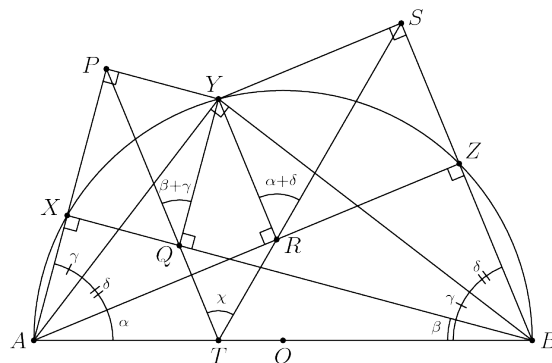
Solution 0 (play like a God)

Angles are directed mod 180 don't worry (take so that the solution can be applied to any configuration) Let the intersection of pq and rs is M .

$m\angle ARM = m\angle ZRS = m\angle ZYS = m\angle YZR = m\angle YZA = m\angle YXP = m\angle XYQ = m\angle XPM = m\angle APM$ This implies $APRM$ is cyclic So: $m\angle PMR = m\angle PAR = m\angle XAZ = \frac{m\angle XOZ}{2}$ and we are done. For any concern mail on harshsahu1098@gmail.com

Solution 1

Let $\alpha = \angle BAZ$, $\beta = \angle ABX$. Since XY is a chord of the circle with diameter AB , $\angle XAY = \angle XBY = \gamma$. From the chord YZ , we conclude $\angle YAZ = \angle YBZ = \delta$.



Triangles BQY and APY are both right-triangles, and share the angle γ , therefore they are similar, and so the ratio $PY : YQ = AY : YB$. Now by **Thales' theorem** the angles $\angle AXB = \angle AYB = \angle AZB$ are all right-angles. Also, $\angle PYQ$, being the fourth angle in a quadrilateral with 3 right-angles is again a right-angle. Therefore $\triangle PYQ \sim \triangle AYB$ and $\angle YQP = \angle YBA = \gamma + \beta$. Similarly, $RY : YS = AY : YB$, and so $\angle YRS = \angle YAB = \alpha + \delta$.

Now RY is perpendicular to AZ so the direction RY is α counterclockwise from the vertical, and since $\angle YRS = \alpha + \delta$ we see that SR is δ clockwise from the vertical. (Draw an actual vertical line segment if necessary.)

Similarly, QY is perpendicular to BX so the direction QY is β clockwise from the vertical, and since $\angle YQP$ is $\gamma + \beta$ we see that QY is γ counterclockwise from the vertical.

Therefore the lines PQ and RS intersect at an angle $\chi = \gamma + \delta$. Now by the central angle theorem $2\gamma = \angle XOY$ and $2\delta = \angle YOZ$, and so $2(\gamma + \delta) = \angle XOZ$, and we are done.

Note that $RTQY$ is a quadrilateral whose angles sum to 360° ; can you find a faster approach using this fact?

Footnote We can prove a bit more. Namely, the extensions of the segments RS and PQ meet at a point on the diameter AB that is vertically below the point Y .

Since $YR = AY \sin(\delta)$ and is inclined α counterclockwise from the vertical, the point S is $AY \sin(\delta) \sin(\alpha)$ horizontally to the right of Y .

Now $AS = AY \cos(\delta)$, so S is $AS \sin(\alpha) = AY \cos(\delta) \sin(\alpha)$ vertically above the diameter AB . Also, the segment SR is inclined δ clockwise from the vertical, so if we extend it down from S towards the diameter AB it will meet the diameter at a point which is $AY \cos(\delta) \sin(\alpha) \tan(\delta) = AY \sin(\delta) \sin(\alpha)$ horizontally to the left of S . This places the intersection point of RS and AB vertically below Y .

Similarly, and by symmetry the intersection point of PQ and AB is directly below Y on AB , so the lines through PQ and RS meet at a point T on the diameter that is vertically below Y .

Footnote to the Footnote The Footnote's claim is more easily proved as follows.

Note that because $\angle QPY$ and $\angle YAB$ are both complementary to $\beta + \gamma$, they must be equal. Now, let PQ intersect diameter AB at T' . Then $PYT'A$ is cyclic and so $\angle YT'A = 180^\circ - \angle APY = 90^\circ$. Hence $T'YSB$ is cyclic as well, and so we deduce that $\angle YST' = \angle YBT' = 90^\circ - \alpha - \delta = \angle YSR$. Hence S, R, T' are collinear and so $T = T'$. This proves the Footnote.

Footnote to the Footnote to the Footnote The Footnote's claim can be proved even more easily as follows.

Drop an altitude from Y to AB at point T . Notice that P, Q, T are collinear because they form the Simson line of $\triangle AXB$ from Y . Also notice that P, Q, T are collinear because they form the Simson line of $\triangle AZB$ from Y . Since T is at the diameter AB , lines PQ and SR must intersect at the diameter.

Footnote to the Fourth Power There is another, simpler solution using Simson lines. Can you find it?

Operation Diagram Of course, as with any geometry problem, DRAW A HUGE DIAGRAM spanning at least one page. And label all your right angles, noting rectangles $PXQY$ and $YSZR$. It looks like there are a couple of key angles we need to diagram. Let's take $\angle ZAB = \alpha$, $\angle XBA = \beta$, $\angle YAZ = \angle YBZ = \delta$. From there $\angle XOZ = 180^\circ - \angle XOY - \angle ZOY = 180 - 2(\beta + \alpha)$.

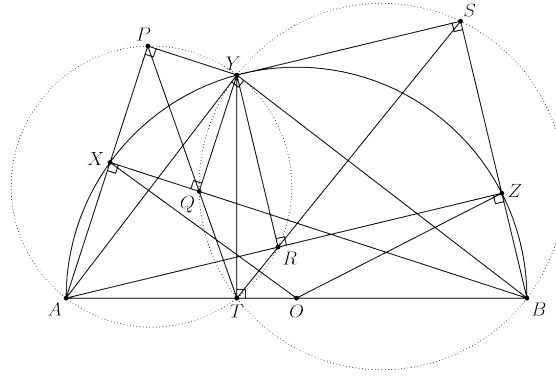
Move on to the part about the intersection of PQ and RS . Call the intersection J . Note that by Simson Lines from point Y to $\triangle ABX$ and $\triangle AZB$, YJ is perpendicular to AB and J lies on AB . Immediately note that we are trying to show that $\angle PJS = 90 - \beta - \alpha$.

It suffices to show that referencing quadrilateral QRJ , where J represents the intersection of XB, AZ , we have $\angle QJR + \angle BQJ + \angle ARJ = 270^\circ + \alpha + \beta$. Note that the reflex angle is $180^\circ + \angle AX = 180^\circ + (90^\circ - \angle XA^*) = 270^\circ - ((90^\circ - \beta) - \alpha) = 180^\circ + \alpha + \beta$, therefore it suffices to show that $\angle BQJ + \angle ARJ = 90^\circ$. To make this proof more accessible, note that via (cyclic) rectangles $PXQY$ and $YSZR$, it suffices to prove $\angle YPJ + \angle YSJ = 90^\circ$.

Note $\angle YPJ = \angle YPQ = \angle YXQ = \angle YXB = \angle YAB = \alpha + \delta$. Note $\angle YSJ = \angle YSR = \angle YZR = \angle YZA = \angle YBA = \angle YBX + \angle XBA = ((90^\circ - \alpha) - \delta - \beta) + \beta = 90^\circ - \alpha - \delta$, which completes the proof.

Footnote to Operation Diagram For reference/feasibility records: took expiLnCalc 56 minutes (consecutively). During the problem expiLnCalc realized that the inclusion of δ was necessary when trying to show that $\angle YSJ + \angle YPJ = 90^\circ$. Don't be afraid to attempt several different strategies, and always be humble!

Solution 2



Let T be the foot of the perpendicular from Y to $\overline{AB}$, let O be the center of the semi-circle.

Since we have a semi-circle, if we were to reflect it over $\overline{AB}$, we would have a full circle, with $\triangle AXB$ and $\triangle AZB$ inscribed in it. Now, notice that Y is a point on that full circle, so we can say that T lies on the Simson Line $\overline{PQ}$ from Y to $\triangle AXB$ and that it also lies on the Simson line $\overline{RS}$ from Y to $\triangle AZB$. Thus, T lies on two distinct lines in a plane, which means that $T = \overline{PQ} \cap \overline{RS}$. Therefore, it suffices to show that $\angle PTS = \frac{1}{2}\angle XOZ$.

Since $m\angle YTA + m\angle YPA = 90^\circ + 90^\circ = 180^\circ$ and $m\angle YTB + m\angle YSB = 90^\circ + 90^\circ = 180^\circ$, we know that $TAPY$ and $TBSY$ are cyclic quadrilaterals.

We use this fact to get

$$\angle PTS = \angle PTY + \angle YTS = \angle PAY + \angle YBS = \angle XAY + \angle YBZ. (1)$$

Now note that $\angle XAY$ is the inscribed angle of minor arc $\widehat{PY}$, and $\angle XOY$ is the central angle of minor arc $\widehat{AB}$, so $\angle XAY = \frac{\widehat{PY}}{2} = \frac{\angle XOY}{2}$. Similarly, $\angle YBZ = \frac{\widehat{YZ}}{2} = \frac{\angle YOZ}{2}$. Thus we can say

$$\angle XAY + \angle YBZ = \frac{\angle XOY}{2} + \frac{\angle YOZ}{2} = \frac{\angle XOY + \angle YOZ}{2} = \frac{\angle XOZ}{2}. (2)$$

Combining statements (1) and (2), we can say that $\angle PTS = \frac{\angle XOZ}{2}$, as desired. $\square$

2. (*USAMO 2010*) There are n students standing in a circle, one behind the other. The students have heights $h_1 < h_2 < \dots < h_n$. If a student with height h_k is standing directly behind a student with height h_{k-2} or less, the two students are permitted to switch places. Prove that it is not possible to make more than $\binom{n}{3}$ such switches before reaching a position in which no further switches are possible.

Solution 1

We adopt the usual convention that $\binom{i}{j} = 0$ unless $0 \leq j \leq i$. With this, the binomial coefficients are defined for all integers via the recursion:

$$\binom{0}{0} = 1, \quad \binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

It is clear that the circle is oriented and all the students are facing in same direction (clockwise or counterclockwise). We'll call this direction *forward*.

In any switch consider the taller student to have moved *forward* and the shorter student to have remained *stationary*. No backward motion is allowed. With this definition of forward motion, the first two students with heights h_1 and h_2 are always stationary, while other students potentially move past them.

For $k > 2$, the student with height h_k can never switch places with the student with height h_{k-1} , and the former can make at most $k-2$ more forward moves than the latter (when all the students of heights $h_1, \dots, h_{k-2}$ are between h_k and h_{k-1} in the forward direction).

Therefore, if the $(k-1)^{\text{st}}$ student can make s_{k-1} forward steps, the k^{th} student can make at most $s_{k-1} + k - 2$ steps. With $s_1 = s_2 = 0$ and $s_3 = 0 + (3 - 2) = 1$, and a constant second difference of 1, we quickly see that $s_k = \binom{k-1}{2}$.

With n students in all, the total number of steps is therefore at most $\sum_{i=3}^n \binom{i-1}{2} = \binom{n}{3}$, by the **Hockey Stick Theorem**.

Solution 2

WLOG, let $h_k = k$. Now, we find the end arrangement with the most switches possible. We claim that the arrangement will be $n, n-1, n-2, \dots, 1$, where the left to right direction is the "backwards" direction. To prove this makes the most switches, we show that there is always at least one more switch that can be done for any other arrangement. This is elementary to show. There will always be one height x such that the number to its right is 2 less than x , unless every number has $x-1$ to the right of x (other than 2 and 1). The exception occurs at our claim, so our claim is proven. Now we want to find the maximum ways we can "undo" our arrangement. But undoing a switch is just doing a switch from our arrangement in the opposite direction. So, the start arrangement with the most possible switches is the reverse of the end arrangement or $1, 2, 3, \dots, n$. We want to find how many switches must be done to get from the start arrangement to the end arrangement. We start by switching n around until it cannot be switched anymore. We find that we can switch n $n-2$ times. When we are switching (or, in other words, moving to the right) the number k , we can switch it $k-2$ times before it is to the left of $k-1$. But then we can also switch each of $k+1, k+2, \dots, n$ (in that order) $k-2$ times before they get stopped again. Let $x = n - k + 2$. Then, the sum of the switches is

$$\sum_{x=2}^{n-1} (n-x)(x-1) = \sum_{x=1}^n (n-x)(x-1).$$

Rearranging and summing, we get

$$\sum_{x=1}^n x(n+1) - \sum_{x=1}^n n - \sum_{x=1}^n x^2 = \frac{n(n+1)^2}{2} - n^2 - \frac{n(n+1)(2n+1)}{6} = \frac{3n(n^2+1)}{6} - \frac{n(n+1)(2n+1)}{6} = \frac{n}{6}(n^2+1-2n-1) = \frac{n}{6}(n^2-2n) = \frac{n}{6}n(n-2) = \frac{n^2(n-2)}{6}$$

Comment This process of switching is very similar to Bubble Sort, except that the terms must be in consecutive decreasing order, and wraps around $2, 1, n, n-1, \dots$. This answer seems to disagree, though, because the worst case efficiency is $\mathcal{O}(n^2)$, not $\mathcal{O}(n^3)$.

https://en.wikipedia.org/wiki/Bubble_sort

3. (*USAMO 2010*) The 2010 positive numbers $a_1, a_2, \dots, a_{2010}$ satisfy the inequality $a_i a_j \leq i + j$ for all distinct indices i, j . Determine, with proof, the largest possible value of the product $a_1 a_2 \dots a_{2010}$.

Solution

The largest possible value is

$$\prod_{i=1}^{1005} (4i-1) = 3 \times 7 \times \dots \times 4019.$$

Proof No larger value is possible, since for each consecutive pair of elements: (a_{2i-1}, a_{2i}) , $1 \leq i \leq 1005$, the product is at most $(2i-1) + 2i = 4i-1$, and so the product of all the pairs is at most:

$$\prod_{i=1}^{1005} (4i-1).$$

If we can demonstrate a sequence in which for all $1 \leq i \leq 1005$ the product $a_{2i-1} a_{2i} = 4i-1$, and all the inequalities are satisfied, the above upper bound will be achieved and the proof complete.

We will construct sequences of an arbitrarily large even length $2n \geq 4$, in which:

$$\begin{cases} a_i a_j = i + j = 2i + 1 & j = i + 1, \\ a_i a_j = i + j = 2n - 2 & j = i + 2 = 2n, \\ a_i a_j < i + j & i \neq 2n - 2, \text{ and } i < j - 1. \end{cases}$$

Given a_1 , from the equations $a_i a_{i+1} = 2i + 1$, $1 \leq i \leq 2n - 1$, we obtain the whole sequence recursively: $a_1 = a_1$, $a_2 = 3/a_1$, $a_3 = 5/a_2 = 5a_1/3$, $a_4 = 7/a_3 = (3 \cdot 7)/(5a_1) \dots$. And as a result:

$$a_{i+2} = a_i \cdot \frac{2i+3}{2i+1} \quad 1 \leq i \leq 2n-2.$$

The same equations $a_i a_{i+1} = 2i + 1$ can be used to compute the whole sequence from any other known term.

We will often need to compare fractions in which the numerator and denominator are both positive, with fractions in which a positive term is added to both. Suppose p, q, r are three positive real numbers, then:

$$\begin{array}{ll}
 p > q \implies & \frac{p+r}{q+r} < \frac{p}{q} \\
 p < q \implies & \frac{p+r}{q+r} > \frac{p}{q} \\
 \text{and always} & \frac{p+r}{q} > \frac{p}{q}
 \end{array}$$

Returning to the problem in hand, for $i < j$, $a_i a_j \leq i + j \implies a_i a_{j+2} < i + j + 2$. If it were otherwise, we would have for some $i < j$:

$$\frac{a_{j+2}}{a_j} \geq \frac{i+j+2}{i+j} > \frac{j+j+2}{j+j} = \frac{2j+2}{2j} > \frac{2j+3}{2j+1} = \frac{a_{j+2}}{a_j},$$

so our assumption is impossible.

Therefore, we need only verify inequalities with an index difference of 1 or 2, as these imply the rest.

Now, when the indices differ by 1 we have ensured equality (and hence the desired inequalities) by construction. So, we only need to prove the inequalities for successive even index and successive odd index pairs, i.e. for every index $i > 2$, prove $a_{i-2} a_i \leq 2i - 2$.

We now compare $a_i a_{i+2}/(2i+2)$ with $a_{i+2} a_{i+4}/(2i+6)$. By our recurrence relations:

$$\begin{aligned}
 \frac{a_{i+2} a_{i+4}}{2i+6} &= \frac{a_i a_{i+2}}{2i+2} \cdot \frac{2i+2}{2i+6} \cdot \frac{a_{i+4}}{a_{i+2}} \cdot \frac{a_{i+2}}{a_i} \\
 &= \frac{a_i a_{i+2}}{2i+2} \cdot \frac{i+1}{i+3} \cdot \frac{2i+7}{2i+5} \cdot \frac{2i+3}{2i+1} \\
 &= \frac{a_i a_{i+2}}{2i+2} \cdot \frac{(i+1)(2i+7)(2i+3)}{(i+3)(2i+5)(2i+1)} \\
 &= \frac{a_i a_{i+2}}{2i+2} \cdot \frac{4i^3 + 24i^2 + 41i + 21}{4i^3 + 24i^2 + 41i + 15} \\
 &= \frac{a_i a_{i+2}}{2i+2} \cdot \left(1 + \frac{6}{4i^3 + 24i^2 + 41i + 15} \right) \\
 &> \frac{a_i a_{i+2}}{2i+2}.
 \end{aligned}$$

So, for both odd and even index pairs, the strict inequality $a_i a_{i+2} < 2i + 2$ follows from $a_{i+2} a_{i+4} \leq 2i + 6$ and we need only prove the inequalities $a_{2n-3} a_{2n-1} \leq 4n - 4$ and $a_{2n-2} a_{2n} \leq 4n - 2$, the second of which holds (as an equality) by construction, so only the first remains.

We have not yet used the equation $a_{2n-2} a_{2n} = 4n - 2$, with this we can solve for the last three terms (or equivalently their squares) and thus compute the whole sequence. From the equations:

$$\begin{aligned}
 a_{2n-1} a_{2n} &= 4n - 1 \\
 a_{2n-2} a_{2n} &= 4n - 2 \\
 a_{2n-2} a_{2n-1} &= 4n - 3
 \end{aligned}$$

multiplying any two and dividing by the third, we get:

$$\begin{aligned} a_{2n}^2 &= \frac{(4n-2)(4n-1)}{4n-3} \\ a_{2n-1}^2 &= \frac{(4n-3)(4n-1)}{4n-2} \\ a_{2n-2}^2 &= \frac{(4n-3)(4n-2)}{4n-1} \end{aligned}$$

from which,

$$a_{2n-3}^2 = \frac{(4n-5)^2}{a_{2n-2}^2} = \frac{(4n-5)^2(4n-1)}{(4n-3)(4n-2)}$$

With the squares of the last four terms in hand, we can now verify the only non-redundant inequality:

$$\begin{aligned} a_{2n-3}^2 a_{2n-1}^2 &= \frac{(4n-5)^2}{a_{2n-2}^2} a_{2n-1}^2 \\ &= \left(\frac{(4n-5)(4n-1)}{4n-2} \right)^2 \\ &= \left(\frac{16n^2 - 24n + 5}{4n-2} \right)^2 \\ &< \left(\frac{16n^2 - 24n + 8}{4n-2} \right)^2 \\ &= (4n-4)^2. \end{aligned}$$

The inequality above follows because the numerator and denominator are both positive for $n > 1$.

This completes the construction and the proof of all the inequalities, which miraculously reduced to just one inequality for the last pair of odd indices.

Additional observations If we choose a different first term, say $a'_1 = M \cdot a_1$, the sequence a'_i will have the form:

$$\begin{aligned} a'_{2i-1} &= M a_{2i-1} \\ a'_{2i} &= \frac{a_{2i}}{M} \end{aligned}$$

the same holds if we have a longer sequence, at every index of the shorter sequence, the longer sequence will be a constant multiple (for all the odd terms) or dividend (for all the even terms) of the corresponding term of shorter sequence.

We observe that our solution is not unique, indeed for any $k > 0$, the same construction with $2n + 2k$ terms, truncated to just the first $2n$ terms, yields a sequence a'_i which also satisfies all the required conditions, but in this case $a'_{2n-2} a'_{2n} < 4n - 2$.

We could have constructed this alternative solution directly, by replacing the right hand side in the equation $a_{2n-2}a_{2n} = 4n - 2$ with any smaller value for which we still get $a_{2n-3}a_{2n-1} \leq 4n - 4$.

In the modified construction, for some constant $M > 1$, we have:

$$\begin{aligned} a_{2n}^2 &= \frac{1}{M} \cdot \frac{(4n-2)(4n-1)}{4n-3} \\ a_{2n-1}^2 &= M \cdot \frac{(4n-3)(4n-1)}{4n-2} \\ a_{2n-2}^2 &= \frac{1}{M} \cdot \frac{(4n-3)(4n-2)}{4n-1} \end{aligned}$$

and so:

$$\begin{aligned} a_{2n-3}^2 a_{2n-1}^2 &= \frac{(4n-5)^2}{a_{2n-2}^2} a_{2n-1}^2 \\ &= M^2 \left(\frac{(4n-5)(4n-1)}{4n-2} \right)^2 \\ &= M^2 \left(\frac{16n^2 - 24n + 5}{4n-2} \right)^2 \\ &= M^2 \left(\frac{16n^2 - 24n + 5}{16n^2 - 24n + 8} \right)^2 \cdot \left(\frac{16n^2 - 24n + 8}{4n-2} \right)^2 \\ &= M^2 \left(\frac{16n^2 - 24n + 5}{16n^2 - 24n + 8} \right)^2 \cdot (4n-4)^2. \end{aligned}$$

which satisfies the required inequality provided:

$$\begin{aligned} M^2 &\leq \frac{16n^2 - 24n + 8}{16n^2 - 24n + 5} \\ &= \frac{(4n-4)(4n-2)}{(4n-5)(4n-1)} = M_{\max}^2. \end{aligned}$$

The ratio $M_{\max}$, between the largest and smallest possible value of a_{2n-3} is in fact the ratio between the largest and smallest values of a_1 that yield a sequence that meets the conditions for at least $2n$ terms.

In the $n = 2$ case, the equation for a_{2n-3} gives: $a_1^2 = \frac{21}{10}$. We will next consider what happens to a_1^2 , and the sequence of squares in general, as n increases.

Let $A_{n,2i-1}, A_{n,2i}$ denote the i^{th} odd and even terms, respectively, of the unique sequence which satisfies our original equations and has $2n$ terms in total. Let $A_{n+1,2i-1}, A_{n+1,2i}$ be the odd and even terms of the solution with $2n+2$ terms. We already noted that there must exist a constant M_n (that depends on n , but not on i), such that:

$$\begin{cases} A_{n+1,2i-1} = M_n \cdot A_{n,2i-1}, & 1 \leq i \leq n \\ A_{n+1,2i} = \frac{A_{n,2i}}{M_n}, & 1 \leq i \leq n \end{cases}$$

This constant is found explicitly by comparing the squares of the last term $A_{n,2n}$ of the solution of length $2n$ with the square of the third last term $A_{n+1,2n}$ of the solution of length $2n+2$:

$$\begin{aligned} (1/M_n)^2 &= \frac{A_{n+1,2n}^2}{A_{n,2n}^2} \\ &= \frac{(4n+1)(4n+2)}{4n+3} \cdot \frac{4n-3}{(4n-2)(4n-1)} \\ &= \frac{32n^3 - 14n - 3}{32n^3 - 14n + 3} \end{aligned}$$

Clearly $M_n > 1$ for all positive n , and so for fixed i , the odd index terms $A_{n,2i-1}$ strictly increase with n , while the even index terms $A_{n,2i}$ decrease with n .

Therefore, for $n \geq 2$,

$$A_{n,1}^2 = \frac{21}{10} \prod_{k=2}^{n-1} M_k^2 = \frac{21}{10} \cdot \prod_{k=2}^{n-1} \frac{(k + \frac{3}{4})(k - \frac{1}{2})(k - \frac{1}{4})}{(k - \frac{3}{4})(k + \frac{1}{2})(k + \frac{1}{4})}$$

The product converges to a finite value even if taken infinitely far, and we can conclude (by a simple continuity argument) that there is a unique infinite positive sequence A_ω , in which $A_{\omega,i} A_{\omega,i+1} = 2i+1$, that satisfies all the inequalities $A_{\omega,i} A_{\omega,j} < i+j$, $i \leq j-2$. The square of the first term of the infinite sequence is:

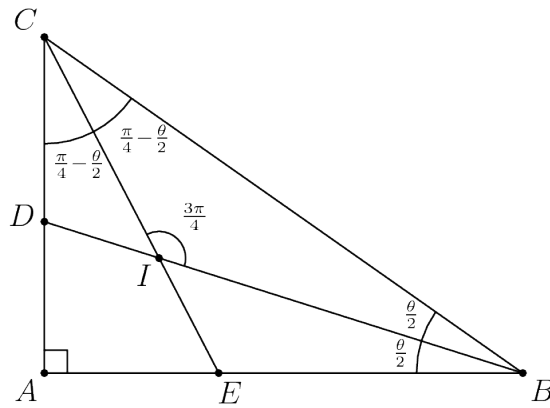
$$\begin{aligned} A_{\omega,1}^2 &= \frac{21}{10} \cdot \lim_{n \rightarrow \infty} \prod_{k=2}^{n-1} \frac{(k + \frac{3}{4})(k - \frac{1}{2})(k - \frac{1}{4})}{(k - \frac{3}{4})(k + \frac{1}{2})(k + \frac{1}{4})} \\ &= \frac{21}{10} \cdot \lim_{n \rightarrow \infty} \left(\frac{\Gamma(\frac{5}{4})\Gamma(\frac{5}{2})\Gamma(\frac{9}{4})}{\Gamma(\frac{11}{4})\Gamma(\frac{3}{2})\Gamma(\frac{7}{4})} \cdot \frac{\Gamma(n + \frac{3}{4})\Gamma(n - \frac{1}{2})\Gamma(n - \frac{1}{4})}{\Gamma(n - \frac{3}{4})\Gamma(n + \frac{1}{2})\Gamma(n + \frac{1}{4})} \right) \\ &= \frac{1}{4} \cdot \frac{\Gamma^2(\frac{1}{4})}{\Gamma^2(\frac{3}{4})} \cdot \lim_{n \rightarrow \infty} \frac{\Gamma(n + \frac{3}{4})\Gamma(n - \frac{1}{2})\Gamma(n - \frac{1}{4})}{\Gamma(n - \frac{3}{4})\Gamma(n + \frac{1}{2})\Gamma(n + \frac{1}{4})} \\ &= \frac{1}{4} \cdot \frac{\Gamma^2(\frac{1}{4})}{\Gamma^2(\frac{3}{4})} \cdot 1 \quad (\text{Stirling's approximation}) \\ &= \frac{\Gamma^4(\frac{1}{4})}{8\pi^2} = \frac{\pi}{\text{AGM}^2(\sqrt{2}, 1)} \approx 2.1884396152264766 \dots \end{aligned}$$

In summary, if we set $a_1 = \frac{\sqrt{\pi}}{\text{AGM}(\sqrt{2}, 1)}$, and then recursively set $a_{i+1} = (2i+1)/a_i$, we get an infinite sequence that, for all $n \geq 1$, yields the maximum possible product $a_1 a_2 \cdots a_{2n}$, subject to the conditions $a_i a_j \leq i+j$, $1 \leq i < j \leq 2n$.

4. (USAMO 2010) Let ABC be a triangle with $\angle A = 90^\circ$. Points D and E lie on sides AC and AB , respectively, such that $\angle ABD = \angle DBC$ and $\angle ACE = \angle ECB$. Segments BD and CE meet at I . Determine whether or not it is possible for segments AB, AC, BI, ID, CI, IE to all have integer side lengths.

Solution

We know that angle $BIC = 135^\circ$, as the other two angles in triangle BIC add to 45° . Assume that only AB, AC, BI , and CI are integers. Using the **Law of Cosines** on triangle BIC ,



$BC^2 = BI^2 + CI^2 - 2BI \cdot CI \cdot \cos 135^\circ$. Observing that $BC^2 = AB^2 + AC^2$ is an integer and that $\cos 135^\circ = -\frac{\sqrt{2}}{2}$, we have

$$BC^2 - BI^2 - CI^2 = BI \cdot CI \cdot \sqrt{2}$$

and therefore,

$$\sqrt{2} = \frac{BC^2 - BI^2 - CI^2}{BI \cdot CI}$$

The LHS ($\sqrt{2}$) is irrational, while the RHS is the quotient of the division of two integers and thus is rational. Clearly, there is a contradiction. Therefore, it is impossible for AB, AC, BI , and CI to all be integers, which invalidates the original claim that all six lengths are integers, and we are done.

Solution 2

The answer is no.

Suppose otherwise. It is easy to see (through simple angle chasing) that $\angle DIC = 45^\circ$. Also, since I is the incenter, we have $\angle IAC = 45^\circ$. Using the Law of Cosines, we have

$$CD^2 = IC^2 + ID^2 - \sqrt{2}(IC)(ID),$$

so that CD is irrational. But $\triangle IAC \sim \triangle DIC$, thus $IC^2 = CD \cdot AC$, implying that CD is rational, contradiction. ■

Solution 3

The result can be also proved without direct appeal to trigonometry, via just the angle bisector theorem and the structure of Pythagorean triples. (This is a lot more work).

A triangle in which all the required lengths are integers exists if and only if there exists a triangle in which AB and AC are relatively-prime integers and the lengths of the segments BI, ID, CI, IE are all rational (we divide all the lengths by the $\gcd(AB, AC)$ or conversely multiply all the lengths by the least common multiple of the denominators of the rational lengths).

Suppose there exists a triangle in which the lengths AB and AC are relatively-prime integers and the lengths IB, ID, CI, IE are all rational.

Since CE is the bisector of $\angle ACB$, by the angle bisector theorem, the ratio $IB : ID = CB : CD$, and since BD is the bisector of $\angle ABC$, $CB : CD = (AB + BC) :$

AC . Therefore, $IB : ID = (AB + BC) : AC$. Now $IB : ID$ is by assumption rational, so $(AB + BC) : AC$ is rational, but AB and AC are assumed integers so $BC : AC$ must also be rational. Since BC is the hypotenuse of a right-triangle, its length is the square root of an integer, and thus either an integer or irrational, so BC must be an integer.

With AB and AC relatively-prime, we conclude that the sidelengths of $\triangle ABC$ must be a Pythagorean triple: $(2pq, p^2 - q^2, p^2 + q^2)$, with $p > q$ relatively-prime positive integers and $p + q$ odd.

Without loss of generality, $AC = 2pq$, $AB = p^2 - q^2$, $BC = p^2 + q^2$. By the angle bisector theorem,

$$AE = \frac{AB \cdot AC}{AC + CB} = \frac{2pq(p^2 - q^2)}{p^2 + q^2 + 2pq} = \frac{2pq(p - q)}{p + q}$$

Since $\triangle CAE$ is a right-triangle, we have:

$$\begin{aligned} CE^2 &= AC^2 + AE^2 = 4p^2q^2 + \left(\frac{2pq(p - q)}{p + q} \right)^2 = 4p^2q^2 \left[1 + \left(\frac{p - q}{p + q} \right)^2 \right] \\ &= \frac{4p^2q^2}{(p + q)^2} \left[(p + q)^2 + (p - q)^2 \right] = \frac{4p^2q^2}{(p + q)^2} (2p^2 + 2q^2) \end{aligned}$$

and so CE is rational if and only if $2p^2 + 2q^2$ is a perfect square.

Also by the angle bisector theorem,

$$AD = \frac{AB \cdot AC}{AB + BC} = \frac{2pq(p^2 - q^2)}{p^2 + q^2 + p^2 - q^2} = \frac{q(p^2 - q^2)}{p}$$

and therefore, since $\triangle DAB$ is a right-triangle, we have:

$$\begin{aligned} BD^2 &= AB^2 + AD^2 = (p^2 - q^2)^2 + \left(\frac{q(p^2 - q^2)}{p} \right)^2 \\ &= (p^2 - q^2)^2 \left[1 + \frac{q^2}{p^2} \right] = \frac{(p^2 - q^2)^2}{p^2} (p^2 + q^2) \end{aligned}$$

and so BD is rational if and only if $p^2 + q^2$ is a perfect square.

Combining the conditions on CE and BD , we see that $2p^2 + 2q^2$ and $p^2 + q^2$ must both be perfect squares. If it were so, their ratio, which is 2, would be the square of a rational number, but $\sqrt{2}$ is irrational, and so the assumed triangle cannot exist.

Solution 4

We proceed by contradiction.

FTSOC, let AB, BI, ID have integer lengths. Then $BD = BI + ID \in \mathbb{Z}$ as well. By trigonometry,

$$BD = \frac{AB}{\cos(\frac{\angle ABC}{2})}.$$

Rearranging we find $\cos(\frac{\angle ABC}{2}) = \frac{AB}{BD} \in \mathbb{Q}$. As $0 < \angle ABC < \frac{\pi}{2}$ so $0 < \frac{\angle ABC}{2} < \frac{\pi}{4}$ but there are no possible angles in the interval that result in a rational cosine by Niven's theorem. So we have contradiction.

5. (USAMO 2010) Let $q = \frac{3p-5}{2}$ where p is an odd prime, and let

$$S_q = \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{5 \cdot 6 \cdot 7} + \cdots + \frac{1}{q \cdot (q+1) \cdot (q+2)}.$$

Prove that if $\frac{1}{p} - 2S_q = \frac{m}{n}$ for relatively prime integers m and n , then $m - n$ is divisible by p .

Solution

Since p is an odd prime, $p = 2r+1$, for a suitable positive integer r , and consequently $q = 3r-1$.

The partial-fraction decomposition of the general term of S_q is:

$$\begin{aligned} \frac{1}{(3k-1)3k(3k+1)} &= \frac{1}{2} \left(\frac{1}{3k-1} - \frac{2}{3k} + \frac{1}{3k+1} \right) \\ &= \frac{1}{2} \left(\frac{1}{3k-1} + \frac{1}{3k} - \frac{1}{k} + \frac{1}{3k+1} \right) \\ &= \frac{1}{2} \left[\left(\frac{1}{3k-1} + \frac{1}{3k} + \frac{1}{3k+1} \right) - \frac{1}{k} \right], \end{aligned}$$

therefore

$$\begin{aligned} \frac{1}{p} - 2S_q &= \frac{1}{2r+1} - \left(\sum_{k=2}^{3r+1} \frac{1}{k} - \sum_{k=1}^r \frac{1}{k} \right) \\ &= \frac{1}{2r+1} - \left[\left(\sum_{k=r+1}^{3r+1} \frac{1}{k} \right) - 1 \right] \\ &= 1 - \left(\sum_{k=r+1}^{2r} \frac{1}{k} + \sum_{k=2r+2}^{3r+1} \frac{1}{k} \right) \\ &= 1 - \sum_{k=1}^r \left(\frac{1}{(2r+1)-k} + \frac{1}{(2r+1)+k} \right) \\ &= 1 - \sum_{k=1}^r \frac{2p}{(p-k)(p+k)} \\ &= 1 - \frac{a}{b} = \frac{b-a}{b} \end{aligned}$$

with a and b positive relatively-prime integers.

Since $r < p$ and p is a prime, in the final sum all the denominators are relatively prime to p , but all the numerators are divisible by p , and therefore the numerator a of the reduced fraction $\frac{a}{b}$ will be divisible by p . Since the sought difference $m - n = (b - a) - b = -a$, we conclude that p divides $m - n$ as required.

Alternative Calculation We can obtain the result in a slightly different way:

$$\begin{aligned}
\frac{1}{p} - 2S_q &= \frac{1}{2r+1} - \left(\sum_{k=2}^{3r+1} \frac{1}{k} - \sum_{k=1}^r \frac{1}{k} \right) \\
&= \frac{1}{2r+1} - \left(\sum_{k=r+1}^{3r+1} \frac{1}{k} - 1 \right) \\
&= 1 - \left(\sum_{k=r+1}^{2r} \frac{1}{k} + \sum_{k=2r+2}^{3r+1} \frac{1}{k} \right)
\end{aligned}$$

In the above sum the denominators of the fractions represent each non-zero remainder $(\text{mod } p)$ exactly once. Multiplying all the denominators yields a number N that is $p-1 \pmod{p}$. The numerator $(\text{mod } p)$ is N times the sum of the $(\text{mod } p)$ inverses of each non-zero remainder, and since this sum is $0 \pmod{p}$, the numerator is $0 \pmod{p}$. The rest of the argument is as before.

6. (*USAMO 2010*) A blackboard contains 68 pairs of nonzero integers. Suppose that for each positive integer k at most one of the pairs (k, k) and $(-k, -k)$ is written on the blackboard. A student erases some of the 136 integers, subject to the condition that no two erased integers may add to 0. The student then scores one point for each of the 68 pairs in which at least one integer is erased. Determine, with proof, the largest number N of points that the student can guarantee to score regardless of which 68 pairs have been written on the board.

Solution

Answer: 43

Attainability: Consider 8 distinct positive numbers. Let there be 5 pairs for each of the numbers including 2 clones of that number. Let there also be 28 pairs that include the negatives of those numbers such that each negative associates with another negative once and exactly once (in graph theoretic terms, a K_8). Let x be the number of positives chosen out of these 8 (assume the other chosen numbers are negatives) for erasing; then, the number of points the student scores is $28 + 5x - \binom{x}{2}$, which is maximized at $x=5$ and $x=6$, and the maximum value is 43. Choosing the first 5 numbers as positive and the other three as negative attains this. Hence, 43 is a possible maximum possible score.

Bounding: We use expected values. WLOG all the pairs of numbers with both numbers identical have only positive values. Consider flipping a weighted coin whether to choose the positive number or its negation for each positive number; it chooses positive with probability p . The pairs with both numbers same are chosen with probability p , the pairs $(k, -k)$ are chosen with probability 1 , and the pairs (x, y) for distinct x and y that sum to a nonzero number are chosen with probability $1 - p^2$. We are trying to minimize the expected value, so we can assume that no pairs $(k, -k)$ exist. Let A be the number of (k, k) pairs, and $68-A$ be the number of (x, y) pairs. The expected number of points scored is $Ap + (68-A)(1-p^2)$. We want to prove this is larger than 42 at all times for some choice of p . If $A < 36$, $1/2$ works for p to give this bound. If $A > 36$, $5/8$ works for p for p to give the desired bound. If $A = 36$, we can use $3/5$ for p to get the desired bound. Hence, in any case, the expected value for the number of points scored across all of these weighted processes is larger than 42, so there exists some case that gives a score of 43. Hence, bounding is complete. We are done with both parts. Q.E.D.

-Solution by thanosaops

§40 USAMO 2011

1. (*USAMO 2011*) Let a, b, c be positive real numbers such that $a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4$. Prove that

$$\frac{ab+1}{(a+b)^2} + \frac{bc+1}{(b+c)^2} + \frac{ca+1}{(c+a)^2} \geq 3.$$

Solution 1

Since

$$\begin{aligned} (a+b)^2 + (b+c)^2 + (c+a)^2 &= 2(a^2 + b^2 + c^2 + ab + bc + ca) \\ &= a^2 + b^2 + c^2 + (a+b+c)^2, \end{aligned}$$

it is natural to consider a change of variables:

$$\begin{aligned} \alpha &= b+c \\ \beta &= c+a \\ \gamma &= a+b \end{aligned}$$

with the inverse mapping given by:

$$\begin{aligned} a &= \frac{\beta + \gamma - \alpha}{2} \\ b &= \frac{\alpha + \gamma - \beta}{2} \\ c &= \frac{\alpha + \beta - \gamma}{2} \end{aligned}$$

With this change of variables, the constraint becomes

$$\alpha^2 + \beta^2 + \gamma^2 \leq 4,$$

while the left side of the inequality we need to prove is now

$$\begin{aligned} &\frac{\gamma^2 - (\alpha - \beta)^2 + 4}{4\gamma^2} + \frac{\alpha^2 - (\beta - \gamma)^2 + 4}{4\alpha^2} + \frac{\beta^2 - (\gamma - \alpha)^2 + 4}{4\beta^2} \geq \\ &\frac{\gamma^2 - (\alpha - \beta)^2 + \alpha^2 + \beta^2 + \gamma^2}{4\gamma^2} + \frac{\alpha^2 - (\beta - \gamma)^2 + \alpha^2 + \beta^2 + \gamma^2}{4\alpha^2} + \frac{\beta^2 - (\gamma - \alpha)^2 + \alpha^2 + \beta^2 + \gamma^2}{4\beta^2} = \\ &\frac{2\gamma^2 + 2\alpha\beta}{4\gamma^2} + \frac{2\alpha^2 + 2\beta\gamma}{4\alpha^2} + \frac{2\beta^2 + 2\gamma\alpha}{4\beta^2} = \\ &\frac{3}{2} + \frac{\alpha\beta}{2\gamma^2} + \frac{\beta\gamma}{2\alpha^2} + \frac{\gamma\alpha}{2\beta^2}. \end{aligned}$$

Therefore it remains to prove that

$$\frac{\alpha\beta}{2\gamma^2} + \frac{\beta\gamma}{2\alpha^2} + \frac{\gamma\alpha}{2\beta^2} \geq \frac{3}{2}.$$

We note that the product of the three (positive) terms is $1/8$, therefore by AM-GM their mean is at least $1/2$, and thus their sum is at least $3/2$ and we are done.

Solution 2

Rearranging the condition yields that

$$a^2 + b^2 + c^2 + ab + bc + ac \leq 2$$

Now note that

$$\frac{2ab + 2}{(a + b)^2} \geq \frac{2ab + a^2 + b^2 + c^2 + ab + bc + ac}{(a + b)^2} = \frac{(a + b)^2 + (c + a)(c + b)}{(a + b)^2}$$

Summing this for all pairs of $\{a, b, c\}$ gives that

$$\sum_{\text{cyc}} \frac{2ab + 2}{(a + b)^2} \geq 3 + \sum_{\text{cyc}} \frac{(c + a)(c + b)}{(a + b)^2} \geq 6$$

By AM-GM. Dividing by 2 gives the desired inequality.

Solution 3

Given $a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4$, prove that

$$\frac{ab + 1}{(a + b)^2} + \frac{bc + 1}{(b + c)^2} + \frac{ac + 1}{(a + c)^2} \geq 3.$$

From the given constraint $a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4$, we can expand to obtain:

$$a^2 + b^2 + c^2 + a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \leq 4$$

$$2(a^2 + b^2 + c^2) + 2(ab + bc + ca) \leq 4$$

$$\frac{a^2 + b^2 + c^2}{2} + \frac{ab + bc + ca}{2} \leq 1.$$

Now, observe that we can rewrite the left-hand side of our inequality as:

$$\begin{aligned} & \sum_{\text{cyc}} \frac{ab + 1}{(a + b)^2} \\ &= \sum_{\text{cyc}} \frac{ab + \frac{a^2 + b^2 + c^2}{2} + \frac{ab + bc + ca}{2}}{(a + b)^2} \\ &= \sum_{\text{cyc}} \left[\frac{ab + \frac{a^2 + b^2}{2}}{(a + b)^2} + \frac{\frac{c^2}{2} + \frac{bc + ca}{2}}{(a + b)^2} \right] \\ &= \sum_{\text{cyc}} \left[\frac{ab + \frac{a^2 + b^2}{2}}{(a + b)^2} \right] + \sum_{\text{cyc}} \left[\frac{\frac{a^2 + c^2}{2}}{(a + c)^2} + \frac{\frac{ac + bc}{2}}{(a + b)^2} \right]. \end{aligned}$$

We now evaluate each part separately. Note that

$$\frac{ab + \frac{a^2 + b^2}{2}}{(a + b)^2} = \frac{\frac{(a + b)^2}{2}}{(a + b)^2} = \frac{1}{2}.$$

Therefore,

$$\sum_{\text{cyc}} \frac{ab + \frac{a^2 + b^2}{2}}{(a + b)^2} = \frac{3}{2}.$$

By the RMS-AM inequality, we have $(a+c)^2 \leq 2(a^2+c^2)$, which gives $\frac{1}{(a+c)^2} \geq \frac{1}{2(a^2+c^2)}$. Thus,

$$\begin{aligned} \sum_{\text{cyc}} \frac{\frac{a^2+c^2}{2}}{(a+c)^2} &\geq \sum_{\text{cyc}} \frac{\frac{a^2+c^2}{2}}{2(a^2+c^2)} \\ &= \sum_{\text{cyc}} \frac{1}{4} = \frac{3}{4}. \end{aligned}$$

The remaining terms can be written as:

$$\begin{aligned} \sum_{\text{cyc}} \frac{\frac{ac+bc}{2}}{(a+b)^2} &= \frac{1}{2} \sum_{\text{cyc}} \frac{c(a+b)}{(a+b)^2} \\ &= \frac{1}{2} \sum_{\text{cyc}} \frac{c}{a+b} \\ &= \frac{1}{2} \left(\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \right). \end{aligned}$$

By Nesbitt's inequality,

$$\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2},$$

so this part contributes at least $\frac{3}{4}$. Adding all three parts together:

$$\frac{ab+1}{(a+b)^2} + \frac{bc+1}{(b+c)^2} + \frac{ac+1}{(a+c)^2} \geq \frac{3}{2} + \frac{3}{4} + \frac{3}{4} = 3.$$

Therefore, the inequality is proven. $\square$

2. (*USAMO 2011*) An integer is assigned to each vertex of a regular pentagon so that the sum of the five integers is 2011. A turn of a solitaire game consists of subtracting an integer m from each of the integers at two neighboring vertices and adding $2m$ to the opposite vertex, which is not adjacent to either of the first two vertices. (The amount m and the vertices chosen can vary from turn to turn.) The game is won at a certain vertex if, after some number of turns, that vertex has the number 2011 and the other four vertices have the number 0. Prove that for any choice of the initial integers, there is exactly one vertex at which the game can be won.

Solution

Let $\mathbb{F}_5$ be the field of positive residues modulo 5. We label the vertices of the pentagon clockwise with the residues 0, 1, 2, 3 and 4. For each $i \in \mathbb{F}_5$ let n_i be the integer at vertex i and let $r_i \in \mathbb{F}_5$ be defined as:

$$r_i \equiv 4n_{i+1} + 3n_{i+2} + 2n_{i+3} + n_{i+4} \equiv \sum_{k \in \mathbb{F}_5} (i-k)n_k \pmod{5}.$$

Let $s = \sum_{i \in \mathbb{F}_5} n_i$. A move in the game consists of

$$(n_i, n_{i+1}, n_{i+2}, n_{i+3}, n_{i+4}) \mapsto (n_i + 2m, n_{i+1}, n_{i+2} - m, n_{i+3} - m, n_{i+4})$$

for some vertex $i \in \mathbb{F}_5$ and integer m . We immediately see that s is an invariant of the game. After our move the new value of r_i is decreased by $3m + 2m \equiv 0$

(mod 5) as a result of the change in the $3n_{i+2}$ and $2n_{i+3}$ terms. So r_i does not change after a move at vertex i .

For all $i \in \mathbb{F}_5$ we have:

$$r_{i+1} - r_i \equiv \sum_{k \in \mathbb{F}_5} ((i+1-k)n_k - (i-k)n_k) \equiv \sum_{k \in \mathbb{F}_5} n_k \equiv s \pmod{5}.$$

Therefore, the r_i form an arithmetic progression in $\mathbb{F}_5$ with a difference of s . Since r_k is unchanged by a move at vertex k , so are all the remaining r_i as the differences are constant.

Provided $s \not\equiv 0 \pmod{5}$, we see that the mapping $i \mapsto r_i$ is a bijection $\mathbb{F}_5 \rightarrow \mathbb{F}_5$ and exactly one vertex will have $r_i \equiv 0 \pmod{5}$. As r_i is an invariant, a winning vertex must have $r_i \equiv 0$, since in the final state each n_k with $k \neq i$ is zero. So, for $s \not\equiv 0 \pmod{5}$, if a winning vertex exists, it is the unique vertex with $r_i \equiv 0$.

Without loss of generality, it remains to show that if $r_0 \equiv 0 \pmod{5}$, then 0 must be a winning vertex. To prove this, we perform the following moves:

$$\begin{aligned} (n_0, n_1, n_2, n_3, n_4) &\mapsto (n_0 - n_1, 0, n_2, n_3 + 2n_1, n_4), & (i, m) &= (3, n_1) \\ &\mapsto (n_0 - n_1 - n_4, 0, n_2 + 2n_4, n_3 + 2n_1, 0) & (i, m) &= (2, n_4) \end{aligned}$$

We designate the new state $(n'_0, 0, n'_2, n'_3, 0)$. Since $r_0 \equiv 0$ is an invariant, and $n'_1 = n'_4 = 0$, we now have $r_0 \equiv n'_3 - n'_2 = 5p$, for some integer p . Our final set of moves is:

$$\begin{aligned} (n'_0, 0, n'_2, n'_3, 0) &\mapsto (n'_0 + p, p, n'_2, n'_3 - 2p, 0), & (i, m) &= (3, -p) \\ &\mapsto (n'_0 + p, 0, n'_2 - p, n'_3 - 2p, 2p), & (i, m) &= (4, p) \\ &\mapsto (n'_0 - p, 0, n'_2 + 3p, n'_3 - 2p, 0), & (i, m) &= (2, 2p) \\ &\mapsto (n'_0 + 2n'_2 + 5p, 0, 0, n'_3 - n'_2 - 5p = 0, 0) & (i, m) &= (0, n'_2 + 3p) \end{aligned}$$

Now our chosen vertex 0 is the only vertex with a non-zero value, and since s is invariant, that value is s as required. Since a vertex i with $r_i \equiv 0 \pmod{5}$ is winnable, and with $s = 2011 \not\equiv 0 \pmod{5}$ we always have a unique such vertex, we are done.

3. (*USAMO 2011*) In hexagon $ABCDEF$, which is nonconvex but not self-intersecting, no pair of opposite sides are parallel. The internal angles satisfy $\angle A = 3\angle D$, $\angle C = 3\angle F$, and $\angle E = 3\angle B$. Furthermore, $AB = DE$, $BC = EF$, and $CD = FA$. Prove that diagonals $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$ are concurrent.

Solution 1

Let $\angle D = \alpha$, $\angle F = \gamma$, and $\angle B = \beta$, $AB = DE = p$, $BC = EF = q$, $CD = FA = r$. Define the vectors:

$$\begin{aligned} \vec{u} &= \vec{AB} + \vec{DE} \\ \vec{v} &= \vec{BC} + \vec{EF} \\ \vec{w} &= \vec{CD} + \vec{FA} \end{aligned}$$

Clearly, $\vec{u} + \vec{v} + \vec{w} = \mathbf{0}$.

Let AB intersect DE at X . Note that $\angle X = 360^\circ - \angle D - \angle C - \angle B = 360^\circ - \alpha - 3\gamma - \beta = 180^\circ - 2\gamma$. Define the points M and N on lines AB and DE respectively so that $\vec{MX} = \vec{AB}$ and $\vec{XN} = \vec{DE}$. Then $\vec{u} = \vec{MN}$. As XMN is isosceles with

$XM = XN = p$, the base angles are both γ . Thus, $|\vec{u}| = 2p \cos \gamma$. Similarly, $|\vec{v}| = 2q \cos \alpha$ and $|\vec{w}| = 2r \cos \beta$.

Next we will find the angles between $\vec{u}$, $\vec{v}$, and $\vec{w}$. As $\angle MNX = \gamma$, the angle between the vectors $\vec{u}$ and $\vec{NE}$ is γ . Similarly, the angle between $\vec{EF}$ and $\vec{v}$ is α . Since the angle between $\vec{NE}$ and $\vec{EF}$ is $\angle E = 3\beta$, the angle between $\vec{u}$ and $\vec{v}$ is $360^\circ - \gamma - 3\beta - \alpha = 180^\circ - 2\beta$. Similarly, the angle between $\vec{v}$ and $\vec{w}$ is $180^\circ - 2\gamma$, and the angle between $\vec{w}$ and $\vec{u}$ is $180^\circ - 2\alpha$.

And since $\vec{u} + \vec{v} + \vec{w} = \vec{0}$, we can arrange the three vectors to form a triangle, so the triangle with sides of lengths $2p \cos \gamma$, $2q \cos \alpha$, and $2r \cos \beta$ has opposite angles of $180^\circ - 2\gamma$, $180^\circ - 2\alpha$, and $180^\circ - 2\beta$, respectively. So by the law of sines:

$$\frac{2p \cos \gamma}{\sin 2\gamma} = \frac{2q \cos \alpha}{\sin 2\alpha} = \frac{2r \cos \beta}{\sin 2\beta}$$

$$\frac{p}{\sin \gamma} = \frac{q}{\sin \alpha} = \frac{r}{\sin \beta},$$

and the triangle with sides of length p , q , and r has corresponding angles of γ , α , and β . It follows by SAS congruency that this triangle is congruent to FAB , BCD , and DEF , so $FD = p$, $BF = q$, and $BD = r$, and D , F , and B are the reflections of the vertices of triangle ACE about the sides. So AD , BE , and CF concur at the orthocenter of triangle ACE .

Solution 2

We work in the complex plane, where lowercase letters denote their corresponding point's position. Let P denote hexagon $ABCDEF$. Since $AB = DE$, the condition $AB \not\parallel DE$ is equivalent to $a - b + d - e \neq 0$.

Construct a "phantom hexagon" $P' = A'B'C'D'E'F'$ as follows: let $A'C'E'$ be a triangle with $\angle A'C'E' = \angle F$, $\angle C'E'A' = \angle B$, and $\angle E'A'C' = \angle D$ (this is possible since $\angle B + \angle D + \angle F = 180^\circ$ by the angle conditions), and reflect A' , C' , E' over its sides to get points D' , F' , B' , respectively. By rotation and reflection if necessary, we assume $A'B' \parallel AB$ and P' , P have the same orientation (clockwise or counterclockwise), i.e. $\frac{b-a}{b'-a'} \in \mathbb{R}^+$. It's easy to verify that $\angle X' = \angle X$ for $X \in \{A, B, C, D, E, F\}$ and opposite sides of P' have equal lengths. As the corresponding sides of P and P' must then be parallel, there exist positive reals r, s, t such that $r = \frac{a-b}{a'-b'} = \frac{d-e}{d'-e'}$, $s = \frac{b-c}{b'-c'} = \frac{e-f}{e'-f'}$, and $t = \frac{c-d}{c'-d'} = \frac{f-a}{f'-a'}$. But then $0 \neq a - b + d - e = r(a' - b' + d' - e')$, etc., so the non-parallel condition "transfers" directly from P to P' and

$$\begin{aligned} 0 &= (a - b + d - e) + (b - c + e - f) + (c - d + f - a) \\ &= r(a' - b' + d' - e') + s(b' - c' + e' - f') + t(c' - d' + f' - a') \\ &= (r - t)(a' - b' + d' - e') + (s - t)(b' - c' + e' - f'). \end{aligned}$$

If $r - t = s - t = 0$, then P must be similar to P' and the conclusion is obvious.

Otherwise, since $a' - b' + d' - e' \neq 0$ and $b' - c' + e' - f' \neq 0$, we must have $r - t \neq 0$ and $s - t \neq 0$. Now let $x = \frac{a'+d'}{2}$, $y = \frac{c'+f'}{2}$, $z = \frac{e'+b'}{2}$ be the feet of the altitudes in $\triangle A'C'E'$; by the non-parallel condition in P' , x, y, z are pairwise distinct. But $\frac{z-x}{z-y} = \frac{s-t}{r-t} \in \mathbb{R}$, whence x, y, z are three distinct collinear points, which is clearly impossible. (The points can only be collinear when $\triangle A'C'E'$ is a right triangle, but in this case two of x, y, z must coincide.)

Alternatively (for the previous paragraph), WLOG assume that $(A'C'E')$ is the unit circle, and use the fact that $b' = a' + c' - \frac{a'c'}{e'}$, etc. to get simple expressions for $a' - b' + d' - e'$ and $b' - c' + e' - f'$.

Solution 3

We work in the complex plane to give (essentially) a complete characterization when the parallel condition is relaxed.

WLOG assume a, b, c are on the unit circle. It suffices to show that a, b, c uniquely determine d, e, f , since we know that if we let E be the reflection of B over AC , D be the reflection of A over CE , and F be the reflection of C over AE , then $ABCDEF$ satisfies the problem conditions. (*)

It's easy to see with the given conditions that

$$\begin{aligned} (a-b)(c-d)(e-f) &= (b-c)(d-e)(f-a) \iff f = \frac{(a-b)(c-d)e + (c-b)(e-d)a}{(a-b)(c-d) + (c-b)(e-d)} \\ \frac{(e-a)(c-b)}{(a-b)(c-d) + (c-b)(e-d)} &= \frac{f-e}{d-e} = \left(\frac{c-b}{a-b}\right)^2 \overline{\left(\frac{a-b}{c-b}\right)} = \frac{c-b}{a-b} \cdot \frac{c}{a} \iff d = \frac{c[(a-b)c + (c-b)a]}{c[(a-b)c + (c-b)a]} \\ \frac{(a-b)(c-d) + (c-b)(e-d)}{(a-e)(c-d)} &= \frac{b-a}{f-a} = \left(\frac{e-d}{c-d}\right)^2 \overline{\left(\frac{c-d}{e-d}\right)}. \end{aligned}$$

Note that

$$\frac{e-d}{c-d} = \frac{(a-b)[c(e-c) + a(e-a)]}{c(c-e)(c-b) - a(a-e)(a-b)},$$

so plugging into the third equation we have

$$\begin{aligned} \frac{a(a-b)(2b-a-c)}{c(c-e)(c-b) - a(a-e)(a-b)} &= \frac{(a-b) + (c-b) \frac{(a-b)[c(e-c) + a(e-a)]}{c(c-e)(c-b) - a(a-e)(a-b)}}{(a-e)} \\ &= \left(\frac{(a-b)[c(e-c) + a(e-a)]}{c(c-e)(c-b) - a(a-e)(a-b)} \right)^2 \overline{\left(\frac{c(c-e)(c-b) - a(a-e)(a-b)}{(a-b)[c(e-c) + a(e-a)]} \right)} \\ &= \left(\frac{(a-b)[c(e-c) + a(e-a)]}{c(c-e)(c-b) - a(a-e)(a-b)} \right)^2 \frac{\frac{1}{c} \left(\bar{e} - \frac{1}{c} \right) \frac{b-c}{bc} - \frac{1}{a} \left(\bar{e} - \frac{1}{a} \right) \frac{b-a}{ba}}{\frac{b-a}{ab} \left(\frac{1}{c} \left(\frac{1}{c} - \bar{e} \right) + \frac{1}{a} \left(\frac{1}{a} - \bar{e} \right) \right)} \\ &= \left(\frac{(a-b)[c(e-c) + a(e-a)]}{c(c-e)(c-b) - a(a-e)(a-b)} \right)^2 \frac{c^3(a\bar{e} - 1)(a-b) - a^3(c\bar{e} - 1)(c-b)}{c(a-b)[a^2(c\bar{e} - 1) + c^2(a\bar{e} - 1)]} \\ &= \frac{(a-b)[c(e-c) + a(e-a)]^2}{[c(c-e)(c-b) - a(a-e)(a-b)]^2} \frac{c^3(a\bar{e} - 1)(a-b) - a^3(c\bar{e} - 1)(c-b)}{c[a^2(c\bar{e} - 1) + c^2(a\bar{e} - 1)]} \end{aligned}$$

Simplifying, this becomes

$$\begin{aligned} &ac(2b-a-c)[c(c-e)(c-b) - a(a-e)(a-b)][a^2(c\bar{e} - 1) + c^2(a\bar{e} - 1)] \\ &= [c(e-c) + a(e-a)]^2 [c^3(a\bar{e} - 1)(a-b) - a^3(c\bar{e} - 1)(c-b)]. \end{aligned}$$

Of course, we can also "conjugate" this equation – a nice way to do this is to note that if

$$x = \frac{(a-b)[c(e-c) + a(e-a)]}{c(c-e)(c-b) - a(a-e)(a-b)},$$

then

$$\frac{a(2b-a-c)}{c(e-c) + a(e-a)} = \frac{x}{\bar{x}} = \overline{\left(\frac{c(e-c) + a(e-a)}{a(2b-a-c)} \right)} = \frac{\frac{1}{c} \left(\bar{e} - \frac{1}{c} \right) + \frac{1}{a} \left(\bar{e} - \frac{1}{a} \right)}{\frac{1}{a} \left(\frac{2}{b} - \frac{1}{a} - \frac{1}{c} \right)},$$

whence

$$ac(2b - a - c)[2ac - b(a + c)] = b[c(e - c) + a(e - a)][a^2(c\bar{e} - 1) + c^2(a\bar{e} - 1)].$$

If $a + c \neq 0$, then eliminating $\bar{e}$, we get

$$e \in \left\{ a + c - \frac{ac}{b}, a + \frac{2c(c - b)}{a + c}, c + \frac{2a(a - b)}{a + c} \right\}.$$

The first case corresponds to (*) (since a, b, c, e uniquely determine d and f), the second corresponds to $AB \parallel DE$ (or equivalently, since $AB = DE$, $a - b = e - d$), and by symmetry, the third corresponds to $CB \parallel FE$.

Otherwise, if $c = -a$, then we easily find $b^2e = a^4\bar{e}$ from the first of the two equations in $e, \bar{e}$ (we actually don't need this, but it tells us that the locus of working e is a line through the origin). It's easy to compute $d = e + \frac{a(a-b)}{b}$ and $f = e + \frac{a(a+b)}{b}$, so $a - c = 2a = f - d \implies c - d = a - f \implies CD \parallel AF$, and we're done.

Comment. It appears that taking (ABC) the unit circle is nicer than, say $e = 0$ or (ACE) the unit circle (which may not even be reasonably tractable).

4. (*USAMO 2011*) Consider the assertion that for each positive integer $n \geq 2$, the remainder upon dividing 2^{2^n} by $2^n - 1$ is a power of 4. Either prove the assertion or find (with proof) a counterexample.

Solution

We will show that $n = 25$ is a counter-example.

Since $2^n \equiv 1 \pmod{2^n - 1}$, we see that for any integer k , $2^{2^n} \equiv 2^{(2^n - kn)} \pmod{2^n - 1}$. Let $0 \leq m < n$ be the residue of $2^n \pmod{n}$. Note that since $m < n$ and $n \geq 2$, necessarily $2^m < 2^n - 1$, and thus the remainder in question is 2^m . We want to show that $2^m \pmod{2^n - 1}$ is an odd power of 2 for some n , and thus not a power of 4.

Let $n = p^2$ for some odd prime p . Then $\varphi(p^2) = p^2 - p$. Since 2 is co-prime to p^2 , we have

$$2^{\varphi(p^2)} \equiv 1 \pmod{p^2}$$

and thus

$$2^{p^2} \equiv 2^{(p^2 - p) + p} \equiv 2^p \pmod{p^2}.$$

Therefore, for a counter-example, it suffices that $2^p \pmod{p^2}$ be odd. Choosing $p = 5$, we have $2^5 = 32 \equiv 7 \pmod{25}$. Therefore, $2^{25} \equiv 7 \pmod{25}$ and thus

$$2^{2^{25}} \equiv 2^7 \pmod{2^{25} - 1}.$$

Since 2^7 is not a power of 4, we are done.

Solution 2

Lemma (useful for all situations): If x and y are positive integers such that $2^x - 1$ divides $2^y - 1$, then x divides y . Proof: $2^y \equiv 1 \pmod{2^x - 1}$. Replacing the 1 with a 2^x and dividing out the powers of two should create an easy induction proof which will be left to the reader as an Exercise.

Consider $n = 25$. We will prove that this case is a counterexample via contradiction.

Because $4 = 2^2$, we will assume there exists a positive integer k such that $2^{2^n} - 2^{2k}$ divides $2^n - 1$ and $2^{2k} < 2^n - 1$. Dividing the powers of 2 from LHS gives $2^{2^n-2k} - 1$ divides $2^n - 1$. Hence, $2^n - 2k$ divides n . Because $n = 25$ is odd, $2^{24} - k$ divides 25. Euler's theorem gives $2^{24} \equiv 2^4 \equiv 16 \pmod{25}$ and so $k \geq 16$. However, $2^{2k} \geq 2^{32} > 2^{25} - 1$, a contradiction. Thus, $n = 25$ is a valid counterexample.

5. (USAMO 2011) Let P be a given point inside quadrilateral $ABCD$. Points Q_1 and Q_2 are located within $ABCD$ such that $\angle Q_1BC = \angle ABP$, $\angle Q_1CB = \angle DCP$, $\angle Q_2AD = \angle BAP$, $\angle Q_2DA = \angle CDP$. Prove that $\overline{Q_1Q_2} \parallel \overline{BA}$ if and only if $\overline{Q_1Q_2} \parallel \overline{CD}$.

Solution 1

First, if $\overline{AB} \parallel \overline{CD}$, then the proof is trivial. Assume $\overleftrightarrow{AB} \cap \overleftrightarrow{CD} = X$. We will show that $\overleftrightarrow{Q_1Q_2}$ passes through X , so $\overline{Q_1Q_2} \parallel \overline{AB}, \overline{CD}$. Note that Q_1 is the isogonal conjugate of P with respect to $\triangle BCX$. Similarly, Q_2 is the isogonal conjugate of P with respect to $\triangle ADX$. Thus, Q_1, Q_2, X lie on the reflection of XP over the angle bisector of $\angle X$, as desired.

-sillybone

-This proof is extremely similar to the one in Evan Chen's <https://web.evanchen.cc/exams/USAMO-2011-notes.pdf>. Please don't sue me.

Solution 2

Lemma. If AB and CD are not parallel, then AB, CD, Q_1Q_2 are concurrent.

Proof. Let AB and CD meet at R . Notice that with respect to triangle ADR , P and Q_2 are isogonal conjugates (this can be proven by dropping altitudes from Q_2 to AB, CD , and AD or BC depending on where R is). With respect to triangle BCR , P and Q_1 are isogonal conjugates. Therefore, Q_1 and Q_2 lie on the reflection of RP in the angle bisector of $\angle DRA$, so R, Q_1, Q_2 are collinear. Hence, AB, CD, Q_1Q_2 are concurrent at R .

Now suppose $Q_1Q_2 \parallel AB$ but Q_1Q_2 is not parallel to CD . Then AB and CD are not parallel and thus intersect at a point R . But then Q_1Q_2 also passes through R , contradicting $Q_1Q_2 \parallel AB$. A similar contradiction occurs if $Q_1Q_2 \parallel CD$ but Q_1Q_2 is not parallel to AB , so we can conclude that $Q_1Q_2 \parallel AB$ if and only if $Q_1Q_2 \parallel CD$.

Solution 3

First note that $\overline{Q_1Q_2} \parallel \overline{AB}$ if and only if the altitudes from Q_1 and Q_2 to $\overline{AB}$ are the same, or $|Q_1B| \sin \angle ABQ_1 = |Q_2A| \sin \angle BAQ_2$. Similarly $\overline{Q_1Q_2} \parallel \overline{CD}$ iff $|Q_1C| \sin \angle DCQ_1 = |Q_2D| \sin \angle CDQ_2$.

If we define $S = \frac{|Q_1B| \sin \angle ABQ_1}{|Q_2A| \sin \angle BAQ_2} \times \frac{|Q_2D| \sin \angle CDQ_2}{|Q_1C| \sin \angle DCQ_1}$, then we are done if we can show that $S=1$.

By the law of sines, $\frac{|Q_1B|}{|Q_1C|} = \frac{\sin \angle Q_1CB}{\sin \angle Q_1BC}$ and $\frac{|Q_2D|}{|Q_2A|} = \frac{\sin \angle Q_2AD}{\sin \angle Q_2DA}$.

So, $S = \frac{\sin \angle ABQ_1}{\sin \angle BAQ_2} \cdot \frac{\sin \angle CDQ_2}{\sin \angle DCQ_1} \cdot \frac{\sin \angle BCQ_1}{\sin \angle CBQ_1} \cdot \frac{\sin \angle DAQ_2}{\sin \angle ADQ_2}$

By the terms of the problem, $S = \frac{\sin \angle PBC}{\sin \angle PCB} \cdot \frac{\sin \angle PDA}{\sin \angle PAD} \cdot \frac{\sin \angle PCD}{\sin \angle PCB} \cdot \frac{\sin \angle PAB}{\sin \angle PDC}$. (If two subangles of an angle of the quadrilateral are equal, then their complements at that quadrilateral angle are equal as well.)

Rearranging yields $S = \frac{\sin \angle PBC}{\sin \angle PCB} \cdot \frac{\sin \angle PDA}{\sin \angle PAD} \cdot \frac{\sin \angle PCD}{\sin \angle PDC} \cdot \frac{\sin \angle PAB}{\sin \angle PBA}$.

Applying the law of sines to the triangles with vertices at P yields $S = \frac{|PC|}{|PB|} \frac{|PA|}{|PD|} \frac{|PD|}{|PC|} \frac{|PB|}{|PA|} = 1$.

Solution 4

Case 1 The lines AB and CD are not parallel. Denote $E = AB \cap CD$.

$$\angle Q_1 BC = \angle ABP, \angle Q_1 CB = \angle DCP \implies$$

Point Q_1 isogonal conjugate of a point P with respect to a triangle $\triangle EBC \implies EP$ and EQ_1 are isogonals with respect to $\angle BEC$.

Similarly point Q_2 isogonal conjugate of a point P with respect to a triangle $\triangle EAD \implies$

EP and EQ_2 are isogonals with respect to $\angle BEC$.

Therefore points E, Q_1, Q_2 lies on the isogonal EP with respect to $\angle BEC \implies Q_1 Q_2$ is not parallel to AB or CD .

Case 2 $AB \parallel CD$. We use **The isogonal theorem in case parallel lines** and get

$$AB \parallel Q_1 Q_2 \parallel CD. \blacksquare$$

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6. (USAMO 2011) Let A be a set with $|A| = 225$, meaning that A has 225 elements. Suppose further that there are eleven subsets $A_1, \dots, A_{11}$ of A such that $|A_i| = 45$ for $1 \leq i \leq 11$ and $|A_i \cap A_j| = 9$ for $1 \leq i < j \leq 11$. Prove that $|A_1 \cup A_2 \cup \dots \cup A_{11}| \geq 165$, and give an example for which equality holds.

Solution

Existence Note that $\binom{11}{3} = 165$, and so it is natural to consider placing one element in each intersection of three of the 11 sets. Since each pair of sets is in 9 3-way intersections—one with each of the 9 remaining sets—any two sets will have 9 elements in common. Since $\binom{10}{2} = 45$, each set is in 45 triples and thus will have 45 elements. We can now throw in 60 more elements outside the union of the A_i and we are done.

Minimality As in the proof of PIE, let I be a finite set. Let $U = \bigcup_{i \in I} A_i$. For $i \in I$, let χ_i be the *characteristic function* of A_i , that is, for $\alpha \in U$,

$$\chi_i(\alpha) = \begin{cases} 1 & \alpha \in A_i \\ 0 & \alpha \notin A_i \end{cases}$$

For each $\alpha \in U$ let $n_\alpha = \sum_{i \in I} \chi_i(\alpha)$, that is the number of subsets A_i of which α is an element.

If $S \subset I$, let $A_S = \bigcap_{i \in S} A_i$. Then the characteristic function of A_S is $\prod_{i \in S} \chi_i$. The number of elements of A_S is simply the sum of its characteristic function over all the elements of U :

$$|A_S| = \sum_{\alpha \in U} \prod_{i \in S} \chi_i(\alpha).$$

For $0 \leq k \leq |I|$, consider the sum N_k of $|A_S|$ over all $S \subset I$ with $|S| = k$. This is:

$$N_k = \sum_{\substack{S \subset I \\ |S|=k}} \sum_{\alpha \in U} \prod_{i \in S} \chi_i(\alpha) = \sum_{\substack{S \subset I \\ |S|=k}} |A_S|$$

reversing the order summation, as an element α that appears in n_α of the A_i , will appear in exactly $\binom{n_\alpha}{k}$ intersections of k subsets, we get:

$$\sum_{\substack{S \subset I \\ |S|=k}} |A_S| = \sum_{\alpha \in U} \sum_{\substack{S \subset I \\ |S|=k}} \prod_{i \in S} \chi_i(\alpha) = \sum_{\alpha \in U} \binom{n_\alpha}{k}.$$

Applying the above with $I = \{1, 2, \dots, 11\}$, and $k = 1$, since each of the 11 A_i has 45 elements, we get:

$$N_1 = \sum_{\alpha \in U} n_\alpha = 11 \cdot 45,$$

and for $k = 2$, since each of the 55 pairs $A_i \cap A_j$ has 9 elements, we get:

$$N_2 = \sum_{\alpha \in U} \binom{n_\alpha}{2} = 9 \cdot 55 = 11 \cdot 45.$$

Therefore

$$\begin{aligned} s_1 &= \sum_{\alpha \in U} n_\alpha = N_1 = 11 \cdot 45 \\ s_2 &= \sum_{\alpha \in U} n_\alpha^2 = 2N_2 + N_1 = 3 \cdot 11 \cdot 45. \end{aligned}$$

Let $n = |U|$ be the number of elements of U . Since for any set of real numbers the mean value of the squares is greater than or equal to the square of the mean value, we have:

$$\frac{s_2}{n} \geq \left(\frac{s_1}{n} \right)^2 \implies n \geq \frac{s_1^2}{s_2} = \frac{11^2 \cdot 45^2}{3 \cdot 11 \cdot 45} = \frac{11 \cdot 45}{3} = 165.$$

Thus $|U| \geq 165$ as required.

Solution 2

We will count the number of ordered triples, (A_i, A_j, x) , where $1 \leq i, j \leq 11$ and $x \in A_i \cap A_j$. We know this is equal to $990 = 11 \cdot 10 \cdot 9$. We can also find that this is $\sum_{k=1}^{225} b_k^2 - b_k$, where b_k is the number of the 11 subsets the k^{th} element of A is in. Since $\sum_{k=1}^{225} b_k = 45 \cdot 11 = 495$, we know $\sum_{k=1}^{225} b_k^2 = 990 + 495 = 1485$. Let $n = |A_1 \cup A_2 \cup \dots \cup A_{11}|$. n is equal to the number of $b_k > 0$, where $1 \leq k \leq 225$. By

the QM-AM inequality, we know $\frac{\sum_{k=1}^{225} b_k^2}{n} \geq \left(\frac{\sum_{k=1}^{225} b_k}{n} \right)^2 \implies \frac{1485}{n} \geq \left(\frac{495}{n} \right)^2 \implies n \geq 165$ and that equality occurs when $b_1 = b_2 = b_3 = \dots = b_{165} = 3, b_{166} = b_{167} = \dots = b_{225} = 0$. $\square$

Solution by randomdude10807

Solution 3

Define x_i to be the amount of elements in A such that the element is contained exactly i times among the 11 subsets A_j . We are trying to prove that $\sum_{i=1}^{11} x_i \geq 165$. Now, note that $\sum_{i=1}^{11} i x_i$ is equivalent to the total amount of not necessarily distinct elements among the subsets A_j , thus $\sum_{i=1}^{11} i x_i = 11|A_j| = 495$. Furthermore, note that $\sum_{i=1}^{11} \binom{i}{2} x_i$ is equivalent to the total amount of not necessarily distinct pairs of subsets such that the subsets share an element. In other words, each subset pair should be counted 9 times in this sum, and there are $\binom{11}{2}$ total distinct pairs of

subsets, so $\sum_{i=1}^{11} \binom{i}{2} x_i = 9 * \binom{11}{2} = 495$. Noting that $2\binom{i}{2} + i = i^2$ for all nonnegative integers, we note

$$\sum_{i=1}^{11} i^2 x_i = 2 \left(\sum_{i=1}^{11} \binom{i}{2} x_i \right) + \left(\sum_{i=1}^{11} i x_i \right) = 495 * 2 + 495 = 495 * 3$$

Finally, CS inequality gives us

$$\begin{aligned} \left(\sum_{i=1}^{11} i^2 x_i \right) \left(\sum_{i=1}^{11} x_i \right) &\geq \left(\sum_{i=1}^{11} i x_i \right)^2 \\ (495 * 3) \left(\sum_{i=1}^{11} x_i \right) &\geq (495^2) \end{aligned}$$

$$\boxed{\sum_{i=1}^{11} x_i \geq 165}$$

as desired.

One equality case occurs when $x_i = 0$ if i is not 3, and $x_3 = 165$. Choose any 165 elements, and represent each with a unique combination of 3 of the 11 subsets ($\binom{11}{3} = 165$). It is then obvious that each subset has 45 distinct elements, because after each subset is chosen, there are $\binom{10}{2} = 45$ ways to choose the other two subsets to produce a unique element. Similarly, each pair of subsets shares exactly 9 elements, because after choosing the two subsets, we can choose one of the remaining 9 subsets to produce the unique element. Hence, proved. ■

§41 USAMO 2012

1. (USAMO 2012) Find all integers $n \geq 3$ such that among any n positive real numbers $a_1, a_2, \dots, a_n$ with

$$\max(a_1, a_2, \dots, a_n) \leq n \cdot \min(a_1, a_2, \dots, a_n),$$

there exist three that are the side lengths of an acute triangle.

Solution

Without loss of generality, assume that the set $\{a\}$ is ordered from least to greatest so that the bounding condition becomes $a_n \leq n \cdot a_1$. Now set $b_i \equiv \frac{a_i}{a_1}$, and since a triangle with sidelengths from $\{a\}$ will be similar to the corresponding triangle from $\{b\}$, we simply have to show the existence of acute triangles in $\{b\}$. Note that $b_1 = 1$ and for all i , $b_i \leq n$.

Now three arbitrary sidelengths x, y , and z , with $x \leq y \leq z$, will form a valid triangle if and only if $x + y > z$. Furthermore, this triangle will be acute if and only if $x^2 + y^2 > z^2$. However, the first inequality can actually be inferred from the second, since $x + y > z \iff x^2 + y^2 + 2xy > z^2$ and $2xy$ is trivially greater than 0. So we just need to find all n such that there is necessarily a triplet of b 's for which $b_i^2 + b_j^2 > b_k^2$ (where $b_i < b_j < b_k$).

We now make another substitution: $c_i \equiv b_i^2$. So $c_1 = 1$ and for all i , $c_i \leq n^2$. Now we examine the smallest possible sets $\{c\}$ for small n for which the conditions of

the problem are not met. Note that by smallest, we mean the set whose greatest element is as small as possible. If $n = 3$, then the smallest possible set, call it $\{s_3\}$, is trivially $\{1, 1, 2\}$, since c_1 and c_2 are obviously minimized and c_3 follows as minimal. Using this as the base case, we see inductively that in general $\{s_n\}$ is the set of the first n Fibonacci numbers. To show this note that if $\{s_n\} = \{F_0, F_1, \dots, F_n\}$, then $\{s_{n+1}\} = \{F_0, F_1, \dots, F_n, c_{n+1}\}$. The smallest possible value for c_{n+1} is the sum of the two greatest values of $\{s_n\}$ which are F_{n-1} and F_n . But these sum to F_{n+1} so $\{s_{n+1}\} = \{F_0, F_1, \dots, F_{n+1}\}$ and our induction is complete.

Now since we know that the Fibonacci set is the smallest possible set which does not satisfy the conditions of the problem, then any set $\{c\}$ whose greatest term is less than F_{n-1} must satisfy the conditions. And since $\{c\}$ is bounded between 1 and n^2 , then the conditions of the problem are met if and only if $F_{n-1} > n^2$. The first n for which this restriction is satisfied is $n = 13$ and the exponential behavior of the Fibonacci numbers ensure that every n greater than 13 will also satisfy this restriction. So the final solution set is $\boxed{\{n \geq 13\}}$.

Solution 2

Outline:

1. Define the Fibonacci numbers to be $F_1 = F_2 = 1$ and $F_k = F_{k-1} + F_{k-2}$ for $k \geq 3$.
 2. If the chosen n is such that $F_n \leq n^2$, then choose the sequence a_n such that $a_k = \sqrt{F_k}$ for $1 \leq k \leq n$. It is easy to verify that such a sequence satisfies the condition that the largest term is less than or equal to n times the smallest term. Also, because for any three terms $x = \sqrt{F_a}, y = \sqrt{F_b}, z = \sqrt{F_c}$ with $a < b < c$, $x^2 + y^2 = F_a + F_b \leq F_{b-1} + F_b = F_{b+1} \leq F_c = z^2$, x, y, z do not form an acute triangle. Thus, all n such that $F_n \leq n^2$ do not work.
 3. It is easy to observe via a contradiction argument that all n such that $F_n > n^2$ produce an acute triangle. (If, without loss of generality, a_n is an increasing sequence, such that no three (in particular, consecutive) terms form an acute triangle, then $a_2^2 \geq F_1 a_1^2, a_3^2 \geq a_2^2 + a_1^2 \geq F_2 a^2$, and by induction $a_n^2 > F_n a_1^2$, a contradiction to the condition's inequality.)
 4. Note that $F_{12} = 144 = 12^2$ and $F_{13} = 233 > 169 = 13^2$. It is easily to verify through strong induction that all n greater than 12 make $F_n > n^2$. Thus, $\boxed{n \geq 13}$ is the desired solution set.
2. (USAMO 2012) A circle is divided into 432 congruent arcs by 432 points. The points are colored in four colors such that 108 points are colored red, 108 points are colored green, 108 points are colored blue and the remaining 108 points are colored yellow. Prove that one can choose three points of each color in such a way that the four triangles formed by the chosen points of the same color are congruent.

Solution

If you rotate the red points 431 times, they will overlap with blue points 108×108 times, for an average of $\frac{108 \times 108}{431}$ per rotation. Note that this average is slightly greater than 27. Therefore at some point 28 red points overlap with blue points. In other words, there exist 28 red and blue points such that the convex 28-gons formed by them are congruent.

Rotate these 28 red points 431 times. They will overlap with green points 108×28 times, for an average of $\frac{108 \times 28}{431}$ per rotation. This average is slightly greater than 7,

so at some point 8 of those red points overlap with green points. In other words, there exist 8 red points, 8 blue points, and 8 green points such that the convex octagons formed by them are congruent.

Rotate these 8 red points 431 times. They will overlap with yellow points 108×8 times, for an average of $\frac{108 \cdot 8}{431}$ per rotation. This average is slightly greater than 2, so at some point 3 of those red points will overlap with yellow points. In other words, there exist 3 red points, 3 blue points, 3 green points, and 3 yellow points such that the triangles formed by them are congruent.

Note Huge bash of Pigeonhole Principle:

$$108 \longrightarrow 28 \longrightarrow 8 \longrightarrow 3$$

$$\left\lceil \left\lceil \left\lceil 108 \cdot \frac{108}{431} \right\rceil \cdot \frac{108}{431} \right\rceil \cdot \frac{108}{431} \right\rceil = \left\lceil \left\lceil 28 \cdot \frac{108}{431} \right\rceil \cdot \frac{108}{431} \right\rceil = \left\lceil 8 \cdot \frac{108}{431} \right\rceil = 3$$

3. (*USAMO 2012*) Determine which integers $n > 1$ have the property that there exists an infinite sequence $a_1, a_2, a_3, \dots$ of nonzero integers such that the equality

$$a_k + 2a_{2k} + \dots + na_{nk} = 0$$

holds for every positive integer k .

Solution that involves a non-elementary result

(Since Bertrand's is well known and provable using elementary techniques, I see nothing wrong with this-tigershark22)

For $n = 2$, $|a_1| = 2|a_2| = \dots = 2^m|a_{2^m}|$ implies that for any positive integer m , $|a_1| \geq 2^m$, which is impossible.

We proceed to prove that the infinite sequence exists for all $n \geq 3$.

First, one notices that if we have $a_{xy} = a_x a_y$ for any integers x and y , then it is suffice to define all a_x for x prime, and one only needs to verify the equation (*)

$$a_1 + 2a_2 + \dots + na_n = 0$$

for the other equations will be automatically true.

To proceed with the construction, I need the following fact: for any positive integer $m > 2$, there exists a prime p such that $\frac{m}{2} < p \leq m$.

To prove this, I am going to use **Bertrand's Postulate** ([1]) without proof. The Theorem states that, for any integer $n > 1$, there exists a prime p such that $n < p \leq 2n - 1$. In other words, for any positive integer $m > 2$, if $m = 2n$ with $n > 1$, then there exists a prime p such that $\frac{m}{2} < p < m$, and if $m = 2n - 1$ with $n > 1$, then there exists a prime p such that $\frac{m+1}{2} < p \leq m$, both of which guarantees that for any integer $m > 2$, there exists a prime p such that $\frac{m}{2} < p \leq m$.

Go back to the problem. Suppose $n \geq 3$. Let the largest two primes not larger than n are P and Q , and that $n \geq P > Q$. By the fact stated above, one can conclude that $2P > n$, and that $4Q = 2(2Q) \geq 2P > n$. Let's construct a_n :

Let $a_1 = 1$. There will be three cases: (i) $Q > \frac{n}{2}$, (ii) $\frac{n}{2} \geq Q > \frac{n}{3}$, and (iii) $\frac{n}{3} \geq Q > \frac{n}{4}$.

Case (i): $2Q > n$. Let $a_x = 1$ for all prime numbers $x < Q$, and $a_{xy} = a_x a_y$, then (*) becomes:

$$Pa_P + Qa_Q = C_1$$

Case (ii): $2Q \leq n$ but $3Q > n$. In this case, let $a_2 = -1$, and $a_x = 1$ for all prime numbers $2 < x < Q$, and $a_{xy} = a_x a_y$, then (*) becomes:

$$Pa_P + Qa_Q - Qa_{2Q} = C_2$$

or

$$Pa_P - Qa_Q = C_2$$

Case (iii): $3Q \leq n$. In this case, let $a_2 = 3$, $a_3 = -2$, and $a_x = 1$ for all prime numbers $3 < x < Q$, and $a_{xy} = a_x a_y$, then (*) becomes:

$$Pa_P + Qa_Q + 3Qa_{2Q} - 2Qa_{3Q} = C_3$$

or

$$Pa_P + Qa_Q = C_3$$

In each case, by Bezout's Theorem, there exists non zero integers a_P and a_Q which satisfy the equation. For all other primes $p > P$, just let $a_p = 1$ (or any other non-zero integer).

This construction is correct because, for any $k > 1$,

$$a_k + 2a_{2k} + \cdots + na_{nk} = a_k(1 + 2a_2 + \cdots + na_n) = 0$$

Since Bertrand's Theorem is not elementary, we still need to wait for a better proof.

–**Lightest** 21:24, 2 May 2012 (EDT)

4. (*USAMO 2012*) Find all functions $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ (where $\mathbb{Z}^+$ is the set of positive integers) such that $f(n!) = f(n)!$ for all positive integers n and such that $m - n$ divides $f(m) - f(n)$ for all distinct positive integers m, n .

Solution 1

By the first condition we have $f(1) = f(1!) = f(1)!$ and $f(2) = f(2!) = f(2)!$, so $f(1) = 1$ or 2 and similarly for $f(2)$. By the second condition, we have

$$n \cdot n! = (n+1)! - n! \mid f(n+1)! - f(n)! \quad (1)$$

for all positive integers n .

Suppose that for some $n \geq 2$ we have $f(n) = 1$. We claim that $f(k) = 1$ for all $k \geq n$. Indeed, from Equation (1) we have $f(n+1)! \equiv 1 \pmod{n \cdot n!}$, and this is only possible if $f(n+1) = 1$; the claim follows by induction.

We now divide into cases:

Case 1: $f(1) = f(2) = 1$

This gives $f(n) = 1$ always from the previous claim, which is a solution.

Case 2: $f(1) = 2, f(2) = 1$

This implies $f(n) = 1$ for all $n \geq 2$, but this does not satisfy the initial conditions. Indeed, we would have

$$3 - 1 \mid f(3) - f(1)$$

and so $2 \mid -1$, a contradiction.

Case 3: $f(1) = 1, f(2) = 2$

We claim $f(n) = n$ always by induction. The base cases are $n = 1$ and $n = 2$. Fix $k > 1$ and suppose that $f(k) = k$. By Equation (1) we have that

$$f(k+1)! \equiv k! \pmod{k \cdot k!}.$$

This implies $f(k+1) < 2k$ (otherwise $f(k+1)! \equiv 0 \pmod{k \cdot k!}$). Also we have

$$(k+1) - 1 \mid f(k+1) - f(1)$$

so $f(k+1) \equiv 1 \pmod{k}$. This gives the solutions $f(k+1) = 1$ and $f(k+1) = k+1$. The first case is obviously impossible, so $f(k+1) = k+1$, as desired. By induction, $f(n) = n$ for all n . This also satisfies the requirements.

Case 4: $f(1) = f(2) = 2$

We claim $f(n) = 2$ by a similar induction. Again if $f(k) = 2$, then by (1) we have

$$f(k+1) \equiv 2 \pmod{k \cdot k!}$$

and so $f(k+1) < 2k$. Also note that

$$k+1-1 \mid f(k+1) - 2$$

and

$$k+1-2 \mid f(k+1) - 2$$

so $f(k+1) \equiv 2 \pmod{k(k-1)}$. Then the only possible solution is $f(k+1) = 2$. By induction, $f(n) = 2$ for all n , and this satisfies all requirements.

In summary, there are three solutions: $\boxed{f(n) = 1, f(n) = 2, f(n) = n}$.

Summary Three functions: $f(x) = x$ since $x! = x!$, $1! = 1$ and $2! = 2$, fixed points on the $x!$ function.

No elegant mod argument needed; $f(m) - f(n) = m - n$ so of course $m - n \mid f(m) - f(n)!$

Solution 2

Let $f(n) = c$ and notice $c! = c \implies (c-1)! = 1 \implies c = 1, 2$, hence the polynomials $f(x) = 1, 2$ work.

Now, we find all other functions which are nonconstant. Assume $f(x) = a_d x^d + \dots + a_0$. Then, $f(n!) = a_d (n!)^d + \dots + a_0$. Also, $(f(n))! = (f(n))(f(n)-1)(\dots)(1)$. Hence the equality

$$f(n!) = a_d (n!)^d + \dots + a_0 = (f(n))(f(n)-1)(\dots)(1)$$

holds. The leading coefficients satisfy $a_d = a_d^n \implies a_d = 1$.

However, the degree of the left side is d and the degree of the right is $d!$ so we have $d = d!$, and that only works if $1 = (d-1)!$, so $d = 1, 2$. If $d = 1$, it is easy to see the polynomial $f(x) = x$ works. If $d = 2$, then $(n!)^2 + b(n!) + c =$

$(n^2 + bn + c)(n^2 + bn + c - 1)(\dots)(1)$. Checking degree however we see that $2 \neq m$ where $m > 2$ is the degree of the RHS. Hence, a contradiction.

The only three polynomials are from the family

$$\boxed{f(x) = 1, f(x) = 2, f(x) = x} \quad \forall x$$

.

It easily follows that $m - n \mid f(m) - f(n)$ $\square$.

Remark:

One can assume that $f(n)$ is a polynomial here rather than an arbitrary function as

- the statement $f(n!) = (f(n))!$ is nested
- $f(x)$ has the property for all m, n , $m - n \mid f(m) - f(n)$.

The second condition is true for all polynomials, and the **second uniqueness lemma**, makes the first condition true. If $f(x)$ was not a polynomial, then it follows that both properties cannot hold simultaneously, particularly $m - n \mid f(m) - f(n)$ as I can construct a non-continuous function that satisfies rule 1 but breaks rule 2 or vice versa.

5. (USAMO 2012) Let P be a point in the plane of triangle ABC , and γ a line passing through P . Let A', B', C' be the points where the reflections of lines PA, PB, PC with respect to γ intersect lines BC, AC, AB , respectively. Prove that A', B', C' are collinear.

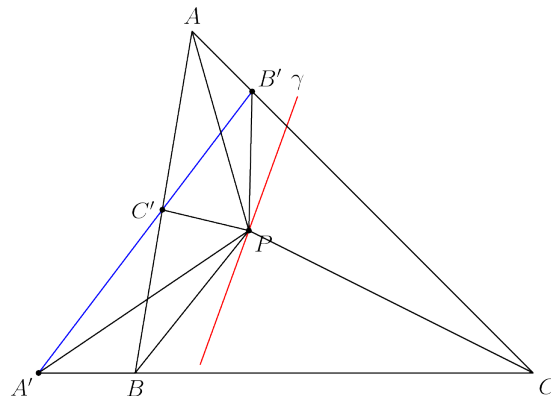
Solution

By the **sine law** on triangle $AB'P$,

$$\frac{AB'}{\sin \angle APB'} = \frac{AP}{\sin \angle AB'P},$$

so

$$AB' = AP \cdot \frac{\sin \angle APB'}{\sin \angle AB'P}.$$



Similarly,

$$B'C = CP \cdot \frac{\sin \angle CPB'}{\sin \angle CB'P},$$

$$\begin{aligned}
CA' &= CP \cdot \frac{\sin \angle CPA'}{\sin \angle CA'P}, \\
A'B &= BP \cdot \frac{\sin \angle BPA'}{\sin \angle BA'P}, \\
BC' &= BP \cdot \frac{\sin \angle BPC'}{\sin \angle BC'P}, \\
C'A &= AP \cdot \frac{\sin \angle APC'}{\sin \angle AC'P}.
\end{aligned}$$

Hence,

$$\begin{aligned}
&\frac{AB'}{B'C} \cdot \frac{CA'}{A'B} \cdot \frac{BC'}{C'A} \\
&= \frac{\sin \angle APB'}{\sin \angle AB'P} \cdot \frac{\sin \angle CB'P}{\sin \angle CPB'} \cdot \frac{\sin \angle CPA'}{\sin \angle CA'P} \cdot \frac{\sin \angle BA'P}{\sin \angle BPA'} \cdot \frac{\sin \angle BPC'}{\sin \angle BC'P} \cdot \frac{\sin \angle AC'P}{\sin \angle APC'}.
\end{aligned}$$

Since angles $\angle AB'P$ and $\angle CB'P$ are supplementary or equal, depending on the position of B' on AC ,

$$\sin \angle AB'P = \sin \angle CB'P.$$

Similarly,

$$\begin{aligned}
\sin \angle CA'P &= \sin \angle BA'P, \\
\sin \angle BC'P &= \sin \angle AC'P.
\end{aligned}$$

By the reflective property, $\angle APB'$ and $\angle BPA'$ are supplementary or equal, so

$$\sin \angle APB' = \sin \angle BPA'.$$

Similarly,

$$\begin{aligned}
\sin \angle CPA' &= \sin \angle APC', \\
\sin \angle BPC' &= \sin \angle CPB'.
\end{aligned}$$

Therefore,

$$\frac{AB'}{B'C} \cdot \frac{CA'}{A'B} \cdot \frac{BC'}{C'A} = 1,$$

so by **Menelaus's theorem**, A' , B' , and C' are collinear.

Solution 2, Barycentric (Modified by Evan Chen)

We will perform barycentric coordinates on the triangle PCC' , with $P = (1, 0, 0)$, $C' = (0, 1, 0)$, and $C = (0, 0, 1)$. Set $a = CC'$, $b = CP$, $c = C'P$ as usual. Since A , B , C' are collinear, we will define $A = (p : k : q)$ and $B = (p : \ell : q)$.

Claim: Line γ is the angle bisector of $\angle APA'$, $\angle BPB'$, and $\angle CPC'$. This is proved by observing that since $A'P$ is the reflection of AP across γ , etc.

Thus B' is the intersection of the isogonal of B with respect to $\angle P$ with the line CA ; that is,

$$B' = \left(\frac{p}{k} \frac{b^2}{\ell} : \frac{b^2}{\ell} : \frac{c^2}{q} \right).$$

Analogously, A' is the intersection of the isogonal of A with respect to $\angle P$ with the line CB ; that is,

$$A' = \left(\frac{p}{\ell} \frac{b^2}{k} : \frac{b^2}{k} : \frac{c^2}{q} \right).$$

The ratio of the first to third coordinate in these two points is both $b^2pq : c^2kl$, so it follows A' , B' , and C' are collinear.

Solution 3, Cartesian

We will use Cartesian coordinates. Without loss of generality, let $P = (0, 0)$ and let γ be the line $x = 0$. Let $A = (x_1, y_1)$, $B = (x_2, y_2)$, and $C = (x_3, y_3)$. Then A' is the intersection of the lines

$$y = -\frac{y_1}{x_1}x$$

$$y - y_2 = \frac{y_2 - y_3}{x_2 - x_3}(x - x_2).$$

Solving the system of equations, we see it is

$$\left(\frac{x_1(x_2y_3 - y_2x_3)}{x_1y_2 + y_1x_2 - x_1y_3 - x_3y_1}, \frac{y_1(y_2x_3 - x_2y_3)}{x_1y_2 + y_1x_2 - x_1y_3 - x_3y_1} \right).$$

To check collinearity, we need the determinant of the matrix

$$\begin{bmatrix} x_1(x_2y_3 - y_2x_3) & y_1(y_2x_3 - x_2y_3) & x_1y_2 + y_1x_2 - x_1y_3 - x_3y_1 \\ x_2(x_3y_1 - y_3x_1) & y_2(y_3x_1 - x_3y_1) & x_2y_3 + y_2x_3 - x_2y_1 - x_1y_2 \\ x_3(x_1y_2 - y_1x_2) & y_3(y_1x_2 - x_1y_2) & x_3y_1 + y_3x_1 - x_3y_2 - x_2y_3 \end{bmatrix}$$

to be zero, which is true because the columns sum to zero.

6. (USAMO 2012) For integer $n \geq 2$, let $x_1, x_2, \dots, x_n$ be real numbers satisfying

$$x_1 + x_2 + \dots + x_n = 0, \quad \text{and} \quad x_1^2 + x_2^2 + \dots + x_n^2 = 1.$$

For each subset $A \subseteq \{1, 2, \dots, n\}$, define

$$S_A = \sum_{i \in A} x_i.$$

(If A is the empty set, then $S_A = 0$.)

Prove that for any positive number λ , the number of sets A satisfying $S_A \geq \lambda$ is at most $2^{n-3}/\lambda^2$. For what choices of $x_1, x_2, \dots, x_n$, λ does equality hold?

Solution 1

For convenience, let $N = \{1, 2, \dots, n\}$.

Note that $2 \sum_{1 \leq i < j \leq n} x_i x_j = (\sum_{i=1}^n x_i)^2 - (\sum_{i=1}^n x_i^2) = -1$, so the sum of the x_i taken two at a time is $-1/2$. Now consider the following sum:

$$\sum_{A \subseteq N} S_A^2 = 2^{n-1} \left(\sum_{j=1}^n x_j^2 \right) + 2^{n-1} \left(\sum_{1 \leq i < j \leq n} x_i x_j \right) = 2^{n-2}.$$

Since $S_A^2 \geq 0$, it follows that at most $2^{n-2}/\lambda^2$ sets $A \subseteq N$ have $|S_A| \geq \lambda$.

Now note that $S_A + S_{N/A} = 0$. It follows that at most half of the S_A such that $|S_A| \geq \lambda$ are positive. This shows that at most $2^{n-3}/\lambda^2$ sets $A \subseteq N$ satisfy $S_A \geq \lambda$.

Note that if equality holds, every subset A of N has $S_A \in \{-\lambda, 0, \lambda\}$. It immediately follows that $(x_1, x_2, \dots, x_n)$ is a permutation of $(\lambda, -\lambda, 0, 0, \dots, 0)$. Since we know that $\sum_{i=1}^n x_i^2 = 1$, we have that $\lambda = 1/\sqrt{2}$.

Solution 2

Let $x_i = x_1, x_2, \dots, x_n$. It is evident that $x_i = \frac{(-1)^i}{\sqrt{n}}$ for evens because of the second equation and $x_i = \frac{(-1)^i}{\sqrt{n-1}}$ for odds (one term will be 0 to maintain the first condition). We may then try and get an expression for the maximum number of sets that satisfy this which occur when $\lambda = \frac{1}{\sqrt{n}}$ since it will be

$$x_1 + x_2 + \dots + x_n$$

for any choice of A we pick, it will have to be greater than $\frac{1}{\sqrt{n}}$ which means we can either pick 0 negative x_m or $j-1$ negatives for j positive terms. Since we also have that there are $\frac{n}{2}$ positive and negative terms for evens. Which then gives us:

$$\sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \sum_{i=0}^{k-1} \binom{\frac{n}{2}}{i}$$

and

$$\sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \left(2^{\frac{n}{2}} - \sum_{i=k}^{\frac{n}{2}} \binom{\frac{n}{2}}{i} \right)$$

For odd values, let it be the same as the last even valued sequence where n is even (i.e. the same as the sequence before it but with an extra 0 in one of the spots). Then, the following is apparent:

$$\sum_{k=1}^{\lfloor \frac{n}{2} \rfloor} \binom{\lfloor \frac{n}{2} \rfloor}{k} \left(2^{\lfloor \frac{n}{2} \rfloor} - \sum_{i=k}^{\lfloor \frac{n}{2} \rfloor} \binom{\lfloor \frac{n}{2} \rfloor}{i} \right) \leq n2^{n-3}$$

Thus, we may say that this holds to be true for all $n \geq 2$ since $n2^{n-3}$ grows faster than the sum. Note that equality holds when $S_A \in \{\lambda, 0, -\lambda\}$ for all i which occurs when $x_i = \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, \dots$ and $\lambda = \frac{1}{\sqrt{2}}$ since $S_A = \frac{1}{\sqrt{2}}$ is the only choice for $S_A \geq \lambda$ ■

Note: The proof to the last inequality is as follows: First we may rewrite this as being:

$$\sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \left(2^{\frac{n}{2}} - \sum_{i=k}^{\frac{n}{2}} \binom{\frac{n}{2}}{i} \right) \leq n2^{n-3}$$

Thus,

$$2^n - 2^{\frac{n}{2}} - \sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \sum_{i=k}^{\frac{n}{2}} \binom{\frac{n}{2}}{i} \leq n2^{n-3}$$

(the second equation is because the sum of the binomial coefficient is $2^{\frac{n}{2}} - 1$)

$$2^n - 2^{\frac{n}{2}} \leq n2^{n-3} + \sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \sum_{i=k}^{\frac{n}{2}} \binom{\frac{n}{2}}{i}$$

Since $2^n - 2^{\frac{n}{2}} \leq n2^{n-3}$ for all $n \geq 8$ and $\sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \sum_{i=k}^{\frac{n}{2}} \binom{\frac{n}{2}}{i} \geq 0$ for all k and i , it is apparent that:

$$\sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \sum_{i=0}^{k-1} \binom{\frac{n}{2}}{i} \leq n2^{n-3}$$

must be true for all $n \geq 8$ (because if we rewrite this we get $2^{\frac{n}{2}}(8-n) \leq 8$.) For all $2 \leq n < 8$ however, we may use some logic to first layout a plan. Since for $n = 6, n = 4$, and $n = 2$, $2^n - 2^{\frac{n}{2}} = n2^{n-3} + 1$, we may say that whole sum will be less than $n2^{n-3}$ because $\sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \sum_{i=k}^{\frac{n}{2}} \binom{\frac{n}{2}}{i} \geq 1$ for all $n \geq 2$ Plugging this inequality back in gives us:

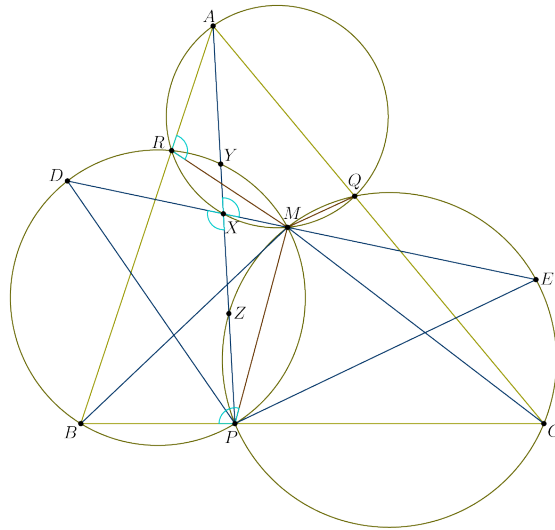
$$2^n - 2^{\frac{n}{2}} - \sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \sum_{i=k}^{\frac{n}{2}} \binom{\frac{n}{2}}{i} \leq 2^n - 2^{\frac{n}{2}} - 1 = n2^{n-3}$$

because of the fact that $-\sum_{k=1}^{\frac{n}{2}} \binom{\frac{n}{2}}{k} \sum_{i=k}^{\frac{n}{2}} \binom{\frac{n}{2}}{i} \leq -1$

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- (USAMO 2013) In triangle ABC , points P, Q, R lie on sides BC, CA, AB respectively. Let $\omega_A, \omega_B, \omega_C$ denote the circumcircles of triangles AQR, BRP, CPQ , respectively. Given the fact that segment AP intersects $\omega_A, \omega_B, \omega_C$ again at X, Y, Z respectively, prove that $YX/XZ = BP/PC$.

Solution 1



In this solution, all lengths and angles are directed.

Firstly, it is easy to see by that $\omega_A, \omega_B, \omega_C$ concur at a point M (the Miquel point). Let XM meet ω_B, ω_C again at D and E , respectively. Then by Power of a Point, we have

$$XM \cdot XE = XZ \cdot XP \quad \text{and} \quad XM \cdot XD = XY \cdot XP$$

Thusly

$$\frac{XY}{XZ} = \frac{XD}{XE}$$

But we claim that $\triangle XDP \sim \triangle PBM$. Indeed,

$$\angle XDP = \angle MDP = \angle MBP = -\angle PBM$$

and

$$\angle DXP = \angle MXY = \angle MXA = \angle MRA = 180^\circ - \angle MRB = \angle MPB = -\angle BPM$$

Therefore, $\frac{XD}{XP} = \frac{PB}{PM}$. Analogously we find that $\frac{XE}{XP} = \frac{PC}{PM}$ and we are done.
courtesy v__enhance, minor clarification by integralarefun

Solution 2

Diagram Refer to the Diagram link.

By Miquel's Theorem, there exists a point at which $\omega_A, \omega_B, \omega_C$ intersect. We denote this point by M . Now, we angle chase:

$$\begin{aligned}\angle YMX &= 180^\circ - \angle YXM - \angle XYM \\ &= 180^\circ - \angle AXM - \angle PYM \\ &= (180^\circ - \angle ARM) - \angle PRM \\ &= \angle BRM - \angle PRM \\ &= \angle BRP = \angle BMP.\end{aligned}$$

In addition, we have

$$\begin{aligned}\angle ZMX &= 180^\circ - \angle MZY - \angle ZYM - \angle YMX \\ &= 180^\circ - \angle MZP - \angle PYM - \angle BMP \\ &= 180^\circ - \angle MCP - \angle PBM - \angle BMP \\ &= (180^\circ - \angle PBM - \angle BMP) - \angle MCP \\ &= \angle BPM - \angle MCP \\ &= 180^\circ - \angle MPC - \angle MCP \\ &= \angle CMP.\end{aligned}$$

Now, by the Ratio Lemma, we have

$$\begin{aligned}\frac{XY}{XZ} &= \frac{MY}{MZ} \cdot \frac{\sin \angle YMX}{\sin \angle ZMX} \\ &= \frac{\sin \angle YZM}{\sin \angle ZYM} \cdot \frac{\sin \angle BMP}{\sin \angle CMP}\end{aligned}$$

(by the Law of Sines in $\triangle MZY$)

$$\begin{aligned}&= \frac{\sin \angle PZM}{\sin \angle PYM} \cdot \frac{\sin \angle BMP}{\sin \angle CMP} \\ &= \frac{\sin \angle PCM}{\sin \angle PBM} \cdot \frac{\sin \angle BMP}{\sin \angle CMP} \\ &= \frac{MB}{MC} \cdot \frac{\sin \angle BMP}{\sin \angle CMP}\end{aligned}$$

(by the Law of Sines in $\triangle MBC$)

$$= \frac{PB}{PC}$$

by the Ratio Lemma. The proof is complete.

Solution 3

Use directed angles modulo π .

Lemma. $\angle XRY \equiv \angle XQZ$.

Proof.

$$\angle XRY \equiv \angle XRA - \angle YRA \equiv \angle XQA + \angle YRB \equiv \angle XQA + \angle CPY = \angle XQA + \angle AQZ = \angle XQZ.$$

Now, it follows that (now not using directed angles)

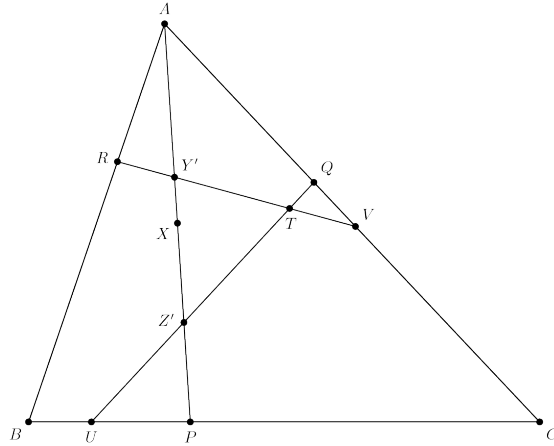
$$\frac{XY}{YZ} = \frac{\frac{XY}{\sin \angle XRY}}{\frac{YZ}{\sin \angle XQZ}} = \frac{\frac{RY}{\sin \angle RXY}}{\frac{QZ}{\sin \angle QXZ}} = \frac{BP}{PC}$$

using the facts that ARY and APB , AQZ and APC are similar triangles, and that $\frac{RA}{\sin \angle RXA} = \frac{QA}{\sin \angle QXA}$ equals twice the circumradius of the circumcircle of AQR .

Solution 4

We will use some construction arguments to solve the problem. Let $\angle BAC = \alpha$, $\angle ABC = \beta$, $\angle ACB = \gamma$, and let $\angle APB = \theta$. We construct lines through the points Q , and R that intersect with $\triangle ABC$ at the points Q and R , respectively, and that intersect each other at T . We will construct these lines such that $\angle CQV = \angle ARV = \theta$.

Now we let the intersections of AP with RV and QU be Y' and Z' , respectively. This construction is as follows.



We know that $\angle BRY' = 180^\circ - \angle ARY' = 180^\circ - \theta$. Hence, we have,

$$\begin{aligned} \angle BRY' + \angle BPY' &= 180^\circ - \theta + \theta \\ &= 180^\circ. \end{aligned}$$

Since the opposite angles of quadrilateral $RY'PB$ add up to 180° , it must be cyclic. Similarly, we can also show that quadrilaterals $CQZ'P$, and $AQTR$ are also cyclic. Since points Y' and Z' lie on AP , we know that,

$$Y' = \omega_B \cap AP$$

and that

$$Z' = \omega_C \cap AP.$$

Hence, the points Y' and Z' coincide with the given points Y and Z , respectively. Since quadrilateral $AQTR$ is also cyclic, we have,

$$\angle Y'TZ' = 180^\circ - \angle RTQ$$

$$\begin{aligned}
&= 180^\circ - (180^\circ - \angle RAQ) \\
&= \angle RAQ \\
&= \alpha.
\end{aligned}$$

Similarly, since quadrilaterals $CQZ'P$, and $AQTR$ are also cyclic, we have,

$$\begin{aligned}
\angle TY'Z' &= 180^\circ - \angle RY'P \\
&= 180^\circ - (180^\circ - \angle RBP) \\
&= \angle RBP \\
&= \beta,
\end{aligned}$$

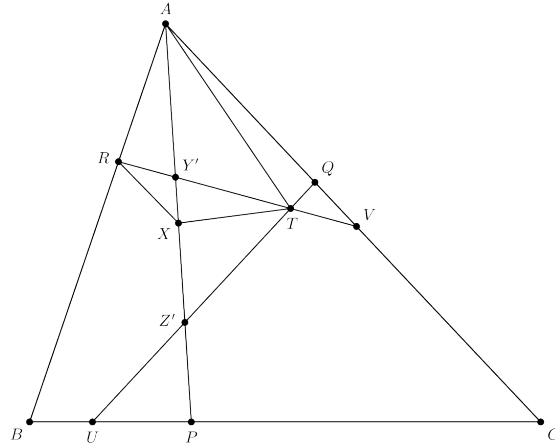
and,

$$\begin{aligned}
\angle Y'Z'T &= 180^\circ - \angle PZ'Q \\
&= 180^\circ - (180^\circ - \angle PCQ) \\
&= \angle PCQ \\
&= \gamma.
\end{aligned}$$

Since these three angles are of $\triangle TY'Z'$, and they are equal to corresponding angles of $\triangle ABC$, by AA similarity, we know that $\triangle TY'Z' \sim \triangle ABC$.

We now consider the point $X = \omega_c \cap AC$. We know that the points A , Q , T , and R are concyclic. Hence, the points A , T , X , and R must also be concyclic.

Hence, quadrilateral $AQTX$ is cyclic.



Since the angles $\angle ART$ and $\angle AXT$ are inscribed in the same arc $\widehat{AT}$, we have,

$$\begin{aligned}
\angle AXT &= \angle ART \\
&= \theta.
\end{aligned}$$

Consider by this result, we can deduce that the homothety that maps ABC to $TY'Z'$ will map P to X . Hence, we have that,

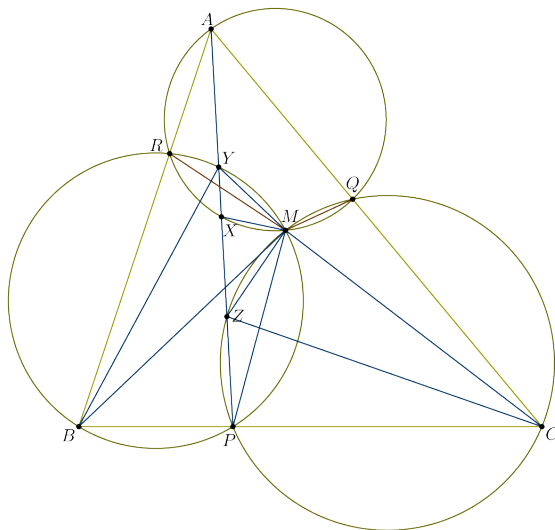
$$Y'X/XZ' = BP/PC.$$

Since $Y' = Y$ and $Z' = Z$ hence,

$$YX/XZ = BP/PC,$$

as required.

Solution 5 (Simple Rotational Homotethy)



We begin again by noting that the three circumcircles intersect at point M by Miquel's theorem. In addition, we state that the angle $\angle MQC = \alpha$, hence $\angle MPC = \angle MZC = 180 - \alpha$, as well as $\angle AQM$, from which follows that $\angle ARM = \alpha$, so $\angle BRM = 180 - \alpha$, and $\angle BPM = \alpha$. We shall prove that the triangles $\triangle MZC$ and $\triangle XMP$ and $\triangle YMB$ are similar, which will imply a rotational homotethy with angle $\angle APB$ about the point M , that takes Y, X, Z to B, P, C , thus proving the problem. (In essence, just imagine we rotate $YMZX$ around M and lengthen things out and get $MCPB$ - the ratios will remain identical.)

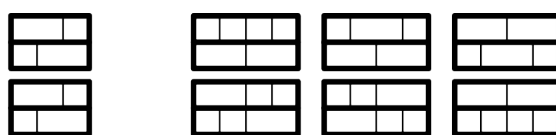
We do this by angle chasing. Denote angle $\angle ZMP = \beta$. From the angles labeled before, we now know $\angle ZPB = \alpha - \beta = \angle ZMC$. In addition, $\angle ZPM = \angle ZCM$. So the angles in triangle $\triangle ZMC$ are $\alpha - \beta$, β and thus $180 - \alpha$, with $\alpha - \beta$ at the point M . In addition, $\angle YRM = \angle YPM = \beta$, so $\angle ARY = \alpha - \beta$, so $\angle YMB = \alpha - \beta$. Since $\angle YBM = \angle YPM = \beta$, the triangle $\triangle YMB$ has angles $\alpha - \beta, \beta, 180 - \alpha$ also, with $\alpha - \beta$ at M . Finally, In triangle $\triangle XPM$, we already know the angle $\angle XPM$ to be β ; we also can find that $\angle AQM = 180 - \alpha$, so $\angle AXM = \alpha$, so $\angle MXP = 180 - \alpha$, so $\angle XMP = \alpha - \beta$. Thus, the three triangles are similar have a common point M , which proves that there is a rotational homotethy around M that maps Y, X, Z to B, P, C as desired.

Solution by SimilarTriangle.

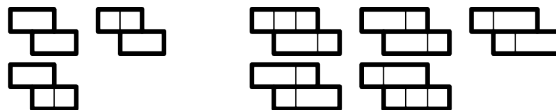
2. (USAMO 2013) For a positive integer $n \geq 3$ plot n equally spaced points around a circle. Label one of them A , and place a marker at A . One may move the marker forward in a clockwise direction to either the next point or the point after that. Hence there are a total of $2n$ distinct moves available; two from each point. Let a_n count the number of ways to advance around the circle exactly twice, beginning and ending at A , without repeating a move. Prove that $a_{n-1} + a_n = 2^n$ for all $n \geq 4$.

Solution 1

We label the points in clockwise order as $1, 2, 3, \dots, n$, where point A is the same as point 1. We start and end at point 1, and we must cross over it, either by visiting it again, or else by making the move from point n to point 2. We interpret each of these cases in terms of tiling. In each move, we either move one or two points clockwise, so we can think of each move as a 1×1 or 1×2 tile. If the point 1 is visited in the middle, then the first cycle around the circle can be thought of as a tiling of a $1 \times n$ board, and the second cycle around the circle can also be thought of as a $1 \times n$ board. We place this second board directly below the first board. Therefore, in this first case, we wish to find the number of tilings of two $1 \times n$ boards, and to guarantee that no move is repeated, we cannot have two tiles of the same type lying directly atop each other in the $2 \times n$ board. Suppose there are u_n such tilings. It can easily be computed that $u_3 = 2$ and $u_4 = 6$ as shown below.



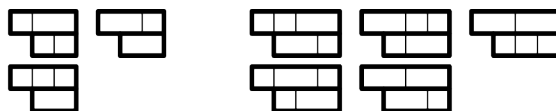
In the second case, where we pass by point 1 by moving from point n to point 2, we can similarly think about it in terms of tiling two rows of $1 \times n$ boards, but we remove the last square in the first row and the first square in the second row to make sure that we jump from point n to point 2. Suppose that we can tile such boards in s_n ways. It can be easily computed that $s_3 = 3$ and $s_4 = 5$ as shown below.



Since these are the only two possible cases, we see that

$$a_n = u_n + s_n. \quad (1)$$

For the sake of convenience in determining recurrence relations, we define another type of board with two $1 \times n$ boards where a specific corner is removed (without loss of generality, we place this in the lower left hand corner). Let t_n be the number of ways to tile such a board without placing two of the same type of tile atop each other. Once again, we compute that $t_3 = 3$ and $t_4 = 5$ as shown below.



We can determine recurrence relations for s_n , t_n , and u_n in terms of each other. For s_n , note that a tiling can end in one of the three following ways such that the rest of the board can be tiled without restriction (the placed tiles are shaded).



In the first, second, and third cases, we see that we can tile the rest of the board in t_{n-1} , t_{n-2} , and s_{n-2} ways, respectively. Hence for $n \geq 5$, we see that

$$s_n = t_{n-1} + t_{n-2} + s_{n-2}. \quad (2)$$

For t_n , note that a tiling can end in one of the three following ways such that the rest of the board can be tiled without restriction.



In the first, second, and third cases, we see that we can tile the rest of the board in s_{n-1} , s_{n-2} , and t_{n-2} ways, respectively. Hence for $n \geq 5$, we see that

$$t_n = s_{n-1} + s_{n-2} + t_{n-2}. \quad (3)$$

For u_n , note that a tiling can end in one of the two following ways such that the rest of the board can be tiled without restriction.



In either case, we are left with a board with a corner removed, hence we can tile the rest of the board in t_{n-1} ways in each case. Hence for $n \geq 4$, we see that

$$u_n = 2t_{n-1}. \quad (4)$$

Subtracting (3) from (2), we find that

$$s_n - t_n = s_{n-1} - t_{n-1}.$$

Therefore, if $s_{n-1} = t_{n-1}$, then $s_n = t_n$. Since $s_3 = t_3$ and $s_4 = t_4$, we see that $s_n = t_n$ for all $n \geq 3$. Therefore, (2) can be rewritten as

$$s_n = s_{n-1} + 2s_{n-2}, \quad (5)$$

and (4) can be rewritten as

$$u_n = 2s_{n-1}. \quad (6)$$

Now by (1), we know that

$$a_n + a_{n-1} = u_n + s_n + u_{n-1} + s_{n-1}. \quad (7)$$

In particular, $a_4 + a_3 = 6 + 5 + 2 + 3 = 2^4$, so the statement is true for $n = 4$. Then by (7), and then substituting (5) and (6) (where these are valid for $n \geq 5$), we find

$$\begin{aligned} a_n + a_{n-1} &= u_n + s_n + u_{n-1} + s_{n-1} \\ &= 2s_{n-1} + (s_{n-1} + 2s_{n-2}) + u_{n-1} + s_{n-1} \\ &= 4s_{n-1} + 2s_{n-2} + u_{n-1}. \end{aligned} \quad (8)$$

But then by (5), and then substituting $4s_{n-3} = 2u_{n-2}$ and $2s_{n-2} = u_{n-1}$ (by (6), and these are valid for $n \geq 6$), we find

$$2s_{n-1} = 2s_{n-2} + 4s_{n-3}$$

$$= u_{n-1} + 2u_{n-2}.$$

We substitute this into (8), finding that for $n \geq 6$,

$$\begin{aligned} a_n + a_{n-1} &= 2s_{n-1} + 2s_{n-2} + u_{n-1} + (u_{n-1} + 2u_{n-2}) \\ &= 2(s_{n-1} + s_{n-2} + u_{n-1} + u_{n-2}) \\ &= 2(a_{n-1} + a_{n-2}). \end{aligned} \tag{9}$$

Therefore, if $a_{n-1} + a_{n-2} = 2^{n-1}$, then $a_n + a_{n-1} = 2^n$. We already know that $a_4 + a_3 = 2^4$. Also, we can compute that

$$\begin{aligned} s_5 &= s_4 + 2s_3 = 5 + 2 \cdot 3 = 11 \\ u_5 &= 2s_4 = 2 \cdot 5 = 10. \end{aligned}$$

Hence $a_5 = s_5 + u_5 = 21$. So as $a_4 = s_4 + u_4 = 11$, we find that $a_5 + a_4 = 2^5$. Then we can use (9) for $n \geq 6$ to find by induction that $a_n + a_{n-1} = 2^n$ for all $n \geq 4$.

Solution 2

We choose the tilings for the first and second pass around the circle simultaneously. As we tile both passes, our state is defined by whether or not we are in a continuation of a length 2 block on the first and second pass. We can't be in a continuation of a length 2 block on both passes, since that would mean we used the same move on the same block on both passes. Thus we have three states: (non-continuation, non-continuation), (non-continuation, continuation), and (continuation, non-continuation), where the elements of each tuple refer to the first and second pass respectively. Listing out the state transitions (while following the rule that we can't use the same move on both passes, and that a continuation must

become a non-continuation), we get the transition matrix $T = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = J - I$,

where J is the ones matrix, and I is the identity. We have that $a_n = (T^n)_{11} + (T^n)_{23}$. The first term represents the case where we end the first pass at point A , and the second term represents the case where we jump over point A when finishing the first pass. Thus it simply remains to compute T^n . We have

$$\begin{aligned} T^n &= (J - I)^n \\ &= \sum_{k=0}^n J^k (-1)^{n-k} \binom{n}{k} \\ &= J \sum_{k=1}^n 3^{k-1} (-1)^{n-k} \binom{n}{k} + (-1)^n I \\ &= \frac{1}{3} J (2^n - (-1)^n) + (-1)^n I \end{aligned}$$

Thus $a_n = (T^n)_{11} + (T^n)_{23} = \frac{2}{3} (2^n - (-1)^n) + (-1)^n$. Thus, $a_{n-1} + a_n = 2^n$, as desired.

Solution 3

We split this into two cases. Case 1: The marker lands on A thrice. This implies that A is touched after one full cycle. Place any nodes not touched in the first

cycle in the set K . Lemma 1: Any unique configuration of K such that [Work In Progress] SigmaPiE

3. (*USAMO 2013*) Let n be a positive integer. There are $\frac{n(n+1)}{2}$ marks, each with a black side and a white side, arranged into an equilateral triangle, with the biggest row containing n marks. Initially, each mark has the black side up. An *operation* is to choose a line parallel to the sides of the triangle, and flipping all the marks on that line. A configuration is called *admissible* if it can be obtained from the initial configuration by performing a finite number of operations. For each admissible configuration C , let $f(C)$ denote the smallest number of operations required to obtain C from the initial configuration. Find the maximum value of $f(C)$, where C varies over all admissible configurations.

Solution

Solution by inconsistent

This problem is headache-inducing... but also fun. Ouch.

Let $n = 4k + b$ where $b \in 1, 2, 3, 4$. Then I claim the answer is $6k + 1, 6k + 2, 6k + 3, 6k + 6$ in these four cases. First, note that the base cases $n = 1, 2, 3$ are trivial by checking to be $1, 2, 3$. Now note the case $n = 4$ needs at least 6 to construct a black triangle with a single white mark at its center by checking. Call this size 4 triangle the evil triangle.

To show the lower bound, notice we can take the top n rows of a $n + 4$ equilateral triangle to be the limiting case for n and put two evil triangles on the bottom left and bottom right, respectively: these force 4 new rows to be occupied since each evil subcase requires at least two rows in every direction, which is a claim easy verified.

To show the upper bound, we reparameterize the triangles in terms of their rows. Notably, we can start counting rows from each corner, and give each mark a barycentric coordinate (a, b, c) where a, b, c are integers denoting the i th row in a given direction the point lies in, and this satisfies $a + b + c = 2n + 1$ by Viviani's theorem.

Now, consider the configuration C that minimizes $f(C)$. We place the rows in boxes: odd rows of each direction, and even rows of each direction, to make six boxes total. Call the numbers of odd rows a, b, c and the even rows d, e, f such that (a, b, c) and (d, e, f) correspond to the same directions respectively in that order. Notice no row is operated on twice since row operations are commutative. Assume for sake of contradiction our claim is wrong, then we can construct a solution for the case one higher than our claim.

We now perform casework: for $b = 4$, it follows one of the quantities $a + b + d + e, b + c + e + f, c + a + f + d$ is greater than n . In this case, invert the rows chosen in those four boxes (i.e. remove the ones chosen and choose the ones unchosen). This fixes the configuration and decreases $f(C)$, giving contradiction.

Now, consider the case $b = 1$. Like the above paragraph, we know $a + d, b + e, c + f$ are equal to $2k, 2k + 1, 2k + 1$ in some order since if any pair sums to more than $4k + 1$ then we reach a contradiction. But then $2k + 1 + 2k + 1 > 4k + 2$, giving contradiction.

Now, for the two remaining cases, we show a lemma: the xor of all the odd rows in two directions and all the even row in a third direction, or the sum of all the

even rows in all directions are both 0. This result follows immediately from our earlier result via Vivani's theorem (it cannot be that all three claims are true, and it cannot be the case that two of the claims are false and one is true by parity).

Thus in the $b = 2$ case it follows that $\min(a+b+f, b+c+d, c+a+e, d+e+f) \geq \frac{12k+6}{4}$ so one of the triples is at least $3k+2$. However inverting the three involved boxes, that total to $6k+3$ rows and don't affect the configuration when all three are inverted, decreases the $f(C)$, giving contradiction.

Finally, in the $b = 3$ it follows that $(a+b+f) + (b+c+d) + (c+a+e) + (d+e+f) = 12k+8$. In particular, the odd/odd/even triples consist of $\frac{n+1}{2} + \frac{n+1}{2} + \frac{n-1}{2} = 6k+5$ rows each, while the even/even/even triple consists of $3 \cdot \frac{n-1}{2} = 6k+3$ rows. For each of the triples to avoid decreases $f(C)$ after being inverted, it needs to satisfy either $\leq 3k+2$ or $\leq 3k+1$ depending on the two cases, so it follows that $LHS \leq 12k+7$, giving contradiction.

Thus we have reached contradiction in all cases of counterexamples, proving our desired claim.

4. (USAMO 2013) Find all real numbers $x, y, z \geq 1$ satisfying

$$\min(\sqrt{x+xyz}, \sqrt{y+xyz}, \sqrt{z+xyz}) = \sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1}.$$

Solution (Cauchy or AM-GM)

The key Lemma is:

$$\sqrt{a-1} + \sqrt{b-1} \leq \sqrt{ab}$$

for all $a, b \geq 1$. Equality holds when $(a-1)(b-1) = 1$.

This is proven easily.

$$\sqrt{a-1} + \sqrt{b-1} = \sqrt{a-1}\sqrt{1} + \sqrt{1}\sqrt{b-1} \leq \sqrt{(a-1+1)(b-1+1)} = \sqrt{ab}$$

by Cauchy.

Equality then holds when $a-1 = \frac{1}{b-1} \implies (a-1)(b-1) = 1$.

Now assume that $x = \min(x, y, z)$. Now note that, by the Lemma,

$$\sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1} \leq \sqrt{x-1} + \sqrt{yz} \leq \sqrt{x(yz+1)} = \sqrt{xyz+x}$$

. So equality must hold in order for the condition in the problem statement to be met. So $(y-1)(z-1) = 1$ and $(x-1)(yz) = 1$. If we let $z = c$, then we can easily compute that $y = \frac{c}{c-1}$, $x = \frac{c^2+c-1}{c^2}$. Now it remains to check that $x \leq y, z$.

But by easy computations, $x = \frac{c^2+c-1}{c^2} \leq c = z \iff (c^2-1)(c-1) \geq 0$, which is obvious. Also $x = \frac{c^2+c-1}{c^2} \leq \frac{c}{c-1} = y \iff 2c \geq 1$, which is obvious, since $c \geq 1$.

So all solutions are of the form $\left(\frac{c^2+c-1}{c^2}, \frac{c}{c-1}, c \right)$, and all permutations for $c > 1$.

Remark: An alternative proof of the key Lemma is the following: By AM-GM,

$$(ab - a - b + 1) + 1 = (a-1)(b-1) + 1 \geq 2\sqrt{(a-1)(b-1)}$$

$$ab \geq (a-1) + (b-1) + 2\sqrt{(a-1)(b-1)}$$

. Now taking the square root of both sides gives the desired. Equality holds when $(a-1)(b-1) = 1$.

Solution 2

WLOG, assume that $x = \min(x, y, z)$. Let $a = \sqrt{x-1}$, $b = \sqrt{y-1}$ and $c = \sqrt{z-1}$. Then $x = a^2 + 1$, $y = b^2 + 1$ and $z = c^2 + 1$. The equation becomes

$$(a^2 + 1) + (a^2 + 1)(b^2 + 1)(c^2 + 1) = (a + b + c)^2.$$

Rearranging the terms, we have

$$(1 + a^2)(bc - 1)^2 + [a(b + c) - 1]^2 = 0.$$

Therefore $bc = 1$ and $a(b + c) = 1$. Express a and b in terms of c , we have $a = \frac{c}{c^2+1}$ and $b = \frac{1}{c}$. Easy to check that a is the smallest among a , b and c . Then $x = \frac{c^4+3c^2+1}{(c^2+1)^2}$, $y = \frac{c^2+1}{c^2}$ and $z = c^2 + 1$. Let $c^2 = t$, we have the solutions for (x, y, z) as follows: $(\frac{t^2+3t+1}{(t+1)^2}, \frac{t+1}{t}, t+1)$ and permutations for all $t > 0$.

–J.Z.

5. (USAMO 2013) Given positive integers m and n , prove that there is a positive integer c such that the numbers cm and cn have the same number of occurrences of each non-zero digit when written in base ten.

Solution 1

This solution is adopted from the [official solution](#). Both the problem and the solution were suggested by Richard Stong.

Without Loss of Generality, suppose $m \geq n \geq 1$. By prime factorization of n , we can find a positive integer c_1 such that $c_1 n = 10^s n_1$ where n_1 is relatively prime to 10. If a positive k is larger than s , then $(10^k c_1 m - c_1 n) = 10^s t$, where $t = 10^{k-s} c_1 m - n_1 > 0$ is always relatively prime to 10.

Choose a k large enough so that t is larger than $c_1 m$. We can find an integer $b \geq 1$ such that $10^b - 1$ is divisible by t , and also larger than $10c_1 m$. For example, let $b = \varphi(t)$ and use Euler's theorem. Now, let $c_2 = (10^b - 1)/t$, and $c = c_1 c_2$. We claim that c is the desired number.

Indeed, since both $c_1 m$ and n_1 are less than t , we see that the decimal expansion of both the fraction $(c_1 m)/t = (cm)/(c_2 t) = (cm)/(10^b - 1)$ and $n_1/t = (c_2 n_1)/(10^b - 1)$ are repeated in b -digit. And we also see that $10^k(c_1 m)/t = (t + c_1 n)/t = 1 + 10^s(n_1/t)$, therefore the two repeated b -digit expansions are cyclic shift of one another.

This proves that cm and $c_2 n_1$ have the same number of occurrences of non-zero digits. Furthermore, $cn = c_2 c_1 n = 10^s c_2 n_1$ also have the same number of occurrences of non-zero digits with $c_2 n_1$.

Solution 2

This is a rephrasing of the above solution.

It is enough to solve the problem when m, n are replaced by km, kn for any positive integer k . In particular, by taking $k = 2^a 5^b$ for appropriate values of a, b , we may assume $n = 10^s n_1$ where n_1 is relatively prime to 10.

Furthermore, adding or removing trailing zeros from m and n doesn't affect the claim, so we may further assume $\gcd(n, 10) = 1$ and that m has a xillion trailing

zeros (enough to make m way bigger than n , and also so that m has at least one trailing zero).

For clarity of exposition, we will also multiply m, n by a small number to make the units digit of n be 1 (though this is not necessary for the solution to work).

The point is that, for any positive integer X , most nonzero digits appear the same number of times in X and $X + 999 \dots 999$ if there are enough 9s; In particular, if the units digits of X is 1, then all nonzero digits appear the same number of times as long as there are at least as many 9s as digits in X .

So we will pick c to satisfy:

- $mc = nc + 999 \dots 999$ where the number of 9s is more than the number of digits of nc
- The units digit of nc is 1.

Because we made the units of n to be 1, the second condition is equivalent to making the units digit of c to be 1.

The first condition is equivalent to $(m - n)c = 999 \dots 999$. Because m has at least one trailing 0, the units digit of $m - n$ is 9, so $\gcd(m - n, 10) = 1$ and there is some c so that $(m - n)c = 999 \dots 999$, and the units digit of c must be 1 which agrees with the other condition.

Finally, as $m \gg n$, $(m - n)c \approx mc \gg nc$ so it is possible to make the number of 9s more than the number of digits in nc .

6. (USAMO 2013) Let ABC be a triangle. Find all points P on segment BC satisfying the following property: If X and Y are the intersections of line PA with the common external tangent lines of the circumcircles of triangles PAB and PAC , then

$$\left(\frac{PA}{XY}\right)^2 + \frac{PB \cdot PC}{AB \cdot AC} = 1.$$

Solution

Let circle PAB (i.e. the circumcircle of PAB), PAC be ω_1, ω_2 with radii r_1, r_2 and centers O_1, O_2 , respectively, and d be the distance between their centers.

Lemma. $XY = \frac{r_1 + r_2}{d} \sqrt{d^2 - (r_1 - r_2)^2}$.

Proof. Let the external tangent containing X meet ω_1 at X_1 and ω_2 at X_2 , and let the external tangent containing Y meet ω_1 at Y_1 and ω_2 at Y_2 . Then clearly X_1Y_1 and X_2Y_2 are parallel (for they are both perpendicular O_1O_2), and so $X_1Y_1Y_2X_2$ is a trapezoid.

Now, $X_1X^2 = XA \cdot XP = X_2X^2$ by Power of a Point, and so X is the midpoint of X_1X_2 . Similarly, Y is the midpoint of Y_1Y_2 . Hence, $XY = \frac{1}{2}(X_1Y_1 + X_2Y_2)$. Let X_1Y_1, X_2Y_2 meet O_1O_2 at Z_1, Z_2 , respectively. Then by similar triangles and the Pythagorean Theorem we deduce that $X_1Z_1 = \frac{r_1\sqrt{d^2 - (r_1 - r_2)^2}}{d}$ and $\frac{r_2\sqrt{d^2 - (r_1 - r_2)^2}}{d}$. But it is clear that Z_1, Z_2 is the midpoint of X_1Y_1, X_2Y_2 , respectively, so $XY = \frac{(r_1 + r_2)}{d} \sqrt{d^2 - (r_1 - r_2)^2}$, as desired.

Lemma 2. Triangles O_1AO_2 and BAC are similar.

Proof. $\angle AO_1O_2 = \frac{\angle PO_1A}{2} = \angle ABC$ and similarly $\angle AO_2O_1 = \angle ACB$, so the triangles are similar by AA Similarity.

Also, let O_1O_2 intersect AP at Z . Then obviously Z is the midpoint of AP and AZ is an altitude of triangle AO_1O_2 . Thus, we can simplify our expression of XY :

$$XY = \frac{AB + AC}{BC} \cdot \frac{AP}{2h_a} \sqrt{BC^2 - (AB - AC)^2},$$

where h_a is the length of the altitude from A in triangle ABC . Hence, substituting into our condition and using $AB = c, BC = a, CA = b$ gives

$$\left(\frac{2ah_a}{(b+c)\sqrt{a^2 - (b-c)^2}} \right)^2 + \frac{PB \cdot PC}{bc} = 1.$$

Using $2ah_a = 4[ABC] = \sqrt{(a+b+c)(a+b-c)(a-b+c)(-a+b+c)}$ by Heron's Formula (where $[ABC]$ is the area of triangle ABC , our condition becomes

$$\frac{(a+b+c)(-a+b+c)}{(b+c)^2} + \frac{PB \cdot PC}{bc} = 1,$$

which by $(a+b+c)(-a+b+c) = (b+c)^2 - a^2$ becomes

$$\frac{PB \cdot PC}{bc} = \frac{a^2bc}{(b+c)^2}.$$

Let $PB = x$; then $PC = a - x$. The quadratic in x is

$$x^2 - ax + \frac{a^2bc}{(b+c)^2} = 0,$$

which factors as

$$\left(x - \frac{ab}{b+c} \right) \left(x - \frac{ac}{b+c} \right) = 0.$$

Hence, $PB = \frac{ab}{b+c}$ or $\frac{ac}{b+c}$, and so the P corresponding to these lengths are our answer.

§43 USAMO 2014

1. (USAMO 2014) Let a, b, c, d be real numbers such that $b - d \geq 5$ and all zeros x_1, x_2, x_3 , and x_4 of the polynomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ are real. Find the smallest value the product

$$(x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1)$$

can take.

Solution

Using the hint we turn the equation into $\prod_{k=1}^4 (x_k - i)(x_k + i) \implies P(i)P(-i) \implies (b - d - 1)^2 + (a - c)^2 \implies \boxed{16}$. This minimum is achieved when all the x_i are equal to 1.

A more detailed version goes as follows:

Observe that $x^2 + 1 = x^2 - (-1) = x^2 - i^2 = (x - i)(x + i)$. Now, notice that:

$$\prod_{k=1}^4 (x_k - i)(x_k + i) = \prod_{k=1}^4 (x_k - i) \cdot \prod_{k=1}^4 (x_k + i).$$

The definition of $P(x)$ is $(x-x_1)(x-x_2)(x-x_3)(x-x_4)$ where the leading coefficient is 1 since P is a monic polynomial as given in the problem. Then, substituting in i , we have:

$$P(i) = (i-x_1)\cdots(i-x_4) = (-1)^4(x_1-i)\cdots(x_4-i).$$

This is exactly our first product. Our second product can be found as follows:

$$P(-i) = (-i-x_1)\cdots(-i-x_4) = (-1)^4(x_1+i)\cdots(x_4+i).$$

Hence, what we want to find is $P(i)P(-i)$. But substituting in i into our expression $x^4 + ax^3 + bx^2 + cx + d$ gets us $P(i) = (1-b+d) + (c-a)i$, and $P(-i) = (1-b+d) + (a-c)i$. Hence:

$$P(i)P(-i) = ((1-b+d)+(c-a)i)((1-b+d)+(a-c)i) = (1-b+d)^2 - (a-c)^2 = (-1)^2(b-d-1)^2 - (a-c)^2$$

where equality holds at $a = c$ and $b - d = 5 \Rightarrow b = d + 5$. Hence, the minimum is 16.

To finish this off we find a construction for this minimum. We know that $a = c$ and $b = d + 5$. Hence

$$P(x) = x^4 + cx^3 + (d+5)x^2 + cx + d.$$

We set $d = 1$ to get as much symmetry as possible within our polynomial. This leads to $x^4 + cx^3 + 6x^2 + cx + 1$. However, note that $\binom{4}{2} = 6$ and $\binom{4}{0} = \binom{4}{4} = 1$, so with some wishful thinking this leads us to think about the binomial theorem. We can try $(x \pm 1)^4$ and realize that those solutions do work.

Hence, 16 is obtained when all of x_i are equal to 1 or all equal to -1 .

2. (USAMO 2014) Let $\mathbb{Z}$ be the set of integers. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$xf(2f(y) - x) + y^2f(2x - f(y)) = \frac{(f(x))^2}{x} + f(yf(y))$$

for all $x, y \in \mathbb{Z}$ with $x \neq 0$.

Solution

Note: This solution is kind of rough. I didn't want to put my 7-page solution all over again. It would be nice if someone could edit in the details of the expansions.

Lemma 1: $f(0) = 0$. Proof: Assume the opposite for a contradiction. Plug in $x = 2f(0)$ (because we assumed that $f(0) \neq 0$), $y = 0$. What you get eventually reduces to:

$$4f(0) - 2 = \left(\frac{f(2f(0))}{f(0)} \right)^2$$

which is a contradiction since the LHS is divisible by 2 but not 4.

Then plug in $y = 0$ into the original equation and simplify by Lemma 1. We get:

$$x^2f(-x) = f(x)^2$$

Then:

$$x^6f(x) = x^4(x^2f(x))$$

$$\begin{aligned}
&= x^4((-x)^2 f(-(-x))) \\
&= x^4(-x)^2 f(-(-x)) \\
&= x^4 f(-x)^2 \\
&= f(x)^4
\end{aligned}$$

Therefore, $f(x)$ must be 0 or x^2 .

Now either $f(x)$ is x^2 for all x or there exists $a \neq 0$ such that $f(a) = 0$. The first case gives a valid solution. In the second case, we let $y = a$ in the original equation and simplify to get:

$$xf(-x) + a^2 f(2x) = \frac{f(x)^2}{x}$$

But we know that $xf(-x) = \frac{f(x)^2}{x}$, so:

$$a^2 f(2x) = 0$$

Since a is not 0, $f(2x)$ is 0 for all x (including 0). Now either $f(x)$ is 0 for all x , or there exists some $m \neq 0$ such that $f(m) = m^2$. Then m must be odd. We can let $x = 2k$ in the original equation, and since $f(2x)$ is 0 for all x , stuff cancels and we get:

$$y^2 f(4k - f(y)) = f(yf(y))$$

for $k \neq 0$. Now, let $y = m$ and we get:

$$m^2 f(4k - m^2) = f(m^3)$$

Now, either both sides are 0 or both are equal to m^6 . If both are m^6 then:

$$m^2(4k - m^2)^2 = m^6$$

which simplifies to:

$$4k - m^2 = \pm m^2$$

Since $k \neq 0$ and m is odd, both cases are impossible, so we must have:

$$m^2 f(4k - m^2) = f(m^3) = 0$$

Then we can let k be anything except 0, and get $f(x)$ is 0 for all $x \equiv 3 \pmod{4}$ except $-m^2$. Also since $x^2 f(-x) = f(x)^2$, we have $f(x) = 0 \Rightarrow f(-x) = 0$, so $f(x)$ is 0 for all $x \equiv 1 \pmod{4}$ except m^2 . So $f(x)$ is 0 for all x except $\pm m^2$. Since $f(m) \neq 0$, $m = \pm m^2$. Squaring, $m^2 = m^4$ and dividing by m , $m = m^3$. Since $f(m^3) = 0$, $f(m) = 0$, which is a contradiction for $m \neq 1$. However, if we plug in $x = 1$ with $f(1) = 1$ and y as an arbitrary large number with $f(y) = 0$ into the original equation, we get $0 = 1$ which is a clear contradiction, so our only solutions are $f(x) = 0$ and $f(x) = x^2$.

Solution 2

Given that the range of f consists entirely of integers, it is clear that the LHS must be an integer and $f(yf(y))$ must also be an integer, therefore $\frac{f(x)^2}{x}$ is an integer. If x divides $f(x)^2$ for all integers $x \neq 0$, then x must be a factor of $f(x)$, therefore $f(0) = 0$. Now, by setting $y = 0$ in the original equation, this simplifies to $xf(-x) = \frac{f(x)^2}{x}$. Assuming $x \neq 0$, we have $x^2 f(-x) = f(x)^2$. Substituting in $-x$

for x gives us $x^2 f(x) = f(-x)^2$. Substituting in $\frac{f(x)^2}{x^2}$ in for $f(-x)$ in the second equation gives us $x^2 f(x) = \frac{f(x)^4}{x^4}$, so $x^6 f(x) = f(x)^4$. In particular, if $f(x) \neq 0$, then we have $f(x)^3 = x^6$, therefore $f(x)$ is equivalent to 0 or x^2 for every integer x . Now, we shall prove that if for some integer $t \neq 0$, if $f(t) = 0$, then $f(x) = 0$ for all integers x . If we assume $f(y) = 0$ and $y \neq 0$ in the original equation, this simplifies to $xf(-x) + y^2 f(2x) = \frac{f(x)^2}{x}$. However, since $x^2 f(-x) = f(x)^2$, we can rewrite this equation as $\frac{f(x)^2}{x} + y^2 f(2x) = \frac{f(x)^2}{x}$, $y^2 f(2x)$ must therefore be equivalent to 0. Since, by our initial assumption, $y \neq 0$, this means that $f(2x) = 0$, so, if for some integer $y \neq 0$, $f(y) = 0$, then $f(x) = 0$ for all integers x . The contrapositive must also be true, i.e. If $f(x) \neq 0$ for all integers x , then there is no integral value of $y \neq 0$ such that $f(y) = 0$, therefore $f(x)$ must be equivalent for x^2 for every integer x , including 0, since $f(0) = 0$. Thus, $f(x) = 0, x^2$ are the only possible solutions.

Solution 3

Let's assume $f(0) \neq 0$. Substitute $(x, y) = (2f(0), 0)$ to get

$$2f(0)^2 = f(2f(0))^2 / 2f(0) + f(0)$$

$$2f(0)^2(2f(0) - 1) = f(2f(0))^2$$

This means that $2(2f(0) - 1)$ is a perfect square. However, this is impossible, as it is equivalent to 2 (mod 4). Therefore, $f(0) = 0$. Now substitute $x \neq 0, y = 0$ to get

$$xf(-x) = \frac{f(x)^2}{x} \implies x^2 f(-x) = f(x)^2.$$

Similarly,

$$x^2 f(x) = f(-x)^2.$$

From these two equations, we can find either $f(x) = f(-x) = 0$, or $f(x) = f(-x) = x^2$. Both of these are valid solutions on their own, so let's see if there are any solutions combining the two.

Let's say we can find $f(x) = x^2, f(y) = 0$, and $x, y \neq 0$. Then

$$xf(-x) + y^2 f(2x) = f(x)^2 / x.$$

$$y^2 f(2x) = x - x^3.$$

(NEEDS FIXING: $f(x)^2/x = x^4/x = x^3$, so the RHS is 0 instead of $x - x^3$.)

If $f(2x) = 4x^2$, then $y^2 = \frac{x-x^3}{4x^2} = \frac{1-x^2}{4x}$, which is only possible when $y = 0$. This contradicts our assumption. Therefore, $f(2x) = 0$. This forces $x = \pm 1$ due to the right side of the equation. Let's consider the possibility $f(2) = 0, f(1) = 1$. Substituting $(x, y) = (2, 1)$ into the original equation yields

$$0 = 2f(0) + 1f(2) = 0 + f(1) = 1,$$

which is impossible. So $f(2) = f(-2) = 4$ and there are no solutions "combining" $f(x) = x^2$ and $f(x) = 0$.

Therefore our only solutions are $\boxed{f(x) = 0}$ and $\boxed{f(x) = x^2}$.

Solution 4

Let the given assertion be $P(x, y)$. We try $P(x, 0)$ and get $xf(2c-x) = f(x)^2/x + c$, where $f(0) = c$. We plug in $x = c$ and get $cf(c) = f(c)^2/c + c$. Rearranging and

solving for c^2 gives us $c^2 = \frac{f(c)^2}{f(c)-1}$. Obviously, the only c that works such that the RHS is an integer is $c = 0$, and thus $f(0) = 0$.

We use this information on assertion $P(x, 0)$ and obtain $xf(-x) = f(x^2)/x$, or $f(-x) = \frac{f(x)^2}{x^2}$. Thus, $f(x)$ is an even function. It follows that $f(x) = 0, x^2$ for each x . We now prove that $f(x) = x^2$, $f(x)=0$ are the only solutions. [in progress] *SigmaPiE*

3. (USAMO 2014) Prove that there exists an infinite set of points

$$\dots, P_{-3}, P_{-2}, P_{-1}, P_0, P_1, P_2, P_3, \dots$$

in the plane with the following property: For any three distinct integers a, b , and c , points P_a, P_b , and P_c are collinear if and only if $a + b + c = 2014$.

Solution (Group Theory)

Consider an elliptic curve with a generator g , such that g is not a root of 0. By repeatedly adding g to itself under the standard group operation, we can build $g, 2g, 3g, \dots$ as well as $-g, -2g, -3g, \dots$. If we let

$$P_k = (3k - 2014)g$$

then we can observe that collinearity between P_a, P_b , and P_c occurs only if $P_a + P_b + P_c = 0$ (by definition of the group operation), which is equivalent to $(3a - 2014)g + (3b - 2014)g + (3c - 2014)g = (3a + 3b + 3c - 3 \cdot 2014)g = 0$, or $3a + 3b + 3c = 3 \cdot 2014$, or $a + b + c = 2014$. We know that all these points P_k exist because $3k - 2014$ is never 0 for integer k , so that none of these points need to be point at infinity (the identity element of the group).

Solution 2 (Function Theory)

Consider letting P_x be the point $(x, f(x))$, where $f(x) = x^3 - 2014x^2$. Then if three points P_a, P_b, P_c are on the same line $y = mx + p$, they must be the solutions to the equation $x^3 - 2014x^2 = mx + p$ (i.e. the intersection of f and the line). By Vieta's Formulas, the sum of the roots to this equation is 2014. Because each value of x corresponds to a different one of a, b , and c by definition of P_x , $a + b + c = 2014$. Conversely, if $a + b + c = 2014$, they must be the solutions to $(x - a)(x - b)(x - c) = x^3 - 2014x^2 - mx - p = 0$ for some real m and p . Clearly, then, P_a, P_b, P_c must all lie on the line $y = mx + p$. Hence, our setting $P_x = f(x)$ produces a valid infinite set of points.

Note: We could have let $f(x) = ax^3 - 2014ax^2 + bx + c$, where a, b , and c are arbitrary constants. (a is nonzero.)

Solution 3

Consider $a \mapsto a - 671$, $b \mapsto b - 671$, $c \mapsto c - 671$, so we find all cases that satisfy $a + b + c = 1$. We prove infinitely many points satisfy $P(n) = (n, Q(n))$ for a polynomial $Q(x)$. Then, notice that if $(a, Q(a)), (b, Q(b)), (c, Q(c))$ are collinear, then

$$(b - c)Q(a) + (c - a)Q(b) + (a - b)Q(c) = 0$$

.

Lemma 1: $(b - c)Q(a) + (c - a)Q(b) + (a - b)Q(c)$ is divisible by $(1 - a - b - c)(a - b)(b - c)(c - a)$.

Proof of Lemma 1:

This is trivial by the fact whenever $a + b + c = 1$ the equality to 0 holds. ■

Indeed, we rewrite as

$$(1 - a - b - c)(a - b)(b - c)(c - a)R(a, b, c)$$

for a multivariable function $R(a, b, c)$.

Lemma 2: We can assume $R(a, b, c) = 1$.

Proof of Lemma 2:

We already can say that $R(x) = k$ which is a constant and now we have $kO(x)$ where $O(x)$ is a polynomial, hence if $O(x) = 0$ whenever $a + b + c = 1$ then one can see that $O(x) = kO(x) = 0$ regardless, hence assume $R(x) = 1$ ■.

Therefore,

$$(b - c)Q(a) + (c - a)Q(b) + (a - b)Q(c) = (1 - a - b - c)(a - b)(b - c)(c - a)$$

.

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Here, a^3 occurs on the RHS and $(b - c)a^3$ is the term. These are equivalent, hence solutions exist only if $Q(x)$ is monic. Then, let $Q(x) = x^3 + R(x)$ where $R(x)$ is a polynomial of degree at most 2. Hence we have that, after reduction,

$$(b - c)R(a) + (c - a)R(b) + (a - b)R(c) = (a - b)(b - c)(c - a)$$

.

Comparing leading coefficients we see that $R(x)$ has leading coefficient -1 and is quadratic. It is easy to see $R(x) = -x^2$ works, hence $Q(x) = x^3 - x^2$, and there exist infinitely many $P(x)$ that satisfy the coordinates:

$$P(n) = (n + 671, n^3 - n^2) \quad \square$$

4. (USAMO 2014) Let k be a positive integer. Two players A and B play a game on an infinite grid of regular hexagons. Initially all the grid cells are empty. Then the players alternately take turns with A moving first. In his move, A may choose two adjacent hexagons in the grid which are empty and place a counter in both of them. In his move, B may choose any counter on the board and remove it. If at any time there are k consecutive grid cells in a line all of which contain a counter, A wins. Find the minimum value of k for which A cannot win in a finite number of moves, or prove that no such minimum value exists.

Solution

We claim that the minimum k such that A cannot create a k in a row is $\boxed{6}$.

It is easy to verify that player A can create a 5 in a row.

Let A place two counters anywhere, and B take off one of them. Then, A should create a "triangle of hexagons" by placing two adjacent counters also next to the unremoved one, so that no matter what B does there will be a line of two counters on adjacent hexagons.

Note: A diagram and more conciseness is required in the following solution. Better maneuvers are always appreciated.

Then, it is advisable for A to make a 4-in-a-row by placing 2 counters at the end of the line. B to prevent a 5-in-a-row must counter by removing a middle counter (a counter not on the edge of the 4-in-a-row). Player A then could reinstate the threat by placing a

counter there and one adjacent to that hexagon (but not adjacent to the other middle counter). It is not in B's best interest to then remove the other middle counter, for then A can "add" a counter to both hexagons (adjacent to the removed counter) on the same side of the originally placed hexagon while preserving the threat. Now, we have two different threats B cannot both counter. Thus, assume B does not remove the other middle counter. Eventually, by "adding" counters and preserving the threat player A can create a web of hexagons surrounding the chosen middle counter. As we have proven, it will be disastrous for B to then remove the other middle counter; thus, B has to remove the chosen middle counter.

Now, player A can administer the decisive winning maneuver as follows. Let X be two adjacent counters of the web, one of which is adjacent to the initial edge counter. Let Y be the other two counters that satisfy the condition. The line X should not have more than two counters on it. Player A should place his two counters on the first two hexagons not adjacent to any counter in X (or the web), so that B is forced to remove the outermost counter A played. Then, A should place two counters adjacent to his remaining counter in a direction parallel to line Y . B has to expunge the outermost one; otherwise A can place two counters to complete a 5-in-a-row. Let A place two more counters in the same direction; then, B has to remove the middle counter played. Now, turn in a direction parallel to line X ; let A place a counter on the first two spaces adjacent to counters already placed. B will remove the second counter played. A's knockout blow occurs when he reinstates the lost counter and a counter adjacent to that counter and the remaining one he played last turn, and it is easy to verify he is guaranteed a 5-in-a-row. Therefore, A can (after an elongated siege) obtain a 5-in-a-row.

To prove that 6-in-a-row is impossible, tile the hexagon board in an A-B-C-A-B-C-A-B-C format. Define an "A tile" to be a tile engraved with the letter A. To prevent player A from obtaining a 6 in a row, all player B has to do is to remove counters placed on A tiles by player A. Because no two A tiles are adjacent, player A can play at most one counter on an A tile at a time; thus, there can at most be one counter on A tiles any time of the game. Clearly, then, a 6 in a row, which requires A-B-C-A-B-C (or two counters on A tiles), is impossible, completing the proof.

5. (USAMO 2014) Let ABC be a triangle with orthocenter H and let P be the second intersection of the circumcircle of triangle AHC with the internal bisector of the angle $\angle BAC$. Let X be the circumcenter of triangle APB and Y the orthocenter of triangle APC . Prove that the length of segment XY is equal to the circumradius of triangle ABC .

Solution 1

Let O_1 be the center of (AHP) , O be the center of (ABC) . Note that (O_1) is the reflection of (O) across AC , so $AO = AO_1$. Additionally

$$\angle AYC = 180 - \angle APC = 180 - \angle AHC = \angle B$$

so Y lies on (O) . Now since XO, OO_1, XO_1 are perpendicular to AB, AC , and their bisector, XOO_1 is isosceles with $XO = OO_1$, and $\angle XO_1O = 180 - \angle A$. Also

$$\angle AOY = 2\angle ACY = 2(90 - \angle PAC) = 2(90 - \frac{A}{2}) = 180 - \angle A = \angle XO_1O$$

But $YO = OA$ as well, and $\angle YOX = \angle AOO_1$, so $\triangle OYX \cong \triangle OAO_1$. Thus $XY = AO_1 = AO$.

Solution 2

Since $AHPC$ is a cyclic quadrilateral, $\angle AHC = \angle APC$. $\angle AHC = 90^\circ + \angle ABC$ and $\angle APC = 90^\circ + \angle AYC$, we find $\angle ABC = \angle AYC$. That is, $ABYC$ is a cyclic quadrilateral. Let D be mid-point of $\overline{AB}$. O, X, D are collinear and $OX \perp AB$. Let M be second intersection of AP with circumcircle of the triangle ABC . Let $YP \cap AC = E$, $YM \cap AB = F$. Since M is mid-point of the arc BC , $OM \perp BC$. Since $AYMC$ is a cyclic quadrilateral, $\angle CYM = \angle CAM = \angle BAC/2$. Since Y is the orthocenter of triangle APC , $\angle PYC = \angle CAP = \angle BAC/2$. Thus, $\angle PYM = \angle BAC$ and $AEYF$ is a cyclic quadrilateral. So, $YF \perp AB$ and $OX \parallel MY$. We will prove that $XYMO$ is a parallelogram.

<https://wiki-images.artofproblemsolving.com//7/7b/Usamo2014-5.png> (figure link)

We see that YPM is an isosceles triangle and $YM = YP$. Also $XB = XP$ and $\angle BXP = 2\angle BAP = \angle BAC = \angle PYM$. Then, $BXP \sim MYP$. By spiral similarity, $BPM \sim XPY$ and $\angle XYP = \angle BMP = \angle BCA$. Hence, $\angle XYP = \angle BCA$, $XY \perp BC$. Since $OM \perp BC$, we get $XYMO$ is a parallelogram. As a result, $OM = XY$.

(Lokman GÖKÇE)

6. (USAMO 2014) Prove that there is a constant $c > 0$ with the following property: If a, b, n are positive integers such that $\gcd(a + i, b + j) > 1$ for all $i, j \in \{0, 1, \dots, n\}$, then

$$\min\{a, b\} > c^n \cdot n^{\frac{n}{2}}.$$

Solution

The following solution is due to Gabriel Dospinescu and v__Enhance (also known as Evan Chen).

Let $N = n + 1$ and assume N is (very) large. We construct an $N \times N$ with cells (i, j) where $0 \leq i, j \leq n$ and in each cell place a prime p dividing $\gcd(a + i, b + j)$.

The central claim is at least 50% of the primes in this table exceed $0.001n^2$. We count the maximum number of squares they could occupy:

$$\sum_p \left\lceil \frac{N}{p} \right\rceil^2 \leq \sum_p \left(\frac{N}{p} + 1 \right)^2 = N^2 \sum_p \frac{1}{p^2} + 2N \sum_p \frac{1}{p} + \sum_p 1.$$

Here the summation runs over primes $p \leq 0.001n^2$.

Let $r = \pi(0.001n^2)$ denote the number of such primes. Now we apply the three bounds:

$$\sum_p \frac{1}{p^2} < \frac{1}{2}$$

which follows by adding all the primes directly with some computation,

$$\sum_p \frac{1}{p} < \sum_{k=1}^r \frac{1}{k} = O(\ln r) < o(N)$$

using the harmonic series bound, and

$$\sum_p 1 < r \sim O\left(\frac{N^2}{\ln N}\right) < o(N^2)$$

via Prime Number Theorem. Hence the sum in question is certainly less than $\frac{1}{2}N^2$ for N large enough, establishing the central claim.

Hence some column $a + i$ has at least one half of its primes greater than $0.001n^2$. Because this is greater than n for large n , these primes must all be distinct, so $a + i$ exceeds their product, which is larger than

$$(0.001n^2)^{N/2} > c^n \cdot n^n$$

where c is some constant.

§44 USAMO 2015

1. (USAMO 2015) Solve in integers the equation

$$x^2 + xy + y^2 = \left(\frac{x+y}{3} + 1\right)^3.$$

Solution

We first notice that both sides must be integers, so $\frac{x+y}{3}$ must be an integer.

We can therefore perform the substitution $x + y = 3t$ where t is an integer.

Then:

$$(3t)^2 - xy = (t+1)^3$$

$$9t^2 + x(x-3t) = t^3 + 3t^2 + 3t + 1$$

$$4x^2 - 12xt + 9t^2 = 4t^3 - 15t^2 + 12t + 4$$

$$(2x-3t)^2 = (t-2)^2(4t+1)$$

$4t+1$ is therefore the square of an odd integer and can be replaced with $(2n+1)^2 = 4n^2 + 4n + 1$

By substituting using $t = n^2 + n$ we get:

$$(2x - 3n^2 - 3n)^2 = [(n^2 + n - 2)(2n + 1)]^2$$

$$2x - 3n^2 - 3n = \pm(2n^3 + 3n^2 - 3n - 2)$$

$$x = n^3 + 3n^2 - 1 \text{ or } x = -n^3 + 3n + 1$$

Using substitution we get the solutions: $(n^3 + 3n^2 - 1, -n^3 + 3n + 1) \cup (-n^3 + 3n + 1, n^3 + 3n^2 - 1)$

Solution 2

Let $n = \frac{x+y}{3}$. Thus, $x + y = 3n$. We have

$$x^2 + xy + y^2 = \left(\frac{x+y}{3} + 1\right)^3 \implies (x+y)^2 - xy = \left(\frac{x+y}{3} + 1\right)^3$$

Substituting n for $\frac{x+y}{3}$, we have

$$9n^2 - x(3n - x) = (n+1)^3$$

Treating x as a variable and n as a constant, we have

$$9n^2 - 3nx + x^2 = (n+1)^3,$$

which turns into

$$x^2 - 3nx + (9n^2 - (n+1)^3) = 0,$$

a quadratic equation. By the quadratic formula,

$$x = \frac{1}{2} \left(3n \pm \sqrt{9n^2 - 4(9n^2 - (n+1)^3)} \right)$$

which simplifies to

$$x = \frac{1}{2} \left(3n \pm \sqrt{4(n+1)^3 - 27n^2} \right)$$

Since we want x and y to be integers, we need $4(n+1)^3 - 27n^2$ to be a perfect square. We can factor the aforementioned equation to be

$$(n-2)^2(4n+1) = k^2$$

for an integer k . Since $(n-2)^2$ is always a perfect square, for $(n-2)^2(4n+1)$ to be a perfect square, $4n+1$ has to be a perfect square as well. Since $4n+1$ is odd, the square root of the aforementioned equation must be odd as well. Thus, we have $4n+1 = a^2$ for some odd a . Thus,

$$n = \frac{a^2 - 1}{4},$$

in which by difference of squares it is easy to see that all the possible values for n are just $n = p(p-1)$, where p is a positive integer. Thus,

$$x + y = 3n = 3p(p-1).$$

Thus, the general form for

$$x = \frac{1}{2} \left(3p(p-1) \pm \sqrt{4(p(p-1)+1)^3 - 27(p(p-1))^2} \right)$$

for a positive integer p . (This is an integer since $4(p(p-1)+1)^3 - 27(p(p-1))^2$ is an even perfect square (since $4(p(p-1)+1)$ is always even, as well as $27(p(p-1))^2$ being always even) as established, and $3p(p-1)$ is always even as well. Thus, the whole numerator is even, which makes the quantity of that divided by 2 always an integer.) Since $y = 3n - x$, the general form for y is just

$$y = 3p(p-1) - \frac{1}{2} \left(3p(p-1) \pm \sqrt{4(p(p-1)+1)^3 - 27(p(p-1))^2} \right)$$

(This is an integer since $4(p(p-1)+1)^3 - 27(p(p-1))^2$ is an even perfect square (since $4(p(p-1)+1)$ is always even, as well as $27(p(p-1))^2$ being always even) as established, and $3p(p-1)$ is always even as well. Thus, the whole numerator is even, which makes the quantity of that divided by 2 always an integer, which thus trivially makes

$$3p(p-1) - \frac{1}{2} \left(3p(p-1) \pm \sqrt{4(p(p-1)+1)^3 - 27(p(p-1))^2} \right)$$

an integer.) for a positive integer p . Thus, our general in integers (x, y) is

$$\left(\frac{1}{2} \left(3p(p-1) \pm \sqrt{4(p(p-1)+1)^3 - 27(p(p-1))^2} \right), 3p(p-1) - \frac{1}{2} \left(3p(p-1) \pm \sqrt{4(p(p-1)+1)^3 - 27(p(p-1))^2} \right) \right)$$

□

-fidgetboss_4000

2. (USAMO 2015) Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

Solution 1

We claim the M lies on the circle centered at the midpoint of $\overline{AO}$ that passes through P .

Proceed via complex numbers. $a = 1, b = -1$ and s, t are free parameters. Then,

$$x = \frac{1}{2}(s + 1 + t - s\bar{t}) \text{ and } \Re(p) = \Re(x) \implies p + \bar{p} = x + \bar{x}$$

Putting it together,

$$\begin{aligned} \left| \frac{1}{2} - \left(\frac{s+t}{2} \right) \right| = \left| \frac{1}{2} - p \right| &\iff (s+t-1)(\bar{s}+\bar{t}-1) = 5 - 2(p+\bar{p}) = 5 - 2(x+\bar{x}) \\ &\iff 3 + s\bar{t} + \bar{s}t - s - t - \bar{s} - \bar{t} = 5 - (s+1+t-s\bar{t} + \bar{s}+1+\bar{t}-\bar{s}t) \end{aligned}$$

which is clear by expanding. Thus, the proof is complete.

-sillybone

Solution 2

We will use coordinate geometry.

Without loss of generality, let the circle be the unit circle centered at the origin,

$$A = (1, 0), P = (1 - a, b), Q = (1 - a, -b)$$

, where $(1 - a)^2 + b^2 = 1$.

Let angle $\angle XAB = A$, which is an acute angle, $\tan A = t$, then $X = (1 - a, at)$.

Angle $\angle BOS = 2A$, $S = (-\cos(2A), \sin(2A))$. Let $M = (u, v)$, then $T = (2u + \cos(2A), 2v - \sin(2A))$.

The condition $TX \perp AX$ yields: $(2v - \sin(2A) - at)/(2u + \cos(2A) + a - 1) = \cot A$. (E1)

Use identities $(\cos A)^2 = 1/(1 + t^2)$, $\cos(2A) = 2(\cos A)^2 - 1 = 2/(1 + t^2) - 1$, $\sin(2A) = 2 \sin A \cos A = 2t^2/(1 + t^2)$, we obtain $2vt - at^2 = 2u + a$. (E1')

The condition that T is on the circle yields $(2u + \cos(2A))^2 + (2v - \sin(2A))^2 = 1$, namely $v \sin(2A) - u \cos(2A) = u^2 + v^2$. (E2)

M is the mid-point on the hypotenuse of triangle STX , hence $MS = MX$, yielding $(u + \cos(2A))^2 + (v - \sin(2A))^2 = (u + a - 1)^2 + (v - at)^2$. (E3)

Expand (E3), using (E2) to replace $2(v \sin(2A) - u \cos(2A))$ with $2(u^2 + v^2)$, and using (E1') to replace $a(-2vt + at^2)$ with $-a(2u + a)$, and we obtain $u^2 - u - a + v^2 = 0$, namely $(u - \frac{1}{2})^2 + v^2 = a + \frac{1}{4}$, which is a circle centered at $(\frac{1}{2}, 0)$ with radius $r = \sqrt{a + \frac{1}{4}}$.

Solution 3

Let the midpoint of AO be K . We claim that M moves along a circle with radius KP .

We will show that $KM^2 = KP^2$, which implies that $KM = KP$, and as KP is fixed, this implies the claim.

$$KM^2 = \frac{AM^2 + OM^2}{2} - \frac{AO^2}{4} \text{ by the median formula on } \triangle AMO.$$

$$KP^2 = \frac{AP^2 + OP^2}{2} - \frac{AO^2}{4} \text{ by the median formula on } \triangle APO.$$

$$KM^2 - KP^2 = \frac{1}{2}(AM^2 + OM^2 - AP^2 - OP^2).$$

As $OP = OT$, $OP^2 - OM^2 = MT^2$ from right triangle OMT . (1)

$$\text{By (1), } KM^2 - KP^2 = \frac{1}{2}(AM^2 - MT^2 - AP^2).$$

Since M is the circumcenter of $\triangle XTS$, and MT is the circumradius, the expression $AM^2 - MT^2$ is the power of point A with respect to (XTS) . However, as $AX \cdot AS$ is also the power of point A with respect to (XTS) , this implies that $AM^2 - MT^2 = AX \cdot AS$. (2)

$$\text{By (2), } KM^2 - KP^2 = \frac{1}{2}(AX \cdot AS - AP^2)$$

Finally, $\triangle APX \sim \triangle ASP$ by AA similarity ($\angle XAP = \angle SAP$ and $\angle APX = \angle AQP = \angle ASP$), so $AX \cdot AS = AP^2$. (3)

By (3), $KM^2 - KP^2 = 0$, so $KM^2 = KP^2$, as desired. *QED*

Solution 4(synthetic)

To begin with, we connect $\overline{AT}$ and we construct the nine-point circle of $\triangle AST$ centered at N_9 .

Lemma 1: $AX \cdot AS = AP^2$. We proceed on a directed angle chase. We get $\angle ASP = \angle AQP = \angle QPA$, so $\triangle PAS \sim \triangle XAP$ and the desired result follows by side length ratios.

Lemma 2: The locus of N_9 as X moves along $\overline{PQ}$ is a circle centered about A . We add the midpoint of $\overline{AS}$, N , and let the circumradius of $\triangle AST$ be R . Taking the power of A with respect to (N_9) , we get

$$AN_9^2 - \left(\frac{1}{2}R\right)^2 = \text{Pow}_{(N_9)}A = AX \cdot AN = \frac{1}{2}AX \cdot AS = \frac{1}{2}AP^2.$$

Hence, $AN_9 = \sqrt{\frac{1}{4}R^2 + \frac{1}{2}AP^2}$, which remains constant as X moves.

Next, consider the homothety of scale factor $\frac{2}{3}$ about O mapping N_9 to G . This means that the locus of G is a circle as well.

Finally, we take a homothety of scale factor $\frac{3}{2}$ about A mapping G to M . Hence, the locus of M is a circle, as desired. - Spacesam

3. (USAMO 2015) Let $S = \{1, 2, \dots, n\}$, where $n \geq 1$. Each of the 2^n subsets of S is to be colored red or blue. (The subset itself is assigned a color and not its individual elements.) For any set $T \subseteq S$, we then write $f(T)$ for the number of subsets of T that are blue.

Determine the number of colorings that satisfy the following condition: for any subsets T_1 and T_2 of S ,

$$f(T_1)f(T_2) = f(T_1 \cup T_2)f(T_1 \cap T_2).$$

Solution

Define function: $C(T) = 1$ if the set T is colored blue, and $C(T) = 0$ if T is colored red. Define the Core = intersection of all T where $C(T) = 1$.

The empty set is denoted as $\emptyset$, $\cap$ denotes intersection, and $\cup$ denotes union. Let $S_n = \{n\}$ are one-element subsets.

Let $m_{c_k} = \binom{m}{k} = \frac{m!}{k!(m-k)!}$ denote m choose k .

(Case I) $f(\emptyset) = 1$. Then for distinct m and k , $f(S_m \cup S_k) = f(S_m)f(S_k)$, meaning only if S_m and S_k are both blue is their union blue. Namely $C(S_m \cup S_k) = C(S_m)C(S_k)$.

Similarly, for distinct m, n, k , $f(S_m \cup S_k \cup S_n) = f(S_m \cup S_k)f(S_n)$, $C(S_m \cup S_k \cup S_n) = C(S_m)C(S_k)C(S_n)$. This procedure of determination continues to S . Therefore, if $T = \{a_1, a_2, \dots, a_k\}$, then $C(T) = C(S_{a_1})C(S_{a_2}) \cdots C(S_{a_k})$. All colorings thus determined by the free colors chosen for subsets of one single elements satisfy the condition. There are 2^n colorings in this case.

(Case II.) $f(\emptyset) = 0$.

(Case II.1) Core = $\emptyset$. Then either (II.1.1) there exist two nonintersecting subsets A and B , $C(A) = C(B) = 1$, but $f(A)f(B) = 0$, a contradiction, or (II.1.2) all subsets has $C(T) = 0$, which is easily confirmed to satisfy the condition $f(T_1)f(T_2) = f(T_1 \cap T_2)f(T_1 \cup T_2)$. There is one coloring in this case.

(Case II.2) Core = a subset of 1 element. WLOG, $C(S_1) = 1$. Then $f(S_1) = 1$, and subsets containing element 1 may be colored blue. $f(S_1 \cup S_m)f(S_1 \cup S_n) = f(S_1 \cup S_m \cup S_n)$, namely $C(S_1 \cup S_m \cup S_n) = C(S_m \cup S_1)C(S_n \cup S_1)$. Now S_{n-1} functions as the $\emptyset$ in case I, with $n - 1$ elements to combine into a base of $n - 1$ two-element sets, and all the other subsets are determined. There are 2^{n-1} colorings for each choice of core. However, there are $nC1 = n$ such cores. Hence altogether there are $n2^{n-1}$ colorings in this case.

(Case II.3) Core = a subset of 2 elements. WLOG, let $C(S_1 \cup S_2) = 1$. Only subsets containing the core may be colored blue. With the same reasoning as in the preceding case, there are $(nC2)2^{n-2}$ colorings.

...

(Case II.n+1) Core = S . Then $C(S) = 1$, with all other subsets $C(T) = 0$, there is $1 = \binom{n}{n}2^0$

Combining all the cases, we have $1 + \left[2^n + \binom{n}{1}2^{n-1} + \binom{n}{2}2^{n-2} + \cdots + \binom{n}{n}2^0 \right] =$

$\boxed{1 + 3^n}$ colorings. sponsored by ALLEN

4. (USAMO 2015) Steve is piling $m \geq 1$ indistinguishable stones on the squares of an $n \times n$ grid. Each square can have an arbitrarily high pile of stones. After he finished piling his stones in some manner, he can then perform stone moves, defined as follows. Consider any four grid squares, which are corners of a rectangle, i.e. in positions $(i, k), (i, l), (j, k), (j, l)$ for some $1 \leq i, j, k, l \leq n$, such that $i < j$ and $k < l$. A stone move consists of either removing one stone from each of (i, k) and (j, l) and moving them to (i, l) and (j, k) respectively, or removing one stone from each of (i, l) and (j, k) and moving them to (i, k) and (j, l) respectively.

Two ways of piling the stones are equivalent if they can be obtained from one another by a sequence of stone moves.

How many different non-equivalent ways can Steve pile the stones on the grid?

Solution

Let the number of stones in row i be r_i and let the number of stones in column i be c_i . Since there are m stones, we must have $\sum_{i=1}^n r_i = \sum_{i=1}^n c_i = m$

Lemma 1: If any 2 pilings are equivalent, then r_i and c_i are the same in both pilings $\forall i$.

Proof: We suppose the contrary. Note that r_i and c_i remain invariant after each move, therefore, if any of the r_i or c_i are different, they will remain different.

Lemma 2: Any 2 pilings with the same r_i and $c_i \forall i$ are equivalent.

Proof: Suppose piling 1 and piling 2 not the same piling. Call a stone in piling 1 wrong if the stone occupies a position such that there are more stones in that position in piling 1 than piling 2. Similarly define a wrong stone in piling 2. Let a wrong stone be at (a, b) in piling 1. Since c_b is the same for both pilings, we must have a wrong stone in piling 2 at column b , say at (c, b) , such that $c \neq a$. Similarly, we must have a wrong stone in piling 1 at row c , say at (c, d) where $d \neq b$. Clearly, making the move $(a, b); (c, d) \implies (c, b); (a, d)$ in piling 1 decreases the number of wrong stones in piling 1. Therefore, the number of wrong stones in piling 1 must eventually be 0 after a sequence of moves, so piling 1 and piling 2 are equivalent.

Lemma 3: Given the sequences g_i and h_i such that $\sum_{i=1}^n g_i = \sum_{i=1}^n h_i = m$ and $g_i, h_i \geq 0 \forall i$, there is always a piling that satisfies $r_i = g_i$ and $c_i = h_i \forall i$.

Proof: We take the lowest i, j , such that $g_i, h_j > 0$ and place a stone at (i, j) , then we subtract g_i and h_j by 1 each, until g_i and h_i become 0 $\forall i$, which will happen when m stones are placed, because $\sum_{i=1}^n g_i$ and $\sum_{i=1}^n h_i$ are both initially m and decrease by 1 after each stone is placed. Note that in this process $r_i + g_i$ and $c_i + h_i$ remains invariant, thus, the final piling satisfies the conditions above.

By the above lemmas, the number of ways to pile is simply the number of ways to choose the sequences r_i and c_i such that $\sum_{i=1}^n r_i = \sum_{i=1}^n c_i = m$ and $r_i, c_i \geq 0 \forall i$. By stars and bars, the number of ways is $\binom{n+m-1}{m}^2$.

Solution by Shaddoll

5. (USAMO 2015) Let a, b, c, d, e be distinct positive integers such that $a^4 + b^4 = c^4 + d^4 = e^5$. Show that $ac + bd$ is a composite number.

Solution 1

Note: This solution is definitely not what the folks at MAA intended, but it works!

Look at the statement $a^4 + b^4 = e^5$. This can be viewed as a special case of Beal's Conjecture (http://en.wikipedia.org/wiki/Beal%27s_conjecture), stating that the equation $A^x + B^y = C^z$ has no solutions over positive integers for $\gcd(a, b, c) = 1$ and $x, y, z > 2$. This special case was proven in 2009 by Michael Bennet, Jordan Ellenberg, and Nathan Ng, as $(x, y, z) = (2, 4, n)$. This case $a^4 + b^4 = e^5$ is obviously contained under that special case, so a and b must have a common factor greater than 1.

Call the greatest common factor of a and b f . Then $a = f \cdot a_1$ for some a_1 and likewise $b = f \cdot b_1$ for some b_1 . Then consider the quantity $ac + bd$.

$$ac + bd = f \cdot a_1 \cdot c + f \cdot b_1 \cdot d = f \cdot (a_1 \cdot c + b_1 \cdot d).$$

Because c and d are both positive, $(a_1 \cdot c + b_1 \cdot d) > 1$, and by definition $f > 1$, so $ac + bd$ is composite.

Solution 2

A more conventional approach, using proof by contradiction:

Without loss of generality, assume $a > d$

Since $a^4 + b^4 = c^4 + d^4$, It is obvious that $b < c$

We construct the equation $(a^4 + b^4)c^2d^2 - (c^4 + d^4)a^2b^2 = (a^2c^2 - b^2d^2)(a^2d^2 - b^2c^2)$ (the right side is the factorization of the left) Which factors into: $e^5(cd - ab)(cd + ab) = (ac - bd)(ac + bd)(ad - bc)(ad + bc)$ (by using $a^4 + b^4 = c^4 + d^4 = e^5$ and factoring)

If $ac - bd$ or $ad - bc$ equal zero, then $a : b$ equals $c : d$ or $d : c$ respectively, which is impossible because $a^4 + b^4 = c^4 + d^4$ and because a, b, c, d, e are distinct. Therefore the right side of the equation above is non-zero and the left side must be divisible by $ac + bd$

If $ac + bd$ were prime,

$$\text{We have } ac + bd - (cd + ab) = (a - d)(c - b) > 0$$

Which means that $ac + bd > cd + ab > cd - ab$ and neither $cd + ab$ nor $cd - ab$ can be multiples of $ac + bd$ meaning e must be a multiple of $ac + bd$ and $e = k(ac + bd)$ for some integer k

But clearly this is impossible since $(k(ac + bd))^5 > a^4 + b^4$.

Therefore, by contradiction, $ac + bd$ is composite.

6. (USAMO 2015) Consider $0 < \lambda < 1$, and let A be a multiset of positive integers. Let $A_n = \{a \in A : a \leq n\}$. Assume that for every $n \in \mathbb{N}$, the set A_n contains at most $n\lambda$ numbers. Show that there are infinitely many $n \in \mathbb{N}$ for which the sum of the elements in A_n is at most $\frac{n(n+1)}{2}\lambda$. (A multiset is a set-like collection of elements in which order is ignored, but repetition of elements is allowed and multiplicity of elements is significant. For example, multisets $\{1, 2, 3\}$ and $\{2, 1, 3\}$ are equivalent, but $\{1, 1, 2, 3\}$ and $\{1, 2, 3\}$ differ.)

Solution

Proposed by mengmeng142857.

We prove this by contradiction. Suppose that the number of times that an integer a appears in A is k_a . Let the sum of the elements in A_n be $S_n = \sum_{i=1}^n i \cdot k_i$. If there are only finitely many $n \in \mathbb{N}$ such that $S_n \leq \frac{n(n+1)}{2}\lambda$, then by the Well Ordering Principle, there must be a largest $m \in \mathbb{N}$ such that $S_m \leq \frac{m(m+1)}{2}\lambda$, and for all $n > m$, $S_n > \frac{n(n+1)}{2}\lambda$.

Now, observe that for some $n > m$,

$$\frac{1}{n} \cdot S_n + \frac{1}{n(n-1)} \cdot S_{n-1} + \frac{1}{(n-1)(n-2)} \cdot S_{n-2} + \dots + \frac{1}{2 \cdot 1} \cdot S_1 = \sum_{i=1}^n k_i \leq n\lambda$$

$$\text{Also, } \frac{S_n}{n} > \frac{n+1}{2}\lambda, \frac{S_{n-1}}{n(n-1)} > \frac{1}{2}\lambda, \frac{S_{n-2}}{(n-1)(n-2)} > \frac{1}{2}\lambda, \dots, \frac{S_{m+1}}{(m+2)(m+1)} > \frac{1}{2}\lambda, \frac{S_m}{(m+1)m} \leq \frac{1}{2}\lambda.$$

Now, for any $n \in \mathbb{N}$, we let $\frac{S_n}{(n+1)n} - \frac{1}{2}\lambda = d_n$, then

$$\frac{1}{n} \cdot S_n + \sum_{i=1}^{n-1} \frac{S_i}{i(i+1)} = n\lambda + (n+1) \cdot d_n + \sum_{i=m+1}^{n-1} d_i + \sum_{i=1}^m d_i \leq n\lambda$$

Let $\sum_{i=1}^m d_i = -c$ (where c is positive, otherwise we already have a contradiction), then

$$n\lambda - c < n\lambda + (n+1) \cdot d_n + \sum_{i=m+1}^{n-1} d_i - c = \sum_{i=1}^n k_i \leq n\lambda$$

Dividing both sides by n , $\lambda - c/n < \frac{\sum_{i=1}^n k_i}{n} \leq \lambda$. Taking the limit yields $\lim_{n \rightarrow \infty} \frac{\sum_{i=1}^n k_i}{n} = \lambda$. Observe that this would imply that

$$\lim_{n \rightarrow \infty} k_n = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n k_i - \sum_{i=1}^{n-1} k_i \right) = \lim_{n \rightarrow \infty} \left(n \cdot \frac{\sum_{i=1}^n k_i}{n} - (n-1) \cdot \frac{\sum_{i=1}^{n-1} k_i}{n-1} \right) = \lambda$$

However, as k_n is an integer, it is impossible for it to converge to λ . This yields the desired contradiction.

§45 USAMO 2016

1. (USAMO 2016) Let $X_1, X_2, \dots, X_{100}$ be a sequence of mutually distinct nonempty subsets of a set S . Any two sets X_i and X_{i+1} are disjoint and their union is not the whole set S , that is, $X_i \cap X_{i+1} = \emptyset$ and $X_i \cup X_{i+1} \neq S$, for all $i \in \{1, \dots, 99\}$. Find the smallest possible number of elements in S .

Solution 1

The answer is that $|S| \geq 8$.

First, we provide a inductive construction for $S = \{1, \dots, 8\}$. Actually, for $n \geq 4$ we will provide a construction for $S = \{1, \dots, n\}$ which has $2^{n-1} + 1$ elements in a line. (This is sufficient, since we then get 129 for $n = 8$.) The idea is to start with the following construction for $|S| = 4$:

$$34 \quad 1 \quad 23 \quad 4 \quad 12 \quad 3 \quad 14 \quad 2 \quad 13 \quad .$$

Then inductively, we do the following procedure to move from n to $n+1$: take the chain for n elements, delete an element, and make two copies of the chain (which now has even length). Glue the two copies together, joined by $\emptyset$ in between. Then place the element $n+1$ in alternating positions starting with the first (in particular, this hits $n+1$). For example, the first iteration of this construction gives:

$$\begin{array}{cccccccc} 345 & 1 & 235 & 4 & 125 & 3 & 145 & 2 & 5 \\ 34 & 15 & 23 & 45 & 12 & 35 & 14 & 25 & \end{array}$$

Now let's check $|S| \geq 8$ is sufficient. Consider a chain on a set of size $|S| = 7$. (We need $|S| \geq 7$ else $2^{|S|} < 100$.) Observe that there are sets of size ≥ 4 can only be neighbored by sets of size ≤ 2 , of which there are $\binom{7}{1} + \binom{7}{2} = 28$. So there are ≤ 30 sets of size ≥ 4 . Also, there are $\binom{7}{3} = 35$ sets of size 3. So the total number of sets in a chain can be at most $30 + 28 + 35 = 93 < 100$.

Solution 2

My proof that $|S| \geq 8$ is basically the same as the one above. Here is another construction for $|S| = 8$ that I like because it works with remainders and it's pretty intuitive. The basic idea is to assign different subsets to different remainders when divided by particular numbers, and then to use the Chinese Remainder Theorem to show that all of the subsets are distinct. The motivation for this comes from the fact that we want X_i and X_{i+1} to always be disjoint, so remainders are a great way to systematically make that happen, since i and $i + 1$ do not have the same remainder modulo any positive integer greater than 1. Anyway, here is the construction:

Let $S = \{1, 2, \dots, 8\}$. For $i = 1, 2, \dots, 98$, we will choose which elements of the set $\{1, 2, 3, 4\}$ belong to X_i based on the remainder of i modulo 9, and we will choose which elements of the set $\{5, 6, 7, 8\}$ belong to X_i based on the remainder of i modulo 11. We do this as follows:

$$\begin{array}{ll}
 \emptyset & i \equiv 0 \pmod{9} \\
 \{1, 2\} & i \equiv 1 \pmod{9} \\
 \{3\} & i \equiv 2 \pmod{9} \\
 \{1, 4\} & i \equiv 3 \pmod{9} \\
 X_i \cap \{1, 2, 3, 4\} = \{2\} & i \equiv 4 \pmod{9} \\
 \{3, 4\} & i \equiv 5 \pmod{9} \\
 \{1\} & i \equiv 6 \pmod{9} \\
 \{2, 3\} & i \equiv 7 \pmod{9} \\
 \{4\} & i \equiv 8 \pmod{9}
 \end{array}$$

$$\begin{array}{ll}
 \emptyset & i \equiv 0 \pmod{11} \\
 \{5, 6\} & i \equiv 1 \pmod{11} \\
 \{7, 8\} & i \equiv 2 \pmod{11} \\
 \{5\} & i \equiv 3 \pmod{11} \\
 \{6, 8\} & i \equiv 4 \pmod{11} \\
 X_i \cap \{5, 6, 7, 8\} = \{5, 7\} & i \equiv 5 \pmod{11} \\
 \{6\} & i \equiv 6 \pmod{11} \\
 \{5, 8\} & i \equiv 7 \pmod{11} \\
 \{6, 7\} & i \equiv 8 \pmod{11} \\
 \{8\} & i \equiv 9 \pmod{11} \\
 \{7\} & i \equiv 10 \pmod{11}
 \end{array}$$

Finally, we specially define $X_{99} = \{1, 2, 3\}$ and $X_{100} = \{5, 6, 7\}$.

It is relatively easy to see that this configuration satisfies all of the desired conditions. We see that $X_{98} = \{4, 7\}$, so X_{98} and X_{99} are disjoint, as are X_{99} and X_{100} . The remainder configuration above takes care of the rest, so any two consecutive sets are disjoint. Then, by the Chinese Remainder Theorem, no two integers from 1 to 98 have the same combination of residues modulo 9 and modulo 11, so all of the sets X_i are distinct for $i = 1, 2, \dots, 98$. It is also easy to verify that none of these match X_{99} or X_{100} , since they all have at most two elements of $\{1, 2, 3, 4\}$ and at most two elements of $\{5, 6, 7, 8\}$, whereas X_{99} and X_{100} do not satisfy this; hence all of the sets are distinct. Finally, notice that, for any pair of consecutive sets, at least one of them has at most 3 elements, while the other has at most 4. Thus, their union always has at most 7 elements, so $X_i \cup X_{i+1} \neq S$ for all $i = 1, 2, \dots, 99$.

All of the conditions are satisfied, so this configuration works. We thus conclude that $\min(|S|) = 8$.

2. (USAMO 2016) Prove that for any positive integer k ,

$$(k^2)! \cdot \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}$$

is an integer.

Solution 1

Note that the expression that we're trying to prove is an integer, call it N , is rational.

We will show that, for every prime p , the **p-adic valuation** of N , written as $v_p(N)$ is non-negative. This implies that N is an integer.

$v_p(N) = \sum_{i=1}^{\infty} \left\lfloor \frac{k^2}{p^i} \right\rfloor + \sum_{j=0}^{k-1} \sum_{i=1}^{\infty} \left\lfloor \frac{j}{p^i} \right\rfloor - \sum_{j=k}^{2k-1} \sum_{i=1}^{\infty} \left\lfloor \frac{j}{p^i} \right\rfloor$, by **Legendre's Formula**.
 $\left\lfloor \frac{x}{p} \right\rfloor = \frac{x - (x \bmod p)}{p}$, and $\sum_{i=0}^{k-1} (i \bmod m) \leq \sum_{i=k}^{2k-1} (i \bmod m)$. (We take out groups of m which are cyclic permutations of numbers 1 to m until there are less than m left, then we have m distinct values, and the minimum possible sum of leftovers is attained on the left hand side as $\sum_{i=0}^{(k-1) \bmod m} i$).

Thus, $v_p(N) = \sum_{m=p^i, i \in \mathbb{N}_+} -\frac{k^2}{m} + \left\lfloor \frac{k^2}{m} \right\rfloor - \frac{\sum_{i=0}^{k-1} r(i, m) - \sum_{i=k}^{2k-1} r(i, m)}{m} \geq \sum_{m=p^i, i \in \mathbb{N}} \left\lceil -\frac{k^2}{m} + \left\lfloor \frac{k^2}{m} \right\rfloor \right\rceil \geq 0$, as the term in each summand is a sum of floors also and is clearly an integer.

Solution 2 (Controversial)

Consider an $k \times k$ grid, which is to be filled with the integers 1 through k^2 such that the numbers in each row are in increasing order from left to right, and such that the numbers in each column are in increasing order from bottom to top. In other words, we are creating an $k \times k$ standard Young tableaux.

The Hook Length Formula is the source of the controversy, as it is very powerful and trivializes this problem. The Hook Length Formula states that the number of ways to create this standard Young tableaux (call this N for convenience) is:

$$N = \frac{(k^2)!}{\prod_{1 \leq i, j \leq k} (i+j-1)}.$$

Now, we do some simple rearrangement:

$$\begin{aligned} N &= (k^2)! \cdot \prod_{j=1}^k \prod_{i=1}^k \frac{1}{i+j-1} = (k^2)! \cdot \prod_{j=1}^k \frac{(j-1)!}{(j+k-1)!} \\ &= (k^2)! \cdot \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}. \end{aligned}$$

This is exactly the expression given in the problem! Since the expression given in the problem equals the number of distinct $k \times k$ standard Young tableaux, it must be an integer, so we are done.

Solution 3 (Induction, Incorrect?)

Note: This is incorrect. (Why?)

Define

$$A(k) = (k^2)! \cdot \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}.$$

Clearly, $A(1) = 1$ and $A(2) = 2$.

Then

$$\frac{A(k+1)}{A(k)} = \frac{(k^2 + 2k + 1)! \cdot \prod_{j=0}^k \frac{j!}{(j+k+1)!}}{(k^2)! \cdot \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}}.$$

Lots of terms cancel, and we are left with

$$\frac{A(k+1)}{A(k)} = \frac{(k^2 + 1)(k^2 + 2) \cdots (k^2 + 2k + 1)}{2(2k + 1)}.$$

The numerator has $2k + 1$ consecutive positive integers, so one of them must be divisible by $(2k + 1)$. Also, there are $2k$ terms left, k of which are even. We can choose one of these to cancel out the 2 in the denominator. Therefore, the ratio between $A(k + 1)$ and $A(k)$ is an integer. By our inductive hypothesis, $A(k)$ is an integer. Therefore, $A(k + 1)$ is as well, and we are done.

Solution 4

Let us work in the ring of polynomials $\mathbb{Z}[q]$. Define

$$[n]_q = 1 + q + \cdots + q^{n-1}, \quad [n]_q! = \prod_{i=1}^n [i]_q.$$

Since

$$[n]_q! = \prod_{i=1}^n \frac{1 - q^i}{1 - q} = \prod_{d=1}^n \frac{\Phi_d(q)^{\lfloor n/d \rfloor}}{1 - q},$$

we have

$$\nu_d([n]_q!) = \lfloor n/d \rfloor.$$

Set

$$P_k(q) = [k^2]_q! \prod_{j=0}^{k-1} \frac{[j]_q!}{[j+k]_q!}.$$

Then

$$\begin{aligned} \nu_1(P_k(q)) &= 0 \\ \nu_d(P_k(q)) &= \lfloor k^2/d \rfloor + \sum_{j=0}^{k-1} \lfloor j/d \rfloor - \sum_{j=0}^{k-1} \lfloor (j+k)/d \rfloor. \end{aligned}$$

Combine the last two sums:

$$\sum_{j=0}^{k-1} \lfloor j/d \rfloor - \sum_{j=0}^{k-1} \lfloor (j+k)/d \rfloor = - \sum_{j=0}^{k-1} (\lfloor (j+k)/d \rfloor - \lfloor j/d \rfloor).$$

Hence

$$\nu_d(P_k(q)) = \left\lfloor \frac{k^2}{d} \right\rfloor - \sum_{j=0}^{k-1} \left(\left\lfloor \frac{j+k}{d} \right\rfloor - \left\lfloor \frac{j}{d} \right\rfloor \right).$$

But for each $0 \leq j < k$, the difference $\lfloor \frac{j+k}{d} \rfloor - \lfloor \frac{j}{d} \rfloor$ counts the multiples of d in the interval $(j, j+k]$. As j runs from 0 to $k-1$, these k intervals lie inside $\{1, 2, \dots, k^2\}$, so

$$\sum_{j=0}^{k-1} \left(\left\lfloor \frac{j+k}{d} \right\rfloor - \left\lfloor \frac{j}{d} \right\rfloor \right) \leq \left\lfloor \frac{k^2}{d} \right\rfloor.$$

It follows that $\nu_d(P_k(q)) \geq 0$, hence $P_k(q) \in \mathbb{Z}[q]$. Evaluating at $q = 1$ completes the proof.

Note: This solution is fundamentally different from the first, which works purely in the integers. Here, we work in the integer polynomial ring $\mathbb{Z}[q]$ —a graded algebra—which allows us to test integrality by tracking the multiplicities of better organized elements like $\Phi_d(q)$ for all integers $1 \leq d \leq k^2$. In contrast, working in $\mathbb{Z}$ —a non-graded algebra—requires using p -adic valuations via Legendre’s formula, which only considers primes which are less organized. The graded structure of $\mathbb{Z}[q]$ simplifies the analysis, making it a powerful strategy to lift problems into a graded algebra whenever possible.

Solution 5 (Structural Proof without polynomials)

This is the same proof strategy as Solution 4, but without the polynomial machinery, and doesn’t rely on unstated theorems of **cyclotomic polynomials**, and proves the hard part of the problem: the interval lemma that Solution 1 proves but Solution 4 merely states.

Let U be the expression (not the numerical value) given in the problem, with the numerator and denominator separated:

$$U = \frac{(k^2)! \cdot \prod_{j=0}^{k-1} j!}{\prod_{j=0}^{k-1} (j+k)!}$$

Let $M_E(d)$ represents the *net count* of multiples of an integer d in an multiplicative expression E (not a number itself!)

$$M_U(d) = \left(\left\lfloor \frac{k^2}{d} \right\rfloor + \sum_{j=0}^{k-1} \left\lfloor \frac{j}{d} \right\rfloor \right) - \sum_{j=0}^{k-1} \left\lfloor \frac{j+k}{d} \right\rfloor$$

To prove N is an integer, it suffices to show that for every prime p , the exponent of p in the prime factorization of N , denoted $v_p(U)$, is non-negative.

Equivalently, $v_p(\text{numerator}(U)) \geq_p (\text{denominator}(U))$

Step 1: Decomposing the Valuation

By Legendre’s Formula, the exponent of a prime p in $n!$ is $v_p(n!) = \sum_{i=1}^{\infty} \lfloor \frac{n}{p^i} \rfloor$. However, we won’t sum these directly. Instead, we each prime power p^i separately count the number of factors that are multiples.

Applying this to our expression, the total exponent $v_p(U)$ is the sum of unit contributions from each power of p :

$$v_p(U) = \sum_{i=1}^{\infty} M(p^i)$$

Step 2: The Structural Inequality

Instead of analyzing the infinite sum for $v_p(U)$ directly, we will prove a stronger condition: that **each individual term** $M_U(d)$ is non-negative for any integer $d \geq 1$.

We rewrite the sum terms in $M_U(d)$:

$$\sum_{j=0}^{k-1} \left\lfloor \frac{j}{d} \right\rfloor - \sum_{j=0}^{k-1} \left\lfloor \frac{j+k}{d} \right\rfloor = - \sum_{j=0}^{k-1} \left(\left\lfloor \frac{j+k}{d} \right\rfloor - \left\lfloor \frac{j}{d} \right\rfloor \right)$$

Thus, the expression for $M_U(d)$ becomes:

$$M_U(d) = \left\lfloor \frac{k^2}{d} \right\rfloor - \sum_{j=0}^{k-1} \left(\left\lfloor \frac{j+k}{d} \right\rfloor - \left\lfloor \frac{j}{d} \right\rfloor \right)$$

Step 3: Algebraic Verification Let $k = qd + r$, where $0 \leq r < d$.

The term inside the summation, $\left\lfloor \frac{j+k}{d} \right\rfloor - \left\lfloor \frac{j}{d} \right\rfloor$, counts how many multiples of d lie in the half-open interval $(j, j+k]$. Since the interval has length k , this count is either q (if no *extra* multiple fits) or $q+1$ (if an extra multiple fits).

Specifically, the *extra* multiple ($q+1$) occurs exactly when the interval *wraps around* a multiple of d in the remainder arithmetic. Thus:

$$\sum_{j=0}^{k-1} \left(\left\lfloor \frac{j+k}{d} \right\rfloor - \left\lfloor \frac{j}{d} \right\rfloor \right) = k \cdot q + \max(0, 2r - d)$$

$M_U(d)$'s leading term $\left\lfloor \frac{k^2}{d} \right\rfloor$ is:

$$\left\lfloor \frac{k^2}{d} \right\rfloor = \left\lfloor \frac{(qd+r)^2}{d} \right\rfloor = q^2d + 2qr + \left\lfloor \frac{r^2}{d} \right\rfloor = k \cdot q + qr + \left\lfloor \frac{r^2}{d} \right\rfloor$$

Thus the condition $\left[M_U(d) = \left(k \cdot q + qr + \left\lfloor \frac{r^2}{d} \right\rfloor \right) - (k \cdot q + \max(0, 2r - d)) \geq 0 \right]$ is equivalent to:

$$qr + \left\lfloor \frac{r^2}{d} \right\rfloor \geq \max(0, 2r - d)$$

We analyze two cases:

- **Case 1** ($2r \leq d$): The RHS is 0. Since $q, r, d \geq 0$, the LHS is clearly ≥ 0 . The inequality holds.
- **Case 2** ($2r > d$): The RHS is $2r - d$. We need to show $\left\lfloor \frac{r^2}{d} \right\rfloor \geq 2r - d$.

Since the LHS is an integer, this is equivalent to showing $\frac{r^2}{d} \geq 2r - d$, or $r^2 \geq 2rd - d^2$. Rearranging gives:

$$(r - d)^2 \geq 0$$

This is the Trivial Inequality.

Step 4: Conclusion

We have proven that $M(d) \geq 0$ for **every** integer d . Therefore, for any prime p , the total valuation is a sum of non-negative terms:

$$v_p(N) = \sum_{i=1}^{\infty} M(p^i) \geq 0$$

Since $v_p(N) \geq 0$ for all primes p , N is an integer.

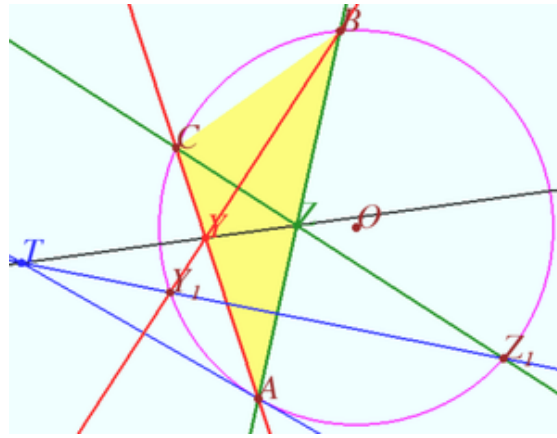
3. (USAMO 2016) Let $\triangle ABC$ be an acute triangle, and let I_B, I_C , and O denote its B -excenter, C -excenter, and circumcenter, respectively. Points E and Y are selected on $\overline{AC}$ such that $\angle ABY = \angle CBY$ and $\overline{BE} \perp \overline{AC}$. Similarly, points F and Z are selected on $\overline{AB}$ such that $\angle ACZ = \angle BCZ$ and $\overline{CF} \perp \overline{AB}$. Lines $\overleftrightarrow{I_B F}$ and $\overleftrightarrow{I_C E}$ meet at P . Prove that $\overline{PO}$ and $\overline{YZ}$ are perpendicular.

Solution

This problem can be proved in the following two steps.

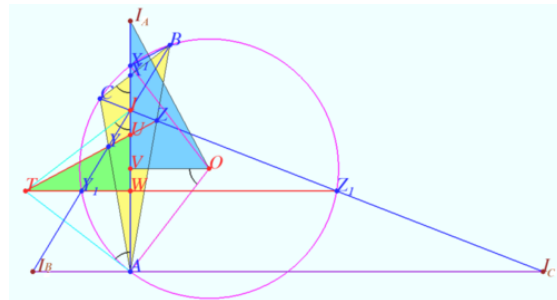
1. Let I_A be the A -excenter, then I_A, O , and P are colinear. This can be proved by the Trigonometric Form of Ceva's Theorem for $\triangle I_A I_B I_C$.
2. Show that $I_A Y^2 - I_A Z^2 = OY^2 - OZ^2$, which implies $\overline{OI_A} \perp \overline{YZ}$. This can be proved by multiple applications of the Pythagorean Thm.

Solution 2



We find point T on line YZ , we prove that $TY \perp OI_A$ and state that P is the point $X(24)$ from ENCYCLOPEDIA OF TRIANGLE, therefore $P \in OI_A$.

Let ω be circumcircle of $\triangle ABC$ centered at O . Let Y_1 , and Z_1 be crosspoints of ω and BY , and CZ , respectively. Let T be crosspoint of YZ and $Y_1 Z_1$. In accordance the Pascal theorem for pentagon $AZ_1 B C Y_1$, AT is tangent to ω at A .



Let I_A, I_B, I_C be A, B , and C -excenters of $\triangle ABC$. Denote

$$a = BC, b = AC, c = AB, 2\alpha = \angle CAB, 2\beta = \angle ABC, 2\gamma = \angle ACB,$$

$$\psi = 90^\circ - \gamma + \beta, X = AI_A \cap \omega, X_1 = BC \cap AI_A,$$

$$I = BI_B \cap CI_C, U = YZ \cap AI_A, W = Y_1 Z_1 \cap AI_A,$$

V is the foot of perpendicular from O to AI_A .

I is orthocenter of $\triangle I_A I_B I_C$ and incenter of $\triangle ABC$.

ω is the Nine-point circle of $\triangle I_A I_B I_C$.

Y_1 is the midpoint of II_B , Z_1 is the midpoint of II_C in accordance with property of Nine-point circle $\implies$

$$Y_1 Z_1 \parallel I_B A I_C \parallel VO, IW = AW \implies TW \perp AI.$$

$$\angle AXC = 180^\circ - 2\gamma \sim \alpha = 90^\circ - \gamma + \beta = \psi.$$

$$\angle TAI = \angle VOA = 2\beta + \alpha = 90^\circ - \gamma + \beta = \psi.$$

$$I_A X_1 = IX_1 = BX_1 = 2R \sin \alpha \implies$$

$$\cot \angle OI_A A = \frac{VI_A}{VO} = \frac{R \sin \psi + 2R \sin \alpha}{R \cos \psi} = \tan \psi + \frac{2 \sin \alpha}{\cos \psi}.$$

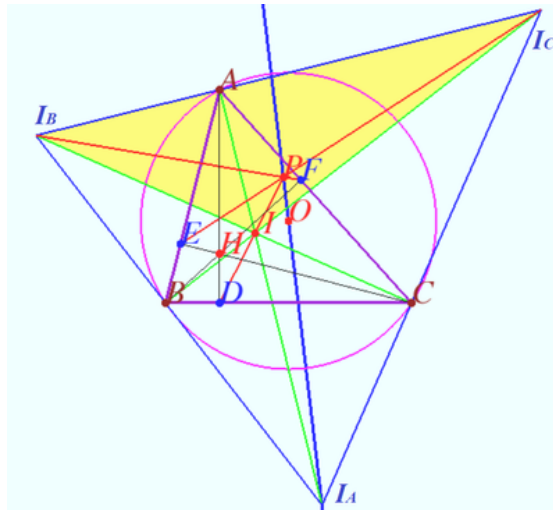
$$CX = \frac{ab}{b+c} \implies \frac{AI}{IX} = \frac{AC}{CX} = \frac{b+c}{a} \implies AI = AX \frac{b+c}{a+b+c},$$

$$AW = \frac{AI}{2}, UW = AU \sim AW,$$

In $\triangle ABC$ segment YZ cross segment $AX \implies \frac{AU}{UX} = \frac{m+nk}{k+1}$, where $n = \frac{a}{b}$, $m = \frac{a}{c}$, $k = \frac{b}{c}$,

$$\frac{AU}{UX} = \frac{2a}{b+c} \implies AU = AX \cdot \frac{b+c}{2a+b+c}.$$

$$\frac{AU \sim AW}{AW} = \frac{b+c}{2a+b+c}.$$



$$\begin{aligned} \cot \angle UTW &= \frac{TW}{UW} = \frac{AW \cdot \tan \psi}{AU \sim AW} = \tan \psi \cdot \frac{2a+b+c}{b+c} = \\ &= \tan \psi \cdot \frac{2a}{b+c} + \tan \psi = \frac{2 \sin \alpha}{\sin \psi} \tan \psi + \tan \psi = \tan \psi + \frac{2 \sin \alpha}{\cos \psi} \end{aligned}$$

$$\implies \angle UTW = \angle OI_A A.$$

$$TW \perp AI_A \implies TYZ \perp OI_A.$$

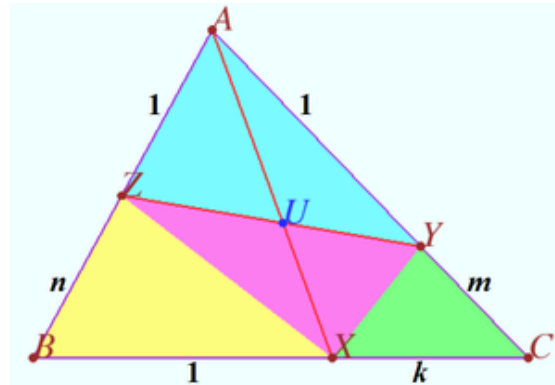
Let $\triangle I I_B I_C$ be the base triangle with orthocenter I_A , center of Nine-points circle $O \Rightarrow O I_A$ be the Euler line of $\triangle I I_B I_C$.

$\triangle ABC$ is orthic triangle of $\triangle I I_B I_C$,

$\triangle DEF$ is orthic-of-orthic triangle.

P is perspector of base triangle and orthic-of-orthic triangle.

Therefore P is point $X(24)$ of ENCYCLOPEDIA OF TRIANGLE CENTERS which lies on Euler line of the base triangle. [\[1\]](#)



Claim

$$\frac{AU}{UX} = \frac{m + nk}{k + 1}.$$

Proof

$$\begin{aligned} \frac{[AYZ]}{[ABC]} &= \frac{AZ \cdot AY}{AB \cdot AC} = \frac{1}{(n+1) \cdot (m+1)}, \\ \frac{[BXZ]}{[ABC]} &= \frac{BZ \cdot BX}{AB \cdot BC} = \frac{n}{(n+1) \cdot (k+1)}, \\ \frac{[CXY]}{[ABC]} &= \frac{CY \cdot CX}{AC \cdot BC} = \frac{mk}{(m+1) \cdot (k+1)}, \\ \frac{[XYZ]}{[ABC]} &= 1 - \frac{[AYZ]}{[ABC]} - \frac{[BXZ]}{[ABC]} - \frac{[CXY]}{[ABC]} = \frac{m + nk}{(m+1) \cdot (k+1) \cdot (n+1)}, \\ \frac{AU}{UX} &= \frac{[AYZ]}{[XYZ]} = \frac{m + nk}{k + 1}. \end{aligned}$$

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4. (USAMO 2016) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real numbers x and y ,

$$(f(x) + xy) \cdot f(x - 3y) + (f(y) + xy) \cdot f(3x - y) = (f(x + y))^2.$$

Solution 1

Step 1: Set $x = y = 0$ to obtain $f(0) = 0$.

Step 2: Set $x = 0$ to obtain $f(y)f(-y) = f(y)^2$.

In particular, if $f(y) \neq 0$ then $f(y) = f(-y)$.

In addition, replacing $y \rightarrow -t$, it follows that $f(t) = 0 \Rightarrow f(-t) = 0$ for all $t \in \mathbb{R}$.

Step 3: Set $x = 3y$ to obtain $[f(y) + 3y^2] f(8y) = f(4y)^2$.

In particular, replacing $y \rightarrow t/8$, it follows that $f(t) = 0 \implies f(t/2) = 0$ for all $t \in \mathbb{R}$.

Step 4: Set $y = -x$ to obtain $f(4x) [f(x) + f(-x) - 2x^2] = 0$.

In particular, if $f(x) \neq 0$, then $f(4x) \neq 0$ by the observation from Step 3, because $f(4x) = 0 \implies f(2x) = 0 \implies f(x) = 0$. Hence, the above equation implies that $2x^2 = f(x) + f(-x) = 2f(x)$, where the last step follows from the first observation from Step 2.

Therefore, either $f(x) = 0$ or $f(x) = x^2$ for each x .

Looking back on the equation from Step 3, it follows that $f(y) + 3y^2 \neq 0$ for any nonzero y . Therefore, replacing $y \rightarrow t/4$ in this equation, it follows that $f(t) = 0 \implies f(2t) = 0$.

Step 5: If $f(a) = f(b) = 0$, then $f(b - a) = 0$.

This follows by choosing x, y such that $x - 3y = a$ and $3x - y = b$. Then $x + y = \frac{b-a}{2}$, so plugging x, y into the given equation, we deduce that $f\left(\frac{b-a}{2}\right) = 0$. Therefore, by the third observation from Step 4, we obtain $f(b - a) = 0$, as desired.

Step 6: If $f \not\equiv 0$, then $f(t) = 0 \implies t = 0$.

Suppose by way of contradiction that there exists a nonzero t with $f(t) = 0$. Choose x, y such that $f(x) \neq 0$ and $x + y = t$. The following three facts are crucial:

1. $f(y) \neq 0$. This is because $(x + y) - y = x$, so by Step 5, $f(y) = 0 \implies f(x) = 0$, impossible.
2. $f(x - 3y) \neq 0$. This is because $(x + y) - (x - 3y) = 4y$, so by Step 5 and the observation from Step 3, $f(x - 3y) = 0 \implies f(4y) = 0 \implies f(2y) = 0 \implies f(y) = 0$, impossible.
3. $f(3x - y) \neq 0$. This is because by the second observation from Step 2, $f(3x - y) = 0 \implies f(y - 3x) = 0$. Then because $(x + y) - (y - 3x) = 4x$, Step 5 together with the observation from Step 3 yield $f(3x - y) = 0 \implies f(4x) = 0 \implies f(2x) = 0 \implies f(x) = 0$, impossible.

By the second observation from Step 4, these three facts imply that $f(y) = y^2$ and $f(x - 3y) = (x - 3y)^2$ and $f(3x - y) = (3x - y)^2$. By plugging into the given equation, it follows that

$$(x^2 + xy)(x - 3y)^2 + (y^2 + xy)(3x - y)^2 = 0.$$

But the above expression miraculously factors into $(x + y)^4$! This is clearly a contradiction, since $t = x + y \neq 0$ by assumption. This completes Step 6.

Step 7: By Step 6 and the second observation from Step 4, the only possible solutions are $f \equiv 0$ and $f(x) = x^2$ for all $x \in \mathbb{R}$. It's easy to check that both of these work, so we're done.

Alternative Solution 1 From steps 1 and 2 of Solution 1 we have that $f(0) = 0$, and $f(-y) \cdot f(y) = (f(y))^2$. Therefore, if $f(y) \neq 0$, then $f(y) = f(-y)$. Furthermore, setting $y = -x$ gives us $(f(x) - x^2) \cdot f(4x) + (f(-x) - x^2) \cdot f(4x) = f(0) = 0$. The LHS can be factored as $(f(x) + f(-x) - 2x^2) \cdot f(4x) = 0$. In particular, if $f(4x) \neq 0$, then we have $f(x) + f(-x) = 2x^2$. However, since we have from step 2 that $f(x) = f(-x)$, assuming $f(x) \neq 0$, the equation becomes $2f(x) = 2x^2$, so

for every x , $f(x)$ is equivalent to either 0 or x^2 . From step 6 of Solution 1, we can prove that $f(x) = 0$, and $f(x) = x^2$ are the only possible solutions.

Solution 2

Step 1: $x = y = 0 \implies f(0) = 0$

Step 2: $x = 0 \implies f(y)f(-y) = f(y)^2$. Now, assume $y \neq 0$. Then, if $f(y) = 0$, we substitute in $-y$ to get $f(y)f(-y) = f(-y)^2$, or $f(y) = f(-y) = 0$. Otherwise, we divide both sides by $f(y)$ to get $f(y) = f(-y)$. If $y = 0$, we obviously have $f(0) = f(0)$. Thus, the function is even..Step 3: $y = -x \implies 2f(4x)(f(x) - x^2) = 0$. Thus, $\forall x$, we have $f(4x) = 0$ or $f(x) = x^2$.

Step 4: We now assume $f(x) \neq 0$, $x \neq 0$. We have $f(\frac{x}{4}) = \frac{x^2}{16}$. Now, setting $x = y = \frac{x}{4}$, we have $f(\frac{x}{2}) = \frac{x^2}{4}$ or $f(\frac{x}{2}) = 0$. The former implies that $f(x) = 0$ or x^2 . The latter implies that $f(x) = 0$ or $f(x) = \frac{x^2}{2}$. Assume the latter. $y = -2x \implies -\frac{3y^2}{2}f(7y) - 2y^2f(5y) = \frac{x^2}{2}$. Clearly, this implies that $f(x)$ is negative for some m . Now, we have $f(\frac{m}{4}) = \frac{m^2}{16} \implies f(\frac{m}{2}) = 0$, $\frac{m^2}{4} \implies f(m) \geq 0$, which is a contradiction. Thus, $\forall x f(x) = 0$ or $f(x) = x^2$.

Step 5: We now assume $f(x) = 0$, $f(y) = y^2$ for some $x, y \neq 0$. Let m be sufficiently large integer, let $z = |4^m x|$ and take the absolute value of y (since the function is even). Choose c such that $3z - c = y$. Note that we have $\frac{c}{z} \rightarrow 3$ and $\frac{y}{z} \rightarrow 0$. Note that $f(z) = 0$. Now, $x = z, y = c \implies$ LHS is positive, as the second term is positive and the first term is nonnegative and thus the right term is equal to $(z + c)^4 - 256z^4$. Now if $f(z - 3c) = 0$, the second term of the LHS/RHS clearly $\rightarrow 0$ as $m \rightarrow \infty$. if $f(z - 3c) \neq 0$, then we have LHS/RHS $\rightarrow 0$, otherwise, we have LHS/RHS $\rightarrow \frac{8^2 \cdot 3z^4}{256z^4} = \frac{3}{4}$, a contradiction, as we're clearly not dividing by 0, and we should have LHS/RHS=1.

5. (USAMO 2016) An equilateral pentagon $AMNPQ$ is inscribed in triangle ABC such that $M \in \overline{AB}$, $Q \in \overline{AC}$, and $N, P \in \overline{BC}$. Let S be the intersection of $\overleftrightarrow{MN}$ and $\overleftrightarrow{PQ}$. Denote by ℓ the angle bisector of $\angle MSQ$.

Prove that $\overline{OI}$ is parallel to ℓ , where O is the circumcenter of triangle ABC , and I is the incenter of triangle ABC .

Solution 1

Let D be the intersection of line AI and the circumcircle of $\triangle ABC$ (other than A), then $OD \perp BC$. Let R be the point such that $NPQR$ is a rhombus. It follows that $OD \perp QR$.

Since $AM = AQ$, $AI \perp MQ$, or $DI \perp MQ$. It follows that $\angle ODI = \angle RQM$.

Since $BO = OD$, $MA = AQ$, $\angle BOD = \angle MAQ$, it follows that $\triangle BOD \sim \triangle MAQ$, so $AQ/MQ = OD/BD$.

It is given that $AQ = NP = RQ$, and by basic properties of the incenter, $ID = BD$. Therefore, $RQ/MQ = OD/ID$, so $\triangle RQM \sim \triangle ODI$. Since the rotation between the two triangles is 90 degrees, $OI \perp MR$. However, ℓ is parallel to the bisector of $\angle MNR$, which is perpendicular to MR , so we are done.

Solution 2

Write $\angle JKL = K$ for all J, K, L chosen as distinct vertices of triangle ABC . Define a, b, c as sides opposite to angles A, B , and C , respectively. Place the triangle in the Euclidean plane with A at the origin and C on the positive x-axis. Assume without loss of generality that C is acute.

Consider the sides of the pentagon as vectors and note that

$$\overrightarrow{AM} + \overrightarrow{MN} + \overrightarrow{NP} = \overrightarrow{AQ} + \overrightarrow{QP} \quad (1)$$

Define δ and γ as the angles made between the positive x-axis and $\overrightarrow{MN}$ and $\overrightarrow{QP}$, respectively. Considering the x and y coordinates of the vectors in (1), it follows that

$$\cos \delta - \cos \gamma = 1 - \cos A - \cos C \quad (2)$$

$$\sin \delta - \sin \gamma = \sin C - \sin A \quad (3)$$

Suppose $\sin C - \sin A = 0$. Then $A = C$, and the triangle is isosceles. In this case, it is clear by symmetry that $\overrightarrow{OI}$ is vertical. Further, since point S exists, $\delta \neq \gamma$, so $\delta + \gamma = 180$ and $\overrightarrow{MN} + \overrightarrow{QP}$ must be vertical as well.

For the remainder of the proof, assume $\sin C \neq \sin A$. Note that

$$\frac{\cos y - \cos x}{\sin x - \sin y} = \frac{(\cos y - \cos x)(\sin x + \sin y)}{(\sin x - \sin y)(\sin x + \sin y)} = \frac{\sin x + \sin y}{\cos x + \cos y}$$

whenever $x, y \in \mathbb{R}$ and $\sin x \neq \sin y$. Note further that the slope of the line defined by the vector formed by summing vectors $(\cos x, \sin x)$ and $(\cos y, \sin y)$ is this expression. Since ℓ is parallel to $\overrightarrow{MN} + \overrightarrow{QP}$, the slope of ℓ can be formed by dividing expressions in (2) and (3) and inverting the sign:

$$\frac{\cos A + \cos C}{\sin C - \sin A} \quad (4)$$

Determine the coordinates of I by drawing perpendiculars from I to the sides and vertices of the triangle. By exploiting congruence between pairs of right triangles that share a vertex, one can partition a, b, c into $p + q, p + r, q + r$ where p, q, r are bases of these triangles that lie on the sides of triangle ABC . From here it is clear that $I = \left(\frac{b+c-a}{2}, \frac{b+c-a}{2} \tan(A/2) \right)$.

To find the coordinates of O , note that $\angle OJK + \angle OJL = \angle J$ and that $\angle OJK = \angle OKJ$ in any acute triangle JKL . It easily follows that $\angle OJK = 90^\circ - L$. Note also that the perpendicular from O to $\overline{JK}$ bisects $\overline{JK}$. Hence,

$$O = \left(\frac{b}{2}, \frac{b}{2} \cot B \right) \quad (5)$$

if triangle ABC is acute.

If triangle JKL is obtuse at $\angle J$, then it can be similarly shown that $\angle OKL = \angle OLK = 90^\circ - L$ but that the remaining angles of this form are still $90^\circ - L$ and $90^\circ - K$. It easily follows that (5) holds if $\angle A$ is obtuse. If $\angle B$ is obtuse then $\angle OAC = 90^\circ - B$ and the y coordinate of O is $-\frac{b}{2} \tan B - 90$. From this, (5) follows in this case as well.

We can conclude the slope of $\overrightarrow{OI}$ is

$$\left(\frac{b \cot B (b + c - a) \tan(A/2)}{a - c} \right) = \left(\frac{\cos B \tan(A/2) \sin B}{\sin A - \sin C} \right) + \tan(A/2) \quad (6)$$

by the Law of Sines and rearrangement.

Setting (6) = (4) is equivalent to

$$1 - \cos A - \cos C = \cos B + \tan(A/2)(\sin A - \sin B - \sin C)$$

Since $\tan(A/2) = \frac{\sin A}{1 + \cos A}$, this equation is equivalent to

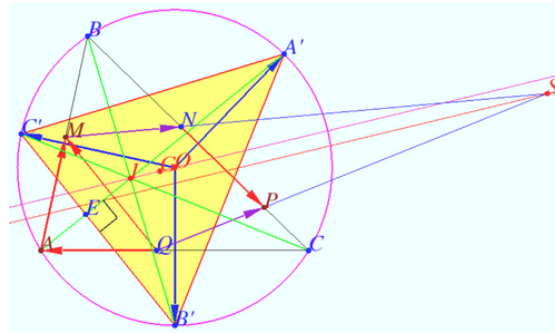
$$(1 + \cos A)(\cos A + \cos B + \cos C) = (\sin A)(\sin B + \sin C - \sin A)$$

This equation is equivalent to

$$\cos(A + B) + \cos(A + C) + \cos B + \cos C = 0$$

which is evident.

Solution 3



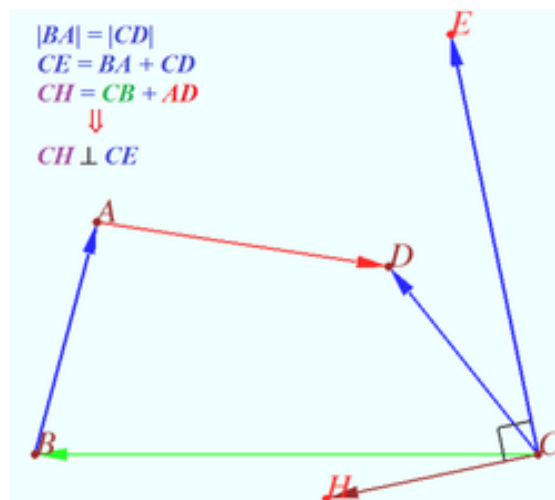
Let A', B' , and C' be the arc midpoints of BC, CA, AB , respectively. Let E be crosspoint of AI and $B'C'$.

Therefore O is the circumcenter of triangle $A'B'C'$.

Points A, I, E , and A' are collinear. $\angle A'EB' = \frac{\widehat{AC'} + \widehat{B'C} + \widehat{CA'}}{2} = 90^\circ \implies AA' \perp B'C' \implies I$ is orthocenter of $\triangle A'B'C' \implies$

IO is the Euler line of $\triangle A'B'C'$.

Let G be the centroid of $\triangle A'B'C' \implies G$ lies on line $IO \implies \overline{OG} = \frac{\overline{OA'} + \overline{OB'} + \overline{OC'}}{3}$ is parallel to $\overline{OI}$. $\overline{OB'} \perp \overline{AC} \implies \overline{OB'} \perp \overline{QA}$.



Similarly $\overline{OC'} \perp \overline{AM}$, $\overline{OA'} \perp \overline{NP}$, rotation from $\overline{OA'}$ to $\overline{NP}$, from $\overline{OB'}$ to $\overline{QA}$, and from $\overline{OC'}$ to $\overline{AM}$ is in clockwise direction, $|\overline{QA}| = |\overline{AM}| = |\overline{NP}|$, $|\overline{OA'}| = |\overline{OB'}| = |\overline{OC'}| \implies$

$\overline{QA} + \overline{AM} + \overline{NP} = \overline{QM} + \overline{NP}$ is perpendicular to $\overline{OI}$.

$|\overline{MN}| = |\overline{QP}|$ therefore in accordance with **Claim** $\overline{MN} + \overline{QP}$ is parallel to $\overline{OI}$.

This sum is parallel to ℓ , so we are done.

Claim

Let $|\overline{BA}| = |\overline{CD}|$, $\overline{CB} + \overline{BA} + \overline{AD} = \overline{CD}$. Then $(\overline{BA} + \overline{CD}) \perp (\overline{CB} + \overline{AD})$.

Proof

$$(\overline{BA} + \overline{CD}) \cdot (\overline{CB} + \overline{AD}) = (\overline{BA} + \overline{CD}) \cdot (\overline{CB} + \overline{CD} \sim \overline{CB} \sim \overline{BA}) = |\overline{CD}|^2 \sim |\overline{BA}|^2 = 0.$$

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6. (USAMO 2016) Integers n and k are given, with $n \geq k \geq 2$. You play the following game against an evil wizard.

The wizard has $2n$ cards; for each $i = 1, \dots, n$, there are two cards labeled i . Initially, the wizard places all cards face down in a row, in unknown order.

You may repeatedly make moves of the following form: you point to any k of the cards. The wizard then turns those cards face up. If any two of the cards match, the game is over and you win. Otherwise, you must look away, while the wizard arbitrarily permutes the k chosen cards and turns them back face-down. Then, it is your turn again.

We say this game is *winnable* if there exist some positive integer m and some strategy that is guaranteed to win in at most m moves, no matter how the wizard responds.

For which values of n and k is the game winnable?

Solution

Case I: $n > k$ We first prove that the game is winnable whenever $n > k$ by demonstrating a winning strategy in this case.

On the i th move, choose the k cards in positions i through $i + k - 1$. Assuming that you do not win on any earlier move, repeat this for $1 \leq i \leq 2n - k + 1$.

Assume that you did not win on any of the first $2n - k + 1$ moves, as described above. Let j be an integer such that $1 \leq j \leq 2n - k$. On the j th move, the wizard revealed the cards in positions j through $j + k - 1$, so you know the labels of all of these cards (just not necessarily in the right order). Then, on the $(j + 1)$ th move, the wizard revealed the cards in positions $j + 1$ through $j + k$, which means that you get to see all of the cards that were moved to positions $j + 1$ through $j + k$. This means that you can uniquely determine the label on card j , since you knew all of the labels from j through $j + k - 1$, and the card in position j could not have moved anywhere else since your last move.

It follows that, after the sequence of $2n - k + 1$ moves described above, you know the labels on the first $2n - k$ cards. Since $n > k$, we have $2n - k \geq n + 1$, so there must be a pair of cards with matching labels in this group of $2n - k$ cards, by the Pigeonhole Principle. On your next move, you can pick a group of k cards that includes that pair of matching cards, and you win.

We have created a strategy that is guaranteed to win in at most $m = 2n - k + 2$ moves. Thus, the game is winnable for all $n > k$.

Case II: $n = k$ We now prove that the game is not winnable if $n = k$. We will say that the game is in a state S if your knowledge about the card labels is of the following form:

There exists a group of n cards for which you know that those n cards have all of the labels $1, 2, \dots, n$ (i.e. you know that they have all distinct labels) in some order, but you know nothing about which of those n cards have which labels. (Call this group of cards Group A .)

Suppose that the game is in such a state S . We will now show that, regardless of your next move, you cannot guarantee victory or an escape from state S .

Clearly, the n cards that are not in Group A must also have all of the labels $1, 2, \dots, n$. (You might know something about which cards have which labels, or you might not.) Call this other collection of cards Group B .

If, on the next move, you pick all of the cards from Group A or all of the cards from Group B , then you clearly will not get a matching pair. The wizard will then arbitrarily permute those cards. Thus, for those n chosen cards, you know their labels are all distinct, but you know nothing about which cards have which labels. Thus, you are back in state S .

Now, suppose you pick x cards from Group A and $n - x$ cards from Group B , where x is an integer and $1 \leq x \leq n - 1$. Then, the cards chosen from Group B will form a set of labels $P \subset Z_n$, where $Z_n = \{1, 2, \dots, n\}$ and $|P| = n - x$. However, you know nothing about which cards in Group A have which labels. Thus, there is no way for you to prevent the x cards from Group A to form the exact set of labels $Q = Z_n \setminus P$. In such a case, there will be no matching cards, so you will not win. Furthermore, the wizard will then arbitrarily permute these n cards, so you will know that they have all n distinct labels, but you will know nothing about which cards have which labels. Therefore, you are again in state S .

We have covered all cases, so it follows that, once you enter state S , you cannot guarantee escape from state S or victory.

Now, look at the very first move you make. Obviously, you cannot guarantee victory on the first move, as you know nothing about which cards have which labels. Assuming that you do not win on the first move, the n cards you chose have all distinct labels. The wizard then permutes the n cards you chose, so you now know that those n cards have all distinct labels but know nothing about which cards have which labels. Therefore, if you do not win on your first move, then the game enters state S , and we have already proven that you cannot guarantee victory from this point.

We therefore conclude that the game is not winnable if $n = k$. We proved earlier that the game is winnable if $n > k$, so the game is winnable if and only if $n > k \geq 2$.

Solution 2

We claim that the game is winnable if and only if $n > k$. Suppose after the first step, the cards 1 to n are shuffled around. Notice that we have n cards that we don't know the position of (which are all cards from 1 to n). Now, suppose we pick p known cards. Note that the p cards are all different (since the known cards are the cards from 1 to n), and there is still a possibility that the other cards from the

unknown cards complement and cause 1 to n . Therefore, we are in the same state as before, and the game is unwinnable.

Now, suppose $n > k$. Denote the i th card counting from the left. We pick cards 1 to k , keeping track of the set of values of the cards. Then, we pick cards 2 to $k + 1$, adding the value of the $k + 1$ th card into the set of value of cards. We keep doing this, until we pick cards $2n - k$ to $2n - 1$, at which point we know the exact number on the $2n$ th card. Now, we go back to 1 through k , and repeat this process, until we reveal the $2n - 1$ th card (unless we win during the process). This process terminates only when there are less or equal to k cards that we don't know the exact numbers on or if we somehow win, clearly, as otherwise we're still revealing new information by picking cards from 1 through k . Note that we now know the exact values on $2n - k$ of the cards. By the Pigeonhole Principle, since $k < n$, 2 of them are the same, and we pick those 2 cards and $k - 2$ other random cards and we win.

§46 USAMO 2017

1. (*USAMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Prove that there are infinitely many distinct pairs (a, b) of relatively prime positive integers $a > 1$ and $b > 1$ such that $a^b + b^a$ is divisible by $a + b$.

Solution 1

Let $n = a + b$. Since $\gcd(a, b) = 1$, we know $\gcd(a, n) = 1$. We can rewrite the condition as

$$\begin{aligned} a^{n-a} + (n-a)^a &\equiv 0 \pmod{n} \\ a^{n-a} &\equiv -(-a)^a \pmod{n} \end{aligned}$$

Assume a is odd. Since we need to prove an infinite number of pairs exist, it suffices to show that infinitely many pairs with a odd exist.

Then we have

$$\begin{aligned} a^{n-a} &\equiv a^a \pmod{n} \\ 1 &\equiv a^{2a-n} \pmod{n} \end{aligned}$$

We know by Euler's theorem that $a^{\varphi(n)} \equiv 1 \pmod{n}$, so if $2a - n = \varphi(n)$ we will have the required condition.

This means $a = \frac{n + \varphi(n)}{2}$. Let $n = 2p$ where p is a prime, $p \equiv 1 \pmod{4}$. Then $\varphi(n) = 2p * \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{p}\right) = p - 1$, so

$$a = \frac{2p + p - 1}{2} = \frac{3p - 1}{2}$$

Note the condition that $p \equiv 1 \pmod{4}$ guarantees that a is odd, since $3p - 1 \equiv 2 \pmod{4}$

This makes $b = \frac{p+1}{2}$. Now we need to show that a and b are relatively prime. We see that

$$\begin{aligned} \gcd\left(\frac{3p-1}{2}, \frac{p+1}{2}\right) &= \frac{\gcd(3p-1, p+1)}{2} \\ &= \frac{\gcd(p-3, 4)}{2} = \frac{2}{2} = 1 \end{aligned}$$

By the Euclidean Algorithm.

Therefore, for all primes $p \equiv 1 \pmod{4}$, the pair $\left(\frac{3p-1}{2}, \frac{p+1}{2}\right)$ satisfies the criteria, so infinitely many such pairs exist.

Solution 2

Take $a = 2n - 1, b = 2n + 1, n \geq 2$. It is obvious (use the Euclidean Algorithm, if you like), that $\gcd(a, b) = 1$, and that $a, b > 1$.

Note that

$$a^2 = 4n^2 - 4n + 1 \equiv 1 \pmod{4n}$$

$$b^2 = 4n^2 + 4n + 1 \equiv 1 \pmod{4n}$$

So

$$\begin{aligned} a^b + b^a &= a(a^2)^n + b(b^2)^{n-1} \equiv \\ a \cdot 1^n + b \cdot 1^{n-1} &\equiv a + b = 4n \equiv 0 \pmod{4n} \end{aligned}$$

Since $a + b = 4n$, all such pairs work, and we are done.

Solution 3

Let x be odd where $x > 1$. We have $x^2 - 1 = (x - 1)(x + 1)$, so $x^2 - 1 \equiv 0 \pmod{2x + 2}$. This means that $x^{x+2} - x^x \equiv 0 \pmod{2x + 2}$, and since x is odd, $x^{x+2} + (-x)^x \equiv 0 \pmod{2x + 2}$, or $x^{x+2} + (x + 2)^x \equiv 0 \pmod{2x + 2}$, as desired.

Solution 4

I claim that the ordered pair $(2^n - 1, 2^n + 1)$ satisfies the criteria for all $n \geq 2$. Proof: It is easy to see that the order modulo 2^{n+1} of $(2^n - 1)^k$ is 2, since $(2^n - 1)^2 = 2^{2n} - 2 \cdot 2^n + 1 \equiv 1 \pmod{2^{n+1}}$, and we can assert and prove similarly for the order modulo 2^{n+1} of $(2^n + 1)^k$. Thus, the remainders modulo 2^{n+1} that the sequences of powers of $2^n - 1$ and $2^n + 1$ generate are 1 for even powers and $2^n - 1$ and $2^n + 1$ for odd powers, respectively. Since $2^n + 1$ and $2^n - 1$ are both odd for $n \geq 2$,

$$(2^n + 1)^{2^n - 1} + (2^n - 1)^{2^n + 1} \equiv 0 \pmod{2^{n+1}}.$$

Since there are infinitely many powers of 2 and since all ordered pairs $(2^n - 1, 2^n + 1)$ contain relatively prime integers, we are done. $\square$

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Solution 5 (Motivation for Solution)

Note that

$$a^b + b^a = a^b - a^a + a^a + b^a.$$

To get rid of the $a^a + b^a \pmod{a+b}$, we can use the sum of powers factorization. However, a must be odd for us to do this. If we assume that a is odd,

$$a^b - a^a + a^a + b^a \equiv a^b - a^a + (a+b)(\text{an integer}) \equiv a^b - a^a \equiv a^a (a^{b-a} - 1) \pmod{a+b}.$$

Because a and b are relatively prime, $a+b$ cannot divide a^a . Thus, we have to show that there exists an integer b such that for odd a ,

$$a^{b-a} \equiv 1 \pmod{a+b}.$$

Suppose that $a = 2n - 1$. To keep the powers small, we try $b = 2n + k$ for small values of k . We can find that $b = 2n$ does not work. $b = 2n + 1$ works though, as $a + b = 4n$ and

$$(2n - 1)^{2n+1-2n+1} \equiv (2n - 1)^2 \equiv 4n^2 - 4n + 1 \equiv 4n(n - 1) + 1 \equiv 1 \pmod{4n}.$$

Because a is odd, $b = a + 2$ is relatively prime to a . Thus,

$$(a, b) = (2n - 1, 2n + 1)$$

is a solution for positive $n \geq 2$. There are infinitely many possible values for n , so the proof is complete. ■

2. (*USAMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $m_1, m_2, \dots, m_n$ be a collection of n positive integers, not necessarily distinct. For any sequence of integers $A = (a_1, \dots, a_n)$ and any permutation $w = w_1, \dots, w_n$ of $m_1, \dots, m_n$, define an A -inversion of w to be a pair of entries w_i, w_j with $i < j$ for which one of the following conditions holds:

$$a_i \geq w_i > w_j,$$

$$w_j > a_i \geq w_i,$$

or

$$w_i > w_j > a_i.$$

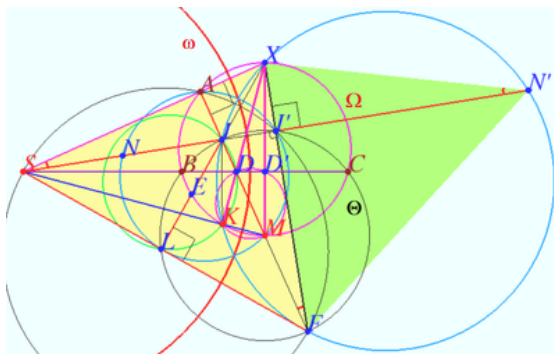
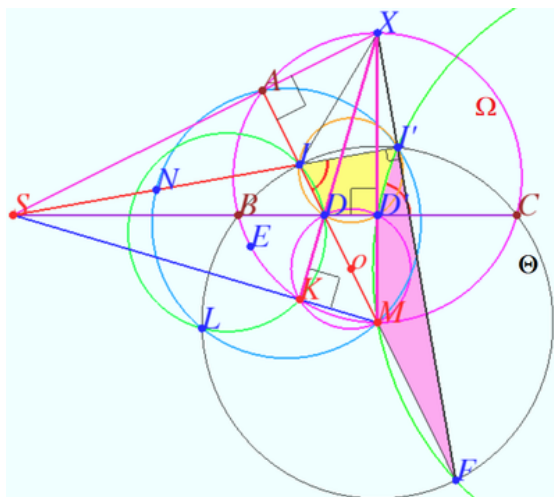
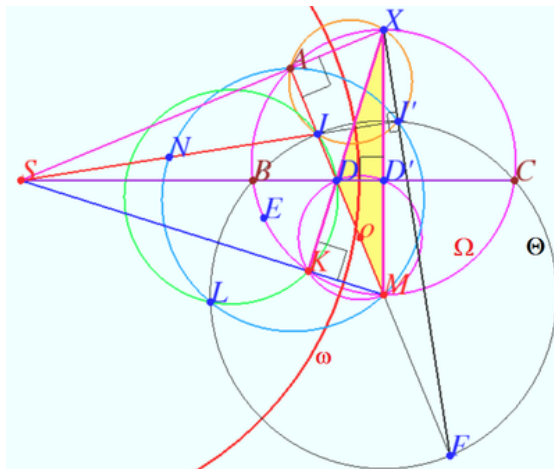
Show that, for any two sequences of integers $A = (a_1, \dots, a_n)$ and $B = (b_1, \dots, b_n)$, and for any positive integer k , the number of permutations of $m_1, \dots, m_n$ having exactly k A -inversions is equal to the number of permutations of $m_1, \dots, m_n$ having exactly k B -inversions.

3. (*USAMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let ABC be a scalene triangle with circumcircle Ω and incenter I . Ray AI meets $\overline{BC}$ at D and meets Ω again at M ; the circle with diameter $\overline{DM}$ cuts Ω again at K . Lines MK and BC meet at S , and N is the midpoint of $\overline{IS}$. The

circumcircles of $\triangle KID$ and $\triangle MAN$ intersect at points L_1 and L_2 . Prove that Ω passes through the midpoint of either $\overline{IL_1}$ or $\overline{IL_2}$.

Solution



Let X be the point on circle Ω opposite M . This means $\angle MAX = 90^\circ, BC \perp XM$.

$\angle XKM = \angle DKM = 90^\circ \implies$ the points X, D , and K are collinear.

Let $D' = BC \cap XM \implies DD' \perp XM \implies$

S is the orthocenter of $\triangle DMX \implies$ the points X, A , and S are collinear.

Let ω be the circle centered at S with radius $R = \sqrt{SK \cdot SM}$.

We denote I_ω inversion with respect to ω .

Note that the circle Ω has diameter MX and contain points A, B, C , and K .

$I_\omega(K) = M \implies \text{circle } \Omega \perp \omega \implies C = I_\omega(B), X = I_\omega(A)$.

$I_\omega(K) = M \implies \text{circle } KMD \perp \omega \implies D' = I_\omega(D) \in KMD \implies \angle DD'M = 90^\circ \implies \text{the points } X, D', \text{ and } M \text{ are collinear.}$

Let $F \in AM, MF = MI$. It is well known that $MB = MI = MC \implies$

$\Theta = BICF$ is circle centered at M . $C = I_\omega(B) \implies \Theta \perp \omega$.

Let $I' = I_\omega(I) \implies I' \in \Theta \implies \angle I'I'M = 90^\circ$. $I' = I_\omega(I), X = I_\omega(A) \implies AII'X$ is cyclic.

$\angle XII' = \angle XAI = 90^\circ \implies \text{the points } X, I', \text{ and } F \text{ are collinear.}$

$I'IDD'$ is cyclic $\implies \angle I'D'M = \angle I'D'C + 90^\circ = \angle I'D + 90^\circ, \angle XFM = \angle I'FI = 90^\circ \sim \angle I'IF = 90^\circ \sim \angle I'ID \implies$

$\angle XFM + \angle I'D'M = 180^\circ \implies I'D'MF$ is cyclic.

Therefore point F lies on $I_\omega(IDK)$.

$FA \perp SX, SI' \perp FX \implies I$ is orthocenter of $\triangle FSX$.

N is midpoint SI, M is midpoint FI, I is orthocenter of $\triangle FSX, A$ is root of height $FA \implies AMN$ is the nine-point circle of $\triangle FSX \implies I' \in AMN$.

Let $N' = I_\omega(N) \implies R^2 = SN \cdot SN' = SI \cdot SI' \implies$

$$\frac{SN'}{SI'} = \frac{SI}{SN} = 2 \implies$$

$\angle XN'I' = \angle XSI' = 90^\circ \sim \angle AXI' = \angle IFX \implies N'XIF$ is cyclic.

Therefore point F lies on $I_\omega(AMN) \implies I_\omega(F) = L \implies$

The points F, L , and S are collinear, $AXFL$ is cyclic.

Point I is orthocenter $\triangle FSX \implies XI \perp SF, \angle ILS = \angle SI'F = 90^\circ \implies$ The points X, I, E , and L are collinear.

$AXFL$ is circle $\implies AI \cdot IF = IL \cdot XI \implies$

$AI \cdot \frac{IF}{2} = \frac{IL}{2} \cdot IX \implies AI \cdot IM = EI \cdot IX \implies AEMX$ is cyclic.

$$E \in \Omega.$$

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Contact Contact v_Enhance at <https://www.facebook.com/v.Enhance>.

4. (USAMO 2017) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $P_1, P_2, \dots, P_{2n}$ be $2n$ distinct points on the unit circle $x^2 + y^2 = 1$, other than $(1, 0)$. Each point is colored either red or blue, with exactly n red points and n blue points. Let $R_1, R_2, \dots, R_n$ be any ordering of the red points. Let B_1 be the nearest blue point to R_1 traveling counterclockwise around the circle starting from R_1 . Then let B_2 be the nearest of the remaining blue points to R_2 travelling

counterclockwise around the circle from R_2 , and so on, until we have labeled all of the blue points $B_1, \dots, B_n$. Show that the number of counterclockwise arcs of the form $R_i \rightarrow B_i$ that contain the point $(1, 0)$ is independent of the way we chose the ordering $R_1, \dots, R_n$ of the red points.

Solution

I define a sequence to be, starting at $(1, 0)$ and tracing the circle counterclockwise, and writing the color of the points in that order - either R or B. For example, possible sequences include $RB, RBBR, BRRRRB, BRBRRBBR$, etc. Note that choosing an R_1 is equivalent to choosing an R in a sequence, and B_1 is defined as the B closest to R_1 when moving rightwards. If no B s exist to the right of R_1 , start from the far left. For example, if I have the above example $RBBR$, and I define the 2nd R to be R_1 , then the first B will be B_1 . Because no R or B can be named twice, I can simply remove R_1 and B_1 from my sequence when I choose them. I define this to be a move. Hence, a possible move sequence of $BRRRRB$ is: $BRR_1RRB_1 \implies B_2BRR_2 \implies B_3R_3$

Note that, if, in a move, B_n appears to the left of R_n , then $\widehat{R_n B_n}$ intersects $(1, 0)$. Now, I define a commencing B to be a B which appears to the left of all R s, and a terminating R to be a R which appears to the right of all B s. Let the amount of commencing B s be j , and the amount of terminating R s be k , I claim that the number of arcs which cross $(1, 0)$ is constant, and it is equal to $\max(j, k)$. I will show this with induction.

Base case is when $n = 1$. In this case, there are only two possible sequences - RB and BR . In the first case, $\widehat{R_1 B_1}$ does not cross $(1, 0)$, but both j and k are 0, so $\max(j, k) = 0$. In the second example, $j = 1, k = 1$, so $\max(j, k) = 1$. $\widehat{R_1 B_1}$ crosses $(1, 0)$ since B_1 appears to the left of R_1 , so there is one arc which intersects. Hence, the base case is proved.

For the inductive step, suppose that for a positive number n , the number of arcs which cross $(1, 0)$ is constant, and given by $\max(j, k)$ for any configuration. Now, I will show it for $n + 1$.

Suppose I first choose R_1 such that B_1 is to the right of R_1 in the sequence. This implies that $\widehat{R_1 B_1}$ does not cross $(1, 0)$. But, neither R_1 nor B_1 is a commencing B or terminating R . These numbers remain constant, and now after this move we have a sequence of length $2n$. Hence, by assumption, the total amount of arcs is $0 + \max(j, k) = \max(j, k)$.

- Here is a counter-case. $BRR_1B_1RR : j = 1, k = 2 \Rightarrow BRRR : j = 1, k = 3$. These numbers may not remain constant.

Thus, this solution probably doesn't work.

Now suppose that R_1 appears to the right of B_1 , but B_1 is not a commencing B . This implies that there are no commencing B s in the series, because there are no B s to the left of B_1 , so $j = 0$. Note that this arc does intersect $(1, 0)$, and R_1 must be a terminating R . R_1 must be a terminating R because there are no B s to the right of R_1 , or else that B would be B_1 . The $2n$ length sequence that remains has 0 commencing B s and $k - 1$ terminating R s. Hence, by assumption, the total amount of arcs is $1 + \max(0, k - 1) = 1 + k - 1 = k = \max(j, k)$.

Finally, suppose that R_1 appears to the right of B_1 , and B_1 is a commencing B . We know that this arc will cross $(1, 0)$. Analogous to the previous case, R_1 is a

terminating R , so the $2n$ length sequence which remains has $j - 1$ commencing B s and $k - 1$ terminating R s. Hence, by assumption, the total amount of arcs is $1 + \max(j - 1, k - 1) = 1 + \max(j, k) - 1 = \max(j, k)$.

There are no more possible cases, hence the induction is complete, and the number of arcs which intersect $(1, 0)$ is indeed a constant which is given by $\max(j, k)$.

-william122

Solution 2

Lemma: If we switch the ordering of two consecutive R_k, R_{k+1} , the number of arcs crossing $(1, 0)$ stays invariant.

Proof: There are two situations. If the two arcs don't cross this is simple because the actual arcs stay the same, and only the number order of the arcs change. Otherwise, if B_k is to the left of R_{k+1} , the two arcs will have some overlap. The order of the points counterclockwise must be $RRBB$ or some rotation of it. Notice how in both the old and the new ordering, the arc between the second R and the first B is counted twice and the arcs between the same colors are only counted once. Thus no matter where $(1, 0)$ is the number of arcs containing it will stay invariant.

Lemma 2: We can swap the place of any two points R_i and R_j while keeping the number of crossings invariant.

Proof: By Lemma 1 we can use consecutive arc moves to swap R_i with R_{i+1} , then the new R_{i+1} with R_{i+2} and so on until it swaps with R_j . Now, we can do the reverse swaps to move the new R_{j-1} to R_{j-2} , and so on into the original position of R_i . Since the other points were moved once in one direction and once in the opposite, all the other points are kept in place.

Now, let I be the sequence where R_1 through R_n is numbered counterclockwise. We show that any ordering has the same number of crossings as I . We can swap the current place of R_1 with the desired position in I . Now we can ignore R_1 and swap the current place of R_2 with the place of R_2 in I , and so on until all the points are in the desired position. Thus, all orderings of R_1 to R_n have the same number of crossings as I .

-tigershark22

5. (USAMO 2017) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $\mathbf{Z}$ denote the set of all integers. Find all real numbers $c > 0$ such that there exists a labeling of the lattice points $(x, y) \in \mathbf{Z}^2$ with positive integers for which: only finitely many distinct labels occur, and for each label i , the distance between any two points labeled i is at least c^i .

Solution 1 (Evan Chen, Calvin Deng)

See page 11 of this PDF: <https://web.evanchen.cc/exams/USAMO-2017-notes.pdf>

Solution 2 (INCOMPLETE)

For $c \leq 1$, we can label every lattice point 1. For $c \leq \sqrt[4]{2}$, we can make a "checkerboard" labeling, i.e. label (x, y) with 1 if $x + y$ is even and 2 if $x + y$ is odd. One can easily verify that these labelings satisfy the required conditions. Therefore, a labeling as desired exists for all $0 < c \leq \sqrt[4]{2}$.

An iterated version of the checkerboard labeling can actually work for all values $c < \sqrt{2}$. For convenience, define the original lattice grid to be the set of all lattice points in the coordinate plane. Define a modified lattice grid of size x to be a structure similar to the lattice points on the coordinate plane, but with the minimum separation between any two points equaling x (as opposed to 1).

On the first step, assign a label 1 to half of the points in a checkerboard arrangement. One can see that the points that have not yet been labeled form a modified lattice grid of size $\sqrt{2}$ (this lattice grid is also rotated by 45° from the original lattice grid). At this point, for the second step, assign a label 2 to half of the points, again in a checkerboard arrangement. At this point, the points that have not yet been labeled form a modified lattice grid of size 2 (and again, it is rotated 45° from the modified lattice grid after the first step). One then continues in this fashion. For the N^{th} step, the points we are labeling are separated by at least $\sqrt{2} \times (\sqrt{2})^{N-1} = (\sqrt{2})^N > c^N$, so we know that our labeling at each step is acceptable.

After the N^{th} step (where N is a natural number), the points that have not yet been labeled form a modified lattice grid with size $(\sqrt{2})^N$. Since $c < \sqrt{2}$, we will eventually have $(\sqrt{2})^N > c^{N+1}$ for some sufficiently large N . At this point, we can label all remaining points in the original lattice grid $N + 1$, and this produces a labeling of all of the lattice points in the plane that satisfies all of the conditions. Therefore, a labeling as desired exists for all $c < \sqrt{2}$.

We now prove that no labeling as desired exists for any $c \geq 2$. To do this, we will prove that labeling a 2^k -by- 2^k square grid of lattice points requires at least $k + 3$ distinct labels for all natural numbers k ; hence for a sufficiently large section of the lattice plane the number of distinct labels required grows arbitrarily large, so the entire lattice plane cannot be labeled with finitely many distinct labels. We will prove this using induction.

For the base case, $k = 1$, we have four points in a square of side length 1. The maximum distance between any two of these points is $\sqrt{2} < c^1$ for all $c \geq 2$, so all four points must have different labels. This completes the base case.

Now, for the inductive step, suppose that labeling a 2^k -by- 2^k square grid of lattice points requires at least $k + 3$ distinct labels for some natural number k . We will now prove that labeling a 2^{k+1} -by- 2^{k+1} square grid of lattice points requires at least $k + 4$ distinct labels.

Take a 2^{k+1} -by- 2^{k+1} square grid of lattice points. Divide this grid into four quadrants, A, B, C , and D . By the inductive hypothesis, A requires at least $k + 3$ distinct labels. At least one of these labels must be $k + 3$ or greater; take one such label and call it L .

The largest distance between any two points in the entire grid is $\sqrt{2}(2^{k+1} - 1) < c^{k+3}$ for all $c \geq 2$. Therefore, the label L cannot be used anywhere else in the grid. However, B, C , and D each require at least $k + 3$ distinct labels as well by the inductive hypothesis. Thus, they must use at least one label that is not used in A . It follows that the entire grid requires at least $k + 4$ distinct labels. This completes the inductive step, and thus we conclude that no labeling as desired exists for any $c \geq 2$.

I have heard from others that the actual boundary is $\sqrt{2}$. This makes intuitive

sense, since the iterated checkerboard labeling outlined above just breaks down at this value (you will be able to get closer and closer to labeling all of the lattice points, but you can never get there, since you will never have $(\sqrt{2})^N > c^{N+1}$). The inductive argument above seems fairly loose, so I think that it can be sharpened to bring the upper bound down to $\sqrt{2}$, but I am not sure yet how exactly to do so. I think the way to do it is to somehow force 2 new labels (instead of just 1) each time you double the side length of the square grid.

Solution 3

We claim that for all $c < \sqrt{2}$, there exists such a configuration.

Lemma 1a: If there exists a configuration for $c = c_1$, then for any $c_2 \leq c_1$, there also exists a configuration for $c = c_2$. Note that any labeling that can be done for $c = c_1$ exists for $c = c_2$. Thus, we may copy the configuration from $c = c_1$ to $c = c_2$.

Lemma 1b: If there does not exist a configuration for $c = c_1$, then for any $c_2 \geq c_1$, there does not exist a configuration for $c = c_2$. Note that any labeling which may be done for $c = c_2$ should also be available for $c = c_1$. However, we already stated that $c = c_1$ has no configuration, and so $c = c_2$ doesn't either.

We now prove that $c = \sqrt{2}$ does not work. WLOG, label the origin of our lattice plan with label 1. WLOG, we can cover every point that we can with label 1 because not labeling any such possible point does not affect the distance between points that aren't allowed to be labelled.

Name the state of the lattice grid before using label 1 state S . After covering every point possible, we are left with only lattice points that have coordinates that sum to an odd integer. Notice, however, that if we discard the covered points and rotate all the new points by $\frac{\pi}{4}$ radians counterclockwise, we end up with the same grid as state S , only all the distances are multiplied by scalar $\sqrt{2}$. However, before placing label 2s, note that we are in the same state S as before, because our minimum distance is also multiplied by scalar $\sqrt{2}$. Thus, we will always be in an infinite loop, stuck in state S , and thus it is impossible to complete the labeling.

It suffices to show that any $c = 2^k$ works given $k < 0.5$. Note that the two smallest distances between any two points in the starting grid are 1 and $\sqrt{2}$. When we are currently in state S using label I , it is obvious that if $2ki - i \leq 1$, we can break out of state S (Our effective minimum length would be at most 1, and we can cover every remaining point). Because $k < 0.5$, there does exist an I where this occurs. Thus, any $c < \sqrt{2}$ works, hence proved. ■ SigmaPiE

6. (USAMO 2017) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Find the minimum possible value of

$$\frac{a}{b^3 + 4} + \frac{b}{c^3 + 4} + \frac{c}{d^3 + 4} + \frac{d}{a^3 + 4}$$

given that a, b, c, d are nonnegative real numbers such that $a + b + c + d = 4$.

Solution

See here: https://artofproblemsolving.com/community/c5t211539f5h1434574_looks_like_mount_ineq
or:

§47 USAMO 2018

1. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a, b, c be positive real numbers such that $a + b + c = 4\sqrt[3]{abc}$. Prove that

$$2(ab + bc + ca) + 4\min(a^2, b^2, c^2) \geq a^2 + b^2 + c^2.$$

Solution 1

https://wiki-images.artofproblemsolving.com//6/69/IMG_8946.jpg

-srisainandan6

Solution 2

Similarly to Solution 2, we will prove homogeneity but we will use that to solve the problem differently. Let $f(a, b, c) = a + b + c - 4\sqrt[3]{abc}$. Note that $f(a, b, c) = f(ka, kb, kc)$, thus proving homogeneity.

WLOG, we can scale down all variables such that the lowest one is 1. WLOG, let this be $a = 1$. We now have $1 + b + c = 4\sqrt[3]{bc}$, and we want to prove $2bc + 2b + 2c + 4 \geq 1 + b^2 + c^2$. Adding $2bc$ to both sides and subtracting $2b + 2c$ gives us $4bc + 4 \geq 1 + (b + c)(b + c - 2)$, or $4bc + 3 \geq (b + c)(b + c - 2)$. Let $\sqrt[3]{bc} = x$. Now, we have

$$4x^3 + 3 \geq (4x - 1)(4x - 3)$$

$$4x^3 - 16x^2 + 16x \geq 0$$

$$4x^2 - 16 + 16 \geq 0$$

$$4(x - 2)^2 \geq 0$$

By the trivial inequality, this is always true. Since all these steps are reversible, the proof is complete. SigmaPiE

Solution 3

WLOG, let $a \leq b$ and $a \leq c$ and add $2(ab + bc + ca)$ to both sides to make the left side a square.

$$4(ab + bc + ca) + 4a^2 \geq (a + b + c)^2$$

$$4bc + 4a(a + b + c) \geq (a + b + c)^2$$

Now we perform the substitution for $a + b + c$.

$$4bc + 4a(4(abc)^{1/3}) \geq 16(abc)^{2/3}$$

Multiply both sides by a because we want all the terms to have a factor of $(abc)^{1/3}$.

$$4abc + 16a^2(abc)^{1/3} \geq 16a(abc)^{2/3}$$

Divide both sides by $4(abc)^{1/3}$.

$$(abc)^{2/3} + 4a^2 \geq 4a(abc)^{1/3}$$

$$((abc)^{1/3} - 2a)^2 \geq 0$$

This is true by the trivial inequality. Lastly, all the steps are reversible so the given inequality has been proved.

-Themathcanadian

2. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f\left(x + \frac{1}{y}\right) + f\left(y + \frac{1}{z}\right) + f\left(z + \frac{1}{x}\right) = 1$$

for all $x, y, z > 0$ with $xyz = 1$.

Solution

Obviously, the output of f lies in the interval $(0, 1)$. Define $g : (0, 1) \rightarrow (0, 1)$ as $g(x) = f\left(\frac{1}{x} - 1\right)$. Then for any $a, b, c \in (0, 1)$ such that $a + b + c = 1$, we have $g(a) = f\left(\frac{1}{a} - 1\right) = f\left(\frac{1-a}{a}\right) = f\left(\frac{b+c}{a}\right)$. We can transform $g(b)$ and $g(c)$ similarly:

$$g(a) + g(b) + g(c) = f\left(\frac{c}{a} + \frac{b}{a}\right) + f\left(\frac{a}{b} + \frac{c}{b}\right) + f\left(\frac{b}{c} + \frac{a}{c}\right)$$

Let $x = \frac{c}{a}$, $y = \frac{a}{b}$, $z = \frac{b}{c}$. We can see that the above expression is equal to 1. That is, for any $a, b, c \in (0, 1)$ such that $a + b + c = 1$, $g(a) + g(b) + g(c) = 1$.

(To motivate this, one can start by writing $x = \frac{a}{b}$, $y = \frac{b}{c}$, $z = \frac{c}{a}$, and normalizing such that $a + b + c = 1$.)

For convenience, we define $h : \left(-\frac{1}{3}, \frac{2}{3}\right) \rightarrow \left(-\frac{1}{3}, \frac{2}{3}\right)$ as $h(x) = g\left(x + \frac{1}{3}\right) - \frac{1}{3}$, so that for any $a, b, c \in \left(-\frac{1}{3}, \frac{2}{3}\right)$ such that $a + b + c = 0$, we have

$$h(a) + h(b) + h(c) = g\left(a + \frac{1}{3}\right) - \frac{1}{3} + g\left(b + \frac{1}{3}\right) - \frac{1}{3} + g\left(c + \frac{1}{3}\right) - \frac{1}{3} = 1 - 1 = 0.$$

Obviously, $h(0) = 0$. If $|a| < \frac{1}{3}$, then $h(a) + h(-a) + h(0) = 0$ and thus $h(-a) = -h(a)$. Furthermore, if a, b are in the domain and $|a + b| < \frac{1}{3}$, then $h(a) + h(b) + h(-(a + b)) = 0$ and thus $h(a + b) = h(a) + h(b)$.

At this point, we should realize that h should be of the form $h(x) = kx$. We first prove this for some rational numbers. If n is a positive integer and x is a real number such that $|nx| < \frac{1}{3}$, then we can repeatedly apply $h(a + b) = h(a) + h(b)$ to obtain $h(nx) = nh(x)$. Let $k = 6h\left(\frac{1}{6}\right)$, then for any rational number $r = \frac{p}{q} \in \left(0, \frac{1}{3}\right)$ where p, q are positive integers, we have $h(r) = 6p * h\left(\frac{1}{6q}\right) = \frac{6p}{q} * h\left(\frac{1}{6}\right) = kr$.

Next, we prove it for all real numbers in the interval $\left(0, \frac{1}{3}\right)$. For the sake of contradiction, assume that there is some $x \in \left(0, \frac{1}{3}\right)$ such that $h(x) \neq kx$. Let $E = h(x) - kx$, then obviously $0 < |E| < 1$. The idea is to "amplify" this error until it becomes so big as to contradict the bounds on the output of h . Let $N = \left\lceil \frac{1}{|E|} \right\rceil$, so that $N \geq 2$ and $|NE| \geq 1$. Pick any rational $r \in \left(\frac{N-1}{N}x, x\right)$, so that

$$0 < x - r < N(x - r) < x < \frac{1}{3}.$$

All numbers and sums are safely inside the bounds of $\left(-\frac{1}{3}, \frac{1}{3}\right)$. Thus

$$h(N(x-r)) = Nh(x-r) = N(h(x)+h(-r)) = N(h(x)-h(r)) = kN(x-r) + NE,$$

but picking any rational number $s \in \left(N(x-r), \frac{1}{3}\right)$ gives us $|kN(x-r)| < |ks|$, and since $ks = h(s) \in \left(-\frac{1}{3}, \frac{2}{3}\right)$, we have $kN(x-r) \in \left(-\frac{1}{3}, \frac{2}{3}\right)$ as well, but since $NE \geq 1$, this means that $h(N(x-r)) = kN(x-r) + NE \notin \left(-\frac{1}{3}, \frac{2}{3}\right)$, giving us the desired contradiction.

We now know that $h(x) = kx$ for all $0 < x < \frac{1}{3}$. Since $h(-x) = -h(x)$ for $|x| < \frac{1}{3}$, we obtain $h(x) = kx$ for all $|x| < \frac{1}{3}$. For $x \in \left(\frac{1}{3}, \frac{2}{3}\right)$, we have $h(x) + h\left(-\frac{x}{2}\right) + h\left(-\frac{x}{2}\right) = 0$, and thus $h(x) = kx$ as well. So $h(x) = kx$ for all x in the domain. Since $h(x)$ is bounded by $-\frac{1}{3}$ and $\frac{2}{3}$, we have $-\frac{1}{2} \leq k \leq 1$. It remains to work backwards to find $f(x)$.

$$\begin{aligned} h(x) &= kx \\ g(x) &= kx + \frac{1-k}{3} \\ f(x) &= \frac{k}{1+x} + \frac{1-k}{3} \quad \left(-\frac{1}{2} \leq k \leq 1\right). \end{aligned}$$

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3. (USAMO 2018) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

For a given integer $n \geq 2$, let $\{a_1, a_2, \dots, a_m\}$ be the set of positive integers less than n that are relatively prime to n . Prove that if every prime that divides m also divides n , then $a_1^k + a_2^k + \dots + a_m^k$ is divisible by m for every positive integer k .

Solution (NO FORMATTING)

<https://maa.org/sites/default/files/pdf/AMC/usamo/2018/2018USAMO.pdf> The integer m in the statement of the problem is $\phi(n)$, where ϕ is the Euler totient function. Throughout our proof we write $ps \parallel m$, if s is the greatest power of p that divides m . We begin with the following lemma: Lemma 1. If p is a prime and ps divides n for some positive integer s , then $1^k + 2^k + \dots + n^k$ is divisible by $ps-1$ for any integer $k \geq 1$. 2018 USAMO – Solutions 4 Proof. Let $a_1, a_2, \dots, a_m$ be a complete reduced residue set modulo ps and $m = ps-1$ ($p-1$). First we prove by induction on s that for any positive integer k , $a_1^k + a_2^k + \dots + a_m^k$ is divisible by $ps-1$. The base case $s = 1$ is true. Suppose the statement holds for some value of s . Consider the statement for $s+1$. Note that $a_1, \dots, a_m, ps+a_1, \dots, ps+a_m, \dots, ps(p-1)+a_1, \dots, ps(p-1)+a_m$ is a complete reduced residue set modulo $ps+1$. Therefore, the desired sum of k -th powers is equal to $a_1^k + \dots + a_m^k + \dots + (ps(p-1)+a_1)^k + \dots + (ps(p-1)+a_m)^k \equiv p(ak_1 + \dots + ak_m) \equiv 0 \pmod{ps}$, where we have used the induction hypothesis for the second congruence. This gives the induction step. $2k + \dots + (ps)k$ is divisible by $ps-1$ for any positive integer k . We use induction on s . The statement is true for $s=1$. Assume the statement is true for $s-1$. The sum $1^k + 2^k + \dots + (ps)k = ak_1 + ak_2 + \dots + ak_m + pk_1 + 2k + \dots + (ps-1)k$ is divisible by $ps-1$, because $ps-1 \mid ak_1 + \dots + ak_m$ and by the induction hypothesis $ps-1 \mid 1k + 2k + \dots + (ps-1)k$. Now we proceed to prove a second lemma, for

Lemma 2. Suppose p is a prime dividing n . Let $a_1, \dots, a_m$ be a complete reduced residue set modulo n , and define $f(k) = a_1^k + a_2^k + \dots + a_m^k$ for any integer $k \geq 1$. *Proof.* We fix p , and use induction on the number of prime factors of n (counted by multiplicity) that are different from p . If there are no prime factors other than p , then $n = p^s$, $m = p^s(p-1)$, and we proved in Lemma 1 that $a_1^k + \dots + a_m^k$ is divisible by p^s . Now suppose the statement is true for n . We show that it is true for nq , where q is a prime not equal to p . *Case 1.* q divides n . We have $p^s \mid \phi(n)$ and $p^s \mid \phi(nq)$, because $\phi(nq) = q\phi(n)$. If $a_1, a_2, \dots, a_m$ is a complete reduced residue set modulo n , then $a_1, \dots, a_m, n+a_1, \dots, n+a_m, \dots, n(q-1)+a_1, \dots, n(q-1)+a_m$ is a complete reduced residue set modulo nq . The new sum of k -th powers is equal to $a_1^k + \dots + a_m^k + \dots + (n(q-1)+a_1)^k + \dots + (n(q-1)+a_m)^k = mnk1^k + \dots + (q-1)^k + 2018 \text{ USAMO - Solutions } 5k1nk-11k-1 + \dots + (q-1)^{k-1}(a_1 + \dots + a_m) + \dots + q(a_1^k + \dots + a_m^k)$. This sum is divisible by p^s because $p^s \mid m$ and $p^s \mid a_j^1 + a_j^2 + \dots + a_j^m$ for any positive integer j . *Case 2.* q doesn't divide n . Suppose $p^b \mid q-1$, where $b \geq 0$. Note that $\phi(nq) = \phi(n)(q-1)$, so $p^s \mid \phi(n)$ and $p^{s+b} \mid \phi(nq)$. Let $a_1, \dots, a_m$ be a complete reduced residue set modulo n . The complete reduced residue set modulo nq consists of the m q numbers $a_1, \dots, a_m, n+a_1, \dots, n+a_m, \dots, n(q-1)+a_1, \dots, n(q-1)+a_m$ with the m elements $qa_1, qa_2, \dots, qa_m$ removed. The new sum of k -th powers is equal to $a_1^k + \dots + a_m^k + \dots + (n(q-1)+a_1)^k + \dots + (n(q-1)+a_m)^k - q(a_1^k + \dots + a_m^k) = mnk1^k + \dots + (q-1)^k + k1nk-11k-1 + \dots + (q-1)^{k-1}(a_1 + \dots + a_m) + \dots + kq-1n(1 + \dots + (q-1)^{k-1})(a_1^{k-1} + \dots + a_m^{k-1}) + q(a_1^k + \dots + a_m^k) - qk(a_1^k + \dots + a_m^k)$. Each term $kjn-k-j1k-j + \dots + (q-1)^{k-j}(a_j^1 + \dots + a_j^m)$, for $0 \leq j \leq k-1$, is divisible by p^{s+b} because $p \mid nk-j$, $p^s \mid a_j^1 + \dots + a_j^m$, and $p^{b-1} \mid 1k-j + \dots + (q-1)^{k-j}$ by Lemma 1. Also $(qk-q)(a_1^k + \dots + a_m^k)$ is divisible by p^{s+b} because $p^b \mid q-1 \mid qk-q$ and $p^s \mid a_1^k + \dots + a_m^k$. Thus p^{s+b} divides our sum and our proof is complete. *Remark.* In fact, one can also show the converse statement: if $a_1, a_2, \dots, a_m$ is as defined in the problem and $a_1^k + a_2^k + \dots + a_m^k$ is divisible by m for every positive integer k , then every prime that divides m also divides n .

4. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let p be a prime, and let $a_1, \dots, a_p$ be integers. Show that there exists an integer k such that the numbers

$$a_1 + k, a_2 + 2k, \dots, a_p + pk$$

produce at least $\frac{1}{2}p$ distinct remainders upon division by p .

Solution

Lemma: For fixed $i \neq j$, where $i, j \in \{1, 2, \dots, p\}$, the statement $a_i + ik \equiv a_j + jk \pmod{p}$ holds for exactly one $k \in 1, 2, \dots, p$.

Proof: Notice that the left side minus the right side is congruent to $(a_i - a_j) + (i - j)k$ modulo p . For this difference to equal 0, there is a unique solution for k modulo p given by $k \equiv (a_j - a_i)(i - j)^{-1} \pmod{p}$, where we have used the fact that every nonzero residue modulo p has a unique multiplicative inverse. Therefore,

there is exactly one $k \in 1, 2, \dots, p$ that satisfies $a_i + ik \equiv a_j + jk \pmod{p}$ for any fixed $i \neq j$. **End Lemma**

Suppose that you have p graphs $G_1, G_2, \dots, G_p$, and graph G_k consists of the vertices (i, k) for all $1 \leq i \leq p$. Within any graph G_k , vertices (i_1, k) and (i_2, k) are connected by an edge if and only if $a_{i_1} + i_1k \equiv a_{i_2} + i_2k \pmod{p}$. Notice that the number of disconnected components of any graph G_k equals the number of distinct remainders when divided by p given by the numbers $a_1 + k, a_2 + 2k, \dots, a_p + pk$.

These p graphs together have exactly one edge for every unordered pair of elements of $\{1, 2, \dots, p\}$, so they have a total of exactly $\frac{p(p-1)}{2}$ edges. Therefore, there exists at least one graph G_k that has strictly fewer than $\frac{p}{2}$ edges, meaning that it has more than $\frac{p}{2}$ disconnected components. Therefore, the collection of numbers $\{a_i + ik : 1 \leq i \leq p\}$ for this particular value of k has at least $\frac{p}{2}$ distinct remainders modulo p . This completes the proof.

Note: This is the same problem as 2018 USAJMO Problem 5.

Solution 2

We consider the Lemma and its proof as in the above solution.

For every k , let x_k be the number of distinct residues that $a_i + ik$ takes on. Further for each residue r_i , let n_i be the number of times it is achieved by an $a_j + jk$ for $i \in [1, x_k]$. Note that $\sum_{i=1}^{x_k} n_i = p$.

The number of pairs (a_i, a_j) s.t. $(a_i + ik) - (a_j + jk) \equiv 0$ for a given k is,

$$\begin{aligned} \sum_{i=1}^{x_k} \binom{n_i}{2} &\geq x_k \binom{(\sum_{i=1}^{x_k} n_i)/x_k}{2} \\ &= \frac{1}{2} p \left(\frac{p}{x_k} - 1 \right). \end{aligned}$$

The first equality is given by a well known theorem, which can be proven by C-S or Jensen's.

Summing over all k , the number of times a pair (a_i, a_j) has $(a_i + ik) - (a_j + jk) \equiv 0$ is,

$$\sum_{k=1}^p \frac{1}{2} p \left(\frac{p}{x_k} - 1 \right).$$

Alternatively, every pair (a_i, a_j) has a unique k s.t. $(a_i + ik) - (a_j + jk) \equiv 0$ by the Lemma so we double-count as,

$$\begin{aligned} \binom{p}{2} &\geq \sum_{k=1}^p \frac{1}{2} p \left(\frac{p}{x_k} - 1 \right). \\ 2p - 1 &\geq \sum_{k=1}^p \frac{p}{x_k}. \end{aligned}$$

By AM-HM inequality,

$$\begin{aligned} \sum_{k=1}^p \frac{p}{x_k} &\geq \frac{p^2}{\sum_{k=1}^p \frac{x_k}{p}} \\ 2p - 1 &\geq \frac{p^3}{\sum_{k=1}^p x_k} \end{aligned}$$

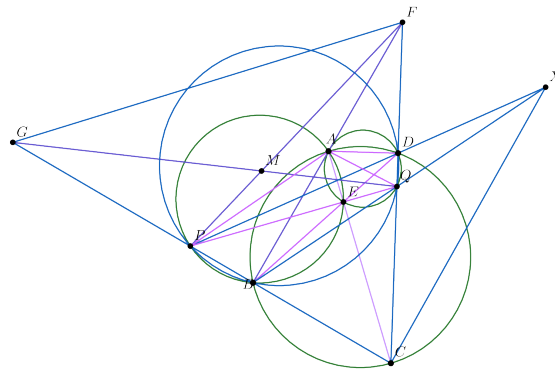
$$\sum_{k=1}^p x_k \geq \frac{p^3}{2p-1} \geq \frac{p^2}{2}.$$

So it is clear that $\exists k$ s.t $x_k \geq \frac{p}{2}$.

5. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

In convex cyclic quadrilateral $ABCD$, we know that lines AC and BD intersect at E , lines AB and CD intersect at F , and lines BC and DA intersect at G . Suppose that the circumcircle of $\triangle ABE$ intersects line CB at B and P , and the circumcircle of $\triangle ADE$ intersects line CD at D and Q , where C, B, P, G and C, Q, D, F are collinear in that order. Prove that if lines FP and GQ intersect at M , then $\angle MAC = 90^\circ$.

Solution



$$\begin{aligned} & \angle DEQ + \angle AED + \angle AEP \\ &= \angle DAQ + \angle AQD + \angle AEP \\ &= 180 - \angle ADC + \angle AEP \\ &= 180 - \angle ADC + \angle ABP \\ &= \angle ABP + \angle ABC \\ &= 180 \end{aligned}$$

so P, E, Q are collinear. Furthermore, note that $DQBP$ is cyclic because:

$$\angle EDQ = \angle BAE = \angle BPE.$$

Notice that since A is the intersection of (EDQ) and (BPE) , it is the Miquel point of $DQBP$.

Now define X as the intersection of BQ and DP . From Pappus's theorem on $BFPDGQ$ that A, M, X are collinear. It's a well known property of Miquel points that $\angle EAX = 90$, so it follows that $MA \perp AE$, as desired. ■ AopsUser101

6. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled

diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a_n be the number of permutations $(x_1, x_2, \dots, x_n)$ of the numbers $(1, 2, \dots, n)$ such that the n ratios $\frac{x_k}{k}$ for $1 \leq k \leq n$ are all distinct. Prove that a_n is odd for all $n \geq 1$.

Solution

Write out the mapping of each k to x_k as follows:

$$\begin{array}{ccccccc} 1 & 2 & 3 & \dots & n \\ x_1 & x_2 & x_3 & \dots & x_n \end{array}$$

Now, consider what happens when the two rows are swapped (and the top-bottom pairs are reordered so that the top reads $(1, 2, 3, \dots, n)$). This will result in a new permutation if and only if $x_k = j$ does not imply $x_j = k$ for all k, j in $(1, 2, 3, \dots, n)$. I will denote this new permutation as $(y_1, y_2, y_3, \dots, y_n)$. Notice that $(y_1, y_2, y_3, \dots, y_n)$ is a valid permutation if and only if $(x_1, x_2, x_3, \dots, x_n)$ is valid, because each fraction $\frac{y_k}{k}$ is the reciprocal of $\frac{x_k}{k}$. This means that we need only consider the parity of number of cases in which $(y_1, y_2, y_3, \dots, y_n) = (x_1, x_2, x_3, \dots, x_n)$, as there will be an even number of cases where $(y_1, y_2, y_3, \dots, y_n) \neq (x_1, x_2, x_3, \dots, x_n)$. Let the number of valid permutations where $(y_1, y_2, y_3, \dots, y_n) = (x_1, x_2, x_3, \dots, x_n)$ be b_n .

Notice that the only permutations that have the property $x_{x_k} = k$ (which is an equivalent statement to the one given above) are those that are formed by taking pairs of elements and swapping their positions having the maximum number of pairs possible and having no two pairs both contain the same element. Some more necessary conditions will be outlined below after we split n into two cases.

Case 1: n is odd. If n is odd then there must be exactly one k such that $x_k = k$. This yields $b_n = n \cdot b_{n-1}$ which has the same parity as b_{n-1} , so we need only consider the parity of b_n for odd n . (**Note: This is wrong, as b_n is only the number of valid involutions for $1, 2, \dots, n$, not $1, 2, \dots, n+1$ with one number removed.**)

Case 2: n is even. If n is even then can be no k so that $x_k = k$, or else there must be at least two distinct numbers k and j so that $x_k = k$ and $x_j = j$ which violates the given conditions. Denote a pair of numbers that are swapped to arrive at the final permutation as the pair (a, b) . Then if a permutation yields an invalid arrangement there must be two pairs (a, b) and (c, d) such that $\frac{a^2}{b^2} = \frac{c^2}{d^2}$. But notice that the two pairs (a, c) and (b, d) will also result in an invalid arrangement. So, there must be an even number of invalid arrangements (**Note: This proof doesn't work, as no specific pairing of invalid arrangements is constructed. Indeed, when three pairs all with the same fraction are present, this doesn't work.**), meaning the parity we desire is just the number of ways to separate n objects into $\frac{n}{2}$ pairs! However, this is quite simply just $(n-1)(n-3)(n-5) \cdots (3)(1)$, which is clearly the product of odd numbers. So we conclude that there are an odd number of valid permutations $(x_1, x_2, x_3, \dots, x_n)$. ■

§48 USAMO 2019

1. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled

diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $\mathbb{N}$ be the set of positive integers. A function $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfies the equation

$$\underbrace{f(f(\dots f(n)\dots))}_{f(n) \text{ times}} = \frac{n^2}{f(f(n))}$$

for all positive integers n . Given this information, determine all possible values of $f(1000)$.

Solution

Let $f^r(x)$ denote the result when f is applied to $f^{r-1}(x)$, where $f^1(x) = f(x)$.

If $f(p) = f(q)$, then $f^2(p) = f^2(q)$ and $f^{f(p)}(p) = f^{f(q)}(q)$

$$\implies p^2 = f^2(p) \cdot f^{f(p)}(p) = f^2(q) \cdot f^{f(q)}(q) = q^2$$

$$\implies p = \pm q$$

$$\implies p = q \text{ since } p, q > 0.$$

Therefore, f is injective. It follows that f^r is also injective.

Lemma 1: If $f^r(b) = a$ and $f(a) = a$, then $b = a$.

Proof:

$f^r(b) = a = f^r(a)$ which implies $b = a$ by injectivity of f^r .

Lemma 2: If $f^2(m) = f^{f(m)}(m) = m$, and m is odd, then $f(m) = m$.

Proof:

Let $f(m) = k$. Since $f^2(m) = m$, $f(k) = m$. So, $f^2(k) = k$.

$$f^2(k) \cdot f^{f(k)}(k) = k^2.$$

Since $k \neq 0$, $f^{f(k)}(k) = k$

$$\implies f^m(k) = k$$

$$\implies f^{\gcd(m,2)}(k) = k$$

$$\implies f(k) = k$$

This proves Lemma 2.

I claim that $f(m) = m$ for all odd m .

Otherwise, let m be the least counterexample.

Since $f^2(m) \cdot f^{f(m)}(m) = m^2$, either

(1) $f^2(m) = k < m$, contradicted by Lemma 1 since k is odd and $f^2(k) = k$.

(2) $f^{f(m)}(m) = k < m$, also contradicted by Lemma 1 by similar logic.

(3) $f^2(m) = m$ and $f^{f(m)}(m) = m$, which implies that $f(m) = m$ by Lemma 2. This proves the claim.

By injectivity, $f(1000)$ is not odd. I will prove that $f(1000)$ can be any even number, x . Let $f(1000) = x$, $f(x) = 1000$, and $f(k) = k$ for all other k . If n is equal to neither 1000 nor x , then $f^2(n) \cdot f^{f(n)}(n) = n \cdot n = n^2$. This satisfies the given property.

If n is equal to 1000 or x , then $f^2(n) \cdot f^{f(n)}(n) = n \cdot n = n^2$ since $f(n)$ is even and $f^2(n) = n$. This satisfies the given property.

2. (USAMO 2019) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $ABCD$ be a cyclic quadrilateral satisfying $AD^2 + BC^2 = AB^2$. The diagonals of $ABCD$ intersect at E . Let P be a point on side $\overline{AB}$ satisfying $\angle APD = \angle BPC$. Show that line PE bisects $\overline{CD}$.

Solution

Let $PE \cap DC = M$. Also, let N be the midpoint of AB . Note that only one point P satisfies the given angle condition. With this in mind, construct P' with the following properties:

- (1) $AP' \cdot AB = AD^2$
- (2) $BP' \cdot AB = CD^2$

Claim: $P = P'$

Proof: The conditions imply the similarities $ADP \sim ABD$ and $BCP \sim BAC$ whence $\angle APD = \angle BDA = \angle BCA = \angle CPB$ as desired. $\square$

Claim: PE is a symmedian in AEB

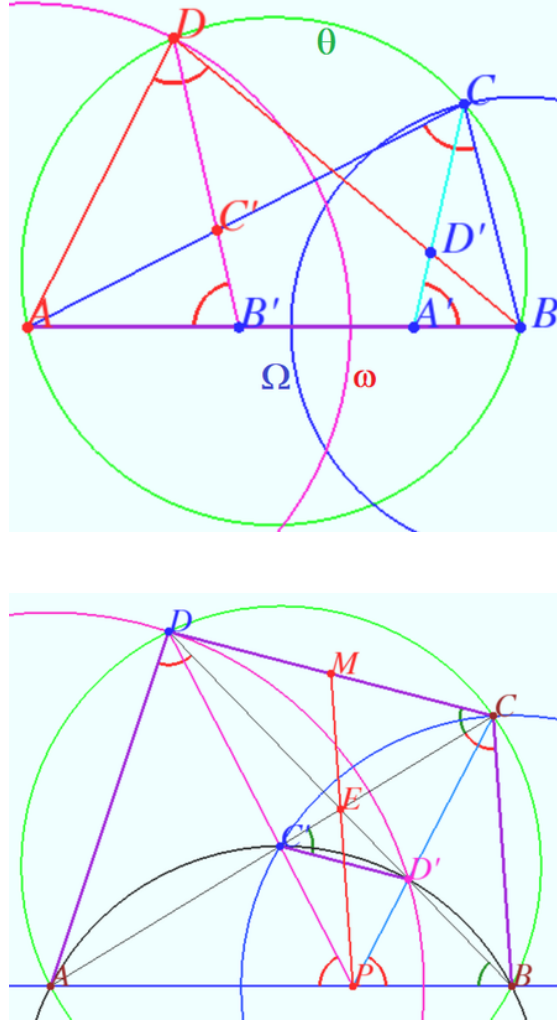
Proof: We have

$$\begin{aligned} AP \cdot AB = AD^2 &\iff AB^2 \cdot AP = AD^2 \cdot AB \\ &\iff \left(\frac{AB}{AD}\right)^2 = \frac{AB}{AP} \\ &\iff \left(\frac{AB}{AD}\right)^2 - 1 = \frac{AB}{AP} - 1 \\ &\iff \frac{AB^2 - AD^2}{AD^2} = \frac{BP}{AP} \\ &\iff \left(\frac{BC}{AD}\right)^2 = \left(\frac{BE}{AE}\right)^2 = \frac{BP}{AP} \end{aligned}$$

as desired. $\square$

Since P is the isogonal conjugate of N , $\angle PEA = \angle MEC = \angle BEN$. However $\angle MEC = \angle BEN$ implies that M is the midpoint of CD from similar triangles, so we are done. $\square$

Solution 2



Let ω be the circle centered at A with radius AD .

Let Ω be the circle centered at B with radius BC .

We denote I_ω and I_Ω inversion with respect to ω and Ω , respectively.

$$B' = I_\omega(B), C' = I_\omega(C), D = I_\omega(D) \implies$$

$$AB' \cdot AB = AD^2, \angle ACB = \angle AB'C'.$$

$$A' = I_\Omega(A), D' = I_\Omega(D), C = I_\Omega(C) \implies$$

$$BA' \cdot AB = BC^2, \angle BDA = \angle BA'D'.$$

Let θ be the circle $ABCD$.

$I_\omega(\theta) = B'C'D$, straight line, therefore

$$\angle AB'C' = \angle AB'D' = \angle ACB.$$

$I_\Omega(\theta) = A'D'C$, straight line, therefore

$$\angle BA'D' = \angle BA'C = \angle BDA.$$

$ABCD$ is cyclic $\implies \angle BA'C = \angle AB'D$.

$$AB' + BA' = \frac{AD^2 + BC^2}{AB} = AB \implies$$

points A' and B' are coincide.

Denote $A' = B' = Q \in AB$.

Suppose, we move point Q from A to B . Then $\angle AQD$ decreases monotonically, $\angle BQC$ increases monotonically. So, there is only one point where

$$\angle AQD = \angle BQC \implies P = Q.$$

$$B = I_\omega(P), D' = I_\omega(D'), C' = I_\omega(C), A = I_\omega(\infty) \implies \\ I_\omega(CD'P) = AC'D'B \text{ is cyclic.}$$

$$\angle ACD = \angle ABD = \angle CC'D \implies C'D' \parallel CD \implies$$

$C'D'CD$ is trapezoid.

It is known that the intersection of the diagonals, intersection point of the lines containing the lateral sides of the trapezoid and the midpoints of two parallel sides are collinear.

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3. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let K be the set of all positive integers that do not contain the digit 7 in their base-10 representation. Find all polynomials f with nonnegative integer coefficients such that $f(n) \in K$ whenever $n \in K$.

Solution

I claim the only such polynomials are of the form $f(n) = k$ where $k \in K$, or $f(n) = an + b$ where a is a power of 10, $b \in K$, and $b < a$. Obviously, these polynomials satisfy the conditions. We now prove that no other polynomial works. That is, we prove that for any other polynomial f with nonnegative coefficients, there is some $n \in K$ such that $f(n) \notin K$.

We first prove the result for monomials, polynomials in which only one coefficient is nonzero. This is obvious for constant polynomials $f(n) = k \notin K$. The next simplest case is $f(n) = an$ with a not a power of 10, and hence $\lg a$ is irrational. By Dirichlet's approximation theorem, the set of multiples of $\lg a$ is dense mod 1, and thus contains an element with fractional part in the interval $[\lg 7, \lg 8)$. In other words, there is a power of a whose decimal expansion starts with a 7. Let a^x be the smallest power of a that is not in K . Obviously, $x > 0$, so letting $n = a^{x-1}$ completes the proof of this part.

We have now proven that for any a that is not a power of 10, there is some $n \in K$ such that $an \notin K$. We proceed to the case where $f(n) = an^x$ for $x > 1$. This splits into 2 cases. If ax is not a power of 10, then pick $m \in K$ such that $axm \notin K$. For any y , we have

$$f(10^y + m) = a10^{yx} + axm * 10^{y(x-1)} + \dots + am^x$$

If we choose y to be large enough, then the terms in the expression above will not interfere with each other, and the resulting number will contain a 7 in the decimal expansion, and thus not be in K .

On the other hand, if ax is a power of 10, then a and x are both powers of 10, and $x \geq 10$. Obviously, $\frac{1}{2}ax(x-1)$ is not a power of 10. By the previous step, which establishes the result for $x = 2$, we can pick $m \in K$ such that $\frac{1}{2}ax(x-1)m^2 \notin K$. Then, for any y ,

$$f(10^y + m) = a10^{yx} + axm * 10^{y(x-1)} + \frac{1}{2}ax(x-1)m^2 * 10^{y(x-2)} + \dots + am^x$$

Similarly, picking a sufficient large y settles this case.

Now, we extend our proof to general polynomials. If a polynomial $f(n) = a_0 + a_1n + a_2n^2 + \dots + a_xn^x$ satisfies the conditions of the problem, then for any $m, y > 0$:

$$f(m * 10^y) = a_xm^x * 10^{yx} + \dots + a_1m * 10^y + a_0$$

Similarly, choosing y to be sufficiently large results in the terms not interfering with each other. If f contains any monomials that do not satisfy the conditions of the problem, then picking suitable m and sufficiently large y causes $f(m * 10^y)$ to not be in K . Thus, f is a linear polynomial of the form $ax + b$ where a is 0 or a power of 10, and $b \in K$. It suffices to rule out those polynomials where $a > 0$ and $b > a$. If this is the case, then since the digit of b corresponding to a is not 7, there must be a single-digit number n such that the digit of $f(n)$ corresponding to a is 7. Therefore, we are done.

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4. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let n be a nonnegative integer. Determine the number of ways that one can choose $(n+1)^2$ sets $S_{i,j} \subseteq \{1, 2, \dots, 2n\}$, for integers i, j with $0 \leq i, j \leq n$, such that: for all $0 \leq i, j \leq n$, the set $S_{i,j}$ has $i + j$ elements; and $S_{i,j} \subseteq S_{k,l}$ whenever $0 \leq i \leq k \leq n$ and $0 \leq j \leq l \leq n$.

Solution 1

Note that there are $(2n)!$ ways to choose $S_{1,0}, S_{2,0}, \dots, S_{n,0}, S_{n,1}, S_{n,2}, \dots, S_{n,n}$, because there are $2n$ ways to choose which number $S_{1,0}$ is, $2n - 1$ ways to choose which number to append to make $S_{2,0}$, $2n - 2$ ways to choose which number to append to make $S_{3,0}$, etc. After that, note that $S_{n-1,1}$ contains the $n - 1$ in $S_{n-1,0}$ and 1 other element chosen from the 2 elements in $S_{n,1}$ not in $S_{n-1,0}$ so there are 2 ways for $S_{n-1,1}$. By the same logic there are 2 ways for $S_{n-1,2}$ as well so 2^n total ways for all $S_{n-1,j}$, so doing the same thing $n - 1$ more times yields a final answer of $(2n)! \cdot 2^{(n^2)}$.

-Stormersyle

Solution 2

There are $\frac{(2n)!}{2^n}$ ways to choose $S_{0,0}, S_{1,1}, \dots, S_{n,n}$. Since, there are $\binom{2n}{2}$ ways to choose $S_{1,1}$, and after that, to generate $S_{2,2}$, you take $S_{1,1}$ and add 2 new elements, getting you $\binom{2n-2}{2}$ ways to generate $S_{2,2}$. And you can keep going down the line, and you get that there are $\frac{(2n)!}{2^n}$ ways to pick $S_{0,0}, S_{1,1}, \dots, S_{n,n}$. Then we can fill out the rest of the grid. First, let's prove a lemma.

Lemma Claim: If we know what $S_{a,b}$ is and what $S_{a+1,b+1}$ is, then there are 2 choices for both $S_{a,b+1}$ and $S_{a+1,b}$.

Proof: Note $a \leq a+1$ and $b \leq b+1$, so $S_{a,b} \subseteq S_{a+1,b+1}$. Let A be a set that contains all the elements in $S_{a+1,b+1}$ that are not in $S_{a,b}$. $A = S_{a+1,b+1} - S_{a,b}$. We know $S_{a+1,b+1}$ contains total $a+b+2$ elements. And $S_{a,b}$ contains total $a+b$ elements. That means A contains only 2 elements since $(a+b+2) - (a+b) = 2$. Let's call these 2 elements m, n . $A = \{m, n\}$. $S_{a,b+1}$ contains 1 elements more than $S_{a,b}$ and 1 elements less than $S_{a+1,b+1}$. That 1 elements has to select from A . It's easy to see $S_{a,b+1} = S_{a,b} \cup \{m\}$ or $S_{a,b+1} = S_{a,b} \cup \{n\}$, so there are 2 choice for $S_{a,b+1}$. Same thing applies to $S_{a+1,b}$.

Filling in the rest of the grid We used our proved lemma, and we can fill in $S_{0,1}, S_{1,2}, \dots, S_{n-1,n}$ then we can fill in the next diagonal, until all $S_{i,j}$ are filled, where $i \leq j$. But, we haven't finished everything! Fortunately, filling out the rest of the diagonals in a similar fashion is pretty simple. And, it's easy to see that we have made $n(n+1)$ decisions, each with 2 choices, when filling out the rest of the grid, so there are $2^{n(n+1)}$ ways to finish off.

Finishing off To finish off, we have $\frac{(2n)!}{2^n} \cdot 2^{n(n+1)}$ ways to fill in the grid, which gets us $\boxed{(2n)! \cdot 2^{n^2}}$

-Alexlikemath

Solution 3

Let C_j represent the set of sets of the form S_{ij} for $1 \leq i \leq n$, a_{ij} denote $S_{(i+1)j} \setminus S_{ij}$, and $b_{ij} = S_{i(j+1)} \setminus S_{ij}$. Begin by considering C_0 and $S_{00} = \emptyset$. Then given S_{i0} we can create $S_{(i+1)0}$ by adding one element (a_{i0}). Using this, the number of ways to form the sequence of $S_{00}, S_{10}, \dots, S_{n0}$ are $(2n)(2n-1) \cdots (n+1)$ where we successively add one of the remaining elements of $[2n]$ to get consecutive terms in the sequence.

Now consider when we are given C_j and we need to find C_{j+1} . So far, there have been $n+j$ chosen distinct elements (via S_{nj}). After finding C_{j+1} we will have $n+j+1$ distinct elements and so in this process we only add one unique element to sets among C_{j+1} . There are $2n - (n+j) = n-j$ ways to chose such a new element called x .

Now notice that $b_{0j}, a_{0(j+1)}, a_{1(j+1)}, \dots, a_{(n-1)(j+1)}$ is a permutation of $x, a_{0j}, \dots, a_{(n-1)j}$ by noting $b_{nj} = x$ and,

$$S_{0j} + b_{0j} + a_{0(j+1)} + a_{1(j+1)} + \cdots + a_{(n-1)(j+1)} = S_{n(j+1)} = S_{0j} + a_{0j} + a_{1j} + \cdots + a_{(n-1)j} + b_{nj}$$

Furthermore,

$$S_{0j} + b_{0j} + a_{0(j+1)} = S_{1(j+1)} = S_{0j} + a_{0j} + b_{1j}$$

$$S_{1j} + b_{1j} + a_{1(j+1)} = S_{2(j+1)} = S_{1j} + a_{1j} + b_{2j}$$

...

$$S_{(n-1)j} + b_{(n-1)j} + a_{(n-1)(j+1)} = S_{n(j+1)} = S_{(n-1)j} + a_{(n-1)j} + b_{nj}.$$

Therefore $b_{ij}, a_{i(j+1)}$ is a permutation of $a_{ij}, b_{(i+1)j}$ for $i < n$. Now let k be the first i such that $b_{ij} = x$. By definition, $b_{(k+1)j}, \dots = x$. Then the number of ways to order $x, a_{0j}, \dots, a_{(n-1)j}$ is 2^{k-1} as there are 2 permutations for each pair before $\{b_{ij}, a_{i(j+1)}\} = \{a_{ij}, b_{(i+1)j}\}$ and each pair after is determined by $b_{ij} = b_{(i+1)j} = k, a_{i(j+1)} = a_{ij}$. For $k = 0$, the permutation is completely determined so there is one way.

Overall the number of ways to add the $j + 1$ 'th row is,

$$(n - j)(1 + \sum_{k=1}^n 2^{k-1}) = 2^n(n - j).$$

In total, there are $(2n)(2n - 1) \cdots (n + 1)$ ways to find C_0 and for each C_j there are $2^n(n - j + 1)$ ways for $1 \leq j \leq n$. So the answer is,

$$\frac{2n!}{n!}(2^n n)(2^n(n - 1)) \cdots (2^n 1) = \boxed{2^{n^2} \cdot (2n)!}.$$

5. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Two rational numbers $\frac{m}{n}$ and $\frac{n}{m}$ are written on a blackboard, where m and n are relatively prime positive integers. At any point, Evan may pick two of the numbers x and y written on the board and write either their arithmetic mean $\frac{x+y}{2}$ or their harmonic mean $\frac{2xy}{x+y}$ on the board as well. Find all pairs (m, n) such that Evan can write 1 on the board in finitely many steps.

Solution

We claim that all odd m, n work if $m + n$ is a positive power of 2.

Proof: We first prove that $m + n = 2^k$ works. By weighted averages we have that $\frac{n(\frac{m}{n}) + (2^k - n)\frac{n}{m}}{2^k} = \frac{m+n}{2^k} = 1$ can be written, so the solution set does indeed work. We will now prove these are the only solutions.

Assume that $m + n \neq 2^k$, so then $m + n \equiv 0 \pmod{p}$ for some odd prime p . Then $m \equiv -n \pmod{p}$, so $\frac{m}{n} \equiv \frac{n}{m} \equiv -1 \pmod{p}$. We see that the arithmetic mean is $\frac{-1+(-1)}{2} \equiv -1 \pmod{p}$ and the harmonic mean is $\frac{2(-1)(-1)}{-1+(-1)} \equiv -1 \pmod{p}$, so if 1 can be written then $1 \equiv -1 \pmod{p}$ and $2 \equiv 0 \pmod{p}$ which is obviously impossible, and we are done.

-Stormersyle

6. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Find all polynomials P with real coefficients such that

$$\frac{P(x)}{yz} + \frac{P(y)}{zx} + \frac{P(z)}{xy} = P(x - y) + P(y - z) + P(z - x)$$

holds for all nonzero real numbers x, y, z satisfying $2xyz = x + y + z$.

Solution

If $P(x) = c$ for a constant c , then $\frac{c(x + y + z)}{xyz} = 3c$. We have $2c = 3c$. Therefore $c = 0$.

Now consider the case of non-constant polynomials. First we have

$$xP(x) + yP(y) + zP(z) = xyz(P(x - y) + P(y - z) + P(z - x))$$

for all nonzero real numbers x, y, z satisfying $2xyz = x + y + z$. Both sides of the equality are polynomials (of x, y, z). They have the same values on the 2-dimensional surface $2xyz = x + y + z$, except for some 1-dimensional curves in it. By continuity, the equality holds for all points on the surface, including those with $z = 0$. Let $z = 0$, we have $y = -x$ and $x(P(x) - P(-x)) = 0$. Therefore P is an even function.

(Here is a sketch of an elementary proof. Let $z = \frac{x+y}{2xy-1}$. We have

$$xP(x) + yP(y) + \frac{x+y}{2xy-1}P\left(\frac{x+y}{2xy-1}\right) = xy\frac{x+y}{2xy-1}\left(P(x-y) + P\left(y - \frac{x+y}{2xy-1}\right) + P\left(\frac{x+y}{2xy-1} - x\right)\right).$$

This is an equality of rational expressions. By multiplying $(2xy-1)^N$ on both sides for a sufficiently large N , they become polynomials, say $A(x, y) = B(x, y)$ for all real x, y with $x \neq 0, y \neq 0, x+y \neq 0$ and $2xy-1 \neq 0$. For a fixed x , we have two polynomials (of y) having same values for infinitely many y . They must be identical. Let $y = 0$, we have $x^{N+1}(P(x) - P(-x)) = 0$.)

Notice that if $P(x)$ is a solution, then is $cP(x)$ for any constant c . For simplicity, we assume the leading coefficient of P is 1:

$$P(x) = x^n + a_{n-2}x^{n-2} + \cdots + a_2x^2 + a_0,$$

where n is a positive even number.

Let $y = \frac{1}{x}$, $z = x + \frac{1}{x}$. we have

$$xP(x) + \frac{1}{x}P\left(\frac{1}{x}\right) + \left(x + \frac{1}{x}\right)P\left(x + \frac{1}{x}\right) = \left(x + \frac{1}{x}\right)\left(P\left(x - \frac{1}{x}\right) + P(-x) + P\left(\frac{1}{x}\right)\right).$$

Simplify using $P(x) = P(-x)$,

$$\left(x + \frac{1}{x}\right)\left(P\left(x + \frac{1}{x}\right) - P\left(x - \frac{1}{x}\right)\right) = \frac{1}{x}P(x) + xP\left(\frac{1}{x}\right).$$

Expand and combine like terms, both sides are of the form

$$c_{n-1}x^{n-1} + c_{n-3}x^{n-3} + \cdots + c_1x + c_{-1}x^{-1} + \cdots + c_{-n+1}x^{-n+1}.$$

They have the same values for infinitely many x . They must be identical. We just compare their leading terms. On the left hand side it is $2nx^{n-1}$. There are two cases for the right hand sides: If $n > 2$, it is x^{n-1} ; If $n = 2$, it is $(1 + a_0)x$. It does not work for $n > 2$. When $n = 2$, we have $4 = 1 + a_0$. therefore $a_0 = 3$.

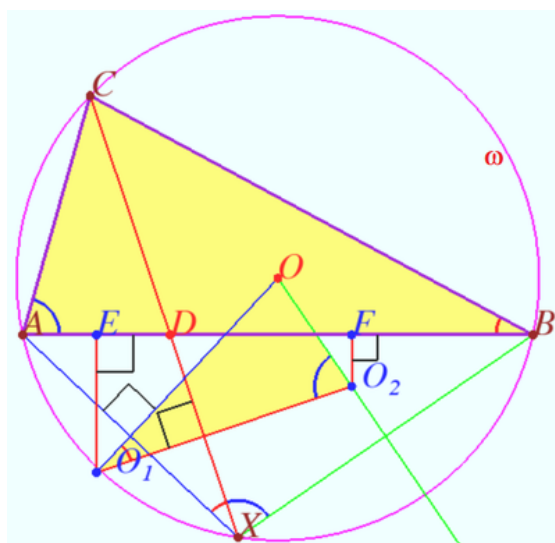
The solution: $P(x) = c(x^2 + 3)$ for any constant c .

-JZ

§49 USAMO 2020

1. (USAMO 2020) Let ABC be a fixed acute triangle inscribed in a circle ω with center O . A variable point X is chosen on minor arc AB of ω , and segments CX and AB meet at D . Denote by O_1 and O_2 the circumcenters of triangles ADX and BDX , respectively. Determine all points X for which the area of triangle OO_1O_2 is minimized.

Solution



Let E be midpoint AD . Let F be midpoint $BD \implies$

$$EF = ED + FD = \frac{AD}{2} + \frac{BD}{2} = \frac{AB}{2}.$$

E and F are the bases of perpendiculars dropped from O_1 and O_2 , respectively. Therefore $O_1O_2 \geq EF = \frac{AB}{2}$.

$$CX \perp O_1O_2, AX \perp O_1O \implies \angle OO_1O_2 = \angle AXC$$

$$\angle AXC = \angle ABC (AXBC \text{ is cyclic}) \implies \angle OO_1O_2 = \angle ABC.$$

$$\text{Similarly } \angle BAC = \angle OO_2O_1 \implies \triangle ABC \sim \triangle O_2O_1O.$$

The area of $\triangle OO_1O_2$ is minimized if $CX \perp AB$ because

$$\frac{[OO_1O_2]}{[ABC]} = \left(\frac{O_1O_2}{AB} \right)^2 \geq \left(\frac{EF}{AB} \right)^2 = \frac{1}{4}.$$

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2. (USAMO 2020) An empty $2020 \times 2020 \times 2020$ cube is given, and a 2020×2020 grid of square unit cells is drawn on each of its six faces. A beam is a $1 \times 1 \times 2020$ rectangular prism. Several beams are placed inside the cube subject to the following conditions:

- The two 1×1 faces of each beam coincide with unit cells lying on opposite faces of the cube. (Hence, there are $3 \cdot 2020^2$ possible positions for a beam.)
- No two beams have intersecting interiors.
- The interiors of each of the four 1×2020 faces of each beam touch either a face of the cube or the interior of the face of another beam.

What is the smallest positive number of beams that can be placed to satisfy these conditions?

Solution

Take one vertex of the cube as origin and establish 3D coordinates along the cube's edges.

Define a beam as $x - dir$ if its long edge is parallel to x-axis. Similarly for $y - dir$ and $z - dir$.

Define a beam's location as (direction, $(1 \times 1$ face's location in 2D coordinate).

For example, $(y, 2, 4)$ indicates the beam with vertex $(1, 0, 3)$ and $(2, 2020, 4)$

Apparently x beam needs the other x or y beams to touch its xy faces, x or z beams to touch its xz faces. Similarly for y and z beam.

If there are only 1-dir or 2-dirs beams, it is easy to approve that 2020×2020 is the minimal number beams.

(for example, if only use $x - dir$ and $y - dir$ beams, all the $x - dir$ beams's xz faces can only be touched by $x - dir$ beams. In the other word, $2020x - dir$ beams will be stacked to meet xz faces touch requirements in that xz layer)

Consider cases with all 3 - dirs beams. WLOG there is a $x - dir$ beam and it needs $x - dir$ or $y - dir$ beams to touch its xy faces (unless it touches the cube surface).

And this $y - dir$ beam also needs a $x - dir$ or $y - dir$ to touch it's xy faces. And so on until one which touches cube surface. So from xy face perspective, it needs 2020 beams.

Similarly from xz and yz face perspective, it also needs 2020 and 2020 beams.

Consider one beam has four 1×2020 faces and it can be counted twice. So there should be at least $2020 \times 3 \div 2 = 3030$ beams.

Here is one solution with 3030 beams.

$(x, 1, 1), (y, 1, 2), (z, 2, 2),$
 $\dots,$
 $(x, (2n + 1), (2n + 1)), (y, (2n + 1), (2n + 2)), (z, (2n + 2), (2n + 2)),$
 $\dots,$
 $(x, (2019, 2019)), (y, 2019, 2020), (z, 2020, 2020)$

3. (USAMO 2020) Let p be an odd prime. An integer x is called a *quadratic non-residue* if p does not divide $x - t^2$ for any integer t .

Denote by A the set of all integers a such that $1 \leq a < p$, and both a and $4 - a$ are quadratic non-residues. Calculate the remainder when the product of the elements of A is divided by p .

Solution

We will be using finite field theory and the Frobenius endomorphism freely. Also we will just give a sketch and leave out details like it's out of fashion.

Case 1: $p \equiv 3 \pmod{4}$. Work in $\mathbb{F}_{p^2} = \mathbb{F}_p[i]$. Suppose $a \in A$. Then, $a = -x^2$ for some $x \in \mathbb{F}_p$ and $4 - a = y^2$ for some $y \in \mathbb{F}_p$, so we have

$$x^2 + y^2 = -4.$$

This factors as

$$(x + iy)(x - iy) = (2i)^2,$$

so we have $x + iy = 2it$ and $x - iy = 2i/t$ for some $t \in \mathbb{F}_p[i]$, so

$$x = i(t + 1/t) \quad \text{and} \quad y = t - 1/t.$$

For both of these to be in $\mathbb{F}_p$, we need $x^p = x$ and $y^p = y$, which when we work out with Frobenius, tells us that $t^{p+1} + 1 = 0$.

Thus, if $a \in A$, we must have $a = (t + 1/t)^2$ for some $t \in \mathbb{F}_p[i]$ such that $t^{p+1} + 1 = 0$. Actually it is easy to see that if t satisfies $t^{p+1} + 1 = 0$, then both a and $4 - a$ are QNRs (just reverse the Frobenius calculation above). Thus,

$$A = \{(t + 1/t)^2 : t \in \mathbb{F}_p[i], t^{p+1} + 1 = 0\}.$$

Note $(t + 1/t)^2 = (t' + 1/t')^2$ if and only if $t' \in \{t, -t, 1/t, -1/t\}$, and also $t^{p+1} + 1 = 0 \iff (t')^{p+1} + 1 = 0$. Note that

$$t^{p+1} + 1 \mid t^{p^2} - t$$

as $2(p+1) \mid p^2 - 1$, so $t^{p+1} + 1$ has distinct roots in $\mathbb{F}_p[i]$. We see that

$$t^{p+1} + 1 = (t^{\frac{p+1}{2}} + i)(t^{\frac{p+1}{2}} - i),$$

and s a root of $t^{\frac{p+1}{2}} - i$ if and only if $1/s$ is a root. Thus, it is easy to see that

$$A = \{(t + 1/t)^2 : t \in \mathbb{F}_p[i], t^{\frac{p+1}{2}} - i = 0\}.$$

In this parameterization, only t and $-t$ produce the same value (and these never coincide as $t \neq 0$), so we actually have

$$T := \prod_{a \in A} = (-1)^{\frac{p+1}{4}} \prod_{\substack{t \in \mathbb{F}_p[i] \\ t^{\frac{p+1}{2}} - i = 0}} (t + 1/t).$$

This becomes

$$T = (-1)^{\frac{p+1}{4}} \prod_{\substack{t \in \mathbb{F}_p[i] \\ t^{\frac{p+1}{2}} - i = 0}} \frac{(t + i)(t - i)}{t}.$$

Letting $f(x) = x^{\frac{p+1}{2}} - i$, we see that this product becomes

$$T = (-1)^{\frac{p+1}{4}} \frac{f(-i)f(i)}{f(0)}.$$

It is tedious but straightforward to see that this always gives 2.

Case 2: $p \equiv 1 \pmod{4}$. Let ℓ be some QNR mod p (sadly we don't have any free QNRs like -1 is for $3 \pmod{4}$ primes). Let ω be a formal square root of ℓ , and work in $\mathbb{F}_{p^2} = \mathbb{F}_p[\omega]$. Then, if $a \in A$, we have

$$\omega^2 a = x^2 \quad \text{and} \quad \omega^2(4 - a) = y^2,$$

so $x^2 + y^2 = 4\omega^2$, so factoring and doing the same as before, we get

$$x = \omega(t + 1/t) \quad \text{and} \quad y = \frac{\omega}{i}(t - 1/t)$$

for some $t \in \mathbb{F}_p[\omega]$. These are in $\mathbb{F}_p$ if and only if $t^{p-1} + 1 = 0$ (using Frobenius). We see $a = (t + 1/t)^2$, so

$$A = \{(t + 1/t)^2 : t \in \mathbb{F}_p[\omega], t^{p-1} + 1 = 0\}.$$

Doing a similar analysis as before, the desired product becomes

$$T := \prod_{a \in A} = (-1)^{\frac{p-1}{4}} \prod_{\substack{t \in \mathbb{F}_p[\omega] \\ t^{\frac{p-1}{2}} - i = 0}} \frac{(t+i)(t-i)}{t},$$

and letting $f(x) = x^{\frac{p-1}{2}} - i$, we get

$$T = (-1)^{\frac{p-1}{4}} \frac{f(-i)f(i)}{f(0)},$$

which again is always 2.

So the answer is $\boxed{2}$.

4. (USAMO 2020) Suppose that $(a_1, b_1), (a_2, b_2), \dots, (a_{100}, b_{100})$ are distinct ordered pairs of nonnegative integers. Let N denote the number of pairs of integers (i, j) satisfying $1 \leq i < j \leq 100$ and $|a_i b_j - a_j b_i| = 1$. Determine the largest possible value of N over all possible choices of the 100 ordered pairs.

Solution

Let's start off with just $(a_1, b_1), (a_2, b_2)$ and suppose that it satisfies the given condition. We could use $(1, 1), (1, 2)$ for example. We should maximize the number of conditions that the third pair satisfies. We find out that the third pair should equal $(a_1 + a_2, b_1 + b_2)$:

We know this must be true:

$$|a_1 b_2 - a_2 b_1| = 1$$

So

$$a_1 b_2 - a_2 b_1 = 1$$

We require the maximum conditions for (a_3, b_3)

$$|a_3 b_2 - a_2 b_3| = 1$$

$$|a_3 b_1 - a_1 b_3| = 1$$

Then one case can be:

$$a_3 b_2 - a_2 b_3 = 1$$

$$a_3 b_1 - a_1 b_3 = -1$$

We try to do some stuff such as solving for a_3 with manipulations:

$$a_3 b_2 a_1 - a_2 b_3 a_1 = a_1$$

$$a_3 b_1 a_2 - a_1 b_3 a_2 = -a_2$$

$$a_3(a_1 b_2 - a_2 b_1) = a_1 + a_2$$

$$a_3 = a_1 + a_2$$

$$\begin{aligned}a_3b_2b_1 - a_2b_3b_1 &= b_1 \\a_3b_1b_2 - a_1b_3b_2 &= -b_2 \\b_3(a_1b_2 - a_2b_1) &= b_1 + b_2 \\b_3 &= b_1 + b_2\end{aligned}$$

We showed that 3 pairs are a complete graph; however, 4 pairs are not a complete graph. We will now show that:

$$\begin{aligned}a_4 &= a_1 + 2a_2 \\b_4 &= b_1 + 2b_2 \\|a_1b_1 + 2a_2b_1 - a_1b_1 - 2a_1b_2| &= 1 \\2|a_2b_1 - a_1b_2| &= 1\end{aligned}$$

This is clearly impossible because 1 is not even and also $|a_2b_1 - a_1b_2| = 1$. The answer is as follows:

$$0 + 1 + 2 + \dots + 2$$

a_1 has 0 subtractions that follow condition while a_2 has 1 and then the rest has 2. There are n terms, so our answer be $2n - 3$ and in case of $n = 100$ that means

$$\boxed{N = 197}.$$

Lopkiloim

Solution 2

We claim the answer is 197.

Study the points $(0, 0), (a_i, b_i), (a_j, b_j)$. If we let these be the vertices of a triangle, applying shoelace theorem gives us an area of $\frac{1}{2}|0 \times b_i + a_i \times b_j + b_i \times 0 - 0 \times a_i - b_i \times a_j - b_j \times 0| = \frac{1}{2}|a_i b_j - a_j b_i| = \frac{1}{2}$. Therefore, the triangle formed by the points $(0, 0), (a_i, b_i), (a_j, b_j)$ must have an area of $\frac{1}{2}$.

Two cases follow. Case 1: Both $(a_i, b_i), (a_j, b_j)$ have exactly one coordinate equal to 0. Here, one point must be on the x axis and the other on the y axis in order for the triangle to have a positive area. For the area of the triangle to be $\frac{1}{2}$, it follows that the points must be $(1, 0), (0, 1)$ in some order.

Case 2: At least one of $(a_i, b_i), (a_j, b_j)$ does not have exactly one coordinate equal to 0. Define $S[l]$ to be a list of lines such that each line in the list has some two lattice points that, with $(0, 0)$, form a triangle with area $\frac{1}{2}$. Note that for any such line that passes through such two lattice points, we may trivially generate infinite lattice points on the line that have nonnegative coordinates.

Note that lines $y = 1$ and $x = 1$ are included in $S[l]$, because the points $(1, 1), (2, 1)$ serve as examples for $y = 1$ and $(1, 1), (1, 2)$ serve as examples for $x = 1$. For the optimal construction, include the points $(1, 0)$ and all the points $(0, 1), (0, 2), (0, 3), \dots, (0, 99)$, in that order. In this case, every adjacent pair of points would count (98), as well as picking $(0, 1)$ and a nonadjacent point (99), so this would be $98 + 99 = 197$.

To prove that this is the maximum, consider the case where some n number of points were neither on $x = 1$ nor on $y = 1$. In this case, we would be removing n adjacent pairs and n options to choose from after choosing $(0, 0)$, resulting in a net

loss of $2n$. By having n points on some other combination of lines in $S[l]$, we would trivially have a maximum gain of $n - 1$ pairs of points on the lines such that there are no lattice points between those pairs. Because these points are not on $x = 1$ or $y = 1$, the altitude from a given point to the line formed by $(0, 0)$ and $(0, 1)$ and $(1, 0)$ is not 1, and so the area of the triangle cannot be $\frac{1}{2}$. Thus, by not having all points on lines $x = 1$ and $y = 1$, we cannot exceed the maximum of 197. Thus, 197 is our answer.

5. (USAMO 2020) A finite set S of points in the coordinate plane is called *overdetermined* if $|S| \geq 2$ and there exists a nonzero polynomial $P(t)$, with real coefficients and of degree at most $|S| - 2$, satisfying $P(x) = y$ for every point $(x, y) \in S$.

For each integer $n \geq 2$, find the largest integer k (in terms of n) such that there exists a set of n distinct points that is not overdetermined, but has k overdetermined subsets.

Solution

The answer is $2^{n-1} - n$. To construct this, have $n - 1$ points on a horizontal line and 1 point not on it. Then, any subset that does not include the last point is overdetermined.

For the bound, the main idea of the problem is the following claim.

Claim: If S and T are overdetermined subsets with $|S| = |T| = k$ but S and T only differ by one element (one is swapped out for another), then $S \cup T$ is overdetermined.

Consider the minimal polynomial of S . It has degree at most $k - 2$ since S is overdetermined, and similarly with the minimal polynomial of T . However, there is a unique polynomial of degree at most $k - 2$ passing through $S \cap T$ since it has $k - 1$ points. Thus, this unique polynomial also passes through S and T . Thus, this polynomial passes through all points of $S \cup T$, as desired.

We now use this claim to inductively bound the number of overdetermined subsets with k elements. Let m_k denote the number of overdetermined subsets of size k . Clearly, $m_n = 0$.

Claim 2: We have

$$m_k \leq \frac{m_{k+1}(k+1) + \binom{n}{k+1} - m_{k+1}}{n-k}.$$

Proof: Each overdetermined subset of size $k + 1$ can have up to $k + 1$ overdetermined subsets of size k , but by the previous claim each non-overdetermined subset of size $k + 1$, which there are $\binom{n}{k+1} - m_{k+1}$ of, can have at most one. Each subset of size k feeds into $n - k$ subsets of size $k + 1$. Note that this is increasing in m_{k+1} .

Claim 3: We have

$$m_k \leq \binom{n-1}{k}.$$

Proof: Use induction. Note that the expression in Claim 2 is nondecreasing in m_{k+1} , and verify that

$$\frac{\binom{n-1}{k+1}(k+1) + \binom{n}{k+1} - \binom{n-1}{k+1}}{n-k} = \binom{n-1}{k}.$$

Thus, we have that the number of overdetermined subsets is at most

$$\sum_{k=2}^n \binom{n-1}{k} = 2^{n-1} - n,$$

as desired.

6. (USAMO 2020) Let $n \geq 2$ be an integer. Let $x_1 \geq x_2 \geq \cdots \geq x_n$ and $y_1 \geq y_2 \geq \cdots \geq y_n$ be $2n$ real numbers such that

$$0 = x_1 + x_2 + \cdots + x_n = y_1 + y_2 + \cdots + y_n$$

$$\text{and } 1 = x_1^2 + x_2^2 + \cdots + x_n^2 = y_1^2 + y_2^2 + \cdots + y_n^2.$$

Prove that

$$\sum_{i=1}^n (x_i y_i - x_i y_{n+1-i}) \geq \frac{2}{\sqrt{n-1}}.$$

Solution

Let $u_i = x_i - x_{n+1-i}$ and $v_i = y_i - y_{n+1-i}$ for $1 \leq i \leq n$.

Since $x_1 \geq x_2 \geq \cdots \geq x_n$ and $y_1 \geq y_2 \geq \cdots \geq y_n$, both (u_i) and (v_i) are non-increasing sequences satisfying $u_{n+1-i} = -u_i$ and $v_{n+1-i} = -v_i$.

We can express the target sum in terms of u_i and v_i :

$$\sum_{i=1}^n (x_i y_i - x_i y_{n+1-i}) = \sum_{i=1}^n x_i v_i = \frac{1}{2} \sum_{i=1}^n u_i v_i$$

By the Rearrangement Inequality, because (y_i) is non-increasing and (y_{n+1-i}) is non-decreasing:

$$\sum_{i=1}^n y_i y_{n+1-i} \leq \frac{1}{n} \left(\sum_{i=1}^n y_i \right)^2 = 0$$

This gives a lower bound on the sum of squares of v_i :

$$\sum_{i=1}^n v_i^2 = 2 \sum_{i=1}^n y_i^2 - 2 \sum_{i=1}^n y_i y_{n+1-i} \geq 2$$

Similarly, $\sum_{i=1}^n u_i^2 \geq 2$.

Applying summation by parts along with the Cauchy-Schwarz inequality for ordered sequences subject to $\sum x_i = \sum y_i = 0$ and $\sum x_i^2 = \sum y_i^2 = 1$ yields:

$$\frac{1}{2} \sum_{i=1}^n u_i v_i \geq \frac{2}{\sqrt{n-1}}$$

Equality is achieved when $x_1 = y_1 = \frac{1}{\sqrt{2}}$, $x_n = y_n = -\frac{1}{\sqrt{2}}$, and all remaining middle variables are equal to 0.

§50 USAMO 2021

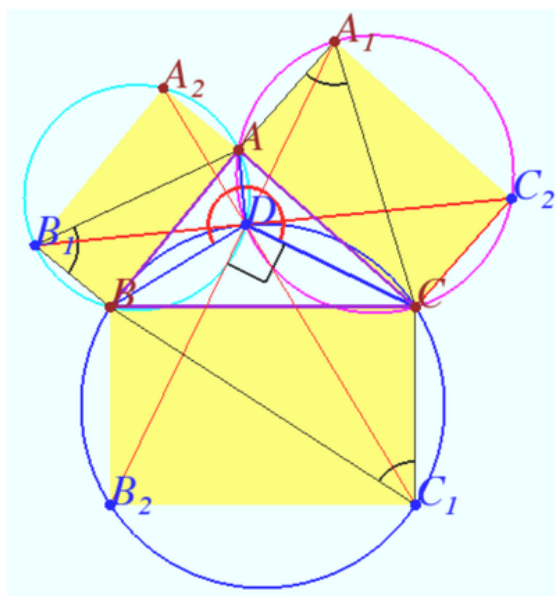
1. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Rectangles BCC_1B_2 , CAA_1C_2 , and ABB_1A_2 are erected outside an acute triangle ABC . Suppose that

$$\angle BC_1C + \angle CA_1A + \angle AB_1B = 180^\circ.$$

Prove that lines B_1C_2 , C_1A_2 , and A_1B_2 are concurrent.

Solution



Let D be the second point of intersection of the circles (AB_1B) and (AA_1C) . Then:

$$\begin{aligned}\angle ADB &= 180^\circ - \angle AB_1B, & \angle ADC &= 180^\circ - \angle AA_1C \\ \angle BDC &= 360^\circ - \angle ADB - \angle ADC \\ &= 360^\circ - (180^\circ - \angle AB_1B) - (180^\circ - \angle AA_1C) \\ &= \angle AB_1B + \angle AA_1C\end{aligned}$$

$$\angle BDC + \angle BC_1C = 180^\circ$$

Therefore, BDC_1B_2 is cyclic with diameters BC_1 and CB_2 , and thus $\angle CDB_2 = 90^\circ$. Similarly, $\angle CDA_1 = 90^\circ$, meaning points A_1 , D , and B_2 are collinear.

Similarly, the points A_2 , D , C_1 and C_2 , D , B_1 are collinear.

(After USAMO 2021 Solution Notes – Evan Chen)

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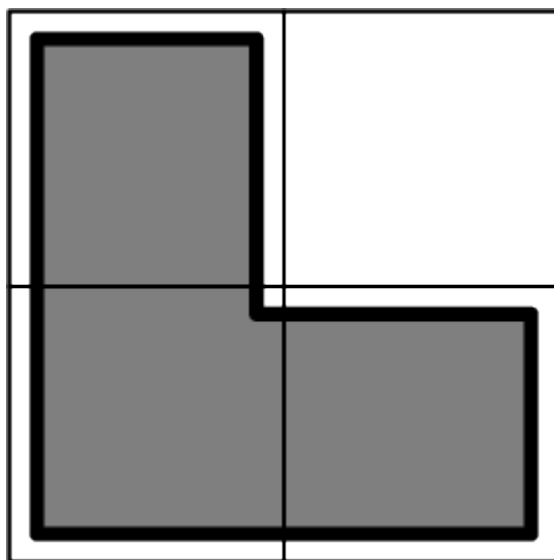
2. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

The Planar National Park is a subset of the Euclidean plane consisting of several trails which meet at junctions. Every trail has its two endpoints at two different junctions whereas each junction is the endpoint of exactly three trails. Trails only intersect at junctions (in particular, trails only meet at endpoints). Finally, no trails begin and end at the same two junctions.

A visitor walks through the park as follows: she begins at a junction and starts walking along a trail. At the end of that first trail, she enters a junction and turns left. On the next junction she turns right, and so on, alternating left and right turns at each junction. She does this until she gets back to the junction where she started. What is the largest possible number of times she could have entered any junction during her walk, over all possible layouts of the park?

3. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 2$ be an integer. An $n \times n$ board is initially empty. Each minute, you may perform one of three moves: If there is an L-shaped tromino region of three cells without stones on the board (see figure; rotations not allowed), you may place a stone in each of those cells. If all cells in a column have a stone, you may remove all stones from that column. If all cells in a row have a stone, you may remove all stones from that row.



For which n is it possible that, after some non-zero number of moves, the board has no stones?

Solution 1

Label the bottom right cell as ω^0 where $\omega = e^{i2\pi/3}$. Then label each cell with ω^d where d is the *Manhattan distance* of the current cell from the bottom right cell.

No matter where an L-shaped tromino is placed, the sum of the labels in the cells is,

$$\omega^0 + \omega^1 + \omega^2 = \frac{\omega^3 - 1}{\omega - 1} = \frac{0}{\omega - 1} = 0$$

If $n = 3k + 1$. Then the sum of all the labels in the cells in a row or column is,

$$k(\omega^0 + \omega^1 + \omega^2) + \omega^0 = 1$$

If $n = 3k + 2$. Then the sum of all the labels in cells in a row or column is,

$$k(\omega^0 + \omega^1 + \omega^2) + \omega^0 + \omega^1 = 1 + \omega$$

Let S be the sum of all the labels of cells containing a stone. In either above case, placing an L-shaped Tromino leaves S invariant but removing a row or column of stone causes $\text{Re}(S)$ to decrease (by 1 or $1/2$ in each case). However the empty grid has $S = 0$, so this can't be reached.

Hence $n = 3k$. We now provide a sequence of moves which empties the grid. Divide the grid into 3×3 grids and do the following to every sub-grid. (When I refer to placing an L-shaped tromino on a cell, I am referring to the position of the bottom left stone)

- Add an L-shaped tromino to the bottom left cell and middle cell
- Remove the middle row
- Add an L-shaped tromino to the middle left cell
- Remove the left and middle columns

and we're done, the grid is empty

4. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

A finite set S of positive integers has the property that, for each $s \in S$, and each positive integer divisor d of s , there exists a unique element $t \in S$ satisfying $\gcd(s, t) = d$. (The elements s and t could be equal.)

Given this information, find all possible values for the number of elements of S .

5. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 4$ be an integer. Find all positive real solutions to the following system of $2n$ equations:

$$\begin{aligned} a_1 &= \frac{1}{a_{2n}} + \frac{1}{a_2}, & a_2 &= a_1 + a_3, \\ a_3 &= \frac{1}{a_2} + \frac{1}{a_4}, & a_4 &= a_3 + a_5, \\ a_5 &= \frac{1}{a_4} + \frac{1}{a_6}, & a_6 &= a_5 + a_7 \\ &\vdots & &\vdots \end{aligned}$$

$$a_{2n-1} = \frac{1}{a_{2n-2}} + \frac{1}{a_{2n}}, \quad a_{2n} = a_{2n-1} + a_1$$

Solution

The unique positive real solution is $a_{2k-1} = 1$ and $a_{2k} = 2$ for all $1 \leq k \leq n$.

First, substitute $a_{2k-1} = \frac{1}{a_{2k-2}} + \frac{1}{a_{2k}}$ into $\sum_{k=1}^n a_{2k-1}^2$:

$$\sum_{k=1}^n a_{2k-1}^2 = \sum_{k=1}^n a_{2k-1} \left(\frac{1}{a_{2k-2}} + \frac{1}{a_{2k}} \right) = \sum_{k=1}^n \frac{a_{2k-1} + a_{2k+1}}{a_{2k}}$$

Using $a_{2k} = a_{2k-1} + a_{2k+1}$, this simplifies to:

$$\sum_{k=1}^n a_{2k-1}^2 = \sum_{k=1}^n \frac{a_{2k}}{a_{2k}} = n$$

Next, expand $\sum_{k=1}^n a_{2k-1}a_{2k}$ in two ways:

$$\sum_{k=1}^n a_{2k-1}a_{2k} = \sum_{k=1}^n a_{2k} \left(\frac{1}{a_{2k-2}} + \frac{1}{a_{2k}} \right) = n + \sum_{k=1}^n \frac{a_{2k}}{a_{2k-2}}$$

$$\sum_{k=1}^n a_{2k-1}a_{2k} = \sum_{k=1}^n a_{2k-1}(a_{2k-1} + a_{2k+1}) = \sum_{k=1}^n a_{2k-1}^2 + \sum_{k=1}^n a_{2k-1}a_{2k+1} = n + \sum_{k=1}^n a_{2k-1}a_{2k+1}$$

Equating the two expressions gives:

$$\sum_{k=1}^n a_{2k-1}a_{2k+1} = \sum_{k=1}^n \frac{a_{2k}}{a_{2k-2}}$$

By AM-GM and Cauchy-Schwarz:

$$\sum_{k=1}^n \frac{a_{2k}}{a_{2k-2}} \geq n \sqrt[n]{\prod_{k=1}^n \frac{a_{2k}}{a_{2k-2}}} = n$$

$$\sum_{k=1}^n a_{2k-1}a_{2k+1} \leq \sqrt{\sum_{k=1}^n a_{2k-1}^2} \sqrt{\sum_{k=1}^n a_{2k+1}^2} = n$$

Thus $n \leq \sum \frac{a_{2k}}{a_{2k-2}} = \sum a_{2k-1}a_{2k+1} \leq n$, forcing equality everywhere:

- Equality in AM-GM gives $a_2 = a_4 = \cdots = a_{2n} = E$.
- Equality in Cauchy-Schwarz gives $a_1 = a_3 = \cdots = a_{2n-1} = O$.

From $a_2 = a_1 + a_3$ and $a_1 = \frac{1}{a_{2n}} + \frac{1}{a_2}$, we get $E = 2O$ and $O = \frac{2}{E}$.

Solving $2O^2 = 2$ yields $O = 1$ and $E = 2$.

Hence, $a_{2k-1} = 1$ and $a_{2k} = 2$ for all $1 \leq k \leq n$.

6. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let $ABCDEF$ be a convex hexagon satisfying $\overline{AB} \parallel \overline{DE}$, $\overline{BC} \parallel \overline{EF}$, $\overline{CD} \parallel \overline{FA}$, and

$$AB \cdot DE = BC \cdot EF = CD \cdot FA.$$

Let X, Y , and Z be the midpoints of $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$. Prove that the circumcenter of $\triangle ACE$, the circumcenter of $\triangle BDF$, and the orthocenter of $\triangle XYZ$ are collinear.

Solution 1

Let M_1, M_2 , and M_3 be the midpoints of CE, AE, AC and N_1, N_2 , and N_3 be the midpoints of DF, BF , and BD . Also, let H be the orthocenter of XYZ . Note that we can use parallel sides to see that X, Z , and M_3 are collinear. Thus we have

$$\text{Pow}(M_3, (XYZ)) = M_3Z \cdot M_3X = \frac{1}{4}AB \cdot DE$$

by midlines. Applying this argument cyclically, and noting the condition $AB \cdot DE = BC \cdot EF = CD \cdot FA$, $M_1, M_2, M_3, N_1, N_2, N_3$ all lie on a circle concentric with (XYZ) .

Next, realize that basic orthocenter properties imply that the circumcenter O_1 of (ACE) is the orthocenter of $\triangle M_1M_2M_3$, and likewise the circumcenter O_2 of (BDF) is the orthocenter of $\triangle N_1N_2N_3$.

The rest is just complex numbers; toss on the complex plane so that the circumcenter of $\triangle XYZ$ is the origin. Then we have

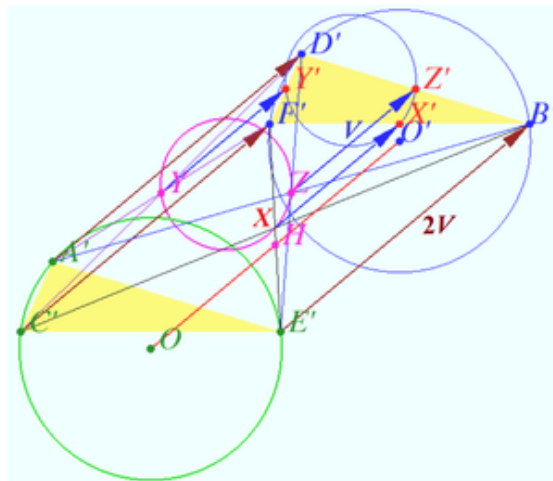
$$o_1 = m_1 + m_2 + m_3 = (c + e)/2 + (a + e)/2 + (a + c)/2 = a + c + e$$

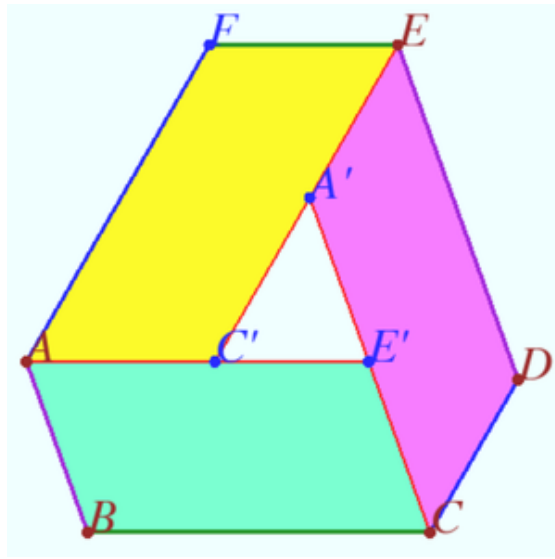
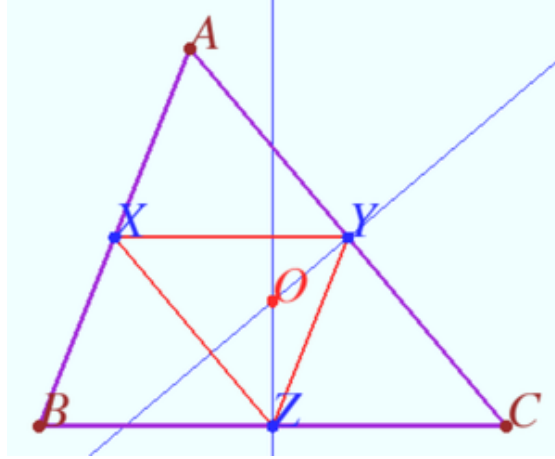
$$o_2 = n_1 + n_2 + n_3 = (b + d)/2 + (d + f)/2 + (b + f)/2 = b + d + f$$

$$h = x + y + z = (a + d)/2 + (b + e)/2 + (c + f)/2 = (a + b + c + d + e + f)/2.$$

Note that from the above we have $h = \frac{o_1 + o_2}{2}$, so H is the midpoint of segment O_1O_2 . In particular, H, O_1 , and O_2 are collinear, as required.

Solution 2





We construct two equal triangles, prove that triangle XYZ is the same as medial triangle of both this triangles. We use property of medial triangle and prove that circumcenters of constructed triangles coincide with given circumcenters.

Denote $A' = C + E \smile D, B' = D + F \smile E, C' = A + E \smile F, D' = F + B \smile A, E' = A + C \smile B, F' = B + D \smile C$. Then $A' \smile D' = C + E \smile D \smile (F + B \smile A) = (A + C + E) \smile (B + D + F)$.

Denote $D' \smile A' = 2\vec{V}$.

Similarly we get $B' \smile E' = F' \smile C' = D' \smile A' \implies \triangle A'C'E' = \triangle D'F'B'$.

The translation vector maps $\triangle A'C'E'$ into $\triangle D'F'B'$ is $2\vec{V}$. $X = \frac{A+D}{2} = \frac{(A+E \smile F) + (D+F \smile E)}{2} = \frac{C'+B'}{2} = \frac{E'+F'}{2}$,

so X is midpoint of $AD, B'C'$, and $E'F'$. Similarly Y is the midpoint of $BE, A'F'$, and $C'D'$, Z is the midpoint of $CF, A'B'$, and $D'E'$. $Z + V = \frac{A'+B'}{2} + \frac{D' \smile A'}{2} = \frac{B'+D'}{2} = Z'$ is the midpoint of $B'D'$.

Similarly $X' = X + V$ is the midpoint of $B'F'$, $Y' = Y + V$ is the midpoint of $D'F'$.

Therefore $\triangle X'Y'Z'$ is the medial triangle of $\triangle B'D'F'$.

$\triangle XYZ$ is $\triangle X'Y'Z'$ translated on $\smile \vec{V}$.

It is known (see diagram) that circumcenter of triangle coincide with orthocenter of the medial triangle. Therefore orthocenter H of $\triangle XYZ$ is circumcenter of $\triangle B'D'F'$ translated on $\vec{V}$.

It is the midpoint of segment OO' connected circumcenters of $\triangle B'D'F'$ and $\triangle A'C'E'$.

According to the definition of points A', C', E' , quadrangles $ABCE', CDEA'$, and $AFEC'$ are parallelograms. Hence

$$AC' = FE, AE' = BC, CE' = AB, CA' = DE, EA' = CD, EC' = AF \implies$$

$$AC' \cdot AE' = CE' \cdot CA' = EA' \cdot EC' = AB \cdot DE \implies$$

Power of points A, C, and E with respect circumcircle $\triangle A'C'E'$ is equal, hence distances between these points and circumcenter of $\triangle A'C'E'$ are the same. Therefore circumcenter $\triangle ACE$ coincide with circumcenter $\triangle A'C'E'$.

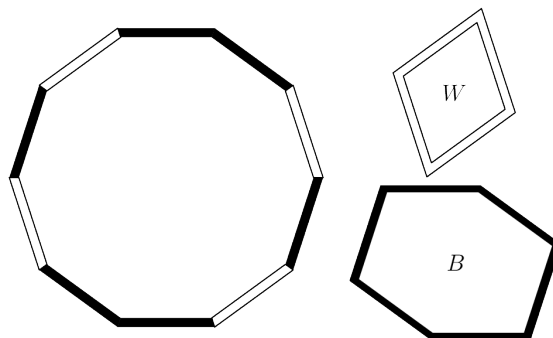
Similarly circumcenter of $\triangle BDF$ coincide with circumcenter of $\triangle B'D'F'$.

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§51 USAMO 2022

- (USAMO 2022) Let a and b be positive integers. The cells of an $(a+b+1) \times (a+b+1)$ grid are colored amber and bronze such that there are at least $a^2 + ab - b$ amber cells and at least $b^2 + ab - a$ bronze cells. Prove that it is possible to choose a amber cells and b bronze cells such that no two of the $a+b$ chosen cells lie in the same row or column.
- (USAMO 2022) Let $b \geq 2$ and $w \geq 2$ be fixed integers, and $n = b + w$. Given are $2b$ identical black rods and $2w$ identical white rods, each of side length 1.

We assemble a regular $2n$ -gon using these rods so that parallel sides are the same color. Then, a convex $2b$ -gon B is formed by translating the black rods, and a convex $2w$ -gon W is formed by translating the white rods. An example of one way of doing the assembly when $b = 3$ and $w = 2$ is shown below, as well as the resulting polygons B and W .

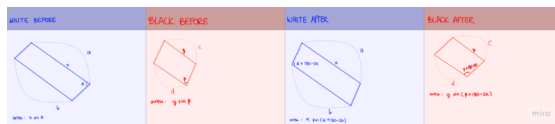


Prove that the difference of the areas of B and W depends only on the numbers b and w , and not on how the $2n$ -gon was assembled.

Solution 1

First notice that the black rods and the white rods form polygons iff in the original $2n$ -gon, if a side is a color x , then the side that is parallel to that side in the original $2n$ -gon is also the color x .

We can prove that the difference in areas is only affected by the values of b and w by showing that for any valid arrangement of $2b$ rods and $2w$ rods, we may switch any two adjacent black and white rods (and their "parallel pairs"), and end up with the same area difference.



In the figure above (click to expand), after the switch, we can see that after removing the mutually congruent parts, we are left with two parallelograms from each color. Let x , α , y , and β be defined as shown. Notice that if we angle chase, the sides of the other parallelogram are the same, but if the angles of the original $2n$ -gon all have measure $2k$, the angles of the new parallelograms are $\alpha + 180 - 2k$ and $\beta + 180 - 2k$, as shown. We must prove that the differences between the areas are the same.

Using area formulas, the change in the difference of areas is $x \sin \alpha - y \sin \beta + y \sin (2k - \beta) - x \sin (2k - \alpha)$, which is equal to $x(\sin k(2 \cos^2 k) - 2 \sin k \cos k \cos \alpha) - y(\sin k(2 \cos^2 k)) + y(\sin k \cos k \cos \beta)$, or $2 \cos k(x \sin (\alpha - k) + y \sin (k - \beta))$. Since $\cos k$ is not 0 because $n \geq 3$, we are left with proving that $x \sin (\alpha - k) + y \sin (k - \beta) = 0$.

Now we rotate the polygon so that the vertex between the two sides that we switched is at the point $(0, 0)$, the angle bisector of that vertex is $y = 0$, and the black side is in the positive y -direction. Now think of all the sides as vectors, all pointing in the clockwise direction of the $2n$ -gon.

Notice the part labeled a in the black polygons. We have that the vector labeled x is really just the sum of all of the vectors in the part labeled a - or all the vectors in the $2n$ -gon that are in the positive y -direction excluding the one that was interchanged. Also notice that the angle of this vector x has a signed angle of $\alpha - k$ with $y = 0$ and has length x - meaning that the vertical displacement of the vector x from $y = 0$ is equal to $x \sin (\alpha - k)$! Similarly, we get that the vertical displacement of the vector y is equivalent to $y \sin (k - \beta)$.

Adding these two together, we get that $x \sin (\alpha - k) + y \sin (k - \beta)$ is simply the vertical displacement of the sum of the vectors x and y . Since the sum of the vectors x and y is equivalent to the sum of the vectors in the positive half of the polygon minus the sum of the black vector that would be switched with the white vector (the leftmost vector in the positive half of the polygon) and the rightmost vector in the positive half (which is the parallel pair of the white vector that would be interchanged later), and we know that this sum happens to have a vertical displacement of 0, along with the fact that the positive half of the polygon summed together also has a vertical displacement of 0, we get that the total vertical displacement is 0, meaning that $x \sin (\alpha - k) + y \sin (k - \beta) = 0$, and we are done.

Solution 2

Pick a pair of parallel sides of the regular $2n$ -gon and flip their color. Let those two lines be the horizontal. The area of the polygon formed by rods of the original

color decreases by a parallelogram with height equal to the sum of the vertical heights of the rods of that color divided by 2 and base length 1. Similarly, the area of the polygon formed by rods of the new color increases by the sum of the vertical heights of the rods of the new color divided by 2. The sum of the heights of all the rod is equal to twice the height of the $2n$ -gon, so the difference between the areas of B and W changes by the height of the $2n$ -gon, which is fixed when n is fixed.

If we pair two pairs different-colored pairs of parallel sides of the $2n$ -gon and flip the colors of both pairs, then b and w and the difference in the areas of B and W will remain unchanged. Thus, the difference in the areas of B and W depends only on b and w .

3. (USAMO 2022) Let $\mathbb{R}_{>0}$ be the set of all positive real numbers. Find all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that for all $x, y \in \mathbb{R}_{>0}$ we have

$$f(x) = f(f(f(x)) + y) + f(xf(y))f(x + y).$$

Solution

Denote the n th iteration of the function $f(x)$ as $f^{(n)}(x)$.

First we prove that, for any $x, y > 0$ such that $f(x) \leq f(y)$, we have $f^{(2)}(x) \geq y$. (1)

Otherwise, $y > f^{(2)}(x)$, so $y - f^{(2)}(x) > 0$; we call this value c , and since $c > 0$, we can substitute y with c in the equation given in the problem, which gives

$$f(x) = f(f(f(x)) + c) + f(xf(c))f(x + c)$$

Recall that $c = y - f^{(2)}(x)$, which means $f(f(x)) + c = y$; so $f(x) = f(y) + f(xf(c))f(x + c)$. However, since $f(x)$ is defined on $\mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$, we know $f(xf(c)) > 0$, $f(x + c) > 0$; so that $f(xf(c))f(x + c) > 0$, and $f(x) = f(y) + f(xf(c))f(x + c) > f(y)$, which is a contradiction.

Now, since $f(x) \leq f(x)$, we know $f^{(2)}(x) \geq x$. Therefore, $f^{(4)}(x) = f^{(2)}(f^{(2)}(x)) \geq f^{(2)}(x)$.

Also, we know $f^{(3)}(x) = f^{(2)}(f(x)) \geq f(x)$, and hence $f^{(5)}(x) = f^{(2)}(f^{(3)}(x)) \geq f^{(3)}(x)$.

This means $f(x) \leq f^{(5)}(x)$, which is $f(x) \leq f(f^{(4)}(x))$; from (1) we know $f^{(2)}(x) \geq f^{(4)}(x)$. Since $f^{(2)}(x) \geq f^{(4)}(x)$, while we also obtained $f^{(4)}(x) \geq f^{(2)}(x)$, we know $f^{(2)}(x) = f^{(4)}(x)$.

Then we try to prove that $f(x)$ is injective.

If $f(x)$ is not injective, this means there exists some $a < b$ such that $f(a) = f(b)$.

First consider substituting x with $f^{(2)}(x)$ and y with a , hence we obtain

$$f(f^{(2)}(x)) = f(f(f(f^{(2)}(x))) + a) + f(f^{(2)}(x)f(a))f(f^{(2)}(x) + a)$$

which means $f^{(3)}(x) = f(f^{(4)}(x) + a) + f(f^{(2)}(x)f(a))f(f^{(2)}(x) + a) = f(f^{(2)}(x) + a) + f(f^{(2)}(x)f(a))f(f^{(2)}(x) + a)$, and that obtains $f^{(3)}(x) = f(f^{(2)}(x) + a)(1 + f(f^{(2)}(x)f(a)))$.

Symmetrically, if we substitute x with $f^{(2)}(x)$ and y with b , we shall obtain $f^{(3)}(x) = f(f^{(2)}(x) + b)(1 + f(f^{(2)}(x)f(b)))$, so that

$$f(f^{(2)}(x) + a)(1 + f(f^{(2)}(x)f(a))) = f(f^{(2)}(x) + b)(1 + f(f^{(2)}(x)f(b)))$$

However, as $f(a) = f(b)$, we have $(1 + f(f^{(2)}(x)f(a))) = (1 + f(f^{(2)}(x)f(b)))$; then we know

$$f(f^{(2)}(x) + a) = f(f^{(2)}(x) + b).$$

Now we substitute only y with a , to obtain

$$f(x) = f(f(f(x)) + a) + f(x(f(a))f(x + a))$$

which is $f(x) = f(f^{(2)}(x) + a) + f(x(f(a))f(x + a))$. Symmetrically substituting only y with b , we get $f(x) = f(f^{(2)}(x) + b) + f(x(f(b))f(x + b))$, so finally

$$f(f^{(2)}(x) + a) + f(x(f(a))f(x + a)) = f(f^{(2)}(x) + b) + f(x(f(b))f(x + b))$$

But we know $f(f^{(2)}(x) + a) = f(f^{(2)}(x) + b)$, and as $f(a) = f(b)$, $f(x(f(a))) = f(x(f(b)))$, so finally we know $f(x + a) = f(x + b)$.

This means $f(x)$ is a periodic function with $b - a$ as a period. We can always take an arbitrarily large integer n , so that $f^{(2)}(a) < a + n(b - a)$.

Recall the statement of (1), whose contrapositive is "if $f^{(2)}(x) < y$, we have $f(x) > f(y)$ ". By comparing $f^{(2)}(a) < a + n(b - a)$, we know $f(a) > f(a + n(b - a)) = f(a)$, which is a contradiction!

Hence, we finished the proof that $f(x)$ is injective.

Then, as $f^{(4)}(x) = f^{(2)}(f^{(2)}(x)) = f^{(2)}(x)$, we can conclude that $f^{(2)}(x) = x$.

Recall the original equation, which $f(x) = f(f(f(x)) + y) + f(xf(y))f(x + y) = f(f^{(2)}(x) + y) + f(xf(y))f(x + y)$, and it further means

$$f(x) = f(x + y) + f(xf(y))f(x + y)$$

.

Substituting x with 1 gives $f(1) = f(1 + y) + f(1 \cdot f(y))f(1 + y) = f(1 + y) + f(f(y))f(1 + y) = f(1 + y) + yf(1 + y)$, so $f(1) = f(1 + y)(1 + y)$ and thus $f(1 + y) = \frac{f(1)}{1 + y}$. Denote $f(1)$ as d , so we have proved that for all $z = 1 + y > 1$, it is true that $f(z) = \frac{d}{z}$.

For $z \leq 1$, we substitute y with $f(z)$, which gives $f(x) = f(x + f(z)) + f(xf(f(z)))f(x + f(z)) = f(x + f(z)) + f(xz)f(x + f(z))$. As the choice for x is arbitrary, we can always take some $x > \frac{1}{z}$. Since $z \leq 1$, we know $x > 1$, and $xz > \frac{1}{z} \cdot z = 1$. So for the equation $f(x) = f(x + f(z)) + f(xz)f(x + f(z))$, we have $x > 1$, $x + f(z) > x > 1$, and $xz > 1$. Then we can write

$$\frac{d}{x} = \frac{d}{x + f(z)} + \frac{d}{xz} \cdot \frac{d}{x + f(z)}$$

Multiplying $\frac{xz(x + f(z))}{d}$ on both sides give $z(x + f(z)) = xz + d$, so $f(z) = \frac{d}{z}$.

We have finally proved that, for all z , it is true that $f(z) = \frac{d}{z}$; and we thus conclude our final answer, that

$$\boxed{f(x) = \frac{d}{x}}$$

where $d \in \mathbb{R}_{>0}$.

4. (USAMO 2022) Find all pairs of primes (p, q) for which $p - q$ and $pq - q$ are both perfect squares.

Solution 1

Since $q(p - 1)$ is a perfect square and q is prime, we should have $p - 1 = qb^2$ for some positive integer b . Let $a^2 = p - q$. Therefore, $q = p - a^2$, and substituting that into the $p - 1 = qb^2$ and solving for p gives

$$p = \frac{a^2b^2 - 1}{b^2 - 1} = \frac{(ab - 1)(ab + 1)}{b^2 - 1}.$$

Notice that we also have

$$p = \frac{a^2b^2 - 1}{b^2 - 1} = a^2 + \frac{a^2 - 1}{b^2 - 1}$$

and so $b^2 - 1 \mid a^2 - 1$. We run through the cases

- $a = 1$: Then $p - q = 1$ so $(p, q) = (3, 2)$, which works.
- $a = b$: This means $p = a^2 + 1$, so $q = 1$, a contradiction.
- $a > b$: This means that $b^2 - 1 < ab - 1$. Since $b^2 - 1$ can be split up into two factors F_1, F_2 such that $F_1 \mid ab - 1$ and $F_2 \mid ab + 1$, we get

$$p = \frac{ab - 1}{F_1} \cdot \frac{ab + 1}{F_2}$$

and each factor is greater than 1, contradicting the primality of p .

Thus, the only solution is $(p, q) = (3, 2)$.

Solution 2

Let $p - q = a^2$, $pq - q = b^2$, where a, b are positive integers. $b^2 - a^2 = pq - q - (p - q) = pq - p$. So,

$$b^2 - a^2 = p(q - 1) \tag{1}$$

- For $q = 2$, $p = b^2 - a^2 = (b - a)(b + a)$. Then $b - a = 1$ and $b + a = p$. $a = \frac{p - 1}{2}$ and $p - q = a^2$. Thus, $p - 2 = \left(\frac{p - 1}{2}\right)^2 \implies p^2 - 6p + 9 = 0$ and we find $p = 3$. Hence $(p, q) = (3, 2)$.

• For $q = 4k + 3$, ($k \geq 0$ integer), by (1), $p(4k + 2) = b^2 - a^2$. Let's examine in mod 4, $b^2 - a^2 \equiv 2 \pmod{4}$. But we know that $b^2 - a^2 \equiv 0, 1$ or $3 \pmod{4}$. This is a contradiction and no solution for $q = 4k + 3$.

• For $q = 4k + 1$, ($k > 0$ integer), by (1), $p(4k) = b^2 - a^2$. Let $k = m \cdot n$, where $m \geq n \geq 1$ and m, n are integers. Since $p > q$, we see $p > 4k$. Thus, by (1), $(b - a)(b + a) = 4p \cdot m \cdot n$. $b - a$ and $b + a$ are same parity and $4p \cdot m \cdot n$ is even integer. So, $b - a$ and $b + a$ are both even integers. Therefore,

$\begin{cases} b + a = 2pn \\ b - a = 2m \end{cases}$ or $\begin{cases} b + a = 2pm \\ b - a = 2n \end{cases}$ Therefore, $a = pn - m$ or $a = pm - n$. For each case, $p - q = p - 4mn - 1 < a$. But $p - q = a^2$, this gives a contradiction. No solution for $q = 4k + 1$.

We conclude that the only solution is $(p, q) = (3, 2)$.

(Lokman GÖKÇE)

Solution 3

We claim that the only solution is $(p, q) = (3, 2)$. First, this clearly works, since $3 - 2 = 1^2$ and $3 \cdot 2 - 2 = 2^2$.

Start as in Solution 2 - assign positive integers m, n , such that $p - q = m^2$ and $pq - q = n^2$. Then, $pq - p = n^2 - m^2$, or $p(q - 1) = (n - m)(n + m)$.

First, $p = q$ is impossible, since $p^2 - p = p(p - 1)$ is not a perfect square. Now notice that $m < p$ and $n < \max(p, q - 1)$. But $p - q > 0$, so $p \geq q - 1$. Now, $m, n < p$.

Therefore, $n - m$ can't ever be a factor of p , and $n + m < 2p$, so $n + m = p$. Then, $n - m = q - 1$, so $p - q = 2m - 1$. But $p - q = m^2$, so $m^2 = 2m - 1$, or $m = 1$. Therefore, one of p and q must be 2 (and 1 is not a prime).

Thus, we conclude that our claim is true.

(grape1984)

5. (USAMO 2022) A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *essentially increasing* if $f(s) \leq f(t)$ holds whenever $s \leq t$ are real numbers such that $f(s) \neq 0$ and $f(t) \neq 0$.

Find the smallest integer k such that for any 2022 real numbers $x_1, x_2, \dots, x_{2022}$, there exist k essentially increasing functions $f_1, \dots, f_k$ such that

$$f_1(n) + f_2(n) + \dots + f_k(n) = x_n \quad \text{for every } n = 1, 2, \dots, 2022.$$

Solution

Answer: $k = 1011$.

Explanation:

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is essentially increasing if $f(s) \leq f(t)$ whenever $s \leq t$ and $f(s) \neq 0, f(t) \neq 0$.

We want the smallest k such that for any $x_1, \dots, x_{2022}$, we can write

$$f_1(n) + f_2(n) + \dots + f_k(n) = x_n \quad \text{for all } n.$$

Lower bound:

Consider $x_1 > x_2 > \dots > x_{2022}$.

Each essentially increasing function can contribute nonzero values

on at most 2 positions (otherwise monotonicity is violated).

$$\Rightarrow k \geq \left\lceil \frac{2022}{2} \right\rceil = 1011.$$

Upper bound:

Group indices as $(1, 2), (3, 4), \dots, (2021, 2022)$.

For each pair, define one function that matches x_{2i-1}, x_{2i}

and is zero elsewhere. This satisfies the condition.

$$\Rightarrow k \leq 1011.$$

6. (*USAMO 2022*) There are 2022 users on a social network called Mathbook, and some of them are Mathbook-friends. (On Mathbook, friendship is always mutual and permanent.)

Starting now, Mathbook will only allow a new friendship to be formed between two users if they have *at least two* friends in common. What is the minimum number of friendships that must already exist so that every user could eventually become friends with every other user?

Solution 1

(but 4044 seems to be the wrong answer?) To answer this question, we need to consider how friendships on Mathbook are formed. If two users have at least two friends in common, then they will be able to become friends with each other. This means that the minimum number of friendships that must already exist in order for every user to eventually become friends with every other user is the smallest number of friendships that guarantees that each pair of users has at least two friends in common.

One way to guarantee that each pair of users has at least two friends in common is to create a complete graph, where each user is friends with every other user. In this case, each pair of users has exactly $2022 - 2 = 2020$ friends in common. However, this is not the minimum number of friendships required, since some pairs of users may have more than two friends in common without forming a complete graph.

To find the minimum number of friendships required, we can use the fact that a friendship between two users implies that they share all of their mutual friends. In other words, if two users are friends, then they must have at least two friends in common. This means that if we want to guarantee that each pair of users has at least two friends in common, we can do so by ensuring that each user has at least two friends.

Therefore, the minimum number of friendships that must already exist in order for every user to eventually become friends with every other user is $2 \cdot 2022 = \boxed{4044}$.

§52 USAMO 2023

1. (USAMO 2023) In an acute triangle ABC , let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM . Suppose the circumcircle of triangle ABP intersects line BC at two distinct points B and Q . Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.

Solution 1

Let X be the foot from A to $\overline{BC}$. By definition, $\angle AXM = \angle MPC = 90^\circ$. Thus, $\triangle AXM \sim \triangle CPM$, and $\triangle BMP \sim \triangle AMQ$.

From this, we have $\frac{MP}{MX} = \frac{MC}{MA} = \frac{MP}{MQ}$, as $MC = MB$. Thus, M is also the midpoint of XQ .

Now, $NB = NC$ if N lies on the perpendicular bisector of $\overline{BC}$. As N lies on the perpendicular bisector of $\overline{XQ}$, which is also the perpendicular bisector of $\overline{BC}$ (as M is also the midpoint of XQ), we are done. Martin2001 SomebodyST (minor edits)

Solution 2

We are going to use barycentric coordinates on $\triangle ABC$. Let $A = (1, 0, 0)$, $B = (0, 1, 0)$, $C = (0, 0, 1)$, and $a = BC$, $b = CA$, $c = AB$. We have $M = (0, \frac{1}{2}, \frac{1}{2})$ and $P = (x : 1 : 1)$ so $\overrightarrow{CP} = (\frac{x}{x+2}, \frac{1}{x+2}, \frac{1}{x+2} - 1)$ and $\overrightarrow{AM} = (-1, \frac{1}{2}, \frac{1}{2})$. Since $\overrightarrow{CP} \perp \overrightarrow{AM}$, it follows that

$$a^2 \left(\frac{1}{2} \cdot \frac{1}{x+2} + \frac{1}{2} \left(\frac{1}{x+2} - 1 \right) \right) + b^2 \left(\frac{1}{2} \cdot \frac{x}{x+2} - \left(\frac{1}{x+2} - 1 \right) \right) + c^2 \left(\frac{1}{2} \cdot \frac{x}{x+2} - \frac{1}{x+2} \right) = 0.$$

Solving this gives

$$x = \frac{2b^2 - 2c^2}{a^2 - 3b^2 - c^2}$$

so

$$P = \left(\frac{b^2 - c^2}{a^2 - 2b^2 - 2c^2}, \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2}, \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} \right).$$

The equation for (ABP) is

$$-a^2yz - b^2zx - c^2xy + ux + vy + wz = 0.$$

Plugging in A and B gives $u = v = 0$. Plugging in P gives

$$-a^2 \left(\frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} \right)^2 - b^2 \cdot \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} \cdot \frac{b^2 - c^2}{a^2 - 2b^2 - 2c^2} - c^2 \cdot \frac{b^2 - c^2}{a^2 - 2b^2 - 2c^2} \cdot \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} + w \cdot \frac{a^2 - 3b^2 - c^2}{2a^2 - 4b^2 - 4c^2} = 0$$

so

$$w = \frac{2b^4 - 2c^4 + a^4 - 3a^2b^2 - a^2c^2}{2a^2 - 4b^2 - 4c^2} = \frac{a^2}{2} - \frac{b^2}{2} + \frac{c^2}{2}.$$

Now let $Q = (0, t, 1 - t)$ where

$$-a^2t(1 - t) + w(1 - t) = 0$$

$$\implies t = \frac{w}{a^2}$$

so $Q = \left(0, \frac{w}{a^2}, 1 - \frac{w}{a^2}\right)$. It follows that $N = \left(\frac{1}{2}, \frac{w}{2a^2}, 1 - \frac{w}{2a^2}\right)$. It suffices to prove that $\overrightarrow{ON} \perp \overrightarrow{BC}$. Setting $\vec{O} = 0$, we get $\vec{N} = \left(\frac{1}{2}, \frac{w}{2a^2}, 1 - \frac{w}{2a^2}\right)$. Furthermore we have $\overrightarrow{CB} = (0, 1, -1)$ so it suffices to prove that

$$\begin{aligned} a^2 \left(-\frac{w}{2a^2} + \frac{1}{2} - \frac{w}{2a^2}\right) + b^2 \left(-\frac{1}{2}\right) + c^2 \left(\frac{1}{2}\right) &= 0 \\ \implies w &= \frac{a^2}{2} - \frac{b^2}{2} + \frac{c^2}{2} \end{aligned}$$

which is valid. $\square$

Solution 3

Let X be the foot from A to BC . Note that $\angle AXC = \angle APC$, so $(AXPC)$ is cyclic. Furthermore, AP is the radical axis of (ABP) and $(AXPC)$, so since M lies on AP , we must have that $BM \cdot MQ = CM \cdot MX$, so M is the midpoint of QX . Therefore, $MN \parallel AX \perp BC$, so we are done. krithikrokcs

2. (*USAMO 2023*) Let $\mathbb{R}^+$ be the set of positive real numbers. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that, for all $x, y \in \mathbb{R}^+$,

$$f(xy + f(x)) = xf(y) + 2$$

Solution 1

Make the following substitutions to the equation:

1. $(x, 1) \rightarrow f(x + f(x)) = xf(1) + 2$
2. $(1, x + f(x)) \rightarrow f(x + f(x) + f(1)) = f(x + f(x)) + 2 = xf(1) + 4$
3. $(x, 1 + \frac{f(1)}{x}) \rightarrow f(x + f(x) + f(1)) = xf\left(1 + \frac{f(1)}{x}\right) + 2$

It then follows from (2) and (3) that $f(1 + \frac{f(1)}{x}) = f(1) + \frac{2}{x}$, so we know that this function is linear for $x > 1$. Substitute $f(x) = ax + b$ and solve for a and b in the functional equation; we find that $f(x) = x + 1 \forall x > 1$.

Now, we can let $x > 1$ and $y \leq 1$. Since $f(x) = x + 1$, $xy + f(x) > x > 1$, so $f(xy + f(x)) = xy + x + 2 = xf(y) + 2$. It becomes clear then that $f(y) = y + 1$ as well, so $f(x) = x + 1$ is the only solution to the functional equation.

3. (*USAMO 2023*) Consider an n -by- n board of unit squares for some odd positive integer n . We say that a collection C of identical dominoes is a maximal grid-aligned configuration on the board if C consists of $(n^2 - 1)/2$ dominoes where each domino covers exactly two neighboring squares and the dominoes don't overlap: C then covers all but one square on the board. We are allowed to slide (but not rotate) a domino on the board to cover the uncovered square, resulting in a new maximal grid-aligned configuration with another square uncovered. Let $k(C)$ be the number of distinct maximal grid-aligned configurations obtainable from C by repeatedly sliding dominoes. Find all possible values of $k(C)$ as a function of n .

Solution

this is solution to jmo3. someone add a solution to amo3

We claim the answer is $(\frac{n+1}{2})^2$.

First, consider a checkerboard tiling of the board with 4 colors: R, G, B, Y. Number each column from 1 to n from left to right and each row from 1 to n from top to bottom. We color a tile R if its row and column are odd, a tile G if its row is even but its column is odd, a tile B if its row and column is even, and a tile Y if its row is odd but its column is even.

Lemma 1: Throughout our moves, the color of the uncolored tile stays an invariant.

Consider that a domino can either only change rows or can only change columns. Therefore, sliding a domino into the hole and creating a new one has two possible colors. Of these, note that the new hole will always trivially be two tiles away from the old hole, meaning that the parity of both the row and column number stays the same. Thus, the lemma holds.

Lemma 2: There are more red tiles than any other color. Because each color is uniquely defined by the parity of a pair of column and row number, it satisfies to show that given an odd integer n , there are more odd positive integers less than or equal to n than even ones. Obviously, this is true, and so red will have more tiles than any other color.

Lemma 3: For any starting configuration C and any blank tile B such that the blank tile's color matches the blank tile's color of C , there is no more than one unique configuration C' that can be produced from C using valid moves.

We will use proof by contradiction. Assume there exists two different C' . We can get from one of these C' to another using moves. However, we have to finish off with the same hole as before. Before the last move, the hole must be two tiles away from the starting hole. However, because the domino we used to move into the blank tile's spot is in the way, that hole must be congruent to the hole produced after the first move. We can induct this logic, and because there is a limited amount of tiles with the same color, eventually we will run out of tiles to apply this to. Therefore, having two distinct C' with the same starting hole B is impossible with some C .

We will now prove that $(\frac{n+1}{2})^2$ is the answer. There are $\frac{n+1}{2}$ rows and $\frac{n+1}{2}$ columns that are odd, and thus there are $(\frac{n+1}{2})^2$ red tiles. Given lemma 3, this is our upper bound for a maximum. To establish that $(\frac{n+1}{2})^2$ is indeed possible, we construct such a C :

In the first column, leave the first tile up blank. Then, continuously fill in vertically oriented dominos in that column until it reaches the bottom. In the next $n - 1$ columns, place $\frac{n-1}{2}$ vertically oriented dominos in a row starting from the top. At the bottom row, starting with the first unfilled tile on the left, place horizontally aligned dominos in a row until you reach the right.

Obviously, the top left tile is red. It suffices to show that any red tile may be uncovered. For the first column, one may slide some dominos on the first column until the desired tile is uncovered. For the bottom row, all the first dominos may be slid up, and then the bottom dominos may be slid to the left until the desired red tile is uncovered. Finally, for the rest of the red tiles, the bottom red tile in the same color may be revealed, and then vertically aligned dominos in the same column may be slid down until the desired tile is revealed. Therefore, this configuration may produce $(\frac{n+1}{2})^2$ different configurations with moves.

Hence, we have proved that $(\frac{n+1}{2})^2$ is the maximum, and we are done. ■

Do not change the question yourself, SigmaPiE. The question was not maximum, it was all possible values. I changed it back to all possible values. The answer is $\{1, 2, \dots, (\frac{n-1}{2})^2\} \cup \{(\frac{n+1}{2})^2\}$.

4. (*USAMO 2023*) A positive integer a is selected, and some positive integers are written on a board. Alice and Bob play the following game. On Alice's turn, she must replace some integer n on the board with $n + a$, and on Bob's turn he must replace some even integer n on the board with $n/2$. Alice goes first and they alternate turns. If on his turn Bob has no valid moves, the game ends.

After analyzing the integers on the board, Bob realizes that, regardless of what moves Alice makes, he will be able to force the game to end eventually. Show that, in fact, for this value of a and these integers on the board, the game is guaranteed to end regardless of Alice's or Bob's moves.

Solution 1

The contrapositive of the claim is somewhat easier to conceptualize: If it is not guaranteed that the game will end (i.e. the game could potentially last forever), then Bob is not able to force the game to end (i.e. Alice can force it to last forever). So we want to prove that, if the game can potentially last indefinitely, then Alice can force it to last indefinitely. Clearly, if there is 1 number on the board initially, all moves are forced. This means the claim is true in this specific case, because if the game "potentially" can last forever, this means it must last forever, since the game can only be played in one way. Ergo Alice can "force" this to occur because it is guaranteed to occur. Now we look at all cases where there is more than 1 number on the board.

Case 1: $v_2(a) = 0$

The game lasts forever here no matter what. This is true because, if the game ends, it means the board was in some position P , Alice added a to some number on the board, and all the numbers now on the board are odd. If there are only odd numbers on the board in position P , Alice will add a to some number on the board, making it even, meaning this cannot have been the state P of the board. If at least one number n on the board is even, Alice can add a to a number other than n , meaning there is still at least 1 even number on the board, meaning this also cannot have been the state P of the board. This covers all possible boards when $v_2(a) = 0$, so we're done.

Case 2: $v_2(a) = 1$

If there is at least one number n on the board that is even, the game can also last forever. On any move, Alice will add a to this number n if and only if $v_2(n) = 1$. This way, the new number $n' = n + a$ satisfies $v_2(n') \geq 2$. If Bob divides n' until $v_2(n) = 1$, Alice will again add a to n resulting in $v_2(n') \geq 2$. This means that Alice can always keep an even number on the board for Bob to divide no matter how Bob plays.

If there is no even number on the board, then the game can clearly not last forever. No matter what Alice does, Bob will have no even number to divide after her move.

General Case: $v_2(a) = x$

In general, it seems to be the case that the game can last indefinitely for some given a if and only if there exists some number n on the board such that $v_2(n) \geq v_2(a)$, so we should aim to prove this.

1. "If"

Alice can apply a similar strategy to the strategy used in case 2. On any move, Alice will add a to n if and only if $v_2(n) = v_2(a)$. If she does this addition, then $v_2(n') \geq v_2(a) + 1$, keeping an even number on the board. Even if Bob divides n' until $v_2(n) = v_2(a)$, Alice will apply the same strategy and keep $v_2(n') \geq v_2(a) + 1$. Alice's use of this strategy ensures that there always exists some number n on the board such that $v_2(n) \geq v_2(a)$, ensuring there always exists an even number n on the board.

2. "Only If"

If $v_2(n) < v_2(a)$ for all n on the board, this means that Alice can never change the value of $v_2(n)$ for any n on the board. Only Bob can do this, and Bob will subtract 1 from each $v_2(n)$ until they are all equal to 0 (all odd), ending the game.

We've shown that the game can last indefinitely iff there exists some number n on the board such that $v_2(n) \geq v_2(a)$, and have shown that Alice can ensure the game lasts forever in these scenarios using the above strategy. This proves the contrapositive, proving the claim.

5. (*USAMO 2023*) Let $n \geq 3$ be an integer. We say that an arrangement of the numbers $1, 2, \dots, n^2$ in a $n \times n$ table is row-valid if the numbers in each row can be permuted to form an arithmetic progression, and column-valid if the numbers in each column can be permuted to form an arithmetic progression. For what values of n is it possible to transform any row-valid arrangement into a column-valid arrangement by permuting the numbers in each row?

Solution 1

The answer is all $\boxed{\text{prime } n}$.

Proof that primes work Suppose $n = p$ is prime. Then, let the arithmetic progressions in the i th row have least term a_i and common difference d_i . For each cell with integer k , assign a monomial x^k . The sum of the monomials is

$$x(1 + x + \dots + x^{n^2-1}) = \sum_{i=1}^n x^{a_i} (1 + x^{d_i} + \dots + x^{(n-1)d_i}),$$

where the LHS is obtained by summing over all cells and the RHS is obtained by summing over all rows. Let S be the set of p th roots of unity that are not 1; then, the LHS of the above equivalence vanishes over S and the RHS is

$$\sum_{p \nmid d_i} x^{a_i}.$$

Reducing the exponents $(\text{mod } p)$ in the above expression yields

$$f(x) := \sum_{p \nmid d_i} x^{a_i} \pmod{p} = 0$$

when $x \in S$. Note that $\prod_{s \in S} (x - s) = 1 + x + \dots + x^{p-1}$ is a factor of $f(x)$, and as f has degree less than p , f is either identically 0 or $f(x) = 1 + x + \dots + x^{p-1}$.

- If f is identically 0, then p never divides d_i . Thus, no two elements in each row are congruent $(\text{mod } p)$, so all residues are represented in each row. Now we can rearrange the grid so that column i consists of all numbers $i \pmod{p}$, which works.

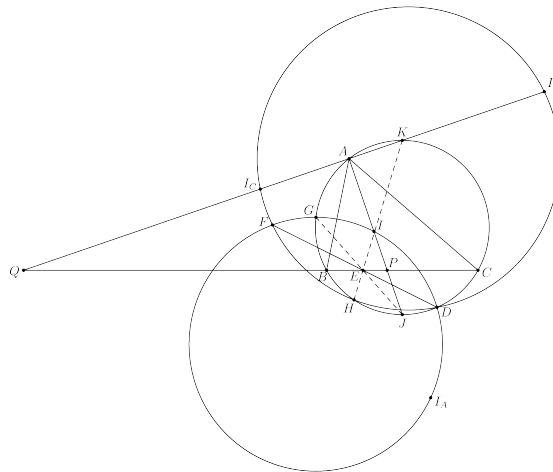
- If $f(x) = 1 + x + \dots + x^{p-1}$, then p always divides d_i . It is clear that each d_i must be p , so each row represents a single residue $(\text{mod } p)$. Thus, we can rearrange the grid so that column i contains all consecutive numbers from $1 + (i - 1)p$ to ip , which works.

All in all, any prime n satisfies the hypotheses of the problem.

Proof that composites do not work Let $n = ab$. Look at the term $a^2b + ab$; we claim it cannot be part of a column that has cells forming an arithmetic sequence after any appropriate rearranging. After such a rearrangement, if the column it is in has common difference $d < ab = n$, then $a^2b + ab - d$ must also be in its column, which is impossible. If the column has difference $d > ab = n$, then no element in the next row can be in its column. If the common difference is $d = ab = n$, then $a^2b + ab - 2d = a^2b - ab$ and $a^2b + ab - d = a^2b$, which are both in the row above it, must both be in the same column, which is impossible. Therefore, the grid is not column-valid after any rearrangement, which completes the proof.

6. (USAMO 2023) Let ABC be a triangle with incenter I and excenters I_a, I_b, I_c opposite A, B , and C , respectively. Given an arbitrary point D on the circumcircle of $\triangle ABC$ that does not lie on any of the lines II_a, I_bI_c , or BC , suppose the circumcircles of $\triangle DII_a$ and $\triangle DI_bI_c$ intersect at two distinct points D and F . If E is the intersection of lines DF and BC , prove that $\angle BAD = \angle EAC$.

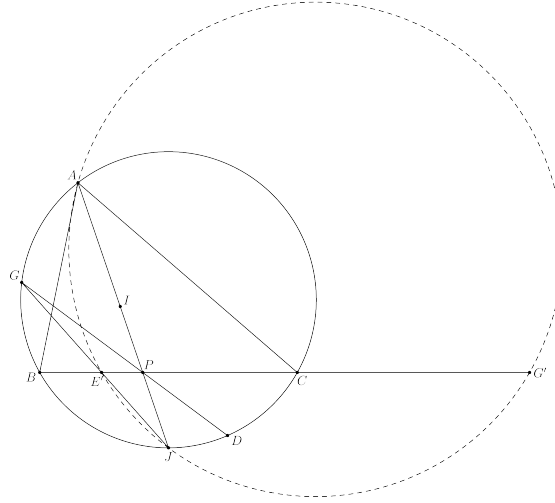
Solution 1



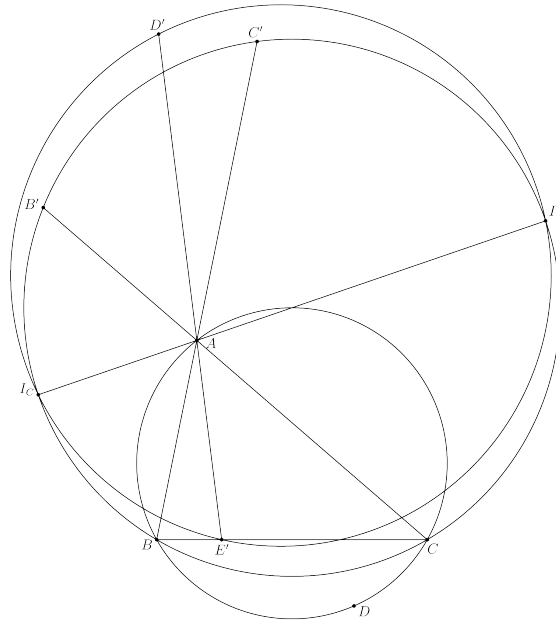
Consider points G, H, J, K, P , and Q such that the intersections of the circumcircle of $\triangle ABC$ with the circumcircle of $\triangle DII_a$ are D and G , the intersections of the circumcircle of $\triangle ABC$ with the circumcircle of $\triangle DI_bI_c$ are D and H , the intersections of the circumcircle of $\triangle ABC$ with line $\overline{II_a}$ are A and J , the intersections of the circumcircle of $\triangle ABC$ with line $\overline{I_bI_c}$ are A and K , the intersection of lines $\overline{II_a}$ and $\overline{BC}$ is P , and the intersection of lines $\overline{I_bI_c}$ and $\overline{BC}$ is Q .

Since IBI_aC is cyclic, the pairwise radical axes of the circumcircles of $\triangle DII_a, \triangle ABC$, and IBI_aC concur. The pairwise radical axes of these circles are $\overline{GD}, \overline{II_a}$, and $\overline{BC}$, so G, P , and D are collinear. Similarly, since BCI_bI_c is cyclic, the pairwise radical axes of the circumcircles of $\triangle DI_bI_c, \triangle ABC$, and BCI_bI_c concur. The pairwise radical axes of these circles are $\overline{HD}, \overline{I_bI_c}$, and $\overline{BC}$, so H, Q , and D are collinear. This means that $-1 = (Q, P; B, C) \stackrel{D}{=} (H, G; B, C)$, so the tangents to the circumcircle of $\triangle ABC$ at G and H intersect on $\overline{BC}$. Let this intersection be

X . Also, let the intersection of the tangents to the circumcircle of $\triangle ABC$ at K and J be a point at infinity on $\overline{BC}$ called Y and let the intersection of lines $\overline{KG}$ and $\overline{HJ}$ be Z . Then, let the intersection of lines $\overline{GJ}$ and $\overline{HK}$ be E' . By Pascal's Theorem on $GGJHHK$ and $GJJHKK$, we get that X, E' , and Z are collinear and that E', Y , and Z are collinear, so E', X , and Y are collinear, meaning that E' lies on $\overline{BC}$ since both X and Y lie on $\overline{BC}$.

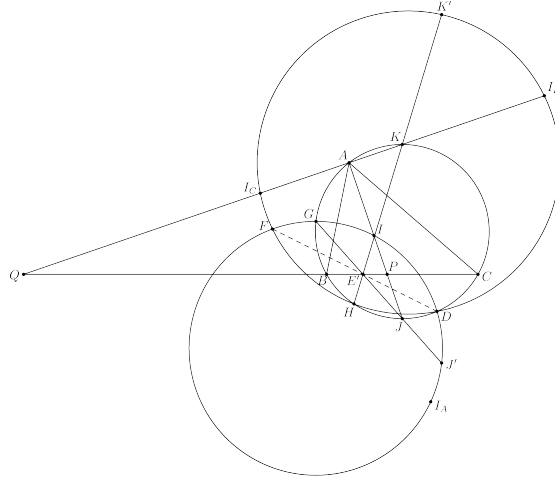


Consider the transformation which is the composition of an inversion centered at A and a reflection over the angle bisector of $\angle CAB$ that sends B to C and C to B . We claim that this sends D to E' and E' to D . It is sufficient to prove that if the transformation sends G to G' , then $AE'JG'$ is cyclic. Notice that $\triangle AGB \sim \triangle ACG'$ since $\angle GAB = \angle G'AC$ and $\frac{AG'}{AC} = \frac{AB \cdot AC}{AG \cdot AC} = \frac{AB}{AG}$. Therefore, we get that $\angle AG'E' = \angle ABG = \angle AJE'$, so $AE'JG'$ is cyclic, proving the claim. This means that $\angle BAE' = \angle CAD$.



We claim that $\angle I_B E' I_C + \angle I_B D I_C = 180^\circ$. Construct D' to be the intersection of line $\overline{AE'}$ and the circumcircle of $\triangle E' I_B I_C$ and let B' and C' be the intersections

of lines $\overline{AC}$ and $\overline{AB}$ with the circumcircle of $\triangle BI_BI_C$. Since B' and C' are the reflections of B and C over $\overline{I_BI_C}$, it is sufficient to prove that A, B', C', D' are concyclic. Since $\overline{B'C}, \overline{D'E'}$, and $\overline{I_BI_C}$ concur and D', E', I_B, I_C and I_B, I_C, B', C' are concyclic, we have that B', C, D', E' are concyclic, so $\angle B'D'A = \angle ACE' = \angle AC'B'$, so A, B', C', D' are concyclic, proving the claim. We can similarly get that $\angle IE'I_A = \angle IDI_A$.



Let line $\overline{E'J}$ intersect the circumcircle of $\triangle DII_A$ at G and J' . Notice that J is the midpoint of $\overline{II_A}$ and $\angle IE'I_A = \angle IDI_A = \angle IJ'I_A$, so $IE'I_AJ'$ is a parallelogram with center J , so $\frac{EJ}{E'J'} = \frac{1}{2}$. Similarly, we get that if line $\overline{E'K}$ intersects the circumcircle of $\triangle DI_BI_C$ at H and K' , we have that $\frac{EK}{E'K'} = \frac{1}{2}$, so $\overline{KJ} \parallel \overline{K'J'}$, so $\angle HGJ' = \angle HGJ = \angle HKJ = \angle HK'J'$, so G, H, J', K' are concyclic. Then, the pairwise radical axes of the circumcircles of $\triangle DII_A, \triangle DI_BI_C$, and $GHJ'K'$ are $\overline{DF}, \overline{HK'}$, and $\overline{GJ'}$, so $\overline{DF}, \overline{HK'}$, and $\overline{GJ'}$ concur, so $E = E'$. We are then done since $\angle BAE' = \angle CAD$.

Solution 2

Set $\triangle ABC$ as the reference triangle, and let $D = (x_0, y_0, z_0)$ with homogenized coordinates. To find the equation of circle DII_A , we note that $I = (a : b : c)$ (not homogenized) and $I_A = (-a : b : c)$. Thus, for this circle with equation $a^2yz + b^2xz + c^2xy = (x + y + z)(u_1x + v_1y + w_1z)$, we compute that

$$\begin{aligned} u_1 &= bc, \\ v_1 &= \frac{bc^2x_0}{bz_0 - cy_0}, \\ w_1 &= \frac{b^2cx_0}{cy_0 - bz_0}. \end{aligned}$$

For circle DI_BI_C with equation $a^2yz + b^2xz + c^2xy = (x + y + z)(u_2x + v_2y + w_2z)$, we find that

$$\begin{aligned} u_2 &= -bc, \\ v_2 &= \frac{bc^2x_0}{bz_0 + cy_0}, \\ w_2 &= \frac{b^2cx_0}{cy_0 + bz_0}. \end{aligned}$$

The equation of the radical axis is $ux + vy + wz = 0$ with $u = u_1 - u_2$, $v = v_1 - v_2$, and $w = w_1 - w_2$. We want to consider the intersection of this line with line $\overline{BC}$, so set $x = 0$. The equation reduces to $vy + wz = 0$. We see that $v = \frac{2bc^3x_0y_0}{b^2z_0^2 - c^2y_0^2}$ and $w = \frac{2b^3cx_0z_0}{c^2y_0^2 - b^2z_0^2}$, so

$$\frac{v}{w} = \frac{b^2z_0}{c^2y_0},$$

which is the required condition for $\overline{AD}$ and $\overline{AE}$ to be isogonal.

§53 USAMO 2024

1. (USAMO 2024) Find all integers $n \geq 3$ such that the following property holds: if we list the divisors of $n!$ in increasing order as $1 = d_1 < d_2 < \cdots < d_k = n!$, then we have

$$d_2 - d_1 \leq d_3 - d_2 \leq \cdots \leq d_k - d_{k-1}.$$

Solution (Explanation of Video Solution)

We can start by verifying that $n = 3$ and $n = 4$ work by listing out the factors of $3!$ and $4!$. We can also see that $n = 5$ does not work because the terms 15, 20, and 24 are consecutive factors of $5!$. Also, $n = 6$ does not work because the terms 6, 8, and 9 appear consecutively in the factors of $6!$.

Note that if we have a prime number $p > n$ and an integer $k > p$ such that both k and $k + 1$ are factors of $n!$, then the condition cannot be satisfied.

If $n \geq 7$ is odd, then $(2)(\frac{n-1}{2})(n-1) = n^2 - 2n + 1$ is a factor of $n!$. Also, $(n-2)(n) = n^2 - 2n$ is a factor of $n!$. Since $2n < n^2 - 2n$ for all $n \geq 7$, we can use **Bertrand's Postulate** to show that there is at least one prime number p such that $n < p < n^2 - 2n$. Since we have two consecutive factors of $n!$ and a prime number between the smaller of these factors and n , the condition will not be satisfied for all odd $n \geq 7$.

If $n \geq 8$ is even, then $(2)(\frac{n-2}{2})(n-2) = n^2 - 4n + 4$ is a factor of $n!$. Also, $(n-3)(n-1) = n^2 - 4n + 3$ is a factor of $n!$. Since $2n < n^2 - 4n + 3$ for all $n \geq 8$, we can use Bertrand's Postulate again to show that there is at least one prime number p such that $n < p < n^2 - 4n + 3$. Since we have two consecutive factors of $n!$ and a prime number between the smaller of these factors and n , the condition will not be satisfied for all even $n \geq 8$.

Therefore, the only numbers that work are $n = 3$ and $n = 4$.

As listed above, $n = 3$ and $n = 4$ work but $n = 5$ and $n = 6$ don't. Assume $n > 6$, let p be the smallest positive integer that doesn't divide $n!$. It is easy to see p is the smallest prime greater than n . It is easy to see that both $p - 1$ and $p + 1$ are divisors of $n!$. Let q be the smallest positive integer greater than p that is 2 mod 6. We see that either $q = p + 3$ or $q = p + 1$ but $q, q + 1, q + 2$ must all be divisors of $n!$ giving a difference of 1 after a difference of 2. Therefore, all $n > 6$ fail. mathphobia

Solution 2

We claim only $n = 3$ and $n = 4$ are the only two solutions. First, it is clear that both solutions work.

Next, we claim that $n < 5$. For $n \geq 5$, let x be the smallest x such that $x + 1$ is not a factor of $n!$. Let the smallest factor larger than x be $x + k$.

Now we consider $\frac{n!}{x-1}$, $\frac{n!}{x}$ and $\frac{n!}{x+k}$. Since $\frac{n!}{x-1} > \frac{n!}{x} > \frac{n!}{x+k}$, if n were to satisfy the conditions, then $\frac{n!}{x-1} - \frac{n!}{x} \geq \frac{n!}{x} - \frac{n!}{x+k}$. However, note that this is not true for $x \geq 5$ and $k > 1$.

Note that the inequality is equivalent to showing $\frac{1}{x(x-1)} \geq \frac{k}{x(x+k)}$, which simplifies to $x + k \geq kx - k$, or $\frac{x}{x-2} \geq k \geq 2$. This implies $x \geq 2x - 4$, $x \leq 4$, a contradiction, since the set of numbers $\{1, 2, 3, 4, 5\}$ are all factors of $n!$, and the value of x must exist. Hence, no solutions for $n \geq 5$.

2. (USAMO 2024) Let $S_1, S_2, \dots, S_{100}$ be finite sets of integers whose intersection is not empty. For each non-empty $T \subseteq \{S_1, S_2, \dots, S_{100}\}$, the size of the intersection of the sets in T is a multiple of the number of sets in T . What is the least possible number of elements that are in at least 50 sets?

Solution 1

Let's determine the smallest possible number of elements that appear in at least 50 of the sets.

First, we establish some notation. For any subset $T \subseteq \{S_1, S_2, \dots, S_{100}\}$, let $\cap T$ denote the intersection of all sets in T . By the problem statement, $|\cap T|$ is divisible by $|T|$.

We'll use a binary encoding approach. For each element x in our universe, define its "signature" as a binary string of length 100, where the i -th bit is 1 if $x \in S_i$ and 0 otherwise.

For any signature v with exactly k ones, corresponding elements must appear in exactly k sets. The problem condition requires that for any subset of sets corresponding to positions where 1's appear in some signature u , the number of elements having signature v that contains all 1's from u must be divisible by $|u|$.

Let's denote by $f(v)$ the number of elements with signature v . The problem condition translates to: for any signature u , $|u|$ divides $\sum_{v \supseteq u} f(v)$.

Key claim: For any signature u with $|u| = 50$, we need $f(u) \geq 50$.

Proof: By the divisibility condition, $\sum_{v \supseteq u} f(v)$ must be divisible by 50. For any v strictly containing u , if we sum over all such u of size 50, each $f(v)$ with $|v| > 50$ gets counted multiple times. To satisfy the divisibility condition for each individual u , we need $f(u) \geq 50$ for each signature u with exactly 50 ones.

Since there are $\binom{100}{50}$ different ways to choose 50 positions for ones in the signature, and each such signature must correspond to at least 50 elements, the minimum number of elements appearing in at least 50 sets is $50 \binom{100}{50}$.

Therefore, the answer is $50 \binom{100}{50}$.

3. (USAMO 2024) Let m be a positive integer. A triangulation of a polygon is m -balanced if its triangles can be colored with m colors in such a way that the sum of the areas of all triangles of the same color is the same for each of the m colors. Find all positive integers n for which there exists an m -balanced triangulation of a regular n -gon.

Note: A triangulation of a convex polygon $\mathcal{P}$ with $n \geq 3$ sides is any partitioning of $\mathcal{P}$ into $n - 2$ triangles by $n - 3$ diagonals of $\mathcal{P}$ that do not intersect in the polygon's interior.

Solution 1

The answer is if and only if m is a proper divisor of n .

We represent the vertices of the regular n -gon using complex numbers. Let $\omega = \exp(2\pi i/n)$ be a primitive n -th root of unity, and label the vertices as $1, \omega, \omega^2, \dots, \omega^{n-1}$.

Key Lemma:

The triangle with vertices $\omega^k, \omega^{k+a}, \omega^{k+b}$ has signed area:

$$T(a, b) := \frac{(\omega^a - 1)(\omega^b - 1)(\omega^{-b} - \omega^{-a})}{4i}$$

Proof of Lemma:

We can rotate by ω^{-k} to assume without loss of generality that $k = 0$. Then we apply the complex shoelace formula to the triangle with vertices $1, \omega^a, \omega^b$:

$$\frac{i}{4} \det \begin{bmatrix} 1 & 1 & 1 \\ \omega^a & \omega^{-a} & 1 \\ \omega^b & \omega^{-b} & 1 \end{bmatrix} = \frac{i}{4} \det \begin{bmatrix} 0 & 0 & 1 \\ \omega^a - 1 & \omega^{-a} - 1 & 1 \\ \omega^b - 1 & \omega^{-b} - 1 & 1 \end{bmatrix}$$

which simplifies to the expression for $T(a, b)$ above.

Construction (Sufficiency):

To construct a valid triangulation and coloring when m divides n , we draw all the diagonals from vertex 1, and then color the resulting triangles with the m colors in cyclic order.

For example, when $n = 9$ and $m = 3$, we get a coloring with red, green, blue as shown in the image, repeating cyclically.

To prove this works, we fix a residue $r \pmod m$ corresponding to one of the colors. Then the total area of triangles with this color is:

$$\sum_{\substack{0 \leq j < n \\ j \equiv r \pmod m}} \text{Area}(\omega^0, \omega^j, \omega^{j+1}) = \sum_{\substack{0 \leq j < n \\ j \equiv r \pmod m}} T(j, j+1)$$

After algebraic manipulation:

$$= \frac{\omega - 1}{4i} \left(\frac{n}{m} (1 + \omega^{-1}) + \sum_{\substack{0 \leq j < n \\ j \equiv r \pmod m}} (\omega^{-1-j} - \omega^j) \right)$$

When m divides n , we can show that the inner sum vanishes, and the total area of each color equals:

$$\sum_{\substack{0 \leq j < n \\ j \equiv r \pmod m}} \text{Area}(\omega^0, \omega^j, \omega^{j+1}) = \frac{n}{m} \cdot \frac{\omega - \omega^{-1}}{4i}$$

Since the right-hand side does not depend on the residue r , this shows all colors have equal area.

Necessity:

To prove that m must divide n , we note that if there were a valid triangulation and coloring, the total area of each color would equal:

$$S := \frac{n}{m} \cdot \frac{\omega - \omega^{-1}}{4i}$$

We then prove the following claim:

The number $4i \cdot S$ is not an algebraic integer when $m \nmid n$.

This can be shown using the fact that $K = \mathbb{Q}(\omega)$ is a number field with ring of integers $\mathcal{O}_K = \mathbb{Z}[\omega]$. We demonstrate that:

$$\omega \cdot 4i \cdot S = \frac{n}{m} \omega^2 - \frac{n}{m}$$

is not an algebraic integer when m doesn't divide n because $\frac{n}{m} \notin \mathbb{Z}$.

However, for any triangle in our construction, $4i \cdot T(a, b)$ is always an algebraic integer. Since a finite sum of algebraic integers is also an algebraic integer, the sum of expressions of the form $4i \cdot T(a, b)$ can never equal $4i \cdot S$ unless m divides n .

We can also show this directly by noting that $T(a, b)$ is always divisible by $\frac{\omega - \omega^{-1}}{4i}$, which means we need $\frac{n}{m}$ to be an algebraic integer. A rational number is an algebraic integer if and only if it's a rational integer, so $\frac{n}{m}$ is an algebraic integer exactly when m divides n .

Therefore, a valid triangulation and coloring exists if and only if m is a proper divisor of n .

4. (*USAMO 2024*) Let m and n be positive integers. A circular necklace contains mn beads, each either red or blue. It turned out that no matter how the necklace was cut into m blocks of n consecutive beads, each block had a distinct number of red beads. Determine, with proof, all possible values of the ordered pair (m, n) .

Solution 1

We need to determine all possible positive integer pairs (m, n) such that there exists a circular necklace of mn beads, each colored red or blue, satisfying the following condition:

No matter how the necklace is cut into m blocks of n consecutive beads, each block has a distinct number of red beads.

Necessary Condition:

1. Maximum Possible Distinct Counts: In a block of n beads, the number of red beads can range from 0 to n . Therefore, there are $n + 1$ possible distinct counts of red beads in a block. Since we have m blocks, the maximum number of distinct counts must be at least m . Thus, we must have:

$$m \leq n + 1$$

Sufficient Construction:

We will show that for all positive integers m and n satisfying $m \leq n + 1$, such a necklace exists.

1. Construct Blocks: Create m blocks, each containing n beads. Assign to each block a unique number of red beads, ranging from 0 to $m - 1$.
2. Design the Necklace: Arrange these m blocks in a fixed order to form the necklace. Since the necklace is circular, cutting it at different points results in cyclic permutations of the blocks.
3. Verification: In any cut, the sequence of blocks (and thus the counts of red beads) is a cyclic shift of the original sequence. Therefore, in each partition, the blocks will have distinct numbers of red beads.

Example:

Let's construct a necklace for $m = 3$ and $n = 2$:

Blocks: Block 1: 0 red beads (BB) Block 2: 1 red bead (RB) Block 3: 2 red beads (RR)

Necklace Arrangement: Place the blocks in order: BB-RB-RR.

Verification: Any cut of the necklace into 3 blocks of 2 beads will have blocks with red bead counts of 0, 1, and 2.

Conclusion:

All ordered pairs (m, n) where $m \leq n + 1$ satisfy the condition. Therefore, the possible values of (m, n) are all positive integers such that $m \leq n + 1$.

Final Answer:

Exactly all positive integers m and n with $m \leq n + 1$; these are all possible ordered pairs (m, n) .

5. (USAMO 2024) Point D is selected inside acute triangle ABC so that $\angle DAC = \angle ACB$ and $\angle BDC = 90^\circ + \angle BAC$. Point E is chosen on ray BD so that $AE = EC$. Let M be the midpoint of BC . Show that line AB is tangent to the circumcircle of triangle BEM .

Solution 1

Let $\angle DBT = \alpha$ and $\angle BEM = \beta$. Extend AD intersects BC at point T , then $TC = TA$, TE is perpendicular to AC

Thus, AB is the tangent of the circle BEM

Then the question is equivalent as the $\angle ABT$ is the auxillary angle of $\angle BEM$.

continued

Solution 2

We notice that $\angle DAC = \angle ACB$, which implies symmetry about the perpendicular bisector of $\overline{AC}$. Therefore, we can extend $\overline{AD}$ to intersect the circumcircle at Q , and by symmetry about the perpendicular bisector, $ABQC$ is an isocles trapezoid. Now, we realize that the midpoint condition is annoying, which really suggests that to simplify things, we take a homothety at B sending M to C , and we just need to prove that the circumcircle of $\triangle BE'C$ is tangent to line $\overline{AB}$, where E' is the resultant point from a homothety at E . However, funnily enough, E lies on the perpendicular bisector of $\overline{AC}$ because $\overline{AE} = \overline{EC}$, and the perpendicular bisector gets sent to an altitude from Q under homothety, which means E' is simply the intersection of $\overline{BD}$ and the altitude from Q onto $\overline{BC}$! Call that point

P . We can now ignore points E and M . Note that we have not used the $\angle BDC$ condition yet, which means we should probably try angle chasing now. Call $\angle BAC = a, \angle ABC = b$. By angle condition, $\angle BDC = 90 + a, \angle CDP = 90 - a$. Now, $\angle QCA = \angle BAC = a$, so $\angle PQC = 90 - \angle ACQ = 90 - a$. A miracle occurs! $\angle CDP = \angle ACQ$, so quadrilateral $QDPC$ is cyclic! $\angle ABC = \angle AQC = b$ from isosceles trapezoid, and from cyclicity $\angle DPC = 180 - b = \angle BPC$. Now we have $\angle ABC = b$ and $\angle BPC = 180 - b$, which looks a lot like tangency condition. We realize that it simply is the tangency condition if we extend line $\overline{AB}$ to get the supplementary angle, so we are done.

6. (USAMO 2024) Let $n > 2$ be an integer and let $\ell \in \{1, 2, \dots, n\}$. A collection $A_1, \dots, A_k$ of (not necessarily distinct) subsets of $\{1, 2, \dots, n\}$ is called ℓ -large if $|A_i| \geq \ell$ for all $1 \leq i \leq k$. Find, in terms of n and ℓ , the largest real number c such that the inequality

$$\sum_{i=1}^k \sum_{j=1}^k x_i x_j \frac{|A_i \cap A_j|^2}{|A_i| \cdot |A_j|} \geq c \left(\sum_{i=1}^k x_i \right)^2$$

holds for all positive integers k , all nonnegative real numbers $x_1, \dots, x_k$, and all ℓ -large collections $A_1, \dots, A_k$ of subsets of $\{1, 2, \dots, n\}$.

Note: For a finite set S , $|S|$ denotes the number of elements in S .

Solution

The largest real number c is $\frac{n-2\ell+\ell^2}{n(n-1)}$.

Proof For each $1 \leq i \leq k$, let $\mathbf{1}_{A_i} : \{1, 2, \dots, n\} \rightarrow \{0, 1\}$ denote the indicator function of the set A_i . We can rewrite the intersection size in the numerator as:

$$|A_i \cap A_j|^2 = \left(\sum_{m=1}^n \mathbf{1}_{A_i}(m) \mathbf{1}_{A_j}(m) \right)^2 = \sum_{m=1}^n \sum_{m'=1}^n \mathbf{1}_{A_i}(m) \mathbf{1}_{A_i}(m') \mathbf{1}_{A_j}(m) \mathbf{1}_{A_j}(m').$$

Substitute this into the original double sum and rearrange the order of summation:

$$\sum_{i=1}^k \sum_{j=1}^k x_i x_j \frac{|A_i \cap A_j|^2}{|A_i| \cdot |A_j|} = \sum_{m=1}^n \sum_{m'=1}^n \left(\sum_{i=1}^k \frac{x_i}{|A_i|} \mathbf{1}_{A_i}(m) \mathbf{1}_{A_i}(m') \right)^2.$$

Define a symmetric $n \times n$ matrix W where the entry in the m -th row and m' -th column is:

$$W_{m,m'} = \sum_{i=1}^k \frac{x_i}{|A_i|} \mathbf{1}_{A_i}(m) \mathbf{1}_{A_i}(m').$$

The expression we wish to bound is the sum of the squares of all entries in W , which we can split into diagonal and off-diagonal sums:

$$\sum_{m,m'} W_{m,m'}^2 = \sum_{m=1}^n W_{m,m}^2 + \sum_{m \neq m'} W_{m,m'}^2.$$

By the Cauchy-Schwarz inequality, the sum of squares of N elements is bounded below by the square of their sum divided by N . Applying this to the n diagonal

elements and the $n(n-1)$ off-diagonal elements separately yields:

$$\sum_{m=1}^n W_{m,m}^2 \geq \frac{1}{n} \left(\sum_{m=1}^n W_{m,m} \right)^2$$

$$\sum_{m \neq m'} W_{m,m'}^2 \geq \frac{1}{n(n-1)} \left(\sum_{m \neq m'} W_{m,m'} \right)^2$$

Next, we evaluate the sums of these entries. The sum of the diagonal elements (the trace) is:

$$\sum_{m=1}^n W_{m,m} = \sum_{i=1}^k \frac{x_i}{|A_i|} \sum_{m=1}^n \mathbf{1}_{A_i}(m) = \sum_{i=1}^k \frac{x_i}{|A_i|} |A_i| = \sum_{i=1}^k x_i.$$

The sum of all entries in the matrix is:

$$\sum_{m,m'} W_{m,m'} = \sum_{i=1}^k \frac{x_i}{|A_i|} \left(\sum_{m=1}^n \mathbf{1}_{A_i}(m) \right)^2 = \sum_{i=1}^k x_i |A_i|.$$

Subtracting the trace from the total sum gives the sum of the off-diagonal elements:

$$\sum_{m \neq m'} W_{m,m'} = \sum_{i=1}^k x_i |A_i| - \sum_{i=1}^k x_i = \sum_{i=1}^k x_i (|A_i| - 1).$$

Because the collection is ℓ -large, we know $|A_i| \geq \ell \geq 1$. Given $x_i \geq 0$, we have:

$$\sum_{m \neq m'} W_{m,m'} \geq \sum_{i=1}^k x_i (\ell - 1) = (\ell - 1) \sum_{i=1}^k x_i \geq 0.$$

Let $S = \sum_{i=1}^k x_i$. Substituting these into our Cauchy-Schwarz bounds:

$$\sum_{m,m'} W_{m,m'}^2 \geq \frac{1}{n} S^2 + \frac{1}{n(n-1)} (\ell-1)^2 S^2 = \left(\frac{n-1 + (\ell-1)^2}{n(n-1)} \right) S^2 = \frac{n-2\ell+\ell^2}{n(n-1)} S^2.$$

This establishes that $c = \frac{n-2\ell+\ell^2}{n(n-1)}$ is a valid lower bound.

To show this bound is achievable, consider the collection of all $k = \binom{n}{\ell}$ distinct subsets of size ℓ , with $x_i = 1$ for all i . By symmetry, all n diagonal entries of W are identical, and all $n(n-1)$ off-diagonal entries are identical. Thus, the equality condition holds for both applications of Cauchy-Schwarz, confirming the maximum constant is exactly $\frac{n-2\ell+\ell^2}{n(n-1)}$.

§54 USAMO 2025

1. (USAMO 2025) Let k and d be positive integers. Prove that there exists a positive integer N such that for every odd integer $n > N$, the digits in the base-2n representation of n^k are all greater than d .

Solution 1

We define a remainder operation $a \bmod b$ to be the remainder when a is divided by b . Also, let $\lfloor x \rfloor$ be the usual floor function.

Base- $(2n)$ Representation:

$$n^k = a_{k-1}(2n)^{k-1} + a_{k-2}(2n)^{k-2} + \dots + a_1(2n) + a_0,$$

where each a_i satisfies $0 \leq a_i < 2n$. Hence, the base- $(2n)$ representation of n^k is $a_{k-1} a_{k-2} \dots a_1 a_0$.

Leading Digit:

$$a_{k-1} = \left\lfloor \frac{n^k}{(2n)^{k-1}} \right\rfloor = \left\lfloor \frac{n}{2^{k-1}} \right\rfloor.$$

General Digit Formula: For $0 \leq i < k$,

$$a_i = \left\lfloor \frac{n^k \bmod (2n)^{i+1}}{(2n)^i} \right\rfloor.$$

Because n is odd, one can show

$$n^k \bmod (2n)^{i+1} \geq n^{i+1},$$

which implies

$$a_i \geq \left\lfloor \frac{n^{i+1}}{(2n)^i} \right\rfloor = \left\lfloor \frac{n}{2^i} \right\rfloor \geq 2^{k-1-i} \left\lfloor \frac{n}{2^{k-1}} \right\rfloor.$$

As n grows large, $\lfloor n/2^{k-1} \rfloor$ becomes arbitrarily big, so each digit a_i eventually exceeds any fixed d . Hence there exists an integer N such that for all odd $n > N$, the digits a_i in the base- $(2n)$ representation of n^k are all $> d$.

2. (USAMO 2025) Let n and k be positive integers with $k < n$. Let $P(x)$ be a polynomial of degree n with real coefficients, nonzero constant term, and no repeated roots. Suppose that for any real numbers $a_0, a_1, \dots, a_k$ such that the polynomial $a_k x^k + \dots + a_1 x + a_0$ divides $P(x)$, the product $a_0 a_1 \dots a_k$ is zero. Prove that $P(x)$ has a nonreal root.

Solution 1

We proceed by contradiction. Assume that all roots of $P(x)$ are real. Let the distinct roots be $r_1, r_2, \dots, r_n$, all nonzero since the constant term is nonzero.

Consider any subset of k roots $\{r_{i_1}, r_{i_2}, \dots, r_{i_k}\}$ and form the polynomial:

$$Q(x) = \prod_{j=1}^k (x - r_{i_j}) = a_k x^k + a_{k-1} x^{k-1} + \dots + a_0$$

By Vieta's formulas:

$$a_0 = (-1)^k \prod_{j=1}^k r_{i_j} \neq 0, a_{k-1} = -\sum_{j=1}^k r_{i_j} a_{k-2} = \sum_{1 \leq m < n \leq k} r_{i_m} r_{i_n}$$

The given condition requires that $a_0 a_1 \dots a_k = 0$. Since $a_0 \neq 0$, at least one other coefficient must be zero.

Case $k = 2$: For any pair of roots (r_i, r_j) , we have:

$$Q(x) = x^2 - (r_i + r_j)x + r_i r_j$$

The condition implies $-r_i r_j (r_i + r_j) = 0$, so $r_i + r_j = 0$ for all pairs. But with $n \geq 3$, considering three roots r_1, r_2, r_3 gives:

$$r_1 + r_2 = 0 \quad \text{and} \quad r_1 + r_3 = 0 \implies r_2 = r_3$$

contradicting distinct roots. In General: For any k roots, some symmetric sum must be zero. For $k = 3$, this would require:

$$r_i + r_j + r_m = 0 \quad \text{for all triples}$$

which leads to contradictions for any $n \geq 4$ as it would force roots to be equal.

Thus, our initial assumption is false, and $P(x)$ must have at least one nonreal root. Jonathan

Solution 2 (Greater detail and more straightforward)

For the sake of contradiction, assume all roots of $P(x)$ are real. Then, there exists a polynomial $Q(x)$ whose roots are subsets of the roots of $P(x)$, such that $Q(x)$ divides $P(x)$ and one of the coefficients of $Q(x)$ is 0.

Indeed, if distinct $r_1, r_2, \dots, r_n$ are the roots of $P(x)$, some subset of these roots satisfy

$$(x - r_{i_1})(x - r_{i_2}) \dots (x - r_{i_k}) = a_k x^k + a_{k-1} x^{k-1} + \dots + a_0$$

.

Here, $a_k = 1$ by method of undetermined coefficients. In addition, $a_0 \neq 0$ by the nonzero constant given in the statement. That implies some other term is 0.

Let s_k denote the k -th symmetric summation. It follows that

$$a_{k-1} = -s_1$$

$$a_{k-2} = s_2$$

$$\dots$$

$$a_0 = (-1)^k s_k \neq 0 \implies s_k \neq 0$$

.

However, that last claim implies one of $s_1, s_2, s_3, \dots, s_{k-1}$ is 0, which then implies, because $s_k = r_{i_1} r_{i_2} \dots r_{i_k} \neq 0$, none of the roots equal 0. However, notice that then if any one of $s_p : p \leq k - 1$ must equal 0.

If k is even, we can rewrite some of the symmetric sums by groups. For one of these to equal 0, it follows that some $\binom{n}{i}$ of the groups formed, with i as the i th symmetric sum, and n is the number of roots, has the property exactly $\frac{\binom{n}{i}}{2}$ of the sums of elements are direct opposites of the others. However, one quickly realizes that if all terms are relatively prime, it is forced at least one of the roots given is 0, which is a contradiction, or if they are not relatively prime, one can divide by the greatest common factor and obtain some root sequence that is indeed relatively prime, which either forces at least one of the roots to be 0 or two roots to equal one another, both of which are contradictions.

If k is odd however, this implies one of the roots must equalate to 0 by the Zero Product Property, a contradiction.

Because no such k exists, no such subset of k roots can exist, and therefore no such polynomial $Q(x)$ can exist, which implies no such $P(x)$ can exist, and therefore it follows $P(x)$ has at least one nonreal root, and so the proof is complete. $\square$

3. (*USAMO 2025*) Alice the architect and Bob the builder play a game. First, Alice chooses two points P and Q in the plane and a subset $\mathcal{S}$ of the plane, which are announced to Bob. Next, Bob marks infinitely many points in the plane, designating each a city. He may not place two cities within distance at most one unit of each other, and no three cities he places may be collinear. Finally, roads are constructed between the cities as follows: for each pair A, B of cities, they are connected with a road along the line segment AB if and only if the following condition holds: For every city C distinct from A and B , there exists $R \in \mathcal{S}$ such that $\triangle PQR$ is directly similar to either $\triangle ABC$ or $\triangle BAC$. Alice wins the game if (i) the resulting roads allow for travel between any pair of cities via a finite sequence of roads and (ii) no two roads cross. Otherwise, Bob wins. Determine, with proof, which player has a winning strategy.

Note: $\triangle UVW$ is directly similar to $\triangle XYZ$ if there exists a sequence of rotations, translations, and dilations sending U to X , V to Y , and W to Z .

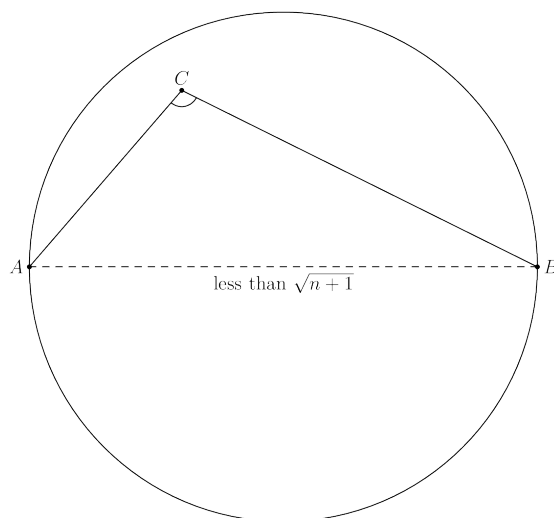
Solution 1

Alice has a winning strategy: Choose $\mathcal{S}$ to be the set of points strictly outside the disk with diameter PQ . Then two cities A and B have a road between them if and only if every other city is outside the closed disk with diameter AB . Note that this condition is equivalent to the angle $\angle ACB$ being acute for every other city C . We need to prove that no matter how Bob chooses cities,

- a) No roads cross, and
- b) All cities are connected.

We first prove (1). Suppose for the sake of contradiction that there are roads between A and C and between B and D , and that these roads cross. Then $ABCD$ is a convex quadrilateral, because its diagonals intersect. Now, recall our condition for the existence of roads between A and C and between B and D : We must have that the angles $\angle ABC$, $\angle BCD$, $\angle CDA$, and $\angle DAB$ are all acute. Because $ABCD$ is a convex quadrilateral, these angles are the interior angles. Then the sum of the interior angles of $ABCD$ is less than 360° , a contradiction with the fact that the sum of the interior angles of a quadrilateral is always 360° . $\rightarrow\leftarrow$

We next prove (2). We use induction on the Euclidean distance between cities. Our base case is that two cities are less than 1 away from each other. If this is the case, then they are the same city, and so the empty path connects them. Now, suppose for the sake of induction that any two cities less than $\sqrt{n}$ away from each other have a finite path between them. We will show the same is true for $\sqrt{n+1}$. Consider any two distinct cities A and B less than $\sqrt{n+1}$ away from each other. If there is a road between them, we are done. Otherwise, there must be some third city C for which $\angle ACB \geq 90^\circ$.



By the Law of Cosines, we get that

$$\overline{AB}^2 \geq \overline{AC}^2 + \overline{CB}^2.$$

Because no two cities are closer than 1 away from each other, both $\overline{AC}$ and $\overline{CB}$ are at least 1. So,

$$\overline{AC}^2 \leq \overline{AB}^2 - \overline{CB}^2 < (n+1) - 1 = n.$$

Hence, by the inductive hypothesis, there is a finite path from A to C . Likewise, there is a finite path from C to B . Joining these paths together, we get a finite path from A to B , as desired.

Hence, any two cities, no matter their distance apart, have a finite path connecting them. $\square$

Note: The graph of roads constructed by Alice's choice of S is called the **Gabriel graph**.

Aside It may be tempting to choose S so that the graph of roads is the **relative neighborhood graph**. This can be done by choosing S to be all points outside of the intersection of the disks centered at A and B with radii $\overline{AB}$. However, this only works for finite sets of cities. When the set of cities is infinite, it is possible that the graph won't be connected. For example, consider the infinite set of cities at $A_n = (2n, 0)$ and $B_n = (2n+1, \sqrt{3})$ for positive integers n . Then perturb these cities slightly so that

$$\overline{B_1A_1} > \overline{A_1A_2} > \overline{A_2B_1} > \overline{B_1B_2} > \overline{B_2A_2} > \overline{A_2A_3} > \overline{A_3B_2} > \overline{B_2B_3} > \overline{B_3A_3} > \dots$$

In this case, there are never any connections between any of the $\{A_n\}$ and the $\{B_n\}$, so the relative neighborhood graph is disconnected.

4. (USAMO 2025) Let H be the orthocenter of acute triangle ABC , let F be the foot of the altitude from C to AB , and let P be the reflection of H across BC . Suppose that the circumcircle of triangle AFP intersects line BC at two distinct points X and Y . Prove that C is the midpoint of XY .

Solution 1

Let AP intersect BC at D . Extend FC to the point E on the circumcircle ω of FAP . Since H is the orthocenter of $\triangle ABC$, we know that $HD = DP$ or $HP = 2HD$, and $AH \cdot HD = CH \cdot HF$. Next we use the power of H in ω :

5. (USAMO 2025) Determine, with proof, all positive integers k such that

$$\frac{1}{n+1} \sum_{i=0}^n \binom{n}{i}^k$$

is an integer for every positive integer n .

Solution 1

https://artofproblemsolving.com/wiki/index.php/File:2025_USAMO_PROBLEM_5_1.jpg

https://artofproblemsolving.com/wiki/index.php/File:2025_USAMO_PROBLEM_5_2.jpg

Solution 2 (combinatorial via Burnside's lemma)

Let

$$X = \{(A_1, \dots, A_k) : A_j \subseteq \{1, \dots, n\}, |A_1| = \dots = |A_k|\}.$$

Then

$$|X| = \sum_{i=0}^n \binom{n}{i}^k.$$

The cyclic group C_{n+1} acts on $\{1, \dots, n+1\}$ by “add 1 mod $n+1$,” and hence diagonally on X .

By Burnside's lemma,

$$|X/C_{n+1}| = \frac{1}{n+1} \sum_{g \in C_{n+1}} |\text{Fix}(g)|.$$

If g has order $d \mid (n+1)$ then $\text{Fix}(g)$ consists of choosing each A_j as a union of the $(n+1)/d$ orbits of g , so

$$|\text{Fix}(g)| = \sum_{i=0}^n \binom{d}{i}^k.$$

Thus

$$(n+1) |X/C_{n+1}| = \sum_{d \mid (n+1)} \varphi\left(\frac{n+1}{d}\right) \sum_{i=0}^n \binom{d}{i}^k.$$

Case 1: $k = 2m$ even

We prove by induction on the divisor-lattice of $n+1$ that for every proper divisor $d < n+1$,

$$d \mid \sum_{i=0}^n \binom{d}{i}^{2m}.$$

Then each term $\varphi((n+1)/d) \sum_i \binom{d}{i}^{2m}$ is divisible by $\varphi((n+1)/d) d$, and since

$$\sum_{d \mid (n+1), d < n+1} \varphi\left(\frac{n+1}{d}\right) d = (n+1) - \varphi(1) \cdot 1 = n+1-1,$$

the entire sum on the right is congruent modulo $n+1$ to

$$\varphi(1) \sum_{i=0}^n \binom{n+1}{i}^{2m} = 1 \cdot \sum_{i=0}^n \binom{n+1}{i}^{2m}.$$

But the left side $(n+1)|X/C_{n+1}| \equiv 0 \pmod{n+1}$, so

$$n+1 \mid \sum_{i=0}^n \binom{n+1}{i}^{2m}.$$

A re-index $n+1 \mapsto n$ yields

$$n+1 \mid \sum_{i=0}^n \binom{n}{i}^{2m},$$

hence

$$A_{2m}(n) = \frac{1}{n+1} \sum_{i=0}^n \binom{n}{i}^{2m} \in \mathbb{Z}.$$

Case 2: k odd

Take $n = 2$. Then

$$A_k(2) = \frac{\binom{2}{1}^k + \binom{2}{2}^k + \binom{2}{3}^k}{3} = \frac{1 + 2^k + 1}{3} = \frac{2 + 2^k}{3}.$$

If k is odd then $2^k \equiv 2 \pmod{3}$, so $2 + 2^k \equiv 4 \equiv 1 \pmod{3}$, whence

$$A_k(2) \notin \mathbb{Z}.$$

Conclusion

$A_k(n) \in \mathbb{Z}$ for all $n \geq 1 \iff k$ even.

6. (USAMO 2025) Let m and n be positive integers with $m \geq n$. There are m cupcakes of different flavors arranged around a circle and n people who like cupcakes. Each person assigns a nonnegative real number score to each cupcake, depending on how much they like the cupcake. Suppose that for each person P , it is possible to partition the circle of m cupcakes into n groups of consecutive cupcakes so that the sum of P 's scores of the cupcakes in each group is at least 1. Prove that it is possible to distribute the m cupcakes to the n people so that each person P receives cupcakes of total score at least 1 with respect to P .

Solution 1

Arbitrarily pick any one person — call her Pip — and her n arcs. The initial idea is to try to apply Hall's marriage lemma to match the n people with Pip's arcs (such that each such person is happy with their matched arc). To that end, construct the obvious bipartite graph $\mathcal{G}$ between the people and the arcs for Pip. We now consider the following algorithm, which takes several steps.

- If a perfect matching of $\mathcal{G}$ exists, we're done!
- We're probably not that lucky. Per Hall's condition, this means there is a *bad* set $\mathcal{B}_1$ of people, who are compatible with fewer than $|\mathcal{B}_1|$ of the arcs. Then delete $\mathcal{B}_1$ and the neighbors of $\mathcal{B}_1$, then try to find a matching on the remaining graph.
- If a matching exists now, terminate the algorithm. Otherwise, that means there's another *bad* set $\mathcal{B}_2$ for the remaining graph. We again delete $\mathcal{B}_2$ and the fewer than $|\mathcal{B}_2|$ neighbors.
- Repeat until some perfect matching $\mathcal{M}$ is possible in the remaining graph, i.e. there are no more bad sets (and then terminate once that occurs).

Since Pip is a universal vertex, it

§55 USAMO 2026

1. (USAMO 2026) Let n be an integer greater than 1. For which real numbers x is

$$\lfloor nx \rfloor - \sum_{k=1}^n \frac{\lfloor kx \rfloor}{k}$$

maximal, and what is the maximal value that this expression can take? Note: $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z .

Solution 1

Let the given function be defined as:

$$f(x) = \lfloor nx \rfloor - \sum_{k=1}^n \frac{\lfloor kx \rfloor}{k}$$

First, we show that $f(x)$ is periodic with period 1. Let $x = y + 1$. Then:

$$\begin{aligned} f(x+1) &= \lfloor n(x+1) \rfloor - \sum_{k=1}^n \frac{\lfloor k(x+1) \rfloor}{k} \\ &= \lfloor nx + n \rfloor - \sum_{k=1}^n \frac{\lfloor kx + k \rfloor}{k} \\ &= \lfloor nx \rfloor + n - \sum_{k=1}^n \left(\frac{\lfloor kx \rfloor}{k} + 1 \right) \\ &= \lfloor nx \rfloor + n - \sum_{k=1}^n \frac{\lfloor kx \rfloor}{k} - n \\ &= f(x) \end{aligned}$$

Because of this periodicity, we can restrict our analysis to $x \in [0, 1)$ and then generalize the result to all real numbers.

We partition the interval $[0, 1)$ into n subintervals of the form $I_m = \left[\frac{m}{n}, \frac{m+1}{n} \right)$ for $m = 0, 1, \dots, n-1$.

Let $x \in I_m$. This implies $m \leq nx < m+1$, so $\lfloor nx \rfloor = m$. Additionally, since $x \geq \frac{m}{n}$ and $k > 0$, we have $kx \geq \frac{km}{n}$, which means $\lfloor kx \rfloor \geq \left\lfloor \frac{km}{n} \right\rfloor$ for all $k \in \{1, \dots, n\}$.

Since $\lfloor nx \rfloor$ remains constant at m throughout I_m and the terms $\lfloor kx \rfloor$ are being subtracted, the maximum of $f(x)$ on I_m is bounded above by its value at the lower endpoint:

$$f(x) = m - \sum_{k=1}^n \frac{\lfloor kx \rfloor}{k} \leq m - \sum_{k=1}^n \frac{\left\lfloor \frac{km}{n} \right\rfloor}{k} = f\left(\frac{m}{n}\right)$$

This proves that the global maximum of $f(x)$ on $[0, 1)$ must be attained at one of the points $x = \frac{m}{n}$.

We now evaluate $f\left(\frac{m}{n}\right)$. Using the identity $\lfloor z \rfloor = z - \{z\}$ (where $\{z\}$ is the fractional part of z), we get:

$$f\left(\frac{m}{n}\right) = m - \sum_{k=1}^n \frac{\frac{km}{n} - \left\{ \frac{km}{n} \right\}}{k} = m - \sum_{k=1}^n \frac{m}{n} + \sum_{k=1}^n \frac{\left\{ \frac{km}{n} \right\}}{k}$$

Since $\sum_{k=1}^n \frac{m}{n} = m$ and $\{\frac{nm}{n}\} = 0$, this simplifies beautifully to:

$$f\left(\frac{m}{n}\right) = \sum_{k=1}^{n-1} \frac{\left\{\frac{km}{n}\right\}}{k} = \frac{1}{n} \sum_{k=1}^{n-1} \frac{km \bmod n}{k}$$

Let $c_k = km \bmod n$. We want to find the integer $m \in \{0, \dots, n-1\}$ that maximizes the sum $\sum_{k=1}^{n-1} \frac{c_k}{k}$.

We claim that the sum is strictly maximized when $m = n-1$. For $m = n-1$, the sequence becomes $c_k = k(n-1) \bmod n = -k \bmod n = n-k$.

To prove that any other m yields a smaller sum, we compare the sequence c_k with the sequence $b_k = n-k$. Let C be the multiset $\{c_1, \dots, c_{n-1}\}$ and B be the set $\{1, 2, \dots, n-1\}$.

Let $d = \gcd(m, n)$. The multiset C consists of the multiples of d from 0 to $n-d$. Specifically, 0 appears $d-1$ times, and each multiple vd (for $v \geq 1$) appears d times.

For any integer $y \in \{1, \dots, n-1\}$, let $N_C(y)$ be the number of elements in C that are $\geq y$. The elements $vd \geq y \implies v \geq \left\lceil \frac{y}{d} \right\rceil$. The number of such valid multipliers v up to $\frac{n}{d} - 1$ is exactly $\frac{n}{d} - \left\lceil \frac{y}{d} \right\rceil$. Since each non-zero multiple appears d times, we have:

$$N_C(y) = d \left(\frac{n}{d} - \left\lceil \frac{y}{d} \right\rceil \right) = n - d \left\lceil \frac{y}{d} \right\rceil$$

Because $d \left\lceil \frac{y}{d} \right\rceil \geq d \left(\frac{y}{d} \right) = y$, we obtain $N_C(y) \leq n - y$. The number of elements in B that are $\geq y$ is precisely $N_B(y) = n - y$. Thus, $N_C(y) \leq N_B(y)$ for all y .

This establishes that the i -th largest element of C is $\leq$ the i -th largest element of B (which is $n-i$). Consequently, for any $j \in \{1, \dots, n-1\}$, the sum of the first j elements of the sequence c_k is bounded by the sum of the j largest elements of C , giving:

$$\sum_{k=1}^j c_k \leq \sum_{k=1}^j (n-k)$$

Let $A_j = \sum_{k=1}^j (n-k-c_k)$. We have proven that $A_j \geq 0$ for all j . Using summation by parts on our sum:

$$\sum_{k=1}^{n-1} \frac{n-k-c_k}{k} = \sum_{j=1}^{n-2} A_j \left(\frac{1}{j} - \frac{1}{j+1} \right) + A_{n-1} \frac{1}{n-1}$$

Since $\frac{1}{j} - \frac{1}{j+1} > 0$ and $A_j \geq 0$, the entire sum is ≥ 0 . This implies $\sum_{k=1}^{n-1} \frac{c_k}{k} \leq \sum_{k=1}^{n-1} \frac{n-k}{k}$. Equality holds if and only if $A_j = 0$ for all j , meaning $c_k = n-k$ for all k , which occurs only when $c_1 = m = n-1$.

Since $m = n-1$ is the unique maximizer among our interval lower bounds, the maximum value of $f(x)$ evaluates to:

$$f\left(\frac{n-1}{n}\right) = \frac{1}{n} \sum_{k=1}^{n-1} \frac{n-k}{k} = \sum_{k=1}^{n-1} \frac{1}{k} - \frac{n-1}{n} = \sum_{k=1}^n \frac{1}{k} - 1$$

To find the continuous values of x where this maximum is achieved, we inspect the interval $I_{n-1} = \left[\frac{n-1}{n}, 1\right)$. For any x in this interval, and for $1 \leq k \leq n$:

$$k - \frac{k}{n} = k \left(\frac{n-1}{n} \right) \leq kx < k$$

Because $1 \leq k \leq n$, the term $\frac{k}{n} \leq 1$, meaning $k-1 \leq k - \frac{k}{n}$. Therefore, $kx \in [k-1, k)$, which locks $\lfloor kx \rfloor$ firmly at $k-1$. Computing $f(x)$ anywhere in this interval yields the constant maximum:

$$f(x) = (n-1) - \sum_{k=1}^n \frac{k-1}{k} = (n-1) - \sum_{k=1}^n \left(1 - \frac{1}{k}\right) = \sum_{k=1}^n \frac{1}{k} - 1$$

Conclusion:

Accounting for the periodicity of 1 established in Step 1, the maximal value is sustained across all intervals shifted by integers.

- Maximal value:

$$\sum_{k=1}^n \frac{1}{k} - 1$$

- Real numbers x that achieve this:

$$x \in \bigcup_{p \in \mathbb{Z}} \left[p - \frac{1}{n}, p \right)$$

Solution 2

After some manipulation (see Solution 1), let $x \in \left[\frac{m}{n}, \frac{m+1}{n}\right)$, where $m \in \{0, 1, 2, \dots, n-1\}$. We proceed by maximizing

$$f(x) \leq f\left(\frac{m}{n}\right) = \frac{1}{n} \sum_{k=1}^n \frac{km \bmod n}{k}$$

The maximal is clearly not achievable when $m = 0$. Let $a_k = km \bmod n$.

Case 1: $\gcd(m, n) = 1$.

We note that:

- 1) For $i, j \in \{0, 1, \dots, n-1\}, i \neq j$, we have $a_i \neq a_j$. To prove, if $a_i = a_j$, then $a_i - a_j \equiv m(i - j) \equiv 0 \pmod{n}$, which yields $i = j$ as a contradiction.
- 2) The sequence $b_k = \frac{1}{k}$ is strictly decreasing for $k = 1, 2, \dots, n$ as the summation occurs. By Rearrangement Inequality, the summation obtains its maximal when $a_1 > a_2 > \dots > a_n$.

We have $a_k \in \{0, 1, 2, \dots, n-1\}$. The sole way for those numbers to be strictly decreasing is

$$n-1 > n-2 > \dots > 0$$

In the case of maximal, this forces $a_k = n - k$.

Thus,

$$a_k \equiv km \equiv -k \pmod{n} \iff$$

$$m \equiv -1 \pmod{n}$$

We have $m = n - 1$. This gives the maximal of f :

$$\frac{1}{n} \sum_{k=1}^n \frac{km \bmod n}{k} \leq \frac{1}{n} \sum_{k=1}^n \frac{n-k}{k} = \sum_{k=2}^n \frac{1}{k}$$

Case 2: $\gcd(m, n) = d > 1$

Let $n = d \cdot q, m = d \cdot b$, where $\gcd(q, b) = 1$. We get $a_k = km \bmod n = d(bk \bmod q)$. This gives a multiset of values of a_k :

$$a_k \in \left\{ \underbrace{n-d, n-d, \dots, n-d}_{d's \ n-d}, \underbrace{n-2d, n-2d, \dots, n-2d}_{d's \ n-2d}, \dots, \underbrace{0, 0, \dots, 0}_{d's \ 0} \right\}$$

By Rearrangement Inequality, let $a_1^* \geq a_2^* \geq \dots \geq a_n^*$ be a rearrangement of the sequence a that yields the maximal. We have

$$\frac{1}{n} \sum_{k=1}^n \frac{a_k}{k} \leq \frac{1}{n} \sum_{k=1}^n \frac{a_k^*}{k}$$

Generally, let $1 \leq t \leq q, 1 \leq s \leq d$. For indices $k = (t-1)d + s$, we have $a_k^* = n - td$. Note that

$$a_k^* = n - td \leq n - td + d - s = n - (t-1)d - s = n - k$$

But the equality holds if and only if $s = d$. We see that $d \nmid k \implies a_k^* < n - k$, which means the upper bound from **Case 1** is **not attainable** for **Case 2**:

$$\frac{1}{n} \sum_{k=1}^n \frac{a_k^*}{k} < \frac{1}{n} \sum_{k=1}^n \frac{n-k}{k} = \sum_{k=2}^n \frac{1}{k}$$

Therefore, by our analysis from **Case 1**, the maximal is achieved when $m = n - 1$:

$$\frac{1}{n} \sum_{k=1}^n \frac{km \bmod n}{k} \leq \sum_{k=2}^n \frac{1}{k}$$

Furthermore, for $x = 1 - \frac{1}{n} + c$, where $0 \leq c < \frac{1}{n}$, since $-1 \leq k \left(c - \frac{1}{n}\right) < 0$, we have

$$\begin{aligned} f(x) &= \lfloor nx \rfloor - \sum_{k=1}^n \frac{\lfloor kx \rfloor}{k} \\ &= \lfloor n - 1 + cn \rfloor - \sum_{k=1}^n \frac{\lfloor k + k \left(c - \frac{1}{n}\right) \rfloor}{k} \\ &= n - 1 - \sum_{k=1}^n \frac{k-1}{k} \\ &= \sum_{k=2}^n \frac{1}{k} \end{aligned}$$

We see that all real $x \in \left[1 - \frac{1}{n}, 1\right)$ yields the maximal.

$f(x)$ has a period of 1. Thus, the real x that attains the maximal satisfies

$$x \in \bigcup_{j \in \mathbb{Z}} \left[j - \frac{1}{n}, j \right)$$

2. (*USAMO 2026*) Annie is playing a game where she starts with a row of positive integers, written on a blackboard, each of which is a power of 2. On each turn, she can erase two adjacent numbers and replace them with a power of 2 that is greater than either of the erased numbers. This shortens the row of numbers, and she continues to take turns until only one number remains. Annie wins the game if the final remaining number is less than 4 times the sum of the original numbers. Is it always possible for Annie to win, regardless of the starting row of numbers?

Solution 1

We start by realizing that the algorithm of combining the smallest term with its smallest neighbor seems to work, and we attempt to prove that this algorithm yields a sum which is $\leq 4 \cdot$ (the original sum).

We proceed with strong induction with the base cases of $n = 1$ and $n = 2$ (if $n = 1$ our result is trivial and if $n = 2$ if the two numbers are x_1 and x_2 , the combined term will be $2 \max(x_1, x_2) \leq 2(x_1 + x_2)$).

Denote the original n integers as $x_1, x_2, \dots, x_n$, and let

$$S = 2x_1 + 4x_2 + 4x_3 + 4x_4 + \dots + 4x_{n-2} + 4x_{n-1} + 2x_n.$$

We attempt to prove that our result is always $\leq k$.

Let k be the first index such that

$$2x_1 + 4x_2 + \dots + 4x_{k-1} + 4x_k = S_k \geq \frac{S}{2},$$

and assume that $k \neq 1$ or n .

WLOG, we can assume that $S_k - 2x_k \leq \frac{S}{2}$, as if not we can just mirror the sequence as both $S_k - 2x_k$ and $S - S_{k-1} - 2x_k$ cannot be greater than $\frac{S}{2}$, which contradicts our initial definition for S .

By our inductive hypothesis, as

$$S_k - 2x_k = 2x_1 + 4x_2 + \dots + 4x_{k-1} + 2x_k,$$

we know we can combine the integers x_1 to x_k to form an integer $\leq \frac{S}{2}$, and similarly, as we know that the terms from x_{k+1} to x_n can be expressed an integer

$$\leq 2x_{k+1} + 4x_{k+2} + \dots + 4x_{n-1} + 2x_n \leq S - S_k \leq \frac{S}{2},$$

implying that the combination of both terms $\leq S$, as desired.

In the case that $k = n$ (or if $k = 1$ we just mirror the entire sequence), we consider the sum of the first $n - 1$ terms, and as we know $S_{n-1} - 2x_{n-1} \leq \frac{S}{2}$, we can combine the first $n - 1$ terms to a value which is $\leq \frac{S}{2}$.

As we know $x_n \leq \frac{S}{2}$ (as $S \geq 2x_n$) we know that we can combine both terms to form a value $\leq S$, as desired.

Solution 2

Let $X = (x_1, x_2, \dots, x_n)$ where $\forall i, x_i \in \{2^k \mid k \in \mathbb{Z}\}$, and let $S = \sum_{i=1}^n x_i$.

Define the adjacent merge operation $\omega(a, b) = 2 \max(a, b)$.

Sequential adjacent applications of ω on $X \iff$ Construction of a full alphabetic (left-to-right) binary tree T .

- $\forall$ leaf $i \in T$: value = x_i , depth = d_i .
- $\forall$ internal node with children u, v : value = $\omega(u, v) = \max(2u, 2v)$.

By induction, the root value R is given by:

$$R = \max_{1 \leq i \leq n} (x_i 2^{d_i})$$

Winning condition $W \iff \exists$ valid tree T such that $R < 4S$.

$\therefore \forall i, x_i \in \{2^k\} \wedge \text{closure of } \omega \implies R \in \{2^k \mid k \in \mathbb{Z}\}$.

Let $M = \max\{2^k \mid 2^k < 4S, k \in \mathbb{Z}\}$.

$$\begin{aligned} \therefore R \in \{2^k\} \implies W &\iff R \leq M \\ &\iff \forall i, x_i 2^{d_i} \leq M \\ &\iff \forall i, 2^{d_i} \leq \frac{M}{x_i} \\ &\iff \forall i, d_i \leq \log_2 M - \log_2 x_i \end{aligned}$$

Let $L_i = \log_2 M - \log_2 x_i$. ($\therefore M, x_i \in \{2^k\} \implies L_i \in \mathbb{Z}$).

$\implies W \iff \exists$ alphabetic binary tree T where $\forall i, d_i \leq L_i$.

Evaluate the Kraft sum constraint for $\{L_i\}$:

$$\sum_{i=1}^n 2^{-L_i} = \sum_{i=1}^n 2^{-\log_2(M/x_i)} = \sum_{i=1}^n \frac{x_i}{M} = \frac{S}{M}$$

By definition of M , we have $M < 4S \leq 2M$.

$$\implies \frac{1}{4} < \frac{S}{M} \leq \frac{1}{2} \implies \sum_{i=1}^n 2^{-L_i} \leq \frac{1}{2}$$

Let $q_i = 2^{-L_i+1} = 2 \cdot 2^{-L_i} \implies \sum_{i=1}^n q_i = 2 \sum_{i=1}^n 2^{-L_i} \leq 1$.

Define $C_1 = 0, C_{i+1} = C_i + q_i$, and intervals $I_i = [C_i, C_{i+1})$.

$\therefore \sum q_i \leq 1 \implies \forall i, I_i \subseteq [0, 1] \wedge (i \neq j \implies I_i \cap I_j = \emptyset)$.

Notice $|I_i| = 2 \cdot 2^{-L_i}$. By the fundamental property of dyadic intervals:

$$\forall I \text{ with } |I| = 2 \cdot 2^{-L}, \exists \text{ sub-interval } J = [k \cdot 2^{-L}, (k+1) \cdot 2^{-L}) \subseteq I, k \in \mathbb{Z}$$

Let $k_i = \lceil C_i \cdot 2^{L_i} \rceil$, and define $J_i = [k_i \cdot 2^{-L_i}, (k_i + 1) \cdot 2^{-L_i})$.

$$\begin{aligned} &\implies J_i \subseteq I_i \\ &\therefore I_i \cap I_j = \emptyset \wedge \text{strictly ordered left-to-right} \implies J_i \cap J_j = \emptyset \wedge \text{strictly ordered} \\ &\therefore J_i = [k_i 2^{-L_i}, (k_i + 1) 2^{-L_i}) \iff \text{Unique binary prefix string of length } L_i \\ &\implies \{J_i\}_{i=1}^n \text{ forms a valid alphabetic prefix code} \\ &\implies \exists \text{ binary tree } T' \text{ with initial nodes } x_i \text{ at strict depth } L_i \end{aligned}$$

Let $T = \text{Prune}(T')$ (recursively eliminate degree-1 internal nodes, redirecting child to parent).

$\implies T$ is a full alphabetic binary tree.

$\therefore$ Pruning strictly decreases or maintains depth without violating left-to-right order.

$\implies \forall i \in T$, final depth $d_i \leq L_i \implies R = \max_{i=1}^n (x_i 2^{d_i}) \leq \max_{i=1}^n (x_i 2^{L_i}) = M < 4S \implies W$ is mathematically guaranteed. $\square$

3. (USAMO 2026) Let ABC be an acute scalene triangle with no angle equal to 60° . Let ω be the circumcircle of ABC . Let $\triangle_B$ be the equilateral triangle with three vertices on ω , one of which is B . Let ℓ_B be the line through the two vertices of $\triangle_B$ other than B . Let $\triangle_C$ and ℓ_C be defined analogously. Let Y be the intersection of AC and ℓ_B , and let Z be the intersection of AB and ℓ_C .

Let N be the midpoint of minor arc BC on ω . Let $\mathcal{R}$ be the triangle formed by ℓ_B , ℓ_C , and the tangent to ω through N . Prove that the circumcircle of AYZ and the incircle of $\mathcal{R}$ are tangent.

Solution

We establish a complex plane with ω being the unit circle, where $|a| = 1, |b| = 1, |c| = 1$ represent points A, B, C , respectively. Let $B_1 = e^{i\frac{2\pi}{3}}b$ and $B_2 = e^{i\frac{4\pi}{3}}b$ be points from $\omega \cap \ell_B$. Also, note that we can assume $n = \sqrt{bc}$ representing N . Let T_N be the tangent of ω through N . Since $B_1 \cdot B_2 = b^2$ and $B_1 + B_2 = -b$, we obtain the following equations:

$$\begin{aligned}\ell_B : z + b^2\bar{z} &= -b \\ \ell_C : z + c^2\bar{z} &= -c \\ AB : z + ab\bar{z} &= a + b \\ AC : z + ac\bar{z} &= a + c \\ T_N : z + bc\bar{z} &= 2\sqrt{bc}\end{aligned}$$

Solving for D, E, F, Y, Z , we have

$$\begin{aligned}d &= \frac{b(c+2n)}{b-c}, \quad e = \frac{c(b+2n)}{c-b}, \quad f = -\frac{bc}{b+c} \\ y &= \frac{b(ab+bc+ca)}{b^2-ac}, \quad z = \frac{c(ab+bc+ca)}{c^2-ab} \\ \bar{y} &= -\frac{a+b+c}{b^2-ac}, \quad \bar{z} = -\frac{a+b+c}{c^2-ab}\end{aligned}$$

We proceed by determining the incenter I of $\mathcal{R}$. Note that

$$|d-f| = \left| \frac{2n(n^2+nc+c^2)}{b^2-c^2} \right| = \left| \frac{2n(n+b+c)}{b^2-c^2} \right|$$

By symmetry, we have $|d-f| = |e-f|$. Also,

$$|d-e| = \left| \frac{2bc+2nc+2nb}{b-c} \right| = \left| \frac{2n(n+b+c)}{b-c} \right|$$

We see that the ratio of side de to the other legs in the same isosceles triangle is $|b+c| = \sqrt{(b+c)(\bar{b}+\bar{c})} = \frac{b+c}{n}$. Thus,

$$I = \frac{d+e+\frac{b+c}{n} \cdot f}{1+1+\frac{b+c}{n}} = \frac{bc}{2n+b+c}$$

Let r be the inradius of the same circle. We have

$$r = \left| \frac{n^2}{2n+b+c} - n \right| = \left| \frac{n+b+c}{2n+b+c} \right|$$

We know that $r = \frac{n+b+c}{2n+b+c}$ because in that case, $r = \bar{r} \in \mathbb{R}$.

Let O be the circumcenter of $\triangle ayz$. We have

$$|O - a| = |O - y| = |O - z| = R$$

Squaring gives $O\bar{O} - a\bar{O} - \bar{a}O + 1 = R^2$. By symmetry, we get

$$(y - a)\bar{O} + (\bar{y} - \bar{a})O = y\bar{y} - 1$$

$$(z - a)\bar{O} + (\bar{z} - \bar{a})O = z\bar{z} - 1$$

Note that

$$b(a+b+c)(ab+bc+ca) + (b^2-ac)^2 = (a^2+ab+b^2)(b^2+bc+c^2)$$

It follows that

$$\begin{aligned} \frac{c(a^2+ab+b^2)}{b^2-ac}\bar{O} - \frac{a^2+ab+b^2}{a(b^2-ac)}O &= -\frac{(a^2+ab+b^2)(b^2+bc+c^2)}{(b^2-ac)^2} \\ \frac{b(a^2+ac+c^2)}{c^2-ab}\bar{O} - \frac{a^2+ac+c^2}{a(c^2-ab)}O &= -\frac{(a^2+ac+c^2)(b^2+bc+c^2)}{(c^2-ab)^2} \end{aligned}$$

Solving yields

$$\bar{O} = -\frac{(a+b+c)(b^2+bc+c^2)}{(b^2-ac)(c^2-ab)}, \quad O = -\frac{a(ab+bc+ca)(b^2+bc+c^2)}{(b^2-ac)(c^2-ab)}$$

Note that

$$(ab+bc+ca)(b^2+bc+c^2) + (b^2-ac)(c^2-ab) = bc(a+b+c)^2$$

The circumradius R evaluates to

$$R = |O - a| = \left| \frac{bc(a+b+c)^2}{(b^2-ac)(c^2-ab)} \right|$$

And

$$R^2 = \frac{bc(a+b+c)^2}{(b^2-ac)(c^2-ab)} \cdot \frac{\frac{1}{bc}(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})^2}{\left(\frac{1}{b^2} - \frac{1}{ac}\right)\left(\frac{1}{c^2} - \frac{1}{ab}\right)} \cdot \frac{a^2b^3c^3}{a^2b^3c^3} = \frac{bc(a+b+c)^2(ab+bc+ca)^2}{(b^2-ac)^2(c^2-ab)^2}$$

Therefore, with

$$\begin{aligned} O &= \frac{-a(b^2+bc+c^2)(ab+bc+ca)}{(b^2-ac)(c^2-ab)}, \quad I = \frac{bc}{2n+b+c}, \\ R &= \frac{n(a+b+c)(ab+bc+ca)}{(b^2-ac)(c^2-ab)}, \quad r = \frac{n+b+c}{2n+b+c}, \end{aligned}$$

it is enough to show that $|O - I| = |R - r| \iff (O - I)(\bar{O} - \bar{I}) = (R - r)^2$.

We make the following substitutions:

$$\begin{aligned} S &= b + c, & U &= a + b + c, & V &= ab + bc + ca, \\ X &= b^2 + bc + c^2, & P &= 2n + b + c, & Q &= (b^2 - ac)(c^2 - ab) \end{aligned}$$

Substituting, we wish to prove that

$$\begin{aligned} \left(-\frac{aXV}{Q} - \frac{n^2}{P}\right) \left(-\frac{UX}{Q} - \frac{1}{P}\right) &= \left(\frac{nUV}{Q} - \frac{n+S}{P}\right)^2 \iff \\ (aXVP + n^2Q)(UXP + Q) &= (nUVP - (n+S)Q)^2 \iff \\ UVP^2(aX^2 - n^2UV) + PQ(X(n^2U + aV) + 2n(n+S)UV) &+ (n^2 - (n+S)^2)Q^2 = 0 \end{aligned} \quad (1)$$

To simplify, we express each variable in terms of a, S, n :

$$\begin{aligned} U &= a + S, & V &= n^2 + aS, & X &= S^2 - n^2 \\ P &= 2n + S, & Q &= b^2c^2 - a(b^3 + c^3) + a^2bc = n^4 - aS^3 + 3an^2S + a^2n^2 \end{aligned}$$

It follows that $n^2 - (n+S)^2 = -2nS - S^2 = -PS$. Also,

$$aX^2 - n^2UV = a(S^4 - 2S^2n^2 + n^4) - n^2(aS^2 + a^2S + n^2S + an^2) = -S(n^4 - aS^3 + 3an^2S + a^2n^2) = -SQ$$

Substituting the two coefficients back to (1) yields

$$\begin{aligned} -SUV P^2 Q + PQ(X(n^2U + aV) + 2n(n+S)UV) - SPQ^2 &= 0 \iff \\ X(n^2U + aV) + (2n^2 + 2nS - PS)UV - SQ &= 0 \iff \\ X(n^2U + aV) + (2n^2 - S^2)UV - SQ &= 0 \end{aligned}$$

Expanding, we have

$$\begin{aligned} X(n^2U + aV) &= 2an^2S^2 + n^2S^3 + a^2S^3 - 2an^4 - n^4S - a^2n^2S \\ (2n^2 - S^2)UV &= 2an^2S^2 + 2n^4S + 2a^2n^2S + 2an^4 - aS^4 - a^2S^3 - n^2S^3 - an^2S^2 \\ -SQ &= -n^4S + aS^4 - 3an^2S^2 - a^2n^2S \end{aligned}$$

Those terms cancel when added, as desired.

4. (*USAMO 2026*) A positive integer n is called solitary if, for any nonnegative integers a and b such that $a + b = n$, either a or b contains the digit 1 when written in base 10. Determine, with proof, the number of solitary integers less than 10^{2026} .

Solution 1

Lemma 1: $n = 2 \cdot 10^k - 1$ is solitary for all $k \in \mathbb{Z}_{\geq 0}$.

Suppose $a + b = n$. WLOG, $a \geq b$. Then, $2a \geq a + b = n \implies a \geq \frac{n}{2}$. Since a, n are integers, and n is odd, this implies $a \geq \frac{n+1}{2} = 10^k$. But, $a \leq n = 2 \cdot 10^k - 1$, so $10^k \leq a \leq 2 \cdot 10^k - 1$, so the leading digit of a is 1. Thus, n is solitary. $\square$

Lemma 2: $m = 2 \cdot 10^k + n, n < 10^k$ is solitary if and only if n is solitary.

First, suppose m is solitary and n is not. We write $n = a + b$ for some a, b with no 1's in their decimal representation. Then, let $a' = a + 2 \cdot 10^k$ and $b' = b$. It follows that

$$a' + b' = a + 2 \cdot 10^k + b = 2 \cdot 10^k + n = m$$

But a' and b' have no 1's in their decimal representation, since $10^k > n \geq a, b$, a contradiction. This proves the only if part.

Now, suppose n is solitary, and m is not. Write $m = a + b$ for some a, b with no 1's in their decimal representation.

Write $a = x \cdot 10^k + a', a' < 10^k$ (the leading digit of a is x). Similarly, $b = y \cdot 10^k + b', b' < 10^k$. By definition, $a, b \neq 1$. But

$$x = y = 0 \implies a + b < 2 \cdot 10^k < n \text{ and } x + y > 2 \implies a + b \geq 3 \cdot 10^k > m$$

Thus, one of x, y is 2, and the other is 0. WLOG $x = 2, y = 0$. Consider a' and b' . We have

$$a' + b' = m - x \cdot 10^k - y \cdot 10^k = m - 2 \cdot 10^k = n$$

But n is solitary, so either a' or b' has a 1 in their decimal representation. But, that implies a or b also has a 1, since $a', b' < 10^k$, a contradiction. This proves the if part. The lemma is complete. $\square$

Lemma 3: For any integer $m \in [10^k, 10^{k+1} - 1]$, if m is not of the form $2 \cdot 10^k - 1$ and not of the form $2 \cdot 10^k + n$ (where $n < 10^k$), then m is not solitary.

By Lemma 2, we can assume the leading digit of m is not 2. Now, take $a = 10^k - 1$ and $b = m - a$.

Let the decimal representations of a, b be $a_{k-1}a_{k-2} \dots a_0$ and $b_kb_{k-1} \dots b_0$, respectively. It follows that $a_i = 9$ for all i .

Also, $m \notin [2 \cdot 10^k - 1, 3 \cdot 10^k - 1] \implies b = m - a \notin [10^k, 2 \cdot 10^k]$, so $b_k \neq 1$.

Now, suppose $b_i = 1, i \in \{0, 1, \dots, k-1\}$. Increase b_i to 2 and lower a_i to 8. This increases b by 10^i while decreasing a by 10^i , so the sum stays the same. Repeating this for all i such that $b_i = 1$, we have a, b such that none of the digits are 1, $a + b = m$, so m is not solitary. $\square$

Now, we use the lemmas to finish. Let s_k be the number of solitary integers $< 10^k$, and let S_k be the set of these integers, so $|S_k| = s_k$. By the lemmas

$$S_{k+1} = \{2 \cdot 10^k - 1\} \cup \{2 \cdot 10^k + n : n \in S_k\} \cup S_k$$

Since no elements are shared, we have $s_{k+1} = 1 + s_k + s_k = 2s_k + 1$. Also, $S_1 = \{1\}$, so $s_1 = 1$. It follows by induction $s_k = 2^k - 1$. Thus, the answer is $\boxed{2^{2026} - 1}$.

Solution 2

Lemma 1: n must contain 1, otherwise it is not solitary.

Proof of Lemma 1:

Definition of solitary, choose $a = 0$ and $b = n$ and the result shows n is not solitary.

Lemma 2:: All such n has the property all digits to the right of "1" are "9".

Proof of Lemma 2:

Consider vertical addition. Let a_n and b_n be the n th place value of such numbers a and b . Then, $a_0 + b_0 = 9$. One can form such numbers using any orientation, either $a_n = 1$ and $b_n = 8$, vice versa, or some non-1 pair. Assume the worst case, where we construct sums of 9 that do not include 1. Then, if the leading digit itself is 1, then notice that one may "truncate" the 19 at the end of the string $1999 \dots 9$. Because 9 is solitary, any single combination of a_0, b_0 result in one value being

greater than 10, hence we must carry a value. Specifically, that implies the length of $a < b$ or vice versa. Hence, if the length $a < b$ is obtained, then notice that then one can let, at the end of the truncation, b start with 9 and a begin with 10, giving the 19 that was truncated at the beginning. This fulfills solitary properties.

Lemma 3: All such n has the property all digits to the left of "1" are either "0" or "2" or a combination of both.

Proof of Lemma 3:

We continue from where we left off in Lemma 2. We realize that b must start with a 0 then move through with 9's. Afterwards, a and b are uniquely determined toward the left side of a 's 1 and b 's 0. Consider forming a sum of 11 instead of 1, and we prove that a number can be formed that is solitary. More importantly, if a_n and b_n do not contain 1 at all yet still sum to 11, then that 1 carries over. Now, we still fulfill the properties of **Lemma 2**, but the number is not solitary! But if we were to carry over that extra 1, then a_{n+1} or b_{n+1} can have either 1 or 9, from which we terminate, or continue analogously. However, when we end, one of a or b has 1, hence the number n is solitary.

2026 USAMO/Q4:

Given the conditions, n is *solitary* if and only if

- n contains 1
- All numbers to the right of the 1 in n are 9
- All numbers to the left of the 1 in n are "2" or "0"

The last part is to count all such n . From any number, the digit 1 can be placed at an index m from the left ($1 \leq m \leq 2026$).

There are exactly $2026 - m$ positions to the right of 1 that are all 9. Otherwise, there are exactly $m - 1$ positions to the left of 1. Those have two choices each, giving 2^{m-1} cases. So, on any given integer with m digits, we may form 2^{m-1} solitary integers.

We sum over all $1 \leq m \leq 2026$, or

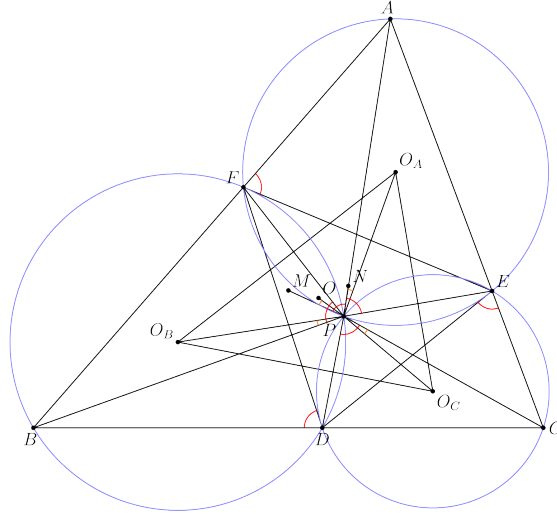
$$\sum_{m=1}^{2026} 2^{m-1} = 1 + 2 + \cdots + 2^{2025} = \boxed{2^{2026} - 1} \quad \square$$

5. (USAMO 2026) Let ABC be a triangle. Points D , E , and F lie on sides BC , CA , and AB , respectively, such that

$$\angle AFE = \angle BDF = \angle CED.$$

Let O_A , O_B , and O_C be the circumcenters of triangles AFE , BDF , and CED , respectively. Let M , N , and O be the circumcenters of triangles ABC , DEF , and $O_A O_B O_C$, respectively. Prove that $OM = ON$.

Solution 1



Let P be the point where the circumcircles of $\triangle AFE$, $\triangle BDF$, $\triangle CED$ intersect, which we know exists because it is the Miquel point of the given configuration.

Claim 1: We have $\triangle ABC \sim \triangle EFD \sim \triangle O_A O_B O_C$.

Proof: For $\triangle ABC \sim \triangle O_A O_B O_C$, we have that $\angle CAB = \angle EAP + \angle PAF = \angle O_C O_A P + \angle P O_A O_B = \angle O_C O_A O_B$ and $\angle BCA = \angle DCP + \angle PCE = \angle O_B O_C P + \angle P O_C O_A = \angle O_B O_C O_A$ from O_A, O_B, O_C being the centers of circumcircles $\triangle AFE, \triangle BDF, \triangle CED$.

For $\triangle ABC \sim \triangle EFD$, we have from $\angle AFE = \angle BDF = \angle CED$ and D, E, F lying on BC, CA, AB respectively that

$$180^\circ = \angle CED + \angle DEF + \angle FEA = \angle AFE + \angle CAB + \angle FEA \implies \angle DEF = \angle CAB$$

and analogously $\angle FDE = \angle BCA$.

Claim 2: P is the center of the spiral similarities s_1, s_2 sending $\triangle ABC$ to $\triangle EFD$ and $\triangle ABC$ to $\triangle O_A O_B O_C$ respectively.

Proof: For s_1 note by cyclic quadrilaterals $AEPF, CDPE, BFPD$ that

$$\angle AFE = \angle BDF = \angle CED = \angle CPD = \angle BPF = \angle EPA,$$

which we call θ , and $\angle AEP = \angle BFP = \angle CDP$. Hence $\triangle AEP \sim \triangle BFP \sim \triangle CDP$ so s_1 rotates $\triangle ABC$ by θ and scales down by the common ratio $\frac{AP}{PE} = \frac{BP}{PF} = \frac{CP}{PD}$ to precisely yield $\triangle EFD$.

For s_2 note that from $\angle PDC = \angle PEA = \angle PFB$ follows $\angle APO_A = \angle BPO_B = \angle CPO_C$ and $\angle PO_A A = \angle PO_B B = \angle PO_C C$, which means $\triangle APO_A \sim \triangle BPO_B \sim \triangle CPO_C$ and similarly suffices to imply that P is the center of s_2 .

To finish, $s_1(M) = N$ and $s_2(M) = O$. Since $s_1(A) = E$ and $s_2(A) = O_A$ while s_1, s_2 share the same center, $\triangle MON \sim \triangle AO_A E$ from which $OM = ON$ because $O_A A = O_A E$.

6. (USAMO 2026) Let a and b be positive integers such that $\varphi(ab+1)$ divides a^2+b^2+1 . Prove that a and b are Fibonacci numbers.

Note: Euler's totient function $\varphi(n)$ is the number of positive integers less than or equal to n that are relatively prime to n .

Solution 1

First, note that since $ab + 1 > 2$, we have $2 \mid \varphi(ab + 1)$. Therefore, $2 \mid a^2 + b^2 + 1$, so one is even and the other is odd.

This implies $a^2 + b^2 + 1 \equiv 2 \pmod{4}$, so $\nu_2(\varphi(ab + 1)) = 1$. To use this, note that $ab + 1$ is odd.

Let $ab + 1 = p_1^{e_1} p_2^{e_2} \dots p_n^{e_n}$, where $p_1, p_2, \dots, p_n$ are odd primes. Then,

$$1 = \nu_2(\varphi(ab + 1)) = \nu_2\left(\prod_{i=1}^n p_i^{e_i-1}(p_i - 1)\right) \geq n$$

Thus, $ab + 1 = p^k$ for some odd prime p and some positive integer k . Furthermore, by the above reasoning, $p \equiv 3 \pmod{4}$.

Case 1: $k \geq 2$

Since $p \nmid a$, we can write

$$ab + 1 \equiv 0 \pmod{p^{k-1}} \implies b \equiv -a^{-1} \pmod{p^{k-1}}$$

Now, $p^{k-1} \mid \varphi(ab + 1) \mid a^2 + b^2 + 1$, so we can substitute like so:

$$a^2 + b^2 + 1 \equiv 0 \pmod{p^{k-1}} \implies a^2 + \frac{1}{a^2} + 1 \equiv 0 \pmod{p^{k-1}}$$

$$\implies 4a^4 + 4a^2 + 4 \equiv 0 \pmod{p^{k-1}} \implies (2a^2 + 1)^2 \equiv -3 \pmod{p^{k-1}} \implies \left(\frac{-3}{p}\right) = 1$$

where $\left(\frac{a}{p}\right)$ is the Legendre symbol. By quadratic reciprocity, using the fact that $p \equiv 3 \pmod{4}$,

$$\left(\frac{-3}{p}\right) = \left(\frac{3}{p}\right) \left(\frac{-1}{p}\right) = -\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right)$$

Thus, $\left(\frac{p}{3}\right) = 1$, so $p \equiv 0 \pmod{3} \implies p = 3$, or $p \equiv 1 \pmod{3}$.

Subcase 1: $p = 3$

Note that $\left(\frac{-3}{9}\right) = -1$, so we must have $k - 1 \leq 1 \implies k = 2$. Thus, $ab + 1 = 9$. It is easy to check that $(a, b) = (1, 8), (8, 1)$ are the only solutions in this case. Since both 1 and 8 are Fibonacci numbers, this subcase is complete.

Subcase 2: $p \equiv 1 \pmod{3}$

Then, $3 \mid p - 1 \mid \varphi(ab + 1) \mid a^2 + b^2 + 1$, so $3 \nmid a, b$. However, $ab + 1 \equiv p^k \equiv 1 \pmod{3} \implies 3 \mid ab$, a contradiction. Thus, this subcase yields no solutions.

Putting all of the above together completes the first case.

Case 2: $k = 1$

Thus, $ab + 1 = p$, so $\varphi(ab + 1) = ab$, and the problem condition is $ab \mid a^2 + b^2 + 1$. Let $\frac{a^2 + b^2 + 1}{ab} = t$, where t is a positive integer. Let (a, b) be the pair with the minimal value of $a + b$. WLOG, $a \geq b$. Then,

$$a^2 - tab + b^2 + 1 = 0$$

We know a is one of the roots of this quadratic. Let x be the other root of this quadratic. Then, $x = tb - a$ and $x = \frac{b^2 + 1}{a}$. The first equation proves that x is an integer, and the second proves that x is positive. Furthermore, if $b^2 + 1 < a^2$,

then $x = \frac{b^2+1}{a} < a$, which gives (x, b) as a valid pair, a contradiction. Thus, we must have $a = b$ for this minimal pair. However, $a^2 \mid 2a^2 + 1 \implies a = 1$, so $(a, b) = (1, 1)$.

This implies that $t = 3$, since otherwise we could follow the above logic to obtain a contradiction by infinite descent.

We claim that all solutions are of the form $(a, b) = (F_i, F_{i-2})$ or (F_{i-2}, F_i) . Proceed by induction on $a + b$. Let S_k be the set of solutions such that $a + b \leq k$. For the base case $k = 2$, $S_k = \{(1, 1)\}$. Now, assume that all solutions in S_{k-1} , $k \geq 3$ are Fibonacci numbers.

Suppose we have a solution such that $a + b = k$. WLOG, $a \geq b$. By the above logic, we have a smaller x such that $a = 3b - x$ and $x = \frac{b^2+1}{a}$. Since $a \geq 2$ and x is an integer, we have $x = \frac{b^2+1}{a} < b$. Furthermore, since $(b, x) \in S_{k-1}$, we have $b = F_i$ and $x = F_{i-2}$. Thus,

$$a = 3b - x = 3F_i - F_{i-2} = 2F_i + F_{i-1} = F_{i+1} + F_i = F_{i+2}$$

Therefore, $(a, b) = (F_{i+2}, F_i)$. This completes the induction.

Putting all this together finishes the second case. Finally, the proof is complete $\square$